MEASURING THE INNOVATION ACTIVITY OF TOURISM BUSINESSES IN BULGARIA

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Abstract: The implementation of innovations in tourism enterprises is a key factor in increasing their competitiveness. A challenge for the tourism businesses in Bulgaria is to develop innovative solutions that will diversify their tourism products, optimize business operations and improve the overall efficiency of the tourism industry.

The **aim** of this research is to systematize and interpret theoretical and empirical statements of innovations in tourism in order to propose and test a model for measuring the innovation activity of tourist enterprises in Bulgaria and, based on this, to formulate conclusions and recommendations. This paper examines the scope of innovation activity of tourism enterprises and measures their innovation activity by means of a proposed model that has been tested in organizations providing tourism-related services. Research methodology includes induction and deduction, method of analysis and synthesis, analogy, content analysis, comparative analysis, survey, etc. As a result of the research, an innovation index is calculated, using the developed Model for measuring innovation activity in tourist enterprises, based on the evaluation of seven groups of qualitative indicators, and conclusions and recommendations are formulated for the promotion of innovation activity in the tourism sector. The tested model shows that it is possible to measure the innovation activity in the tourism sector by covering all enterprises offering the main tourism services.

Key words: innovations, innovation activity, tourist enterprise, tourism.
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Introduction

The tourism industry is a key pillar of the Bulgarian economy, contributing significantly to the economic growth and employment opportunities of the country. However, in order to maintain a competitive edge, the industry must constantly adapt to the changing market requirements and consumer preferences. Innovations play a crucial role in the adaptation process as they can diversify tourism products, optimize business operations, and improve the overall efficiency and competitiveness of the tourism industry.

The aim of this study is, by systematizing and interpreting theoretical and empirical statements of innovations in tourism, to propose and test a model for measuring the innovation activity of tourism enterprises in Bulgaria and, on this basis, to formulate conclusions and recommendations. To achieve the aim, the following research tasks have been set: presenting the main directions of innovation in the tourism businesses; developing and testing a model for measuring the innovation activity in the tourism sector; an empirical study of the innovation activity of Bulgarian tourism enterprises.

The relevance of this research stems from the need for a thorough understanding of innovation processes in the tourism sector, as they can increase the adaptability of tourism enterprises to changing conditions and contribute to increasing the competitiveness of a tourism product.

1. Literature review

The essence of innovation in tourism is expressed in promoting the creation of new tourism products, services, marketing approaches, organizational practices and implementing changes in activities in order to achieve a competitive advantage (Schumpeter, 2017; Ilieva, 2024; (Stanovcic et al., 2015).

The development of an innovative tourism business depends on a number of factors, including the size of the enterprise; the availability of financial resources; human capital; technological capability; managerial culture and external support. (Sin et al., 2020; Stanovcic et al., 2015; Yuzbasioglu et al., 2014; Sushchenko et al., 2024). The development of innovative tourism products and services, the implementation of new technologies, the optimization of business processes and the adaptation to the changing environment enable tourism enterprises to increase their competitiveness and add value to their customers. By implementing such innovations, tourism enterprises diversify their products, improve the efficiency of their operations, and adapt to changing market requirements. (Alonso-Almeida et al., 2016; Sin et al., 2020; Savvopoulos et al., 2019).

The tourism industry has undergone significant transformations over the past decades, but they have been particularly intense in the last four years, driven by rapid advances in technology and the ever-evolving tourists' expectations. Technology is becomings increasingly integrated into the operations of tourism businesses, which has led to a wave of innovative solutions that have the potential to improve the productivity and competitiveness of the industry. (Bhat & Shah, 2014).

The main types of innovations adopted by the tourism businesses for economic viability are the following (Ilieva & Todorova, 2023; Ilieva, 2024; Stoykov, 2023; Ilieva, Petrova & Todorova, 2023; Proenca, 2024):

Technological innovations: These include the implementation of new or significantly improved methods of providing services and delivering tourism products through technology transfer from other industries or innovations specific to the tourism industry. Technological innovations encompass equipment, human resources, working methods or a combination of them all. The innovation process is long-term, with significant investments and hiring not only employees, but often external experts as well.

Product Innovation: Product innovation refers to the introduction of a new product or an improved version of a product that differs from a company's current product offerings. This does not include minor modifications to existing products.

Marketing innovations: This type of innovation includes enterprise initiatives to more effectively commercialize tourism products, build brand loyalty, and manage customer relationships.

Organizational innovations: Organizational innovation is based on business practices for organizing the work process and relationships with other businesses and organizations; as well as the organization of work related to the distribution of responsibilities, decision-making and human resource management.

In addition to innovations aimed at achieving an organization's economic goals, a growing number of tourism organizations are also turning to **socially responsible innovations** in an attempt to respond to the changes in consumer behaviour and expectations. These include social innovations, sustainable innovations, also known as "green innovations," and responsible innovations.

Digital innovations that are becoming widespread as a result of consumer expectations, such as: Internet of Things (IoT) and Internet of Everything (IoE); virtual reality, augmented reality; big data; artificial intelligence; robots; wearables; mobile devices (smartphones and tablets) are also of interest (Popova et al., 2023a; Petrova et al., 2018; Ivchenko et al., 2023; Popova et al., 2023b; Kirilov, 2024; Oryekhov et al., 2024). The implementation of technological innovations is related to the change in the preferences of modern tourists towards the use of digital technologies when choosing a specific tourist destination and tourist site, when planning and organizing their trip (Mileva-Bojanova, 2023). In addition, tourists perceive digital innovations as a means of enriching their experiences and personalizing tourism products and offers.

Another important change in society is related to sustainable development, which requires the implementation of ecological and socially responsible innovations (Miloiu et al., 2023). Ecological innovations are considered to be all those that reduce the use of natural resources, respectively reduce the emission of harmful substances throughout the life cycle of tourism products and services. Socially responsible innovations, on the other hand, focus on building a connection between researchers, businesses, citizens, and educational institutions, with the aim of collaborating in the overall process of creating innovations that best meet the needs of society.

Although the tourism industry is recognized as a sector with significant potential for innovation (Booyens, 2018), enterprises from the sector are not among those surveyed when forming the Innovation Index of the country. This is mainly due to the fact that tourism is part of the service sector and new technologies are integrated and adapted for the purposes of tourism enterprises, i.e. the tourism business innovates by adapting innovative solutions originating from other sectors of the economy. Regardless of this interpretation of innovation activity in the tourism sector, innovation in tourism can take different forms - from the introduction of new tourism products and services to the adoption of new technologies and business models (Aldebert et al., 2011; Ilieva et al., 2023). However, measuring the impact and effectiveness of these innovations is a challenging task, due to the heterogeneous nature of the tourism product, which combines tangible and intangible components.

A possible approach to measuring innovation in tourism is to adapt the Oslo Manual (OECD/Eurostat, 2018) and the European innovation scoreboard, which are widely used in other industries, to the specific context of the tourism sector (Booyens, 2018). This includes collecting data on the types of innovation introduced in a tourism enterprise, the resources and activities involved in the innovation process, and the subsequent impact of the innovation on the final results. This type of enterprise-level innovation research can provide valuable insight into the drivers and barriers to innovation in tourism, as well as the factors that contribute to the success or failure of specific innovation initiatives.

On the other hand, to identify and measure service innovations, an adaptation of a model developed in the report of Working Group II of the EU initiative EPISIS is used. The model consists of three parts (ESIC European Service Innovation Centre, 2014):

 Policy: Regulation, infrastructure, education system and other societal factors, combined with policies, create the framework within which innovation in enterprises takes place;

 Innovation in the service sector: In the model, innovation in the service sector is viewed as three closely related parts: innovation input, actual implementation of the innovation, and market output;

 Result: The structural change that results from the transformative power of service innovation.

Taking a closer look at the definition of service innovation, the model describes the three parts of service innovation as follows (ESIC European Service Innovation Centre, 2014) (Fig. 1):

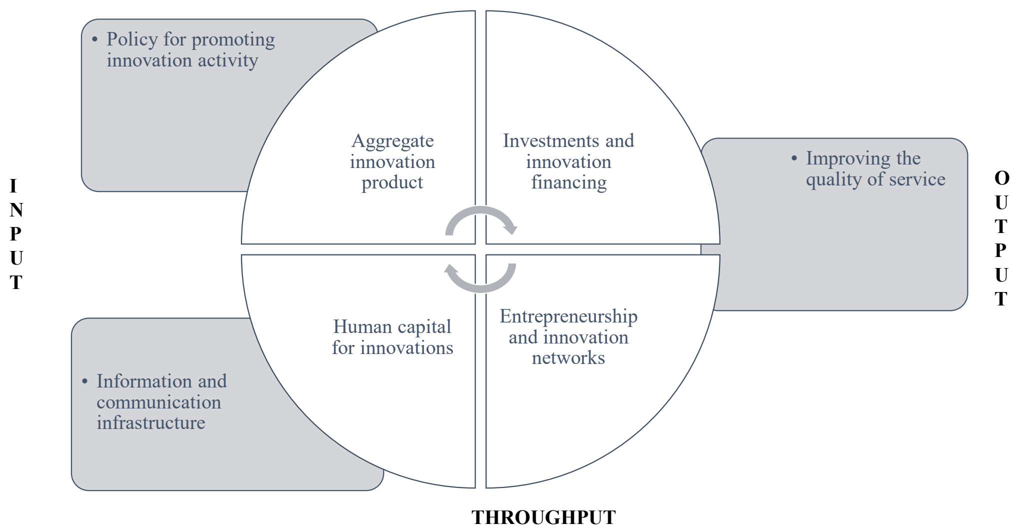

Figure 1. A model for measuring innovation activity in a tourism enterprise

Source: Adapted frmo the Model developed in the report of Working Group II of the EU initiative EPISIS (ESIC European Service Innovation Centre, 2014) and Methodology of the Applied Research and Communications Found (Working Group, 2006)

• An input that represents the targeted development of service innovation. Service innovation is developed with a specific purpose, and the innovation process is goal-directed. The innovation process is therefore purposeful and did not just happen - the latter case can be characterized as evolution rather than innovation;

• Throughput, which represents the aggregate innovation product;

• Output, which is the value creation. It can be the value to both the company and the customer. If no value is created, it will not be considered a service innovation in this analysis.

The proposed model focuses on the processes taking place in companies where knowledge is created and which, together with entrepreneurship, are at the heart of the innovation process. However, the model also shows how other factors influence the ability of enterprises to innovate, most notably how resources are attracted and allocated through human and financial capital and how specific innovations are selected and developed.

The innovation index of an enterprise can be calculated by measuring indicators grouped into seven main groups (Ilieva, 2011) that form its innovation activity (Working Group, 2006):

• **Policy for promoting the innovation activity**. Public policy can influence business activities in direct and indirect ways. The regulatory and enforcement framework influences the way in which tourism businesses can exploit the results of implemented innovations, as well as the numerous relationships and transactions they are engaged in, while the tax system influences the costs of business activities.

• **Determining the aggregate innovation product** (as a result of all types of innovations introduced in an enterprise) is the most important step towards a comprehensive evaluation of the innovation intensity of enterprises, which provides an accurate picture of whether enterprises innovate and how often they do so; whether innovations are focused in one area or are comprehensive; and whether they are gradual or radical. The main areas of innovation are related to technical and technological innovations, product innovations, marketing innovations, organizational innovations and socially responsible innovations.

• The next group of indicators measures **entrepreneurship and innovation networks**, which support the study of the entrepreneurial activity of owners and managers of tourism enterprises. Information is collected on new tourist sites that were put into operation during the year under study; mergers with other tourist enterprises; creation of networks for technology and innovation transfer; signing of contracts with new intermediary structures and acquisition of rights to established innovation practices.

• The fourth group of indicators measures **investment and financing of innovations**. These indicators aim to study the investment policy of the enterprise and the extent to which it is aimed at financing innovation projects. In addition, this group of indicators aims to determine whether the enterprise uses financing from various programs and funds to support innovative initiatives in the tourism sector.

• The fifth set of indicators focuses researchers' attention on **measuring human capital for implementing innovations in the tourism enterprises**. It is assumed that there is a relationship between the age structure of the employees and the innovativeness of the enterprise.

• The sixth set of indicators is aimed at **measuring the access of tourism enterprises to information and communication technologies and their use**. Computer and information systems play a crucial role in facilitating the activities and management of a tourism organization as they provide the opportunity to implement specialized software to organize and monitor various operations, such as reservations, customer relations, and accounting.

• The seventh group of indicators measures the extent to which the **innovations introduced by a tourism enterprise lead to the improvement of service quality and the creation of value for both the enterprise and its customers**. This is a key aspect of service innovation, as innovation must be perceived by target customers as increasing the quality of services provided.

It is appropriate, within a tourism enterprise, after measuring the innovation index, to move on to measuring the economic and financial benefits that the tourism enterprise receives as a result of its innovation efforts. These metrics focus on quantifying the increase in direct revenue, cost reduction, and the market share gain that can be directly attributed to the successful implementation of innovations. By examining these impact metrics, the analysis can provide insight into the overall return on investment in innovation and help tourism businesses see the tangible business results of their innovation initiatives.

To summarize, the proposed **innovation index** for tourism enterprises covers seven key dimensions. By incorporating these different dimensions – from aggregate innovation products to entrepreneurship, networks and human capital – the innovation index aims to provide a multifaceted and nuanced evaluation of the innovation activity of tourism enterprises.

This comprehensive innovation index can serve as a valuable framework for tourism organizations to identify their strengths, weaknesses and areas for improvement in fostering a culture of innovation, allowing them to develop targeting strategies to drive innovation and sustain a competitive advantage in the marketplace.

2. Methodology

The research methodology includes: induction and deduction, method of analysis and synthesis, analogy, content analysis, comparative analysis, survey research, etc. In this study, an innovation index is calculated by using a developed Model for measuring the innovation activity in the tourism enterprises, based on the evaluation of seven groups of qualitative indicators. For the purpose of testing the proposed model, two survey cards have been used. The first questionnaire "Measuring innovation activity and the effect of innovations on tourism" is intended for representatives of tourism businesses, and the second - "Research on consumer preferences for innovations in tourism" is aimed at consumers of tourism services.

The surveys were conducted by using electronic surveys on Microsoft Forms, in the period June - July 2024, and the survey results were processed and analyzed by using Microsoft Excel. The data obtained cover the period 2019-2023.The surveys were conducted by using Microsoft Forms electronic surveys in the period June 6 - July 6, 2024, and the survey results were processed and analyzed by using Microsoft Excel.

In order to measure the scope of the innovation activity in the tourism sector, responses were received from 116 representatives of the tourism sector, of which: hoteliers - 59%, restaurateurs - 14%, tour operators and travel agents - 17%, event organizers - 7% and tourist information centres (TICs) - 3%. Territorially, they are representatives of both the Black Sea destinations and mountain and urban destinations.

The survey method of data collection is appropriate to use in this study because innovation activity is individual and cannot be measured by quantitative methods. The responses regarding the activities carried out according to the specified criteria are given by the entrepreneurs by filling out a questionnaire, which serves the researchers in creating a checklist. The systematization and analysis of information allows for measurement of the innovation activity of an enterprise. The results of measuring the innovation activity using this index can be used to compare the innovation activity of different tourism enterprises, as well as to monitor the process over time for the same enterprise. The use of this index contributes to revealing obstacles to innovation and promoting the innovation activity in tourism.

The positive indicators used with their minimum and maximum values have been upgraded by taking into account the indicators specified by ESIC (European Service Innovation Centre) (ESIC European Service Innovation Centre, 2014) in the proposed Methodology for Measuring the Transformative Power of Service Innovation. The proposed methodology has been upgraded with two new groups of indicators: Policy for encouraging innovation activity and Improving the quality of service, tested in a study of hotel enterprises in rural tourism (Ilieva et al., 2010) and the evaluation of innovation activity in hotel enterprises in the Albena Resort (Ilieva, 2011), where dependencies between the values of the innovation index and the predominantly implemented types of innovations in the research period were highlighted. Each of the indicators included in the methodology receives a score, which forms the overall score of the respective group indicator. The importance of the indicators is proposed as follows: In the absence of the specified operation – 0, in the case of a positive answer regarding the presence of the specified operation during the year – 1. In each group of indicators, a score is given for each of them, taking an average score from the submitted answers, giving the overall score for the group indicator. In such a situation, the possible maximum of points that can be scored, in case a positive answer is received for the seven groups of indicators, is 1, and the possible minimum – 0. After processing the data, an average score for each group indicator is calculated. It is only necessary to divide the total sum of the actually obtained scores for the seven group indicators by their possible maximum number - 7, in order to obtain a single coefficient (or converted into a percentage), which is the value of the innovation index.

3. Results

Measuring the innovation activity of a tourism enterprise provides assessment of its innovation activity. Since innovation operations shape the innovation activity of an enterprise, the degree of innovation also indicates the degree of development of its innovation activity. After reviewing and analyzing scientific publications and empirical research conducted in the field of innovation implementation in the tourism sector, taking into account the opinions of experts, the developed model for measuring innovation activity was applied.

Out of the total 116 tourism enterprises covered, the model was applied to 10 enterprises offering accommodation services, 10 enterprises providing restaurant services and 10 representatives of tour operators and travel agents. The owners of these enterprises have indicated that they implement innovations. As can be seen in Table 1., after measuring the innovation activity, an innovation index of each of the studied tourist enterprises is calculated.

Table 1.
Innovation index of tourism enterprises

Accommodation	1	2	3	4	5	6	7	8	9	10
Policy to promote innovation activity	0.33	0.33	0.33	0.67	0.50	0.50	0.50	0.67	0.33	0.33
Innovation product	0.60	0.73	0.67	1.00	0.80	0.70	0.93	0.87	0.73	0.93
Entrepreneurship and innovation networks	0.40	0.00	0.47	0.47	0.67	0.47	0.67	0.47	0.40	0.57
Investment and financing of innovation	0.56	0.31	0.29	0.69	0.88	0.56	0.42	0.75	0.33	0.75
Human capital for innovation	0.61	0.55	0.35	0.67	0.55	0.23	0.39	0.71	0.23	0.59
Information and communication infrastructure	0.63	0.56	0.56	0.81	0.63	0.56	0.56	0.81	0.56	0.81
Improving the quality of services and increasing value	0.75	0.54	0.75	0.75	0.88	0.67	0.75	0.75	0.42	0.75
Innovation index i	**0.55**	**0.43**	**0.49**	**0.72**	**0.70**	**0.53**	**0.60**	**0.72**	**0.43**	**0.68**
Food and entertainment establishments	1	2	3	4	5	6	7	8	9	10
Policy to promote innovation activity	0.33	0.33	0.33	0.67	0.50	0.50	0.33	0.67	0.33	0.67
Innovation product	0.60	0.63	0.27	0.90	0.73	0.63	0.27	0.93	0.63	0.93
Entrepreneurship and innovation networks	0.13	0.00	0.00	0.47	0.30	0.07	0.00	0.47	0.00	0.47
Investment and financing of innovation	0.63	0.31	0.08	0.69	0.88	0.31	0.08	0.69	0.31	0.69
Human capital for innovation	0.61	0.19	0.19	0.79	0.61	0.19	0.19	0.79	0.19	0.79
Information and communication infrastructure	0.63	0.56	0.56	0.81	0.63	0.56	0.56	0.81	0.56	0.81
Improving the quality of services and increasing value	0.75	0.42	0.50	0.75	0.75	0.42	0.50	0.67	0.42	0.75
Innovation index i	**0.52**	**0.35**	**0.28**	**0.72**	**0.63**	**0.38**	**0.28**	**0.72**	**0.35**	**0.73**
Tour operators and travel agents	1	2	3	4	5	6	7	8	9	10
Policy to promote innovation activity	0.00	0.33	0.33	0.33	0.00	0.33	0.00	0.33	0.00	0.25
Innovation product	0.57	0.40	0.43	0.67	0.40	0.67	0.40	0.43	0.57	0.40
Entrepreneurship and innovation networks	0.30	0.23	0.83	0.40	0.27	0.40	0.27	0.73	0.30	0.23
Investment and financing of innovation	0.38	0.36	0.36	0.31	0.23	0.31	0.23	0.36	0.38	0.36
Human capital for innovation	0.59	0.59	0.43	0.63	0.35	0.63	0.35	0.43	0.59	0.59
Information and communication infrastructure	0.69	0.50	0.81	0.63	0.38	0.81	0.38	0.81	0.69	0.75
Improving the quality of services and increasing value	0.75	0.34	0.50	0.50	0.25	0.50	0.25	0.50	0.75	0.34
Innovation Index i	**0.47**	**0.39**	**0.53**	**0.49**	**0.27**	**0.52**	**0.27**	**0.51**	**0.47**	**0.42**

Source: Author's research

The limits of the measured innovation index are (Ilieva, 2009): 0 < i < 0.33 – low innovation activity, 0.33 < i < 0.66 – medium innovation activity and 0.66 < i < 1 – high innovation activity, which shows that tour operators and travel agents are relatively less innovative (i=0.43) compared to accommodation (i=0.58) and catering and entertainment establishments (i=0.50). However, all three groups of enterprises offering the main tourist activities fall within the limits of medium innovation-active companies.

Despite the averaged coefficients, in the separate groups of enterprises there are also those with above average innovation activity. For example, in the accommodation group there are three hotels with high innovation activity (above 0.70), which is mainly due to the implementation of innovations from the five groups, i.e. the creation of the overall innovation product and the increased quality of services. As can be seen from the table, places of accommodation 4, 5 and 8 stand out from the others both in terms of the innovations implemented and the growth of investments, and in the use of information and communications infrastructure and human capital for the implementation of innovations. The indicators are similar for food and entertainment establishments. Three of the studied restaurants have an index above 0.70, and all three are 4-star category restaurants. The data in the table shows that all three restaurants have the maximum number of points for the implemented innovations, which cover the five groups of innovations that make up the overall innovation product. The high index is also due to the good results on indicators related to the quality of services provided, high levels of investment in innovation, the effective use of information and communications infrastructure and highly qualified human capital in these establishments. These findings indicate that the restaurants in the sample are leaders in innovation and technology adoption in their sector. Despite the low levels of innovation activity among tour operators and travel agents compared to the other two categories of tourism enterprises (hotels and restaurants), they also feature organizations with higher innovation activity. We can single out 3, 6 and 8 companies that are tour operators and create their own products, invest in reservation systems, create partnerships and organize the sale of their products in the value creation channel.

The tested model shows that it is possible to measure the innovation activity in the tourism sector by covering all enterprises offering the main tourism services. The individual indicators can be further developed with additional questions to allow for greater differentiation of the actions taken on the implementation of innovations, the creation of networks and the transfer of knowledge.

4. Discussion

Based on the responses received in this study, it is clear that there is practically no policy to promote innovation in the tourism sector. The few available innovation funding initiatives are mainly focused on resource-saving technologies that have an indirect impact on the quality of tourism services.

It seems that intellectual property and knowledge management are not among the priorities of tourism enterprises, as they do not pay special attention or make investments in this direction. Moreover, there is a lack of a clear strategy and support from the state and local authorities to foster innovation in tourism, which further complicates efforts of the enterprises in this direction.

An issue that was identified in this study is the difficulty tourism businesses experience in networking, joining associations and collaborating with other organizations in implementing innovations. There is a lack of coordination and interaction between state institutions, academic units and businesses with the aim of developing an innovation ecosystem in tourism. This limits the opportunities for the exchange of knowledge, technology and good practices, which is of key importance for increasing the innovation capacity of the tourism sector.

Another serious problem found in the research of innovation activity is the lack of purposeful management of innovations in tourism organizations. This also leads to difficulties in tracking the return and effectiveness of implemented innovations. Without clear processes for managing innovation, tourism businesses often face challenges in evaluating the benefits and impact of their innovation initiatives.

Another problem related to the implementation of innovations is the lack of human capital. A large number of the owners of tourism enterprises do not engage consultants in the development of innovation projects, and the organizations lack employees to deal with this activity. This is the reason why many promising ideas remain undeveloped and tourism businesses miss opportunities to improve their competitive position through innovation.

Another significant problem identified in this study is the difficulty faced by tourism businesses in determining whether the innovations they have implemented really qualify as true innovations. Many of the interviewed entrepreneurs were excluded from the study for this very reason – they simply failed to clearly and categorically define whether the changes they were making in their business activities were truly innovative or not. The challenge of distinguishing between true innovation and mere modifications or incremental improvements is a serious obstacle that prevents tourism businesses from fully adopting and benefiting from innovative practices.

Conclusion

The analysis carried out reveals that the innovation activities of tourism enterprises in Bulgaria are quite limited in scope and are mainly aimed at technological innovations, while non-technological innovations, such as social, sustainable and responsible innovations, are still at an early stage of development.

To solve the outlined problems, specific measures could be implemented to support innovation activities in tourist enterprises:

- developing a comprehensive national innovation strategy for the tourism sector, which is in accordance with the overall economic development goals of the sector and the country;
- creating a system to support innovation in the tourism sector, including financing schemes, tax incentives and advisory services in order to encourage tourism enterprises to invest in innovation projects;
- promoting cooperation and knowledge sharing between tourism enterprises, research and educational institutions and government organizations to foster an innovation ecosystem in the tourism industry;
- investing in building the innovation capacity of tourism enterprises by providing training and resources to develop skills and capabilities for innovation management;
- encouraging tourism enterprises to adopt a more holistic approach to innovation, focusing not only on technological innovations, but also on non-technological innovations that can increase the overall competitiveness and sustainability of the Bulgarian tourism product.

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CONTENTS

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SPACE INSURANCE – INNOVATIONS AND CHALLENGES FOR THE INSURANCE BUSINESS

Irena Markova[1]

Abstract:[2] The study focuses on the theoretical and applied aspects of insurance in the aerospace industry. The author's aspiration is to substantiate the role of space insurance as one of the main methods of increasing corporate security in space companies. The increased insurance needs and interests in the space sector in recent years have stimulated innovations in insurance activity and posed new challenges for insurers. The harmonious combination of insurance with the prevention and limitation of the impact of space risks is a prerequisite for effective corporate risk management in the space sector. The article proposes a model of the space insurance market, which analyses the relationships between its entities and substantiates the interrelationships between the space industry and the insurance business.

Keywords: space insurance, insurance products, space industry, space risks, environmental pollution liability insurance

JEL: G22.

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Introduction

Over the last decade, the space industry has seen a leap in its development, mainly due to high technological progress and the introduction of innovation processes in space activities. The implementation

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of a large part of the space activities is accompanied by a number of risks, as well as the discharge of harmful substances into nature. In addition, there is a possibility of production accidents, the consequences of which are related to environmental pollution. The manifestation of these risks leads to the occurrence of adverse consequences, such as: non-pecuniary damages related to the life, health and working capacity of the crew members of the space mission; property damage to launch complexes; damage from orbital explosions; loss of launch vehicles, spacecraft, satellites, etc.

In the conditions of globalization and a strong competitive environment in the space industry, the insurance business worldwide is facing new challenges and making innovative management decisions related to the adaptation of insurance services and products to the specifics of space missions and activities. In the effort of countries to increase national and international security in combating space risks, insurance is an effective mechanism for compensating for losses from accidental events in the space sector.

The implementation and development of the “space insurance” institution allows to significantly reduce the impact of space risks, reduce state expenses for environmental protection, guarantee the constitutional rights of citizens to a favourable natural environment, as well as to compensate for the damage caused to their health, work capacity and property through insurance protection.

The realization of space risks is very likely to lead to transboundary pollution of the environment. It is possible that the manifestation of the complex impact of some space risks could spread beyond the borders of the country and threaten not only national but also regional security. It is also possible that the manifestation of space risks could lead to an increase in space debris and pollution of the Earth's space environment – an issue related to international security and safety.

The relevance of the studied issue is determined by the need to offer adequate insurance protection, tailored to the demand for insurance by business entities in the space industry. In the Bulgarian insurance literature, there are still not enough publications related to the study of the impact of space risks on the environment and the effects of insurance protection for the space industry. Insurance companies still do not have enough experience and traditions in offering space insurance on the Bulgarian insurance market. The limitations of the study are due to the lack of official statistical data on the national insurance market, and therefore the author refers to information on the international insurance market and on the practice of leading countries in space insurance. The author's aspiration is to delve into this wide-ranging topic by exploring the possibilities of space insurance as a mechanism to promote the sustainable development of the space industry.

In the process of the historical development and growth of the space industry, the functioning of the space insurance market is carried out in response to the manifestation of new insurance needs and interests of different categories of customers. On the one hand, the space industry is one of the “most exciting and promising industries” of our time. On the other hand, the implementation of space activity is accompanied by a number of technological, financial and organizational-legal problems. Despite the fact that today the space industry has reached “high technical maturity”, launching spacecraft and operating them in orbit remains a rather risky activity. In this regard, the space industry implements security measures related to reducing the likelihood of such risks occurring and improving risk management in the sector. Practice shows that one of the most effective measures to combat risk is insurance.

The purpose of the study is to substantiate the importance of insurance for the space industry and outline the prospects for the insurance business in making innovative management decisions in accordance with consumer demand in the space insurance market. In the context of the formulated objective, the article substantiates the structure of the space insurance market and analyses the elements, connections and relationships between its participants. The author offers a model of the space insurance market and emphasizes the relationship between the space and insurance industries.

1. Literature review

Space insurance is carried out in accordance with the insurance legislation of the individual countries in which it is practiced, as well as in accordance with the principles of environmental law, on the one hand, and space law, on the other – as one of the newest areas of international law. The construction of the general conditions of space insurance implies the synchronization of the insurance legislation with the policy of the EU in relation to the protection of the environment and with the space policy of the member states.

A matter of wide public importance, which affects both the space industry, the insurance business and society as a whole, is the question of protecting the Earth's environment and the space in orbit around the Earth from the harmful effects of space activities. According to experts, the problem with the so-called space junk is becoming relevant today, and providing a precise definition of the concept of “space waste” turns out to be a difficult task. When discussing the issue in the Scientific and Technical Subcommittee of the UN Committee on the Peaceful Uses of Outer Space in 1997, it was proposed that the concept of “space debris” should include a non-functioning man-made space object incapable of resuming the performance of its tasks, including its fragments and parts. Space debris can also be created as a result of the explosion and collision of aircraft, from the solid motor debris of rockets, etc. According to *Belova, G.*, “so far, there are no economically viable and acceptable methods for cleaning outer space from the accumulated garbage. Therefore, the main efforts will be directed to measures to control and limit space debris, to prevent orbital explosions, etc.”. The author shares the opinion of a number of experts, according to whom the existing principles for the prevention of outer space pollution need to be supplemented with sanctions, and regulations have to be put into effect no later than 2030 if humanity wants to preserve space for the next generations (Belova, 2010). In this sense, *Solomon Passy*, as early as 2011, raised the question of the need to implement effective mechanisms to control movement in space and to improve “space law (and the instructions for its application) to regulate and guarantee the peaceful use of Space for decades to come” (Passy, 2011).

In her publication, *Popova, R.* believes that “space debris can be considered in the context of the more general concept of “space traffic management” (STM), which is a hot point of debate in the scientific discussion and could lead to proposing a new legal regime (or regional regimes)” (Popova, 2011). The opinion of specialists in space law is that, “international cooperation to protect the Earth's environment from the harmful effects of activities in space can be defined as ineffective in the light of the application of universal international treaties”. In this sense, *Milanov, Al.* and *G. Peychev* consider that “the latter cannot guarantee the interests of humanity due to the following reasons: a) lack of definition of the term “harmful pollution” in the Outer Space Treaty; b) unedited definition of the term “harm” in the Liability Convention; c) poorly balanced and ineffective rules on international liability for environmental damage also in the Liability Convention” (Milanov & Peychev, 2020).

According to *Markova, I.*, damages resulting from environmental pollution from space risks of a technogenic nature can be compensated under the Environmental Pollution Liability Insurance (EPLI) (Azman-Saini et al., 2010), (Misheva, 2015). The issue concerns innovation in the insurance business, which is called upon to adapt to the new risk situation of space objects and entities and to offer effective insurance security. With the help of EPLI, the insurer can protect the property interests of space companies and compensate property, non-property and moral damages to injured third parties. The successful implementation of EPLI in the space industry will depend to a large extent on the harmonization of national legal norms with international legal norms in the areas of space law, environmental law and insurance legislation.

The development of effective product and market strategies in insurance companies for *innovation in the broad and narrow sense* of the word is an essential prerequisite for responding to consumer demand in the space insurance market and *improving business models in the entrepreneurial activity (*Denchev et al., 2020) of insurers.

The increase in revenue in the space industry from activities related to space, infrastructure, ground stations and space-based products ($427.6 billion in 2022 compared to $396.2 billion in the previous year), as well as the largest business – the sale of satellite data for position, navigation and weather (39% of all commercial revenues), is expected to give *impetus to the development of the insurance business* in the near future (Grush & Kendall, 2023). This means that the development of space infrastructure in parallel with the challenges of space data – generation, processing and transmission of data in space – is a prerequisite for innovations in insurance activity. The increasing pace of development in the space industry is *driving advances in space insurance*, which must keep up with the challenges of bringing satellites to market. For example, the production of 40 to 60 satellites per month, or two to three per day, represents a pace never before seen in the space industry, while simultaneously reducing the cost of a satellite to between $400, 000 and $500, 000 (Coykendall et al., 2023). Orbit-based space services based on satellite systems such as Galileo, Copernicus and EO are a prerequisite for offering *new insurance services* adapted to the requirements of the environment. The offering of flexible insurance services and products by the insurers specializing in space insurance will significantly expand the capacity of the two insurance branches – personal insurance and non-life insurance – not only at the national level (Prodanov et al., 2022), but also in the regional aspect (Prodanov et al., 2023).

In the context of the issues, the conclusion is drawn in scientific circles that the discussed problems should be solved by adopting common standards for reducing the level of outer space pollution, which should be adopted with the widest possible participation and cooperation between countries and with the coordination of the various international structures.

2. History, development and regulation of space activity worldwide

The space industry encompasses a wide variety of space activities, operations and types of work that, over several decades, has become "a thriving complex industry with fierce political and technological competition between major industrial powers".

Since the 1970s, outer space has become a field for the manifestation of various economic interests, especially with the advent of telecommunication technologies (telephones, television), and then with the advent of surveillance technologies (Spot) and navigation (GPS, Galileo). There are currently 90 nations operating in space. According to data for 2021, in a ranking of countries' investments in the Space Tech Sector, the following countries fall into the top 10: USA, China, United Kingdom, India, Izrael, Switzerland, Canada, Germany, Sweden, Belgium. The leader in the sector is the USA. The second place is occupied by China and the third position belongs to the United Kingdom (WorldEconomicForum, 2022).

The establishment of the International Space Station (ISS) is the first major intergovernmental cooperative project aimed at carrying out numerous studies in microgravity, and in the long term, towards future space travel to the Moon or Mars. For example, the ARTEMIS manned space program, initiated by the United States, has a primary goal related to manned flights to the Moon.

Today, launching artificial satellites is a common industrial activity. The number of satellites launched by multiple operators – both public and private – is growing every year. The use of smartphones, for example, would not be possible without the GPS positioning provided by constellations of satellites (Zajdenweber, 2017). The current period of the development of the space industry is defined in scientific circles as the "space renaissance", and the sector is the basis for innovations in a number of other branches of economy. In this sense, satellite imagery helps farmers monitor crops, businesses monitor their environmental, social and economic performance, governments monitor CO2 emissions and real-time events on the ground, etc. Data shows that 10 times more satellite images of Earth are being produced today than five years ago, and 10 times more communications bandwidth is being spread around the planet (Forum, 2022).

International legislation in the field of space activities finds expression in a number of instruments and normative acts for the regulation and control of the sector. Among the most important is the Outer Space Treaty of January 27, 1967, which establishes the global legal framework. It states that the exploration and use of outer space, including the Moon and other celestial bodies, must be for the benefit of all countries and specifies that space cannot be the object of national ownership. The Treaty of 27 January 1967 was supplemented by the Convention of 29 March 1972 on International Liability for Damage Caused by Space Objects and the Convention of 1975 on the Registration of Objects Launched into Space.

Spaceflight safety issues are still regulated at the level of each country's government agencies. However, the cooperation between space powers, the emergence of new players (India, Brazil, China) and the emergence of private operators of “space tourism” outline the need to reform and harmonize the existing legislative framework and adapt it to the new reality. The mission of the International Association for the Advancement of Space Security (IAASS) conference is to implement this reform.

The emergence, development and rise of NewSpace gives rise to new legal issues related to the provision of services in Earth orbit and the introduction of new space activities other than traditional activities such as telecommunications, satellite positioning and Earth and space surveillance. SpaceX operations, Blue Origin and other private companies that launch their own rockets and deploy satellite constellations are the leading players in the space industry. Lower costs of space missions “open the door for new start-ups and encourage established aerospace companies to explore new opportunities that once seemed too expensive or difficult” (Brukardt, 2022). Companies that operate in the space value chain and contribute to the growth of the global space sector are specialized in six major areas: (Coykendall et al., 2023) Space data-as-a-service; In-space manufacturing (ISM); Additive manufacturing; Robotics in space; Space sustainability; National security space. Table 1 provides information on the state of the aerospace industry in 2023 and the types of segments responsible for the growth of the sector (Coykendall et al., 2023). Satellite integration has the largest relative share with 21% among the other segments. Continuous technological improvements and ambitious projects in the space industry encourage investment in the space economy.

Table 1.
Segments in driving the growth of the global space sector in 2023 (in %)

rank	1	2	3	4	5
relative share	21	20	17	16	15
segments	Satellite integration	Components	Launch vehicles	Value-added services	Payloads

Source: The Deloitte 2023 Riding the Exponential Growth in Space Survey.

Table 2 provides an update on investments in the sector in 2023 and their role in enhancing national security, facilitating ground-based space activities and activities in Earth orbit.

Table 2.
Investments in driving the growth of the global space sector in 2023, in %

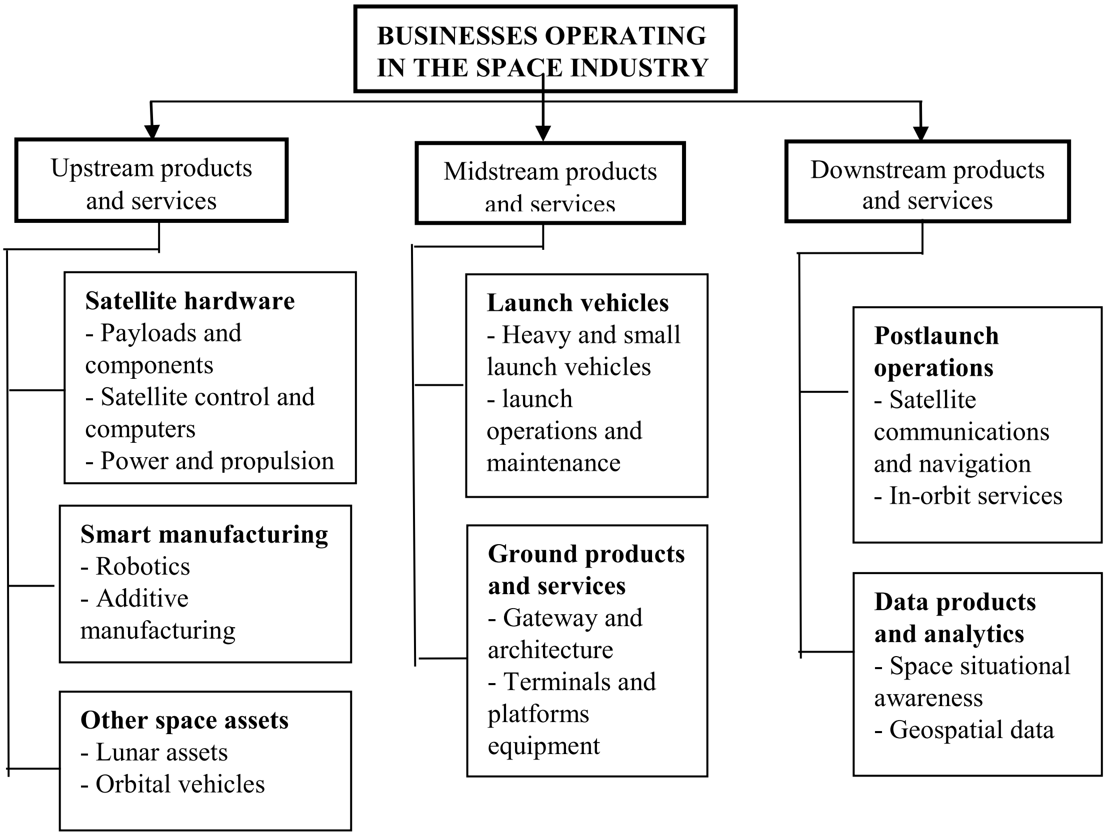
rank	1	2	3	4
relative share	30	24	12	10
investments	National security	Satellite communications	Edge computing and artificial intelligence	Earth observation and remote sensing

Source: The Deloitte 2023 Riding the Exponential Growth in Space Survey.

Figure 1 shows the types of space activities that underpin the growth of space industry and create added value.

The vast potential of space defines the challenges facing the space industry.

The competitive environment, the globalization of the space industry market and the construction of an effective regulatory framework regarding space activities are key prerequisites for the growth and sustainable development of space industry. According to the Space Foundation, the size of the global space economy in 2023 is estimated at $570 billion in 2023, compared to $531 billion in 2022, an increase of 7.4% and evidence of the continued growth of the industry in both the public and private sectors. This growth is in line with the industry’s five-year compound annual growth rate (CAGR) of 7.3% and is almost double the size of the space economy a decade ago.



Source: The scheme was made by the author based on information from Deloitte analysis.

Figure 1. Space activities along the value chain and growth in the space industry

3. Structure and state of the space insurance market

Space insurance marks its beginning in 1965, and only for about 60 years of development, it has become a leading sector of the international insurance market.

The launch of the Early Bird satellite on April 6, 1965 was associated with the conclusion of the first space insurance, the coverage of which was limited to the inclusion of terrestrial risks only. Lloyds of London provides insurance coverage for Itelsat 1 for material damage. In 1968, the Intelsat III series of satellites was insured in the launch phase. In 1975, insurance liability was extended to full risk coverage, i.e. the risks assumed at the time of signing the insurance contract refer to the end of the existence of the satellite in orbit.

The first claim compensation dates back to 1977 with the destruction of the Thor-Delta launch vehicle and the loss of the OTS1 satellite. In 1980, six new commercial satellites were insured. In the early 1990s, the boom in the development of the information industries sector stimulated the construction of satellites and launch vehicles.

In the international market of space insurance, the leading countries in the supply of space insurance for the “until rocket launch” phase are the following: France, Great Britain, USA, Italy, Germany, Bermuda, Austria, Scandinavian countries, Belgium, Japan (Draganov & Neikov, 2000).

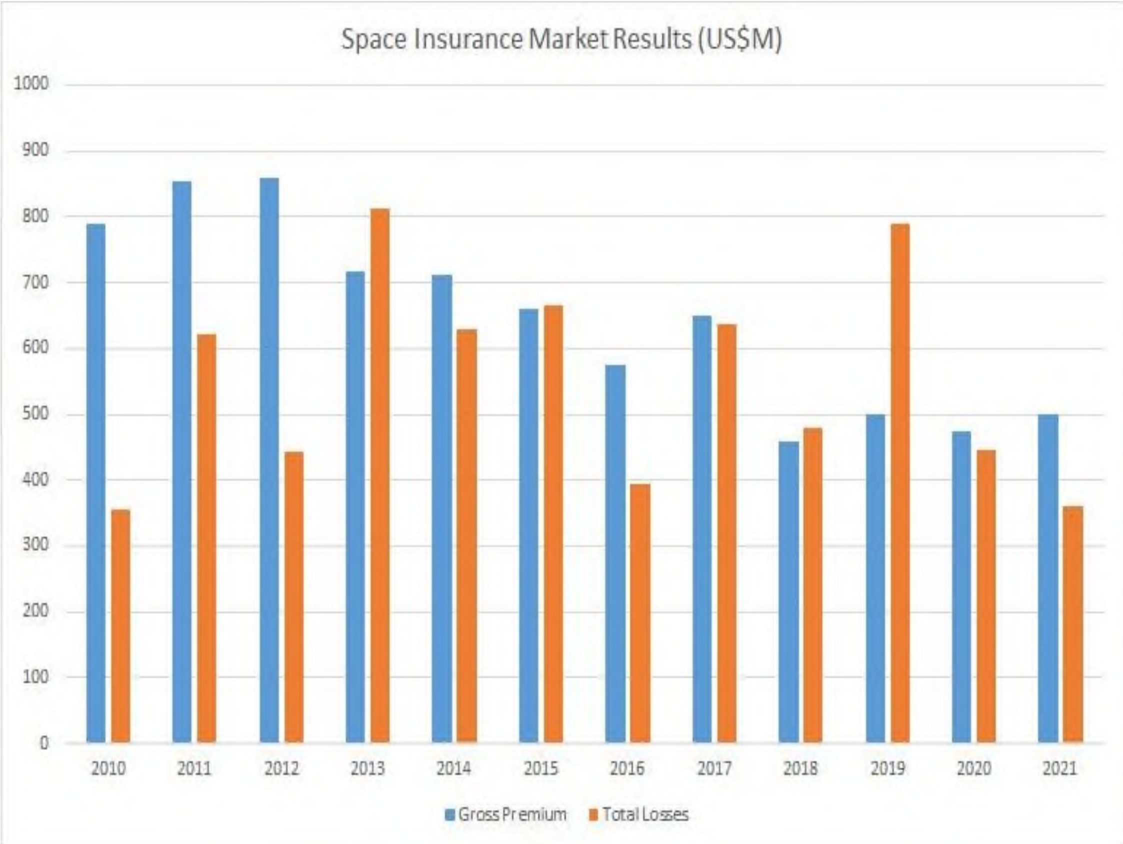
Table 3 presents information on the market share of the leading 10 countries in the space insurance market in 1997.

Table 3.
Market share of countries in space insurance for 1997 (in %)

Country	Market share (in %)
Great Britain	21.94
France	21.83
USA	16.79
Italy	13.54
Germany	10.40
Bermuda	5.42
Austria	3.25
Belgium	3.03
Scandinavian countries	2.71
Japan	1.08

Source: Draganov & Neikov, 2000

Claims made for insurance benefits lead to a decrease in the number of satellites launched and a significant impact on the space insurance market. Collected insurance premiums for the period 1987 – 1997 range between 120 and 820 million dollars. The compensations paid range from 53 to 750 million dollars. In 1998, large losses due to cumulative losses of satellites in orbit halted the development of the space industry. About 15 – 20 years later, the situation in the space insurance market is radically changing. Figure 2 provides information on the state of the international space insurance market through the indicators “premium income” and “claims paid” for the period 2010 – 2021.



Source: Seradata (https://payloadspace-com.translate.goog/the-space-insurance landscape/?_x_tr_sl=en&_x_tr_tl=bg&_x_tr_hl=bg&_x_tr_pto=sc).

Figure 2. Space insurance market dynamics for the period 2010 – 2021

The data show a trend towards an increase in the premium income from space insurance in the range of about 300 to nearly 900 million dollars, with the highest premium income observed at the beginning of the analysed period (2010 – 2013). The largest amount of insurance payments during the period was observed in 2013 and 2019 – nearly 800 million dollars.

At the current stage of development, space insurance accounts for 15 to 20% of the total budget of space programs, and this share is constantly increasing, today occupying third position in the budgets of space operators. Approximately 20 new satellites are insured for launch each year, bringing more than 150 insured satellites launched or currently in orbit (L'assurance spatiale - Atlas Magazine, 2023). The number of satellites in low orbit (LEO) is growing in parallel with the increase in the number of private space operators. This also determines the leap in the development of space insurance, designed to meet new insurance needs, as well as to identify, monitor and analyse new and atypical risks. On average, there are about a hundred satellite launches each year, and of those, only a third are insured. The *Galileo satellites* that failed to enter orbit in 2014 were not insured. But now the market is facing a major tipping point with more and more private operators and smaller satellites for which insurance is a mandatory step in their financing.

Table 4.
Insured satellites in orbit as a market segment in space insurance for 2018

Active satellites	Number of satellites	Satellites in percent	Insured satellites (in %)
GEO satellites	492	22	43
LEO satellites and others	1. 715	78	6
Total satellites	2. 207	100	49

Source: International Union of Aerospace Insurers and author's calculations.

Data show that nearly 50% of active satellites in orbit were insured in 2018. GEO satellites are only 22 in number, but 43% of them are insured, as opposed to LEO satellites and others, of which only 6% are insured. Five years later, in 2023, there are more than 9,000 satellites in LEO – low Earth orbit, of which only about 50 are insured. This shows a more than 5-fold increase in the number of LEO satellites, but little interest from companies in insurance. In 2023, the total number of satellites in orbit reaches 10,000, with only 300 of them are insured and the majority of them are in GEO.[3] The total number of satellites in orbit has also increased by more than 5 times over the last five years (in 2023 compared to 2018).

Forecasts are for the number of low orbit satellites to increase tenfold over the next five years, clarification by Dominic Rora, Head of Space Insurance at Axa XL To address this development, *Axa XL* signs a partnership agreement with French startup *SpaceAble* to improve the coverage of satellite operators by aggregating, processing and modelling spatial data. According to Denis Bousquet (Senior Space Underwriter), the Axa XL partnership will continue the group's development in long-term space insurance. The launch relies on data processing software and The Orbiter inspection satellite. The data provided by the startup will be critical in an increasingly complex environment that operates "several thousand active satellites, along with space junk".

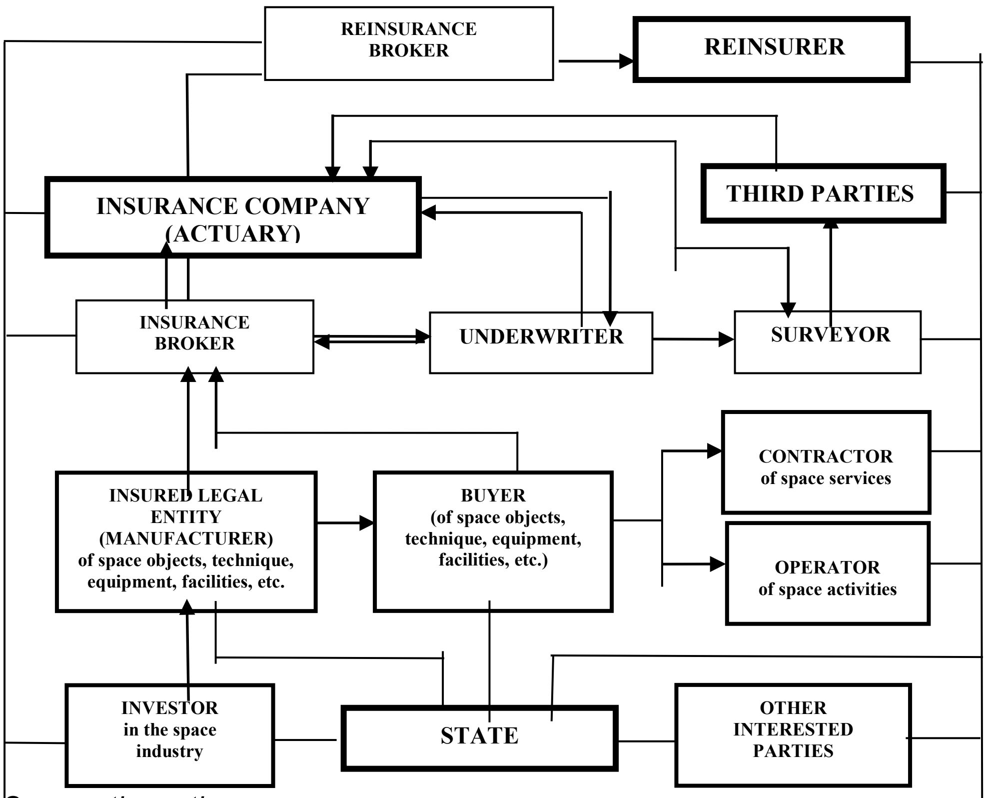
According to a *Technavio report*, the global market – for satellite launch and space insurance – was forecast to reach US$1.3 billion by 2022. Experts from the company indicate that the growth of the market is explained by the increase in the number of satellite launches. According to *The Space Report* 2022 of *the Space Foundation*, the space economy was worth $468 billion in 2021, a 9% increase over the previous year. In addition, more than 1, 000 spacecraft are launched in the first six months of

[3] www.datacenterdynanics.com.

this year, which is more than those launched in the first 52 years of space exploration (WorldEconomicForum, 2022).

The space sector is a key factor for the growth and efficiency of other sectors of the economy. *The European Space Agency* points out that “the deployment of new space infrastructure has brought benefits to sectors such as: meteorology, energy, telecommunications, transport, maritime, aviation, urban development, including insurance”. The report states that more investment in the space industry comes from the private sector rather than the public sector.

A large number of participants in the space industry are also entities in the space insurance market. The participants in the space insurance market and the relationships between them are presented in figure 3.



Source: the author

Figure 3. Space insurance market model

The main participants in the space insurance market are insurance companies and users of insurance services and products, between which specific economic relations arise – insurance relations. *The insurer* and *the insured* are both parties to the insurance legal relationship when signing an insurance contract. *Leading insurance companies* with experience and traditions in the international space insurance market are: Generali, AGF, Mazcham, La Reunion Spatiale, SCOR, INTEC, Munich Re, USAIG, ACE,etc.

Other participants in the space insurance market are *insurance brokers*, whose intermediary activity is the transfer of risk from the client to the insurer. Some of the leading players in insurance mediation in the space insurance market are Insurance brokers *Verspieren International*, *AJ Gallagher,* etc.

The development of space insurance would not be effective without risk equalization through reinsurance. This means seeking financial security for the insurer by transferring part of the original insurance liability to *the reinsurer*, which is the main entity in the space insurance market. *The reinsurance broker* is a specialized intermediary with in-depth general and special training who is at the heart of the relationship between the insurer and the reinsurer regarding the levelling of the risk outside the insurance company and assists in the placement of the reinsurance contract.

For insurance companies, it is important to conduct an adequate risk assessment, carried out by a specialized expert in insurance – *an underwriter*. In terms of space insurance, two-level underwriting is appropriate, given the specifics of space risks (Misheva , 2015). *Actuaries* in insurance companies are directly related to the development of insurance products and the construction of insurance tariffs. In order to price insurance products and ensure the solvency of insurance companies through adequate reserves, actuaries must assess the risk of future insured events (Pavlov & Mihova, 2018) – such as death, illness and disability – that may materialize into personal insurances for space missions in future, for astronauts and tourists during space travel related to space tourism. The international space insurance market currently offers a number of insurance products, such as: life, health and work capacity insurance for astronauts and personnel carrying out space activities; satellite insurance for operation in orbit; spacecraft liability insurance; insurance for damage during construction and installation of launch pads, and others.

A wide range of *consumers* of insurance products participate in the space insurance market, which are governmental and non-governmental segments, respectively a wide range of legal entities and individuals. Figure 3 shows the groups of market segments that determine the demand on the space insurance market, such as: *manufacturers of space objects, buyers of space equipment, space service providers and operators of space activities*. Therefore, this includes the so-called space companies that create products such as launch vehicles and satellites and/or provide services to users in space. *The new space service providers* are startup companies where *engineers* develop reusable components for launch vehicles, reducing costs.

Investors in the space industry are an important player in the space insurance market, interested in insuring their investments and projects. Forecasts indicate that the global space economy will be worth $1.8 trillion by 2035 (Report, 2024). *Investments in research and development* are aimed at further reducing startup costs. Relativity Space, for example, plans to use 3D printing, artificial intelligence and autonomous robotics to build a fleet of fully usable, low-cost rockets. A large part of *the investment* in the space industry *has gone into innovation in space activities*. Driven by robust private and public sector growth, the global space economy is expected to reach $570 billion by 2023, with commercial revenues accounting for 78% and exceeding $445 billion (Space Foundation). In the US, *public agencies*, particularly *NASA*, *the Department of Defense and the intelligence community*, have traditionally provided the most investment in space.

A trend is emerging where commercial funding could replace government funding within 20 years. On the other hand, this trend is expected to lead to mutually beneficial *public-private partnerships* between entities in the space insurance market.

For *manufacturers in the space industry*, increasing demand for satellites is important, as well as reducing costs and obtaining economies of scale by increasing production volume. There are also *other interested parties* involved in the conduct of space insurance, which include: scientific laboratories, innovators, researchers, analysts, academia, the IT sector and other sectors of the economy, application users, as well as other non-space companies affected by space commercialization and services.

Thanks to *researchers,* new technologies such as higher-resolution sensors are being created to improve image capture, data processing, and other functions. Satellites can now collect, analyze and transfer much larger data stores than they could just five years ago. Satellites are becoming more sophisticated every year, allowing researchers to develop new proposals. Many companies now deploy smaller, cheaper LEO satellites to provide better satellite connectivity. SpaceX's Starlink has already launched a LEO constellation and has customers paying for its satellite broadband network. Although much uncertainty remains, *analysts* are optimistic about the space industry and expect it to become a $1 trillion industry thanks to the development of entirely new applications. The new applications can be divided into two broad categories: Space-to-Earth applications, which facilitate ground-based activities, and Space-to-Space applications, which only involve activities that take place in orbit.

Thanks to *innovators,* 3D printing and other innovations are helping to reduce costs by streamlining the manufacturing process and improving supply chains. The cost of LEO satellite launches has dropped from $65,000 per kilogram to $1,500 per kilogram in 2021, a reduction of more than 95% (Brukardt, 2022). The new generation of satellites and applications support major sectors of the economy, including insurance. For the insurance business, better imaging could allow more insurers to assess risk and damage in remote locations, with improved resolution, and more clearly pinpoint problems and eliminate the need for in-person visits (Brukardt, 2022).

Despite the existing difficulties, *stakeholders* in the space industry can consider measures that will help space companies and other economic entities navigate the new situation and take measures to reduce the likelihood of space collisions and space debris.

Conclusion

Despite the complex of reasons that make insurance activities difficult, there are favourable prospects for the development of space insurance.

The application of robotics in space and the use of artificial intelligence allows the space sector to remotely operate and control spacecraft, rovers and other devices for the study of celestial bodies. This is a factor in expanding insurance liability by including *new space technologies* in insurance coverage.

Increasing revenues in the space industry from activities related to space-based products, ground stations as well as the largest business – the sale of Position, Navigation and Time (PNT) satellite data is expected to boost the development of the insurance business in near future. This means that the development of the space infrastructure in parallel with the challenges of space data – generation, processing and transmission of data in space is a prerequisite for innovations in terms of insurance services and products, by including new space activities and risks in the insurance liability.

Improving *the relationship between the parties to the insurance contract* – insurance companies and space companies – is a prerequisite for *adapting the insurance programmes* to the increased requirements of the space sector. On the one hand, the direct financing of preventive measures regarding space risks by insurance companies leads to an increase in corporate security in the space industry. On the other hand, stimulating the preventive activities of space companies through the insurance conditions and tariffs reflects on the price of the insurance protection and leads to its reduction.

The participation of other entities and interested parties in *initiatives related to reducing the impact of space risks* on the Earth's environment and outer space is an important factor in the development of national space insurance markets.

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D. A. Tsenov Academy of Economics, Svishtov

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CONTENTS

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CORPORATE SOCIAL RESPONSIBILITY OF AGRIBUSINESS AS A DETERMINANT FOR ENSURING THE DEVELOPMENT OF RURAL AREAS OF UKRAINE

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Oleksandr Vilenchuk[3],
Larysa Nedilska[4],
Olga Vikarchuk [5]

Abstract: The growth of scientific and practical interest in the problems of corporate social responsibility relates to the public demand for value orientations of modern life, which are based on the principles of democracy, objectivity and justice. The basis of multifunctional rural development should be recognized as the social responsibility of all stakeholders interested in improving the quality of rural residents' lives. The purpose of the study is to conceptually reveal and substantiate the activation processes of corporate social responsibility in the rural development of Ukraine in the conditions of European integration. The methodological basis of the study is based on the use of various methods of scientific knowledge, in particular: abstract-logical, structural-functional, induction and deduction, concretization, and graphic methods. During the conducted research, an assessment of the level of corporate social responsibility was carried out based on the parameters of the increase in the volume of

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social expenses and their structure among business entities in the agricultural sector of Ukraine for 2014-2022. Using consolidated reporting (financial and non-financial) of the leading producers of agricultural products in Ukraine, a correlation-regression model was formed. An assessment of the impact on the social costs' quantity of agricultural holdings of factors such as capital profitability, land profitability and the volume of net income per 1 ha of agricultural land was carried out. Based on the modelling results, it was established that the mentioned agricultural producers have the necessary financial resources to strengthen social activity in rural areas, primarily thanks to the rational use of land capital. The main results of the study are aimed at integrating the corporate social responsibility of agribusiness into all areas of life in rural areas, ensuring a combination of moral and ethical, and socio-economic foundations of rural development.

Keywords: agricultural sector, corporate social responsibility (CSR), rural development

JEL: A13, M14, Q13

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INTRODUCTION

Rural development should be considered as one of the key priorities in solving the problems of global food security, which are increasingly facing the world community. The relevance of this issue for Ukraine is gaining particular importance, considering the potential opportunities of production and export of agricultural products to different countries of the world. Undoubtedly, the military aggression of the Russian Federation causes huge material and non-material losses for the agricultural sector of the national economy. The post-war recovery of Ukraine should be based on the formation of certain multifunctional centres of rural communities, focused on attracting investments, creating a trajectory for social well-being and ensuring the ecological sustainability of the territories. Despite all the complexity and problematic nature of the outlined development goals, there is a need for long-term interaction between business, government and society. Considering the experience of the European Union countries, the specified interaction can be based on the principles of corporate social responsibility.

In the process of forming modern civilizational strategies for humanity, the ideas of social responsibility are gaining more and more importance. In particular, the global Sustainable Development Goals until 2030 (2023) declared by the UN are aimed at: the development of the health care system (Goal 3), the formation of quality education (Goal 4), ensuring decent work and economic growth (Goal 8), rational consumption (Goal 12), and the achievement of peace and justice (Goal 16).

In the countries of the European Union, there is an obvious trend towards the socialization of corporate business. With the adoption of the European Directive 2014/95/EU, large companies submit reports on corporate social responsibility not voluntarily, but in a mandatory form (European Union, 2014). Such legislative transformations at the EU level are due to the need to increase investors' confidence in the competitiveness of companies, the need to increase the transparency of their published information and improve their own reputation (Arraiano & Hategan, 2019). In Ukraine, since 2020, the Concept of the state policy implementation in the field of promoting the development of socially responsible business for the period up to 2030 (2020) has been adopted, the main postulates of which are aimed at the effective interaction of various stakeholders, namely: the state, business, society with the aim of strengthening mutual responsibility of all participants in social relations.

The hypothesis of the study is the assumption that the growth and realization of the financial and economic potential of agricultural holdings will contribute to the strengthening of their social activity and responsibility for making strategic and tactical management decisions, which will have a positive all-encompassing impact on multifaceted rural development in Ukraine.

1. Literature review

A critical review of literary sources gives reasons to single out five common features of corporate social responsibility that regulate the activities of corporations, namely: a) responsibility to society; b) responsibility for society; c) ethical rules and responsibility; d) initiatives for society and the environment; f) management of relations with society (Moon, 2014).

Conceptual principles of social responsibility are based on three generally recognized theories: Carroll (provides economic, legal, ethical and philanthropic responsibility); triple bottom line (aimed at economic, social and environmental responsibility), as well as stakeholder theory (aimed at regulating stakeholders' interests) (Brin & Nehme, 2019). The research of these theories gives reasons to distinguish their common features, specifically: observance of human rights and freedom, coordination of economic, social and ecological interests in society, application of moral and ethical standards of individual behaviour in society.

The modern model of rural development of the EU countries is based on the basic ideas and principles of social responsibility. This model is based on partnership and is aimed at the implementation of integrated rural development to achieve more efficient use of resources and reduce regional social inequality (Permingeat & Vanneste, 2019). We share the point of view that “a socially responsible enterprise is one of the strategic factors in balancing the socio-economic interests of villagers, a factor in the sustainable development of rural areas and a mechanism for ensuring inclusiveness” (Duke, 2020).

The modern scientific interpretation of the “corporate social responsibility” concept should be considered as: a powerful tool for maximizing the positive impact of corporations and commercial activities on the development of human society (Okpa et al., 2020); an opportunity to improve citizens’ well-being through socio-economic interventions (Azizi & Jamali, 2016; Ahijo et al., 2017; Jarmusevica et al., 2019); the way in which companies integrate social, environmental and economic concerns into their values, culture, decision-making, strategy and operations in a transparent and accountable manner, and thus can implement practices within the firm, create wealth and improve society (Mourougan, 2015). It is worth agreeing with Sieminski (2023), who emphasizes the need to integrate socially responsible values into the everyday practice of managing enterprises of various profiles.

In the scientific literature, there are quite polar opinions regarding the role and importance of large-scale corporations in realizing the potential of rural development. For example, Zinchuk & Levkivskyi (2019) emphasize the need “to focus agricultural holdings on the social, economic and ecological components of sustainable development in accordance with the Sustainable Development Goals, which, in addition to obtaining maximum profit for producers, will minimize the negative impact on the environment, ensure the development of rural infrastructure, as well as improve the life and well-being of the rural population.”

At the same time, it is worth emphasizing the existing threatening trends in the activities of agricultural holdings in Ukraine. Such threats are: monopolization of the land resource rental market and the market of agricultural products; absorption of small enterprises, limitation of opportunities for personal and farm development, etc. (Zabynets, 2018). It is believed that the further activities of agricultural holdings should be focused on the growth of their social and environmental mission, especially in the war and post-war (restoration) periods. Therefore, the study of scientific sources shows that the role of corporate social responsibility in business at the national and global levels will only grow. This gives grounds for stating the expediency of scientific research in the given direction.

2. Methodology

2.1. Research methods. The study of the social responsibility of business as a determining criterion of the rural development of Ukraine in the context of European integration necessitated the use of a methodological toolkit based on a systematic approach in the application of methods of scientific knowledge. For a theoretical reflection of the reasons for the current demand in society for corporate social responsibility in a modern society, *an abstract-logical method* was used. The application of *the specification method* made it possible to substantiate the role of socially responsible business in the implementation of the EU common agricultural policy.

The structural-functional method contributed to the formation of practical proposals aimed at systematically strengthening the corporate social responsibility of stakeholders for balanced rural development in Ukraine. *The method of formalization* ensured the possibility of making theoretical generalizations necessary for conclusions and proposals formulation.

Based on the results of analytical research, a correlation-regression model was created to assess the impact of the financial and economic condition of agricultural holdings on their social activity. The basic indicator of the model is the amount of social costs per 1 employee of the agricultural holding (Y is the dependent variable in US dollars per 1 person), the following relative indicators are defined as the main influencing factors: return on capital (X1); land profitability (X2) and the amount of net income per 1 ha of agricultural land (X3). The research was carried out based on the annual reports of agricultural holdings for 2014–2022.

This study is focused on identifying the dependence between the level of social activity of agricultural holdings and their profitability and scale of management. The regression analysis results of the cumulative effect of the selected factors on the activity of the social position of agricultural holdings made it possible to construct an equation of the linear type, on the basis of which the statement about its sufficient statistical significance and reliability was formed. The parameters and main characteristics of the regression equation were estimated, namely: coefficients; standard error; t-statistics and p-values. The developed correlation-regression model provided an opportunity to formulate practical proposals aimed at activating the social position of stakeholders as a determining factor in Ukraine's rural development and its European integration direction.

2.2. The research information base is formed by: normative legal acts of Ukraine (Law of Ukraine, 2017; Cabinet of Ministers of Ukraine, 2020); expert conclusions of the National rating of corporate reputation management quality (2023), the results of the study of business sustainability with the formation of the CSR Index (2023), the consolidated reporting of the leading agricultural holdings of Ukraine for 2014–2022, as well as data from the State Statistics Service of Ukraine. In addition, the scientific works of Ukrainian researchers, scientists from other countries, and information resources from the worldwide Internet are reflected in the presented study.

3. Research results

3.1. The role of CSR in ensuring rural development. Corporate social responsibility is based on philanthropic and people-oriented activities, the strategy of which is aimed at the comprehensive formation of humane and benevolent relations in rural society. Processes of deep multi-vector (economic, intellectual, personal, spiritual, social, political, etc.) transformation over many decades have formed the objective demand of society for responsibility. This conclusion is based on the following arguments.

First, there is a gradual change in social priorities, i.e. along with the desire to obtain material benefits, the question of systematic strengthening social and environmental responsibility for management decisions is being raised more and more loudly. Secondly, the integration and globalization of economic relations in the world give reasons to consider social responsibility as one of the determining factors for ensuring competitive advantages in the market, including agricultural products. Thirdly, compliance with the basic principles of social responsibility (openness, transparency, self-improvement, morality and humanity) will ensure the implementation of a human-centred approach in modern society.

The Common agricultural policy of the European Union is deeply imbued with a sense of social responsibility for management decisions made in the field of rural areas. Such generalizations are based on the analysis of

EU regulations on agricultural policy. In particular, the adopted Communiqué of the European Commission “The Future of Food and Farming” (2017) formulated the priorities of rural development in the following areas: decentralization, innovation, efficiency improvement, development of rural areas, environmental protection, young farmers, food safety.

The multifunctionality of rural area development is primarily focused on preserving and multiplying human, natural and economic potential; promoting social cohesion, sustainability; and defending the true values of life. This involves systemic changes in the formation and functioning of rural areas, socio-economic and cultural-humanitarian development, as well as a gradual increase in the well-being of citizens.

In 2021, the European Commission, as the highest executive body of the European Union, adopted the EU long-term plan for rural development “Towards stronger, connected, resilient and prosperous rural areas by 2040” (European Commission, 2021). This document envisages the implementation of comprehensive transformations aimed at improving the demographic situation in rural areas, the introduction of multi-level management of territories considering climate changes, the balancing of socio-economic development and digitalization. Therefore, the adoption of the strategic plan of the specified transformations in the field of agrarian policy provided an opportunity to form a “road map” of the rural areas functioning in the medium and long term.

3.2. Analysis of the social cost structure in the field of agriculture. According to the results of the conducted research, it was established that the level of social responsibility of business entities is measured in their ability to carry out social expenses on a permanent basis, which traditionally include payments to employees, deductions for social events, project financing, maintenance of institutions and social infrastructure facilities, etc. The volume and structure of the specified costs indicate the dynamics of social responsibility and the social activity directions of business entities, which determines the ability to form sustainable development of rural areas.

In Ukraine, during 2014–2022, there was a significant increase in the amount of social costs of economic entities in the agricultural sector (Table 1) – by 2.92 times, of which the cost of employee salaries increased by 3.28 times, and by 1 .94 times – deductions for social activities (the choice of the research period is determined by the presence of an increase in social activity among agricultural entities and their ability to carry out social expenses.

Table 1.
Increase in the volume of social expenditure and their structure in economic entities of the agricultural sector of Ukraine, %*

Indicator	Increase in the volume of expenses, 2014-2022	Structure of expenses					Change in the structure, v.p.
		2014	2016	2018	2020	2022	
Employee benefits expense of business entities	292.52	100	100	100	100	100	-
large entrepreneurship entities	352.30	10.09	9.80	8.63	12.44	12.16	2.06
medium entrepreneurship entities	266.46	63.75	67.58	65.49	s	58.07	-5.68
small entrepreneurship entities	332.97	26.16	22.62	25.88	s	29.78	3.62
of which micro-entrepreneurship entities	281.25	7.63	7.13	8.06	7.77	7.34	-0.29
Wages and salaries of business entities	328.09	73.40	82.00	82.08	82.15	82.32	8.92
large entrepreneurship entities	397.83	7.46	8.15	7.18	10.31	10.15	2.69
medium entrepreneurship entities	299.94	46.72	55.69	54.02	s	47.91	1.18
small entrepreneurship entities	369.47	19.22	18.16	20.88	s	24.27	5.05
of which micro-entrepreneurship entities	300.29	5.70	5.52	6.30	6.08	5.85	0.15
Social security costs of business entities	194.38	26.60	18.00	17.92	17.85	17.68	-8.92
large entrepreneurship entities	223.32	2.63	1.65	1.45	2.13	2.01	-0.62
medium entrepreneurship entities	174.56	17.02	11.89	11.46	s	10.16	-6.86
small entrepreneurship entities	232.00	6.95	4.47	5.01	s	5.51	-1.44
of which micro-entrepreneurship entities	225.20	1.93	1.61	1.76	1.69	1.49	-0.45

Source: *calculated based on data from the State Statistics Service of Ukraine.*

** According to Ukrainian law, large enterprises are defined as enterprises with annual revenues of more than EUR 50 million and more than 250 employees; small enterprises are defined as enterprises with annual revenues of up to EUR 10 million and up to 50 employees, of which microenterprises are enterprises with annual revenues of up to EUR 2 million and up to 10 employees; the rest of the enterprises are classified as medium-sized.*

From 2023, studies of the mentioned dynamics will be complicated due to limited access to relevant information, which relates to the presence of military operations on the territory of Ukraine). It is important to emphasize that the greatest social activity is shown by large enterprises, which, thanks to the scale of production, can finance large-scale activities and maintain the appropriate level of well-being in rural areas.

In the structure of social expenses, the special place of medium-sized enterprises attracts attention, which represents from 58.2% to 67.9% of the costs of employee wages and from 57.5% to 66.3% of deductions for other social measures. At the same time, the development of large-scale agrarian entrepreneurship contributes to the solution of the food problem at the regional, national and global levels. However, irrational use of land resources leads to their exhaustion, loss of fertility and other irreversible processes. In addition, it is worth noting the possible risks of land market monopolization, displacement threats to small and medium-sized agrarian businesses, creation of significant social and environmental problems. Obviously, the conflict between agricultural enterprises and society is of an economic nature (Sumets et al., 2022). Overcoming this conflict situation is possible thanks to the systematic strengthening of social responsibility of all stakeholders involved in the development of rural areas.

3.3. Assessment of the social activity level of agricultural holdings in Ukraine. According to the National Quality Rating of Corporate Reputation Management (2023), the top five agricultural holdings of Ukraine include Astarta, IMC, Kernel, Nibulon and MHP (Reputational activists, 2023). According to the logic of the presented research, it has become necessary to identify the factors that have the most significant impact on the social activity of agricultural holdings. Based on multifactorial correlation-regression analysis, the effect of key relative financial and economic indicators was evaluated, such as return on capital, land profitability and net income per hectare of agricultural land.

The choice of the research base was grounded on the level of social activity of agricultural holdings and their contribution to the multidimensional development of rural areas. The combination of financial, economic and social responsibility to society was reflected in the presented reporting for the period under study. This created the basis for further modelling and the formation of relevant proposals in the specified direction.

The basic indicator of the correlation-regression model was chosen as the amount of spending on social activities per 1 employee of the agricultural holding as a key indicator of an active social position (Y is the dependent variable in US dollars per 1 person), the determining factors of which were:

• return on capital, % (x1), because of the assumption that the higher the level of profitability, the more net profit of agricultural holdings is directed towards social expenses;

• profitability of land (x2), due to the assumption that the higher the profit an agricultural holding receives for every 1 ha of agricultural land, the more funds it can allocate for social support of local territories;

• amount of net income per 1 ha of agricultural land USD / 1 ha (x3), which is associated with the assumption of significant opportunities for large-scale production to provide sufficient volumes of social spending.

The matrix of paired correlation coefficients (Table 2) shows that all the selected factors have a noticeable and strong influence of the reverse (-) action on increasing the amount of spending on social activities in the ratio per 1 employee of an agricultural enterprise. This indicates an insufficient level of social activity of agricultural holdings in relation to their business scale and profitability (Table 1).

Table 2.
Matrix of paired correlation coefficients (r)

Indicator	Y	x 1	x 2	x 3
The volume of expenses for social activities in the ratio per 1 employee of the agricultural holding, dollars. USA / 1 person (U)	1.0000	-0.7725	-0.3665	-0.4008
Return on capital (x1)	-0.7725	1.0000	0.7276	0.3709
Land profitability (x2)	-0.3665	0.7276	1.0000	0.5912
Amount of net income per 1 ha of agricultural land (x3)	-0.4008	0.3709	0.5912	1.0000

***Source**: own research.*

Based on the results of the regression analysis of the cumulative effect of selected factors on the activity of the social position of agricultural holdings (Table 3), a linear equation was constructed: Y = 25.6289 + (-1.3093) *x1 + 42.5858*x2 + (-3.90417) *x3 with Fisher’s test F = 25.07. Based on which it can be stated that the coefficient of determination of the built model of social activity of agricultural producers (R2=0.7737) is statistically significant, and the equation is statistically reliable.

Table 3.
Evaluation of the parameters and main characteristics of the regression equation

Indicator	Y	x1	x2	x3
Coefficients	25.6289	-1.3093	42.5858	-3.9042
Standard error	4.8904	0.1733	10.7423	1.2813
t-statistics	5.2406	-7.5548	3.9643	-3.0470
p-value	0.0000	0.0000	0.0007	0.0059

***Source**: own research.*

Thus, the determination coefficient of the obtained regression equation R2 = 0.07737 and the weighted determination coefficient R = 0.7428 confirm the proposed hypothesis about the close dependence of the active social position on the factors included in the model, and the constructed equation reflects the following patterns:

a0 = 25.6289 – indicates that under the conditions of stability of the selected factors, the amount of expenses for social activities per 1 employee will be $25.63 (USD);

a1 = (-1.3093) – reflects that, based on the results of modelling, it was established that an increase in the return on capital by 1 percentage point leads to a reduction in the financing of social expenses by $1.3 (USD), other characteristics remaining unchanged. This trend is due to the orientation of agricultural holdings to maximize profit against the background of insignificant social costs;

a2 = 42.5858 – characterizes that, with an increase in the level of land profitability by 1 percentage point, there will be an increase in the amount of social expenses per 1 employee by $42.59 (USD), which proves the close interdependence between the scale of agriculture and the level of social responsibility of business entities;

a3 = (-3.9042) – indicates that an increase of $1 (USD), the amount of net income of agricultural holdings in relation to 1 ha of agricultural land causes a reduction in the amount of social expenses by $3.9042 (USD). Such a regularity highlights the problem of the insignificant level of social activity of agricultural holdings, given the growth of their incomes.

Based on the parameters of the formed regression model of social activity of agricultural holdings, it is worth pointing out that such agricultural producers have significant resources for financing social activities and programs for the development of rural areas. However, their priority focus on profit maximization remains a restraining factor in actively spreading social initiatives and supporting rural development.

A strong reserve for strengthening the social responsibility of agricultural holdings is their close dependence on land capital. According to the simulation results, it is the land capital that acts as the basic potential for the possibility of increasing the financing number of social activities, since an increase in the level of land profitability by 1 percentage point leads to an increase in the amount of social expenses per 1 employee by $42.59 (USD).

4. Discussion

The results of the conducted correlation-regression modelling give grounds for asserting that corporate social responsibility plays a decisive role in the formation of multifunctional rural development. Its successful implementation in the life of rural communities depends on the sequence and intensity of the implementation of interrelated measures:

1. The basis for the formation of a socially responsible business is a relationship of trust between interested parties. Based on the principles of trust, the foundation is created for the optimal combination and rational use of available resources (natural, economic, ecological, technological and social), certain opportunities (that is the choice of the type and direction of professional activity) and tools (legal, institutional and financial) to achieve strategic and tactical goals of entrepreneurial activity in rural areas.

2. Increasing the social responsibility of stakeholders in rural development should be primarily aimed at creating the necessary conditions to ensure the quality of life and socio-economic well-being of the rural population. This will make it possible to more actively attract highly qualified specialists (including citizens who are temporarily abroad) to rural areas with the simultaneous implementation of organizational, infrastructural, technological, environmental and other projects aimed at comprehensive diversification of the potential and existing development opportunities to improve the social life of villagers. In the post-war reconstruction of Ukraine, rural areas should be considered as special locations, created for the restoration and rehabilitation of citizens and the establishment of small and medium-sized agribusinesses by them.

3. Social partnership should be recognized as a creative option for the interaction of participants in rural society. It is based on social dialogue, responsibility, constructivism and mutual respect between all interested stakeholders. The process of broad implementation of social partnership into the agenda of the territorial communities functioning involves: creating

conditions for the development of various forms of ownership in villages, increasing labour productivity thanks to the introduction of innovative methods of organization and business conduct (technological, financial and economic, managerial, etc.) in rural areas, stimulating and motivating villagers towards self-realization, expanding their opportunities to master educational, informational, communication and cultural spaces, forming positive social capital (i.e. public society), as well as promoting inclusive rural development based on the creation of new economic opportunities and equal access to the benefits of civilization for all types of citizens who live in rural areas. Social partnership and social responsibility should be integrated into most business processes of business entities. Such measures will stimulate the reduction of social tension in society and increase moral and ethical norms of behaviour in the process of forming strategic and tactical decisions with the aim of ensuring multi-vector rural development.

4. The social responsibility of business can be realized through the diversification of sources of financial support for the development of rural areas. Streamlining the corporate social responsibility financing system contributes to the sustainable development of the country, increasing the level of education, health care, environmental protection and rural development (Singh et al., 2023). Considering international experience, it is worth following the principle “who can pay”. This principle is the basis of the progressive taxation system. We share the position of Borodina et al. (2020) regarding the feasibility of introducing per-hectare payments for the accumulation of financial resources in the Rural Development Fund. The payers of such payments should mostly be large land users (agroholdings) that use more than 5,000 hectares of agricultural land. In addition, it is worth increasing inclusiveness, that is, the accessibility of rural residents to various financial instruments, including investment and credit resources, as well as microinsurance and insurance. It is worth noting that the trajectory of international financial flows is increasingly directed towards the development of “green” and information and communication technologies, which are implemented in the model of digitalization and social guarantees (Sribna et al., 2023). Thus, the implementation of the outlined measures should contribute to the achievement of spatial and social cohesion in the process of rural development.

5. Over the last decade, there has been a rapid development of the digital economy, associated with the use of the latest technologies, improvement of the information and communication environment, opportunities for innovative decision-making, including the vital activities of

rural areas. Instead, the rapid digitalization of socially significant processes causes the generation of certain risks, and this leads to the objective necessity of strengthening social responsibility in the field of modern technology use. According to Art. 6 of the Law of Ukraine “On the Basic Principles of Ensuring Cybersecurity in Ukraine” (2017), the food production industry and agriculture are classified as critical infrastructure objects, which indicates the need to increase the responsibility of all stakeholders for possible cyber-attacks and cyber threats in rural areas. It is believed that the growth of social responsibility of entities whose professional activity relates to the use of modern information and communication technologies is an adequate and timely solution for effective countermeasures against probable cyber threats in the current period, as well as in the future.

Therefore, further development of rural areas in Ukraine requires increasing social responsibility in almost all spheres of rural livelihoods. In the case of non-implementation (or incomplete implementation) of the proposed measures, destructive processes of an economic, social and ecological nature may intensify, which is undesirable given the complexity of Ukraine’s post-war recovery.

CONCLUSIONS

The results of the conducted research give grounds for asserting that modern civilizational challenges (economic, ecological, political, food, social) have formed an objective demand of society for increased responsibility in making management decisions in the short-term, medium-term and long-term waiting horizon. In the process of research, it was established that the versatility of rural development depends on the social activity of all stakeholders interested in improving the quality of life of rural residents.

In the process of research, a positive trend was established regarding the increase in the amount of social expenses of economic entities in the field of agriculture by almost 2.92 times for the period 2014–2022, which indicates the readiness of agribusiness to make socially responsible decisions in the context of sustainable development of rural areas. Based on the results of correlation-regression modelling, an impact assessment of the financial and economic indicators of large land users on their social activity was carried out. The specified model made it possible to identify the existence of a close relationship between the active social position of agricultural holdings and parameters, such as capital profitability, land profitability and the amount of net income per 1 ha of agricultural land.

On the other hand, it should be noted that there is significant untapped potential for financing social projects in rural areas. This is due to several main reasons: relatively low motivation for social activity; the presence of various risks (institutional, economic, financial, etc.) that objectively restrain the process of implementing social and environmental initiatives in rural areas; military aggression of the Russian Federation against Ukraine, especially the temporarily occupied territories. Research results indicate that land capital should be considered as a significant reserve for further expansion of the financing of social activities. This conclusion is based on the calculations, in particular, that an increase in the level of land profitability by 1 percentage point ensures an increase in the amount of social expenses by $42.59 (USD) per 1 employee.

The deep awareness of the need for multifaceted rural development in Ukraine gives grounds to testify that increasing the social responsibility of interested parties will contribute to solving the following problems, namely: the combination of moral-ethical and socio-economic components of rural development; increasing the quality of life in rural areas; development of social partnership, strengthening digitization processes in rural areas, etc. The expected basic effect of the systemic interaction of the constituent elements of corporate social responsibility should be recognized as ensuring a deep transformation of values aimed at the formation of a socially oriented model of the economy.

Strengthening the role of social responsibility in various spheres of life in rural areas requires diversification of the sources of their financial support. In this context, it is expedient to introduce a system of financial and economic factors for medium and large land users to increase their interest in investing resources in the development of social infrastructure in rural areas. We consider it necessary to further expand financial inclusion for residents of rural areas, that is, access to investment and credit resources, payment instruments, savings and insurance. Therefore, the positive impact of social responsibility on the development of rural areas in Ukraine should contribute to the increase of spatial and territorial justice and, consequently, to the gradual increase in the well-being of rural residents.

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CONTENTS

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PERCEPTIONS ABOUT IMPLEMENTATION OF SERVICE ROBOTS IN HOSPITALITY BUSINESSES IN KAZAKHSTAN AND RUSSIA - COMPARATIVE ANALYSIS

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Alexandr Koichubayev[3]

Abstract: Our goal is to assess the understanding and acceptance of robotization in hospitality, based on research done in the two countries. This research applies a survey method to collect respondents' opinion of hospitality employees vs hospitality customers. The main sampling frames are formed by employees of hotels, restaurants and other hospitality businesses. Five hypotheses are studied and tested for the two categories of respondents in each country. The results from the statistical analysis differ for the studied countries, and between the two studied categories of respondents. The study confirms the expectations of improving the quality and increasing the effectiveness of robotizing the hospitality businesses. This research adds knowledge and understanding of how the hospitality employees and customers in Kazakhstan and Russia consider the implementation of service robots. The statistics show that the findings have value and can be used by hospitality managers when they consider possible robotization.

Keywords: hospitality, perceptions, robotization, quality, effectiveness
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INTRODUCTION

The difference between the consumers' perception in Kazakhstan and Russia about robotization has not been studied much, with a few exceptions (e.g., Ivanov, Webster and Garenko, 2018), although the robotization and AI implementation in general needs to be analyzed because of the different effects and consequences of that process.

The objectives of this study are to analyze on comparative basis the two countries – Kazakhstan and Russia, with the aim to find answers to questions that are important for the management of the hospitality businesses, including:

1. The respondents' (consumers' and hospitality businesses employees') perception about being served by service robots in the hospitality businesses, especially in the hotels and restaurants.
2. The respondents' anxiety of losing their jobs in case of intensive robotization.
3. The changes in business efficiency as perceived by the respondents.
4. The respondents' perception about the quality of services after possible robotization.

It is important to acknowledge that the development and use of human-like robots - both humanoid (resembling humans) and android (machines with human-like movement and speech) - may initially have a mixed impact. While these innovations could positively influence customer perceptions over time, they may also provoke some negative reactions, especially at the early stages of adoption. e.g., Uncanny valley effect (So et al., 2023) as well. This raises questions about the limitations of any research on this topic, including this study.

The contribution of this study of Kazakhstan and Russia is expected to be in answering the following questions: 'Do users/customers of hospitality services perceive implementation of service robots of different types as a positive idea?', 'Will there be improvement of business efficiency and quality of services?' Another important issue for research is whether the implementation of service robots create some level of anxiety among customers. Overall, this study makes the following contributions:

This comparative study aims to clarify the difference in service quality and efficiency, between the traditional human service staff and service robot from the perspective of respondents.

Second, this research aims to examine anxiety as a possible effect of robotization in case that the respondents expect changes in their employment status, as this is a risk factor in the process.

The discussion on that is expected to help managers decide to what extent to invest in robotization, which technically is an excellent substitute for human staff, but can lead to unwanted marketing and economic effects if customers do not approve of such changes.

THEORETICAL BACKGROUND

As already discussed above, the consumers' perception about being served by service robots in different businesses, including hospitality businesses, especially in the hotels and restaurants, has been studied in many publications, e.g., So et al., (2022). The spectrum of publications includes analysis of the active use of artificial intelligence (AI), including augmented reality, virtual reality (e.g., Grundner & Neuhofer, 2021; Verma et al., 2022) and robots used for repetitive, leading to stress and tiredness operations (e.g., Kim et al., 2023) studying hotels, including luggage and room services provided by robots in hotels, and the food and beverage industry (Mende et al., 2019). The results of this research contribute to improving service quality and the service experience.

In 2023 and 2024 only, dozens of excellent research-based papers were published, e.g., Liu et al, 2024; Pitardi et al, 2024; Rasul et al, 2024; Soliman et al, 2024; Tekeli, Kemer & Tekeli, 2024; Wirtz et al, 2024. Most of them analyze developments and trends in accordance with the ideas for mass replacement of the human factor with robots and robotized systems.

So, these papers study the robotization in the hospitality industry, analyzing it from different angles, from improving the quality of services (e.g., Mingotto et al., 2020; Jabeen et al., 2022; Yang and Zhang, 2022), to consumer behaviour toward service robots (e.g., Soliman et al., 2024), the impacts of robot anthropomorphism on consumers' trust, receptivity and satisfaction (e.g., So et al., 2023), and even the possibilities for offering ChatGPT services (AI based) in cultural tourism (Sigala et al., 2024). Addressing the new demand of some of the younger customers to receive modern service, and related customer engagement has been studied as well (e.g., Rasul et al., 2024). Another topic is analyzing the difference between the customers' views on older automated self-service technologies versus service robots (Wirtz et al., 2024). Tekeli et al. (2024) analyze the differences in the employees' attitude to robotization in relation to their education, position and the department where they work. These are all important viewpoints on customers' perception of service robots, as the economic and therefore financial results of robotization are achieved based on the consumer liking or disliking of the new services they get. In many of the recent publications robotization is analyzed from demand-side perspective (e.g. Pizam et al., 2024; Soliman et al., 2024; Moriuchi and Murdy, 2024). Song et al. (2024), Pitardi et al., (2024), and Saputra et al. (2024) go further analyzing the customers' intention to continue to use the services of hotels based on their feelings about the anthropomorphism and perceived intelligence they see, or feel being served by robots. It is clear that some customers will feel anxiety and discomfort when they are served by robots or AI systems, which they define for themselves as robots (e.g., Soliman et al., 2024; Wang et al., 2023). The effect of services done by robots on customers' emotional and behavioral responses (e.g., Bagozzi et al., 2022; Kim et al., 2023), and the metaperception effect on the customer response (Pitardi et al., 2024) have been analyzed in research-based papers. This research extends previous work by exploring the broader influence of general technology perceptions on attitudes towards service robots.

Based on their different education, experience and cultural characteristics the customers understand and interpret the service robots in a different way. The robots, in particular service robots, no matter how defined, e.g., as intelligent physical devices or intelligent physical devices with a certain degree of autonomy, or anthropomorphism-based AI robots (Saputra et al., 2024), are often interpreted by the customers in a relatively simple way.

Previous research indicates that hospitality clients generally accept service robots, hold positive views about them, accept using them and are willing to pay for services from businesses employing robots (e.g., Wang et al., 2022). Tourists perceive robots as innovative and appealing devices which add to service quality (Belanche et al., 2020).

Research has also highlighted the influence of anxiety caused by service by robots and the specific emotions (e.g., Wang et al., 2023) on consumer attitudes and behaviour towards service robots. Although the role of robotic service expectations in determining readiness to pay for robot services has been studied, less research is done on how anxiety and negative emotions influence customer intentions, especially the cognitive and emotional responses.

The researchers have also emphasized the need for further investigation into how robots affect service quality and customer satisfaction, which is related to customer perceptions of added value. Rasul et al. (2024), based on 73 independent studies with a total sample size of 41,757 respondents, underline that customer/tourist experience (CE) has a positive impact on both tourist satisfaction and hedonic value, but it is not clear to what extent, if at all, the service robots lead to tourist satisfaction, and if they add hedonic value, which refers to the emotional state of an individual who experiences pleasure or excitement from the product. Schepers et al. (2022) emphasize that robots have the potential to increase customer satisfaction across various types of services, from low-cost options that rely on mechanical robots to full-service providers. However, this remains a hypothesis for now. Several studies have explored robotization in the two countries (e.g., Gasumova & Porter, 2019; Starovatova, 2023; Shustova, Blagoev & Zhuk, 2024), focusing on the shift from managing human workers to managing organizations using robots. Ivanov, Webster & Garenko (2018) have examined customer perceptions of robots in Russian restaurants and hotels. The authors have found that while preferences for human employees over robots in hotels are relatively high (Mean 4.18 out of 5), it has been found that robotized service is still viewed as an interesting experience (Mean 4.04). The research has also explored how respondent gender and general attitudes towards new technologies influence these preferences. The main tasks where service robots were favoured included luggage handling (Mean 4.32), fulfilling customer requests for items like towels (Mean 4.22), and processing payments (Mean 4.12 for card payments, 4.02 for cash). Robots were also appreciated for providing information about hotel services (Mean 4.06) and local destinations (Mean 3.98).

In terms of service quality, there is a distinction between machine-type robots that perform specific tasks without resembling humans and human-looking (anthropomorphic) robots, specially designed to look and behave more like humans. Such humanoid robots can meaningfully interact with customers, which improves the quality of service. Mende et al. (2019) and others argue that anthropomorphism is a preferred trait in service robots, as it facilitates easier customer interaction, improving their perception of service quality. For instance, Sheehan et al. (2020) have found that anthropomorphism boosts fondness for both the brand and product. However, other researchers warn that robots which look too human-like could provoke negative reactions from customers who sees them as a threat to human identity (Mende et al., 2019). Over time, as humanoid robots become more prevalent, it is expected that people will become accustomed to them and may even prefer such designs. The “uncanny valley” theory (So et al., 2023) suggests that there may be a U-shaped relationship between customer comfort and robots which imitate humans (e.g., Crolic et al., 2022). So et al. (2023), for example, have examined the impact of robot anthropomorphism on consumer satisfaction and while Huang et al. (2021) have argued that customers typically respond positively to more human-like robots, experiencing little discomfort during interactions.

Metaperception

Our respondents in Kazakhstan and Russia have limited personal experience with anthropomorphic service robots in hospitality, but they have good experience with AI services. Therefore, we must analyze the meta perception as a possible drive in forming their perceptions. We have tried to analyze the impact of robots on customers' behaviour and emotional responses (e.g., Bagozzi et al., 2022; Pitardi et al., 2024). In our case, the possible feelings of discomfort (e.g., Liu et al., 2023) are of interest, as our research shows a difficult-to-explain combination of respondents' liking of robotization with their anxiety about losing their jobs as a result of robotization. Metaperception is the process and effect on people evaluating the impressions that others may form about them and their behaviour. Pitardi et all. (2024) claim that when customers experience discomfort, their perception that service robots cannot think or form opinions will form a perception (i.e., metaperceptions) of robots only as service assistants and thus reduce the discomfort experienced, compared to that in case of being served by human beings. Becker et al. (2023) have found that "human-like robots can increase comfort, leading to more positive service outcomes", but this will probably be different for customers who find those robots too human, and because of that "dangerous". If there is a positive metaperception, this will obviously lead to higher intentions of re-use and re-visit.

In conclusion, based on the substantial previous research and the discussions above, it is clear that hospitality clients generally accept service robots, maintain positive views of them and value positively being served by robots and robotized systems. Robots have the potential to increase customer satisfaction across a variety of services, from low-cost options that rely on mechanical robots to full-service providers.

METHODOLOGY

This research adopts a demand-side perspective, e.g. in Moriuchi and Murdy (2024), as well as in Song et al. (2024) and Pizam et al. (2024). The focus on tourists' perceptions is based on the opinion of customers about being served by robots.

With all this in mind, we have decided to analyze the following hypotheses, comparing the findings in the two countries:

H1: The respondents believe that the hospitality customers prefer being served by robots instead of human employees.

H2: The respondents support the intensive implementation of service robots in the hotels and restaurants, as well as in other hospitality businesses.

H3: The respondents are not afraid of losing their jobs after robotization.

H4: The respondents believe that the effectiveness of hospitality businesses will increase based on implementation of service robots.

H5: The respondents believe that the service robots will increase the quality of provided hospitality services.

Research design

Data collection took place between March 2023 and April 2024, using a survey administered via email to respondents within the research population from the two countries under study ‐ Russia and Kazakhstan. A non-probability sampling method was employed. The research population comprised individuals over 18 years old residing in Kazakhstan and Russia. The two samples included 251 respondents in Russia, and 337 in Kazakhstan.

Data collection and analysis

The questionnaire was designed in Google Form. It consists of three main sections – demographics, questions related to the studied hypotheses, and others, e.g. 'Does the company of the respondent use AI and service robots?'. The questionnaire was emailed to the managers, as well as to contact emails of the hotels, restaurants and other hospitality businesses, as well as to alumni, colleagues and friends who were active users of hospitality services. Google Forms was used for collecting the primary data from the respondents.

Different statistics were used to analyze the findings, including Descriptive statistics and Correlation analysis.

Limitations, Reliability, and Validity of the results

The reliability and validity of this study has been limited by the small number of cities (districts) of respondents surveyed – Semey, Astana and Pavlodar in Kazakhstan, and St. Petersburg in Russia. It is clear that these numbers cannot represent correctly enough the opinion of the population in the two countries. In addition, the number of employees in hospitality businesses in the two countries is small (136 in Kazakhstan, and 78 in Russia, and although the statistical results should be taken as indicative, they cannot be considered fully representative for that population. Still, the findings give interesting results, which can be used by managers in case they consider robotization in the hospitality businesses they manage.

Ethics

The questionnaire begins with a consent section that explains the purposes of the survey and the way the responses will be processed. Adding to this the use of Google Forms to process the responses ensures that no ethical rules and principles have been violated, and the findings of this study should be considered as valid.

FINDINGS

The statistical analysis of the survey results is made with the aim to analyze the stated hypotheses.

Hypothesis 1

The first hypothesis is that the respondents believe that **the hospitality customers prefer being served by robots instead of human employees**. The findings of this study show the following results (Table 1):

Table 1.
Respondents working in the hospitality businesses believe that customers prefer to be served by people

CUSTOMERS PREFER HUMAN SERVICE (% employees in hospitality businesses)	**KAZAKHSTAN**	**RUSSIA**
	71.3	**84.3**
Mean	2.14	2.53
Standard Error	0.10	0.08
Median	2	3
Mode	3	3
Standard Deviation	0.84	0.65

As Table 1 shows, the vast majority of hospitality employees believe that their customers prefer to be served by humans rather than by service robots. The results comparing the views of hotel employees with other respondents (Table 2) are similar for Russia and different for Kazakhstan. Over 77% of other respondents in Kazakhstani believe that the customers prefer to be served by humans versus 71.3% of hospitality employees.

Overall, the vast majority of the respondents believe that customers prefer to be served by humans rather than service robots.

Table 2.
Respondents' perspective on hospitality customers' perception of being served by people

KAZAKHSTAN Hospitality employees vs Other respondents			**RUSSIA** Hospitality employees vs Other respondents		
HUMAN services preferred	**Hospitality employees**	**OTHER respondents**	**HUMAN services preferred**	**Hospitality employees**	**OTHER respondents**
	71.3	**77.5**		**84.3**	**82.4**
Mean	2.14	2.33	Mean	2.53	2.47
Standard Error	0.10	0.06	Standard Error	0.08	0.10

Thus, H1 is rejected. The respondents believe that the customers in the hospitality businesses do not expect and do not prefer to be served by service robots.

Hypothesis 2

The second hypothesis H2 is that **the respondents support the intensive implementation of service robots in hotels and restaurants**, as well as in other hospitality businesses. Obviously, this hypothesis should be expected to be rejected according to the findings on H1. However, the survey shows different results (Table 3):

Table 3.
Respondents' view on implementing service robots in the hospitality businesses

KAZAKHSTAN Implement service robots Hospitality employees vs Other respondents			**RUSSIA Implement service robots** Hospitality employees vs Other respondents		
SERVICE ROBOTS to be implemented	**Hospitality employees**	**OTHER respondents**	**SERVICE ROBOTS to be implemented**	**Hospitality employees**	**OTHER respondents**
	62	**59.7**		**60.7**	**64.4**
Mean	1.86	1.79	Mean	1.82	1.93
Standard Error	0.10	0.06	Standard Error	0.10	0.10

The findings of this survey confirm H2, although they totally neglect the findings on H1. The same respondents, who believe that the customers prefer to be served by humans, strongly support the implementation of service robots. The statistics for hospitality employees and other respondents are similar, and in Kazakhstan the hospitality employees support the robotization even more than the other respondents. For the two studied samples (countries) the results do not differ much.

Hypothesis 3

Hypothesis 3 states that **the respondents are not afraid of losing their jobs** as result of robotization.

Table 4.
Respondents' view on the possibility of losing their jobs (in hospitality businesses) as a result of implementing service robots

KAZAKHSTAN Possible losing their jobs after robotization Hospitality employees vs Other respondents			**RUSSIA Possible losing their jobs after robotization** Hospitality employees vs Other respondents		
Lose their jobs after robotization	**Hospitality employees**	**OTHER respondents**	**Lose their jobs after robotization**	**Hospitality employees**	**OTHER respondents**
% agree	45.8	57.2	**% agree**	60.6	50.9
Mean	1.38	1.72	Mean	1.82	1.53
Standard Error	0.08	0.06	Standard Error	0.07	0.10

The findings in Table 4 are different for the two countries. The hospitality employees in Russia are much more afraid of losing their jobs after robotization compared to their colleagues in Kazakhstan (60.6% vs 45.8%), while the findings from the other respondents are different in the other direction (50.9% in Russia vs 57.2% in Kazakhstan). There is no clear explanation for these differences, except for possible differences in the management of hospitality businesses in the two countries. It is possible that in Russia the employees in hospitality businesses are more likely to openly discuss the possible consequences of robotization, while in Kazakhstan they assume that it is an expensive process and customers will exert pressure in favour of human service. Overall, the findings reject Hypothesis 3, as the majority of respondents are afraid of losing their jobs as a result of robotization.

Hypothesis 4

Hypothesis 4 states that **the effectiveness of hospitality businesses will increase** based on implementation of service robots.

The findings of this survey show the following results (Table 5):

Table 5.
Respondents' view on the effectiveness of hospitality businesses as a result of implementing service robots

KAZAKHSTAN Change of the effectiveness Hospitality employees vs Other respondents			**RUSSIA Change of the effectiveness** Hospitality employees vs Other respondents		
EFFECTIVENESS Increased	**Hospitality employees**	**OTHER respondents**	**EFFECTIVENESS Increased**	**Hospitality employees**	**OTHER respondents**
% agree	*65.7*	*68*	**% agree**	71.3	73.1
Mean	1.97	2.04	Mean	2.14	2.19
Standard Error	0.11	0.06	Standard Error	0.07	0.11

Table 6 shows similar results on the respondents' perception of possible increase in effectiveness. The Russian respondents expect higher results, but the difference with the Kazakhstani respondents is not significant. Altogether, H4 is confirmed.

Hypothesis 5

Hypothesis 5 states that the service robots will increase the quality of provided hospitality services.

Table 6.
Respondents' view on the change of quality as a result of implementing service robots

KAZAKHSTAN Change of the quality Hospitality employees vs Other respondents			**RUSSIA Change of the quality** Hospitality employees vs Other respondents		
QUALITY Increased	**Hospitality employees**	**OTHER respondents**	**QUALITY Increased**	**Hospitality employees**	**OTHER respondents**
% agree	*62*	*6.,7*	**% agree**	70.8	73.6
Mean	1.86	1.85	Mean	2.13	2.21
Standard Error	0.10	0.06	Standard Error	0.10	0.10

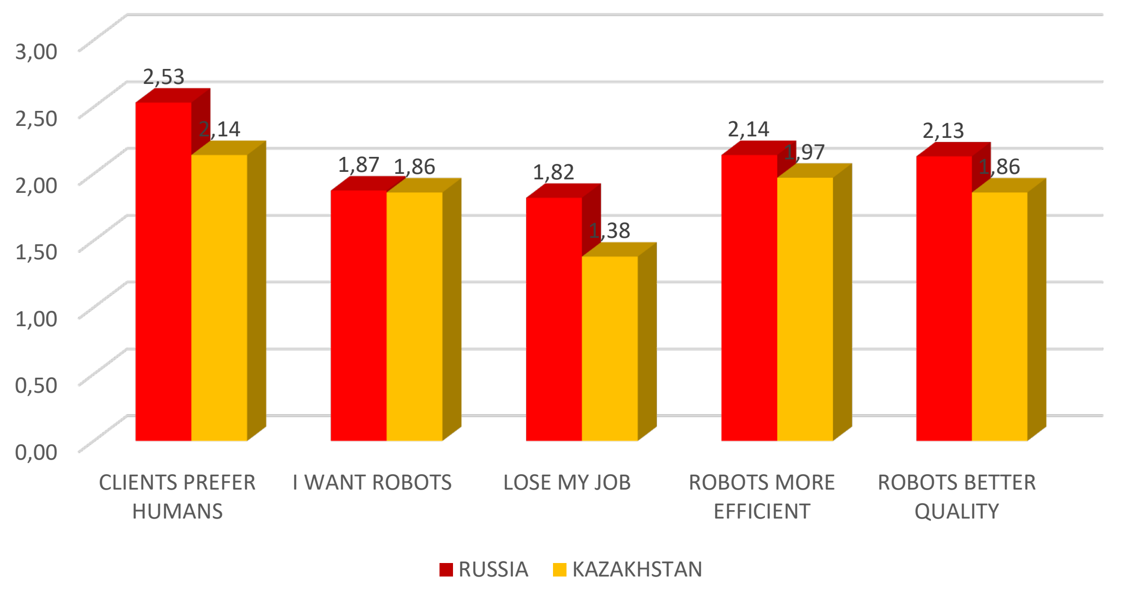
The findings of this study (Table 6) show that generally the respondents support this view, with respondents in Russia showing significantly higher expectations (70.8% in Russia vs 62% in Kazakhstan for the hospitality employees, and 73.6% vs 61.7% for the other respondents). There is no clear explanation what causes this difference. It is possible that the current experience of the respondents with AI-related services shows an increase in service times compared to previous human services, which the respondents interpret as decreased quality of service. Overall, Hypothesis 5 is confirmed. However, we must take into account the fact that about 30% of the respondents in Russia and about 40% of the respondents in Kazakhstan do not expect an increase in the quality of services in the hospitality industry after the implementation of service robots.

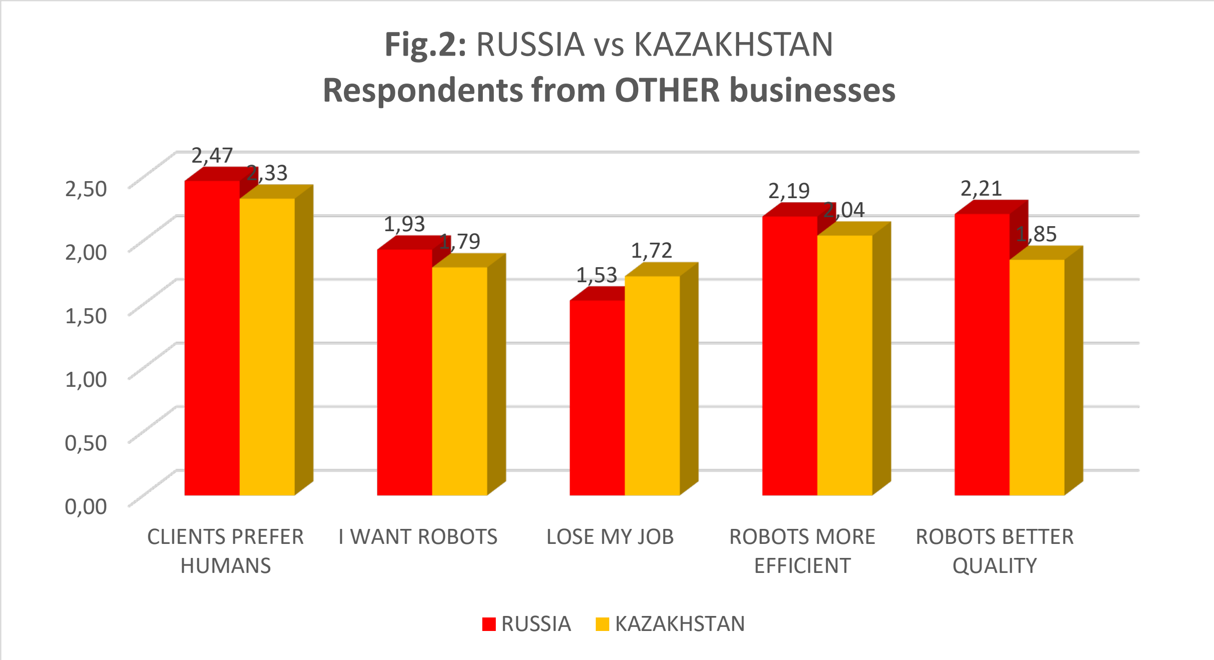
The results of H4 and H5 show that the Russian respondents are more positive about the implementation and use of AI and robots in services in the hospitality businesses.

DISCUSSION

The findings can be illustrated as shown in Fig.1 and Fig. 2. The most interesting element is the differences between the perceptions of hospitality employees and other respondents in the samples in Kazakhstan and Russia. Tables 4, 5 and 6 show a difference in the opinions of these respondents. As discussed above, this may be a result of different management in the two countries, as well as different perceptions of the respondents about the possible influence of the customers who may prefer human service. Finally, we have to take into account the differences between the national cultures.

Fig.1: RUSSIA vs KAZAKHSTAN
Employees in hospitality businesses





The most interesting and significant correlations are between the support for implementation of service robots in the hospitality businesses and the concern about possible job losses. The employees in hospitality in Russia have higher expectations of losing their jobs as result of robotization (Fig.1, mean 1.82 vs 1.38, stand. dev. 0.89 vs 0.70) compared to Kazakhstani respondents.

Conclusions

The research findings can be summarized in the following conclusions:

1. The service robots and AI technologies in general, have entered all areas of business activities, including the hospitality businesses in the two countries studied. This research shows that 42% of the respondents in Kazakhstan report using AI and robots in their companies, while in Russia about 28% claim the same. Obviously, this difference has influenced the responses. It can be said that the respondents in Kazakhstan have a better knowledge about the positive and negative aspects of using robots compared to the respondents in Russia.

2. The first hypothesis - the respondents believe that hospitality customers prefer being served by robots instead of human employees - was rejected (Table 2). The survey shows the opposite results. The respondents believe that the customers in the hospitality businesses do not expect and do not prefer to be served by service robots.

3. The second hypothesis is confirmed - the respondents support the intensive implementation of service robots in hotels and restaurants, as well as in other hospitality businesses (Table 3).

4. Hypothesis 3 states that the respondents are not afraid of losing their jobs as a result of robotization. The results of the survey reject this hypothesis (Table 4). Interestingly, the findings for the employees show 45.8% in Kazakhstan vs 60.6% for Russia, while the results for the clients' perception are opposite: 57.2% in Kazakhstan vs 50.9% in Russia.

5. Hypothesis 4 - the effectiveness of hospitality businesses will increase based on implementation of service robots - is confirmed (Table 5).

6. Hypothesis 5 - the service robots will increase the quality of provided hospitality services, is also confirmed (Table 6).

7. The results on H4 and H5 show that the Russian respondents are more positive about the implementation and use of AI and robots in services in the hospitality businesses.

Based on the results of this research we can say that the use of service robots and AI, in general, will increase the efficiency of companies operating in various sectors of the economy. It is expected that labour costs will be reduced, work processes will be intensified, in service enterprises the quality of services for the population will improve, and in industrial enterprises the quality of products will also improve. Further research on the topic, especially adding other countries, and analyzing the responses of larger samples will add important information to help make appropriate decisions when implementing AI and robots.

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CONTENTS

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THE IMPACT OF DIGITAL TRANSFORMATION, CORPORATE SOCIAL RESPONSIBILITY AND MARKET ORIENTATION ON FIRM PERFORMANCE: EVIDENCE FROM VIETNAMESE COMMERCIAL BANKS

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Abstract: Digital Transformation (DT) has become a critical factor for maintaining competitiveness in the financial sector. However, the interplay between Market Orientation (MO), Corporate Social Responsibility (CSR), DT, and their combined impact on Firm Performance (FP) remains underexplored, particularly in emerging markets such as Vietnam. This study addresses this gap by examining the relationships in Vietnam's financial industry. The structural equation modeling (SEM) approach was applied to analyze 250 valid responses collected from 2,411 commercial bank branches in Southeast Vietnam. Stratified sampling and snowball sampling methods ensured regional representation. The findings reveal that DT directly enhances FP and acts as a key mediator, strengthening the effects of MO and CSR. DT facilitates real-time responsiveness, enhances transparency, and integrates technological advances with market and social strategies. The study highlights the strategic importance of embedding DT into CSR and MO frameworks to optimize efficiency, achieve sustainable development, and create competitive advantages in an increasingly digitized financial sector.

Keywords: CSR, market orientation, digital transformation, firm performance, banking in Vietnam

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1. Introduction

In an era of rapid technological advancement and global integration, Vietnam's banking sector faces increasing pressure to improve operational efficiency and sustainability. As a key pillar of the economy, banks not only provide essential capital but also drive digital transformation across various industries. The adoption of digital technology has advanced significantly, with over 90% of transactions being conducted online and nearly half of customers using electronic payment services by the end of 2023 (State Bank of Vietnam, 2024). The Fourth Industrial Revolution presents both opportunities and challenges, positioning Digital Transformation as a critical requirement for maintaining competitiveness. Although studies highlight DT’s role in optimizing processes and driving business performance, its interplay with Corporate Social Responsibility and Market Orientation remains underexplored. Existing research underscores MO’s focus on market intelligence and strategic responsiveness and CSR’s commitment to ethical and societal goals, yet their combined synergy under DT’s influence has not been comprehensively examined (Kohli, 2017).

This study addresses this gap by exploring how Digital Transformation mediates and amplifies the impact of CSR and market orientation on firm performance. Through a conceptual model based on Resource-Based View and Stakeholder Theory, the study employs Partial Least Squares Structural Equation Modeling (PLS-SEM) to validate hypotheses using data from 250 valid responses gathered from commercial bank branches in Vietnam. By integrating these frameworks, the research offers actionable insights for leveraging DT to enhance both competitive advantage and sustainable development in the rapidly digitizing banking sector.

2. Literature Review

Stakeholder Theory, introduced by Freeman (1984), broadens corporate responsibility beyond shareholders to encompass all stakeholders affected by business activities. This perspective highlights the strategic role of Corporate Social Responsibility in managing stakeholder relationships, thereby fostering customer satisfaction, loyalty, and ultimately, financial performance (Li et al., (2023). CSR initiatives not only enhance ethical and legal compliance, but also strengthen corporate reputation and contribute to long-term competitive advantages through sustainable stakeholder engagement. The Resource-Based View (RBV), proposed by Barney, (1991), underscores the importance of internal resources such as technology, managerial capabilities, and brand equity—in achieving sustainable competitive advantage. The integration of Digital Transformation and Market Orientation (MO) enhances these resources by optimizing operational efficiency, fostering innovation, and improving customer responsiveness (Cheng et al., (2023). These advancements are particularly crucial in highly competitive markets, where differentiation through technology and strategic orientation is key.

CSR, defined as a firm's commitment to balancing economic, legal, ethical, and societal responsibilities (Carroll, 1999), positively influences firm performance by building trust and loyalty among stakeholders. Recent research shows that CSR strengthens brand reputation and financial performance through enhanced stakeholder relationships and ethical practices (Li et al., 2023). MO, which involves generating, disseminating, and responding to market intelligence, significantly improves operational efficiency and competitiveness by aligning organizational strategies with customer needs and market dynamics (Kohli & Jaworski, 1990).

Digital transformation, defined as the integration of digital technologies across all aspects of business operations, drives innovation, operational efficiency, and customer satisfaction (Karimi & Walter, 2015). In banking, Digital transformation plays a transformative role by automating manual operations, streamlining processes, and enhancing fraud detection capabilities through advanced technologies like Artificial Intelligence (AI) and data analytics (Cheng et al., (2023). These technological advancements enable firms to deliver personalized customer experiences, improve decision-making, and maintain competitiveness in a rapidly evolving digital economy.

In Vietnam, rapid industrialization and modernization have positioned Digital transformation as a critical driver of economic progress. The banking and financial sectors have spearheaded this transformation, leveraging mobile banking, e-commerce, and digital payment systems to enhance accessibility and operational efficiency (Pham, 2023; State Bank of Vietnam, 2024). However, challenges such as high investment costs, cybersecurity issues, and inconsistent customer data management hinder the widespread adoption of digital technologies.

Despite these barriers, the synergistic effects of digital transformation, market orientation, and CSR present significant opportunities for enhancing firm performance. Digital transformation acts as a mediator, amplifying the impact of market orientation by enabling real-time responsiveness and aligning customer insights with innovative solutions. Simultaneously, digital transformation optimizes CSR initiatives by improving transparency, automating processes, and fostering stakeholder engagement.

This study addresses a critical research gap by examining how digital transformation mediates the relationship between market orientation, CSR, and firm performance in Vietnam's financial sector. The findings provide actionable insights for leveraging digital transformation as a strategic tool to enhance market responsiveness, stakeholder trust, and sustainable business growth. By integrating MO’s market intelligence, CSR’s societal commitments and DT’s technological advancements, the study offers a comprehensive framework for achieving competitive advantages and driving transformative outcomes in an increasingly digitalized global economy.

3. Hypotheses

3.1 Corporate Social Responsibility Positively Impacts Firm Performance

CSR is recognized as both an ethical obligation and a strategic tool that enhances financial performance and operational efficiency. According to Stakeholder Theory, CSR fosters strong relationships with stakeholders, driving loyalty, reputation and reduced business risks, which directly improve firm performance. Empirical studies, such as Li et al. (2023), demonstrate that companies with robust CSR strategies achieve higher revenue growth and profitability by attracting socially conscious investors and enhancing brand credibility. Additionally, CSR reduces costs linked to legal and ethical risks by ensuring compliance with regulations and societal norms, improving operational efficiency. Research also highlights CSR's role in driving innovation and value creation, as noted by Chernev and Blair (2015), who observed its positive influence on corporate reputation and customer perceptions. Furthermore, CSR enhances internal collaboration and employee motivation, improving productivity and service quality, as emphasized by Weber (2008) and Hsu (2012).

CSR has long been regarded not only as an ethical obligation but also as a strategic tool that delivers financial benefits and sustainable operational efficiency. According to Stakeholder Theory, CSR enables firms to establish strong relationships with stakeholders such as customers, suppliers, and communities. This, in turn, fosters loyalty, enhances reputation, and minimizes business risks factors that directly contribute to improved firm performance.

Empirical studies have demonstrated that CSR enhances financial performance by elevating corporate image and building trust among stakeholders. Li et al., (2023) found that companies with well-defined CSR strategies achieve higher revenue growth and profitability. They attribute this to CSR’s ability to attract socially and environmentally conscious investors while strengthening brand credibility. Additionally, CSR helps companies reduce costs associated with legal and ethical risks. By adhering to regulatory and societal norms, firms can avoid legal complications and improve operational efficiency. Some studies have also highlighted CSR's positive impact on innovation and value creation. For instance, Chernev & Blair, (2015) note that CSR not only enhances corporate reputation but also positively influences customer perceptions of a firm's products and services.

Moreover, CSR serves as a significant driver of performance by fostering internal interaction and collaboration within the organization. As highlighted by Weber, (2008) and Hsu, (2012), CSR motivates employees and strengthens organizational commitment, thereby improving labour productivity and service quality.

Based on these arguments, CSR emerges as not merely an ethical requirement but also a strategic instrument for enhancing firm performance. This forms the basis for proposing hypothesis:

H1: Corporate Social Responsibility positively impacts firm performance.

3.2 Digital Transformation as a Critical Mediator in the Relationship CSR and Firm Performance

According to Stakeholder Theory, CSR enhances brand reputation and strengthens relationships with customers and stakeholders. However, digital transformation optimizes these effects by digitizing processes, increasing transparency, and improving operational efficiency. Studies have shown that firms investing in CSR often adopt digital transformation to enhance interaction capabilities, automate CSR management processes, and leverage digital platforms to increase transparency in CSR activities, such as social responsibility reporting. Digital transformation serves as a crucial mediator in the relationship between CSR and Firm Performance, amplifying the impact of CSR initiatives through process digitalization, increased transparency, and operational efficiency. According to (Cheng et al., 2023), DT enhances CSR outcomes by automating reporting and facilitating stakeholder engagement, thereby improving financial and operational performance. Similarly, (Meng et al., 2022) emphasize that digitalization across value chains enables CSR activities to align more effectively with organizational and societal goals, strengthening their influence on firm performance. Innovation also emerges as a key factor, with (Ruggiero & Cupertino, 2018) highlighting that DT-driven innovation initiatives optimize CSR effectiveness and positively influence financial outcomes. Additionally, Liu & Jung, (2021) argue that digital transformation reinforces CSR authenticity by integrating transparency and traceability, which increases stakeholder trust and enhances performance metrics. In specific sectors like manufacturing and finance, Wang et al., (2022) and Al-Shammari et al., (2022) demonstrate that DT-powered CSR strategies result in cost efficiencies, improved visibility, and enhanced corporate reputation, which collectively contribute to superior firm performance.

The integration of CSR with digital transformation results in reduced operational costs while fostering customer trust and satisfaction, thereby optimizing the relationship between CSR and firm performance. Given these findings, Hypothesis H2 is proposed:

H2: Digital Transformation mediates the relationship between Corporate Social Responsibility and Firm Performance.

3.3 Market Orientation Positive Impact on Firm Performance

Market Orientation is defined as a firm's ability to generate, disseminate and respond to market information, enabling the optimization of business strategies and meeting customer demands (Kohli & Jaworski, 1990). Research has demonstrated that market orientation not only focuses on satisfying current customers but also anticipates future needs, enhancing a firm’s competitiveness and achieving superior performance. (Jaworski & Kohli, 1993) confirm that firms with effective market orientation strategies often achieve higher financial performance through improvements in revenue and profitability. This is further supported by (Chaudhary et al., 2023) who highlight that market orientation, when combined with strategic flexibility, enhances business performance, particularly in less competitive environments. Beyond financial performance, market orientation also fosters innovation. Gotteland et al., (2020) reveal that firms with strong market orientation capabilities are better equipped to develop new products and seize market opportunities, thereby achieve faster growth. Additionally, Lansiluoto et al., (2019) have found that MO-driven firms often implement Performance Measurement Systems (PMS), improving resource management and optimizing operations, which ultimately leads to better financial outcomes.

Apart from financial benefits, Appiah-Nimo & Chovancova, (2020) emphasize that market orientation supports firms in undertaking sustainable activities, meeting customer and societal demands for environmental and social responsibility. Market orientation enables firms to achieve higher performance through internal coordination, swift market responsiveness, and the promotion of innovative and sustainable practices. Based on this foundation, Hypothesis H3 is proposed as follows:

H3: Market Orientation positively impacts firm performance.

3.4 Digital Transformation as a Mediator in the Relationship between Market Orientation and Firm Performance

The pivotal role of digital transformation as a mediator between market orientation and Firm Performance is increasingly substantiated by academic research. Grounded in the RBV theory, organizations with robust market orientation leverage market intelligence to craft strategies and implement digital initiatives, thereby attaining competitive advantages (Barney, 1991). Market orientation facilitates the integration of technology into business operations, enhancing real-time responsiveness to customer needs and market dynamics (Cheng et al., 2023; Kohli & Jaworski, 1990).

As an intermediary, digital transformation significantly boosts organizational performance by automating processes, enhancing productivity and reducing costs (Karimi & Walter, 2015). Moreover, digital transformation fosters operational transparency and fortifies customer relationships in market-oriented firms, yielding improved financial and operational outcomes (Meng et al., 2022). Firms combining DT with MO exhibit superior agility, enabling them to navigate market fluctuations and capitalize on sustainable growth opportunities (Gangi et al., 2019).

Empirical studies reinforce this synergy between MO and DT in driving firm performance. (Abbu & Gopalakrishna, 2021) demonstrate that market orientation enhances internalization and operational processes, which, when amplified by digital transformation, lead to superior financial and customer service outcomes. Similarly, (Yu et al., 2022) highlight DT's mediating role in aligning strategic orientations like market orientation with operational performance, emphasizing the value of integrating market insights with digital innovations. (Joensuu-Salo et al., 2018) confirm that digital tools aligned with market orientation directly improve FP, transforming market data into actionable strategies, particularly within SMEs.

Further supporting evidence comes from (Pan et al., 2021), who have revealed that digital transformation bolsters innovation and new product development within market-driven frameworks. (Rostika, 2023) affirms DT’s mediating role in aligning strategic orientations, including MO, with operational performance, enabling firms to achieve competitive advantages through advanced technologies. Finally, Zhao et al., (2023) have underscored the importance of digital transformation strategies in customer relationship management and process optimization, mediating MO's effects on firm performance, particularly in the financial sector.

These insights collectively highlight the critical role of digital transformation as a bridging mechanism that amplifies the positive impacts of market orientation on firm performance, validating the proposed hypothesis H4 is proposed as follows:

H4: Digital Transformation mediates the relationship between Market Orientation and Firm Performance.

3.5 Digital Transformation and Its Impact on Firm Performance

Digital transformation is defined as the adoption and integration of digital technologies into all aspects of a business to optimize operational processes, enhance decision-making capabilities, and improve overall performance. According to the RBV theory, digital transformation equips firms with the ability to innovate and leverage technology to enhance operational efficiency in a dynamic and highly competitive business environment (Barney, 1991; Prisecaru, 2016). DT facilitates the automation of processes, minimizes errors, and boosts operational efficiency. Karimi & Walter, (2015) emphasize that the integration of digital technologies not only increases productivity, but also enhances data analytics capabilities, enabling firms to make more effective decisions.

Digital technologies, such as Big Data analytics and Artificial Intelligence (AI), empower businesses to better understand customer needs, deliver personalized services and strengthen customer relationships. Gong et al., (2021) highlight that digitalized customer experiences increase customer loyalty and enhance financial performance. Moreover, digital transformation enables firms to adapt swiftly to market changes and seize new growth opportunities. Svahn et al., (2017) underscore that digital transformation facilitates the development of new products and services, thereby driving revenue growth and improving competitive positioning.

Digital transformation serves as a vital factor in enhancing firm performance and optimizing operational effectiveness in competitive markets. Based on the RBV theory, digital transformation leverages internal technological resources to meet dynamic market demands and foster innovation (Barney, 1991). Empirical studies substantiate the impact of DT on firm performance, emphasizing its role in automating processes, improving productivity and reducing operational costs (Karimi & Walter, 2015).

For instance, H. Wang et al., (2022) reveal that digital transformation enhances the performance of Chinese manufacturing firms through low-cost empowerment and innovation mechanisms, demonstrating its strategic value in dynamic industries. Similarly, Wielgos et al., (2021) highlight that digital business capabilities, which encompass digital transformation, significantly influence firm performance across industries by fostering agility and competitive advantage.

Additional evidence from Meng et al., (2022) underscores that digital transformation bolsters transparency and customer relationship management, further enhancing financial and operational performance. In the service sector, Masoud & Basahel, (2023) identify customer experience and IT innovation as key mediators that amplify positive impact of DT on FP.

Furthermore, Yu et al., (2022) demonstrate that digital transformation positively influences firm performance by improving employee efficiency and fostering sustainable growth. This aligns with the findings of Zhao et al., (2023), who indicate that user participation and environmental dynamics augment the contribution of DT to firm performance.

Based on theoretical foundations and empirical evidence, digital transformation emerges not merely as a supporting tool, but also as a critical determinant of firm performance. Thus, the following hypothesis is proposed:

H5: Digital Transformation positively impacts Firm Performance.

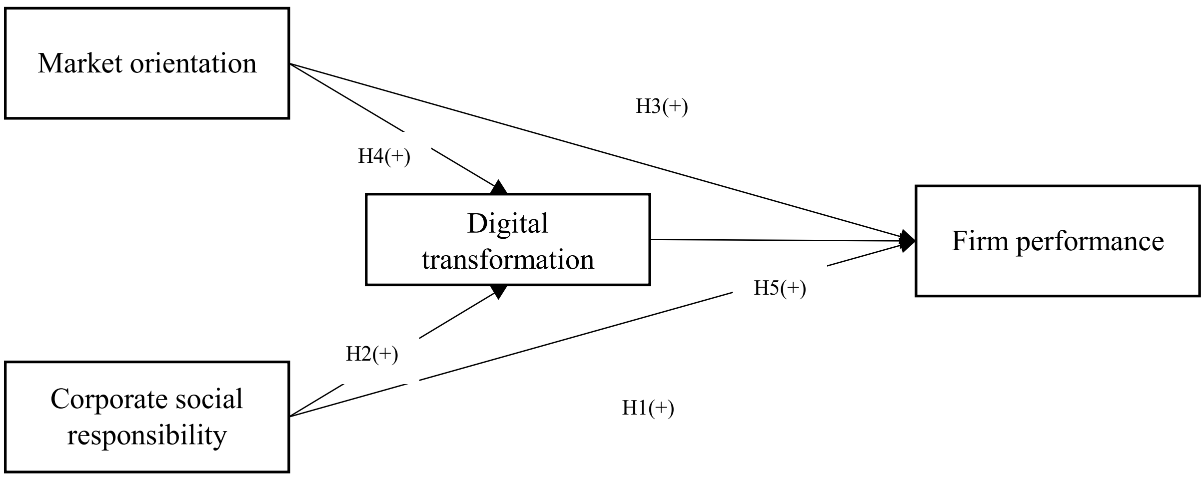

Figure 1. Research Framework

4. Research Methodology

4.1 Sample and data gathering

We conducted a survey targeting Vietnam's banking industry to evaluate the proposed conceptual framework, focusing on banks undergoing digital transformation. The survey covered branches in the Southeast region, including Ho Chi Minh City, Binh Duong, Dong Nai, Tay Ninh, and Ba Ria-Vung Tau provinces. Questionnaires were distributed via email and face-to-face interviews, targeting middle and senior level executives, department heads, and senior employees familiar with their bank’s digital transformation and CSR activities. Respondents were required to have at least three years of experience to ensure accurate and unbiased insights. The study population included 2,411 branches of joint-stock commercial banks, with Ho Chi Minh City accounting for over 70%. Using stratified and snowball sampling methods, the final sample comprised 250 respondents (10% of the population), distributed across the provinces to ensure diversity and representativeness. Snowball sampling further enhanced participation by leveraging referrals within the banking network.

4.2 Measurement

The study utilized four scales to measure the key constructs: CSR, Market Orientation, Digital Transformation, and Firm Performance. These measurement scales were adapted from validated international studies and refined through expert interviews to align with the Vietnamese context.

CSR was assessed in four main dimensions: economic, legal, ethical and philanthropic responsibilities. The scale is based on Carroll's model (1991) and has been validated in previous studies in the business and financial sectors.

The market orientation scale encompasses market intelligence generation, dissemination and responsiveness, aligning with the theoretical framework, established by Kohli & Jaworski, (1990). The MARKOR scale was subsequently adapted and refined by Matsuno et al., (2000).

DT was measured through four dimensions adapted from Blixen-Finecke, (2020): (1) Digital Product Experience, evaluating online presence through website functionality and design; (2) E-commerce, assessing the ability to conduct online sales and optimize shopping experiences; (3) e-CRM, measuring customer interaction and retention via digital tools; and (4) Digital Marketing, focusing on branding and customer engagement through online platforms. These dimensions collectively assess the maturity of a firm's digital transformation and its impact on operational performance.

FP was evaluated through three dimensions: (1) Cost-based performance, focusing on profitability after cost deductions, reflecting resource optimization (Jaworski & Kohli, 1993); (2) Revenue-based performance, highlighting sales and revenue outcomes to demonstrate market opportunity utilization (Wu & Cavusgil, 2006); and (3) Global performance, assessed via managerial comparisons to strategic goals and competitors, shaping competitive positioning (Keh et al., 2007). Indicators such as profitability, market share growth, and sales were utilized to comprehensively measure performance in alignment with modern trends (Homburg & Wielgos, 2022; Z. Wang & Kim, 2017).

The present study employed indicators such as profitability, market share growth, and sales outcomes, integrating these with comparisons with the firm's strategic objectives. This subjective approach, supported by extensive market orientation literature, enabled a comprehensive evaluation of performance, aligned with contemporary trends, including digital data measurement (Homburg & Wielgos, 2022; Z. Wang & Kim, 2017). Additionally, we used a five-point Likert scale that ranged from 1 (indicating strong disagreement) to 5 (indicating strong agreement).

4.3 Analytical approach

Structural Equation Modeling (SEM), encompassing covariance-based SEM (CB-SEM) and partial least squares SEM (PLS-SEM), is widely used in data analysis. While CB-SEM has traditionally been favoured, PLS-SEM has become prominent for its ability to handle complex models and small samples (Hair et al., 2021). In 2020, over 1,400 academic studies utilized PLS-SEM, showcasing its rising adoption.

This study applies PLS-SEM to analyze complex models with a small sample size. The analysis validates lower-order measurement models using the Repeated Indicator Approach and higher-order models via the Disjoint Two-Stage Approach, ensuring convergent and discriminant validity. Hypotheses are tested through Bootstrapping with 5,000 iterations, yielding robust results without assuming normal data distribution.

The research framework was developed from a literature review on CSR, market orientation, digital transformation and firm performance. The measurement tool was created in English, translated into Vietnamese, and pre-tested with 15 bank executives in Ho Chi Minh City. Data collection from March to December 2023 produced 250 valid responses from 260 distributed questionnaires.

Table 1.
Respondents' demographic characteristics

Demographics	Category	Frequency	Percentage
Gender	Male	153	61.20%
	Female	97	38.80%
Age group	<25	28	11.20%
	25 - <35	79	31.60%
	35 - <50	113	45.20%
	≥ 50	30	12.00%
Education	PhD/M.Sc	56	22.40%
	Bachelor	182	72.80%
	Diploma	12	4.80%
Monthly salary	< 15 mil VND	22	8.80%
	15 – <30 mil VND	63	25.20%
	30 – <50 mil VND	128	51.20%
	≥ 50 mil VND	39	15.60%
Location of bank	Ho Chi Minh City	180	72.00%
	Binh Duong	22	8.80%
	Binh Phuoc	6	2.40%
	Dong Nai	19	7.60%
	Ba Ria - Vung Tau	16	6.40%
	Tay Ninh	7	2.80%

5. Results

The reliability of the measurement scales was evaluated using internal consistency methods through Cronbach's Alpha and Composite Reliability (CR). A total of 12 scales with 79 observed variables were assessed, and the results revealed Cronbach's Alpha values ranging from 0.799 to 0.925, indicating high reliability according to the standards of (Nunnally & Bernstein, 1994), with an optimal range of 0.7–0.9. No observed variables exhibited item-total correlations below 0.3, confirming that all variables met the necessary reliability requirements for subsequent analysis (Hair et al., 2020). Additionally, the CR values for the lower-order constructs exceeded 0.7, further reinforcing the stability of the scales. The results in Table 2 demonstrate that the measurement scales employed in the study are reliable and suitable for further analysis.

Table 2.
Reliability and validity test

	Cronbach's Alpha	Composite Reliability (CR-rho-C)	Average Variance Extracted (AVE)
DEP	0.853	0.891	0.577
DMK	0.790	0.857	0.545
ECM	0.784	0.853	0.538
ECR	0.901	0.922	0.629
ECRM	0.822	0.867	0.583
EMR	0.885	0.911	0.594
LR	0.914	0.931	0.660
PR	0.908	0.926	0.610
MID	0.731	0.817	0.528
MIG	0.783	0.841	0.599
MRE	0.827	0.869	0.557
FP	0.787	0.854	0.540

** DEP: Digital Product Experience; DMK: Digital Marketing; ECM: e-Commerce; ECRM: e-CRM; MIG: Market Intelligence Generation; MID: Market Intelligence Dissemination; MRE: Market Responsiveness; EMR: Economic Responsibility; LR: Legal Responsibility; ECR: Ethical Responsibility; PR: Philanthropic Responsibility; FP: Firm Performance*

To assess the convergent validity of the measurement scales, the Average Variance Extracted (AVE) must achieve a minimum threshold of 0.5 (Anderson & Gerbing, 1988). The analysis results indicated that the lower-order constructs all had AVE values equal to or greater than 0.5, demonstrating that the higher-order constructs in this study achieved convergent validity.

The discriminant validity of the scales was evaluated using multiple methods, including Cross-loading, the Fornell and Larcker criterion and the Heterotrait-Monotrait Ratio (HTMT). Among these, the Fornell and Larcker criterion, a widely used approach in research, stipulates that the square root of the AVE of a construct must exceed all correlation coefficients between that construct and other constructs in the model (Fornell & Larcker, 1981). The results confirmed that all lower-order constructs satisfied this criterion. Furthermore, analyses of Cross-loading and HTMT also verified that the lower-order constructs in this study achieved discriminant validity, ensuring the reliability and validity of the measurement scales used in the research.

Table 3.
Fornell and Larcker Discriminant Validity Results – Lower-Order Constructs

	DEP	DMK	ECM	ECR	ECRM	EMR	FP	LR	MID	MIG	MRE	PR
DEP	**0.860**											
DMK	0.633	**0.738**										
ECM	0.605	0.617	**0.733**									
ECR	0.372	0.435	0.370	**0.793**								
ECRM	0.559	0.573	0.525	0.250	**0.695**							
EMR	0.422	0.441	0.378	0.616	0.258	**0.77**						
FP	0.458	0.485	0.426	0.597	0.320	0.594	**0.735**					
LR	0.447	0.485	0.387	0.632	0.259	0.644	0.603	**0.813**				
MID	0.272	0.253	0.207	0.283	0.166	0.303	0.397	0.275	**0.654**			
MIG	0.343	0.331	0.276	0.430	0.219	0.448	0.646	0.446	0.543	**0.631**		
MRE	0.320	0.339	0.294	0.439	0.215	0.439	0.601	0.432	0.486	0.546	**0.676**	
PR	0.419	0.472	0.409	0.618	0.257	0.617	0.616	0.834	0.296	0.422	0.417	**0.781**

The results of the quantitative data analysis at the higher-order structural level were developed based on the foundation of lower-order data analysis. This approach aimed to validate and evaluate the higher-order structural model comprising four key constructs: (1) Corporate Social Responsibility (CSR), (2) Market Orientation (MO), (3) Digital Transformation (DT) and (4) Firm Performance (FP). The analysis of outer loadings in Table 4 revealed that all constructs met the required threshold of ≥ 0.7, ensuring validity in accordance with the recommendations of Hair et al., (2020).

The reliability of the higher-order constructs was assessed using two key indicators: Cronbach's alpha and Composite Reliability (CR). According to Nunnally & Bernstein, (1994), values of Cronbach's alpha and CR between

0.7 and 0.9 are considered optimal. The analysis results demonstrated that all higher-order constructs achieved Cronbach's alpha and CR values exceeding 0.7, ensuring reliability. Additionally, the Average Variance Extracted (AVE) for the constructs within the higher-order structure met the threshold of ≥ 0.5, indicating convergent validity. These findings confirm that the higher-order constructs are reliable and appropriate for use in subsequent analyses.

Table 4.
Outer Loadings Results for Higher-Order Constructs

	CSR	DT	FP	MO
DEP		0.927		
DMK		0.937		
ECM		0.910		
ECRM		0.715		
LR	0.941			
EMR	0.932			
PR	0.930			
ECR	0.928			
FP1			0.720	
FP2			0.766	
FP3			0.768	
FP4			0.748	
FP5			0.767	
MID				0.754
MIG				0.917
MRE				0.896

According to Fornell & Larcker, (1981), a construct is considered to have discriminant validity when the square root of its Average Variance Extracted (AVE) exceeds all its correlations with other constructs in the model. The analysis results in this study, as presented in Table 5, indicate that the square root of the AVE values for the constructs ranges from 0.540 to 0.870. This confirms that all constructs in the study satisfy the discriminant validity criterion proposed by Fornell and Larcker, ensuring the independence of the constructs in the model and the validity of subsequent analytical results.

Table 5.
Fornell-Larcker criterion and Reliability and Convergent Validity – Higher-Order Constructs

	CSR	DT	FP	MO	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
CSR	**0.933**				0.950	0.964	0.870
DT	0.473	**0.877**			0.897	0.929	0.769
FP	0.646	0.488	**0.735**		0.787	0.854	0.540
MO	0.494	0.369	0.655	**0.859**	0.821	0.893	0.737

The study evaluated the structural equation model (SEM) using the coefficient of determination (R^2) to measure variable influence, variance inflation factor (VIF) to check for multicollinearity and Bootstrapping for hypothesis testing. All VIF values were below 5, confirming no multicollinearity and validating the reliability of the measurement scales. These findings support the analysis of direct and indirect relationships within the model. Hypotheses were tested using Bootstrapping with 5,000 iterations, allowing for precise evaluation of path coefficients without assuming normal distribution. Hypotheses with P-values below 0.05 were deemed statistically significant (Figure 2).

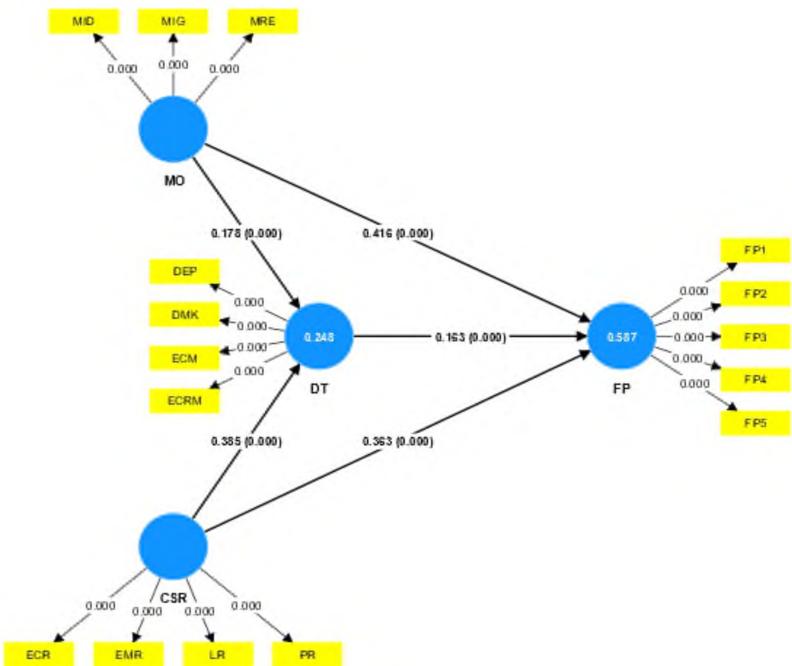

Figure 2. PLS-SEM analysis results of the research framework model

The analysis results indicate that the R² values for the dependent variables in the model are significant, demonstrating varying degrees of impact from the independent variables. Specifically, in Table 6, the R² value for Digital Transformation (DT) is 24.8% (R² = 0.248), and for Firm Performance (FP) - the main dependent variable - is 58.7% (R² = 0.587). These values fall within the range of 0 to 1, where values closer to 1 indicate higher explanatory power, aligning with the roles of each factor within the model (Hair et al., 2020).

The study measured the impact of independent variables on dependent variables using standardized path coefficients, with higher coefficients indicating stronger influence. Bootstrapping validated the significance of both direct and indirect effects, confirming the reliability of the hypotheses. Among the direct effects, Market Orientation (MO) had the strongest impact on Firm Performance (FP) (coefficient: 0.416), emphasizing its key role in enhancing efficiency. Corporate Social Responsibility (CSR) also significantly influenced FP (coefficient: 0.363), improving both ethics and performance. Digital Transformation (DT), while showing the weakest direct effect (coefficient: 0.163), still demonstrated its importance in driving business improvement through technology.

Table 6.
Hypothesis results and decision

Hypothesis	Constructs	Original sample (O)	Standard deviation (STDEV)	P values	Supported hypothesis
H1	CSR -> FP	0.363	0.055	0.000	Supported
H2	CSR -> DT -> FP	0.063	0.019	0.001	Supported
H3	MO -> FP	0.416	0.052	0.000	Supported
H4	MO -> DT -> FP	0.029	0.012	0.013	Supported
H5	DT -> FP	0.163	0.044	0.000	Supported
R^2 Digital transformation = 0.248 R^2 Firm performance = 0.587			Q^2 Digital transformation = 0.186 Q^2 Firm performance =0.308		

The findings reveal statistically significant indirect effects (P-values < 0.05), with CSR influencing FP via DT (coefficient: 0.063) and MO impacting FP through DT with a coefficient of 0.029. These results underscore DT's role in enhancing the effectiveness of CSR and market-oriented strategies. The model's predictive ability, confirmed by out-of-sample predictive relevance (Q^2>0), further supports its robustness (Hair et al., 2017).

These results highlight DT's critical mediating role in amplifying the effects of CSR and MO on FP. Firms should focus on CSR initiatives, invest in DT, and integrate these with market-oriented strategies to maximize performance. Prioritizing impactful factors, such as MO, CSR and DT, enables firms to enhance efficiency and secure a sustainable competitive advantage.

6. Discussion

This study investigates the relationships between Market Orientation, Corporate Social Responsibility and Digital Transformation, and their effects on Firm Performance in Southeast Vietnam’s banking sector using structural equation modeling (SEM). The findings reveal that MO is the most significant driver of FP ($\beta = 0.416$), highlighting the importance of customer needs analysis, competition monitoring and internal coordination in boosting revenue and operational efficiency, while acting as strategic guide for effective business strategies amid digital transformation. CSR has a direct impact on FP ($\beta = 0.363$) and an indirect influence through DT ($\beta = 0.063$), with initiatives such as legal compliance and community development enhancing corporate image, fostering customer trust and increasing financial value, further amplified by DT. DT plays a critical mediating role, influencing FP directly ($\beta = 0.163$) and acting as a bridge linking CSR to FP ($\beta = 0.385$) and MO to FP ($\beta = 0.369$), while optimizing processes and performance through technologies such as AI, Big Data and CRM. The integration of MO, CSR, and DT significantly enhances FP, emphasizing the necessity of strategic alignment to achieve sustained competitiveness in an increasingly demanding market. The study underscores the critical interplay between MO, CSR and DT, with DT serving as both an independent driver and a strategic bridge to optimize business performance, offering actionable insights for banks to develop sustainable and competitive strategies in a digitally evolving landscape.

Theoretical implications

The study developed and validated an integrated theoretical model linking Market Orientation, Corporate Social Responsibility and Digital Transformation to enhance Firm Performance, offering a comprehensive framework for synergizing business strategies, social responsibility and technology to improve performance. DT emerged as a vital intermediary, bridging the relationships between CSR, MO and FP, highlighting the strategic importance of digitalization in optimizing processes, enhancing responsiveness and increasing transparency in CSR activities. The measurement scales, tailored to Vietnam's banking sector, also demonstrated broader applicability, with dimensions such as information dissemination (MO), philanthropic responsibility (CSR) and e-commerce (DT), standardized and validated for future research. This study advances corporate management theory, particularly in digital transformation and social responsibility, providing insights for developing countries to align technological integration with sustainable development goals, offering both theoretical and practical contributions.

Practical implications

Banks should prioritize developing robust systems for collecting information from customers, competitors and market trends, using tools such as Big Data analytics and regular customer surveys. Cross-departmental collaboration should be strengthened through training and meetings, and prompt response mechanisms are essential for addressing customer complaints and adapting to market changes to enhance competitiveness and customer experience.

Corporate Social Responsibility must span economic, legal, ethical and philanthropic dimensions. Banks should aim to maximize shareholder value, ensure legal compliance through transparent audits, uphold ethical practices to foster customer loyalty and engage in community-building initiatives, such as supporting underprivileged groups.

Digital transformation is crucial for boosting operational efficiency and customer satisfaction. Banks should invest in digital marketing, optimize mobile platforms and strengthen e-commerce and CRM systems to reduce costs and enhance convenience, fostering long-term customer relationships.

The integration of Market Orientation, CSR and Digital Transformation is key to achieving sustainable growth. MO helps identify CSR activities and supports accurate forecasting of customer needs, while DT optimizes CSR processes and enhances transparency. Seamless integration of these elements improves performance and builds lasting trust with customers and communities.

To succeed, banks should foster a collaborative culture, train employees on integrating MO, CSR and DT, and use Key Performance Indicators (KPIs) to measure success across financial performance, customer satisfaction and social impact, enabling continuous strategy refinement in a competitive environment.

Limitations and Scope for future research.

Future research should broaden its scope to include other regions in Vietnam or neighbouring Southeast Asian countries, enhancing generalizability and enabling comparisons across regions, countries and industries. Incorporating qualitative methods such as in-depth interviews or focus groups would provide richer insights into the complex relationships among Market Orientation, Corporate Social Responsibility and Digital Transformation, supporting the interpretation of quantitative findings and generating new, practical hypotheses.

Future studies should also consider external factors such as government policies, global economic trends or societal disruptions, which could reshape the dynamics between MO, CSR, DT and Firm Performance. Additional variables, including leadership capabilities, organizational culture and innovation, may further enrich theoretical and practical insights, acting as moderators or amplifiers of these relationships.

Longitudinal studies are crucial for understanding the evolution of these interactions over time in dynamic economic, technological and societal contexts. Advanced analytical tools like AI, machine learning and big data analytics can uncover hidden relationships and enhance the precision of analyses, clarifying the roles of MO, CSR, and DT in improving FP.

These directions not only expand the academic value of future research, but also offer actionable guidance for businesses to develop sustainable and effective strategies by integrating innovative variables, methodologies and real-world contexts.

7. Conclusion

This study provides valuable insights into the interconnected roles of Market Orientation, Corporate Social Responsibility and Digital Transformation in driving Firm Performance in Vietnam’s banking sector. The findings highlight that market orientation serves as the most significant driver of firm performance by enhancing customer intelligence, competitive awareness and internal coordination. CSR contributes to firm performance both directly and indirectly through digital transformation, reinforcing corporate reputation, customer trust and long-term financial value.

Meanwhile, DT acts as both a standalone performance enhancer and a crucial mediator, optimizing operational processes and amplifying the impact of market orientation and CSR through digital solutions such as AI, Big Data and CRM. The integration of these three elements is critical for achieving sustained competitiveness in a rapidly evolving financial landscape.

Beyond Vietnam, these insights hold relevance for other emerging markets facing similar challenges in digital transformation and sustainable business practices. In Southeast Asia, where financial services are expanding rapidly, banks can leverage market orientation and CSR strategies supported by digital transformation to enhance operational efficiency and customer engagement. Similarly, in developing economies with growing fintech ecosystems, integrating digital technologies with market-driven and socially responsible strategies can foster financial inclusion and business resilience. Policymakers and financial institutions in these regions can apply the study's findings to develop frameworks that align digital transformation with market orientation and corporate responsibility, ensuring both competitive advantage and sustainable growth. Future research could explore cross-market comparisons to validate these relationships and refine digital strategies for diverse economic contexts.

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CONTENTS

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APPLICATION OF MACHINE LEARNING ALGORITHMS IN PREDICTING CUSTOMER LOYALTY TOWARDS GROCERY RETAILERS

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Abstract: Retailers strive for customer loyalty in the sense of repeat purchases, but also as a high proportion of purchases (compared to competitors) and willingness to recommend to other customers. This paper examines customer loyalty in the grocery sector as a three-dimensional construct and shows how machine learning techniques can be useful in its study. Price characteristics of the retailer (price level, value for money, price dynamics, price communication and price dispersion) and non-price characteristics of the retailer (general product range, retailer's private label product range, store design and atmosphere, service level and location) are included in the model as predictor variables. Using the data collected through the primary research conducted in Croatia, 433 samples were divided into 10 independent predictor variables and one dependent variable (customer loyalty), a prediction was created using supervised machine learning classification algorithms. The Random Forest classifier proves to be the best choice overall, with ROC_AUC value of 0.790, a high accuracy of 0.915 and an F1 score of 0.954, reflecting both precision and responsiveness. The application of the SHapley Additive exPlanations analysis additionally enables the interpretation of the results, highlighting the influence of features on the accuracy of the prediction. The results indicate that price dynamics and service level are the most important features for the model predictions, followed by value for money and price communication.

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Introduction

Every company strives for customer loyalty, for a long-term relationship with the consumer. The specificity of grocery and food retailing lies in a wide range of factors, including the price and non-price characteristics of the retailer, which can influence the loyalty of customers towards the retailer and its brand as a company. The approach to customer loyalty in retail is more sophisticated and comprehensive than loyalty to a single manufacturer brand. Retailers advertise on price and product availability while providing a certain level of service that consumers continually evaluate. Retailers collect data on consumers through POS systems and loyalty programs, which encourages the use of machine learning (ML) approaches to data analysis. It is also even easier for online retailers to use AI and ML tools (Gauri et al., 2021). The topic is also gaining attention among academics who are exploring the role and possibilities of machine learning, AI and IoT (Popova, & Petrova, 2024; Popova et al., 2024; Petrova et al., 2022) and its predictive capabilities with the aim of increasing customer satisfaction and loyalty (e.g. Andresini et al., 2023; Wang et al., 2021; Rane et al., 2023; Islamgaleyev et al., 2020; Arefin et al., 2024). The aim of this paper is to evaluate the application of ML algorithms to predict loyalty towards retailers in traditional grocery retailing, focusing on the high accuracy and reliability of the prediction results. Particular attention will be paid to analyzing the factors and identifying the key price and non-price factors that contribute to customer loyalty. The aim is to enable more precise targeting and adaptation of marketing strategies. The ML algorithms aim to gain a more detailed and accurate understanding of the factors that influence customer loyalty to grocery retailers. The application of SHapley Additive exPlanations (SHAP) analysis additionally enables the interpretation of the results by highlighting the influence of individual characteristics on the accuracy of the prediction. The research problem includes the following aspects: the identification of key factors, the questions of how accurately different ML algorithms can predict customer loyalty based on the identified factors, how the individual factors behave within the model and how much they contribute to the final prediction. This approach improves prediction performance and provides valuable insights for developing effective strategies to create and maintain customer loyalty.

1. Literature Review

1.1. Customer loyalty and influencing factors

Customer loyalty towards the retailer can encompass more dimensions than the repeat purchase itself, even though it is very often seen as loyalty. Attracting customers to the store is only one step towards the goal of creating a loyal customer. In this paper, customer loyalty is examined as a multidimensional construct, taking into account not only purchase intention but also purchase share and willingness to recommend. Purchase intention refers to the intention to buy again and mainly concerns retailers with which consumers have already had experience. Purchase share is an important dimension as most consumers make their purchases at multiple retailers and through multiple channels (Flavian et al., 2020). The focus of this research is on the share of purchases from different grocery retailers and their physical stores, as online sales of food and fast moving consumer goods (FMCG) still lag behind other retail sectors. Willingness to recommend is an emotional dimension of loyalty and one of the effects of overall consumer satisfaction (Kumar et al., 2017).

In this paper, the prediction of customer loyalty is examined on the basis of price and non-price characteristics of the grocery retailer, which are described very briefly below. Price level (PL) refers to a certain image of how cheap or expensive the retailer is in customers' minds (Graciola et al., 2018). Value for money (VM) can be seen as the result of an appropriate price and quality evaluation (Zielke, 2006), not only of the perceived price-quality ratio of the individual product but of the retailer's overall service. Price dynamics (PDY) refers to frequent and short-term price changes, usually in the form of price promotions (Cacchiarelli i Sorrentino, 2019). Its main purpose is to attract customers into stores. Price communication (PC) can influence customers' perception of price image (Grandi & Cardinali, 2022) and refers to the general information provided to consumers about prices through various communication channels, including the store. Price dispersion (PD) stands for the dispersion of prices within the retailer's range, which largely depends on the size and depth of the range (Hamilton

& Chernev, 2013). As one of the most important non-price characteristics of the retailer that expresses the retailer's identity (Graciola et al., 2018), the total product range (PR) stands out, "a set of products and/or services that a retailer offers to consumers" (Kumar et al., 2017). To further differentiate themselves from the competition (Kumar & Kim, 2014), grocery retailers are intensively developing their private label product range (PLPR). Store design and atmosphere (SDA) refer to factors that influence consumers' shopping mood (Kumar et al., 2017). Retail service level (SL) is a very broad concept. This paper focuses on in-store staff and their contribution to a positive customer experience (O'Cass & Grace, 2008) as well as the overall impression of service level, e.g. product availability. Location (L) remains one of the most important attributes of physical stores. Bell (2014) points out that even in modern retail, despite the development of e-commerce, location is still "everything".

1.2. Supervised machine learning and classification (prediction) algorithms

ML methods can be categorized according to the degree of human control or supervision of the learning process, including supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. Supervised learning aims to develop a model based on labeled data to enable the prediction of future data, with the two most common tasks being classification and regression. These methods form the basis of modern ML thanks to their precision and wide application, and are crucial for the development of advanced analytical and predictive models (Fosic et al., 2022). The Random Forest (RF) algorithm consists of multiple decision trees applied to different subsets of a dataset to improve prediction accuracy over the original dataset. RF combines the predictions of all trees and determines the final result based on a majority decision (Hasan et al., 2014). Support Vector Classifier (SVC) is an advanced ML algorithm used in classification tasks and is based on defining a boundary between different datasets that can be categorized into different classes (Sen et al., 2020). Gaussian Naive Bayes (GNB) supports continuous data that follows a Gaussian normal distribution and relies on Naive Bayes (NB), derived from Bayes' theorem. NB is based on the assumption that features are independent of each other (Naiem et al., 2023). The Stochastic Gradient Descent Classifier (SGD) is a variant of the gradient algorithm used for optimization. SGD uses only a random sample to estimate the gradient in each iteration. This reduces computational complexity and makes the learning process faster and more efficient, especially for large datasets, although gradient estimation may be less precise (Tian et al., 2023). K-nearest neighbors (KNN) is a supervised ML algorithm that classifies results based on the majority class of its nearest neighbors. Its goal is to classify new entries based on samples from the training set. It is based on finding the minimum distance between the observed sample and the samples from the training set to identify the nearest neighbors. The class of the majority of these nearest neighbors is used to predict the class of the observed sample (Hu et al., 2016). A Decision Tree (DT) is a non-parametric supervised learning method suitable for classification and regression. It predicts the value of the target variable by learning simple decision rules based on data features. The DT Classifier takes two data as input: an array of X shapes (n_samples, n_features) containing the training samples and an array of Y integer values (n_samples) containing the class labels for these samples (Decision Trees, n.d.). Gradient Boosting Classifier (GBC) is an algorithm that gradually builds an additive model and enables the optimization of different differentiable loss functions. It efficiently models complex relationships between features and targets, improving accuracy (Czakon, 2024). Classification models have two main goals: 1) they should perform well, i.e. they should predict the output data for new input features as accurately as possible; 2) they should be interpretable, i.e. provide an understanding of the relationship between input features and output data. The most common analysis of classification models is via feature importance, i.e. how important a particular feature is to the performance of a classification model. This is a measure of the individual contribution of the corresponding feature to a particular classifier, regardless of the form or direction of the feature influence. The importance of the features of the input data depends on the corresponding classification model and a feature that is important for one model may be unimportant for another model (Saarela & Jauhiainen, 2021). Shapley Additive Explanation (SHAP) belongs to the family of additive, model-independent feature mapping techniques. It uses game-theoretic concepts to calculate the importance of features in two steps: 1) by training a classification model with all features; 2) by calculating the SHAP value for each feature and ranking it to identify the most important features. The average SHAP value indicates the typical influence of each feature on the predictions of the model over the entire data set, while the absolute SHAP value represents the importance of the feature regardless of its direction (positive or negative). By sorting the features according to their average absolute SHAP values, features with higher SHAP values are identified as more influential on the model predictions (Wang et al., 2024).

2. Research Methodology

2.1. Methodology and research sample

For the empirical study of customer loyalty towards grocery retailers, a survey method was used. It was conducted both web-based with Google Forms and with a questionnaire using pen and paper in 2022. The survey instrument was a highly structured questionnaire based on previous research in the field of retailing and the behavioral aspects of retailing. The respondents were customers over the age of 18 who buy groceries and everyday products for their household in Croatia. A 5-point Likert scale was used for the scales of all variables of the model, which allowed respondents to indicate their level of agreement. 435 questionnaires were collected, two of which were excluded from further processing. 69% of respondents were women, mostly in the age groups 18 – 31 (39%) and 46 – 59 (27%), with a college degree (50%) and employed (62%). Using the collected data consisting of 433 samples divided into 10 independent predictor variables and one dependent variable, a prediction was generated using seven supervised machine learning classification algorithms (RF, SVC, GNB, SGDC, KNN, DT and GBC). The predictor variables were determined by grouping questionnaire questions according to predefined characteristics: retailer price characteristics (PL, VM, PDY, PC and PD) and retailer non-price characteristics (PR, PLPR, SDA, SL and L).

2.2. Data preparation

The data preparation involved several steps. First, the data was cleansed to remove incomplete or incorrect data records. This included the identification and elimination of rows with missing values and the correction or removal of anomalies that could affect the quality of the results. The features of the data set (10 independent features) were functionally determined as the average values of the respondents' statements in this Likert scale for specific questionnaire questions. The dependent variable, representing customer loyalty to retailers, was also determined by calculating the average of respondents' answers to the retailer loyalty questions. For the classification or prediction of the dependent variable by machine learning, the dependent variable was converted to binary values 0 or 1, depending on the average of the values from which the dependent variable was determined. The method of converting the dependent variable into binary form is such that average values less than or equal to 3.0 are labelled with zero (0), while average values of the dependent variable greater than 3.0 are labelled with one unit (1). In this way, a dependent variable in binary form was obtained, where the value 1 indicates the loyalty of customers towards retailers and the value 0 means the lack of loyalty towards retailers. This process of data preparation resulted in an extremely unbalanced data set from the aspect of the dependent variable, where 88% of the dependent variable represented the value 1, i.e. the majority class, and the rest of the 12% of the data represented the minority class, i.e. the value 0 in the dependent variable. To further segment the respondents' loyalty, three dependent sub-variables were identified to determine the specifics of customers' loyalty towards retailers. The determination process was the same, i.e. the average of certain variables, i.e. questions about respondents' loyalty, was calculated. The binary values are also determined according to the same principle as the global dependent variable. This dependent variable also determined the data set as a very unbalanced data set. The process of segmenting the dependent variable was carried out with three groups of questions on customer loyalty towards the retailer, and three different reduced data sets were obtained from the aspect of the dependent variable. After the preparation and creation of four data sets: the base data set, where the dependent variable is represented by the average of all values on the loyalty questions, and three reduced data sets in terms of how the "clustered" dependent variable was created, the data was split into a training and a test data set in a 70:30 ratio to allow the evaluation of model performance on unseen data and to reduce the possibility of overfitting.

3. Results

3.1. Implementation and evaluation of prediction

Python libraries such as scikit-learn and pandas were used to implement the above algorithms. The data was split into a training set and a test set in a 70/30 ratio to enable validation of model performance. Model evaluation was performed using various metrics, including classical accuracy, AUC and F1-score. The success of an ML algorithm is evaluated using various metrics. Classification evaluation is usually done using a single evaluation number that attempts to summarize the specifics of the model. For imbalanced datasets, a common mistake is to rely on the classical classification accuracy score, which can be misleading. The classification of imbalanced datasets refers to the classification problem when the number of samples of the different categories in the dataset is unequal. The solution to such problems depends on the method and the use of data processing algorithms (Zhou et al., 2023). Therefore, the F-beta score and the AUC score were used in this study because they allow a more accurate assessment of model performance in imbalanced datasets. The F-1 score considers the balance between precision and responsiveness, while the AUC score measures the model's ability to discriminate (Fosic et al., 2022). The F-1 score is calculated as 2 * (accuracy * response) / (accuracy + response). It treats accuracy and responsiveness as equally important. One of the most commonly used techniques for evaluating classifiers is the ROC (Receiver Operating Characteristic) curve, a graphical representation of the response measure versus the false positive rate. The classification performance information in the ROC curve can be summarized in a result known as the AUC (Area Under the Curve). This measure is less sensitive to imbalances in the class distribution and represents a compromise between responsiveness and specificity of results. The AUC values range from 0 to 1. An excellent model has an AUC close to 1, which means that it can separate the classes well. A poor model has an AUC value close to 0, which means that it separates the classes very poorly, i.e. it predicts the opposite of the actual values. For a model that cannot separate the classes at all, the AUC value is equal to 0.5. Classical accuracy is measured as the ratio of correct predictions relative to the total number of predictions and is often used as a basic measure of a model's success. However, for imbalanced datasets, classical accuracy can be misleading as it can give high values even if the model does not correctly classify less represented classes (Czakon, 2024; Stapor, 2018). Pseudocode algorithm 1 was used for prediction and classification to find the optimal classifier for the most successful prediction of store loyalty. The results obtained by implementing Algorithm 1 in the Python programming language are shown in Table 1.

Table 1.
Implementation of Algorithm 1 in the Python programming language

MODEL	ACCURACY	ROC_AUC	F1 SCORE
RF	0.915	0.79	0.954
SVC	0.9	0.625	0.946
GNB	0.885	0.691	0.933
SGDC	0.908	0.6	0.95
KNN	0.915	0.662	0.954
DT	0.885	0.674	0.936
GBC	0.915	0.749	0.953

Source: authors' work

Based on the performance metrics shown in Table 1, the Random Forest (RF) classifier proves to be the best choice overall. Not only does it exhibit the highest ROC_AUC value of 0.790, indicating an excellent ability to discriminate between classes, but it also achieves a high accuracy of 0.915 and an F1 score of 0.954, reflecting both precision and responsiveness. Although the Support Vector Classifier (SVC) and Stochastic Gradient Descent Classifier (SGDC) have high accuracy and F1 score, they are less reliable due to their significantly lower ROC_AUC values (0.625 and 0.600). The Gaussian Naive Bayes (GNB) classifier, despite its high ROC_AUC value of 0.691, lags behind the best results in terms of accuracy and F1 score. Even the Decision Tree (DT) classifier with a ROC_AUC value of 0.674 does not achieve the high accuracy and F1 score of RF, KNN and Gradient Boosting Classifier (GBC). Therefore, if the ROC_AUC value is the main focus, RF is the best classifier. If the main focus is on accuracy and F1 score, RF, KNN and GBC are almost equal in their prediction. Based on the capabilities of the RF classifier, which is also the best classifier according to the obtained results shown in Table 1, it can be determined which feature of the dataset is most significant in predicting the results considering the entire dependent variable. The analysis shows that PDY and SL (Figure 1a) are the most important features for the model predictions. VM and PC also contribute significantly, but to a slightly lesser extent. The features PD, PR, SDA and PL are of medium importance, while PLPR and L are the least important.

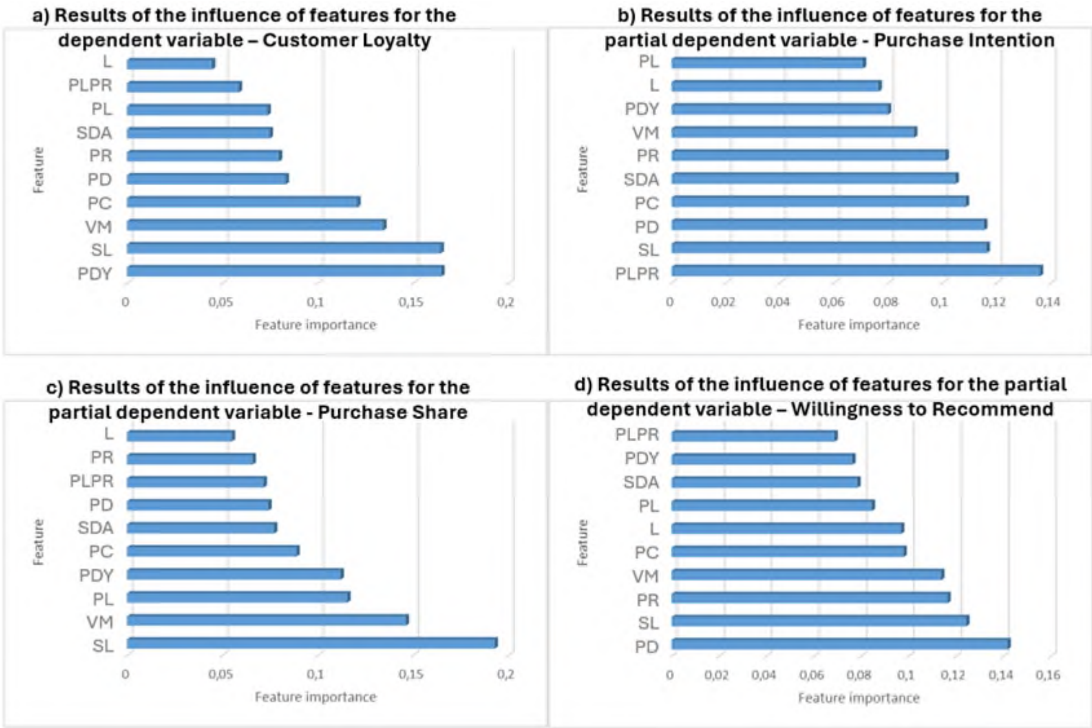

Figure 1. Results of the influence of features for the (partial) dependent variable

Source: authors' work

Since the dependent variable was segmented in the data preparation for a more detailed analysis of respondents' loyalty, three different reduced data sets were obtained from the aspect of the dependent variable. The analysis shows that PLPR is the most important feature for the Purchase Intention (Figure 1b), followed by SL and PD, which are also crucial for the predictions of the model. PC and SDA are moderately important features with a slightly lower influence. PR, VM, PDY, L and PL are less important, with PL being the least important. For the Purchase Share (Figure 1c), the feature importance analysis shows that SL is the most influential feature that significantly affects the model predictions. VM is the second most important feature, while PL and PDY also make an important contribution. PC and SDA are moderately important features, i.e. they play a significant but minor role compared to the most important features. PD, PLPR and PR have lower, but still significant importance, while L is the least important feature and has minimal impact on the model. Finally, for the Willingness to recommend (Figure 1d), the analysis shows that PD is the most influential feature and has the largest impact on the model's predictions. SL, PR and VM also play an important role and contribute significantly to model performance. PC and L have moderate importance, i.e. they are significant but less important than the main features. PL contributes, but to a lesser extent. SDA, PDY and PLPR are the least important features with minimal impact on model predictions. It is recommended to ensure high data quality and representation for PD, SL, PR and VM. Further research on PC and L can help to understand and improve their role. A review of SDA, PDY and PLPR can shed light on whether they add noise or can be improved.

3.2. SHAP analysis

The SHAP analysis was performed for the total and partial dependent variables. In Figure 2, the Y-axis shows features such as VM, SL, PC, PD, PDY, SDA, PL, PR, PLPR and L. The X-axis shows the SHAP values, which represent the influence of each feature on the model output. Positive SHAP values mean that the feature increases the prediction, while negative values mean that it decreases it. For the overall dependent variable (Figure 2a), features such as PC and PD have a wide range of SHAP values, indicating a significant and diverse influence on model performance. These features can both increase and decrease predictions depending on their values. In contrast, features such as L and PLPR have a narrower range of SHAP values, indicating a more consistent and less variable influence on model predictions.

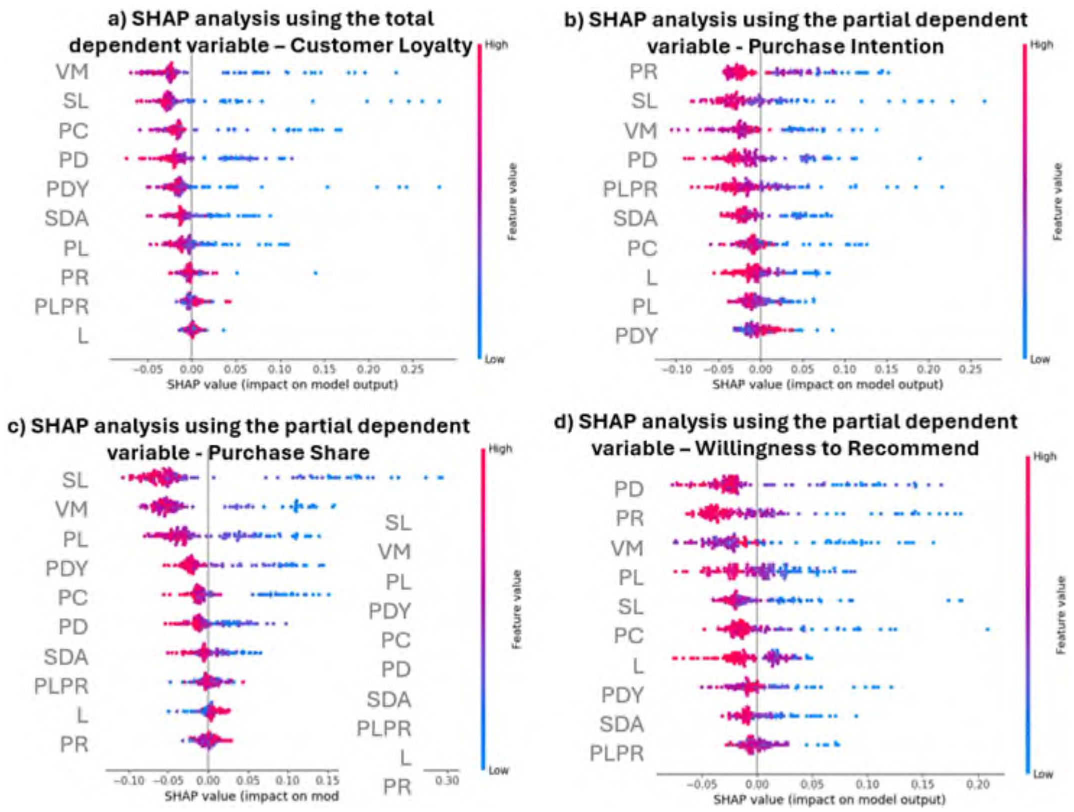

Figure 2. SHAP analysis using the total and partial dependent variable
Source: authors' work

The color of the dots varies from blue to red and stands for features values, with blue for low and red for high values. In the case of PC, for example, high feature values (red dots) are generally associated with positive SHAP values, which means that higher values of this feature improve the prediction of the model. On the other hand, low feature values (blue dots) are associated with negative SHAP values, meaning that lower values of this feature decrease the prediction of the model. In general, features with a wider range of SHAP values, both positive and negative, have a greater impact on model predictions. For example, PC and PD are highly influential due to the wide range of SHAP values. In contrast, features with SHAP values that cluster around zero, such as L, have less influence on model predictions. The color coding helps you understand the relationship between the feature values and their influence because it shows how different values of each feature affect the model output. When classifying (predicting) with the partial dependent variable Purchase Intention (Figure 2b), features with a wider range of SHAP values such as PR, SL, VM and PD have a greater influence on the model predictions. Features PR and SL have a wide range of SHAP values, indicating a significant and diverse influence, while features such as PDY and PL have a narrower range, indicating a more consistent and smaller influence. For the partial dependent variable Purchase Share (Figure 1c), the trait SL has a wide range of SHAP values, indicating a significant and diverse influence on the model output. In contrast, features such as PR and L have a narrower range, indicating a more consistent influence. Finally, for the partial dependent variable Willingness to Recommend (Figure 1d), features such as PD and PR have a wide range of SHAP values, indicating that they have a significant and varied influence on the model output. In contrast, features such as SL and PLPR have a narrower range of SHAP values, suggesting a more consistent influence. High PC and PD values (red dots) increase predictions, while low values (blue dots) decrease predictions.

4. Discussion

If we analyze the importance of the features, we come to the conclusion that PDY and SL are crucial for the predictions of the model, while VM and PC also contribute, but to a lesser extent. The features PD, PR, SDA and PL have a medium importance, while PLPR and L are less important. It is recommended to focus on optimizing PDY and SL to improve data quality, as these features have the greatest impact on the model. Price dynamics confirm the need and impact of keeping customers interested in retailers' offers and creating short-term and frequent excitement through price promotions. The service level, reflected in the availability of products and contact with in-store staff, stands out as an important factor. The importance of value for money highlights that customers are evaluating the overall package of the grocery retailer's offering and efforts - what they get for what they pay. In the context of customer loyalty, being fair is much more important than being cheap (price level). Transparency, clarity and visibility of price communication build on these customers' expectations of being fair. Price dispersion is closely related to product range, brand architecture and variety of offerings - customers prefer a diverse range of products in one place, ranging from economy to premium prices and satisfying different needs.

The SHAP analysis provided an insight into the importance and influence of individual features on the model predictions and highlighted attributes such as PC and PD as highly influential due to the wide range of SHAP values. For features such as PLPR and L, further investigation is recommended to identify their potential negative impact on the model or opportunities for improvement. The private label product range is found to be an important attribute for the partially dependent variable Purchase Intention, but less important for Purchase Share and Willingness to Recommend, suggesting that a deeper customer relationship with the retailers' brands is still developing. Examining interactions between the most important features and other variables can also significantly improve the predictive power of the model. This should be prioritized in future analysis to achieve better accuracy and performance of the model. Based on this analysis, it is possible to improve the prediction by:

- data optimization: focusing on improving the quality of PDY and SL features, as these have the greatest impact on the model;
- investigating less important features: further investigating features such as PLPR and L to determine their potential negative influence on the model.
- improving predictive power: investigate interactions between key features and other variables to improve model accuracy and performance.

5. Conclusion

The application of ML algorithms is superior to standard statistical methods in several key aspects, including predictive accuracy and the ability to process complex data. The limitations of this study arise from the geographically limited area from which the respondents are drawn and the non-representative sample. However, the algorithms used in this paper can significantly improve the understanding of consumer behavior patterns and optimize marketing strategies in the context of predicting customer loyalty. In addition, the research has identified the key predictor variables that have the greatest impact on prediction, which can be very helpful in developing effective strategies to promote and maintain customer loyalty. This paper contributes to the literature by demonstrating how ML techniques can be effective in predicting loyalty towards grocery retailers and by highlighting potential areas for future research and improvement.

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CONTENTS

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FINANCIAL EFFICIENCY OF THE BULGARIAN GENERAL INSURANCE COMPANIES IN THE CONTEXT OF FINANCIAL CONTROLLING

Lydmil Krastev[1], Marin Marinov[2], Valentin Milinov[3]

Abstract: The purpose of the article is to examine the economic efficiency of the leading insurance companies (according to the "gross premium income" criterion) from the "General Insurance" branch. The research methodology is based on leading indicators for assessing the economic efficiency of the insurance company - "operating profit margin" and "return on equity". Economic efficiency indicators are also presented in the form of an index. The main results of the article are presented in figures, with an analysis of the economic efficiency of the leading insurance companies from the "General Insurance" branch. Analysis of the data shows that the leading insurance companies are performing oppositely to the General Insurance sector averages.

Keywords: financial result, financial controlling, economic efficiency, gross premium income, insurance company

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Introduction

The present study assesses the economic efficiency of the insurance company (general insurance). In the insurance sector, efficiency and proper resource utilization are crucial because they directly impact profitability, risk management, and long-term sustainability (Ilkaz & Cebi, 2024). Economic efficiency includes a certain group of indicators directly related to the profitability of the company. These are indicators such as operating profit margin, return on equity, return on invested capital. In the present study, the activities of four of the leading insurance companies in the "General Insurance" branch were evaluated based on economic efficiency indicators.

The article aims to present the target indicators of financial controlling in the company. The main target values of the financial controlling in the company are connected with summarizing indicators of economic efficiency. The article advocates the thesis that the assessment of the insurance company's activity should be given on the basis of economic efficiency indicators. The premium income of the insurer should not be considered a leading indicator, but rather a secondary one.

The relevance of this study is dictated by the need to examine the financial efficiency of the work of insurance companies operating in the "General Insurance" industry. The financial efficiency of the work of insurance companies in the "General Insurance" industry is examined in the context of financial controlling - a problem that is poorly researched in the Bulgarian literature.

The article uses information officially published by general insurance companies. The object of the study is placed solely on the economic efficiency of the work of companies in the "General Insurance" branch. All other problems related to the work of insurance companies remain outside the object of the study.

I. Problem Statement

There are a number of publications on the problems of controlling. In the Bulgarian literature, controlling is considered in the context of management, as it is aimed at coordinating the general activity of the company and achieving the previously set goals (Kamenov & Asenov, Kontroling, 2010). In the Bulgarian literature, special attention has been paid to the adaptive role of controlling in the company. Its role is to coordinate, plan, manage and control the actions of the company's management, adapting them to changes in the environment (Iliev, 2011). The main function of controlling is related to supporting senior management in making decisions related to achieving the set goals (Iliev, 2009). Controlling in the company is primarily oriented towards supporting management in making management decisions (Petrov, 2008).

In foreign literature, controlling is considered as a tool for solving the problems related to the coordination of individual activities in the company (Horvath et al., 2020). Controlling is also seen as a process of mastering the economic situation in the company (Daile, 2001). The functions of the controller (the person involved in controlling issues in the company) are seen as mandatory to the modern style of management - a style related to delegation of authority and management by objectives (Daile, 2001). The objective function of controlling is expressed in directing all management processes towards the achievement of company goals (Shulte, 1996). The tasks of controlling include the introduction of a controlling system and the development of a system of indicators for financial analysis (Shulte, 1996).

Controlling is reflected in all spheres and activities of the company. In Western literature, there is talk of marketing-controlling, logistics controlling, personal controlling (Shulte, 1996). Controlling in the company also affects its financing. In this context, we talk about financial controlling of the company (Daile, 2001). The term "financial controlling" is also found in Bulgarian literature. However, the understanding of financial controlling in the company is broader than that given in the Western literature. In Bulgarian literature, financial controlling in the company is related to financial indicators. These indicators form the basis of the development of company budgets and serve senior management as a basis for formulating and pursuing goals (Krastev, 2022). Financial indicators can be of different nature. They can be presented as absolute values (revenue, profit) and relative values - operating profit margin, return on equity, return on invested capital. The object of research in this article are the indicators "return on equity" and "operating profit margin".

II. Data and methodology

In the process of the research, financial indicators of insurance companies from the "General Insurance" branch were used. The profitability of the first four insurance companies, ranked according to the "premium income" criterion, was investigated. From the standpoint of financial analysis, a coefficient analysis of the profitability of the insurance company was made. Indicators were used to evaluate the operational activity of the insurance company - operating profit margin, as well as a general indicator of economic profitability - return on equity. For the purposes of the analysis, a time horizon of five years was used.

In the analysis of the data, two main financial ratios were used - operating profit margin and return on equity. The operating profit margin is calculated as the ratio between the size of the company's operating profit for the respective year and the size of the insurer's net premium income for the same period. The second indicator - return on equity is calculated as the ratio between the insurance company's net profit for the respective year and the size of the equity for the same period.

An analysis was made of the first four insurance companies in the "General Insurance" industry because they generate nearly half of the gross premium income on the general insurance market in Bulgaria.

A comparative analysis of the economic efficiency of the leading four insurance companies operating in the "General Insurance" branch was made. The comparative analysis includes both a comparison between the individual insurance companies and a comparison with the average values for the "General Insurance" industry for the analyzed five-year period. Data from the website of the Financial Supervision Commission, statistics section, were used.

III. Research findings

Controlling is based on certain target values formulated through financial indicators. These financial indicators are general and represent the overall economic profitability of the developed activity. Controlling is a continuous process, including the phases of planning, control and analysis. The contribution of controlling as a scientific discipline is that it does not consider the mentioned phases in isolation, but on the contrary - connects them in a continuous process. Controlling intervenes both in planning - by formulating the company's goals, and in the control and reporting of the achieved results. Company goals are set at the planning stage, and they form the basis for the development of company budgets. The development of the company budget is subordinated to the main or leading objective pursued by top management. This goal is usually identified with a leading financial indicator that summarizes the profitability of the business. As a summarizing indicator when formulating the company goal and developing company budgets, the return on invested capital is indicated in the literature (Daile, 2001).

The formulation of company goals and the development of company budgets are subject to internal planning in the company. The company's internal plans are confidential and not accessible to the general public. However, the activities of certain companies (such as insurance companies) is publicly available from the point of view of published financial results (financial statements). On the basis of the officially disclosed financial information, it is possible to judge how effective the insurance company's activities are, how it (the company) fares in the insurance market compared to its competitors and compared to the average values for the industry. The financial information and, in particular, the achieved results provide answers to the questions - Are the managers of the insurance company interested in economic efficiency? and How well do they know the instruments of controlling?

In the course of the research, indicators that have a direct relationship to the profitability of insurance companies operating in the "General Insurance" branch will be analyzed. The analysis used data for both individual years and average values - average annual profitability for a period of five years. The profitability values of the analyzed insurance companies are presented in the form of an index, with the average value for the "General Insurance" branch for the analyzed five-year period taken as a base of 100. The object of analysis are the first four insurance companies from the "General Insurance" branch, ranked according to the "gross premium income" criterion. The four analyzed insurance companies - ZK "Lev Ins" AD, ZD "Euroins" AD, ZAD "Bulstrad Vienna Insurance Group" AD, "DZI - General Insurance" EAD hold approximately 45% of the general insurance market by the end of the third quarter of 2023 according to the Financial Supervision Commission (FSC) data.

The first metric to look at is return on equity (ROE). This indicator compares the amount of net income generated by the insurance company at the end of the year to the amount of equity capital. Here we should make a clarification that return on equity is usually calculated on an annual basis. In this way, the managers of the insurance company can give a practically quantitative assessment of the work done during the past calendar year. Again, it should be noted that the financial controlling in the company is based on quantitative values that give a general assessment of the economic efficiency of the insurance company’s performance. Again, we will note that these indicators should be relative, not absolute values. Absolute values can hardly give a real assessment of the efficiency of the insurance company's work. For this reason, the companies that have realized the highest premium income do not always have the highest profitability. On the contrary - it may turn out to be just the opposite - companies that realize a much smaller volume of premium income generate much higher efficiency compared to the leading insurance companies ranked according to the "premium income" criterion. From this point of view, a distinction should be made between volume - revenue and profit and economic efficiency. Therefore, the high volume of premium income of the insurance company is not always a guarantee of high economic efficiency.

Relative indicators of economic efficiency have another important advantage. These indicators allow a comparison both between the insurance companies themselves and against the average values for the analyzed industry. In this way, a true assessment can be given as to whether the analyzed insurance company is doing better than the industry as a whole or vice versa. One more important clarification should be made - the analysis of the economic efficiency of the insurance company can be done both on an annual basis and for a period of five or ten years. In the latter case, averaged data should be used. The main advantage of the average values is that they show the real economic efficiency of the insurance company in the long term. From this point of view, the analysis of annual indicators of economic efficiency should be seen as inaccurate. Long-term economic efficiency should be considered precisely as a sustainable indicator when analyzing the activity of insurance companies.

The following figures (Figures 1-5) present the return on equity of the leading insurance companies in the General Insurance industry according to the criterion "gross premium income".

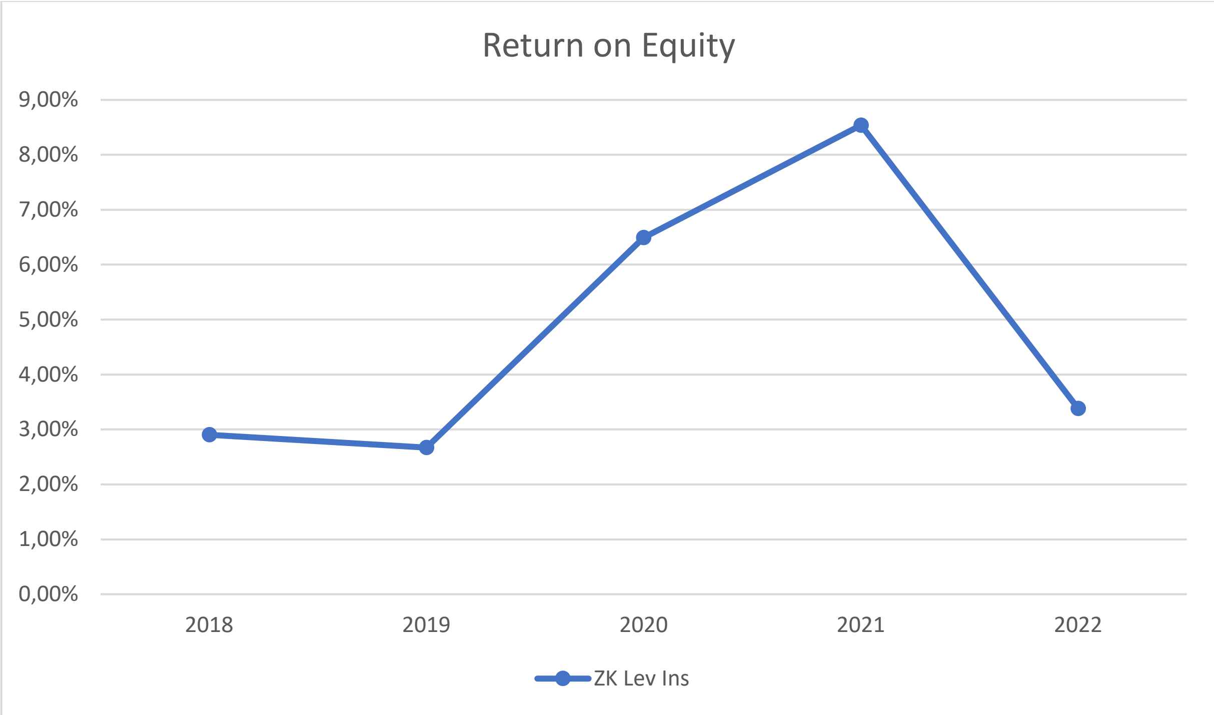

Figure 1. Return on Equity, ZK Lev Ins, years 2018-2022
Source: own calculations based on FSC data

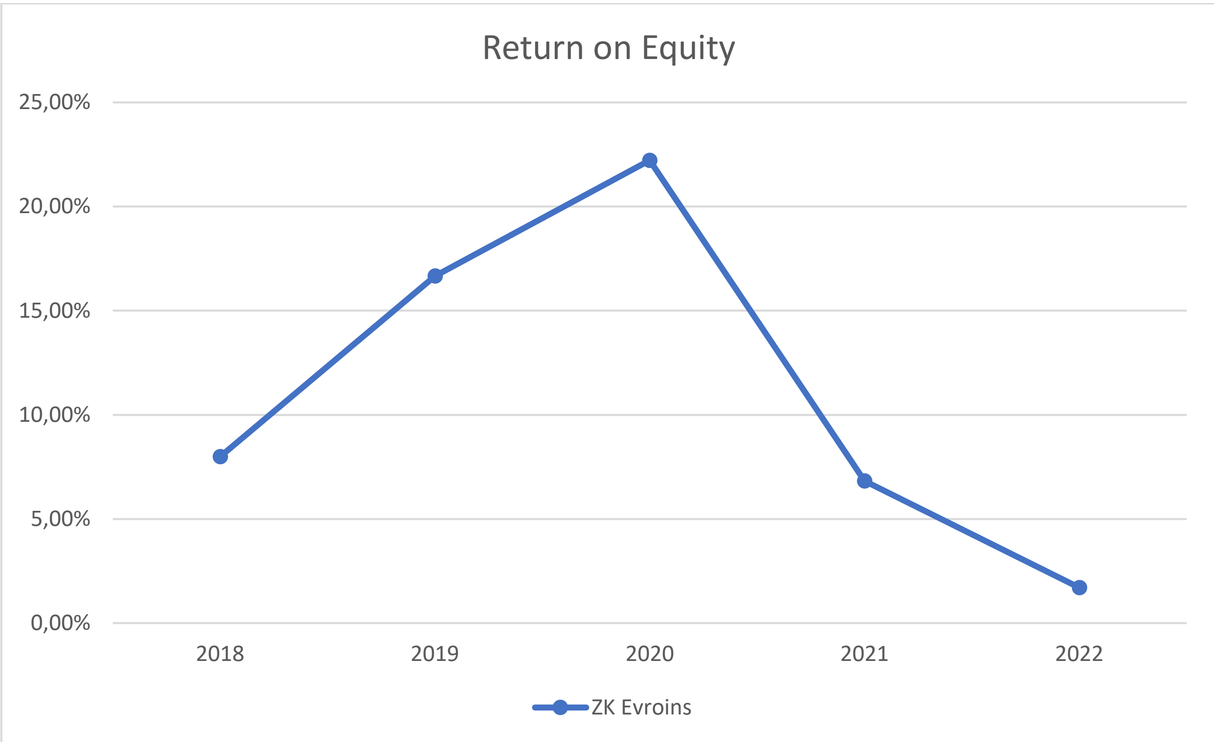

Figure 2. Return on Equity, ZK Evroins, years 2018-2022
Source: own calculations based on FSC data

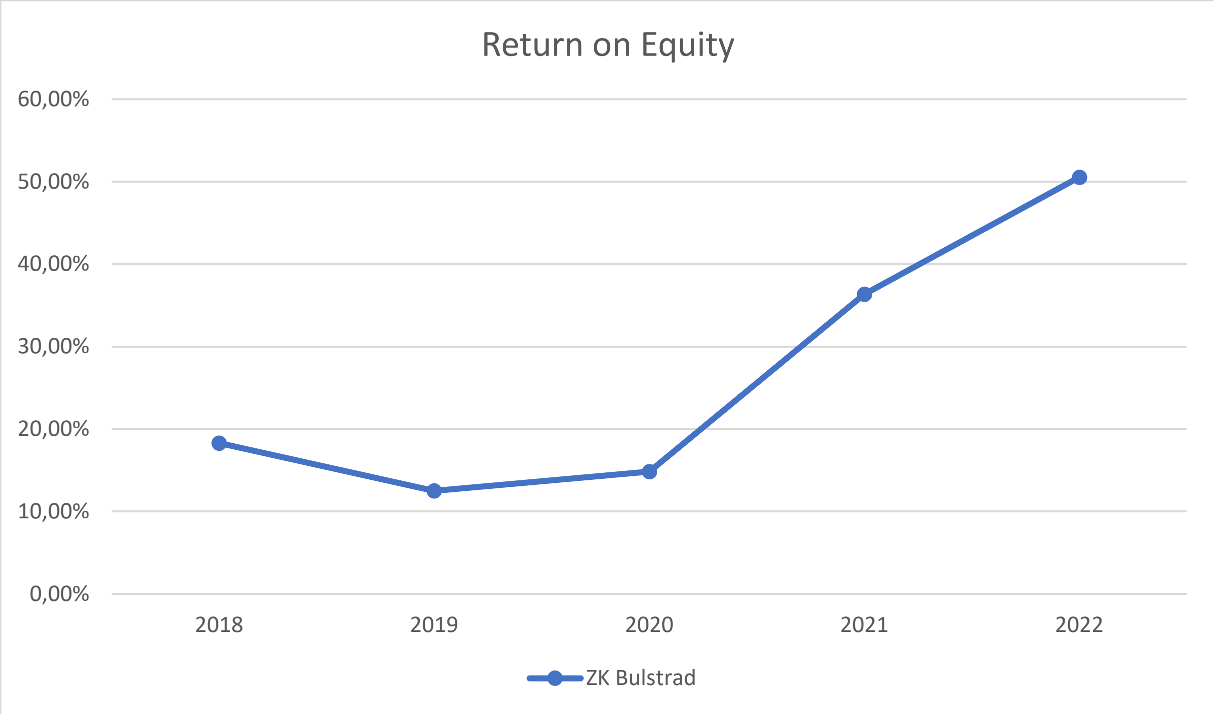

Figure 3. Return on Equity, ZK Bulstrad, years 2018-2022
Source: own calculations based on FSC data

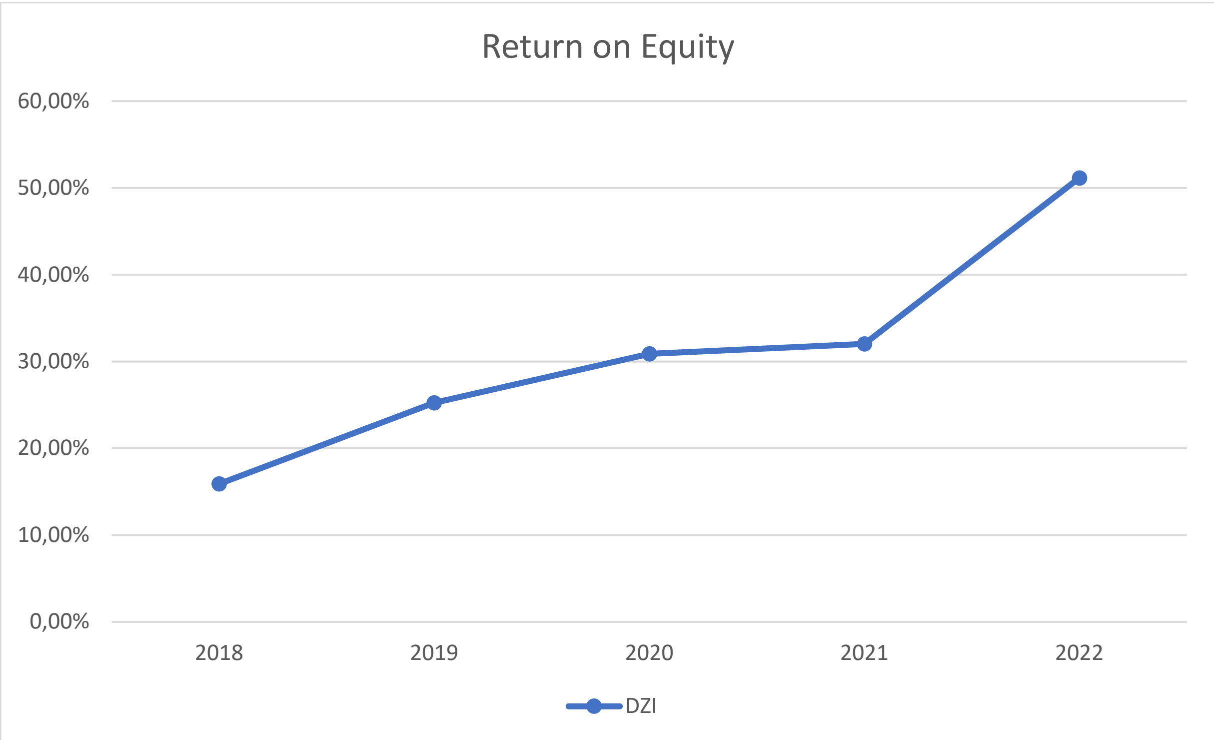

Figure 4. Return on Equity, DZI, years 2018-2022
Source: own calculations based on FSC data

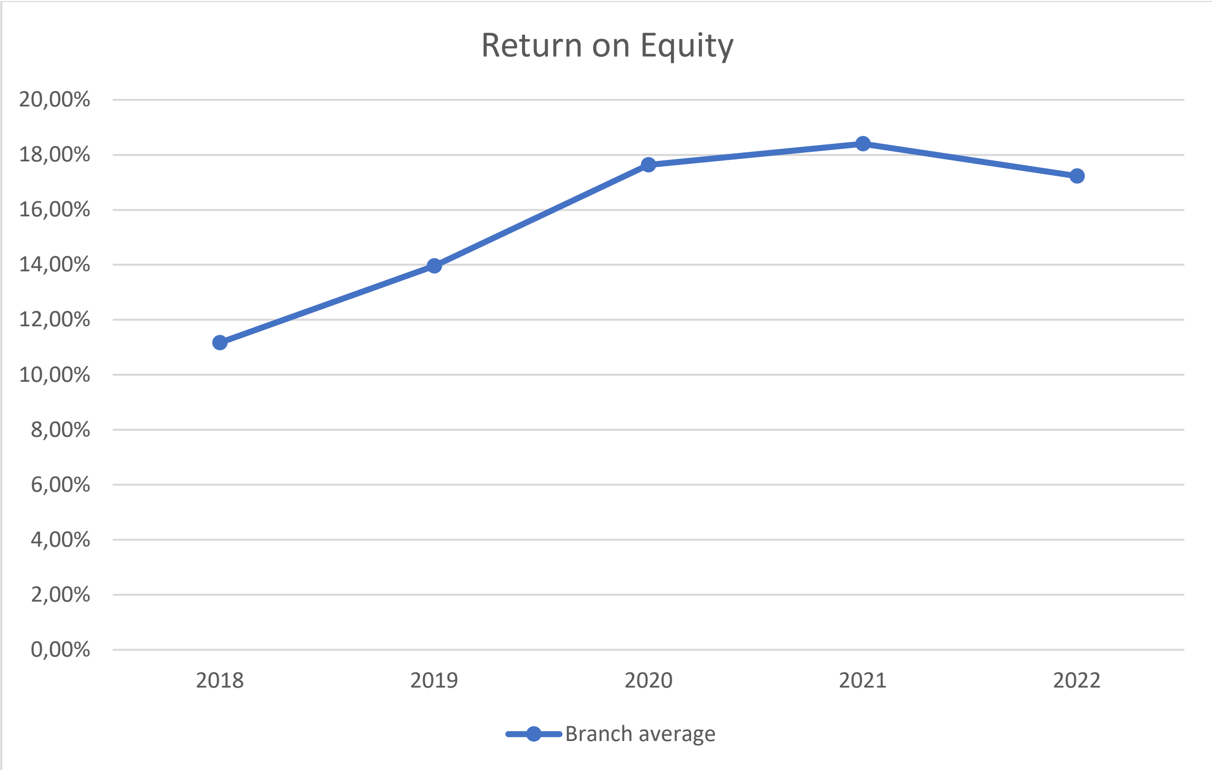

Figure 5. Return on Equity, Branch average, years 2018-2022
Source: own calculations based on FSC data

The return on equity of insurance companies for the analyzed five-year period is not the same. Moreover, this return is not the same for the insurance companies themselves in individual years. There are years when companies realize a return on equity above the general insurance industry average and years when it is below it. The only exceptions are ZK Lev Ins, which over the years realized a return below the average for the industry, and DZI, which, on the contrary, realized a high return on equity - significantly above the average for the "General Insurance" industry. In the case of the other two insurance companies - ZK Euroins and ZK Bulstrad, the return on equity varies compared to the average values for the "General insurance" industry. We will recall again that the financial controlling in the company focuses its goals on general indicators of economic profitability, such as return on equity and return on invested capital. From this point of view, the average annual return on equity of insurance companies for the analyzed five-year period is important. Such information would give us the opportunity to assess to what extent insurance companies from the "General Insurance" branch are interested in economic profitability indicators and whether they apply the controlling tools in the right way. It is possible that the high return on equity can be achieved without applying the instruments of controlling. In this case, however, the high return on equity is the result of haphazard management rather than purposeful management action.

The following figure (Figure 6) presents the return on equity of the four companies analyzed – averaged over a five-year period (2018 – 2022). The return on equity of insurance companies is presented in the form of an index, with the average value for the General Insurance industry having a value of 100.

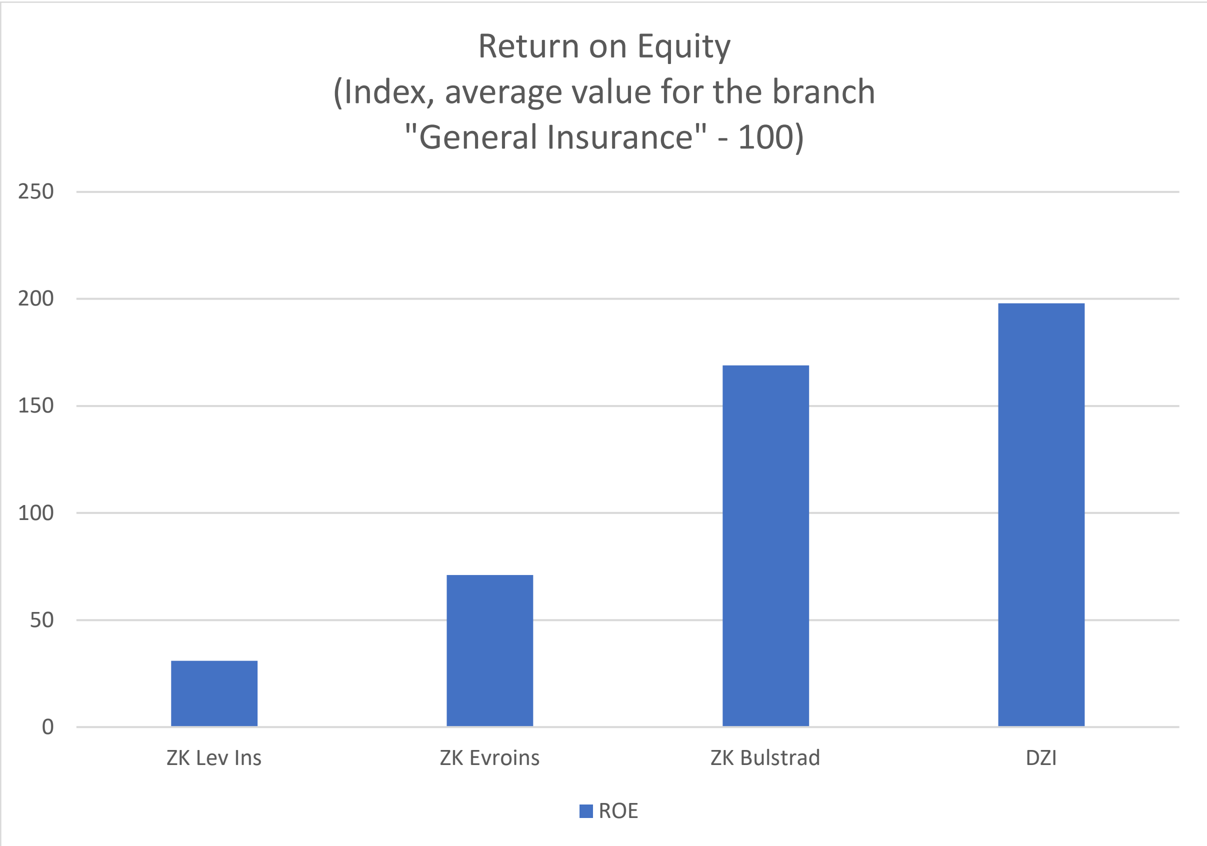

Figure 6. Return on Equity – five-year average, years 2018-2022
Source: own calculations based on FSC data

There are companies that achieve returns above the industry average (ZK "Bulstrad" and ZK "DZI") and those that achieve returns below the average (ZK "Lev Ins" and ZK "Euroins"). It should be noted that for the analyzed five-year period, the insurance company "DZI" shows an average annual return on equity, exceeding almost twice the average return on equity for the "General Insurance" industry. “Bulstrad” Insurance Company shows an average annual return on equity that exceeds 1.5 times the average annual return for the industry. Contrary to the above, the insurance companies "Lev Ins" and "Euroins" show an average annual return on equity, which is significantly lower than the average annual return for the "General Insurance" industry (at ZK "Lev Ins" it is nearly three times lower than the industry average).

The data presented show opposite results (for the analyzed five-year period). The question here is - Why do some companies realize multiple times higher return on equity compared to the general insurance industry averages, while other insurance companies realize multiple times lower returns than the industry averages? In both cases, we are talking about insurance companies that are leaders in the "General Insurance" branch according to the "gross premium income" criterion. The data from the survey support the thesis that the high premium income of the insurance company is not a guarantee of high economic efficiency. In this case, the goals pursued by senior management, as well as the costs associated with the implementation of controlling in the insurance company, are important. On the one hand, the high economic efficiency of certain companies (ZK "Bulstrad" and ZK "DZI") is a signal for setting certain priorities. On the other hand, the low economic efficiency of ZK "Lev Ins" and ZK "Euroins" casts doubt on the goals pursued by the top management.

In the case of publicly traded companies, investors are interested not only in the generated income, but also in the achieved financial results. The financial results provide an objective assessment of the goals achieved by the top management of public companies. These results are monitored by investors and form the market value of equity capital.

An assessment of the results of the insurance activity should be given not only in terms of the growth of premium income and conquered market share, but also in terms of the achieved efficiency of these sales through the "operating profit margin" indicator. The following figures (Figures 7-11) presents the operating profit margin of the leading insurance companies (according to the "gross premium income" criterion) for the analyzed five-year period. Operating profit margin is a ratio between an insurance company's operating profit (earnings before interest and taxes) and net premium income (gross premium income less premiums ceded to the reinsurer).

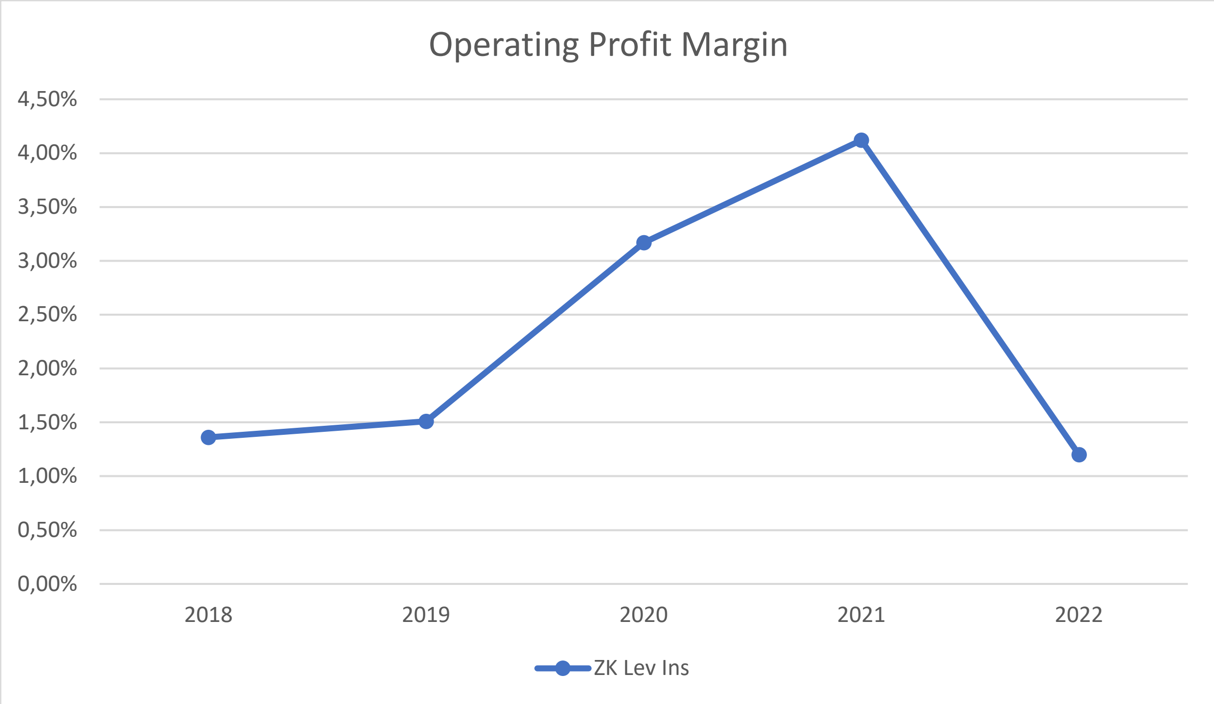

Figure 7. Operating Profit Margin, ZK Lev Ins, years 2018-2022
Source: own calculations based on FSC data

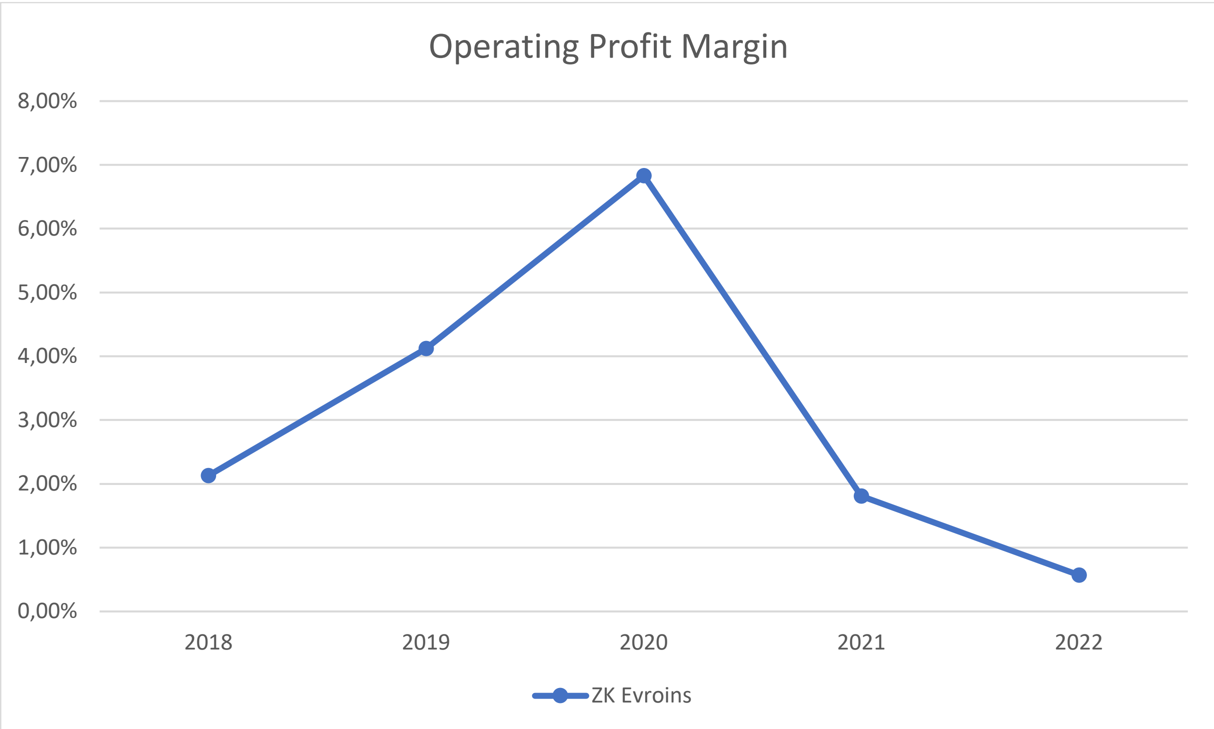

Figure 8. Operating Profit Margin, ZK Evroins, years 2018-2022
Source: own calculations based on FSC data

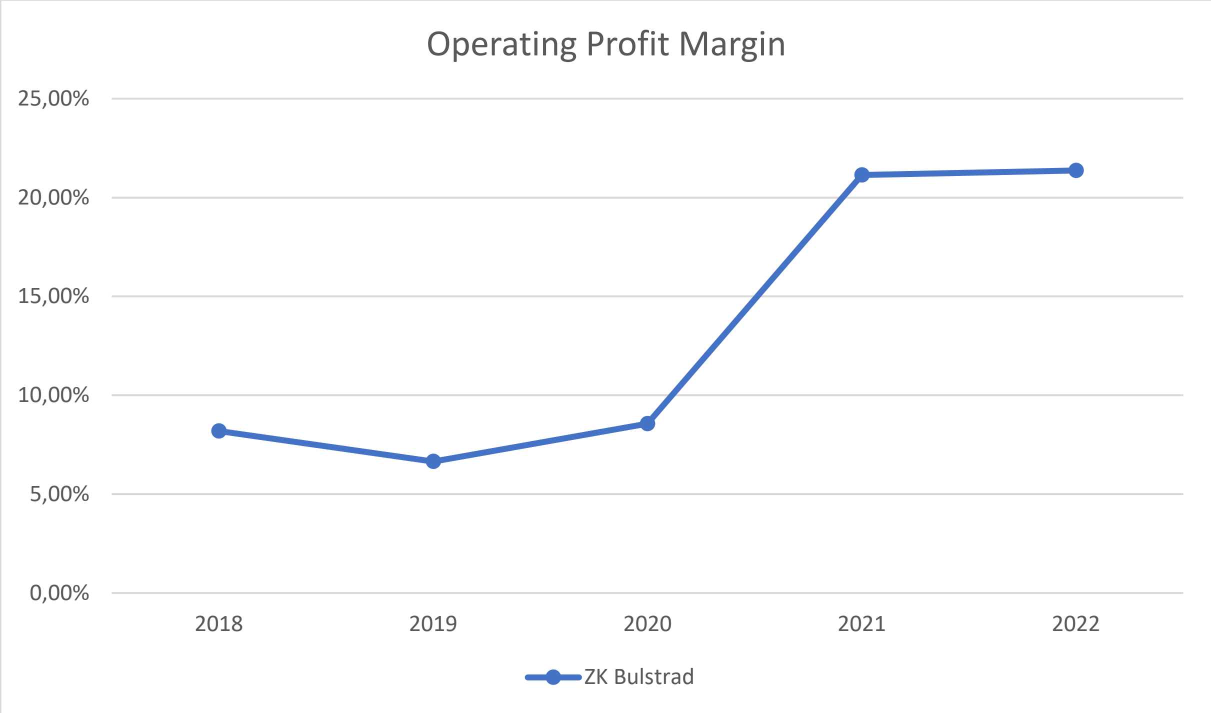

Figure 9. Operating Profit Margin, ZK Bulstrad, years 2018-2022
Source: own calculations based on FSC data

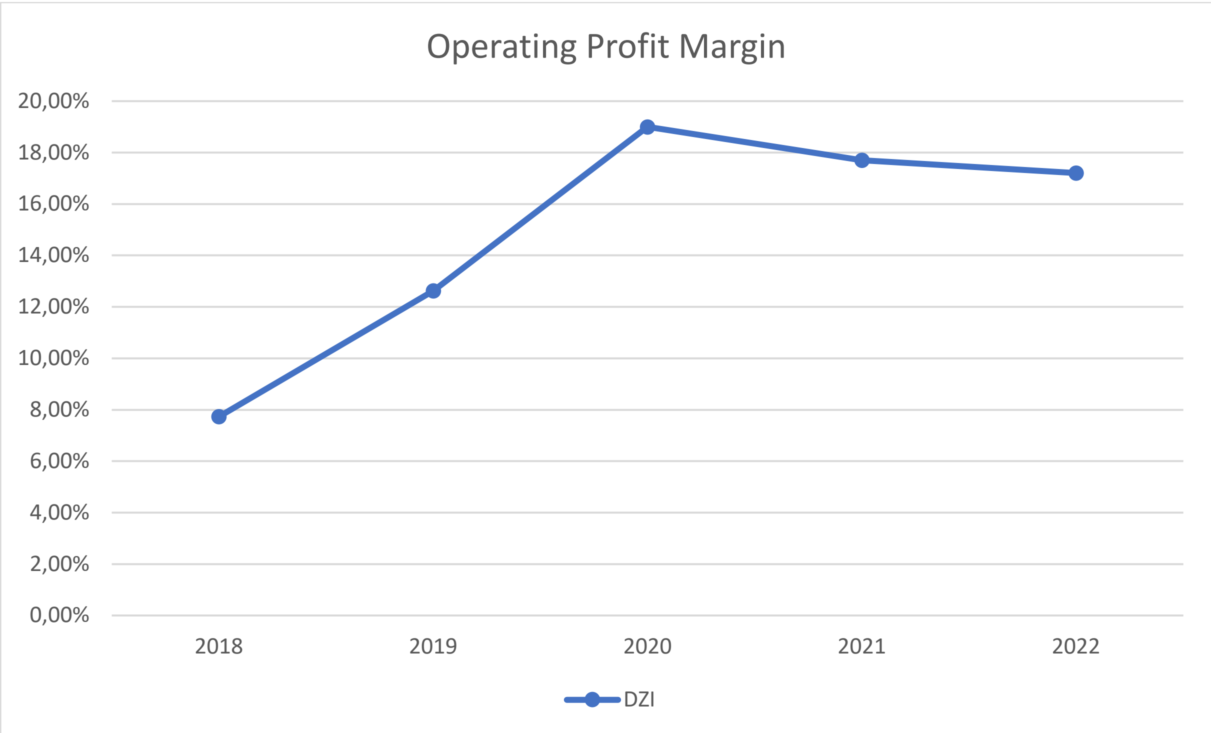

Figure 10. Operating Profit Margin, DZI, years 2018-2022
Source: own calculations based on FSC data

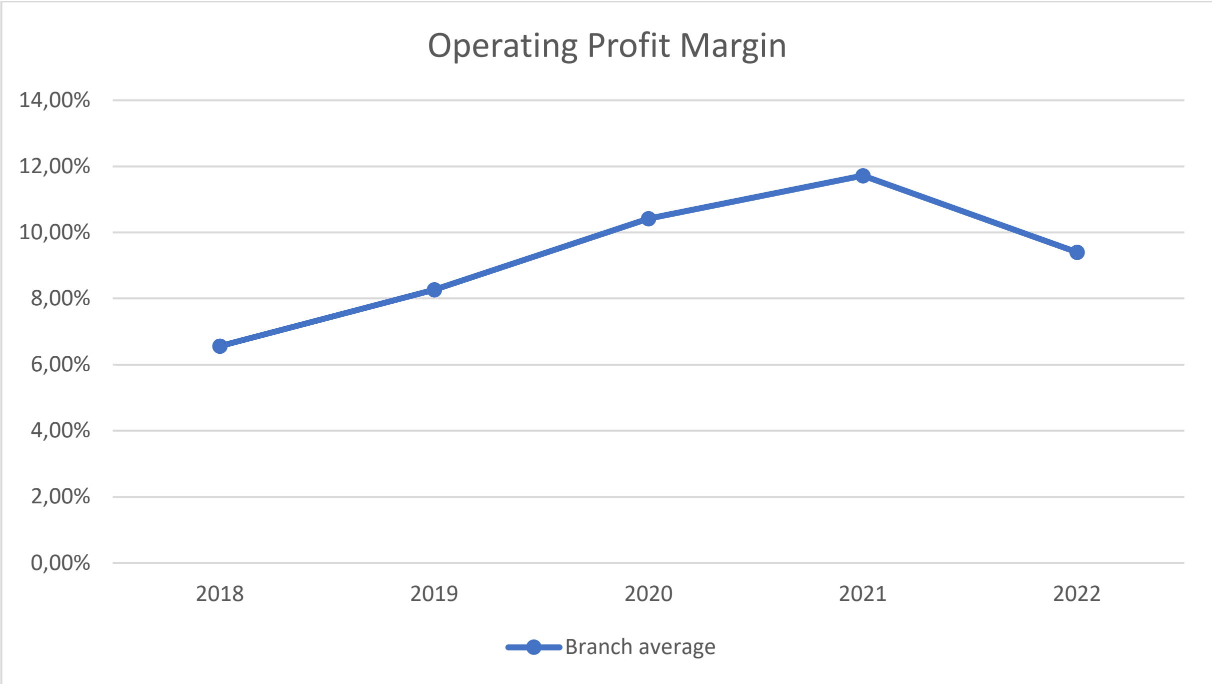

Figure 11. Operating Profit Margin, Branch average, years 2018-2022
Source: own calculations based on FSC data

From the above figure, it is clear that, for the analyzed five-year period, the "General insurance" branch realized a positive trend in the operating profit margin. The value of the operating profit margin for the General Insurance industry is just over 5% for the year 2018, while for the year 2021 this value exceeds 10%. The results shown by the leading insurance companies from the General insurance industry are again ambiguous. There are companies that show an operating profit margin value well above the General Insurance industry average and those that show a value well below it. The data in the indicated figures confirm that a high profitability of sales revenue also leads to a higher return on equity. ZK "Bulstrad" managed to realize an operating profit margin for 2021, exceeding 20%, with an average value for the "General insurance" branch for the same year - slightly above 10%. ZK "DZI" manages to realize an operating profit margin for 2020 of nearly 20%, compared to an average value for the "General Insurance" industry for the same year - nearly 10%. In the other direction are the insurance companies Lev Ins and Euroins. The value of the operating profit margin of the mentioned companies is below the average value for the "General insurance" industry in all five years of the analyzed period.

The operating profit margin can also be presented in the form of an index. The following figure (Figure 12) shows the operating profit margin

(OPM) of the leading insurance companies from the "General insurance" branch - an index with an average value for the branch of 100. The index represents the five-year average return on sales.

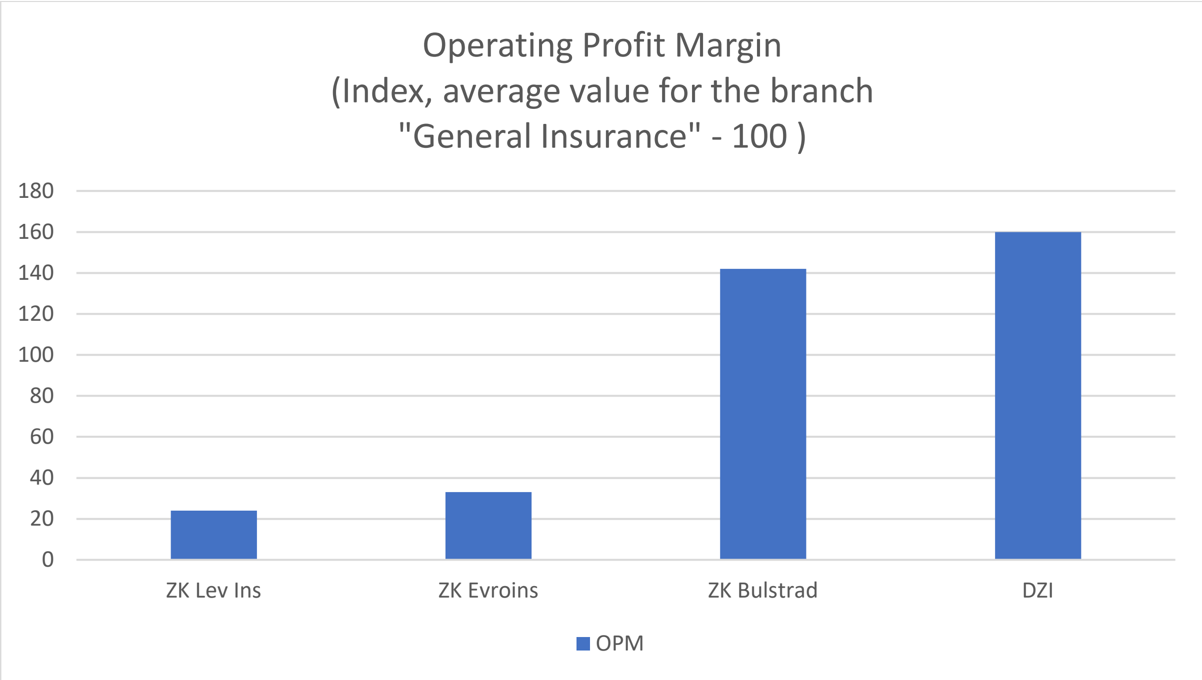

Figure 12. Operating profit margin, five-year average, years 2018-2022
Source: own calculations based on FSC data

The presentation of the operating profit margin for the analyzed five-year period in the form of an index has significant advantages. The index presents average values of the operating profit margin of the indicated companies for the analyzed five-year period. This makes the results sustainable over time. On the other hand, the index enables a direct comparison of sales profitability both between individual companies and against the average for the General Insurance industry. The data in the figure confirm what has already been said above, that for the analyzed five-year period, ZK "Bulstrad" and ZK "DZI" realize an operating profit margin significantly above the average for the "General Insurance" industry. Quite the opposite - ZK "Lev Ins" and ZK "Euroins" realize an operating profit margin significantly below the average for the "General insurance" industry for the analyzed five-year period.

The figure below shows a regression analysis between operating profit margin (independent variable) and return on equity (dependent variable) for the General Insurance industry. Despite the small number of observations (five-year period), the high values of the correlation coefficient and the coefficient of determination are evidence of the existence of a relationship between operating profit margin and return on equity for the General Insurance industry.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0,940584
R Square	0,884699
Adjusted R Square	0,846265
Standard Error	0,011893
Observations	5

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0,003256	0,003256	23,01876	0,01723
Residual	3	0,000424	0,000141		
Total	4	0,00368			

Figure 13. Regression analysis between operating profit margin (independent variable) and return on equity (dependent variable) for the General Insurance industry, years 2018-2022
Source: own calculations based on FSC data

IV. Discussion

The question remains open - what are the reasons for the different economic efficiency achieved by insurance companies? We will recall the fact that, according to the FSC data, the insurance companies mentioned in the study hold approximately 45% of the general insurance market at the end of the third quarter of 2023, and for 2022 their market share exceeds 50%. These are companies that generate gross premium income not in the amount of hundreds of thousands, but in the amount of hundreds of millions of BGN, and in some of them (ZK "Euroins" and ZK "Lev Ins") the gross premium income from insurance activity significantly exceeds four hundred million BGN for 2022. This raises the question - Why do companies with such a large gross premium income achieve economic efficiency below the average for the General Insurance industry?

The answers to the mentioned questions are hidden in the financial statements of the insurance companies. High premium income is not a guarantee of a high financial result, given the fact of the huge amount of premiums ceded to the reinsurer and the amount of expenses related to paid insurance benefits. However, here we should pay attention to the fact that the financial statements reflect the results of the insurance company's activity. They do not provide an explanation for the developed company plans (budgets), the set goals, the way to achieve them and the extent of their implementation. The conflicting results (operating profit margin and return on equity below the industry average) cast doubt on the goals pursued by the insurance company's senior management. The economic efficiency of the developing insurance business is also in doubt. It is also questionable to what extent the senior management knows the controlling tools in the company and whether they are applied in the appropriate way. In this case, the preventive activity of the insurance company, which is related to the control of the incurred expenses, deserves attention (Krastev, 2017).

Conclusions

The above analysis of the activities of the leading insurance companies (according to the "gross premium income" criterion) from the "General Insurance" branch leads to the following conclusions. First, the leading insurance companies show different economic efficiency according to the "return on equity" and "operating profit margin" indicators. One of them shows efficiency significantly above the average for the "General Insurance" branch - ZK "DZI" and ZK "Bulstrad", while others show efficiency below the average for the branch - ZK "Euroins" and ZK "Lev Ins". Second, higher operating profit margin of insurance companies leads to a higher return on equity.

Attention should also be paid to the fact that insurance companies can be traded on the stock exchange. Publicly traded companies are valued in the stock market according to the market value of equity. In publicly traded companies, there is a relationship between the market value of the equity capital and the achieved financial results. The specialized literature advocates the opinion that the current market price of shares of public companies reflects investors' expectations regarding future financial results (Rappaport, 1998). This puts pressure on senior management regarding the results achieved. We will only recall that the leading goal of company management is aimed at increasing the market value of the equity capital of publicly traded companies. From this point of view, the achieved results and economic efficiency are essential for company management.

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CONTENTS

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DEVELOPING AN EQUILIBRIUM MODEL FOR ASSESSING HOUSE PRICE DEVIATION

Dragomir Stefanov[1],
Yana Stoencheva[2],
Petar Ivanov[3]

Abstract: This article deals with the problem of developing a reliable diagnostic tool that could help in reducing the unpredictability of house price cycles that cause undesirable reverberations in the economy. Such type of cycles is among the main drivers of price volatility and crises not only in the real estate sector itself but also for the financial system and the whole economy too. The purpose of this study is to introduce a step-by-step methodology for assessing the true deviations of house prices from their long-term equilibrium levels. Factors with a significant impact on house price dynamics are also determined. The result is a ready-to-use econometric model that detects possible formation of house price bubbles early enough, so that policymakers and regulators could take adequate actions against market overheating in due time.

Key words: house prices, equilibrium, econometric model, methodology, price cycles

JEL: C32, E32, R31.

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Introduction

House prices are historically prone to be *cyclical* in nature as their dynamics goes through constantly repeating phases of *booms* and *bursts*.

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However, while housing booms typically *build up silently* in time, market downturn could surprise by its *sudden and rapid evolution* (Corradin & Fontana, 2013). Moreover, cycles could differ in length and time. A major plummet of house prices could have a detrimental impact not only on the real estate and construction sector, but it could also be transmitted to the financial sector and the whole economy, too. Such a powerful and *contagious effect* is a typical trait of the house price bubbles.

A notorious example of the above is the US *housing bubble* that took place in the 2000s. It originated from the uncontrolled expansion of *high-risk mortgage bonds* in the banking sector. This initially domestic crisis of overvalued houses eventually led to the onset of the *Global Financial Crisis* (GFC) in 2007-2008 and to a widespread economic recession as an aftermath. Eigner & Umlauft (2015) and others consider GFC to be the *worst financial crisis* since the 1930s Great Depression. This is convincing evidence for the threat posed by a house price bubble.

The main *motives* behind the current research are the challenges before the *timely detection* of house price bubbles. Finding an instrument to deal with these challenges will assist *central banks* and other institutions to prevent, or at least weaken, the negative consequences of a bursting bubble by undertaking proper macroeconomic and credit policy measures. In that way the *goal* of the article is to develop an *econometric model* for detecting house price bubbles through measuring the *deviations* of house prices from their *equilibrium levels*. *Abnormal behaviour* of house prices could be an early signal for a bubble formation and hence monitoring is important to be regularly done with the proper toolkit.

There are several ways for measuring deviations of house prices. Our methodology for that is based on building a multivariate *equilibrium econometric model* that is a kind of either a Vector Autoregression (VAR) or Vector Error Correction Model (VECM) – two techniques widely used in related papers (see the literature review section below). The model itself presents a *contemporaneous equilibrium of dependable variables* with the change of the national house price index as one of them and the others are factors with long-term impact on it. In that way the equilibrium is built as a single *dynamic system* of *mutual long-term interaction*. In the case of a VECM, a part with *short-term equilibrium* is also added, in which external factors participate, too.

1. Literature review

A *bibliographic survey* by Li & Li (2022) gives a good idea of the scope of research on house prices and the drivers of their change. Their review of *academic publications* on the subject for the last six decades (from 1960 to 2020) showed a total number of 4,125 relevant publications in the Web of Science database. Articles studying the behaviour of *house prices* would most often recourse to econometric models, especially of the VAR/VECM type and their derivatives. The main motive here is that such dynamic models would help override the limitations intrinsic in any static methods, where the long-term average will be calculated as a historically average, i.e., it will remain fixed in time.

Kotseva & Yanchev (2017) conducted one of the most *comprehensive studies in Bulgaria* on the current topic. They developed a VECM-type model for measuring the deviation of house prices in Bulgaria from their *long-term trend* by studying the cointegration relation between house prices, the total amount of *hypothetical borrowings* and the direct foreign investments (FDIs) into the real estate sector. *Hypothetical borrowing* of money is a coefficient derived by the authors from the interaction between household disposable income and mortgage interest rates. Another qualitative research on the topic, specifically for Bulgaria, is that of Yovkova (2009). The author examines the correlation between the domestic real estate and mortgage markets.

Econometric research on the *house price drivers* is very thorough for the *developed countries* (especially the USA, UK and EU countries). In the last two decades an increasing number of studies have been conducted for the *emerging markets* too, including China, the UAE and the CEE countries (where Bulgaria belongs). They are attractive for the researchers with the price dynamics, characterized by prolonged phases of a booming market. At the same time, however, researchers of those markets encounter several difficulties related to the *quality* and *scope* of the available data. Also, the CEE region has its own specifics for the inflating house price bubble preceding the Global Financial Crisis of 2007-2008, among which are the *increased demand for dwellings from abroad* and the *credit boom* connected with financial globalization and liberalization of credit standards (see Egert, B. & Mihaljek, D. (2007)).

As we see, studies on the current topic are predominantly model based with a wide range of applied methods. For example, Cevik & Naik (2022) apply a relatively innovative approach to analyse house price change in 10 emerging housing markets in Europe. They use a method based on a *panel quantile regression*. Applying this method, they perform segmental price analysis, which allows them to compare the changes that occur in the *upper and lower price ranges* of the housing market. In a study covering the impact of monetary policy on house prices, Lee & Park (2022) use a VAR-type model with *a time-varying component*, or the so-called TVP-VAR (time-varying parameter). This model accounts for the variable influence of interest-rate shocks during the periods of implementation of different monetary policies, in contrast to the standard VAR model, where the influence of the factors is assumed constant.

The model of Beraldi & Zhao (2023) proves that the *elasticity of housing demand* depends on the change in the *interest rate* on *mortgage loans*. The authors empirically find out that a 1% reduction in mortgage rates will theoretically result in rise of house prices by about 10%. Aastveit & Anundsen (2022) and several other authors emphasize on the effects of *monetary policy* of central banks on the house price movement. For example, during the COVID-19 pandemic the *monetary policy easing* has its own contribution to house price appreciation that is intensified by the *shift of household preferences* from renting to owning a home.

The European Central Bank (ECB), for example, has developed and implemented a comprehensive *toolkit for monitoring and detecting* the signals of bubble-inflated house prices. For this purpose, the institution monitors a set of four indicators: on the one hand, deviations from the long-term average of the “price-to-income” and “price-to-rent” ratios, and on the other: econometric estimates obtained by applying inverse demand and asset valuation models[4]. In addition to these, the ECB uses a model called “*Residential real estate price-at-risk model*”, which collects information from the above four indicators and others related to them, including *financial conditions*, the *level of cyclical systemic risk*, the current rate of *house price growth* and *consumer attitudes* (Jarmulska et al.,2022).

As we could *summarize* our findings from the above and from other sources, the use of *econometric models* has become *ubiquitous* in almost any quantitative research, especially when they cover macro-financial indicators. In support of the thesis for *qualitative accumulation* in economic theories is the study of Guo et al. (2024). What is crucial to the persistence of a successful econometric model, however, is the choice of *appropriate variables*. Bojinova et al. (2010) presented a methodology for that in a study

[4] The inverse demand function considers price as a function of quantity, rather than the regular scenario in which the quantity demanded is a function of price.

of public companies in Bulgaria. In our article, we also introduce a *procedure* based on quantitative methods for *sorting out the country-specific factors* most closely connected with the change of house prices.

2. Presenting the model

We adopt an approach in which we *group* diverse factors into separate *cluster groups* based on a common feature. Therefore, the study of the impact on house prices will first begin at the cluster group level, and then will proceed *individually* for each of the factors included in it, i.e., we follow the principle from the general to the specific. As Bespalov et al. (2021) note in their study, *cluster analysis methods* are widely used in economics, especially when comparing objects that have common properties.

We present in detail the methodology for building the so-called by us *Equilibrium Econometric Model of House Prices* (EEMHP) in the next section. This methodology's composition encompasses 5 stages with a total of 17 steps within them. The main dependent variable (Y) is the *House Price Index* (HPI) at the national level in Bulgaria, calculated by the National Statistical Institute (NSI) on a regular basis. The HPI is a quarterly indicator and represents the *change in market prices of dwellings* (both newly built and existing) purchased by households.

The desired equilibrium econometric model will represent a *system of interrelated regression equations*. Each of these equations will yield an outcome variable, which is the change in Y or one of the other endogenous variables closely interacting with Y. The behaviour of the output variables will be programmed so that it depends on the combined impact of the other *endogenous* and *exogenous* variables included in the model, whereby they form *subsystems of long-term and short-term equilibrium*.

3. Stages of compiling the EEMHP

In **stage 1**, it is planned to carry out an initial review of the time series dynamics and their statistical properties. We work with quarterly data. Step one of this stage is to look up the indicators which interact with house prices we intend to measure in the relevant statistical sources. We *derive* their time series with all the observations for the reference period. Step two involves performing *transformations* on the time series where needed. There are three possible transformations, namely: 1) *negative-to-positive transformation*, 2) *logarithmic transformation* (all indicators are expressed in their natural logarithm form except for indicators representing interest rates, growth rates and shares) and 3) *differentiation* (we apply to time series that are not *stationary*).

The *stationarity test* is the third step of stage 1. It is performed visually and statistically. In the latter, we perform unit root tests, which include the *Augmented Dickey-Fuller (ADF), Phillips and Perron (PP) test*, and the *Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test*. The final step of this stage is the fourth, in which we *examine and compare* the behaviour of each of the indicators, as well as its group as a whole, against that of the HPI.

In **stage 2**, we examine the regression dependencies within each of the groups and identify the factors with the strongest influence on house prices. Step one at this stage is to construct regression equations for each group of indicators, in which they would play the role of independent variables (x) with the HPI being the dependent variable (y). We also include a constant term (C) in them. The regression equations will be *double logarithmic* and will take the following form:

$$log\, y_t = \beta_0 + \beta_1\, log\, x_{1,t} + \beta_2\, log\, x_{2,t} + \beta_3\, log\, x_{3,t} + \varepsilon_t \quad (1)$$

The double log form is suitable for the intended purpose as it allows measuring the elasticity of the dependent variable (*y*) with respect to the behaviour of the independent variables (x_i). Thus, the coefficients before *x* will indicate the percentage change in *y* per unit change in *x*. In addition, the logarithmic transformation of all the variables allows us to work with a linear model even if there is no linear dependence between them.

The applicable method for calculating the above type of equations (1) is that of *linear ordinary least squares* (OLS). We examine the readings of each factor and check the direction of their coefficients to see whether it corresponds to preconceived theoretical notions. When we find *discrepancies* between empirical results and theoretical assumptions, we should *reconsider the relationship* between house prices and the specific factor that has caused the disturbance. Finally, we highlight the factors that are *statistically significant* for Y and have high coefficients. These will be among the *potential candidates* to play the role of endogenous or exogenous variables in our econometric model.

We extend the analysis by presenting each of the regression equations in *two variants with different scope*: the first variant would be narrower and would include the period up to the end of 2019 (i.e., before the onset of the COVID-19 pandemic), while the second would cover the *entire period* with available data, e.g., until the end of 2023. The reason behind this division is to *measure the impact of the pandemic* on the existing regression relationships and also to find out whether the economic shock from it causes their disruption. Finally, in this step, we draw conclusions about the joint *influence* of the factors of the particular group on the dependent variable Y by comparing the readings of the *coefficient of determination (R^2) and the Durbin-Watson (DW) autocorrelation statistics* for the whole equation.

Step two would involve conducting pairwise *tests of Granger causality* between the HPI (Y) and each of the factors of the specific group. The *F-statistic and probability (ρ)* of each pair in any of the two directions of causality are compared. We note all cases, in which we reject the null hypothesis of no causality. We single out the factors from the respective pair that exhibit causality in *both directions*—i.e., from the corresponding indicator to Y, as well as from Y to it. Factors meeting this condition together with Y would be candidates for inclusion in the role of *endogenous variables* in our projected model. We check if all other factors that exhibit Granger causality only in *one direction* or *none at all* are statistically relevant in the regression equation from the previous step. In case of a positive answer, we direct the corresponding factor to the group of potential candidates for *exogenous variables*.

The last step three of the second stage involves compiling a summary table based on the conclusions drawn. There, we indicate the *role of each of the considered indicators* – whether it will participate in the model as an endogenous variable, as an exogenous variable, or will not participate at all.

In **stage 3**, we *model the long-term dependence* of the HPI on other variables that we have chosen to be endogenous. By long-term equilibrium we mean a system, in which the variables would *move in synchronization*, i.e., exhibit a form of co-integration. In this step, we will define the model specification type. We will limit the choices to *two classic types* of vector models – *Vector Auto Regression* (VAR) and *Vector Error Correction Model* (VECM). The algorithm for choosing the suitable type of specification is presented in the form of a diagram (see *Fig. 1*).

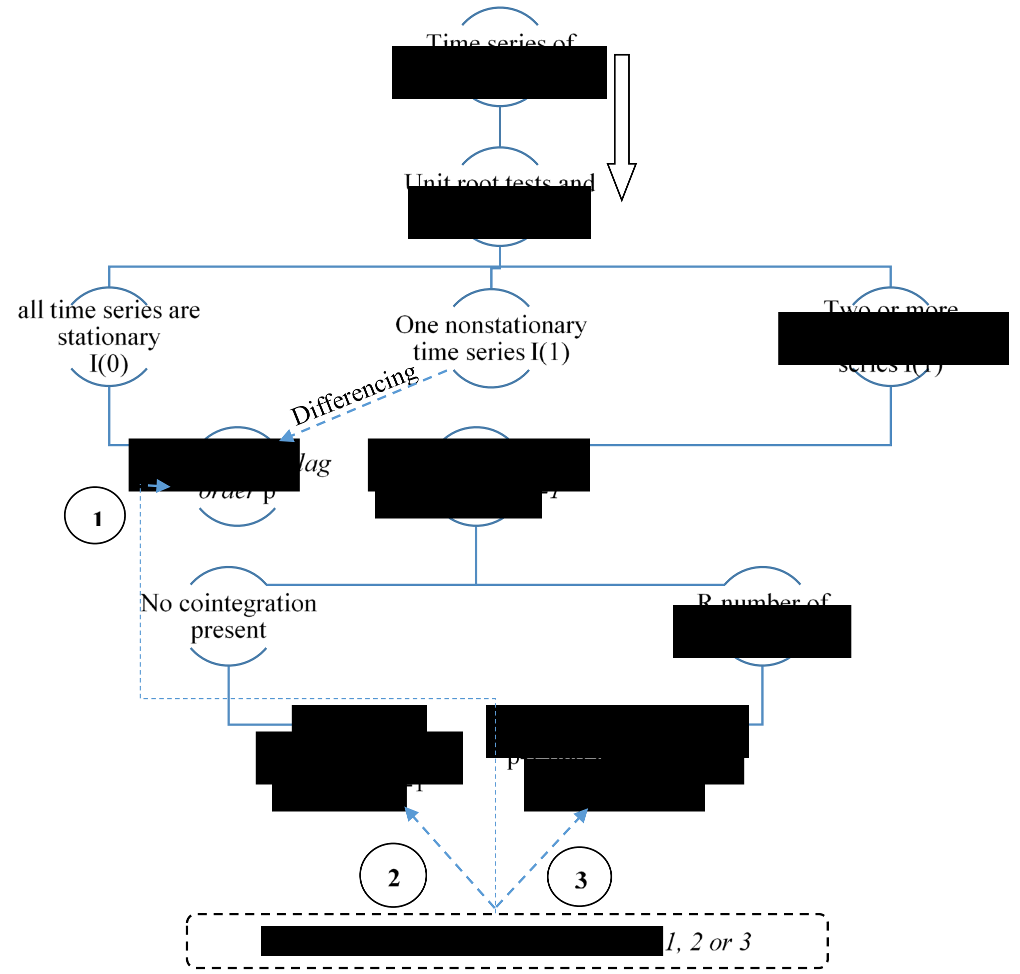

Figure 1. Procedure for choosing an econometric model depending on the statistical properties of the data
Source: own methodology, see Ivanov (2024)

Finding a *stable long-term (co-integration) relationship* between the endogenous variables involved suggests choosing a VECM type model. It would allow to determine the pace with which deviations from the long-run equilibrium are cleared out after a shock. BNB applies a similar type of model to determine the *equilibrium* (long-term) level of house prices in Bulgaria and to *assess the impact* of identified macroeconomic factors on their dynamics (Kotseva & Yanchev, 2017).

Initially, model construction starts with a VAR-type specification, which is expected to subsequently *undergo modifications* depending on the results of the tests conducted. Essentially, a VAR is a system of equations that can be written mathematically in the manner shown in (2). As to the example below, we will assume that we are working with three variables (x, y, z), with each of them participating with 2 lags, i.e., the system would match type VAR (2):

$$x_t = c_1 + \sum_{i=1}^{2} a_{1,i} y_{t-i} + \sum_{i=1}^{2} \beta_{1,i} x_{t-i} + \sum_{i=1}^{2} \gamma_{1,i} z_{t-i} + \epsilon_{x,t}$$
$$y_t = c_2 + \sum_{i=1}^{2} a_{2,i} y_{t-i} + \sum_{i=1}^{2} \beta_{2,i} x_{t-i} + \sum_{i=1}^{2} \gamma_{2,i} z_{t-i} + \epsilon_{y,t}$$
$$z_t = c_3 + \sum_{i=1}^{2} a_{3,i} y_{t-i} + \sum_{i=1}^{2} \beta_{3,i} x_{t-i} + \sum_{i=1}^{2} \gamma_{3,i} z_{t-i} + \epsilon_{z,t} \quad (2)$$

where $c_{1,2,3}$ is a constant (intercept); α, β and γ are coefficients before the variables x, y and z; ϵ is a random error, with t being the current period.

Depending on the results of the *stationarity tests* performed for each time series, the VAR specification may need additional operation for differentiation. The latter would imply differentiating the variables on both sides of the equations, i.e., variables to be represented by their differences (e.g. Δx, Δy, Δz) from each two adjacent periods. This is the reason why the number of lags in this variant shall be reduced by one. If we come across a hypothesis of *non-stationary time series* and *existing co-integration* between these time series, then we should convert the model specification from VAR type to VECM-type with *p-1* number of lags and *r* number of co-integrations. In this case, we have an *error correction component*.

Having considered the theoretical settings, we would move on to list the individual steps needed for the implementation of **stage 3**. The first step thereof will be to determine the *optimal lag* of the model (first task) and possibly to detect *co-integration relationships* between the endogenous variables (second task). *Finding the optimal lag* will require comparing the readings of different types of information criteria. In our case, we choose to work with the following three types: Akaike's criterion, Schwartz's criterion, and Hannan-Quinn's criterion.

The technical choice concerning the ***optimal number of lags*** will be based on the rank, at which the system would have the maximum number of *lowest values* of such information criteria. As long as a smaller number of lags would typically *simplify* the model, we will go for a variant with *fewer lags*, should the loss of information, compared to a model with more lags, is not significant.

Then we would move on to the implementation of the *second task* of the current step: searching for co-integration relations, however, only if the condition is met that at least two of the endogenous variables involved in the model exhibit *non-stationarity of the first integral order* I (1). In this case, we proceed to study the existence of a long-term *co-integration relationship* between them. For this purpose, we conduct *Johansen's co-integration test*, which would either prove or reject the existence of a general trend (i.e., long-term dependence). In the case of a positive result, we find out what the number of *co-integration relationships* is.

At the end of step one of stage 3, what we do is analyse the results of all tests performed and following the diagram in *Fig. 1,* we select a type of model *specification* against one of the three variants proposed. Then we proceed by carrying out a *transformation* of the original VAR specification, if necessary, and in the new version of the model we take into account the proposed number of lags in the previous procedure and possibly the presence of co-integration relationships.

In step two of the third stage, we conduct a series of tests to determine the optimal composition of the model. For this purpose, while we try a *series of combinations* between the endogenous variables involved so far, we monitor their significance and coefficient readings. Moreover, we also compare the coefficients of determination (R^2), DW statistics and information criteria of the different model variants. During this step, some of the selected endogenous variables may *be dropped* if they do not fit optimally into the modified model structure. However, if dropped, they will also be tested whether they would be suitable for *exogenous or not*. In case endogenous drop out, we should again carry out the tests for the detection of an optimal lag and to see whether there is co-integration or not. Where we detect changes, we update the current model specification.

Step three is the last step of the current stage, and here we should make sure that the model constructed so far is both *sustainable and reliable*. For this purpose, we conduct *three types of tests* on the model's *residuals*, namely for the presence of *autocorrelation, heteroscedasticity and normal distribution*. When we are satisfied with their results, we can continue with the model, saving the new settings on it, if we have made any.

Then in **stage 4**, we finalise the model by constructing the *short-run equilibrium* subsystem related to the endogenous variables included. This step is feasible in cases where we are working with a VECM type model (this is expected to be the default case). The *short-run equilibrium* is constructed by adding *exogenous (external) variables* to the system of equations that have a significant influence on the endogenous ones, but without experiencing a counter-effect on themselves. By adding these, the model takes into account the influence of a *wider range* of relevant factors, adjusting better to short-term fluctuations. The result is that the explanatory power of the model increases. The *group of potential exogenous factors* should already be compiled at the end of step three of stage 2, in which, having made a review and analysis, we have determined the *role* of each factor that was proposed at the beginning of the study. To this group, we add the endogenous ones, which during the numerous modifications of the model were *dropped* from it in the previous stages.

In step one of stage 4, we check *whether there is a structural break* in the relationship between the endogenous variables and Y. What we understand under this term is a *sudden change* in the behaviour of a time series from a certain point onwards, which is expressed in a change in the average or some other parameter in the data generating process. Should such a phenomenon not be taken into account, the constructed econometric model would run the risk of not being sustainable and would be likely to produce large deviations in the forecasts. Based on the results of this step, we will determine whether we need to introduce an *additional structural break* variable into the model.

We check for structural breaks by visually inspecting the dynamics of the *logarithmic ratios* between each endogenous variable and Y. To confirm the visual observations, we proceed with conducting a specific *test to check whether there are any multiple structural breaks or not*. Essentially, the test consists of multiple *consecutive* single tests performed according to Bai and Perron's methodology. The detection of a structural break requires the inclusion in the model of *dummy variables* that take only *two possible values*, which are either *0 и 1*, with the border between them being the date of the break. We *add* such *dummy variables* if we have identified the need for them to the *list* of potential exogenous factors to be tested for significance.

In step two of stage 4, we search for the *optimal combination of endogenous and exogenous variables*. Having completed this step, we would arrive at the *final version* of the model, provided that subsequent control tests confirm its strength, stability and reliability.

In case we are using a VECM-type specification, we should perform some *additional settings*. The *first* of them is related to the presence of a *co-integration equation* in the model, the calculation of whose parameters would require that *the first endogenous variable coefficient* (i.e., the Housing Price Index) *be fixed at 1, i.e.,* $b_1 = 1$ in the above example. This is how we will be able to *measure the influence* that each of the remaining endogenous variables exerts on Y in order to maintain the system in long-term equilibrium.

The *second setting* in the VECM specification of the model would be related to the imposition, if necessary, of a weak exogeneity condition in the error correction term of one or more of the model's endogenous ones. While the mentioned *second setting* made to a VECM-type model would not always apply, it will depend on whether the correction coefficients α_m, α_p and α_y are significant, cf. (3).

$$\Delta m_t = \alpha_m (1 * m_{t-1} + b_2 p_{t-1} + b_3 y_{t-1} + b_4) + \lambda_{MM} \Delta m_{t-1} + \lambda_{MP} \Delta p_{t-1} + \lambda_{MY} \Delta y_{t-1} + v_t$$
$$\Delta p_t = \alpha_P (1 * m_{t-1} + b_2 p_{t-1} + b_3 y_{t-1} + b_4) + \lambda_{PM} \Delta m_{t-1} + \lambda_{PP} \Delta p_{t-1} + \lambda_{PY} \Delta y_{t-1} + u_t$$
$$\Delta y_t = \alpha_Y (1 * m_{t-1} + b_2 p_{t-1} + b_3 y_{t-1} + b_4) + \lambda_{YM} \Delta m_{t-1} + \lambda_{YP} \Delta p_{t-1} + \lambda_{YY} \Delta y_{t-1} + \eta_t \quad (3)$$

In this case, by *m, p* and *y* we denote endogenous variables. α_m in particular (the coefficient in front of house prices) is initially important to be statistically significant (*so that its t-statistic is below -1.96*), otherwise the VECM-type model will not be applicable.

This would mean that if we prove that one of the coefficients in the error term is statistically insignificant, then its variable exhibits a property of *weak exogeneity*, i.e., we can no longer treat it as a fully-fledged endogenous variable. It will continue to influence the long-run equilibrium, however it *will not participate with an error correcting coefficient in the short-term settings* of the model, as it responds more slowly to the influence than the other endogenous variables. Therefore, if any of the first differences of the endogenous variables show signs of weak exogeneity, we introduce a *constraint* on its error correction coefficient and fix it to *zero*.

The *short-term equilibrium subsystem* of the model resembles a standard VAR system of equations. When constructing this short-term equilibrium, we *do not need to impose any constraints* as opposed to setting the long-term equilibrium and the error correction term. In particular, *there would be no need to make a weak exogeneity check* as long as what we examine is the exogenous influence of all variables. In the VECM structure, we add the exogenous variables to the second part of the equations of (3) as follows (see underlined):

$$\Delta m_t = [\ldots] + \lambda_{MM} \Delta m_{t-1} + \lambda_{MP} \Delta p_{t-1} + \lambda_{MY} \Delta y_{t-1} + \underline{\lambda_{MEx1} \Delta Ex_{1,t} + \lambda_{MEx2} \Delta Ex_{2,t} + [\ldots]} + v_t$$
$$\Delta p_t = [\ldots] + \lambda_{PM} \Delta m_{t-1} + \lambda_{PP} \Delta p_{t-1} + \lambda_{PY} \Delta y_{t-1} + \underline{\lambda_{PEx1} \Delta Ex_{1,t} + \lambda_{PEx2} \Delta Ex_{2,t} + [\ldots]} + u_t$$
$$\Delta y_t = [\ldots] + \lambda_{YM} \Delta m_{t-1} + \lambda_{YP} \Delta p_{t-1} + \lambda_{YY} \Delta y_{t-1} + \underline{\lambda_{YEx1} \Delta Ex_{1,t} + \lambda_{YEx2} \Delta Ex_{2,t} + [\ldots]} + \eta_t \quad (4)$$

where the content of the second square brackets follows the same pattern with the exogenous variables denoted before it, that is to add the product between an exogenous variable (Ex) and its corresponding coefficient (λ) until all involved exogenous variables are listed.

As we described at the beginning of the step, we are looking for the composition with the best combination of variables both in terms of their *individual contribution* to the variation of Y and in terms of the overall representation of the system of equations. Specifically, we examine the *t-statistic* of each of the outliers with respect to Y in search of values equal to, below (≤) *the cutoff of* -1.96 or above (≥) *the cutoff of 1.96* (equivalent to $p < 0.05$). We remove from the model factors that fail achieving this criterion (which t-statistic is between -1.96 and + 1.96) because they are insignificant for the system.

Moreover, we also monitor the general performance of the system of equations, taking into account the readings of its main indicators: the coefficient of determination, DW statistics and information criteria among others. We also check the coefficients in front of the corresponding variable, whether they correspond to the initially set theoretical assumptions about the *direction of the relationship* (positive or negative correlation) with respect to HPI. In case of drastic discrepancies, we switch to another modification of the model. After all these operations, we single out an optimal variant of a model.

Starting from here through the end of stage 4, we run a series of control tests to validate the model and make final adjustments to it, as appropriate. The first of these is a check for the presence of *multicollinearity* between the exogenous variables. We conduct it as part of the third step of stage 4. We perform the testing by constructing a *correlation matrix* of the exogenous factors and checking the pairs to see whether there is a *strong correlation* or not: that means values either *above 0.7* or *below -0.7* with a possible range of -1 to 1. Based on the results, we should decide *whether to remove* any of the variables in case it causes a strong correlation with another and its exclusion does not lead to significant information loss.

In the last step four of stage 4, we again run the tests for *autocorrelation, heteroscedasticity and normal distribution* on the last modification of the model (see the procedure in step three of stage 3). If the results are satisfactory, then we accept the tested configuration as final and proceed to the *interpretation* of the results obtained. Otherwise, we consider whether to make any new *structural changes* provided the potential to find a more optimal configuration that is still existent. It is possible that the efforts for the latter might go beyond the scope of the present research, so, in this case, we should settle for the last best configuration and proceed with analysing the results thereof.

In the last stage of the current methodology, **stage five**, we *review and interpret* the findings for the *observed interrelations* between the HPI and the other variables in the final variant of the model.

4. Empirical results from applying the model

The methodology was executed with an initial data set of *35 indicators*, which were supposed to influence in a direct or indirect way the nationwide *House Price Index* in Bulgaria. These 35 factors, distributed into *9 cluster groups*, were potential participants in the model, but only a few were chosen after applying the sorting out procedure. Thus, we succeeded to identify the exact *endogenous* and *exogenous factors* to be used together with the HPI in the construction of the EEMHP (Ivanov, 2024). The selected factors are displayed in Table 1.

Table 1.
Endogenous and exogenous variables included in the EEMHP

ENDOGENOUS		EXOGENOUS	
Y	House price index	F02	Consumer price index
F10	Average salary	F03	Index of housing rents
F15	Current account of the balance of payments	F09	Unemployment rate
F21	Volume of new housing loans	F22	Number of registered mortgages in the country

After identifying the variables, we can build the model itself. The focus of the research is that single equation from the dynamic system, in which the *dependable variable* is the *change of house prices* ($\Delta\gamma_t$). We express it with the following notation:

$$\Delta\gamma_t = \varphi_\gamma(\lambda_1\gamma_{t-1} + \lambda_2 F10_{t-1} + \lambda_3 F15_{t-1} + \lambda_4 F21_{t-1} + \lambda_5) + \quad (5)$$

$$+\beta_{\gamma,1}\Delta\gamma_{t-1} + \beta_{\gamma,2}\Delta F10_{t-1} + \beta_{\gamma,3}\Delta F15_{t-1} + \beta_{\gamma,4}\Delta F21_{t-1} + c_\gamma +$$
$$+\beta_{\gamma,5}\Delta F02 + \beta_{\gamma,6}\Delta F03 + \beta_{\gamma,7}\Delta F09 + \beta_{\gamma,8}\Delta F22$$

Running the model with the real values of the above listed variables for the period 2005-2022 brings out the result in Figure 2. What we see on the graph is the *modified equilibrium HPI* (dashed line) suggested by the EEMHP (Y'), *the actual HPI* (solid line, Y) and their *difference* (columns) that is a projection of the house price deviation from the equilibrium. As we see on the graph, a clearly pronounced *real estate bubble* was observed in the period 2007-2008. After it burst out, there was a price plunge, followed by a prolonged period of *stagnation*. Eventually, it turns out that the model *precisely identified* the boom and the burst phase of the first contemporary house price bubble in Bulgaria. This outcome confirms that the model is reliable.

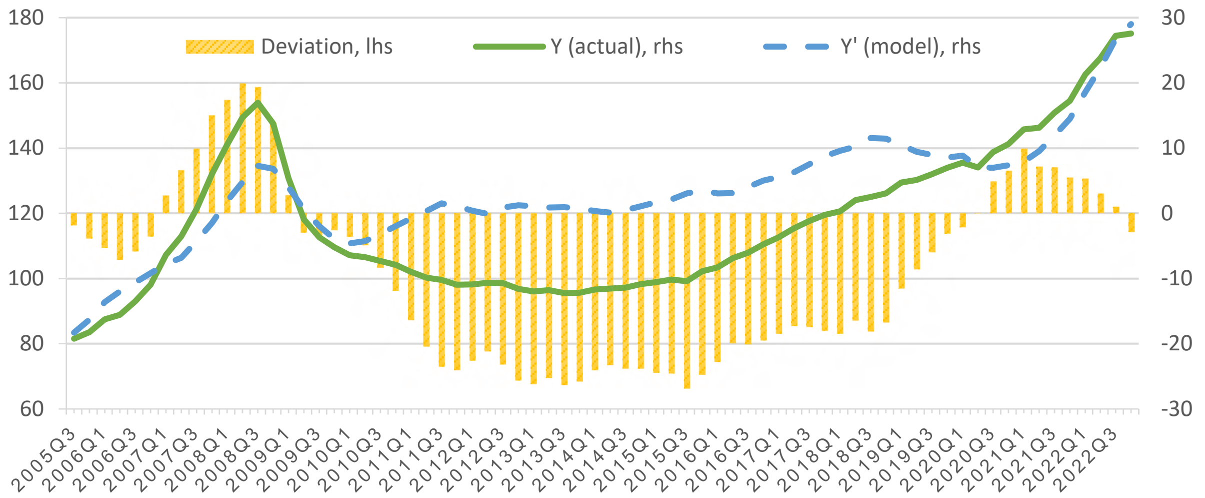

Figure 2. Deviation of house prices from their equilibrium levels according to the EEMHP

Legend: lhs – left hand side axis, rhs – right hand side axis;

The application of the model refutes the prevailing *public speculations* about an *existing housing bubble* that has been inflating since the start of the COVID-19 pandemic. No clear evidence for that was found indeed. Moreover, the model suggests that after some anomalies in house prices in 2020, they were once again returning to their equilibrium. Even at the end of 2022, they *fell* just below the equilibrium level despite the fact they were in an upward trend. The possible explanation of this is that the *surge* of house prices *is not big enough* to catch up the *increase of the other factors* in the system, among which: the growing *inflation* and *salaries*. The model even tells us that some dwellings were *a little underpriced* at the end of the period. To conclude, *no price bubble* seems to be of an immediate threat, at least not for the nationwide HPI. The actual situation, however, could be different for separate cities, especially the capital.

Based on theoretical and empirical evidence, we might well make a link between the probability of house price bubbles formation and the occurrence of *anomalies* in the house price cycle.

Conclusion

House price bubbles occur as a result of *hyperactive behaviour* from the demand side, i.e. the *demand function* approaches a state of *hyper-inelasticity*. Such a phenomenon is typical during the market's booming phase. Its negative consequences, which would often feature a wider economic and financial impact, are associated with the sharp onset of a *severe* and *prolonged phase* of *price correction*.

The procedure for building an *equilibrium econometric model,* presented in this article, reflects the complex behaviour of house prices under the aggregate and dynamic influence of heterogeneous groups of factors. The model has the potential to detect early signals of *house price bubbles*, so that the negative economic and financial effects from its spread could be mitigated with proper actions.

In fact, house prices are often out of *balance*. As the results from the application of the EEMHP point out, *achieving an equilibrium* for house prices is only an *attractive anchor,* but because of their volatile nature, equilibrium is hardly reached and not stable over time. Among the reasons for this are the complex *mechanism of real estate pricing* and the complex *transmission* of effects between *the real and financial sectors*, due to the intertwined interaction of multiple factors. This tends to generate difficulties and challenges in terms of constructing an appropriate econometric model to account for such centrifugal forces.

The model proved its capacity in finding out price abnormalities as it exactly detected the house price bubble in Bulgaria in 2007-2008. Based on a solid methodology, it is capable of bringing out well-argued outcomes. Such an instrument would nicely complement the toolkit of the central banks when determining their policies on credit standards and monetary actions. Nevertheless, almost every econometric model the EEMHP has its *structural drawbacks* and *limitations* for use. That is why, its results should not be taken as absolute. They are just approximations that should be considered in *combination with other methods* and above all to be assessed with an *expert judgement*.

With all that said, the current study accomplished its goal. We successfully constructed, applied and tested the EEMHP in *assessing the deviations of house prices* and for *detecting house price bubbles*. We also presented the empirical results from the application of the EEMHP for the country specific case of Bulgaria. Additional research on the topic could further elaborate the model and supplement it with additional diagnostic tools.

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CONTENTS

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COMMUNICATION FOR THE IMPORTANCE OF THE CREATION AND INTRODUCTION OF INNOVATION IN BULGARIAN SMALL AND MEDIUM ENTERPRISES

Yanica Dimitrova[1]

Abstract: Innovation is becoming an imperative for modern business organizations that want to increase the sustainability of their competitive performance. Communication about the importance of innovation is critical to minimizing stakeholder resistance to innovation endeavors and the perception of the need to implement them. Innovations are also directly related to the partnerships implemented by organizations, corporate culture, and knowledge management. Understanding the importance of professionally executed corporate communications is critical. The communication practices for promoting innovation activity are presented based on an empirical study among 350 Bulgarian enterprises. The study examines how communications are implemented in Bulgarian small and medium-sized enterprises, the innovations they implement, and the relationship of corporate culture with company competitive performance. Relevant conclusions and recommendations have been made.

Key words: corporate culture, innovation, internal communication, integrated marketing communication, motivation, stakeholders.

JEL: M14, O30.

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Introduction

In the conditions of Industry 4.0, any company that would like to survive and achieve a sustainable competitive advantage must implement and introduce various types of innovation. For this purpose, its

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stakeholders—internal and external—in all the conditionality of this division—must be motivated for their implementation. Communications are at the core of good practices related to the company's innovation actions. The corporate culture managed by the organization is essential for their competent and professional implementation.

1. Innovation

According to Freeman and Soete (1997), an innovation is a new product or process that is improved and commercialized. The invention does not have to be something new; it can be an improved product, service, or process implemented in a new way. Innovations are based on knowledge and are designed to provide answers to the demands and needs of stakeholders. Innovations are classified according to:

1. Purpose – products and processes.
2. Degree of change of radical and incremental.
3. Area of impact – technological and administrative (Bessant & Tidd, 2007). Innovation is closed and open.

Innovation is a strategy that helps companies introduce new and improve existing products and services, improve organizational processes, and give them an edge over competitors. (Lim, et.al., 2010). Organizations create and maintain a competitive advantage through innovation based on knowledge, technological skills, and creativity.

The organization's choice of an innovative strategy is influenced by internal and external factors related to the characteristics of the organization's essence and the business environment in which it operates (Dodgson et al., 2008).

Innovation requires openness and willingness to interact with the company's key stakeholders, which indicates the close relationship between innovation and corporate culture, also defined as a culture of innovation (Dimitrova, 2017).

The ability of companies to generate knowledge during the innovation process can be identified as a critical source of competitive advantage (Bierly et al., 2009). Many companies also involve external stakeholders in creating and exchanging knowledge during innovation processes (Hoyer et al., 2010; Mahr et al., 2014). Inclusion occurs through various interaction and competition mechanisms, encouraging them to propose new product design ideas, create additional opportunities, and solve research-related problems (Horn & Brem, 2013; Hoyer et al., 2010).

The innovation process results from diverse sources of knowledge realized through the interactions between diverse stakeholder groups. These sources derive from the firm's human capital, customers, suppliers, strategic partners, and competitors (Collins & Clark, 2003). Companies must constantly scan their environment to access new knowledge through new sources inside and outside them.

Companies' capabilities to successfully manage knowledge help achieve competitive advantage (Decarolis & Deeds, 1999; Levinthal & March, 1993) and enhance financial performance (Wiig, 1997). Numerous studies have shown that innovation gains substantial benefits from access to diverse sources of knowledge (Chesbrough & Brunswicker, 2014; Frenz & Ietto-Gillies, 2009), the process by which knowledge is shared and the extent to which they can retain this knowledge from competitors.

Companies must continuously strive to generate knowledge by integrating external, new knowledge from various sources. The more substantial access a firm must have to sources of knowledge, the more innovative it is. The acquired knowledge should complement what exists in the organization (Nonaka, 1994). Companies that engage in strategic alliances acquire new knowledge from and through their partners. This knowledge is integrated into the company's existing one. It can modify it by becoming the company's stock of knowledge (Van Den Bosch, Volberda, & De Boer, 1999), supporting the creation of new mental models, procedures and processes (Galunic & Rodan, 1997; Rodan & Galunic, 2004). The company's ability to integrate new with existing knowledge favourably affects its problem-solving capabilities (Carneiro, 2000), organizational learning, absorptive capacity (Cohen & Levinthal, 1990), the introduction of new products to the market, and the management of innovation processes. Access to diverse sources ensuring relevant and valuable knowledge is conducive to innovation (Dimitrova, 2018).

According to Chesbrough (2003), open innovation is a paradigm through which an organization can use internal and external knowledge to improve its performance. The concept of open innovation is a two-way process in which the combination of internal and external knowledge generates new knowledge.

External knowledge occurs outside the firm's boundaries, resulting from interactions with customers, suppliers, external partners and other stakeholders (Chesbrough, 2003). Managing this knowledge is critical for businesses. It will only be valuable if incorporated into firms' innovative activities (West & Gallagher, 2006). If there is an increase in internal innovation, the open innovation system will improve. Communication is a critical factor in enhancing employee innovation efforts. Their innovation activity is enhanced through internal and external innovation. Innovative ideas are only generated when internal stakeholders communicate and share. The various communication channels and strategies that managers use to implement internal communication are also important.

Communication is one of the main ways to increase employees' innovative efforts (De Jong & Den Hartog, 2007). In the communication process, ideas are also born and exchanged between employees, which helps promote innovation. Knowledge management must be based on communications.

In the context of SMEs, open innovation is evident from research indicating that companies engaged in open innovation have greater access to ideas, knowledge, and technologies in interactions with stakeholders in their ecosystem (Spithoven et al., 2012; Dimitrova, 2020). Open innovation projects reduce R&D costs, support risk management, and bring innovations to market faster (Chesbrough, 2010).

The smaller a firm is, the more it depends on interactions. Therefore, it is essential to focus on all stakeholder groups in open innovation processes (Darnall & Henriques, 2010)

Companies that implement processes related to open innovation are much more open to communication, creating and maintaining diverse relationships, and engaging with their external stakeholders. They innovate more successfully to meet market trends and customer needs and, in many cases, redefine them.

Strategic alliances, joint ventures, open-source platforms and participation in diverse, specialized professional communities are mechanisms that support innovation processes. Open innovation is also associated with implementing an open strategy. The latter is associated with creating and introducing various policies due to the interaction between internal and external expertise.

The theory related to innovation networks can also illuminate the possibilities of integrating diverse stakeholder groups during innovation processes. In many sectors, the locus of innovation is not in themselves but in the networks in which they participate (Ritter & Gemünden, 2003). Innovation networks combine resources, knowledge, and capabilities (Perks & Moxey, 2011; Ritter & Gemunden, 2003) that are often impossible through market transactions alone. This combination becomes extremely important in the increasing complexity of new products and services (Pullen et al., 2012).

2. Corporate culture

Theorists and practitioners dealing with the issue of corporate culture cannot unite around a single definition of its essence, which in turn highlights the diversity of the construct and its effects on the organisational essence. The diversity of definitions of corporate culture is associated with the different perspectives through which it is studied.

One definition of corporate culture is "a general constellation of beliefs, norms, value systems, behavioral norms and ways of doing business that is unique to each corporation" (Tunstall, 1983). Corporate culture is inextricably linked with strategy, and the organization's structure is mutually determined with leadership.

The research literature has revealed multidirectional approaches and paradigms for the evolution and study of the construct of corporate culture (Smircich, 1983; Martin, 1992; Martin & Frost, 2011). Various models have also been developed to explain the complex nature of the construct. Some of these are Schein (1992), Cameron and Quinn (1999, 2011), Denison (1984, 1990), Denison and Mishra (1995), Fey and Denison (2003), Denison et al. (2004, 2006, 2012), and Johnson and Scholes (1999), among others. The variety of theoretical concepts and empirical studies based on them show us the different dimensions of corporate culture and the efforts required to study and measure it.

Corporate culture is also considered to be strong, weak, or adaptive. According to Kotter and Heskett (1992), the most suitable culture for the successful development of a company is an adaptive corporate culture, one that encourages and enhances innovation, thereby supporting its competitive performance in a dynamic business environment. This type of culture can also be defined as “learning” because it enables the organisation to create and implement mechanisms that help it identify and adapt to changes in the business environment, ultimately leveraging these changes to achieve new levels of competitiveness and sustainability. Adaptive culture requires the presence of communication facilitated by the hierarchical constraints of the organisation’s structure, allowing for continuous interactivity between the organisation's stakeholders and between the organisation and its key stakeholder groups.

Corporate culture is a significant driver of organizational change. While not all organizational change necessarily requires innovation, the introduction of any innovation is a condition for change. Companies with a long-term strategic orientation are more likely to innovate than those with a shorter-term focus.

Corporate culture is a multi-layered construct, the essence of which we will not dwell on in detail in this study. Here, we will emphasize one of the most researched aspects of the domain, namely - its influence on the organization's competitiveness. According to Kotter and Heskett (2011), the organization's adaptability, through its culture, should be promoted precisely to maintain long-term positive financial performance.

Corporate culture is also directly related to innovation (Dimitrova, 2017; Dimitrova, 2018).

Organizations are under constant pressure to compete successfully, and innovation supports them in these efforts. Innovation is a crucial factor in the competitive landscape, as it provides opportunities to respond to the ever-changing demands of customers and contributes to technological development while also increasing the chances of acquisition and positive performance in a highly competitive business environment. Innovation is considered a key factor in implementing change, and corporate culture is an essential condition for the existence and implementation of innovations within a company; in other words, corporate culture is at the heart of innovation.

The research literature presents various studies confirming the positive relationship between corporate culture and company innovation (Hernandez-Mogollon et al., 2010; Dimitrova, 2017, 2018; Graham et al., 2022).

Martins and Terblanche (2003) believe that it is essential to support creativity and innovation in the organisation, enabling employees to understand its vision and mission. If discrepancies arise between these and the reality within the organisation, they should be addressed through the use of creativity and innovation.

The corporate culture managed in the organization predetermines how communications are implemented. In turn, through communications, corporate culture is recreated. Culture is the invisible infrastructure of any organization; therefore, competent communication management supports a culture of innovation and encourages the actions of organizational members to create and implement innovations.

3. Communications

The organization's communications are divided into internal and external, and management's communication practices are included (Welch & Jackson, 2007; Dimitrova, 2013). Modern trends in the development of communication science and related research emphasize the importance of integrating the organization's communications, requiring harmony in the messages sent inside and outside the conventional boundaries of the organization (Christensen & Cheney, 2001; Quirke, 2012).

Innovation-related processes include creating, adopting, introducing, and subsequent routinization of new products or services, processes, and policies. Each of these stages involves communication related to persuasion and engagement. (Crossan & Apaydin, 2010).

About innovation, it is necessary to note that communication should reveal the company's ability to create value (Payne et al., 2006). Through communications with internal stakeholders, it is necessary to emphasize the understanding of the innovation by presenting its essence and meaning, to encourage support for it, and to positivize the aspiration for its realization, as well as the motivation for the sustainability of the attitudes to implement innovations.

After his meta-analysis, Damanpour (1991) postulated that internal communication positively correlates with organizational innovation and assists the organization in disseminating information related to innovation and its accompanying processes.

Communication with top management is significant because it increases employees' knowledge about the organisation's essence, mission, vision, and goals.It supports identification with the organization and contributes to involvement in the various organizational processes. Multiple studies prove that understanding and support from top management affect the diffusion of knowledge in the organization and promote employee-driven innovation (Lin, 2007).

There is an essential relationship between internal communication and organizational change management (Welch & Jackson, 2007). Innovation projects include uncertainty, adaptability, risks, upsides, assumptions, and guesswork. Therefore, communication about implementing innovative projects must be carefully planned to encourage the process.

Internal communication enhances competitive performance by increasing satisfaction, strengthening involvement in the organization's affairs, and strengthening commitment to it because these are necessary prerequisites for achieving identification with the organization and stability of organizational identity.

A weak communication policy regarding innovation projects can increase stress and resistance to change that will occur in connection with the implementation and subsequent introduction of innovations and reduce competitiveness. In this context, innovation communication requires interactivity and continuous feedback.

Regarding external stakeholders, it is necessary to present the company's innovative image and discoveries, which are perceived as beneficial for customers, suppliers, partners, and society.

Regarding innovation, external communication supports creating and managing an innovative image, maintaining trust, and minimizing uncertainty when contacting different stakeholder groups and the organization.

Innovative companies with excellent external communication policies are believed to experience increased sales, increased market niches and concern for stakeholders and meeting their needs (Bonn, 2001). However, the above could only be realized with corporate culture and internal communication, which determine organizational behaviour that promotes dialogues and interactions between company members and external stakeholder groups.

The traditional understanding is that external communications are aimed at the environment surrounding the organization. Several mechanisms are noted in the literature through which the company can capture information from the external environment - interactions with customers and other external stakeholders, benchmarking and inter-organizational project teams, the various communication channels through which communication and the exchange of information and knowledge are carried out interactively.

SMEs must use communications effectively to facilitate knowledge transfer and to generate innovation. According to Pfeffermann (2011), innovation communication is integral to organizational communication programs. It must include perspectives on creating and co-creating value and the various management strategies in this context. It must also be constantly analyzed to define new goals, tools, and tactics.

Effective communications help SMEs share value through diverse communication strategies. Successful firms must learn to implement communication strategies to share value, mission statements, and strategies with internal and external stakeholders. Usually, due to resource constraints, SMEs do not employ PR specialists to manage communications. Competent communication management is necessary for the visibility of the company, its products and services in the markets where it carries out its business activities.

4. Integrated Marketing Communications

Integrated marketing communications (IMC) represent the company's ability to plan, manage and implement the programs set out in the corporate brand management strategy (Pham et al., 2017). The philosophy of integrating the elements of the mix lies in the possibility of applying a common strategy for managing the corporate brand and its related product brands and that of the employer. Kamboj et al. (2015) believe that the above leads to an increase in the company's financial performance. The capabilities of integrated marketing communications help companies identify, communicate and serve their target markets, increasing their competitive performance. (Hao & Song, 2016; Takahashi et al., 2016).

Competitive advantage refers to the specific values of the product or service by which the firm outperforms its competitors (Porter, 1990).

Integrated marketing communications dominate and influence companies' communications and their implementation strategies. Social media complement IMC by offering new communication channels and methods for communicating with stakeholders while allowing users to generate content.

Porcu et al. (2019) defined IMC as the stakeholder-oriented interactive process of cross-functional planning and alignment of organizational, analytical, and communication processes that permit continuous discussion by conveying transparent and consistent messages via all media to foster long-term profitable relations that create value.

Integrated marketing communications is at the heart of stakeholder relationship management. Brands respond to competitors, helping to enhance competitive advantage (Dawar, 2004).

Integrated communication consistency allows experts to coordinate multiple structured messaging sources to manage brand identity across all key stakeholder groups.

Innovation requires constant communication and adaptation between development and production (Teece, 1996). Therefore, communication is essential for successful innovation (Srivastava & Moreland, 2012) and should always be considered when implementing processes related to introducing and implementing innovation (Nordfors, 2006; Wells, 2008).

More and more organizations are using Unified Communications as a Service. The innovation in web-based technologies offers both the members of the organization and the other stakeholders' different channels of communication - blogs, podcasts, wikis - through which the effectiveness of communication is increased, the time of information exchange is shortened, it is easier knowledge management, and overall communication effectiveness is positive.

Social media has become an equal participant as a communication channel for realizing communication within and between the organization and its key stakeholders. Through social sites, blogs, social tagging, bookmark applications, forums, and chat rooms, IM supports creating communities (virtual) and facilitating the sharing of information and knowledge.

Social media is also essential for customer relationship management (CRM). With their help, the organization can attract and retain customers much better than through traditional methods (the use of which is included, combined with social media).

Social media facilitates communication within and outside the organization, creating communities unfettered by hierarchy or bureaucratic procedures. They are handy for exchanging knowledge and quickly obtaining expert opinions on various issues because they are related to the so-called "boundary-spanning" functions—crossing the boundaries, they remove the organization's boundaries, regardless of how conditional they are. The abovementioned factors increase the organization's absorptive capacity and support the generation of new ideas and, thus, innovation.

Communication is vital to the perfect running of all processes in the organization. Through communication, the corporate culture and related constructs are managed and related to the successful adoption and implementation of innovations in companies (Dimitrova, 2018).

5. Empirical Study

The sample is based on the number of companies implementing innovations in Bulgaria. The data is according to the National Statistical Institute (2022).

The profile of the surveyed companies is as follows: the number of respondents - 206 companies, and the owners and managers of the companies were interviewed. The survey is part of the project “Innovation in business and education. Development of a conceptual communication model for innovations in a business-information organizational environment”, Fund of Scientific Research, which is headed by the author of this article. The author pre-prepared the closed-question survey. „Sova Harris“, an agency for sociological, political and marketing research, conducted the research from August to September 2023 as an element of the second stage of the aforementioned project. The results were analysed using the statistical package for the social sciences (SPSS).

The size of the companies is determined according to the European classification. The micro enterprises (up to 10 people) - one hundred and two respondents, small (10-49 people) – sixty-three, medium (50-249) - thirty-three, and large (over 250 employees) - seven. From the demographic profile, we see that the respondents in the survey are mainly representatives of small and medium enterprises. By field of activity: Ten of the respondents in the survey are employed in trade – six are from micro-enterprises, two from small and two from medium. Of the companies engaged in manufacturing, twenty-six are micro firms, thirty-one are small, twenty are medium-sized, and four are large. In the services sector, ninety-three respondents from micro firms, thirty-six from small firms, and nine from medium firms are engaged. In the light and processing industries – two are from micro firms, five from small, and four from medium ones. Respondents in mechanical engineering are divided into eight from micro firms, four from small firms, and two from medium-sized firms. In the sphere of energy – seven are from micro firms, one is from medium firms, and one is from large firms. In the processing industry, there are - one micro firm, five small firms, and four medium-sized firms. In science and education – thirty from micro firms, four from small, one from medium and six from large firms. Participants in the survey, which represented the IT sector, included thirteen from micro firms, fifteen from small firms, and two from medium-sized firms. Other respondents are engaged in transport and tourism – two from micro firm, culture – two from a micro firm, ecology – two from micro; media and advertising – four from micro and one from a small firm; communications – two from micro and three from small firms, financial sector – one from a small firm, health – six from micro firms, construction and design – six from micro and small firms, two from large, and publishing activities – six from micro firms.

Through the presented results, we visualize the relationship between the understanding of the positive influence that corporate culture has on the overall performance of companies and the competitive strategies they implement. The relationship between investments in innovation and the channels for conducting internal communications and the influence of internal communications on the motivation of employees for quality performance of their official duties was examined. The partnerships built by the companies investing in innovation and the applied practices in the context of integrated marketing communications are visualized.

Understanding corporate culture's positive influence on companies' competitive advantages is permanent among Bulgarian businesses. This understanding has been repeatedly confirmed in the author's previous studies (Dimitrova, 2012; Dimitrova, 2017; Dimitrova, 2020). It is necessary to emphasize that the research did not seek an understanding of the essence of the corporate culture construct from the respondents but only the awareness of the positive nature of its relationship with the competitive advantage.

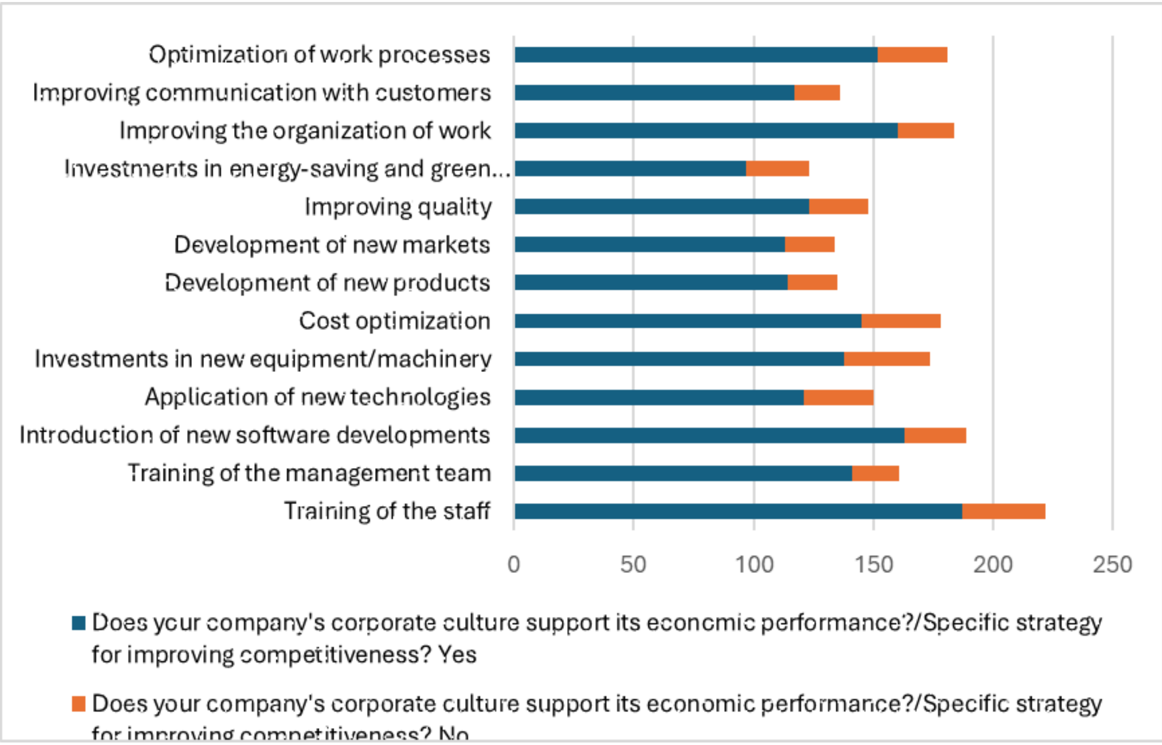

Figure 1. Does your company’s corporate culture support its economic performance?/ Specific strategy for improving competitiveness?

The results of the present empirical research are in harmony with the persistently observed, among Bulgarian businesses, understanding of the positive influence that corporate culture has on the competitiveness of companies. Implementing strategies for increasing competitiveness is in harmony with the perception of the importance of corporate culture for improving performance. Corporate culture is the framework in which innovations are implemented; it can contribute to an environment that tolerates creativity and innovation, can also be a barrier to their introduction, and is essential in applying the concept of open innovation. A corporate culture that is tolerant of experimentation, uncertainty, and risk-taking must be managed in organizations. A culture promoting the processes of organizational learning and, accordingly, unlearning "outdated" practices. A corporate culture must support rapid adaptation and a drive to increase competitive performance through innovation.

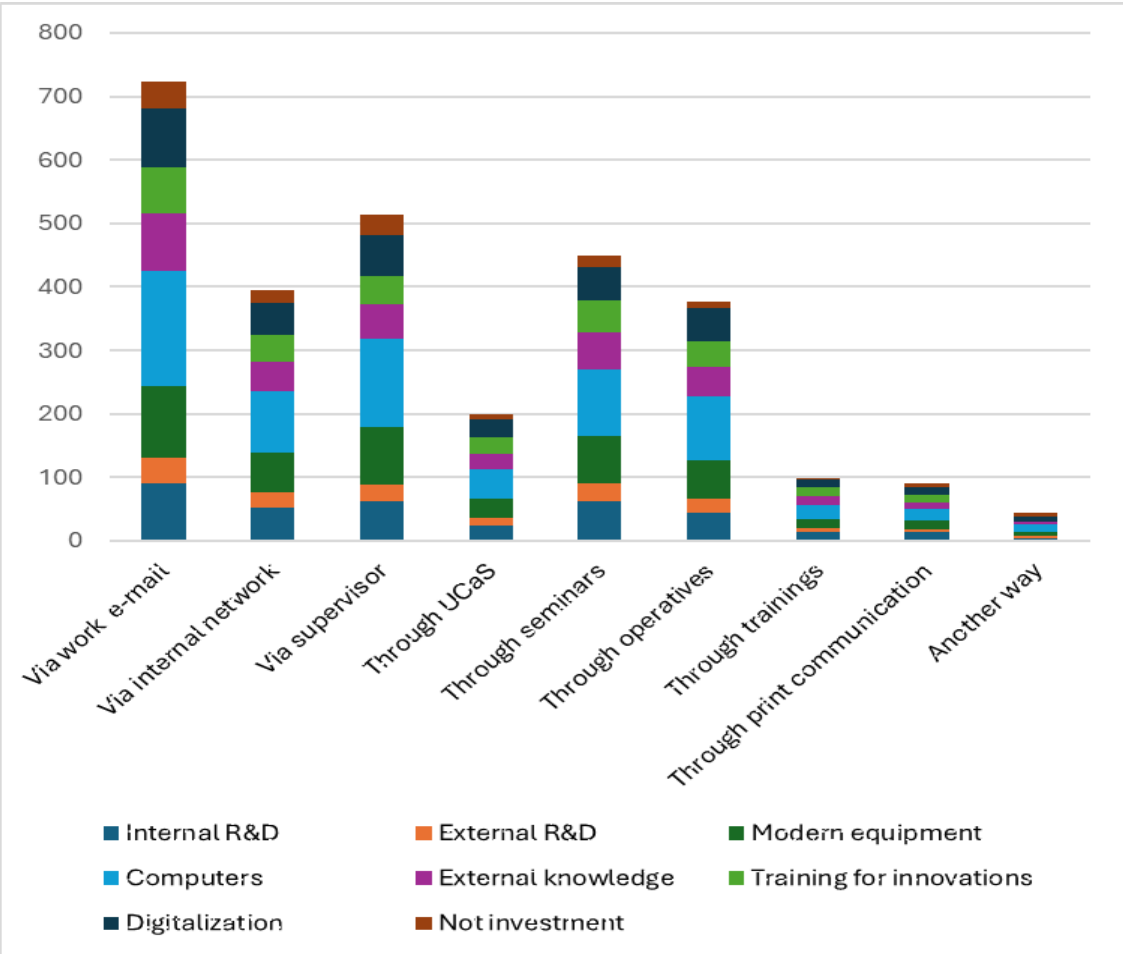

Figure 2. In the last three years, have you invested in:/ How are internal communications carried out in your company?

Respondents who invest in innovation use a variety of channels to implement intra-organizational communication. The main ones used are e-mail, the company's internal network, operatives, seminars, and communication through the direct manager. The use of unified communications as a service is less common.

The widespread use of electronic communication channels – such as e-mail and intranets – is growing due to the development of technology and the possibilities they offer for information dissemination and ease of use (Ruck & Welch, 2012). According to Daft and Lengel (1984), this channel effectively disseminates relevant information that employees can use to generate innovation. According to their theory, face-to-face communication is the most effective communication channel and, in this case, also supports the overall implementation of innovations and related processes.

The structured policy for implementing internal communications is sustainable in Bulgarian SMEs (Dimitrova, 2012; Dimitrova, 2017; Dimitrova, 2020).

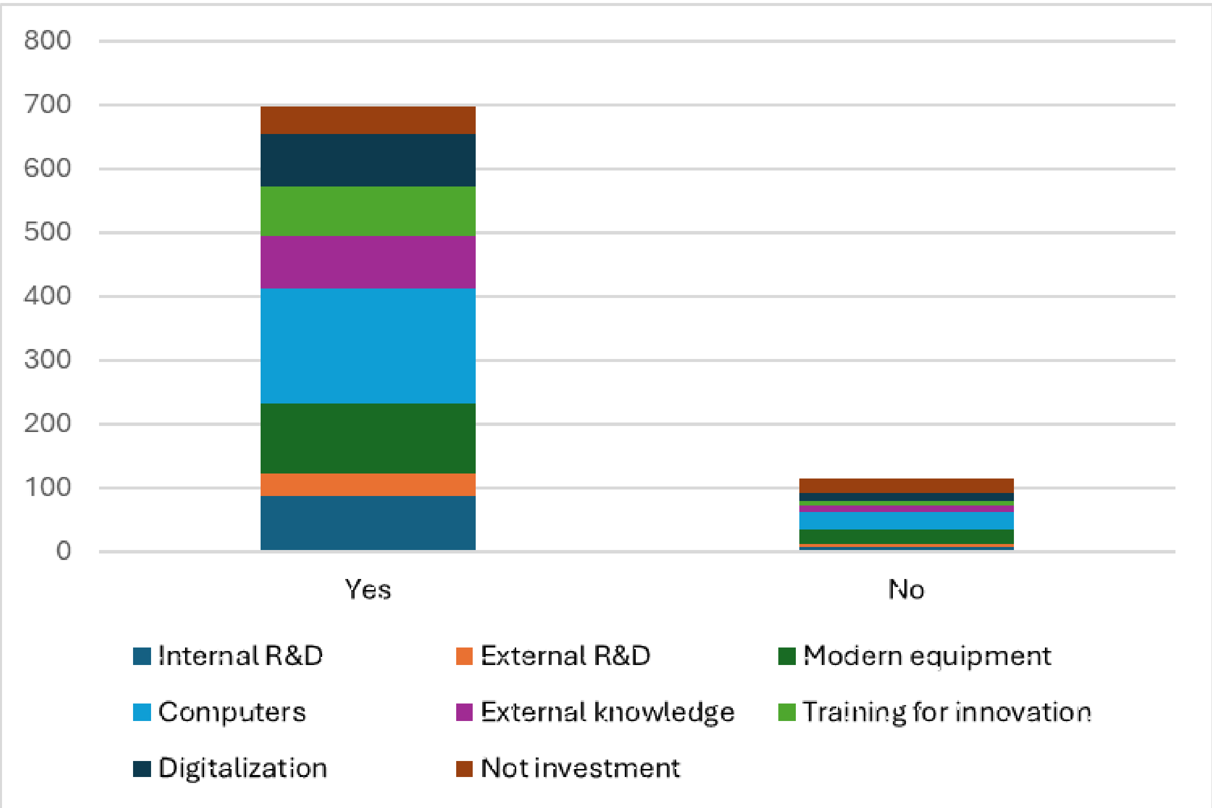

Figure 3. In the last three years, have you invested in:/Does your company's internal communication system motivate employees to fulfil their duties?

Understanding the motivating effect of intra-organizational communication is permanent among Bulgarian businesses (Dimitrova, 2012; Dimitrova, 2013; Dimitrova, 2020). In the present study, we also find that the internal communication system managed by the respondent companies motivates the employees to fulfil their official duties, which shows the consistency of the communication policy. According to the respondents' answers, we can conclude that the importance of a competently managed and accordingly conducted internal communication policy is realized to encourage the actions of the employees in the context of the adequate performance of work tasks, including the overall implementation of the processes in the organization related to the creation and introduction of innovations.

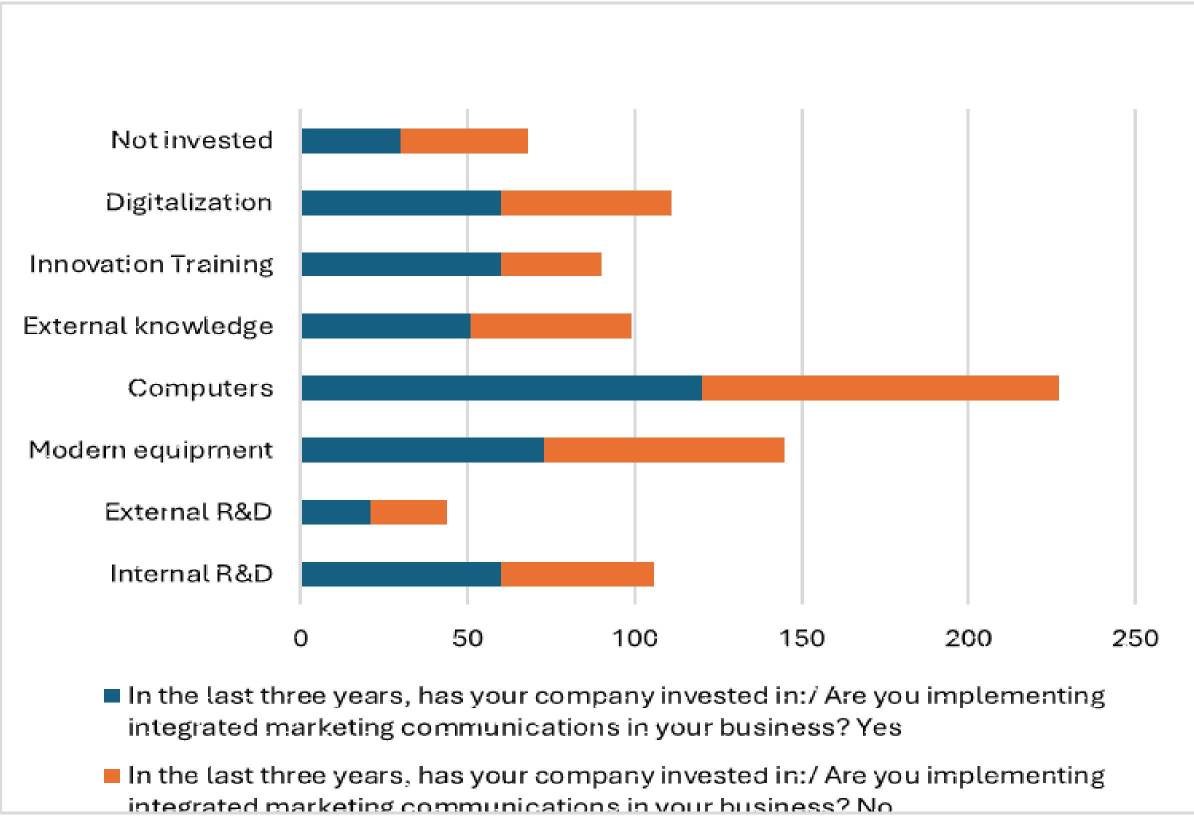

Figure 4. In the last three years, your company has invested in:/ Are you implementing integrated marketing communications in your business?

Integrated marketing communications are yet to be widespread among SMEs operating in Bulgaria. An explanation can be sought in resource limitations; not all enterprises have the budget to include a PR specialist, marketers, and other communication specialists in their structure.

SMEs need to realize the positive effects of introducing IMC for the overall management of the corporate brand and the related brands of the products and services to maintain a positive image of companies dedicated to implementing innovations in front of all their key stakeholders.

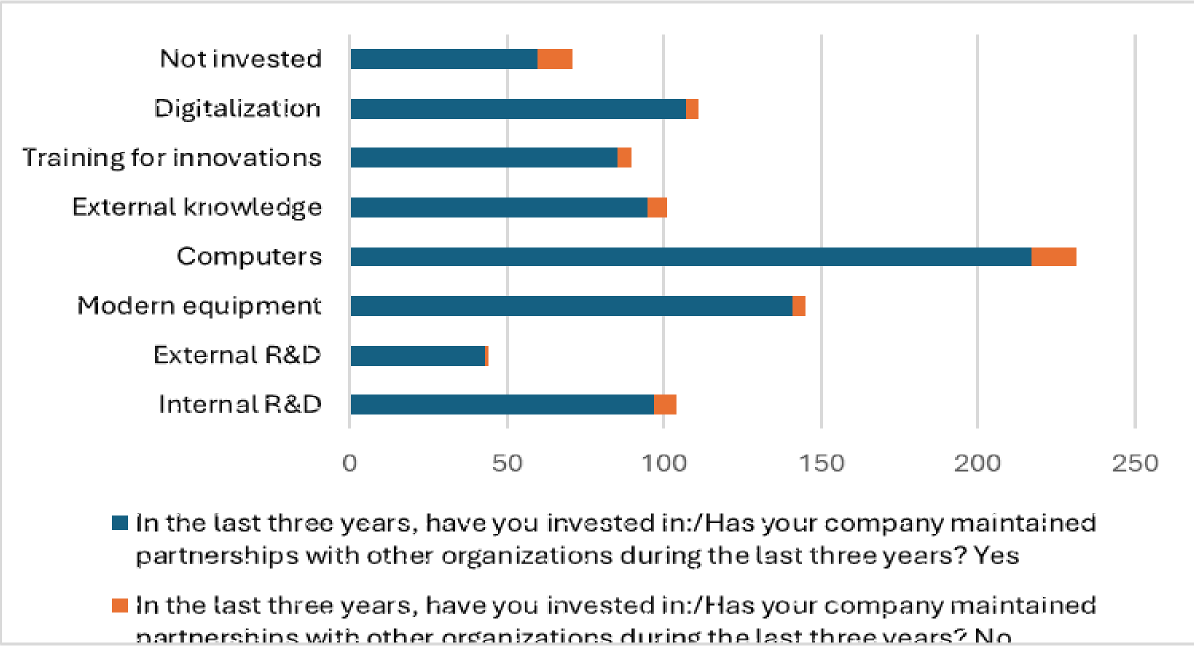

Figure 5. In the last three years, have you invested in:/Has your company maintained partnerships with other organizations for the last three years?

Most of the respondents in the study maintain various types of partner relationships, which indicates the realization of collaborations between organizations of different sizes and spheres of activity. Partnerships support both the use of external knowledge and the development and improvement of internal expertise. The respondents who invest in external research and development activity are the ones who, to the most minor extent, indicate that they do not implement partnerships. The diverse partnerships they develop are critical for small and medium-sized enterprises because they help overcome resource constraints.

Discussion

The results of the empirical research show that Bulgarian businesses realize the importance of a competently implemented internal communication policy. Communication for innovation is successful when employees are motivated to initiate and introduce innovative products, services, and processes. It is said with the presumption that organizations use a variety of channels to communicate with internal stakeholders. Awareness of the importance of competently planned and accordingly implemented internal communication is also permanently observed among Bulgarian businesses (Dimitrova, 2013; Dimitrova, 2017; Dimitrova, 2020).

The communication policy can be optimized through the broader use of social networks and unified communications as a service, which are less prevalent at this stage.

The results of the empirical research show that in some respondent organizations, employees are trained in innovations. These trainings would also be an excellent opportunity to apply creative approaches, such as different types of brainstorming, the Six Thinking Hats method, and other courses offered by lateral thinking.

A good opportunity is using social media to exchange knowledge between different stakeholder groups of companies, which will facilitate crowdsourcing, the generation of new ideas and projects, and the inclusion of all stakeholders in value co-creation processes. In this regard, the importance of creating diverse partnerships with organizations of different natures and essences, which are at the core of innovation networks, has been realized.

The development of communication technologies is also linked to managing open innovations and their associated strategies. In addition to reaching different stakeholder groups, they also help to overcome various communication barriers that are part of the complexity of communication processes.

Integrated marketing communications still need to be used more among Bulgarian SMEs. Their investment will pay off because, when managed professionally, they work with a common strategy and an expected budget to maintain the brand. One of the types of innovation is precisely in marketing. Respondent companies can develop programs in the spirit of guerilla marketing, BTL marketing, ambient marketing, and others in the context of creativity rather than the investment of severe material resources, given that SMEs are much more often resource-constrained compared to large companies. Through marketing innovations and other components of the IMC toolkit, processes, products and services will be managed more effectively and efficiently.

Conclusion

It is essential for the success of any communication program that the company's management understands the importance of communication for its competent implementation. Innovation is a multi-stage process, and effective communication must accompany it from initiation to implementation and subsequent adoption. To understand the need for competently constructed and effectively implemented communication programs that support the adoption and implementation of innovations, it is necessary to adopt and manage a culture of innovation. The culture of innovation fosters innovative behaviour among the organization's members. It guides employees in adopting innovation as a core value within the organisation and encourages their commitment to this value. The creation and implementation of innovations are directly related to enhancing the competitive advantages of organisations, as well as to fostering a corporate culture. Innovation and corporate culture are also closely linked to change. The more actively companies engage in their innovation efforts, the more they will implement effective changes that enhance their competitive advantages. Culture is "transmitted" through communications, and the implementation of communications is directly dependent on the culture existing in the organization. Therefore, the flow of communications within and beyond the conventional boundaries of organizations should be facilitated as much as possible. Communications must be transparent, support employee empowerment, and engage other key stakeholder groups in value co-creation processes that are directly related to innovation.

The difficulties associated with assessing the various aspects of the communication process that affect the successful implementation of innovations should also be noted. In the context of the above, it is important to take into account characteristics such as the size of the company, the degree of intensity of research and development activities, the tendency to acquire external knowledge, the conduct of various pieces of training related to innovations; the demographic characteristics of employees, the experience of staff in the field of innovations should also be taken into account by researchers. It is also necessary to consider the partnerships implemented by the organisation, as well as interactions, dialogues and debates with and between different stakeholder groups, feedback and accountability, and the possibility of joint decision-making and problem-solving related to innovation efforts.

Small and medium-sized enterprises are the backbone of the modern economy. As we have seen, they prevail in the Bulgarian business reality. In principle, this type of enterprise is more flexible and adaptable in responding to changes in the turbulent environment. With the help of the different kinds of innovation they implement, they can overcome the limitations in front of them and achieve new levels of competitive performance. It is essential to realize the benefits of having diverse partnerships and collaborations in innovation networks, supporting the sharing and management of knowledge, facilitating organizational learning, creating and implementing innovations and the absorption of resources dedicated to innovation-related activities. In the context of the above, essential collaborations are those between small and medium-sized enterprises, universities, and various research institutes and centres. Through such collaborations, all participants have shared access to new technologies, information, knowledge, resources and opportunities to solve problems, overcome risks, and acquire new skills.

The above-mentioned can be fully implemented in an environment shaped by a culture of innovation and relevant communication policies, supporting the smooth running of all organizational processes and procedures.

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CONTENTS

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THE IMPACT OF TAXATION ON INVESTMENT FINANCING: THE CASE OF MOROCCAN SMALL AND MEDIUM-SIZED ENTERPRISES

Toufik Marmad[1],
Ritahi Oussama[2],
Echaoui Abdellah[3]

Abstract: Small and medium-sized enterprises (SMEs) are pivotal to investment, employment, and economic growth in developing economies. However, their financing decisions are often influenced by complex tax environments that can either encourage or hinder investment. This study investigates the impact of taxation on the investment financing decisions of Moroccan SMEs, focusing on three fiscal dimensions: tax pressure, fiscal incentives, and the tax treatment of financing modes. A quantitative survey was conducted between March 2024 and October 2024 among 390 SMEs operating in various sectors across Morocco. Data were collected through a structured questionnaire and analyzed using partial least squares structural equation modeling (PLS-SEM). The findings show that all three tax-related factors significantly and positively affect financing decisions, with fiscal incentives and tax treatment exerting the strongest influence. These results emphasize the strategic role of tax policy in shaping SME financial behavior and offer actionable insights for policymakers aiming to enhance private sector investment and support SME development.

Keywords: taxation, investment financing, SMEs, fiscal incentives, tax pressure, Morocco, PLS-SEM, tax policy

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Introduction

Investment financing is a key lever for ensuring the growth, competitiveness, and long-term sustainability of small and medium-sized enterprises (SMEs). In emerging economies, and particularly in Morocco, SMEs represent a significant share of the economic fabric and play a driving role in job creation and regional development. However, these enterprises often face major constraints when it comes to mobilizing the financial resources needed for their investment projects. Among the factors likely to influence their financing capacities, taxation plays a central role—both through its direct impact on business cash flow and its indirect effects on the business climate.

In Morocco, tax reforms implemented over the past decades have aimed to simplify the system, broaden the tax base, and encourage private investment. Nevertheless, despite these efforts, SMEs continue to perceive taxation as a constraint rather than a tool for development. High tax pressure, the complexity of procedures, and the weight of tax and parafiscal charges are frequently cited among the main obstacles to the growth of these enterprises. This raises the central question of whether the current tax system acts as a break or, on the contrary, as a catalyst for investment financing among Moroccan SMEs.

This issue is particularly relevant in an economic context where public authorities are striving to strengthen the national productivity base, especially through SME promotion. The national strategy for improving the business climate and recent targeted tax measures reflect this ambition. However, the actual effectiveness of these initiatives remains open to debate, particularly in light of the disparities between large companies and SMEs in terms of access to financing and fiscal incentives. It is therefore essential to rigorously analyze the impact of tax measures on SMEs' ability to finance investments, innovate, and grow sustainably.

Investment financing remains one of the major challenges facing Moroccan small and medium-sized enterprises (SMEs). These businesses, which form the backbone of the national economy, often struggle to access the resources needed to support their development and strengthen their competitiveness. In this context, taxation plays a decisive role, directly influencing the ability of SMEs to mobilize funds and make investment decisions. However, the nature of this impact is complex and uncertain. On one hand, a heavy tax burden can be a significant constraint, reducing available cash flow and increasing the cost of capital. On the other hand, tax incentive schemes can stimulate investment by reducing the fiscal load and enhancing the attractiveness of business projects. This duality raises a central question: To what extent does taxation influence the investment financing decisions of Moroccan SMEs?

To provide answers to this issue, it is necessary to analyze the different ways through which taxation affects SME financing behavior. This involves examining the overall tax burden borne by these businesses, the effectiveness of fiscal incentives implemented by public authorities, and the differential tax treatment of various financing methods, such as self-financing, debt financing, or capital market funding. Such analysis will help shed light on how fiscal mechanisms shape the financing choices of SME managers, depending on their size, sector of activity, and financial structure.

Accordingly, this article aims to explore the impact of taxation on investment financing in Moroccan SMEs through both empirical and theoretical analysis. The objective is to identify the tax mechanisms that hinder or encourage investment, assess entrepreneurs' perceptions of the tax system, and propose recommendations to improve the efficiency of public policies in this area. Positioned at the intersection of economic and managerial perspectives, this study seeks to contribute to the academic debate on the relationship between taxation and SME development, while providing valuable insights for policymakers and practitioners.

1. Literature review

The impact of taxation on SME investment financing decisions has sparked growing interest in the economic literature, particularly in the context of emerging economies such as Morocco.

A growing body of empirical literature in the past decade has examined how taxes impact business investments and financing, with a mix of findings. Cross-country studies provide broad evidence that higher corporate taxes deter investment and shift financing behavior. According to the OECD, taxation is one of the major global obstacles to SME financing, as it reduces profitability and limits self-financing capacity. This tax burden, which weighs more heavily on SMEs with narrow profit margins, hinders their growth and competitiveness potential (OECD, 2024). The International Monetary Fund supports this analysis by emphasizing that in developing countries—where SMEs rely primarily on internal resources—excessive tax burdens erode those resources and discourage investment initiatives (IMF, 2023).

Moreover, fiscal complexity and uncertainty represent additional barriers, reducing SMEs' willingness to take risks and engage in productive investment projects.

Although findings from individual studies vary and are not always fully aligned with theoretical predictions, a broad consensus emerges from surveys and meta-analyses (Hassett & Hubbard, 2002; Devereux & Maffini, 2007; de Mooij & Ederveen, 2008): higher corporate tax rates—particularly headline rates, effective marginal tax rates (EMTRs), and effective average tax rates (EATRs)—tend to significantly reduce investment levels. This conclusion is further supported by more recent empirical research, such as Bond and Xing (2015). A particularly illustrative case is provided by Maffini, Xing, and Devereux (2019), who investigated the impact of enhanced capital allowances for medium-sized UK firms introduced in 2004. Their analysis found that eligible firms increased their investment rates by 2.1 to 2.5 percentage points compared to ineligible firms. While the literature is complex and does not lend itself to rigid generalizations, Devereux (2021) offers a useful rule of thumb: a 1 percentage point increase in the EATR is associated with a 2.5% decline in foreign direct investment inflows, and a 1 percentage point rise in the EMTR is estimated to reduce overall investment by approximately 7%.

In the same vein, the World Bank notes that tax burden not only restricts growth but also limits SMEs' investment strategies by lowering fixed capital investment and increasing project costs. It recommends lighter tax policies and better access to external financing to foster investment and innovation within these enterprises (World Bank, 2025). Similarly, the European Commission observes that in Europe, SMEs operating in more favorable tax environments tend to invest more and are better integrated into formal financial systems. Conversely, excessive taxation is correlated with reduced investment, particularly among SMEs that lack access to alternative financing options (European Commission, 2024). The report also emphasizes the crucial role of stable and predictable taxation in enabling SMEs to plan their investments over the medium and long term.

On the other hand, contemporary economic literature increasingly highlights the pivotal role of fiscal incentives in promoting investment, especially among SMEs. According to the OECD, instruments such as tax reductions, exemptions, and accelerated depreciation schemes are among the most commonly used levers to encourage private investment. These measures lower the fiscal cost of projects, increase their expected profitability, and encourage firms to allocate resources to productive activities, particularly in industrial, technological, and export-oriented sectors (OECD, 2025). However, the OECD stresses that to maximize their impact, these incentives must be precisely targeted and supported by broader investment-facilitating policies. At the national level, the Casablanca-Settat Regional Investment Center lists several sectoral incentives—such as temporary corporate tax exemptions and tax advantages in industrial zones—designed to improve SMEs' ability to finance projects and stimulate both domestic and foreign investment (Casainvest.ma, 2025).

At the international level, the International Institute for Sustainable Development (IISD) confirms that well-designed tax incentives can effectively steer investment toward strategic sectors while enhancing local business competitiveness. However, it warns of the budgetary costs and risks of poor implementation (IISD, 2024). The World Bank highlights the essential role these tax tools play for SMEs facing financing constraints, as they help reduce initial project costs and improve net investment returns (World Bank, 2025).

In developing countries, the private sector alone cannot drive economic growth and development; the state therefore plays a crucial pioneering role in certain activities. In these contexts, development efforts often require public support through carefully designed tax incentives, particularly to assist those who lack sufficient financial resources. However, it is essential to precisely identify and appropriately target the beneficiaries of such exemptions. This can only be effectively achieved through a robust and well-managed state tax administration. When tax incentives are properly allocated to the right sectors, they can significantly contribute to fostering economic growth and advancing development (Siverekli Demircan, 2003).

By easing financing conditions, tax incentives enable companies to overcome financial barriers and undertake productive projects. The World Bank report further shows that countries combining these measures with structural reforms—such as administrative simplification and governance improvements—achieve stronger private investment growth and broader SME development (World Bank, 2025).

Furthermore, the relationship between the tax treatment of financing methods and investment decisions is attracting increasing attention in recent economic studies. According to the OECD, taxation significantly influences firms' financing choices, favoring instruments like debt through interest deductibility. This phenomenon—known as the "debt bias"—encourages firms to use debt financing over equity, as interest payments reduce taxable income (OECD, 2024). However, the OECD recommends rebalancing these tax incentives to avoid excessive debt dependency, while noting that well-calibrated mechanisms can enhance investment capacity, especially for SMEs.

Recent empirical studies indicate that the responsiveness of firm investment to corporate tax rates is not uniform but varies significantly depending on firm-specific characteristics. Factors such as firm age and industry sector (Schwellnus & Arnold, 2008; Fuest, Peichl & Siegloch, 2018; Federici & Parisi, 2015), financing structure and liquidity constraints (Zwick & Mahon, 2017), degree of market power (Kopp et al., 2019), opportunities for tax planning (Sorbe & Johansson, 2017), and overall profitability (Millot et al., 2020) all contribute to this heterogeneity. These findings highlight the need for a more differentiated analysis of how corporate taxation affects investment.

For instance, Djankov et al. (2010) assembled data on effective corporate tax rates for 85 countries and found a strong negative effect of corporate tax rates on investment rates and entrepreneurship. They estimated that a 10 percentage-point increase in the effective corporate tax rate is associated with about a 2 percentage-point drop in a country's investment-to-GDP ratio. In the same study, the authors also observed that higher corporate tax rates press firms to use significantly more debt relative to equity, confirming the tax shield incentive empirically. This finding – that firms in higher-tax environments have higher debt-to-equity ratios – is consistent with the theoretical debt bias due to taxes.

Klapper et al. (2008) investigated the impact of taxation on corporate financing policy by utilizing the 2001 corporate tax reform in Croatia as a natural experiment. The study found compelling evidence that tax reductions influenced firms' capital structure, leading to increased equity financing and a decline in long-term debt. These results align with the trade-off theory of capital structure, which posits that lower tax rates reduce the attractiveness of debt due to diminished interest deductibility. Similarly, Clemente-Almendros et al. (2014) examined the influence of corporate taxation on financing decisions among firms listed on the Spanish stock exchange during the period 2007–2013, employing panel data analysis over seven consecutive years for each firm. The findings indicated that marginal tax rates significantly shaped the debt policies of Spanish firms, reinforcing the relevance of tax considerations in corporate financing decisions, particularly in light of Spain's specific tax regime.

When focusing on emerging and developing economies, studies often highlight that tax burdens and incentives materially shape SME performance and growth. A common theme is that heavy or complex taxes impede small business expansion, while well-targeted tax incentives can alleviate financing constraints. For example, a recent Nigerian study by Babatayo and Adegbie (2021) examined tax incentives (such as tax holidays and exemptions) for SMEs. Using surveys of firms in Kwara State, they found that tax incentives have a significant positive impact on SME growth indicators, including sales revenue, profit growth, and business expansion. Similarly, Twesigye and Gasheja (2019), studying Rwandan SMEs, report that tax incentives significantly boosted the growth of small enterprises in their sample (Nyagatare district), underscoring the role of fiscal policy in SME development. In Iran, although a bit older, Farzbod's analysis (as cited by a 2022 study) revealed that exorbitant tax rates coupled with low incentives led to low productivity among SMEs, and recommended a more supportive tax system to spur entrepreneurship.

Other empirical evidence suggests a strong inverse relationship between corporate tax rates and industrial investment. Dobbins and Jacob (2016), for instance, found that lowering corporate tax rates stimulates investment activity, especially among firms that depend on internal financing. Similarly, Cristea and Nguyen (2016) highlight that high taxes can undermine managerial confidence and discourage industrial managers from pursuing new investments, primarily due to reduced utilization of machinery and production capacity.

Furthermore, Serrato and Zidar (2016) argue that elevated corporate taxes lead to higher industrial goods prices, which consequently dampen demand and lower sales volumes. This dynamic is particularly evident in capital-intensive sectors such as manufacturing, where investment decisions are more sensitive to tax burdens.

Turning specifically to the Middle East and North Africa (MENA) and similar developing regions, empirical literature in the last decade is somewhat limited but growing. Many MENA countries have introduced SME-specific tax policies (reduced corporate tax rates, simplified tax regimes, etc.), and researchers have been evaluating their effectiveness. Bird and Zolt (2014), in a policy analysis, suggested focusing on broadening tax bases and avoiding excessive special preferences, noting that poorly designed SME tax breaks can sometimes be exploited by larger firms or encourage firms to remain small to stay qualified.

In North African contexts, research has often centered on tax policy as part of the business environment. Chellaf and Chaabita (2023) conducted an econometric study on the impact of tax pressure on investment and growth in Morocco. Using Moroccan quarterly data (2007–2020) and a vector autoregression (VAR) model, they found a statistically significant long-run relationship between the overall tax burden and macroeconomic performance. Interestingly, their model suggested that an overly large drop in the tax-to-GDP ratio could slightly reduce long-run growth (they estimate a 1% decrease in tax pressure might lead to a 0.23% drop in long-term GDP growth). This counter-intuitive result likely reflects the role of public investment and services funded by taxes – implying there is an optimal level of taxation needed to support infrastructure that SMEs rely on.

Amraoui, Jianmu & Bouarara (2018), analyzing 52 Moroccan firms, likewise found that macroeconomic factors, such as the corporate tax rate, had no significant effect on firms' leverage decisions in Morocco. Firm-specific characteristics (profitability, asset structure) were more salient.

This in-depth literature review highlights that taxation plays a crucial role in SME investment financing decisions, operating through multiple channels. On one hand, excessive tax burdens act as a major obstacle, limiting available internal resources and increasing project costs. On the other hand, well-designed tax incentives and favorable tax treatment of financing methods can mitigate these barriers and encourage investment—particularly within emerging economies like Morocco. These findings provide a strong conceptual basis to guide the empirical investigation of this study and to formulate policy recommendations aimed at optimizing tax policy in support of SME growth.

Based on this literature, the following hypotheses were formulated:

H1: Tax pressure (PF) has a significant influence on investment financing decisions (DF)

H2: Tax incentives (IF) positively influence investment financing decisions (DF)

H3: The tax treatment of financing modes (TF) positively affects investment financing decisions (DF)

2. Methods

This study adopted a quantitative research design using a cross-sectional survey approach. The target population consisted of managers of small and medium-sized enterprises (SMEs) operating in Morocco. A intentional sampling technique was used to select firms that had recently undertaken investment projects and were involved in financing decisions influenced by tax considerations. Inclusion criteria required participants to be directly involved in their company's investment financing decisions.

The survey was administered to managers and financial decision-makers of Moroccan SMEs who had recently engaged in investment activities. The questionnaire was prepared in both Arabic and French.

The study adopted a non-probability convenience sampling strategy to identify SME participants, offering flexibility and ease of access to firms from different sectors and regions across Morocco. This approach was deemed appropriate given the practical constraints related to time, limited resources, and availability of a comprehensive list of SMEs and their financial decision-makers.

Following the methodological guidelines of Hair et al. (2013), the final version of the survey was distributed online via email and professional networks. The survey introduction clearly outlined the study's objectives, the confidentiality of responses, and the voluntary nature of participation. Participants provided informed consent before completing the questionnaire.

In the absence of a comprehensive database detailing Moroccan SMEs and their employee information, the study was conducted between March 2024 and October 2024 and a target sample size of 500 was initially set. Following the distribution of the questionnaire, a total of 390 valid responses were collected.

The variables used in this study were primarily derived from established and validated literature related to corporate finance and taxation, with contextual adaptations made to fit the specific case of Moroccan SMEs. A structured questionnaire was developed to measure these variables using a five-point Likert scale, ranging from 'strongly agree' (5) to 'strongly disagree' (1). The aim was to investigate the influence of various tax-related factors—such as tax pressure, fiscal incentives, and tax treatment of financing modes—on investment financing decisions.

The questionnaire was composed of two main sections. Section 1 collected demographic and organizational information about the SMEs and their respondents (e.g., company size, sector, respondent's role). Section 2 comprised of multiple choice items designed to capture perceptions and experiences regarding the fiscal environment and its impact on financing choices. The tax-related dimensions were operationalized as follows: five items for tax pressure, four items for fiscal incentives, and four items for the tax treatment of financing modes. These items were adapted from relevant prior studies and tailored to reflect the Moroccan context. The finalized instrument, detailed in Table 1, was used to ensure a comprehensive evaluation of the study's conceptual framework.

Table 1.
Items of the measurement instrument

Variable	Code	Item (Likert Scale Statement)
Tax Pressure (PF)	PF1	The nominal tax rates applied to my company (legal fixed rates) negatively influence our investment financing decisions.
	PF2	The effective tax rates (after exemptions and deductions) applied to my company have a significant impact on our investment financing decisions.
	PF3	The tax burden related to financing significantly affects my company's financial situation.
	PF4	The current taxable base of my company is considered high relative to its actual capacity.
Tax Incentives (IF)	IF1	The five-year corporate tax exemption for industrial companies provides a significant fiscal advantage for investment.
	IF2	The deductibility of donations made to the COVID-19 relief fund (tax credit) positively impacts the financing of our investments.
	IF3	The 100% tax exemption on distributed dividends encourages investment decisions.
	IF4	The exemption from paying minimum contribution (CM) during the first three years of a company's existence facilitates investment.
	IF5	The VAT exemption on the purchase of capital investment goods positively supports investment decisions.
	IF6	The VAT exemption on imported equipment, machinery, and tools provides financial relief for investment.
Tax Treatment of Financing Modes (TF)	TF1	The tax treatment of self-financing has favorable implications for our investment choices.
	TF2	The fiscal consequences of capital increases affect our decisions on equity financing.
	TF3	Using shareholder current accounts as a financing method provides tax advantages.
	TF4	Bank borrowing leads to significant fiscal implications for our company.
Financing Decision (DF)	DF1	Self-financing using internal reserves is beneficial for our company's investment strategy.
	DF2	Relying on shareholder current accounts to finance the company is advantageous in our case.
	DF3	Bank financing provides relevant advantages for our investment needs.
	DF4	Our company has used leasing (credit-bail) as a means to finance specific investments.

For data analysis, structural equation modeling (SEM) using the partial least squares approach (PLS-SEM) was employed via SmartPLS v.4

software. This method was chosen for its effectiveness in analyzing complex relationships between latent and observed variables, particularly suited to testing the conceptual model developed to examine the influence of tax-related factors on investment financing decisions in Moroccan SMEs.

Validity measures how well an instrument evaluates a particular concept, ensuring the concepts and measurements used by researchers are appropriate. Valid data show no discrepancies between the data reported by the researcher and the data gathered as the object of research. In PLS-SEM, two types of validity are used: convergent and discriminant validity.

Convergent validity assesses the extent to which a measurement correlates positively with alternative measures of the same construct. It is evaluated using two metrics: outer loadings and average variance extracted (AVE). Outer loadings, with a standard of 0.70, indicate how much variation within the item is explained by the construct. If an indicator's outer loadings value is greater than 0.70, it meets convergent validity and is considered reliable. AVE, with a standard of 0.50, implies that the construct explains more than half of the variance of its indicators. Specifically, an AVE greater than 0.50 means that the construct accounts for at least 50% of the variance in its manifest variables.

The formula is as follows:

$$AVE = \frac{\sum \lambda_i^2}{\sum \lambda_i^2 + \sum \theta_i}$$

Where:

λ_i: are the factor loadings of the indicators for a construct.

θ_i: are the error variances of the indicators.

Reliability indicates the stability and consistency of instruments measuring concepts, assessing measurement correctness and error. Cronbach's alpha, a conservative measure of reliability, provides estimates based on the intercorrelation of observed indicator variables. Due to its limitations, composite reliability is also used to measure consistency. For this study, the acceptable range for reliability is a minimum of 0.70 and a maximum of 0.90, with the desired range being between 0.80 and 0.90.

The formula is as follows:

$$\alpha = \frac{N}{N-1}(1 - \frac{\sum_{i=1}^{N} \sigma_i^2}{\sigma_{total}^2})$$

Discriminant validity compares the value of loadings of a parameter with those of other latent variable constructs. It is measured using the

Heterotrait-Monotrait ratio (HTMT) and the Fornell-Larcker criterion. HTMT assesses discriminant validity by evaluating whether correlations between constructs within a measurement model differ significantly from correlations across different constructs. If HTMT is significantly less than 0.9, discriminant validity is suggested; if it approaches or exceeds 1, there may be issues. The Fornell-Larcker criterion compares the square root of the AVE value with the correlation between latent variables.

Structural Model Analysis

Inner model testing, also known as structural model testing, aims to describe the relationships between latent variables. This study considers several factors including collinearity, the coefficient of determination (R^2), effect size (f^2), predictive relevance (Q^2), and path coefficients.

The coefficient of determination (R^2) is a widely used metric for evaluating structural models. It predicts the model and is calculated as the squared correlation between the actual value and the predicted endogenous construct based on the number of connected exogenous constructs. The R^2 value ranges from 0 to 1 and is categorized into three levels: 0.75 (substantial), 0.50 (medium), and 0.25 (weak). Higher R2 values indicate greater predictive accuracy.

Effect size (f2) measures the change in R^2 value before and after removing exogenous constructs from the model, indicating the substantive impact of exogenous constructs on endogenous constructs. The f^2 values are categorized as 0.02 (small), 0.15 (medium), and 0.35 (large). Values smaller than 0.02 suggest no significant effect of the latent exogenous variable.

Path coefficients test the relationships between constructs as part of the research hypothesis. Standard path coefficients range from −1 to +1. Values close to +1 indicate a strong positive relationship, while values close to −1 indicate a strong negative relationship. Values near zero suggest a weakening relationship.

Model fit analysis evaluates how well the proposed model structure aligns with empirical data, helping to identify specification errors. The normalized impact factor (NIF) index is commonly used in fit model analysis to evaluate the quality of previous research journals.

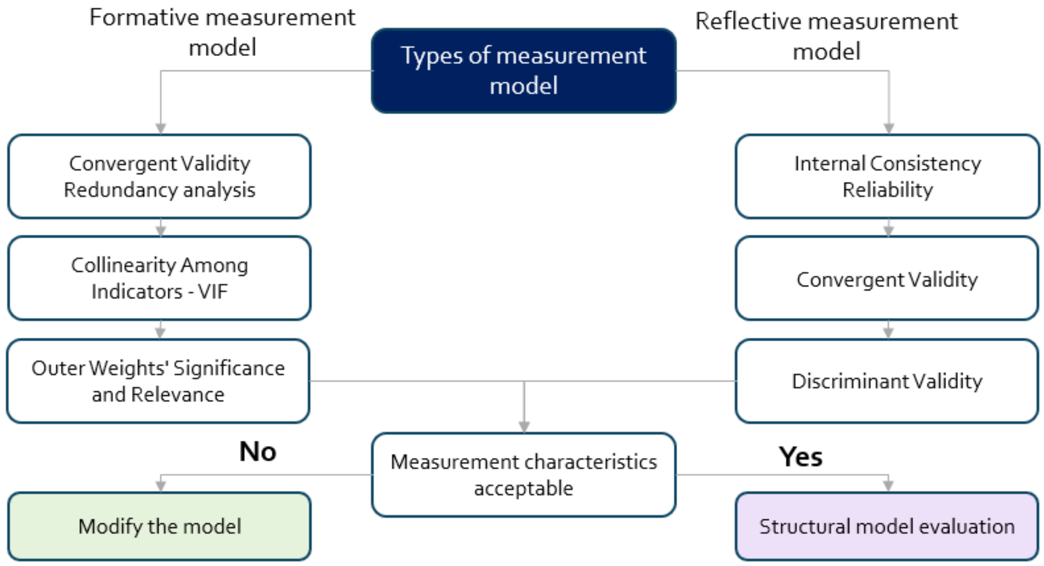

Figure 1. Decision Flowchart for Measurement Model Validation in PLS-SEM
Source: Author's elaboration

After running the PLS-SEM algorithm, researchers use path coefficients to test hypothetical estimates for structural model relationships. The significance of these coefficients depends on the standard error obtained through bootstrapping, which assesses the contribution of indicators to the corresponding construct. The standard error allows for calculations using t-values and p-values for all structural path coefficients. When the t-value exceeds the critical value, the coefficient is considered statistically significant at a certain probability level.

Hypothesis testing reveals whether the estimated hypothesis is accepted or rejected. Critical t-values are used to determine the significance of the coefficient. If the empirical t-value exceeds the critical t-value (typically >1.65 for a significance level of 5% with one-tailed tests), the hypothesis is rejected.

The demographic analysis of the 390 respondents offers insights into the diversity of the SME decision-makers who participated in the study. In terms of gender, 62% of the respondents were male and 38% female. The majority (46%) were between 31 and 45 years old, followed by 28% aged 46 to 60, 18% under 30, and 8% over 60. Regarding educational attainment, 52% held a university degree, 30% had a postgraduate qualification, and 18% had a high school diploma or lower. In terms of professional role, 41% were business owners, 35% were general managers or directors, and 24% held financial or administrative roles.

Table 2.
Demographic analysis of respondents

Demographic Variable	Category	Count	Percentage (%)
Gender	Male	242	62
	Female	148	38
Age Group	Under 30	70	18
	31–45	179	46
	46–60	109	28
	Over 60	32	8
Education Level	High school or less	70	18
	University degree	203	52
	Postgraduate (Master/PhD)	117	30
Professional Role	Business owner	160	41
	General manager/director	137	35
	Financial/administrative staff	93	24
Company size	Less than 5 employees	168	43
	Between 5 and 20 employees	149	38
	Between 20 and 50 employees	67	17
	Between 50 and 100 employees	4	1
	Over 100 employees	2	1
Sector	Agribusiness	124	32
	Industrial	97	25
	Service	169	43

Source: Author's elaboration based on SPSS

3. Results

To ensure the reliability and validity of the questionnaire, the measurement model was evaluated following established guidelines in the PLS-SEM literature. The analysis was carried out using SmartPLS, which facilitates a rigorous assessment of both the latent constructs and the subsequent structural model. This evaluation step is crucial to confirm that the survey items accurately capture the theoretical dimensions they are designed to measure (Becker, 2022).

To begin with, the factorial structure of the model was validated through an analysis of item loadings. As shown in Table 3, all item loadings exceeded the recommended threshold of 0.70, confirming satisfactory convergent validity.

Table 3.
Factor loadings results

	DF	IF	PF	TF
DF1	0.777			
DF2	0.733			
DF3	0.86			
DF4	0.843			
IF1		0.728		
IF2		0.786		
IF3		0.963		
IF4		0.96		
IF5		0.961		
IF6		0.938		
PF1			0.933	
PF2			0.943	
PF3			0.953	
PF4			0.904	
TF1				0.701
TF2				0.763
TF3				0.977
TF4				0.851

Source: Author's elaboration based on SmartPLS

Specifically, for the Financing Decision (DF) construct, item loadings ranged from 0.733 to 0.860, indicating strong contributions of all four items to the underlying factor. The Tax Incentives (IF) construct also demonstrated excellent convergence, with item loadings between 0.728 and 0.963. Similarly, the Tax Pressure (PF) construct revealed high factor loadings (ranging from 0.904 to 0.953), and the Tax Treatment of Financing Modes (TF) construct showed loadings between 0.701 and 0.977.

Table 4.
Internal consistency results

	Cronbach's alpha	Rho_A	Rho_C	Average variance extracted (AVE)
DF	0.767	0.771	0.784	0.593
IF	0.94	0.975	0.953	0.775
PF	0.971	0.952	0.973	0.879
TF	0.963	0.924	0.922	0.704

Source: Author's elaboration based on SmartPLS

Following this, the internal consistency and reliability of the measurement model were assessed using Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE). All constructs displayed Cronbach's alpha values above the 0.70 benchmark, confirming strong internal reliability. In particular, PF and TF demonstrated excellent reliability with alpha values of 0.971 and 0.963, respectively. Composite reliability values (CR) ranged from 0.784 to 0.973, and AVE values were all above the minimum recommended level of 0.50—further confirming convergent validity (see Table 4). The AVE scores were particularly high for PF (0.879) and IF (0.775), indicating that the majority of variance in these constructs is explained by their indicators.

Table 5.
Fornell-Larcker results

	DF	IF	PF	TF
DF	0.702			
IF	0.468	0.88		
PF	0.095	0.281	0.938	
TF	0.117	0.138	0.013	0.839

Source: Author's elaboration based on SmartPLS

After evaluating the internal consistency of the measurement model, it was necessary to examine its discriminant validity. To do so, the Fornell–Larcker criterion was employed (see Table 5). The square roots of AVE (diagonal values) were greater than the inter-construct correlations (off-diagonal values), demonstrating that each construct shared more variance with its own indicators than with others. For example, the square root of AVE for PF (0.938) exceeded its correlation with IF (0.281) and with DF (0.095), satisfying the condition for discriminant validity. The same pattern was observed across all constructs, confirming the distinctiveness of the four latent variables in the model.

To further assess discriminant validity, the HTMT criterion was applied. According to Henseler et al. (2015), HTMT values below the threshold of 0.85 (or more conservatively, 0.90) indicate adequate discriminant validity. As shown in the HTMT matrix (see Table 6), all inter-construct values fall well below the 0.85 threshold.

Table 6.
HTMT criterion results

	DF	IF	PF	TF
DF				
IF	0.526			
PF	0.19	0.294		
TF	0.099	0.111	0.049	

Source: Author's elaboration based on SmartPLS

To evaluate the structural model, multicollinearity diagnostics and the coefficient of determination (R^2) were examined to assess the predictive power and reliability of the model relationships.

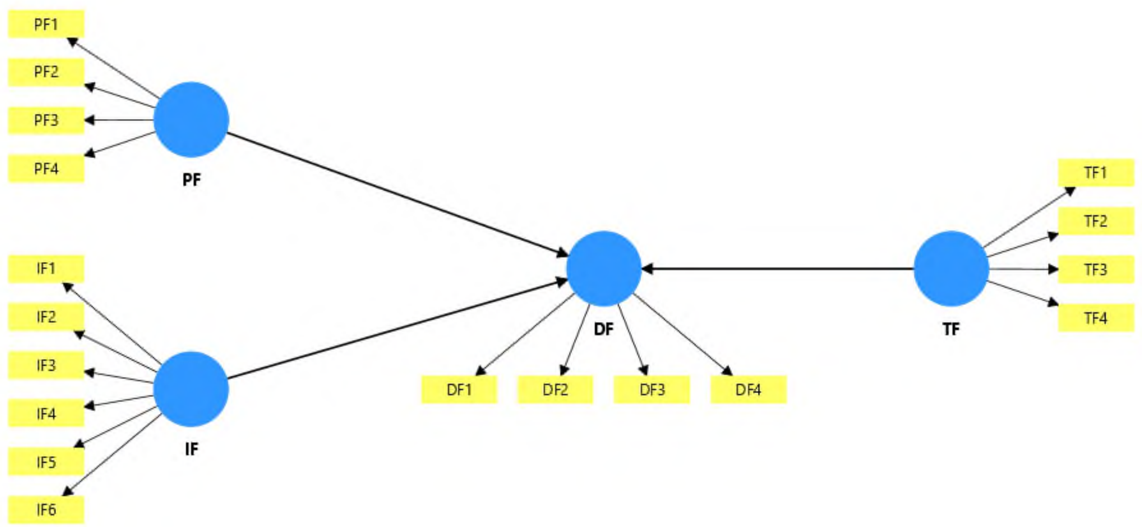

Figure 2. Conceptual model
Source: SmartPLS

Variance Inflation Factor (VIF) values were inspected to detect potential multicollinearity issues among the predictor variables. According to Hair et al. (2017), VIF values below 5 are considered acceptable, and values under 3 are preferred for more conservative assessments. As presented in Table 7, all VIF values in the model range from 1.153 to 4.604, with the highest being IF3 (4.604) and IF5 (3.417). Despite this, all values remain below the critical threshold of 5, indicating no serious multicollinearity problem. Therefore, the latent constructs included in the model do not exhibit redundancy, and the model structure is statistically reliable for further analysis.

Table 7.
Colinearity assessment - VIF

Item	VIF	Item	VIF
DF1	1.226	**IF1**	2.450
DF2	1.153	**IF2**	2.115
DF3	1.616	**IF3**	4.604
DF4	1.544	**IF4**	2.908
PF1	2.974	**IF5**	3.417
PF2	2.093	**IF6**	2.498
PF3	1.942	**TF1**	1.282
PF4	3.370	**TF2**	1.793
TF3	2.104	**TF4**	1.740

Source: Author's elaboration based on SmartPLS

The R^2 value for the endogenous variable "Financing Decision (DF)" is 0.224, indicating that approximately 22.4% of the variance in investment financing decisions among Moroccan SMEs can be explained by the three exogenous constructs: tax pressure (PF), tax incentives (IF), and tax treatment of financing modes (TF). The adjusted R^2 value is 0.199, which accounts for model complexity and suggests a modest explanatory power. According to Chin (1998), R^2 values of 0.19, 0.33, and 0.67 can be classified as weak, moderate, and substantial respectively in PLS-SEM. Based on this, the explanatory power of the model is considered weak to moderate, yet still meaningful given the complex and multidimensional nature of tax policy effects on financing decisions.

Table 8.
R^2 values

	R-square	R-square adjusted
DF	0.224	0.199

Source: Author's elaboration based on SmartPLS

To evaluate the structural relationships proposed in the research model, path coefficients were tested using the bootstrapping procedure with 5,000 subsamples. The significance of each path was assessed based on the coefficient magnitude, standard error, t-statistic, and corresponding p-value. Table X summarizes the results of hypothesis testing.

Table 9.
Research hypothesis results

	Coefficient	Ecart-type	T Statistic	P values
IF -> DF	0.442	0.099	4.465	0.000
PF -> DF	0.392	0.148	2.649	0.009
TF -> DF	0.432	0.124	3.484	0.001

Source: Author's elaboration based on SmartPLS

H1: Tax pressure (PF) has a significant influence on investment financing decisions (DF)

The path coefficient for PF → DF is 0.392, with a t-value of 2.649 and a p-value of 0.009, indicating a statistically significant and positive effect at the 5% level. Therefore, H1 is supported, suggesting that higher perceived tax pressure influences SME financing decisions in a meaningful way.

H2: Tax incentives (IF) positively influence investment financing decisions (DF)

The path coefficient for IF → DF is 0.442, with a t-value of 4.465 and a p-value of 0.000, indicating a highly significant and positive relationship. This supports H2, confirming that fiscal incentives such as exemptions and deductions encourage SMEs to invest by improving financing feasibility.

H3: The tax treatment of financing modes (TF) positively affects investment financing decisions (DF)

The path coefficient for TF → DF is the strongest at 0.432, with a t-value of 3.484 and a p-value of 0.001, signifying a highly significant impact. This validates H3, suggesting that favorable tax treatments linked to self-financing, leasing, or shareholder advances play a critical role in shaping SMEs' financing choices.

4. Discussion

The findings of this study confirm that fiscal variables exert a statistically significant influence on the investment financing decisions of Moroccan SMEs. Specifically, the three independent variables—fiscal pressure (PF), fiscal incentives (IF), and the tax treatment of financing modes (TF)—were all found to have significant positive effects on the dependent variable, financing decision (DF). These results provide robust support for the study’s hypotheses H1, H2, and H3.

First, the positive and significant relationship between fiscal pressure and financing decisions ($\beta = 0.382$, $p = 0.011$) suggests that even though high tax burdens are often perceived as constraints, firms may adapt their financing strategies accordingly. This aligns with earlier studies by Ibrahim et al. (2018), which found that corporate tax rates influence the capital structure choices of firms in Nigeria through increased awareness and strategic planning. Also, fiscal pressure on equity demonstrated a generally small negative influence on asset profitability and investment levels. This outcome may reflect the impact of rising tax obligations and various levies, which can increase operational costs and reduce available resources for reinvestment. In some cases, however, fiscal pressure on equity was also associated with a positive relationship with return on investment (ROI), suggesting that certain companies might respond to higher tax burdens by optimizing their capital structures or improving operational efficiency to sustain returns (Batrancea et al., 2021).

Second, fiscal incentives ($\beta = 0.472$, $p < 0.001$) emerged as a strong predictor of financing decisions, indicating that tax exemptions and deductions play a critical role in stimulating investment among SMEs. These findings are consistent with the results of Afolabi et al. (2023), who demonstrated that incentives such as education tax credits and capital allowances significantly enhance firms’ ability to mobilize capital for expansion. The importance of tax incentives in shaping financing behavior confirms the practical relevance of policy tools designed to lower the effective tax burden. Empirical evidence from Japan further confirms that SMEs using tax incentives experienced a significantly higher capital investment rate and improved productivity, particularly among financially constrained firms, indicating that incentives mitigate financing obstacles (Hosono, 2023). Similarly, studies in China demonstrate that VAT and income tax incentives substantially boost SME asset growth and sales revenue.

Third, the tax treatment of financing methods ($\beta = 0.538$, $p < 0.001$) was also significantly associated with investment decisions, underscoring how favorable fiscal treatment of methods such as leasing, equity contributions, or retained earnings can incentivize firms to pursue particular funding strategies. This corroborates theoretical models that emphasize the impact of differential tax treatments on debt-versus-equity choices, as noted in Schanz and Schanz (2011), who showed how tax deductions for interest payments increase the attractiveness of debt financing compared to equity.

These results challenge traditional perspectives that associate high corporate tax rates solely with reduced investment. Instead, they suggest that when coupled with targeted fiscal incentives and supportive tax treatments, even higher tax regimes can encourage firms to leverage diverse financing sources to maintain or expand investment levels. The positive moderating role of fiscal policy on financing decisions highlights its potential to mitigate the adverse effects of tax burdens and strengthen firms' resilience, as suggested by Ohrn (2018) and Wu and Yue (2009).

Finally, at the macroeconomic level, complementary factors such as firm size, profitability, and sales growth further interact with fiscal measures to shape investment patterns. Larger firms, driven by higher market demand, tend to make substantial investments in productive assets (Ajide, 2017), while sales growth compels additional capital expenditure to support expanding operations (Adelino et al., 2017; Al-Gamrh et al., 2020; Farooq et al., 2021b). Meanwhile, GDP growth provides an enabling environment that further supports investment activity (Ajide, 2017).

Conclusion

This study set out to examine the influence of taxation on the investment financing decisions of Moroccan small and medium-sized enterprises (SMEs), focusing on three key fiscal dimensions: tax pressure, fiscal incentives, and the tax treatment of financing methods. Using a quantitative approach and partial least squares structural equation modeling (PLS-SEM), the results reveal that all three fiscal variables exert a statistically significant and positive influence on SMEs' financing decisions.

The findings demonstrate that tax pressure, although typically seen as a constraint, influences financing behavior, possibly by compelling firms to adopt more strategic financial planning. More notably, fiscal incentives emerged as a powerful determinant, highlighting the effectiveness of tax exemptions and deductions in fostering a more favorable investment environment. Likewise, the tax treatment of financing modes—such as leasing, shareholder advances, and self-financing—was found to play a critical role in shaping financing strategies.

By confirming the relevance of these fiscal factors, the study contributes both theoretically and practically. It reinforces the applicability of financing theories such as the pecking order theory in a tax-sensitive context and provides policymakers with evidence-based insights to refine tax policy for greater SME support.

This study set out to examine the influence of taxation on the investment financing decisions of Moroccan small and medium-sized enterprises (SMEs), focusing on three key fiscal dimensions: tax pressure, fiscal incentives, and the tax treatment of financing methods. Using a quantitative approach and partial least squares structural equation modeling (PLS-SEM), the results reveal that all three fiscal variables exert a statistically significant and positive influence on SMEs' financing decisions.

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Lastly, by investigating sector-specific tax consequences, incorporating qualitative methods to gather managers' perspectives, or carrying out comparison studies, a future study could build on these findings. Also, it is important to highlight that one of the limitations our study is the low R^2 of the explained variable which could be enhanced by including more variables in the conceptual model. Our comprehension of the complex interaction between taxes and SME financing options in the Moroccan context will be enhanced by such work.

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CONTENTS

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THE POSITIVE IMPACTS OF GREEN TOURISM ON LOCAL COMMUNITIES IN TRA VINH PROVINCE OF VIETNAM

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Cuong Nguyen[2],
Khuong Tran[3]

Abstract: This study aims to find out how green tourism positively impacts the local community in Tra Vinh province of Vietnam. The paper uses the questionnaire method to survey the awareness of the local community in Tra Vinh province about the benefits of green tourism, along with statistical and analytical methods to clearly identify the level of impact on tourism. The green calendar brings the community together here. The results show that from a resource and cultural advantages perspective, green tourism has positively impacted the community's environment, socio-culture and economy, but most notably the overall awareness of local green tourism. This study shows that the local community values the environmental, cultural and heritage conservation more than the economic benefits reaped, in promoting and maintaining green tourism activities in the local areas.

Key words: Green Tourism, Positve Impacts, Local Community, Tra Vinh Province, Vietnam.

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Introduction

The significant contributions of tourism to global economic growth, job creation, and infrastructure development have led to adverse environmental and social impacts. This has fostered the emergence of green tourism, characterized by responsible tourism activities that prioritize environmental conservation, cultural and social sensitivity, and economic sustainability (Baloch et al., 2023). The role of local communities is increasingly recognized in promoting tourism, particularly green tourism. Active participation of local communities in planning and management is essential for successfully implementing green tourism activities aimed at cultural preservation, environmental protection, job creation, and income generation. The engagement of local communities depends on their awareness, economic and social benefits, and involvement in the decision-making processes (Barbieri et al., 2020). Located between the Tien and Hau rivers, Tra Vinh is bordered by the East Sea, featuring numerous islets and riverine and coastal sandbars, along with specialized fruit orchards. These characteristics position Tra Vinh as a prime location for developing beach tourism, riverine eco-tourism, and garden-resort tourism. Its natural environment, still largely unspoiled due to limited exploitation, is well-suited for international tourists seeking discovery and experiences—distinctive traits of the Northern region of Vietnam (Van et al., 2024). Furthermore, Tra Vinh is home to a convergence of three cultural traditions from the Kinh, Khmer, and Hoa ethnic groups, creating unique indigenous cultural values. These constitute significant potential and advantages for Tra Vinh to develop various forms of tourism, especially green garden-community tourism associated with cultural-historical tourism, with a prominent focus on the cultural elements of the Khmer ethnic community. The local people are noted for their honesty, hospitality, and preservation of the Southwestern Vietnamese character. Their vibrant and diverse culture includes unique languages, traditions, and beliefs. Despite challenges such as poverty and illiteracy, the local community has opportunities to preserve and promote their cultural heritage through education, tourism, and collaboration with other ethnic groups. Achieving this goal requires recognizing the importance of developing green tourism as a means to generate income, create green jobs, and lay the foundation for environmentally-friendly economic development within Tra Vinh's communities, thereby promoting green growth and poverty alleviation (Nguyen, 2018). Developing green tourism at the local level will help protect the natural landscape and biodiversity of conservation areas. The findings of this study will provide insights into the positive impacts of green tourism on the local communities of Tra Vinh province. The significant contributions of tourism to global economic growth, job creation, and infrastructure development have led to adverse environmental and social impacts. This has fostered the emergence of green tourism, characterized by responsible tourism activities that prioritize environmental conservation, cultural and social sensitivity, and economic sustainability (Baloch et al., 2023). The role of local communities is increasingly recognized in promoting tourism, particularly green tourism. Active participation of local communities in planning and management is essential for successfully implementing green tourism activities aimed at cultural preservation, environmental protection, job creation, and income generation. The engagement of local communities depends on their awareness, economic and social benefits, and involvement in decision-making processes (Barbieri et al., 2020). 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Literature Review

Green tourism

According to The International Ecotourism Society (TIES), green tourism can be defined as "responsible travel to natural areas that conserves the environment, sustains the well-being of local people, and involves education." Green tourism, also referred to as ecotourism, is a form of tourism based on two main pillars: nature and culture. It integrates environmental education with the active participation of local communities to promote conservation and sustainable development of natural resources and ecosystems (Font & Tribe, 2001). The implementation of green tourism can vary depending on the location and context. The overarching goal of green tourism is to achieve a balance between the economic benefits of tourism and the need for environmental conservation, as well as the preservation of local culture and communities for future generations (Pan et al., 2018).

Benefits of green tourism

Economic Benefits: Green tourism generates job opportunities in areas such as hospitality services, ecotourism tour-guides, and environmental education (Amerta et al., 2018). By creating environmentally responsible employment, green tourism stimulates economic growth and supports local businesses and agriculture. It promotes local food consumption, reducing carbon emissions from food transportation, while preserving culinary traditions and cultural heritage, thereby driving economic development (Sims, 2009).

Environmental Benefits: Green tourism contributes to the conservation of natural and cultural resources while supporting environmental preservation efforts. Through sustainable practices and low-impact activities, green tourism minimizes negative environmental impacts. It also promotes cultural heritage by safeguarding historical sites and monuments, and enhances environmental awareness and conservation efforts (Edgell, 2019).

Cultural and Social Benefits: Green tourism fosters cultural exchange, respect for traditions, and mutual understanding between tourists and local communities. It enhances diversity, bridges cultural gaps, and promotes local products while supporting traditional practices. These activities contribute to sustainable tourism benefits for both visitors and local populations (Barna et al., 2021).

Local community awareness of green tourism

The level of awareness about green tourism among local communities varies depending on factors such as education, exposure to tourism, and cultural and social norms. Some communities actively participate in sustainable tourism activities, recognizing the importance of environmental protection and cultural heritage preservation (Li and Hunter, 2015). However, others may have limited awareness due to a lack of information, education, or involvement in tourism development. Cultural and social norms that prioritize economic growth can also hinder the promotion of sustainable tourism. Raising awareness through education, outreach programs, and community participation is essential to foster sustainable green tourism (Okazaki, 2008).

Commitment of local communities to green tourism models

The commitment of local communities to green tourism is influenced by fundamental factors such as awareness of the benefits of sustainable tourism and the limitations of traditional practices. Active involvement of local communities in decision-making processes and ownership of sustainable tourism initiatives enhances commitment. Economic benefits, including job opportunities and support for local businesses, also contribute to stronger engagement (Hassan, 2000). The alignment of cultural and social values further reinforces commitment. Overall, engaging local communities, considering their values, and providing tangible benefits encourage their commitment to green tourism (Barbieri et al., 2020).

Relationship between awareness and commitment to green tourism

The relationship between awareness and commitment to green tourism among local communities is complex. Greater awareness of the benefits of green tourism can lead to increased likelihood of active participation in sustainable tourism activities. Conversely, if local communities perceive that engaging in sustainable tourism initiatives does not provide significant economic benefits or if they face other economic challenges, their participation may decline (Aref, 2011). Therefore, it is crucial not only to raise local community awareness about the benefits of green tourism but also to actively involve them in decision-making processes and ensure they experience tangible benefits from their participation.

Potentials for developing green tourism in Tra Vinh province

Tra Vinh boasts rich potential for cultural, historical, coastal, and riverine garden tourism, with its specialized fruit orchards, coastal sandbars, and the unique cultural identity of its local communities. Additionally, Tra Vinh is home to numerous national historical sites, architectural landmarks, and intangible cultural heritage. The Khmer pagodas of Southern Vietnam, such as Hang Pagoda, Nodol Pagoda, Ang Pagoda, and Vam Ray Pagoda, stand out with their ancient and unique architecture. Ang Pagoda, one of the oldest in the Mekong Delta, is a prominent attraction frequently visited by tourists and locals alike (Van Chat, 2024). Among Tra Vinh's ecological tourism highlights, Ba Om Pond is a key destination, chosen by many visitors. This area features a collection of ponds and centuries-old trees, with a cool climate year-round. Ba Om Pond itself is a clear water body approximately 500 meters long and 300 meters wide. The surrounding landscape includes rolling sand dunes and hundreds of ancient trees such as dipterocarps and hopeas, with massive exposed roots forming intriguing shapes. The Rung Duoc Eco-Tourism Area (in Long Khanh commune, Duyen Hai district) is another must-visit destination, forming part of the protected mangrove ecosystem along the Southern Vietnamese coast. The pristine forest is rich in biodiversity, featuring a variety of mangrove trees like mangroves, thorn palms, and phoenix palms, with the predominant species being mangroves. It is also home to diverse flora and fauna. Visitors to the Mangrove Forest can choose to explore via forest trails or by boat through the dense mangroves. With its vast natural area and unspoiled beauty, this destination offers fresh air, breathtaking landscapes, and captivating local folklore. Chim Islet, covering an area of approximately 60 hectares, offers a rustic charm with half of its land dedicated to rice cultivation and the remainder to vegetables and fruit trees. Visitors to Chim Islet can immerse themselves in rural life and enjoy seasonal local dishes. Its unique approach, "appointment-only seasonal tourism," ensures visitors can experience local production activities by scheduling their visit in advance. Another notable destination is Ho Islet, a "hidden gem" of Tra

Vinh's tourism, with an area of 22 hectares and home to only 21 households. This islet captivates visitors with its lush greenery and the warm, genuine hospitality, characteristic of the Mekong Delta. A standout experience here is "night tourism without electricity," providing an authentic and memorable experience (Nguyen, 2022). Recognizing the vast potential of green tourism and its critical role in local socio-economic development, Tra Vinh province has implemented numerous mechanisms and policies to support tourism growth. These efforts include promoting and advertising tourism, developing state management frameworks for the sector, and focusing on training a skilled tourism workforce to enhance the quality of green tourism services and products. Key initiatives include river tourism (e.g., the Long Binh River and Lang The River routes), community-based tourism at Chim Islet, and agricultural tourism models in Con Ong Hamlet, Dan Thanh Commune, and Duyen Hai Town. Currently, the province is leveraging several green tourism routes, including: The ecological route: "Tra Vinh City – Khmer Cultural and Tourism Village – Ho Islet"; The cultural and ecological route: "Tieu Can – Cau Ke – Tra Cu"; The coastal route: "Tra Vinh City – Cau Ngang – Ba Dong Beach" These flagship routes have been embraced by travel agencies both within and beyond the province, solidifying Tra Vinh as a vital destination in the Mekong Delta.

Research Method

This study employed a survey questionnaire method, with data collected through questionnaires targeting local residents living near tourist sites and attractions in Tra Vinh. The questionnaire was designed to gather insights into the positive impacts of green tourism on the "environment," "socio-cultural aspects," and "economy." Based on the survey results, the study focused on criteria related to socio-cultural, environmental, and economic dimensions. The questionnaire was structured into two parts: the first part collected demographic information, while the second part focused on economic, environmental, and socio-cultural indicators and the degree to which green tourism influences local communities. The research employed a random sampling method, targeting 200 local residents living near tourist sites. All 200 questionnaires were completed and returned, resulting in a 100% response rate.

Table 1
Demographic Characteristics of Survey Participants

Variable	Attribute	Age							
		18-24		**25-34**		**45-54**		**Above 56**	
		Number	%	Number	%	Number	%	Number	%
Gender	**Female**	10	13,2%	32	42,1%	26	34,2%	8	10,5%
	Male	15	12,1%	52	41,9%	41	33,1%	16	12,9%
Education	**Bachelor Degree**	3	8,1%	17	45,9%	16	43,2%	1	2,7%
	College	9	11,1%	36	44,4%	23	28,4%	13	16,0%
	High School Graduate	7	14,0%	19	38,0%	18	36,0%	6	12,0%
	Less than High School	6	18,8%	12	37,5%	10	31,2%	4	12,5%

Results

The analysed results of the impact of green tourism on local communities clearly demonstrate that green tourism activities have brought environmental, socio-cultural, and economic benefits to the local population. These positive impacts have significantly shifted the local community's awareness of green tourism.

Table 2
Indicators and Levels of Impact of Green Tourism on Local Communities

Criteria	**Mean**	**Impact level**
1. Environment	**4.35**	**Positive**
Security in the area	4.40	Positive
Environmental hygiene in the area	4.38	Positive
Waste collection methods in the area	4.31	Positive
Visitors from other regions	4.23	Positive
Social conditions in the area	4.41	Positive
2. Socio-Cultural	**4.38**	**Positive**
Spiritual life	4.34	Positive
Cultural preservation	4.32	Positive
Cultural exploitation	4.42	Positive
Social life	4.45	Positive
3. Economic	**4.33**	**Positive**
Personal income	4.35	Positive
Family income	4.33	Positive
Employment of local residents	4.30	Positive
Expenses for leisure and entertainment	4.35	Positive
Other taxes and fees	4.31	Positive

Statistical survey results indicate that green tourism has had a positive environmental impact (4.35) on the awareness of the local community. The green tourism model, linked to community responsibility, is being actively adopted by businesses in the area. Local households and service providers have ceased generating plastic waste. The province has implemented measures to remind and encourage all tourists not to bring plastic bags or single-use plastic items to eco-tourism sites. It also calls upon residents, business owners, and tourism enterprises at these sites to reject single-use plastic products, replacing them with environmentally friendly alternatives, sorting waste at the source, and participating in waste collection and treatment efforts. The local community in Tra Vinh has benefited from the socio-cultural advantages of green tourism (4.38) by encouraging residents to engage in green tourism models through community-based tourism activities. These activities allow tourists to interact directly with the local community, learn about their customs and traditions, and participate in traditional and local cultural events. This cultural exchange fosters mutual benefits, enabling both tourists and local communities to learn from one another, share experiences, and gain a deeper understanding of each other's cultures (Chan, et al., 2021). Several green tourism and agricultural tourism models have emerged in the locality, each showcasing unique characteristics that contribute to the distinctiveness of green tourism in the region. These models include activities such as reenacting the traditional process of pounding “cốm dẹp”—a traditional Khmer dish—or enabling each household to offer a specific type of service using locally produced goods made on-site. These services include homestay rentals, cooking and baking lessons for tourists, crab fishing, setting traps for shrimp, engaging in folk games, and recreating traditional rural markets. These initiatives have not only helped stabilize the local economy, improve livelihoods, and alleviate poverty for many households, but also preserve and promote the traditional cultural values of the ethnic communities. The local community has increased its awareness of the benefits of green tourism, which has encouraged their involvement in the decision-making process, ensuring they receive tangible benefits from their participation. Doing so can enhance their commitment to implementing sustainable tourism and actively support and promote these initiatives within their communities (Choi, 2013). Finally, statistical evidence from the average scores of the indicators reveals that the economic benefits of green tourism (4.33) have had a lower impact on the local community compared to environmental (4.35) and socio-cultural benefits (4.38). Currently, households only receive a modest share of the economic benefits, primarily from ticket sales, lodging, food services, and employment opportunities during the agricultural off-season. This indicates that the local community in Tra Vinh still prioritizes the preservation of their environment and cultural heritage over economic gains.

Discussion

To continue fostering the positive impacts of green tourism on local communities, Tra Vinh province should implement the following solutions:

Environmental protection:

To sustain and develop green tourism activities, it is essential to consistently emphasize environmental protection and the preservation of natural ecosystems at tourism sites. Environmental protection is a critical responsibility in planning and implementing green tourism projects and strategies within the province, as well as at specific tourism zones and destinations. Therefore, it is necessary to install signs, banners, posters, and other visual aids around green tourism sites to remind and encourage tourists and local residents to adopt green consumption practices in tourism. This includes implementing environmental protection measures in tourism activities, reducing the use of non-biodegradable plastic bags and single-use plastic products, and ensuring proper waste management to prevent littering on streets and waterways.

Enhancing the contribution of green tourism to local communities:

Sharing the benefits of green tourism with local communities has increasingly become a key objective to ensure social equity and optimize tourism’s advantages. This can be achieved by providing greater and more targeted benefits to local residents, such as: offering professional training in tourism services; conducting surveys to gather local opinions and needs for tourism development projects; and investing in infrastructure development. Additionally, supporting households participating in green tourism models to develop unique and distinctive products can attract tourists by leveraging cultural and regional specialties. This approach moves beyond the limited economic benefits previously gained from ticket sales, lodging, and food services, empowering local residents to play a more active and rewarding role in green tourism initiatives.

Enhancing the quality of human resources with knowledge of green tourism:

Encourage businesses to organize on-site training programs for local human resources to support green tourism sites and destinations. Training sessions for residents at green tourism locations should include knowledge about environmental protection in tourism, green tourism principles, and communication skills. These topics should be integrated into educational and awareness-raising programs for the local community during the development of green tourism. Fostering a cultured, civilized, polite, friendly, and welcoming approach to tourism is essential, as the quality of tourism products largely depends on the attitude and competency of the workforce at tourism destinations.

Enhancing the quality of human resources with knowledge of green tourism:

Collaborate with businesses within and outside the province to develop new inter-regional tourism spaces, such as green urban tourism areas, coastal eco-tourism spaces, and garden eco-tourism spaces integrated with ethnic culture. Additionally, enhance the landscape by improving greenery along transportation routes and destinations, and develop plans for green transportation systems. Design programs and plans to develop tourism products that are aligned with ecosystem protection, biodiversity conservation, and the preservation of traditional cultural values. Encourage large-scale tourism investment projects to create standout green tourism products, thereby increasing the number of visitors to the local area.

Conclusions

The results illustrate that from resource and cultural advantages, green tourism has positively impacted the community's environment, socio-culture and economy, but most notably is the increase in awareness of local green tourism. The local communities value the environment, culture and heritage conservation more than the economic benefits in promoting and maintaining green tourism activities. Green tourism is a form of tourism that operates with the goal of minimizing environmental impacts, contributing positively to biodiversity conservation, utilizing renewable energy, and promoting natural and cultural heritage, while developing eco-friendly products. Compared to other provinces in the Mekong Delta region, Tra

Vinh faces challenges in tourism development due to limitations in road infrastructure and geographic location. However, by overcoming these difficulties, Tra Vinh has strategically focused on eco-community tourism linked to indigenous culture as its primary product, aiming to sustainably develop a "green" tourism economy. This study explores the positive impacts of green tourism on local communities in Tra Vinh. The findings highlight the significance of environmental and socio-cultural benefits in shaping the community's awareness and commitment to green tourism development. Cultural exchange has been shown to benefit both tourists and locals, fostering mutual understanding and appreciation of diverse cultures, protecting natural resources, and promoting sustainable environmental conservation. Additionally, the study reveals that local communities prioritize the preservation of natural resources, cultural heritage, and traditional customs over economic gains. This emphasizes the importance of involving communities in the decision-making process and ensuring that they derive tangible benefits from their participation. In doing so, their commitment to sustainable green tourism activities can be strengthened, leading to the promotion and support of these initiatives within their communities. Eco-community tourism models have significantly transformed the lives of local residents in Vietnam (Anh et al., 2024). The study further demonstrates how changes in awareness and actions among the local community regarding green tourism help them recognize its necessity for development, aligning with the advantages and potential of Tra Vinh's local tourism resources. These findings can inform policymakers and tourism stakeholders in developing strategies to actively engage local communities, ensuring their meaningful participation and mutual benefits in the evolving context of modern tourism.

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CONTENTS

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FORECASTING THE EFFECTIVENESS OF NON-STATE PENSION FUND INVESTMENT STRATEGIES: THE CASE OF GEORGIA

Asie Tsintsadze[1],
Sofio Tsetskhladze[2]

Abstract: This paper evaluates the effectiveness of Georgia's non-state pension fund using monthly data for 2022–2023 and develops short-to-medium-term forecasts to 2030. We assemble indicators on pension contributions, investment returns, and the exchange rate, and define a fund-effectiveness measure to capture the system's ability to meet obligations. Ordinary Least Squares (OLS) regression is used to estimate the relationship between effectiveness and its determinants; we verify time-series properties with Augmented Dickey–Fuller tests, assess Granger causality, and construct an ARIMA model to forecast dynamics under current policies.

The results indicate a positive and statistically significant association between contribution inflows and effectiveness, while the link between investment returns and effectiveness is weak or negative during the sample period, consistent with portfolio transition, limited diversification, and market volatility. Exchange-rate movements appear positively associated with effectiveness, reflecting the valuation of foreign assets, but also imply additional risk to beneficiaries' purchasing power. Forecasts suggest that, absent reforms, effectiveness is unlikely to rise materially by 2030.

We discuss implications for policy design and governance. Priority actions include: strengthening risk management and diversification; aligning asset duration with liabilities; instituting foreign exchange risk hedging where appropriate; publishing clearer performance and risk disclosures; and enhancing default portfolio design and member choice architecture. The findings inform the calibration of contribution policy and investment guidelines for Georgia's non-state pension system.

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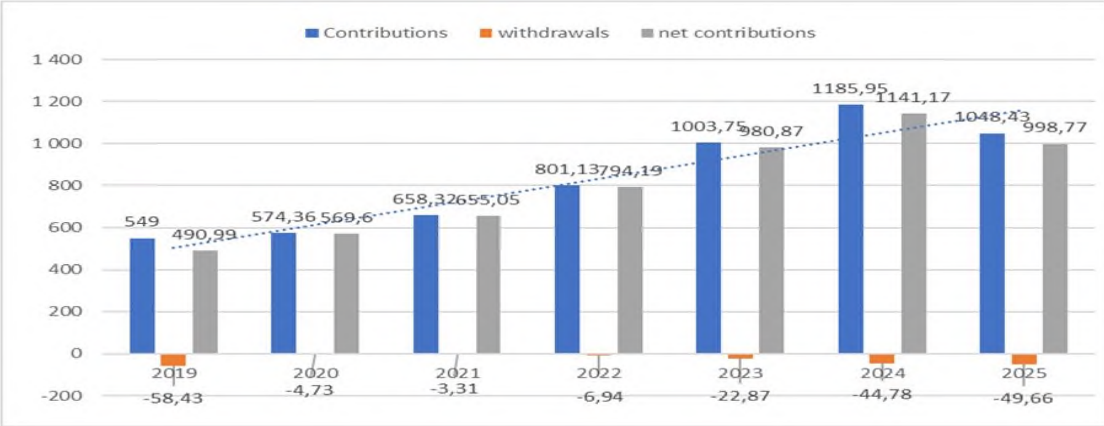
Introduction

The economic crisis of the 1990s disrupted demographic stability in Georgia. The working-age population declined while the ageing index rose and is expected to continue rising. In this context, interest in a private, accumulation-based pension system has grown. The current model is therefore discussed both in terms of its relevance and its risks and has generated strong interest among the economically active population. The relevance of this paper also follows from the high public attention to the newly introduced pension system.

Although Georgia's accumulation-based pension program is relatively young, assessing its efficiency and its role as a motivational tool is important. Comparative literature (e.g., Blank et al., 2016) notes that as state pension generosity falls, younger cohorts are increasingly pushed toward private retirement saving, exposing them to market-based products and commission-driven providers; thus, reform design should aim at fair conditions that protect both young and old. Georgia's model seeks to mitigate shortcomings of the pay-as-you-go solidarity approach by integrating features of an accumulation system.

According to official statistics, about 771,000 people initially joined the fund; within the first four months roughly 38,000 participants, around 5%, opted out, with higher exit rates among those over 40 (Law of Georgia "On Accumulative Pension," 2018; Pension Agency, 2019–2022). A core feature of the system is state co-financing: depending on employee income, the government contributes 0–2% to individual accounts. This creates direct budgetary outlays that have increased with asset growth since the program's launch. Pension Agency data (Fig. 1) show rising contributions and assets; total assets reached GEL 1,185.95 million by December 2024. While this expansion signals uptake and growing contributions, prudent investment strategy remains essential for long-run sustainability. The system's scale can also influence capital-market development and the advancement of domestic financial instruments.



Graf 1. Contributions, withdrawals, net contributions (mn)
Source: Georgian Pension Agency

Aim. This study aims to evaluate the effectiveness and fiscal sustainability of Georgia's accumulation-based pension scheme and to assess its role as a motivational instrument for retirement saving and as a catalyst for capital-market development. Specifically, the study:

Describes the demographic and policy context motivating the reform;

Documents participation dynamics (enrolment, opt-outs) since launch;

Quantifies state co-financing, contributions/withdrawals, and asset growth (2019–2024);

Assesses investment performance and risk with respect to long-term sustainability;

Estimates determinants of participation and net contributions using regression models;

Forecasts fund dynamics using ARIMA to gauge near-term trajectories;

1. Literature Review

Researchers (Barr, 2001) and (Hinz & Holzmann, 2005) widely discussed pension reforms across the world, focusing on how countries transition from state systems to mixed or fully private pension schemes. Their work reflects the historical development, advantages, and challenges of non-state pension systems.

In the study by (Morina & Grima,2022), the impact of pension fund assets on the economies of developing countries is analyzed. Based on econometric analysis, they determine that the development of a country's economy is directly linked to investments made by households in the real sector, in both production and pension funds. This, in turn, influences financial markets by increasing investments in pension funds and leads to sustainable development. There are several studies (Poterba et al. 2007) and (Schieber & Shoven, 1997) that focus on the role of private pension funds in ensuring pension security. They examine the benefits of such systems, which are directly linked to investing pension fund resources to achieve higher income. Their research also highlights the risks associated with investment and market volatility. Non-state pension funds play a significant role in maintaining the living standards of the pension-aged population in developing economies. This, in turn, contributes to the development of financial resource markets and impacts the country's economic growth.

Researchers (Palacios & Whitehouse, 2006) in their studies emphasized the development of private pension systems in developing countries as a means to reduce dependency on the state and improve the sustainability of the economy. (Khidesheli & Chikvadze, 2018) provide a clear overview of the reforms in Georgia's pension system. Their research reflects the processes, conflicts, and challenges of transitioning from the Soviet-era pension system in Georgia to a market-oriented, non-state pension system. The conclusion in their studies emphasizes the sharp issue of financial literacy among the population, which is linked to the negative attitude of the population towards the implementation of the non-state pension system. In this paper, the use of econometric methods in pension fund analysis is aimed at analyzing the state of the pension fund and predicting its future, in order to slightly neutralize the misconceptions about the pension system. (Greenwood-Nimo at al. 2013) and (Stock & Watson, 2007) widely used econometric tools such as time series analysis, data models, and cointegration techniques to study the performance of pension funds. Econometric models are crucial for assessing the impact of various factors on pension outcomes, such as economic growth, inflation, and demographic changes. (**Kumar et al.,** 2024) analyzes various investment approaches and their effects on pension adequacy in transitional economies. It employs econometric modeling. This is important for developing optimal investment strategies for pension funds to gradually improve Georgia’s complex pension situation.

Econometric models have been used to study factors such as investment returns, risks, and the efficiency of non-state pension fund management. For example, (Bikker & Meringa, 2024) use regression models to analyze how factors like fund size, asset allocation, and other variables affect the fund's performance, providing insights into the optimization of pension fund management.

In the study by (Zhao & Liu, 2024), it is found that the quality of fund governance is directly correlated with efficiency. Governance is a sensitive issue for Georgia, as the risks arising from ineffective management are significant and often lead to frequent changes in leadership.

The econometric analysis of non-state pension funds faces challenges related to data availability and accuracy, especially in developing markets. In Georgia, the lack of historical data and the reliability of pension fund operations are challenges for econometric analysis. Georgian researchers (Nutsubidze & Nutsubidze, 2017) analyze pension reform initiatives in Georgia, providing the population with information on how the country introduced private pension funds, their regulatory mechanisms, and the implementation process. (Urotadze, 2020) investigates how the Georgian government's policies have impacted the development of non-state pension funds. In Georgia, as in many other countries, the literature discusses that one of the biggest barriers to the success of non-state pension funds is low public awareness and lack of trust in the system, says in his study (Kbiltsetskhlashvili & Chezhia, 2020). (Smith & Lee, 2024) focus on the financial stability and sustainability of private pension funds during the post-2023 reforms. Using financial data, the study describes developments driven by demographic changes, particularly emphasizing economic instability, which may impact Georgia’s pension sector. (Nguyen & Tran, 2025) use risk analysis models and emphasize diversification.

One of the main obstacles to the implementation of the funded pension system in Georgia is the behavioral attitude of the population, which is reflected in the preference for using financial resources equivalent to today’s pension contributions in the present, rather than allocating them for long-term benefits. (Sanchez & Morales, 2025) combine behavioral economics with pension participation data. They highlight behavioral barriers, which are useful for interacting with the pension community in Georgia. Although many scholars from different countries conduct research on the challenges of non-state pension systems, little attention is paid to countries where such systems have a very short history. Therefore, based on the results achieved in a short period of time, the authors of this study outline a promising perspective through forecasting, which may contribute to the development of the pension system.

2. Research Methodology

2.1 Quantitative Models and Their Justification

The study employs multiple linear regression analysis to quantify the relationships between selected independent variables and the effectiveness of Georgia's non-state pension fund (the dependent variable, Y). This approach is grounded in econometric theory and is particularly suitable for analyzing time-series data with a limited observation window, as is the case here (monthly data over 24 months from 2022 to 2023). The choice of regression modeling is justified by its ability to isolate the marginal effects of individual factors while controlling for others, thereby enabling causal inference under clearly stated assumptions. Similar regression-based approaches have been applied to assess pension fund performance in various contexts, including emerging markets (e.g., Champagne et al., 2017; Yakubu et al., 2023; Danquah et al., 2025).

The general form of the regression model is specified as:

$Y = \beta 1\, x1 + \beta 2x2 + \beta 3x3 + \varepsilon$ where:

Y: Pension fund effectiveness (measured as a composite index of net asset growth adjusted for inflows and outflows),

X1: Size of pension contributions (monthly aggregate inflows),

X2: Investment returns (monthly portfolio yield, net of fees),

X3: Exchange rate (GEL/USD, end-of-month),

β 0: Intercept,

$\beta 1$, $\beta 2$, $\beta 3$: Slope coefficients representing marginal effects,

ε: Stochastic error term, assumed to be normally distributed with zero mean and constant variance.

This specification follows the framework established by Kumen and Wicht (2022), who demonstrated that pension fund performance is systematically influenced by contribution volume, portfolio returns and currency fluctuations in emerging market contexts. Comparable models have been utilized in studies of pension schemes in African emerging economies, such as Nigeria and Ghana, to evaluate risk-adjusted performance and market impacts (Yakubu et al., 2023).

2.2 Data Structure and Time-Series Considerations

The analysis uses monthly frequency panel data from the Pension Agency of Georgia (2019–2024), with the core estimation sample restricted to 2022–2023 (n = 24) to align with the operational launch of income-generating activities in the non-state pension scheme. Although the sample size is modest by macroeconomic standards, it is adequate for short-horizon financial modeling where high-frequency data reduce noise and improve signal detection (Wooldridge, 2019). Monthly observations capture seasonal patterns in contributions (e.g., year-end bonuses) and market volatility more effectively than annual aggregates.

To address potential non-stationarity - a common issue in financial time series - the study applies the Augmented Dickey-Fuller (ADF) test with trend and intercept terms (Dickey & Fuller, 1979). The test equation is:

$$\Delta y = \alpha + \beta_t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta i \ \Delta y_{t-1}$$

where:

$\Delta y\_t = y_t - y_{t-1}$— the change in the variable.

α— intercept (constant).

β_t— time trend (included if a trend is needed).

γy_{t-1}— effect of the lagged level on the current change.

$\sum_{i=1}^{p} \delta i \ \Delta y_{t-1}$— lagged differences capturing short-run effects.

u_t— random error term;

where rejection of the null hypothesis (gamma = 0) indicates stationarity. The lag length p is selected using the Schwarz Information Criterion (SIC) to balance parsimony and residual whiteness. Stationarity is a prerequisite for valid inference in ordinary least squares (OLS) regression, as non-stationary series can produce spurious correlations (Granger & Newbold, 1974).

2.3 Model Estimation and Diagnostic Framework

The model is estimated via Ordinary Least Squares (OLS) using the EViews software platform. OLS is preferred due to its best linear unbiased estimator (BLUE) properties under the Gauss-Markov assumptions:

Linearity in parameters,

Exogeneity of regressors (E[\varepsilon|X] = 0),

No perfect multicollinearity,

Homoscedasticity (Var(\varepsilon|X) = \sigma^2),

No autocorrelation in errors (Cov(\varepsilon_t, \varepsilon_s) = 0 \ \forall t \neq s).

While some assumptions (e.g., no autocorrelation) may be violated in short financial panels, post-estimation diagnostic tests are conducted to assess robustness. This estimation technique aligns with practices in pension fund research, where OLS and its variants are commonly employed to control for multiple factors influencing performance (Reuter & Zitzewitz, 2021).

To evaluate overall model fit, the coefficient of determination (R2) and adjusted R2 are reported. The adjusted R2 penalizes inclusion of irrelevant predictors:

$$R^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1},$$

where n is the sample size and k is the number of regressors.

An F-test is used to assess joint significance of all coefficients (excluding the intercept).

Individual coefficient significance is tested using t-statistics under the null hypothesis H0: = 0. The critical threshold is set at p ≤ 0.05, though marginal significance (p ≤ 0.10) is noted given the small sample.

Addressing Autocorrelation and Multicollinearity

Given the time-series nature of the data, autocorrelation in residuals is diagnosed using the Durbin-Watson (DW) statistic (Durbin & Watson, 1950):

$$\boldsymbol{DW} = \frac{\sum_{t=2}^{n} (e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}$$

Values near 2 indicate no first-order autocorrelation. If DW deviates significantly, Newey-West standard errors are considered in robustness checks to correct for heteroscedasticity and autocorrelation (HAC-consistent covariance matrix).

Multicollinearity is assessed via Variance Inflation Factors (VIF) (O'Brien, 2007):

$$\boldsymbol{VIF_j} = \frac{1}{1 - R_j^2}$$

where Rj^2 is from regressing Xj on all other predictors. VIF > 10 signals harmful collinearity, potentially inflating standard errors and reducing precision. Such diagnostics are essential in fund performance studies, where variables like returns and exchange rates may exhibit interdependence (Meligkotsidou et al., 2009).

Graphical and Predictive Validation

Beyond statistical tests, actual vs. fitted value plots are generated to visually inspect model alignment. Close tracking between observed and predicted Y supports functional form correctness. Residual plots are examined for randomness; systematic patterns would suggest misspecification (e.g., omitted nonlinearity or structural breaks).

2.4 Forecasting model ARIMA

For short-horizon policy projections, we estimate a univariate ARIMA (p,d,q) on Y_t. Orders (p,d,q) are selected by AIC/BIC, supported by ACF/PACF inspection. ARIMA is standard for monthly macro-financial series with limited length.

Validation (reported in Results).

Backtesting: expanding-window forecasts with a fixed hold-out to compute RMSE, MAE, MAPE.

Stability: rolling-window re-estimation; stability of coefficients and forecast errors.

Comparators: naïve random-walk and seasonal naïve baselines to demonstrate incremental value.

Pension contributions (X1): Primary driver of fund size; higher inflows mechanically increase assets under management (AUM) and enable economies of scale in administration.

Investment returns (X2): Core performance metric; reflects asset allocation efficiency and market exposure.

Exchange rate (X3): Critical for funds with foreign securities; GEL depreciation boosts local-currency value of USD-denominated assets.

These variables collectively capture flow, return, and valuation effects, forming a comprehensive yet parsimonious model. Variable selection draws from empirical studies in comparable settings, such as Tanzanian pension schemes where demographic and economic factors are modeled similarly (Masele & Magova, 2017).

Limitations of the Approach

The short data horizon (n = 24) limits degrees of freedom and increases sensitivity to outliers. The model assumes linear relationships; future extensions could incorporate interaction terms (e.g., X_2 \times X_3) or threshold effects. External validity is constrained to the Georgian context during 2022–2023.

This methodological framework ensures transparency, replicability, and scientific rigor, separating model specification and justification from empirical outcomes, which are presented in subsequent sections.

3. Empirical Analysis

A regression analysis was conducted using factors affecting the effectiveness of the pension fund. Since Georgia’s non-state pension fund began generating income in 2022—through diversification of its investment portfolio in both domestic and foreign securities, continuing in subsequent years—data for enhancing the reliability of the model were selected according to these factors, Table 1 presents the monthly investment income, exchange rate, and pension contributions for the years 2022 and 2023 (Data from Pension Agency, 2024).

Table 1
Statistical Data of the Factors Affecting the Pension Fund

	Y	x1	x2	x3
Conditional Period	Pension Fund Efficiency	Pension Contributions	Return on Investment (ROI)	Exchange rate
2022-I	0.73	1781000.00	282.00	3.05
2022-II	3.14	1838000.00	300.00	3.15
2022-IV	3.81	1908000.00	318.00	3.11
2022-V	3.32	1971000.00	329.00	3.05
2022-VI	3.41	2040000.00	329.00	2.95
2022-VI	2.99	2101000.00	336.00	2.92
2022-VII	3.14	2167000.00	346.00	2.76
2022-VIII	2.99	2232000.00	374.00	2.90
2022-IX	2.77	2294000.00	369.00	2.83
2022-X	3.26	2369000.00	397.00	2.77
2022-XI	2.70	2433000.00	418.00	2.71
2022-XII	3.16	2510000.00	438.00	2.70
2023-I	3.18	2591000.00	460.00	2.64
2023-II	2.70	2661000.00	477.00	2.62
2023-IV	3.01	2744000.00	492.00	2.56
2023-V	3.28	2827000.00	510.00	2.49
2023-VI	3.18	2917000.00	556.00	2.59
2023-VI	2.73	3003000.00	628.00	2.61
2023-VII	3.01	3086000.00	680.00	2.64
2023-VIII	3.91	3773000.00	305.60	2.62
2023-IX	4.24	3930000.00	247.05	2.67
2023-X	2.54	4030000.00	260.73	2.70
2023-XI	4.46	4210000.00	295.84	2.71
2023-XII	3.32	4350000.00	431.63	2.68

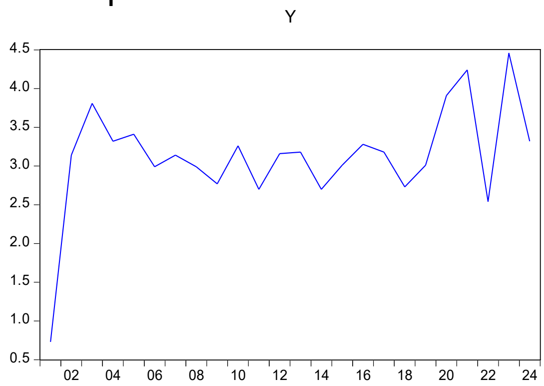
Source: Pension Agency of Georgia (2019-2023)

The relevance of the research findings depends on the stationarity of the data series. To determine this, the following hypotheses were formulated:

(H0): The time series is non-stationary;

(H1): The time series is stationary.

The graphical analysis of factors influencing the efficiency of the pension fund shows that some of the data in the time series do not meet the stationarity requirements, but the overall picture is consistent with the development of events.



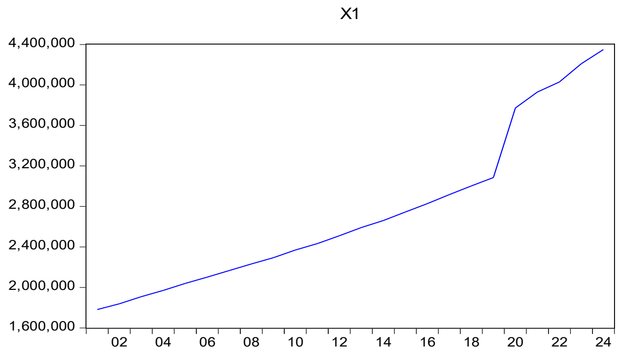
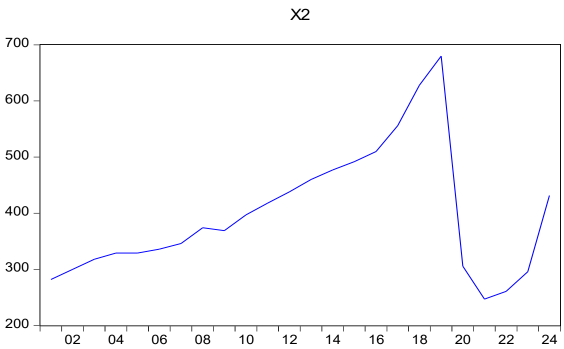


Figure 2. Time Series *Figure 3. Time Series*

Source: Figure 2 and Figure 3 Result of the Eviews computer program



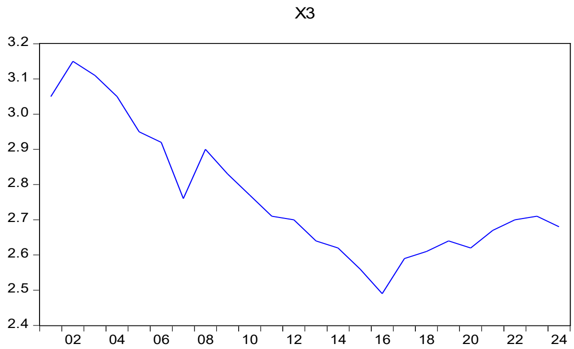


Figure 4. Time Series *gure 5. Time Series*

Source: Figure 4 and Figure 5 Result of the Eviews computer program

For example, pension contributions are increasing over time, and the sharp increase shown on the graph is related to the growth of pension program beneficiaries. Similarly, investment income is also increasing. Given that the time span of the data is small for econometric research, the data were taken over a 2-year period based on monthly data of active fund asset investments.

To strengthen the stationarity of the time series data, an extended Dickey-Fuller test was conducted.

Table 2
Results of the Extended Dickey-Fuller Test

Variable	t Crit	t Stat	P(probability)	F Stat	Akaike info criterion	Schwarz criterion	Durbin-Watson stat
y	2.99806	-3.6.9072	0.00000	4.77105	15.08373	16.07111	2.384639
X1	3.004861	-4.05406	0.00053	16.4354	26.54108	26.64026	2.008563
X2	3.004861	-3.79931	0.00094	14.4347	12.00165	12.10084	1.873354
X3	3.004861	-5.60511	0.00002	14.1733	24.83824	23.84639	1.948942

Source: Result of the Eviews computer program

The values of the t_{stat} and t_{critic} values confirm the stationarity of the variable series, and also the P (probability) is less than 0.05.

The information about the F_{stat} in the table is evidence that there are significant differences between the groups. The higher the F-stat value, the lower the P probability, which allows for the rejection of the null hypothesis with certainty.

The high Schwarz criterion for pension contributions indicates that other factors (demographic conditions, economic policy, employment levels) also influence pension contributions. To determine the autocorrelation between the variables, the Durbin-Watson criterion is provided in the table. From its values, it can be seen that the dependent variable (pension fund efficiency - Y) has a slight negative autocorrelation with the other variables. X1 - the pension contributions test shows zero autocorrelation, while the return on investment - X2 and the exchange rate - X3 variables indicate positive autocorrelation according to the Durbin-Watson criterion. Nevertheless, based on the selected factors, building a forecasting model is feasible. To strengthen the reliability, the variables of the regression model were examined graphically.

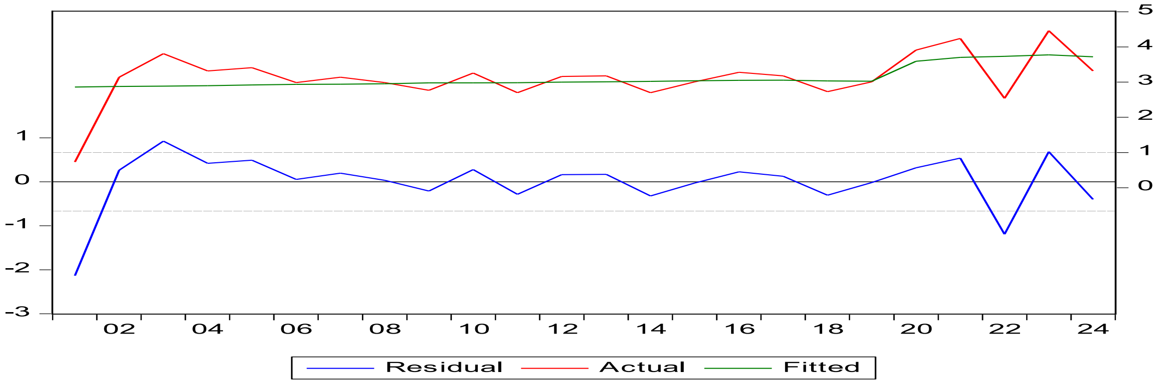


Figure 6. Actual and Predicted Graph of Regression Model Variables
Source: Pension Agency of Georgia (2022-2024)

The situation depicted in the graph shows that the regression model variables have been correctly selected, as the actual and predicted graphs are closely aligned with each other. To confirm the relevance of the research results, a regression analysis was conducted using the Eviews program, which demonstrated:

Table 3
Results of Regression Analysis

Dependent Variable: Y				
Method: Least Squares				
Date: 01/16/25 Time: 14:39				
Sample: 2001 2024				
Included observations: 24				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
X1	3.89E-07	2.63E-07	1.477416	0.0551
X2	-0.000800	0.001717	-0.465901	0.0346
X3	0.063706	1.359854	0.046848	0.0463
C	2.200959	4.828662	0.455811	0.0253
R-squared	0.699069	Mean dependent var		3.124167
Adjusted R-squared	0.778929	S.D. dependent var		0.694781
S.E. of regression	0.666799	Akaike info criterion		2.178355
Sum squared resid	8.892408	Schwarz criterion		2.374697
Log likelihood	-22.14025	Hannan-Quinn criter.		2.230444
F-statistic	1.656975	Durbin-Watson stat		1.799940
Prob(F-statistic)	0.008199			

Source: Result of the Eviews computer program

According to the regression analysis, the marginal effect of X1 (Pension Contributions) on Y is very small, and the t-statistic provides weak evidence against the null hypothesis. Also, the impact of X2 (Return on Investment) on Y is negative, but very small in magnitude, indicating that X2 is not a statistically significant predictor of Y when considering other terms in the model. However, it should be noted that the main criterion for rejecting the null hypothesis, Pprob ≤ 0.05, is met, and the short data period should also be taken into account.

The impact of X3 (Exchange rate) on Y is positive, but the standard error is large compared to the coefficient, which results in an insignificant t-statistic. Here, too, Pprob ≤ 0.05, which confirms that the null hypothesis is rejected and the time series data is considered stationary.

From the regression analysis, R2 = 0.699069, while the adjusted R2 = 0.778929, which is higher than the R-squared value. Typically, it is lower or close to R2. This finding implies a very strong fit, considering the number of predictive factors and the sample size.

The p-value for the F-test ≈ 0.0082 indicates that the model, as a whole, explains a significant amount of the variability in Y, but several individual coefficients have insignificant t-tests. This can occur in cases of multicollinearity or a small sample size (n = 24).

Ultimately, the impact of factors x1, x2, x3 on the effectiveness of the pension fund allows for forecasting.

4. Result

The values of the parameters obtained during the analysis emphasize the significance of the model, and the equation takes the following form:

$$Y = 3{,}89\text{E-}07\ X1 - 0{,}000800\ X2 + 0{,}063706\ X3 \qquad (2)$$

The data obtained from the regression analysis indicates that the following factors statistically influence significantly the efficiency of the pension fund: the volume of pension contributions (X1), the return on investment (X2), and the exchange rate (X3).

A direct relationship was found between pension contributions and the fund's efficiency — the X1 factor in the model is represented positively, and its p-value (< 0.01) indicates high statistical significance. This means that as the volume of contributions increases, the efficiency of the fund increases, reflecting the activity of the participants and the growth of resources in the fund.

A different result was obtained regarding the return on investment component (X2): its coefficient is negative, indicating that an increase in investment returns does not lead to an increase in the fund's efficiency. The cause of this paradox may be related to inadequate investment strategies, lack of asset diversification, or external shocks in the financial market. Therefore, in the fund management process, more attention should be paid to managing investment risks and allocating resources to efficient instruments.

A positive relationship is observed with the exchange rate (X3), which may be due to the fact that the increase in foreign investments is associated with a rise in fund revenues during depreciation of the Georgian Lari. However, this factor requires more caution, as exchange rate fluctuations increase systemic risk and affect the beneficiaries' portfolios. Therefore, foreign exchange risk hedging is necessary in the investment strategy.

To obtain relevant results for the prediction of pension fund efficiency, it is essential to establish the cause-effect relationship between variables using Granger's causality test.

Table 4
Results of Granger's Causality Test

Pairwise Granger Causality Tests			
Date: 01/16/25 Time: 15:02			
Sample: 2001 2024			
Lags: 4			
Null Hypothesis:	Obs	F Statistic	Prob
X1 does not Granger Cause Y	20	0.763910	0,0034
Y does not Granger Cause X1		0.27533	0,8878
X2 does not Granger Cause Y	20	9.94807	0,0012
Y does not Granger Cause X2		0.12997	09682
X3 does not Granger Cause Y	20	1.12759	0,3927
Y does not Granger Cause X3		0.60843	0.6650

Source: Result of the Eviews computer program

Granger's causality test further strengthens the rejection of the null hypothesis, as the probability that the values of X1 will not lead to changes in the values of y in the future is zero. It also rejects the evidence that X2 does not affect future values of y, with a probability of P = 0.0012. As for the opposite case of the national currency exchange rate, this is due to the instability of the exchange rate.

The analysis performed allows for the prediction of dependent and independent variables.

To determine which indicator will lead to changes in the pension fund's efficiency in the future, individual predictions were made for the key factors that have the greatest impact on it. The selected factors include pension contributions, return on investments, and the exchange rate. During the process, other factors were considered, but due to high correlation, these factors were excluded from the analysis (GDP, wage levels, pensions paid).

"The most reliable way to forecast the future is to try to understand the present." – (John Naisbitt, 1929). The stationarity of the time series for the factors analyzed in the pension fund is not strictly established; however, based on the significance of the variables and the results of the Dickey–

Fuller test, the series tends to lean toward stationarity. This allows for forecasting using the ARIMA model. The model parameters (p, d, q) were determined based on the correlogram of each variable.

5. Forecasting model

To select the forecasting model, the key coefficients of various model configurations based on different states of the variables were evaluated according to established criteria: a model is considered appropriate if it has a high adjusted R^2 and low values for the AIC and SBIC information criteria. The coefficients of the five selected models are presented in Table 5. Based on these conditions, the most appropriate model is ARIMA (1, 2, 1). To confirm the model's validity, it is necessary to assess indicators such as t_{statis}, prob and stand.error, which are presented in Table 6. Since the ARIMA (1, 2, 1) model was selected, its reliability is considered sufficient, based on the values of these indicators.

Table 5
ARIMA Forecast Model Data

	Arima (1 1 1)	Arima (1 2 1)	Arima (1 1 3)	Arima (1 1 4)	Arima (1 1 5)
Significant Coefficients	1	2	1	1	1
R2 adj	0.509	0.746	0.404	0.309	0.361
AIC	1.585	0.966	1818	1.966	1.888
SBIC	1.684	1.114	1.966	2.115	2.037

Table 6
Significant Coefficients of the ARIMA Model

	$t_{statistic}$	Prob	Stand.eror	$F_{statistic}$	$Prob_{Fstatistic}$
AR (1)	0.8009	0.0433	0.0812	31.854	0.000001
MA (1)	-3.7525	0.0014	0.4221		

Using the established data, forecasting was conducted.
Y - Forecasting graph of the dependent variable (pension fund efficiency).

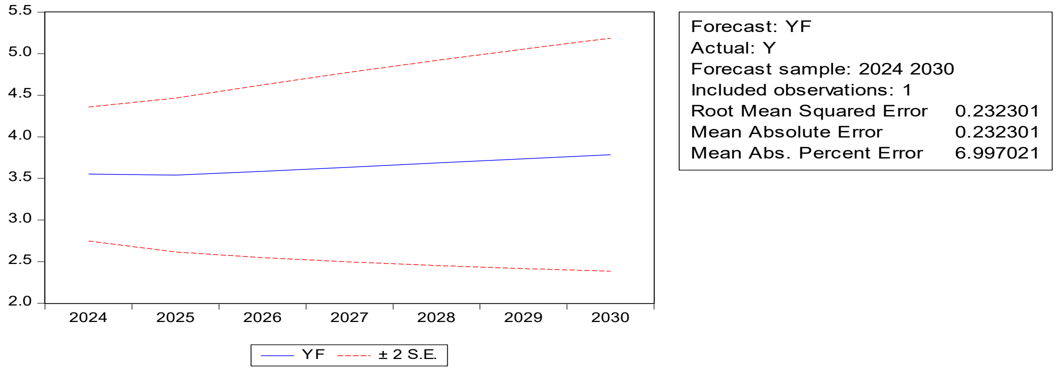


Figure 7. Forecasting results of the dependent variable Y

As the forecasting graph shows, the forecast horizon is set for the year 2030. The data used in the study is based on 2022-2023, but for the efficiency of the statistical program, the two years of data were divided into monthly information.

The more the dependent variable data deviates, the more accurate the forecasting results are. In this case, the RMSE (Root Mean Square Error) is 0.232301, which is considered acceptable because the data for Y fluctuates between 1.5 and 2. The graph also shows that the pension fund's investment efficiency does not significantly increase from 2024 to 2030. Although the evidence is based on a short period of data, the forecasting results should be taken into account to avoid shock events.

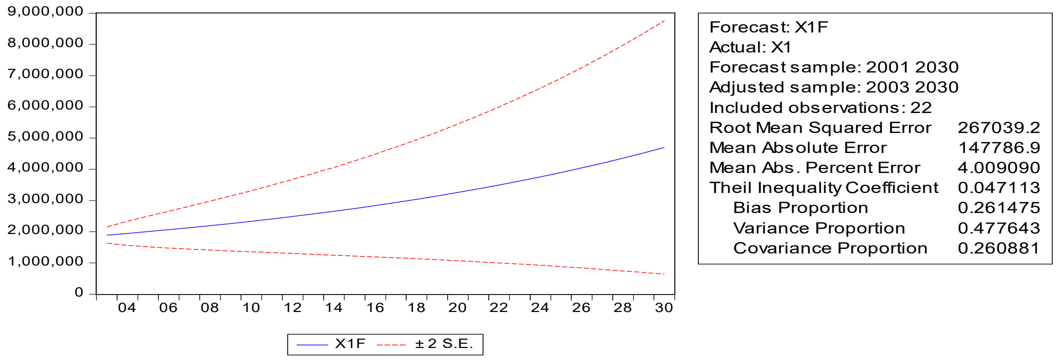


Figure 8. Forecasting Graph for X1 - Independent Variable (Pension Contributions)

By analyzing the results of the forecasting of independent variables, it is possible to outline the directions for managing future processes. Specifically, the volume of pension contributions in millions has been achieved, and the recorded value of RMSE indicates a high quality of forecasting accuracy. This conclusion is supported by the inequality coefficient -0.047113, which shows that the forecast is functioning because it changes from 0 to 1, and the lower the value, the more accurate the forecast is. This represents a preliminary view of the pension fund's expansion, which is clearly reflected in the forecast graph as well. Due to the limited data, the study cannot claim a high-quality forecast, and therefore, the determination coefficient (Rias Proportion-0.261475) suggests that 26.15% of the forecast could indicate excessive or insufficient results. However, based on other coefficients, decisions can be made in the direction of increasing pension income.

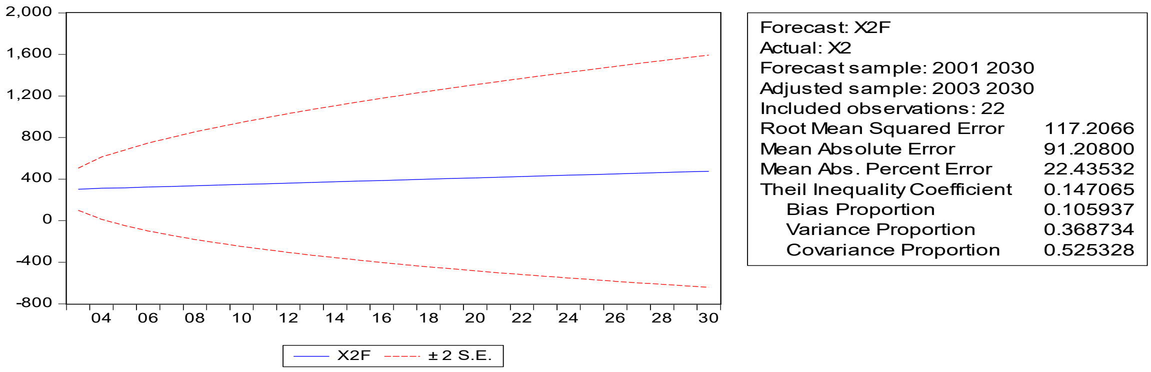


Figure 9. X2 - Independent variable (Investment Return) forecast graph

A key indicator for evaluating the efficiency of a pension fund is the return on investment, which answers many questions related to the fund’s operation, considering the interests of both the fund’s managers and its participants. Based on the graph, the return on investment will hardly change until 2030. This situation raises the question: if pension contributions are increasing, a small change in return should indicate unhealthy investment. The analysis of the forecast data is as follows:

The difference between the maximum and minimum values of the fund’s return is approximately 300 units, measured in thousands. Consequently, an RMSE of 11,720 indicates that, on average, the model’s forecasts deviate from the actual values by 117.20 units. Based on the size of the data, this is considered an acceptable norm of accuracy.

Among the factors affecting the efficiency of the pension fund, the choice of exchange rate is considered significant due to its impact on investment strategies and asset allocation, both in foreign trade and the investment of the pension fund in foreign securities. Specifically, a decline in the national currency exchange rate leads to higher investment income, while an increase in the exchange rate reduces returns. Exchange rate changes affect the financial security of the beneficiaries, particularly in the case of Georgia, where the non-state pension program still raises doubts among participants. This factor may lead to the fund's bankruptcy if participants withdraw their pension contributions prematurely.

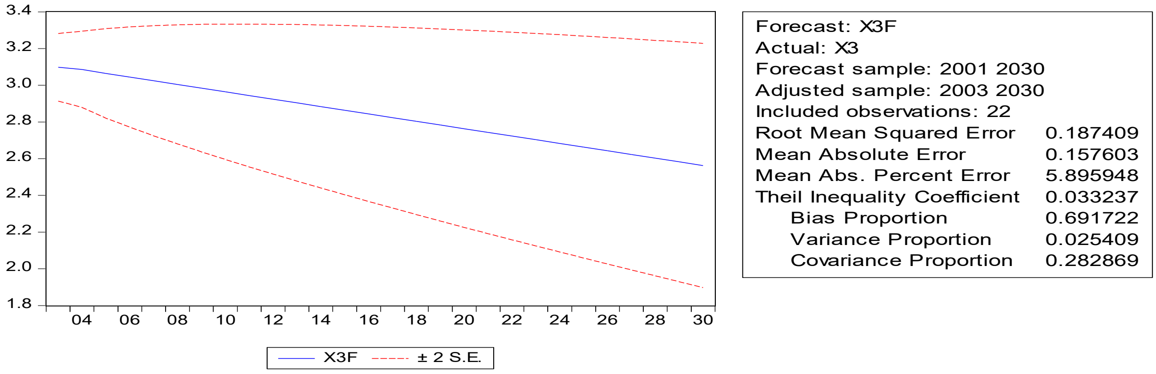


Figure 10. Forecasting Graph of the Independent Variable (Exchange Rate)

The forecasting diagram shows that by 2030, the exchange rate will decrease according to the aforementioned expectations. Specifically, the fall in the exchange rate will increase the value of the pension fund's foreign investments and, consequently, its income. However, it should also be noted that at the same time, this could lead to inflation in pensioners' incomes and a decrease in their purchasing power. It is important to emphasize that in managing the fund, attention should be paid to the results of forecasts when determining investment strategies in order to maintain the fund's stability and fulfill long-term obligations.

The forecast obtained using the ARIMA model determines that the fund's efficiency will not show a sharp increase by 2030, which highlights the need for fundamental reforms in the investment strategy and fund management.

6. *Discussion*

The conducted study, focused on analyzing the effectiveness of Georgia's non-state pension fund, was based on a two-year dataset (2022-2023) and examined three selected determining factors: the volume of pension contributions, investment income, and exchange rate fluctuations. Although the observation period was relatively short, the use of monthly *data made it possible to model and forecast the fund's performance. For the relevance of the research, the stationarity of the data series is essential to ensure that the results of both the regression model and the ARIMA forecasting model are statistically significant. To this end, the stationarity of the time series data was assessed using the Dickey-Fuller and Augmented Dickey-Fuller tests. All three independent variables (X1, X2, X3), as well as the dependent variable, demonstrated stationarity at the 5% significance level, making it possible to construct statistically reliable models. In transition economies such as Georgia, short time series may limit the reliability of the model and the accuracy of forecasts. Nevertheless, the transformation of time series data into stationary series confirms the reliability of the selected econometric model. The regression analysis revealed statistically significant coefficients and a positive relationship between pension contributions (X1) and the effectiveness of the pension fund. According to the obtained coefficient, even a small increase in contributions leads to an improvement in the fund's effectiveness. Specifically, increased contributions expand the volume of the fund's assets, improve its liquidity, and support broader diversification of investments. This result is consistent with the findings of Morina & Grima (2022) [8]and Palacios & Whitehouse (2006) [11], which emphasize the role of sustained contribution inflows in enhancing the fund's performance and stability. However, the analysis identified a negative coefficient for investment income (X2), indicating a weak or even inverse relationship between investment activity and the fund's effectiveness during the analyzed period. This correlation can be explained by several factors:*

•The diversification of the investment portfolio, introduced in 2022, may have involved errors in asset selection;

• Given that the financial market is characterized by high risks, the quality of management may have been insufficient, leading - despite nominal returns - to high volatility or a decline in asset value;

• A possible mismatch between short-term investment returns and the fund's long-term obligations may have reduced the accuracy of the fund's effectiveness indicator.

The findings call for a strategic reassessment of the Pension Agency's investment framework, including aligning asset duration with liability maturities, placing greater emphasis on risk-adjusted returns, and improving governance. Similar studies by Zhao and Liu (2024) [17] emphasize that strong governance positively correlates with pension fund performance, particularly in emerging markets. The positive coefficient for *the exchange rate (X3) suggests that depreciation of the Georgian Lari (GEL) may improve the fund's performance in the short term, likely due to increased income from foreign investments. However, this effect has a dual nature. While returns from foreign investments may rise with the weakening of the lari, beneficiaries' purchasing power could decline if payouts are not indexed to inflation. The Granger causality test further confirmed that pension contributions and investment income can be used to forecast the fund's effectiveness. Specifically, the null hypotheses stating that X1 and X2 do not Granger-cause Y, were both rejected with high statistical significance ($p < 0.005$), affirming the influence of these variables. Interestingly, the exchange rate (X3) did not exhibit Granger causality in either direction, likely due to its volatility and, at this stage, weak structural integration into pension fund operations. These findings indicate that although exchange rate fluctuations affect income, they do not systematically determine the fund's effectiveness through forecastable patterns. The ARIMA (1,2,1) model, selected based on the minimum AIC and SBIC criteria, does not project a significant improvement in fund effectiveness by 2030, pointing to the need for changes in investment strategy and strategic management. In the long term, greater transparency, stricter regulation and enhanced public awareness of the fund's operations are recommended. These measures will help increase public trust and raise participation rates in the fund.*

7. *Conclusions*

The following key findings can be drawn as conclusions: The growth of pension contributions has a significant impact on the efficiency of the non-state pension fund - active participation increases the fund's resources and mobilization ability.

The return on investment is not a sufficient indicator to determine efficiency and requires consideration of accompanying factors — such as the quality of governance and investment decisions. The impact of the exchange rate on the fund's income is significant, however, it contains systemic risks that require careful policy. The forecast does not show a sharp improvement in the fund's efficiency by 2030, which indicates the need for changes in investment and strategic management. In the long term, greater transparency, stricter regulation, and increased public awareness of the fund's activities are recommended, which will increase trust and the participation rate in the fund. The conducted econometric analysis shows that the existing system requires not only technical improvements but also a review of management and communication strategies — which will have a significant impact on the country's social stability and economic development. Taking the findings into account is important when planning pension policies, assessing the demographic situation, because as Basiglio and Oggero note, "Transparent information about pension systems facilitates individuals' retirement planning and decision-making. In addition, future pension policies should pay more attention to the factors causing gender inequality in pensions in order to reduce gender inequality in old age. The absence of appropriate policies aimed at reducing not only wages but also gender inequality in pensions may threaten the financial security of older people".

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CONTENTS

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COMMERCIAL SECRET MANAGEMENT IN TERMS OF SHADOW DIGITAL ECONOMY

Serghei Ohrimenco[1],
Dinara Orlova[2],
Valeriu Cernei[3]

Abstract: The article explores trade secret protection management in the context of the shadow digital economy (SDE). The study aims to identify threats to trade secrets associated with SDE and propose effective protection methods. It emphasizes that the economic value of trade secrets lies in their confidentiality, making them vulnerable to data leaks, cyberattacks, and other risks, particularly in the digital environment.

The authors examine the nature of trade secret in terms of SDE spread. The paper presents methods of trade secret protection, including documentary, economic-organizational, and information-analytical types. Special attention is paid to the combined use of patent protection and encryption techniques.

The study concludes with key findings, highlighting the need for enhanced cybersecurity, international cooperation to combat cyber espionage, and improved organizational measures. This work significantly contributes to the understanding of commercial secret management and offers valuable recommendations for businesses operating in the digital economy.

Keywords: shadow digital economy, entrepreneurial activity, commercial secret
JEL: L1, K0, C8, O17, M15.
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INTRODUCTION

Information is one of the most significant factors of production and management in modern economies. Modern entrepreneurship depends as much on the possession of information about potential customers, trade venues, and technical data as it does on personnel, machinery, and other capital goods. The reliance of industry on rapidly developing technologies amplifies the value of information. However, information is also different from other types of economic resources. It can be divided into an infinite amount of parts and, hence, is very complex. Importantly, it is often impossible to guarantee exclusive possession of this resource.

The importance of informational security for management is especially keen in today's world, where the information highway influences all aspects of social and economic life (Popova, P., Popov, V., Marinova, K., Petrova, M. and K. Shishmanov, 2024); (Petrova, M., Popova, P., Popov, V., Shishmanov, K. and K. Marinova, 2022); (Popova, P., Popov, V., Marinova, K. and M. Petrova, 2023). It becomes more and more important in terms of spreading the SDE. The EU Cybercrime Directive acknowledges the link between cyberattacks and the "loss or alteration of commercially significant confidential information". This means that protecting company data from leaks is a crucial aspect of cybersecurity (Porcedda, 2023). The efficiency of commercial production and other business activities increasingly depends on the ability of a business to use information correctly (Tairov & Petrova, 2022). However, information is only an asset that gives certain companies an edge over the competition if it is an excludable good that is not freely available to all. In an environment of escalating competition and in terms of SDE spread, business profits depend on the secrecy of information about entrepreneurial potential, special production technologies, and other data. For example, if a company's trade secrets were freely available, a competitor could develop a competing product using technological information or even market its own products to the company's clients using confidential client lists.

1. LITERATURE REVIEW

The critical literature review conducted by the authors allows the identification of several groups of important publications, which characterize the following directions:

1. One significant and highly important area of research is the analysis of the legislative framework regulating this field of knowledge. These include works such as (Lang, 2003; Mali, 2019; Maskus, 2012; Pasquale, 2011), and etc.

2. Publications that describe the trade secrets managing practices, its definition as an economic category, and the information constituting trade secrets. In particular (Margoni, 2025; Peskova et al., 2017; Png, 2012; Robertson, 2015; Schneider, 2017; West, 2015).

3. Sources analyzing the shadow digital economy (SDE) as a phenomenon in the development of the modern economy, its impact on actions related to categories such as cyber threats, information leaks (including trade secrets), and other activities constituting criminal activity. This group literature includes (Huateng, 2021; Ohrimenco et al., 2024; Ohrimenco et al., 2023).

2. RESEARCH METHODOLOGY

The methodology of this research is based on a systematic approach to investigate the management and protection of trade secrets in the context of the shadow digital economy (SDE). It includes the formation of a concept and statement of tasks, the collection of analytical information from available sources (scientific and monographic literature, reports of research companies specializing in the field of information security and its economic aspects), processing and analysis of the collected information. The research is structured around both qualitative and quantitative methodologies to explore the legal, organizational, and technological aspects of commercial secret protection in terms of SDE.

In legal and economic frameworks, different types of secrets serve to protect information that is crucial for national security, economic stability, and individual privacy. The most commonly recognized types of secrets include:

- State secrets – classified information related to national security, defense, foreign policy, and intelligence. Unauthorized disclosure may pose a threat to national sovereignty and public safety.
- Commercial (Trade) secrets – confidential business information that provides a company with a competitive advantage, such as production processes, client databases, and marketing strategies.

- Banking secrets – confidential information about a client's financial transactions, bank accounts, and credit history, protected by banking laws to ensure financial privacy.
- Medical secrets – Patient-related data, including diagnoses, treatments, and medical history, safeguarded to uphold doctor-patient confidentiality.
- Personal data protection (Privacy secrets) – information related to an individual's identity, including biometric data, personal correspondence, and communication records, protected under privacy laws.
- Judicial secrets – information related to ongoing investigations, court proceedings, and legal cases, ensuring fairness and integrity in the justice system.
- Professional secrets – confidential information held by professionals such as lawyers, journalists, and priests, ensuring trust and ethical responsibility in various fields.

Among these different types of secrets, commercial secrets hold a unique position due to their direct influence on economic competitiveness and innovation. Unlike state or judicial secrets, which are primarily associated with governance and legal order, or banking and medical secrets, which ensure personal confidentiality, trade secrets drive market success by protecting proprietary knowledge and business strategies. A very important issue relates to personal data protection, which is governed by the profound regulations established by EU legislation (Mali, 2019; Taal, 2021).

Trade secrets are critical for both large corporations and small businesses, as they help maintain market leadership, encourage research and development, and prevent unfair competition.

Given the rapid development of the digital economy (Ohrimenco et al., 2023; Ohrimenco et al., 2024) and increasing risks of industrial espionage and cyberattacks, the importance of commercial secrets continues to grow. Companies must implement comprehensive security measures, including legal safeguards, cybersecurity protocols and employee training, to ensure the confidentiality of their commercially valuable information.

The concept of a commercial (trade) secret was developed to limit access to commercially valuable information.

A "commercial secret" is a piece of confidentially kept information that gives an entrepreneur some economic advantage over competitors. This could range from information about specific production techniques or production formulas to marketing strategies or customer data. In (Lang, 2003), (Cottier, 2005; Robertson, 2015) explained the term as follows:

"Commercial (trade secret)" means information including but not limited to a formula, pattern, compilation, program, method, technique or process, or information contained or embodied in a product, device or mechanism which (1) is, or may be used in an entrepreneurship activity, (2) is not generally known in that trade or business, (3) has economic value from not being generally known and (4) is the subject of efforts that are reasonable under the circumstances to maintain its secrecy.

Trade secrets are an inherent component of a market economy. Allowing managers to protect their economic interests in valuable information facilitates the establishment of the economy model, which in turn stimulates innovative development and entrepreneurship (Sherwood, 2008). Regardless of the form of ownership, trade secrets help ensure the competitiveness of the entrepreneurs that possess them. This is particularly important in transitional economies, which are often dominated by small and medium enterprises that must protect their unique know-how in order to succeed in an open market (Värv et al., 2010).

Nonetheless, while protecting trade secrets can clearly advance a growing economy, doing so also creates a risk that certain shadow economic activities can be hidden under the umbrella of a trade secret. This is particularly the case in emerging economies where shadow economic activities have formed the backbone of the market for years. Un-entrenching these shadow activities even while creating safeguards for entrepreneurs is a daunting problem.

The term 'trade secrecy' points at two major characteristics of the phenomenon: secrecy and a connection with commercial activities (Pasquale, 2011). In other words, the commercial secret doctrine protects the owner's commercial interests by giving the owner the right to preserve the secret of commercially valuable information for unlimited time period. The term 'trade secret' itself has a slightly different meaning and impact based on the point of reference. From a scientific and practical point of view, a commercial secret could be either (i) *information* (i.e., knowledge or data) that is kept secret for commercial reasons or (ii) the concept of *the protection of* the secrecy of this information. From an economics perspective, a commercial secret is generally understood as 'a piece of confidentially kept information that gives an entrepreneur some economic advantage over competitors' (Peskova et al., 2017). In law, a 'trade secret' is a piece of information protected under the law because of measures that have been taken to protect it.

It is important to note that the authors encountered certain challenges in developing the methodology for the present study. First, the absence of official statistical data significantly complicated the information-gathering process. Second, data collection was carried out based on reports issued by leading specialized firms engaged in cybersecurity research. However, it should be emphasized that the statistical data presented were obtained through surveys and, therefore, do not fully capture the complexity of the phenomena under investigation. This represents yet another unresolved issue, which calls for the development of new methodological approaches and measurement techniques, as well as comprehensive sets of economic-mathematical models and advanced statistical data processing tools.

The methodological specificities of the subject area necessitated the construction of the following technological sequence of actions: identifying and verifying reliable information sources; analyzing and systematizing their content; selecting relevant data sets; and subsequently processing and analyzing the collected data.

3. DISCUSSION

It should be mentioned that the concept of commercial secret is more developed in law literature. For instance, the book “Trade Secrets: Law and Practice” by David Quinto and Stuart Singer (Quinto & Singer, 2009) serves as a practical guide to the protection of trade secrets, exploring their legal nature, methods of safeguarding, and strategic approaches in litigation. The authors provide a detailed analysis of what qualifies as a trade secret, examining key elements such as “independent economic value,” “not generally known to others,” and “reasonable efforts to maintain secrecy”.

The trade secrets are often understood in general as a part of intellectual property. In the book "Intellectual Property Culture: Strategies to Foster Successful Patent and Trade Secret Practices in Everyday Business" (Dobrusin & Krasnov, 2012), trade secrets are mentioned in the context of their role in corporate culture. Specifically, emphasis is placed on the importance of creating an intellectual property protection policy, which includes measures for safeguarding trade secrets, such as confidentiality agreements and restrictions on disclosing information. The protection of trade secrets is seen as a crucial element in maintaining competitive advantages, and the book highlights the need to incorporate this aspect into the daily operations of the organization. It also discusses the importance of developing corporate strategies to protect these assets, training employees, and implementing systems that allow for effective information management. SDE escalates the risks of commercial secret loss tremendously.

3.1 Shadow digital economy as a damaging environment for the commercial secret

As noted in (Keshelava, 2017), the topic of the digital economy is highly expansive and has become increasingly popular in recent years. "The excitement surrounding this field, on the one hand, and the lack of a unified conceptual framework, on the other, have led to the emergence of a vast number of seemingly incompatible opinions and, consequently, to the impossibility of a coherent dialogue."

Chinese expert Ma Huateng (Huateng, 2021) presents an important and valid argument that the digital economy represents a new stage of economic and social development following the agrarian and industrial economies. Public understanding of the fundamental principles of the digital economy has evolved gradually. Among the many definitions of the digital economy, one of the most representative is the one proposed within the framework of the *Development and Cooperation of the Digital Economy* project of the G20 and presented at the G20 Summit in Guangzhou in 2016. This initiative defines the digital economy as encompassing a range of economic activities in which digitized knowledge and information serve as key production factors, with modern information networks acting as their primary carrier, and the effective use of information and communication technologies (ICT) as the main driving force for increasing efficiency and optimizing economic structures.

The *shadow economy* refers to the informal sector of a national economy that is not accounted for in official statistics. It encompasses all forms of unregistered and unrecorded economic activities, including the following (Schneider, 2017):

- Transactions that are not prohibited by law (the so-called "gray market");
- Criminal activities explicitly outlawed by legislation (the "black market");
- Non-market activities, where goods and services are produced and consumed within households;
- Barter transactions involving goods and services that do not enter formal market exchanges.

Unlike the *classical* shadow economy, the SDE differs in its composition and structure, which will be briefly outlined below.

The *shadow digital economy* refers to a sector of economic relations encompassing all forms of production and business activities that, by their nature, content, characteristics and structure, contradict legal requirements and are conducted in defiance of state economic regulation and oversight.

At its core, the SDE consists of shadow entrepreneurial activities, which are characterized by:

- A hidden, latent (covert) nature, meaning that these activities are not registered with government authorities and are not reflected in official reports;
- Coverage of all phases of the economic reproduction process (production, distribution, exchange, and consumption);
- A parasitic nature of operations at all levels.

Three economic justifications have traditionally been advanced to support the protection of trade secrets.

First, the protection of trade secrets creates incentives to engage in the creative, inventive process and then to invest in the development of such innovations (West, 2015; Risch, 2011). For example, one study (Png, 2012) finds that technology and manufacturing firms in US states that have passed laws protecting trade secrets tend to spend more on research and development than firms in states without such protections. This is an important justification for protecting trade secrets in developing and transitional countries, where innovation is a key component of economic growth (Maskus, 2012).

Second, when the government or legal process provides protection of such confidential information, businesses themselves do not need to invest in ‘costly measures to prevent breach of security’ (Margoni, 2025). This allows the wealth of innovation to be spread more easily; for example, a firm will not be as likely to feel a need to hire only family members or to pay ridiculously high salaries to prevent employee movement (see (Risch, 2011)). Third, and somewhat contrary to what one might think, protecting trade secrets allows for their dissemination to more people—resulting in consequent learning and spillovers (Margoni, 2025). For example, if a firm knows that it can bring a lawsuit against former employees who divulge a trade secret, the firm might be more willing to share the secret with the employees. Even if the employees never share the specific information they have learned with others, the employees ‘may benefit from their enhanced stock of knowledge' and this more general learning may be passed on to future employers or business associates.

These merits have come under criticism, however. Importantly, trade secrecy laws are meant to protect information that the owner *is already taking steps to protect*; that is, the information is not already in the public domain. This is quite a different type of protection from that offered under patent or trademark laws. Those laws protect information or technology that is already public from being copied by others, and then only for a certain period of time. Upon expiration of the period, the patent or trademark must either be renewed, or the intellectual property becomes a public good (Risch, 2011). The real power of trade secrecy laws does not come from any power of the law itself to prevent disclosure; the owner of the commercial secret is responsible for taking steps to do this. The power of trade secrecy laws is creating an incentive for owners of trade secrets to undertake efforts to protect their own property by (i) not publicly disclosing the confidential data and (ii) alerting employees or business partners who have access to the confidential data of the possibility of becoming subject to legal damages if they misappropriate the information. In short, at most, trade secrecy merely allows firms to obtain damages when a secret is misappropriated and then used to the firm's detriment.

Furthermore, trade secrecy protection continues indefinitely; disclosure is *never* allowed. In fact, a court might only access a commercial secret once the secret has been divulged. In the usual course of business, trade secrets remain part of a business's proprietary information. This means that if a firm were so inclined, it might have an incentive to use the veil of trade secrecy as an excuse to hide all sorts of 'secret information' from the public domain. This is a special area of concern in countries such as Russia, where private firms engage in a wide range of 'quasi-public' activities - from constructing and operating telecommunications to overseeing electronic voting. Often, these operations require advanced technologies that would normally be subject to commercial secret protection. However, when public or at least quasi-public activities are involved, hiding information about these technologies from public scrutiny could create ethical risks.

These industries, such as those in the financial and energy world, are using commercial secret exemptions in open government laws to prevent the public from accessing basic information about the use of taxpayers' money. Governments are funding private-sector research, or even providing the facilities in which the research is conducted, and yet the public is denied access to the results of that research because of trade secrecy doctrine.

3.2 Methods of commercial secret protection

The management of corporate commercial secrets should include their protection from threats such as leaks and loss. One of the key vulnerabilities of commercial secrets is their low resistance to disclosure in the SDE. They can be subject to simple analysis, expert "data pumps," accidental leaks, hacker attacks, and more.

Provisionally, the process of commoditization divides information kept as a commercial secret into two types:

- Inalienable commercial secrets, which ensure the safety and competitive integrity of a company's internal activities.
- Seizable commercial secrets, which gain the status of a commodity.

In practice, inalienable commercial secrets are more widespread, while seizable commercial secrets are better protected by patents to facilitate commercialization.

A comprehensive threat assessment should precede the implementation of a data protection system. The sequence of actions includes: identifying threats (evaluating probability levels, systematization), estimating potential losses, selecting appropriate commercial secret protection methods, and assessing economic efficiency and associated costs. The following classification of threats to commercial secrets can be suggested (Table 1):

Table 1.
Classification of commercial secret threats in SDE

Criteria	Types
Threat Source	External (business espionage, competitors' illegal activities, theft of tangible and intangible assets); Internal (employee disclosures, low motivation, ineffective security services)
Severity of Consequences	High (business shutdown, severe financial losses); Medium (high recovery costs, moderate downtime); Low (potentially disruptive impact)
Probability of Occurrence	Unlikely; Real risk
Business Stage Affected	Initialization phase (most critical); Ongoing operations phase
Object of Infringement	Information, finances, tangible assets, goodwill, etc.
Infringement Subjects	Organized crime, unscrupulous competitors, employees, state authorities
Type of Damage	Direct financial loss, lost profit

Elaborated by authors using (Peskova et al., 2017)

As shown in Table 1, threats to commercial secrets are classified by their source (external and internal), severity of consequences, and likelihood of occurrence. These threats range from disruptive to business-critical, with even unlikely risks requiring attention. Enterprises are especially vulnerable during the initialization phase, and targets often include information, finances and goodwill. Offenders may range from criminal organizations to state authorities, leading to direct financial losses and reduced profits. This highlights the need for a comprehensive and proactive security strategy across all stages of business development.

4. RESULTS

Existing methods for protecting commercial secrets can be grouped based on functional criteria (Figure 1) (Peskova, 2017):

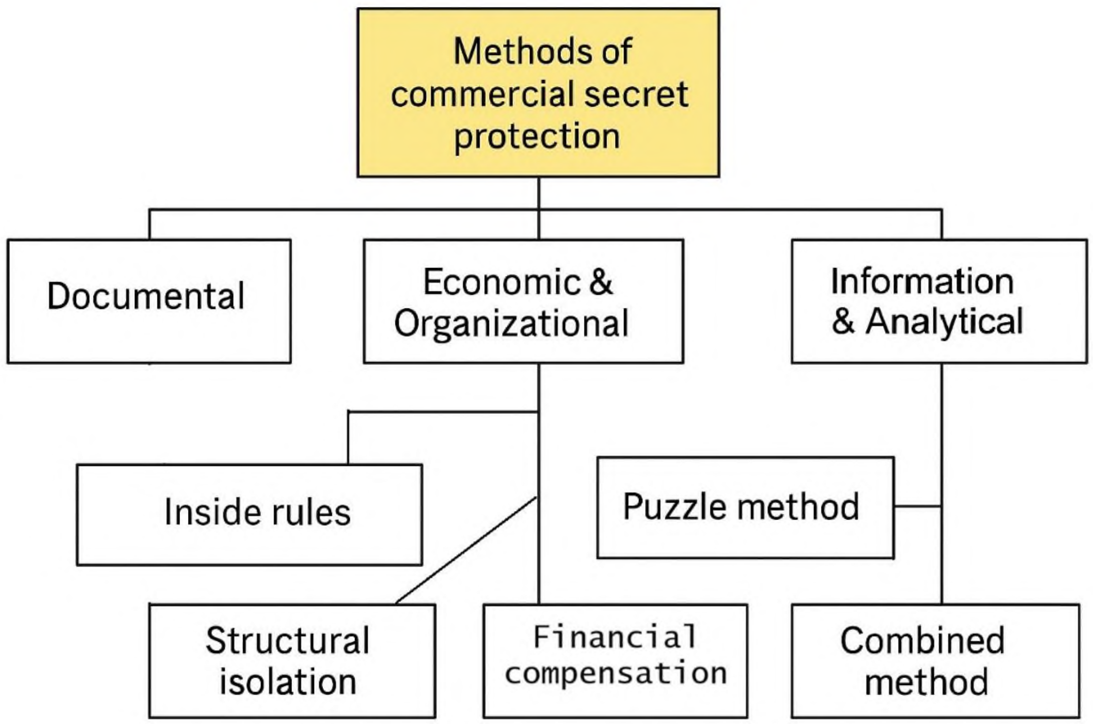

Figure 1. Methods of Commercial Secret Protection
Elaborated by authors using (Peskova, 2017)

4.1 Documental methods

These include regulations governing the circulation of internal documents containing confidential information. A "commercial secret" status is considered established once confidentiality measures are implemented:

- Creating a list of confidential information subject to protection.
- Regulating internal relations concerning the handling of commercial secrets via labour contracts and civil law agreements with business partners.
- Labeling physical media containing confidential data with a "Commercial Secret" designation, specifying the information owner.
- Utilizing technical measures for information protection.

While these methods provide clarity and enforceability, they mainly protect against unintentional leaks. However, judicial measures for enforcing confidentiality often indicate an irreversible loss of secret information.

4.2 Economic and organizational methods

Within the institutional economic framework, an employer can adopt three primary approaches to protect commercial secrets:

1. Establishing corporate policies for handling confidential information.
2. Implementing compensation schemes for employees involved in creating and maintaining commercial secrets.
3. Structuring internal operations to ensure departmental isolation.

a) Inside rules

Companies can enforce strict internal regulations requiring employees to adhere to confidentiality protocols. However, weak enforcement measures may render these rules ineffective. Examples include restricting inter-company communication with competitors and regulating access to sensitive documentation through password systems and physical security.

b) Financial compensation

Providing financial incentives for employees to maintain secrecy can be an effective strategy. This compensation should be proportionate to the potential losses incurred from information leaks. Moral incentives, such as team-building and corporate loyalty programs, can also strengthen adherence to confidentiality requirements.

c) Structural isolation

Restricting access to sensitive areas and information to only authorized personnel minimizes the risk of leaks. This approach enhances protection by narrowing the scope of vulnerability.

4.3 Information and analytical methods

a) Puzzle method

This method involves segmenting commercial secrets into discrete components, with different access conditions for each. Only key personnel have a complete understanding of the full picture, making unauthorized comprehension of the data challenging. However, improper implementation may lead to unintentional information disclosure.

b) Combining patents and Trade secrets

Patent protection can complement trade secret safeguards. Patents offer legal recognition, commercialization opportunities and legitimacy, while trade secrets protect non-patentable but commercially valuable information. The primary disadvantages of patents include limited duration, high costs, and potential exposure of technological innovations to competitors.

Porcedda (Porcedda, 2023) provides the following recommendations for strengthening protection of the commercial secret:

- Enhancing encryption mechanisms and access controls.
- Developing international cooperation in combating cyber espionage.
- Integrating law enforcement approaches to cybercrime and data protection.

Thus, cybersecurity plays a key role in protecting trade secrets.

None of the methods discussed above can provide absolute protection for a company’s trade secrets, especially in the face of an evolving shadow digital economy that continuously adapts to new technological advancements. However, recognizing these challenges opens avenues for further research into more sophisticated commercial secret management strategies, integrating emerging cybersecurity solutions, regulatory frameworks, and corporate governance practices. Future studies could explore the role of artificial intelligence in commercial secret protection, the impact of cross-border legal harmonization, and the ethical implications of balancing transparency with corporate confidentiality. We invite scholars, policymakers, and practitioners to contribute to this critical discourse, fostering a more resilient and adaptive approach to commercial secret security in an increasingly complex digital landscape.

5. EMPIRICAL DATA STUDY

The object of the study will be two sets of data: statistics on data leaks and expert assessments of damage costs. The data used in the study are based on empirical evidence and reflect real-world practical situations.

One of the most serious global issues in the field of IT is information leakage. We will analyze the available empirical data, which primarily characterize data leakage. Information leakage refers to the intentional or accidental disclosure of sensitive, confidential, or personal information (such as identification data, credit card details, usernames, passwords, etc.) and its exposure to third parties, which can have serious consequences for the affected organizations and individuals. Unauthorized transmission of information occurs via web channels, through programs and applications, and on physical media (storage devices, printer outputs, etc.).

The general picture of information leaks in the world from 2013 to 2024 is shown in the following figure (Infowatch, 2025).

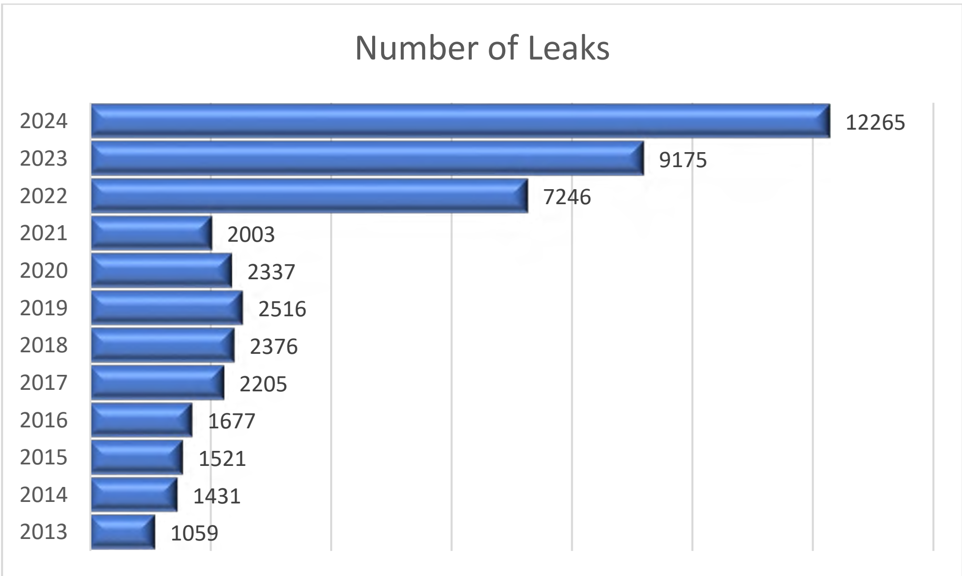

Figure 2. Number of information leaks in the world in 2013-2024
Source: (Infowatch, 2025)

An analysis of the table data leads to the principal conclusion that there has been a quantitative increase in data breaches worldwide over the observed period, with consistently high growth rates. Specifically, the number of breaches recorded in 2024 was 8.6 times higher than in 2013. The peak in the number of breaches occurred in 2023, amounting to 12,265 incidents.

It is important to note that the table reflects quantitative metrics only, encompassing breaches involving personal data records, trade secrets, and other sensitive information.

The primary channel of information leakage remains information and communication technologies, with internet-related breaches accounting for approximately 69% of all incidents. Paper-based document leaks constitute slightly more than 10%. Other significant channels include theft of removable media, loss or theft of equipment, and unauthorized manipulation of email systems.

The following table presents data on the dynamics of information breaches and the number of leaked personal data records in selected countries for the period 2023–2024.

Table 2.
Dynamics of information leaks and number of leaked personal data records

	Data Breaches			Leaked Data Records (bln)		
Country	2023	2024	Change (%)	2023	2024	Change (%)
USA	4823	3310	-31.37	8.18	553	-32.40
Russia	786	778	-1.02	1.2	1.58	31.67
India	542	498	-8.12	4.44	2.34	-47.30
United Kingdom	384	352	-8.33	3.12	0.18	-94.23
Canada	280	266	-5.00	0.2	0.35	75.00
France	313	232	-25.88	0.24	0.45	87.50
Indonesia	245	217	-11.43	4.99	1.29	-74.15
Germany	251	207	-17.53	0.08	0.09	12.50
Italy	220	177	-19.55	0.13	0.22	69.23
Brazil	258	176	-31.78	1.58	0.77	-51.27
Spain	223	166	-25.56	0.05	2.08	4060.00
Australia	204	159	-22.06	0.11	0.15	36.36
China	272	144	-47.06	12.75	3.93	-69.18
Thailand	112	105	-6.25	0.3	0.17	-43.33
Israel	142	95	-33.10	0.09	0.05	-44.44

Source: (Infowatch, 2025)

An analysis of the table's content indicates a decrease in the number of data breaches across all countries without exception. However, the same cannot be said for the number of leaked personal data records, which shows both increases and decreases. For example, Russia experienced a rise of +31.7%, Canada +75%, France +87.5%, Germany +12.5%, Italy +69.22%, Australia +36.4%, and the most significant increase was recorded in Spain, reaching +4060%.

We will expand the analysis of the data presented in the previous table by incorporating information on the compliance with the key provisions of the GDPR. Specifically, we will analyze the overall number of personal data breaches across European countries, drawing on an analytical report (Dlapiper, 2025).

Our focus will be on the data regarding the total number of personal data breaches (both registered and reported), as well as the overall amount of imposed fines.

Table 3.
Total number of violations GDPR (2018-2020)

	Country	Total number of data breaches from May 25, 2018, to January 1, 2020	Total number of data breaches from May 25, 2018, to December 31, 2019	Total number of data breaches from May 25, 2018, to December 31, 2019	Cost of data breaches in 2019 (million EUR)
1	Netherlands	40647	25247	15400	147.2
2	Germany	37636	25036	12600	31.12
3	United Kingdom	22181	11581	10600	17.79
4	Ireland	10516	6716	3800	132.52
5	Denmark	9806	6700	3100	115.43
6	Poland	7478	5278	2200	13.74
7	Sweden	7333	4383	2500	48.14
8	Finland	6535	3983	2500	71.11
9	France	3459	2153	1306	3.2
10	Norway	2844	2004	820	37.31
11	Italy	1886	1276	610	2.05
12	Slovenia	1845	1105	740	5.25
13	Spain	1698	1028	670	2.08
14	Austria	1644	1164	580	12.1
15	Belgium	1332	912	420	7.88
16	Hungary	749	479	270	4.87
17	Czech Republic	720	430	290	4.03
18	Romania	668	408	260	1.9
19	Luxembourg	545	345	200	56.97
20	Iceland	338	313	25	91.15
21	Malta	239	139	100	31
22	Greece	232	162	70	1.5
23	Lithuania	222	118	105	4.18
24	Estonia	132	82	121	9.74
25	Latvia	127	117	57	6.13
26	Cyprus	94	59	35	4.8
27	Liechtenstein	30	15	15	39.18
28	Bulgaria	-	-	-	-
29	Portugal	-	-	-	-
30	Slovakia	-	-	-	132.60

Source: Calculated from DLA Paper GDPR Data breach survey (DLA, 2025)

European countries are ranked in descending order based on the total number of personal data breaches. The Netherlands tops the list with a total of 40647 breaches while Liechtenstein ranks last with 30. A markedly different pattern emerges when considering fines: Germany leads with €54,574,525 followed by Finland (€51,100,000) Austria (€18,107,700) and Italy (€11,500,000). These data highlight the complexity of implementing the General Data Protection Regulation (GDPR).

In addition to the number of data leaks we can include information about the general GDPR fines as of January 2022: Luxembourg €746,299,400; Ireland €226,046,500; Italy €79,144,728; Germany €69,329,916; France €58,580,300; UK €45,350,000; Spain €29,003,000; Austria €24,853,650; Sweden €18,000,000; Netherlands €10,073,000 (Infowatch, 2025). The provided data indicates high amounts and significant fines.

Another important indicator is the channels through which information leakage occurs. Among them, the following should be highlighted: computer networks, removable storage devices, mobile devices, paper documents, email, as well as theft and loss of equipment. These channels are responsible for the leakage of personal data, payment information, trade secrets, and state secrets.

The very notable recent examples of valuable information leakage include two cases:

- Bleeping Computer: The largest data breach in the history of healthcare occurred in the spring of 2024 in the United States. Data on 190 million American patients was stolen as a result of a hack targeting a subsidiary of the insurance company UnitedHealth. The attack was carried out by hackers from the AlphV (BlackCat) group. The company paid $22 million in ransom in hopes of retrieving the data and preventing its dissemination. However, the attackers, having received the money, did not keep their promise. As of 9 months after the incident, in 2024, UnitedHealth estimated the damage at $2.45 billion.

- The Cyber Express: A data leak involving the Italian government leadership and other prominent individuals occurred in Italy in the same 2024 year. It resulted from a conspiracy between hackers and police officers. The accused gained access to confidential information by bribing law enforcement personnel and installing spyware. The obtained data was used for blackmail, allowing the perpetrators to extort over €3 million in total.

The second group of empirical data represents the information of individual experts regarding losses from a wide range of cyber threats, accumulated both by independent researchers and leading computer companies (Vergara & Cakir, 2025), (Miranda et al., 2024), (Correia, 2022).

For example, Steve Morgan, the editor-in-chief of Cybercrime Magazine (Morgan, 2024), presented to experts a material describing the top five most significant facts about cybersecurity:

1. According to forecasts, by 2025, the global damage from cybercrime will amount to $105 trillion per year. If this damage were evaluated as a country, cybercrime, which is predicted to cause damage totaling $6 trillion worldwide in 2021, would be the third-largest economy in the world after the USA and China. Global costs of cybercrime are expected to grow by 15 percent annually over the next five years, reaching $105 trillion per year by 2025, compared to $3 trillion in 2015. Innovations and investments in cybercrime significantly outweigh the damage caused by natural disasters and will be more profitable than global trade in all major illegal products combined (including drugs, pornography, arms trafficking, etc.).

2. Global spending on cybersecurity from 2021 to 2025 will exceed $175 trillion. In comparison, the global cybersecurity market was valued at only $35 billion in 2004, and now it is one of the largest and fastest-growing sectors of the information economy. It is expected that the cybersecurity market will grow by 15 percent annually from 2021 to 2025.

3. According to forecasts, by 2031, the global damage from ransomware will exceed $265 billion. Global losses from ransomware damage are expected to reach $20 billion per year in 2021, compared to $325 million in 2015, which is 57 times higher. In ten years, the costs will exceed $265 billion. The average ransom amount, according to estimates [56], is a significant sum — $220,298 ($220,298 compared to $154,108, which is 43% higher than in Q4 2020). The median ransom amount is $78,398 ($78,398 compared to $49,450, or 59% more than in Q4 2020), which signals a quantitative and qualitative increase in new attacks.

4. By 2025, the total volume of global data storage is expected to exceed 200 zettabytes. This includes data stored in private and public IT infrastructures, private and public cloud data centers, and personal computing devices.

CONCLUSION

This article explores the key aspects of trade secret protection in the context of the SDE. The economic value of trade secrets lies in their confidentiality, making them vulnerable to data leaks, cyberattacks and other risks, especially in the digital environment. In the context of SDE, the risks of leakage become particularly relevant, as many business processes are shifting to informal, state-unregulated sectors, making control and protection more challenging.

The shadow digital economy is characterized by its covert nature and violation of legal frameworks, which increases risks for the security of trade secrets. Modern protection methods - ranging from documentary to information-analytical approaches, including patenting and encryption - offer effective tools for minimizing these threats. However, no method can guarantee absolute protection, especially given the continuous evolution of technologies and methods of cyber espionage.

The conducted study has made it possible to identify a number of significant aspects characterizing the current state of the institution of trade secrecy in the context of the digital shadow economy. However, the presented material does not claim to provide an exhaustive account of all the issues and challenges in this area. On the contrary, the results of the analysis highlight the complexity and multifaceted nature of the phenomenon under consideration. In this regard, the development of a new research methodology, as well as the improvement of approaches to the collection, systematization and analytical processing of relevant empirical data, appears to be necessary.

We consider it necessary to highlight another issue that remains unresolved to this day. Specifically, this concerns the development and implementation of ethical practices in the field of modern cybersecurity. In certain disciplines - such as medicine, engineering, and law - comprehensive codes of ethics and conduct have been established and are widely observed. Unfortunately, a comparable ethical framework is lacking in the cybersecurity domain. This absence should be viewed as a potential threat to individual, societal and state security, as well as a significant factor hindering trade and constraining investment. Addressing this issue remains an urgent and unmet imperative.

Thus, protecting trade secrets in the SDE requires a comprehensive approach that includes not only technical security measures but also legislative initiatives and international cooperation to combat cyber espionage. A key challenge for the future is the development of more advanced trade secret management strategies that integrate new solutions in cybersecurity and regulatory frameworks. Further research into the role of artificial intelligence and the harmonization of international legal norms in the context of trade secret protection will open new opportunities for enhancing business resilience in the digital age.

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CONTENTS

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RESEARCH OF SOME FACTORS AFFECTING DISCRIMINATION IN THE ORGANIZATION AND ITS IMPACT ON CONFLICT SITUATIONS

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Abstract: The study of discrimination in organizations remains significant. Discrimination leads to various processes linked to conflict situations. In many instances, discrimination stems from the leader's management style. It is also important to consider that any form of discrimination in an organization that causes conflict situations affects the psycho-emotional state of employees, negatively affects their health and personal lives, and ultimately disrupts the work-life balance. The mentioned condition directly affects employee performance. We identified the factors that created significant problems in the relationship between leaders and followers in Georgian organizations. The study was conducted in the Ministry of Economy and its subordinate structural units. Data were collected through an anonymous questionnaire. The developed questionnaire included 64 questions. 720 respondents participated in the survey. Hypotheses were developed and tested using the SPSS statistical package during the research process. Based on the data analysis, important recommendations were developed.

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Introduction

The study of discrimination in organizations remains significant. Discrimination leads to various processes linked to conflict situations. In many instances, discrimination stems from the leader's management style (Garzon et al., 2024). It is important to study the decisions made by the manager while performing his/her work (Kalabina et al., 2021; Mushkudiani et al., 2020; Paresashvili et al., 2021). What is the manager's attitude towards the subordinate who uses non-working? What is the manager's attitude towards the subordinate who uses non-working hours to complete the task? Are they compensated differently? Does the subordinate have opportunities for career advancement? Recently, researchers have been studying job satisfaction intensively, specifically in relation to fair compensation. They discuss professional barriers and limited access to labour resources while highlighting the systemic inequality in career advancement opportunities (Wang & Chang, 2025). Research into the relationship between employee satisfaction and career position has also become relevant, as career support has a major impact on employee retention (Ferdiana et al., 2023). Workplace conflicts are prevalent and are often caused by personal differences and managerial relationships. Mentioned conflicts can impact teams and departments, resulting in dismissal, resignations, and absenteeism (Indeed, 2024).

Recognizing that individuals' perceptions of workplace discrimination are shaped by societal views on discrimination and the behavioral norms linked to organizational culture is crucial. Attitudes and beliefs regarding organizational discrimination often reflect perceptions of organizational culture (Kartolo & Kwantes, 2019).

Organizational culture significantly contributes to a stressful environment when attitudes towards discrimination are not assessed accordingly (Gryshova et al., 2019; Kazbekova et al., 2024; Mussapirov et al., 2019; Seitzhanov et al., 2020).

Subordinates perceive conflicts arising from discrimination as acute. The situation is further exacerbated when the supervisor makes an unfair and biased decision in the conflict resolution process (Kharadze & Gulua, 2023). Often, subordinates believe that rights and responsibilities do not correspond to their position and remuneration and consider themselves victims of discrimination. On the other hand, some employees perform less work but receive disproportionately high remuneration. Some problems related to employee discrimination represent a significant challenge, and it is necessary not to allow it to become a social norm. Dealing with this challenge is the prerogative of the leader. It is important for the leader to clearly define rights and responsibilities, study the interests of followers, and make fair decisions (Popova et al., 2023; Popova et al., 2024; Petrova et al., 2018; Petrova et al., 2020; Petrova et al., 2025).

It is also important to consider that any form of discrimination in an organization that causes conflict situations affects the psycho-emotional state of employees, negatively affects their health and personal lives, and ultimately disrupts the work-life balance (Clausen et al., 2022). The mentioned condition directly affects employee performance (McCarthy, 2024; Bernabei & Kabat, 2024). In addition, the constant development and rapidly changing work environment increase the likelihood of psychosocial risks, which poses significant challenges for organizations to protect the psychological well-being of employees. (Amoadu et al., 2024).

The mentioned issues are the subject of our research. We identified the factors that created significant problems in the relationship between leaders and followers in Georgian organizations. The study was conducted in the Ministry of Economy and its subordinate structural units. Data were collected through an anonymous questionnaire. The developed questionnaire included 64 questions. 720 respondents participated in the survey.

Literature Review

The paper “Diversity and Discrimination in Research Organizations” presents a social understanding of discrimination. It also addresses the risk factors that shape discrimination and suggests ways to manage diversity and discrimination (Muller et al., 2022). Studies on various issues often reveal signs of discrimination, and similar studies were related to career management issues (Sulkhanishvil & Kharadze, 2025). The work of scientists is dedicated directly to the discrimination issues in the public sector (Dugladze et al., 2024). Our country is multinational, and we encounter people of different religious beliefs in the workplace. Considering the above, it is interesting to study the facts of discrimination on religious grounds. (Linando, 2023) The study found an increase in workplace discrimination against Muslims. Therefore, seeing what pattern would emerge for our country was interesting. The study offers an interesting perspective on age discrimination (Kunze & Meulenaere, 2024). In this regard, studying discrimination in countries, where respect for older individuals is a core value, is also interesting. In addition, it is noteworthy that the victim of discrimination is not an older person but often a younger employee. The study focuses on interesting topics such as the workplace and internal tension caused by a toxic organizational culture (Deselnicu et al., 2024). When considering discrimination, it is difficult to avoid mentioning nepotism. Undoubtedly, nepotism is frequently condemned; nevertheless, little is known about what employees see in nepotism and why it is regarded as problematic.

In some cases, although the appointment of family members in politics and business is often frowned upon, it is considered a common occurrence (Burhan et al., 2020). An interesting international study was devoted to the preventive role of psychological support among victims of discrimination. (Eltaybani, et al., 2024). It is important to analyze these studies and adapt them to our environment. The employee evaluation system often serves as the basis for discrimination (Tarigan et al., 2025). Although all forms of discrimination require in-depth analysis, our study will focus on the supervisor's behaviour.

METHODOLOGY

The research was conducted in the Ministry of Economy and its subordinate structural units. 760 respondents participated in the survey. The obtained data were processed using the statistical software package SPSS.

Data filtering was utilized to select a certain data group for hypothesis testing. To establish the dependence and presence of a statistical relationship between variables, we employed consumer tables with the included chi-square test and General Linear Model /Univariate Analysis of Variance. Furthermore, Multivariate analysis of variance was used in the data processing process. Various graphical analysis tools were used to represent the data.

Results and Discussion

We formulated the following hypotheses to study the impact of discrimination on conflict situations and stressful environments in an organization.

Hypothesis 1: The following factors influence the frequency and causes of conflict situations arising based on discrimination: the leader's fairness in the conflict resolution process, clearly defining rights and obligations at work, and the labor discrimination acceptance as a social norm in Georgia.

The mentioned hypothesis tests the influence of three factors on two dependent variables. Therefore, we used Multivariate Analysis of Variance (MANOVA) to assess the first hypothesis.

The main table obtained by the multivariate procedure (see Table 1) presents the tests of between-subjects effects. At the end of the table, we obtained an R Squared of .358, which indicates that 35.8% of the variance in the dependent variables is explained by the influence of the independent variables. Table 1 demonstrates which factors affect the dependent variables and which do not.

Table 1
Tests of Between-Subjects Effects

Source	Dependent Variable	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	Q58 How often do conflict situations arise based on discrimination?	279.728[a]	34	8.227	7.416	.000	.358
	Q59 What is the cause of conflict in your organization?	142.192[b]	34	4.182	2.631	.000	.165
Intercept	Q58 How often do conflict situations arise based on discrimination?	1631.967	1	1631.967	1471.009	.000	.765
	Q59 What is the cause of conflict in your organization?	1729.947	1	1729.947	1088.529	.000	.706
Q60	Q58 How often do conflict situations arise based on discrimination?	77.852	2	38.926	35.087	**.000**	.134
	Q59 What is the cause of conflict in your organization?	15.468	2	7.734	4.866	**.008**	.021

Q21	Q58 How often do conflict situations arise based on discrimination?	36.015	2	18.007	16.231	**.000**	.067
	Q59 What is the cause of conflict in your organization?	24.418	2	12.209	7.682	**.001**	.033
q63_4	Q58 How often do conflict situations arise based on discrimination?	32.428	4	8.107	7.307	**.000**	.061
	Q59 What is the cause of conflict in your organization?	14.115	4	3.529	2.220	.066	.019
Q60 * Q21	Q58 How often do conflict situations arise based on discrimination?	27.772	4	6.943	6.258	**.000**	.052
	Q59 What is the cause of conflict in your organization?	16.188	4	4.047	2.546	**.039**	.022
Q60 * q63_4	Q58 How often do conflict situations arise based on discrimination?	16.364	6	2.727	2.458	**.024**	.032
	Q59 What is the cause of conflict in your organization?	12.554	6	2.092	1.317	.248	.017
Q60 * Q21 * q63_4	Q58 How often do conflict situations arise based on discrimination?	18.971	9	2.108	1.900	**.050**	.036
	Q59 What is the cause of conflict in your organization?	14.347	9	1.594	1.003	.437	.020
Error	Q58 How often do conflict situations arise based on discrimination?	502.567	453	1.109			
	Q59 What is the cause of conflict in your organization?	719.931	453	1.589			
Total	Q58 How often do conflict situations arise based on discrimination?	5872.000	488				
	Q59 What is the cause of conflict in your organization?	6414.000	488				
Corrected Total	Q58 How often do conflict situations arise based on discrimination?	782.295	487				
	Q59 What is the cause of conflict in your organization?	862.123	487				

a. R Squared = .358 (Adjusted R Squared = .309)
b. R Squared = .165 (Adjusted R Squared = .102)
Source: Author's findings

Assessment of the influence of factors according to Table 1:

The data in Table 1 demonstrate that R Squared for the first dependent variable is 0.358 while for the second is 0.165. Accordingly, the combined effect of the three factors on the dependent variable - Q58 "How often do conflict situations arise based on discrimination?" - is quite strong. 36% of this variable is explained by the influence of the combination of all three factors, on the second dependent variable - Q59 "What is the cause of conflict in your organization?" - the effect of the influence of all three factors is 17%.

Factor - Q60 "Is the manager's attitude fair in the process of resolving the conflict in cases of discrimination?" - impacts both dependent variables at the 0.01 level of statistical significance. However, it has a greater impact on the variable - Q58 "How often do conflict situations arise based on discrimination?" - F=35.087, the effect size of the influence is high, Eta Squared=0.134. And on the variable of causes (Q59) F=4.866, the effect size of the influence is small, Eta Squared=0.021.

The factor "Q21 Do you have clearly defined rights and obligations at work?" also impacts both dependent variables at the 0.01 level of statistical significance. However, it has a greater impact on the variable - Q58, "How often do conflict situations arise based on discrimination?" F= 16.231, the effect size of the statistical influence is medium, Eta Squared=0.067. And on the variable of causes (Q59) F=7.682, the effect size of the influence is small, Eta Squared=0.033.

Factor Q63_4, “In Georgia, labour discrimination is increasingly perceived as a social norm,” - affects the variable - Q58, "How often do conflict situations arise based on discrimination?" - at the 0.01 level of statistical significance, F= 7.307, the effect size of the influence is medium, Eta Squared=0.061. And on the variable of causes (Q59), factor q63_4 has a very weak effect, $0.05<P<0.1$, F=2.220, and the effect size of the influence is also small, equal to 0.019.

According to the results, we can conclude that the first hypothesis is confirmed under the influence of individual factors.

Both variables are influenced by a combination of factors: Q60 "Is the manager's attitude fair in the process of resolving the conflict in cases of discrimination?" and Q21 "Do you have clearly defined rights and obligations at work?". On the variable Q58, "How often do conflict situations arise based on discrimination?", with a statistical significance level of 0.01, F=6.258, the effect size of the influence is close to the average, Eta Squared=0.052, and on the variable of causes (Q59) at the 0.05 level (P=0.039) F=2.546, the effect size of the influence is small =0.022. It should be noted that all three factors together influence only the first dependent variable: Q58 "How often do conflict situations arise based on discrimination?" at the 0.05 level of statistical significance (P=0.039), F= 1.900.

Hypothesis 2: The stressful situation in the organization affects the employee's psycho-emotional state, health, personal life, and labour productivity.

We utilized a consumer table and the chi-square test to evaluate the second hypothesis. In the mentioned hypothesis, the stressful situation in the organization is a factor, while the other variables are dependent. The study of the mentioned topic was based on the data of those respondents selected as a result of filtering who did not deny the existence of a stressful environment.

By implementing the procedure, we obtained the frequency distributions of the dependent variables in relation to the stressful environment variable (see Table 2) and a table of the chi-square test (see Table 3).

Table 2
Frequency distributions of dependent variables in relation to the stressful environment variable

Dependent variables	Categories	Q54 Do you think there is a stressful environment at your workplace? (Row N %)		
		Yes	Partially	I don't have an answer
Q55 Does discrimination in the workplace cause a deterioration in your psycho-emotional state?	Never	23.8%	71.4%	4.8%
	Sometimes	29.8%	66.0%	4.3%
	Often	40.8%	59.2%	0%
	Always	37.5%	58.9%	3.6%
	I don't have an answer.	14.5%	32.9%	52.6%
Q61 How does discrimination in an organization affect your health?	Significantly negative	51.0%	44.9%	4.1%
	Negative	34.7%	63.4%	2%
	Does not affect	21.1%	73.7%	5.3%
	I don't have an answer	18.8%	56.3%	25%
	There is no discrimination in the organization	14.0%	44.2%	41.9%
Q62 How does discrimination in the organization affect your personal life?	Significantly negative	45.7%	50%	4.3%
	Negative	48%	49.3%	2.7%
	Positive	0%	75%	25%
	Does not affect	23.4%	70.3%	6.3%
	I don't have an answer	9.9%	45.7%	44.4%
Q64 Does discrimination in your organization affect your productivity?	Yes	37.7%	58.5%	3.8%
	No	26.7%	53.3%	20%

Source: Author's findings

Table 3
Chi-square test results

		Q54 Do you think there is a stressful environment at your workplace?
Q55 Does discrimination in the workplace cause a deterioration in your psycho-emotional state?	Chi-square	179.599
	Df	8
	Sig.	**.000**
Q61 How does discrimination in an organization affect your health?	Chi-square	130.766
	Df	8
	Sig.	**.000**
Q62 How does discrimination in the organization affect your personal life?	Chi-square	149.373
	df	8
	Sig.	**.000**
Q64 Does discrimination in your organization affect your productivity?	Chi-square	11.597
	df	2
	Sig.	**.003**

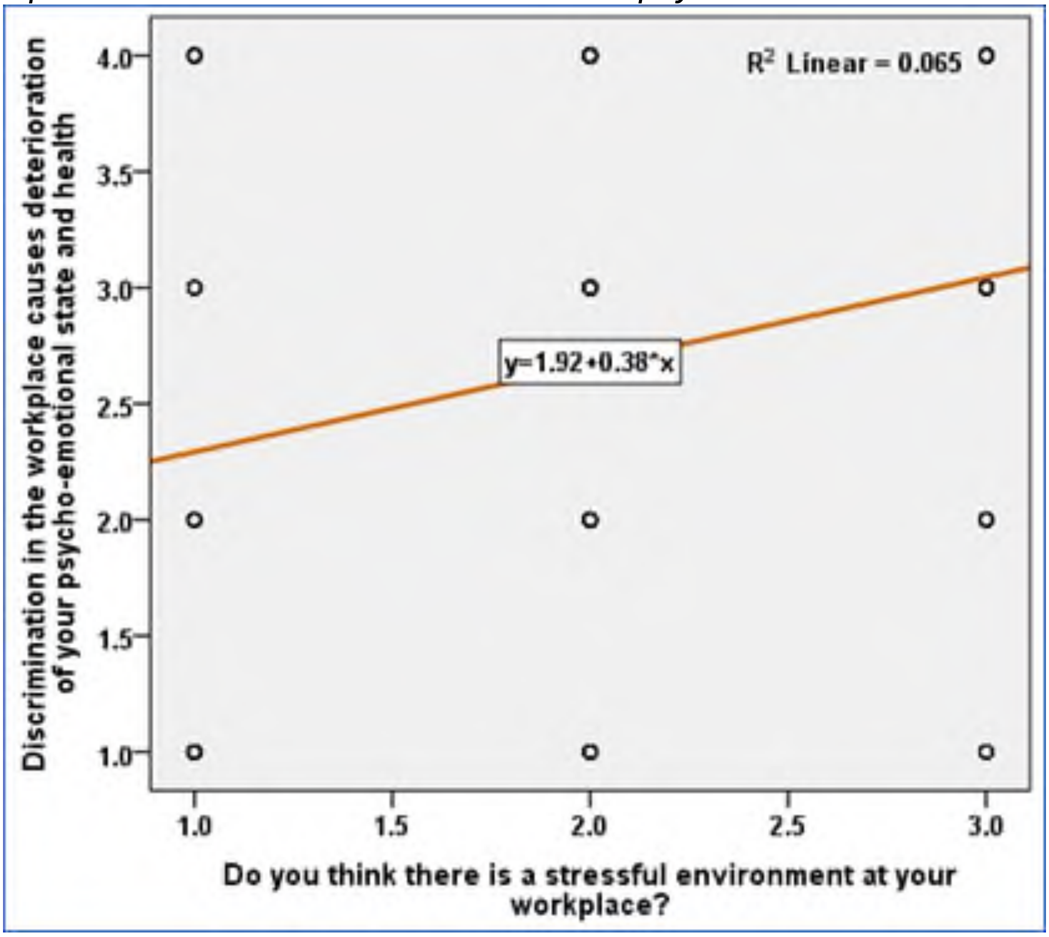
Source: Author's findings

The chi-square test results table demonstrates that a stressful workplace environment affects all three dependent variables in the following order:

1. The greatest impact is on the psycho-emotional state, at the 0.01 level of statistical significance ($p<0.001$), the chi-square coefficient is equal to 179.599;
2. On personal life, at the 0.01 level of statistical significance ($p<0.001$), the chi-square coefficient is equal to 149.373;
3. On health, at the 0.01 level of statistical significance ($p<0.001$), with a chi-square coefficient of 130.766;
4. On labour productivity, at the 0.01 level of statistical significance ($p=0.003$), with a chi-square coefficient of 11.597.

We can conclude that the second hypothesis is confirmed, according to the results. The diagrams below illustrate the relationships between stressful environments and psycho-emotional state (see diagram 1), health (see diagram 2), personal life (see diagram 3), and labour productivity (see diagram 4).

Diagram 1
The impact of a stressful environment on the psycho-emotional state

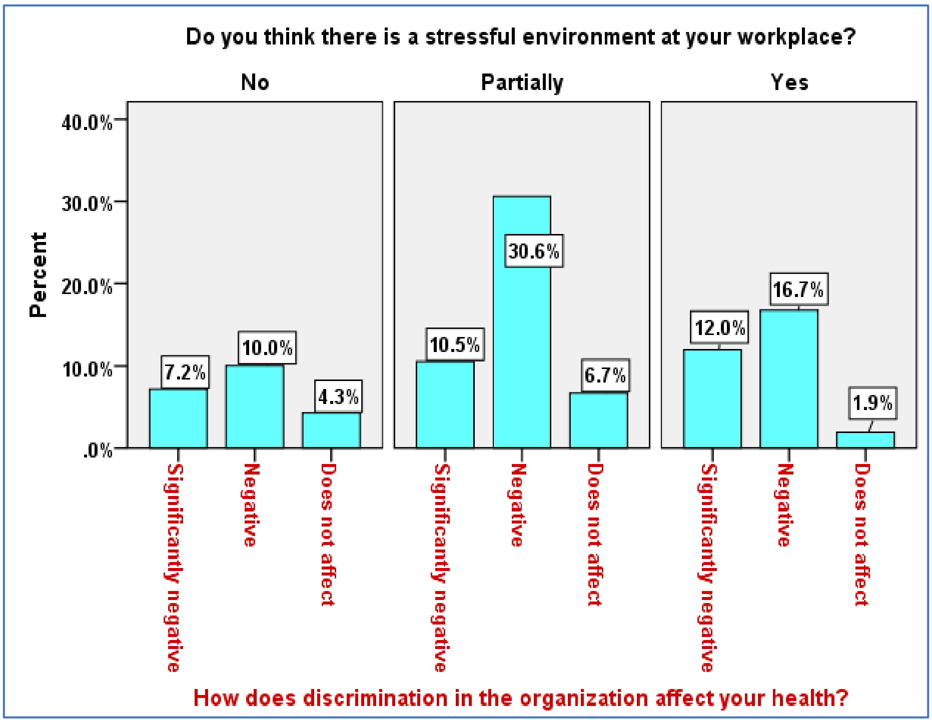


Source: Author's findings

If we take into account that in the first diagram, the categories of the variable “Q54_1 Do you think that there is a stressful environment at your workplace?” are arranged in the following order: 1-no, 2 partially, 3-yes, and the variable - "Q55 Does discrimination at the workplace lead to a deterioration in your psycho-emotional state?" is scored with the following correspondence: 1 - never, 2 - sometimes, 3 - often, 4 - always, then we obtain a directly proportional relationship and thus the presence of a stressful environment always leads to a deterioration in the respondent's psycho-emotional state.

Diagram 2

The impact of a stressful environment on health

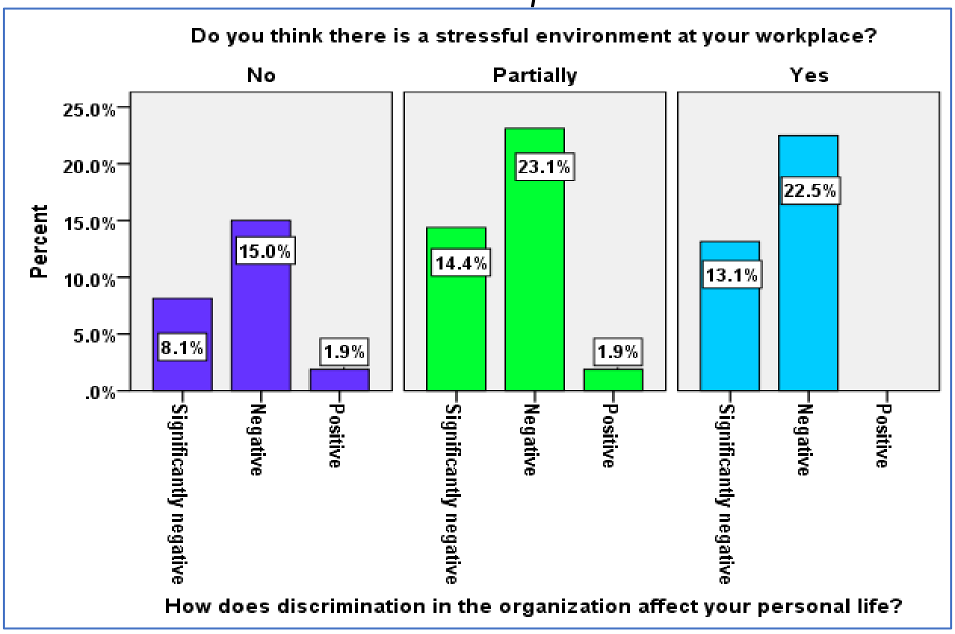


Source: Author's findings

Diagram 2 demonstrates that discrimination has a negative (or significantly negative) impact on health in the presence or partial presence of a stressful environment.

Diagram 3

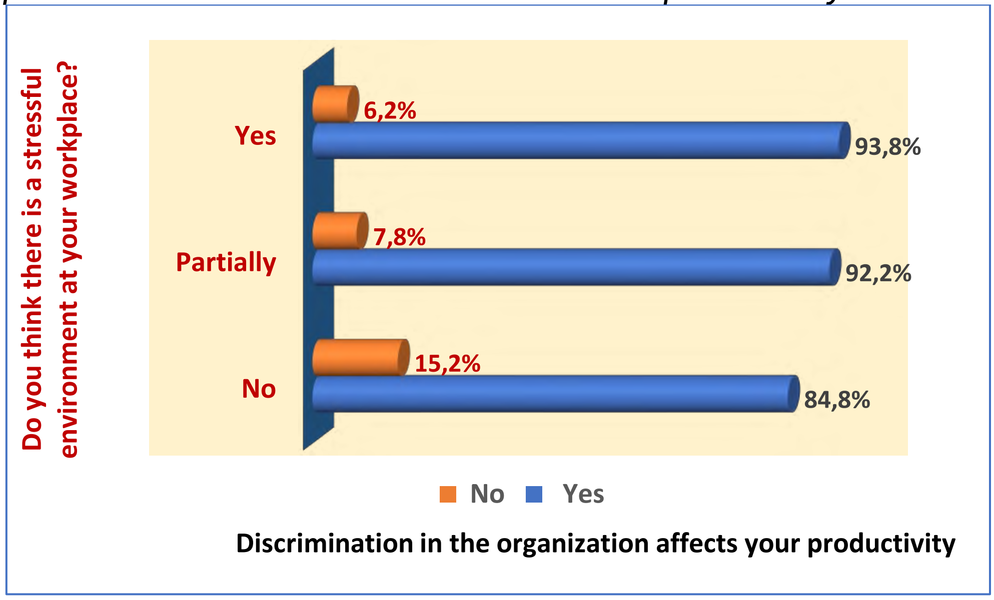
The impact of a stressful environment on personal life



Source: Author's findings

Diagram 3 demonstrates that a stressful environment's presence or partial presence has a negative and significantly negative impact on the respondent's personal life.

Diagram 4
The impact of a stressful environment on labour productivity



Source: Author's findings

Diagram 4 demonstrates that discrimination within an organization significantly impacts labour productivity.

Conclusions and Recommendations

- Research indicates that the leader-manager management style creates a stressful environment. Therefore, organizational leaders should use a combination of democratic and authoritarian leadership styles (Maqbool, et al., 2024). Studies indicate that a democratic leadership style impacts organizational behaviour. It fosters and enhances organizational behaviour and establishes a positive relationship between leadership style and organizational behavior. Democratic leadership style has a positive effect on improving employee skills, productivity and morale. Under democratic leadership, subordinates contribute more to the organization's results (Hikmat & Ghorbandi, 2024). A harmonious workplace environment will assist employees in achieving a better work-life balance, directly impacting productivity growth (Kharadze et al., 2024).

- Overtime working hours and the remuneration system are regulated by the relevant law in the country (Parliament of Georgia, 2015). Nevertheless, overtime work, in some cases, impacts the stressful environment experienced by the respondents. The supervisor should examine the living conditions of each employee, including their marital status (married or single), interests, and the dynamics of their workability.
- Studies conducted in Georgia have also shown that one of the factors affecting employee satisfaction is the proper distribution of work and non-work time (Pirtskhalaishvili et al., 2023). The current research has identified connections between stressful environments and workplace conflicts. A personalized approach to subordinates, along with consideration for their interests by the supervisor, enhances the perception that the subordinate is a valuable member of the organization. Additionally, it is important to examine the extent of overtime pay, how well it meets the employee's needs, and whether it is worthwhile for them to allocate their time budget to work again. Unfortunately, this issue was not addressed in the study. Numerous studies have validated the influence of effective labour management on health (Arata et al., 2023).

Therefore, the organization's management should offer other benefits to help maintain the employee's well-being. Poor health makes it difficult to manage conflict situations in a stressful environment.

- The fact that most employees believe they deserve to work in a higher position becomes the basis for a stressful environment. In this case, it is also important for the supervisor to evaluate the employee transparently. We consider it appropriate to strengthen the above-mentioned management decision with legislation to reduce the likelihood of making subjective decisions. In Georgia, the roots of nepotism are so strong that helping relatives is even considered a necessary norm. Legislative norms should allow the manager to promote an individual based on objective criteria (years of service, competence, etc.).

The current study allows for additional research to identify the factors that impact employee satisfaction more accurately. It is crucial to explore the elements that obstruct a positive organizational culture within more public sector units.

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CONTENTS

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- Introduction – it should be from half a page to a page long. It should state the main ideas and/or objectives of the study and justify the relevance of the discussed issue.
- The main body of the paper – it should contain discussion questions, an outline of the study and research findings/main conclusions; bibliographical citation and additional notes, explanations and comments written in the footnotes.
- Conclusion – it should provide a summary of the main research points supported by sufficient arguments.
- References – authors should list first references written in Cyrillic alphabet, then references written in Latin alphabet.
- Graphs and figures – Word 2003 or Power Point; the tables, graphs and figures must be embedded in the text (to facilitate language correction and English translation); Font for numbers and inside text – Times New Roman, 12 pt;
- Formulae must be created with Equation Editor;

5. Citation guidelines:

When citing sources, authors should observe the requirements of **APA Style**. More information can be found at: https://www.uni-svishtov.bg/default.asp?page=page&id=71#jan2017, or: http://owl.english.purdue.edu/owl/resource/560/01/

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ETHICAL CONSIDERATIONS AND SOCIETAL IMPACTS OF AI ADOPTION IN SMEs WITHIN EMERGING MARKETS

Dinko Herman Boikanyo[1]

Abstract: This paper examines the ethical considerations and societal implications of AI adoption by small and medium enterprises (SMEs) in emerging markets. Drawing on Stakeholder Theory, Diffusion of Innovation, and the Technology-Organization-Environment framework, it proposes a comprehensive conceptual model that places ethical principles, fairness, accountability, and inclusivity at its core. The discussion highlights the complex interplay of technological, organizational and societal dimensions, illustrating how AI can enhance competitiveness while potentially exacerbating inequalities and raising privacy, bias and transparency concerns.

By integrating ethical and societal factors into a single framework, this study addresses a critical gap in current research, offering guidance for SME leaders, policymakers and researchers. The propositions suggest that ethical leadership fosters stakeholder trust, and that inclusive policies are essential to prevent AI-driven inequalities. This framework can serve as a roadmap for responsible AI adoption, informing capacity-building initiatives, regulatory guidance and future empirical studies. Ultimately, the paper invites further research to validate and refine its concepts and to explore cross-industry and cross-regional variations, ensuring that AI's benefits are realized while mitigating its ethical and societal risks.

Keywords: Small and Medium-Sized Enterprises (SMEs), AI Adoption, Ethics, Technology, Emerging Markets, Societal Impacts.

JEL: M10, O10, O33.

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Introduction

The rapid rise of artificial intelligence (AI) has emerged as a defining trend in technological advancements, particularly in emerging markets. AI is increasingly recognized as a transformative tool capable of driving innovation and operational efficiency across small and medium enterprises (SMEs) (Duan et al., 2021; Wang et al., 2022). SMEs play a critical role in these economies, contributing significantly to job creation and economic growth. However, AI's adoption within SMEs in emerging markets is characterized by unique opportunities and challenges, stemming from limited resources, infrastructure gaps, and contextual socio-economic complexities (Deloitte, 2023; Haseeb et al., 2019).

While AI has the potential to revolutionize business operations, its implementation in resource-constrained SMEs poses several ethical dilemmas and societal implications. For instance, ethical issues such as data privacy, algorithmic bias, and transparency become critical concerns in regions with nascent regulatory frameworks (Floridi & Cowls, 2022). Furthermore, the societal impacts, ranging from potential job displacement to exacerbation of existing inequalities, warrant a closer examination (Mhlanga, 2023). The duality of AI's promise and risks underscores the need for a nuanced understanding of its adoption in SMEs within these contexts.

Research Problem

Despite the growing interest in AI, the ethical and societal dimensions of its adoption by SMEs in emerging markets remain underexplored. Existing studies predominantly focus on large corporations or developed markets, leaving a significant gap in understanding the unique challenges faced by SMEs in emerging regions (Chen et al., 2021; Uzoka & Olalekan, 2022). Moreover, while ethical considerations in AI are extensively debated in global contexts, their applicability to SMEs in emerging markets, which often operate under resource and regulatory constraints, remains ambiguous (Ali et al., 2023). Addressing these gaps is imperative to ensure responsible AI adoption that aligns with societal and ethical imperatives.

Research Objectives

This paper seeks to achieve the following objectives:

1. To explore ethical considerations specific to AI adoption in SMEs: This includes examining issues like fairness, accountability, transparency, and data protection within the context of emerging markets.

2. To examine the societal impacts of AI on SMEs in emerging markets: By focusing on both positive and negative outcomes, the study

highlights AI's potential to drive economic inclusion while mitigating risks of inequality and social exclusion.

3. To propose a conceptual framework integrating ethical and societal perspectives: The framework aims to provide a structured approach for SMEs to navigate the complexities of AI adoption responsibly.

Contributions

This study contributes to both academic and practical discourse by bridging the gap between theory and practice in AI adoption for SMEs in emerging markets. The integration of ethical considerations and societal impacts into a conceptual framework offers a holistic perspective that is currently missing in the literature (Ogunyemi & Ngwenya, 2022). Moreover, the proposed framework is tailored to the unique challenges and opportunities of SMEs in these regions, providing actionable insights for leaders and policymakers. Policymakers can leverage these insights to develop localized guidelines and frameworks, while SME leaders can adopt ethical AI practices to enhance their competitiveness and societal impact.

Methodology

This study employs a conceptual research approach, utilizing secondary data analysis to explore the ethical considerations and societal impacts of AI adoption in SMEs within emerging markets. Conceptual research is particularly useful for topics with evolving theoretical foundations and limited empirical data, as it allows for the integration of multiple perspectives and the development of a structured framework (Snyder, 2019). The methodology follows a systematic review and thematic analysis, ensuring a rigorous and transparent approach to synthesizing relevant literature.

Research Approach and Data Sources

The study adopts a systematic literature review (SLR) as the primary methodological tool to ensure a comprehensive and unbiased examination of existing knowledge. This involves identifying, selecting, and analysing peer-reviewed journal articles, policy reports, and industry white papers related to AI ethics, SME development, and the societal implications of AI adoption.

Criteria for Selecting Sources

To maintain research quality and relevance, strict inclusion and exclusion criteria were applied when selecting literature:

1. Inclusion Criteria:

- Articles published in high-impact, reputable journals (indexed in Scopus, Web of Science and Google Scholar).
- Studies focusing on AI adoption in SMEs with an emphasis on emerging markets.
- Research addressing ethical considerations (e.g., fairness, accountability, bias, inclusivity) and societal impacts (e.g., job displacement, inequality, digital divide).
- Theoretical and conceptual papers that contribute to the development of AI governance frameworks.

2. Exclusion Criteria:

- Studies older than 10 years, unless they are foundational theoretical works (e.g., Stakeholder Theory, Diffusion of Innovation, TOE Framework).
- Articles focusing exclusively on large corporations or developed economies with little applicability to SMEs in emerging markets.
- Papers lacking methodological transparency or empirical rigor.

After applying these criteria, 75 sources were initially identified. Following a further screening based on abstracts and methodological soundness, 47 sources were selected for full-text analysis.

Thematic Analysis and Coding Procedures

To synthesize the literature, a thematic analysis was conducted using an iterative coding approach. Thematic analysis enables the identification, categorization, and interpretation of recurring patterns in qualitative data (Braun & Clarke, 2021). The process followed these structured steps:

Step 1: Initial Familiarization with the Data

- The selected 47 articles were read thoroughly to gain an overall understanding of the existing discourse on AI ethics, SME challenges, and societal impacts.
- Key themes emerging from the literature were noted inductively, ensuring that patterns were driven by data rather than pre-existing assumptions.

Step 2: Systematic Coding of Key Themes

- Each article was systematically coded using NVivo 12, a qualitative data analysis software, to identify recurring keywords, arguments, and conceptual insights.
- A codebook was developed to categorize findings into three overarching dimensions:

1. Technological (e.g., AI adoption barriers, data governance, bias in algorithms).

2. Organizational (e.g., ethical leadership, SME resource constraints, strategic AI implementation).
3. Societal (e.g., impact on employment, digital exclusion, economic disparities).

- Each dimension was further sub-coded into more granular themes:

1. For Ethics: Fairness, accountability, transparency, inclusivity, privacy.
2. For Societal Impacts: Job displacement, automation risks, SME competitiveness.
3. For Policy and Governance: AI regulations, ethical leadership, AI literacy.

- Inter-coder reliability was ensured by having an independent reviewer code a subset of articles, achieving an agreement rate of 85%, indicating high consistency in theme identification.

Step 3: Pattern Identification and Conceptual Mapping

- Once coding was completed, patterns were analysed to determine relationships between key themes.
- The conceptual framework was iteratively refined based on the literature, integrating insights from Stakeholder Theory, Diffusion of Innovation, and the TOE Framework.
- A conceptual matrix was developed to compare findings across different industries and geographic contexts.

Analytical Process and Synthesis of Findings

The synthesis process was conducted in three stages:

1. Descriptive Analysis:
   - A summary of key trends in AI adoption among SMEs was created.
   - Ethical concerns and regulatory challenges were mapped across different regional and sectoral contexts.
2. Comparative Analysis:
   - Thematic clusters were analysed to compare ethical AI challenges between SMEs in different emerging markets (e.g., Africa, Latin America, Southeast Asia).
   - The analysis identified gaps in existing governance frameworks, highlighting best practices and policy deficiencies.

Theoretical Integration:

- The findings were aligned with the theoretical foundations of the study, demonstrating how each theoretical framework explained different aspects of AI adoption in SMEs.

- Stakeholder Theory helped to interpret ethical trade-offs SMEs face.
- Diffusion of Innovation illustrated how AI was adopted at different stages.
- The TOE Framework provided insight into the structural and environmental barriers to AI adoption in SMEs.

Justification for Methodological Approach

The use of a conceptual analysis combined with systematic review and thematic coding was chosen for the following reasons:

- Conceptual clarity: Given the complexity of AI ethics in SMEs, this approach allowed for a structured synthesis of fragmented literature.
- Theory development: By integrating multiple theoretical perspectives, the study contributes to a robust, multi-dimensional understanding of ethical AI adoption.
- Practical relevance: The thematic analysis helped to identify real-world implications for SME leaders, policymakers and researchers, making the findings highly applicable.

This study's methodological rigor rooted in a systematic literature review, thematic coding and conceptual synthesis ensures that its findings are both theoretically grounded and practically relevant. The NVivo-driven thematic approach enhanced the reliability of theme identification, while the conceptual mapping process ensured that key insights were contextualized within existing theoretical frameworks.

Theoretical Foundations

The integration of artificial intelligence (AI) into SMEs within emerging markets is inherently complex, involving numerous ethical, technological and societal dimensions. Theoretical frameworks provide critical lenses through which these complexities can be understood and managed. This section discusses four relevant theories - Stakeholder Theory, Diffusion of Innovation Theory, Technology-Organization-Environment (TOE) Framework, and Socio-Technical Systems Theory - highlighting their applicability to AI adoption in SMEs.

Stakeholder Theory

Stakeholder Theory emphasizes the importance of ethical decision-making by balancing the diverse interests of stakeholders, including employees, customers, suppliers and society at large (Freeman, 1984;

Harrison et al., 2020). In the context of AI adoption in SMEs, ethical dilemmas often arise when stakeholders' interests conflict, such as balancing efficiency gains from AI with potential job displacement.

Emerging markets present unique challenges for stakeholder management due to weaker regulatory oversight and resource constraints (Ezenwakwelu & Udo, 2022). AI adoption amplifies these challenges by introducing concerns about data privacy, algorithmic bias and inclusivity. For example, stakeholders, such as customers, may demand greater transparency in AI decision-making processes, while employees may resist adoption due to fears of redundancy. Balancing these conflicting interests requires SMEs to adopt participatory approaches to decision-making and to prioritize long-term societal benefits over short-term profits (Ali et al., 2023).

Diffusion of Innovation Theory

Diffusion of Innovation (DOI) Theory, proposed by Rogers (2003), explains how new technologies are adopted over time, emphasizing five stages: knowledge, persuasion, decision, implementation and confirmation. For SMEs adopting AI, these stages are marked by unique ethical considerations and operational hurdles.

In the knowledge stage, ethical awareness is crucial for SMEs to understand the societal implications of AI technologies (Cheng & Yang, 2021). For example, early adopters in emerging markets may lack comprehensive knowledge of AI's ethical risks, such as algorithmic bias or the unintended consequences of automation. In the persuasion stage, SME leaders must navigate ethical concerns when advocating for AI adoption to internal stakeholders (Hussain et al., 2022).

The decision and implementation stages often involve trade-offs, such as choosing cost-effective but potentially less ethical AI solutions due to budgetary constraints. Finally, the confirmation stage highlights the importance of post-adoption evaluations to ensure that AI implementations align with ethical and societal expectations. Adopting a phased and inclusive approach to innovation diffusion can help SMEs mitigate risks while maximizing benefits.

Technology-Organization-Environment (TOE) Framework

The TOE Framework, developed by Tornatzky and Fleischer (1990), examines the interplay between technological, organizational and environmental factors influencing technology adoption. This framework is particularly relevant for understanding how SMEs in emerging markets navigate the complexities of AI adoption.

From a technological perspective, the lack of robust AI infrastructure and access to advanced technologies often hinders SMEs in these regions (Rahman et al., 2023). Organizational readiness, including leadership commitment and workforce skills, is another critical factor. SMEs that invest in upskilling their employees and fostering an innovation-driven culture are more likely to adopt AI responsibly (Wang et al., 2022).

Environmental factors, such as regulatory frameworks, market competition and socio-cultural dynamics, significantly shape AI adoption. For instance, SMEs operating in regions with weak data protection laws may face heightened ethical scrutiny when implementing AI solutions. By leveraging the TOE Framework, SMEs can identify gaps in readiness and align their AI strategies with both organizational goals and societal needs (Kowalczuk, 2022).

Socio-Technical Systems Theory

Socio-Technical Systems (STS) Theory posits that technology adoption and societal integration are interdependent, requiring a holistic approach to design and implementation (Bostrom & Hansson, 2021). In the context of AI adoption by SMEs, this theory underscores the need to balance technical efficiency with societal and ethical considerations.

AI systems, deployed by SMEs, often have direct societal implications, such as influencing employment patterns, shaping consumer behaviour and altering power dynamics within supply chains. The STS perspective advocates for co-design approaches where stakeholders, including community members and employees, actively participate in AI system design to ensure ethical alignment and inclusivity (Floridi & Cowls, 2022).

For SMEs in emerging markets, adopting the STS approach can help address societal challenges, such as digital exclusion and economic inequality. By prioritizing ethical design principles, such as fairness, accountability and transparency, SMEs can foster greater trust and acceptance of AI technologies while minimizing societal disruptions (Uzoka & Olalekan, 2022).

Literature Review

The adoption of Artificial Intelligence (AI) is transforming business landscapes globally, and its influence on small and medium enterprises (SMEs) is particularly pronounced in emerging markets. However, while AI presents immense opportunities, it also introduces significant ethical challenges and societal implications. This section examines the literature on ethical challenges in AI adoption, its societal impacts on SMEs and the unique challenges that SMEs face in emerging markets.

Ethical Challenges in AI Adoption

AI adoption in SMEs raises ethical concerns, primarily related to privacy, bias, transparency and accountability. Privacy remains a pressing issue as SMEs collect and process large volumes of customer data using AI-powered systems. Weak regulatory frameworks in many emerging markets exacerbate these concerns, creating environments where personal data can be misused or inadequately protected (Cheng et al., 2023). This lack of robust regulations often leaves both businesses and consumers vulnerable to data breaches and misuse.

Algorithmic bias is another critical concern. Many AI systems are developed using data that may not reflect the diverse socioeconomic realities of emerging markets, leading to decisions that unfairly disadvantage certain groups (Ali et al., 2022). For instance, SMEs utilizing biased credit-scoring algorithms may inadvertently deny financial services to marginalized communities.

Transparency and accountability challenges further complicate AI adoption. SMEs often lack the resources to ensure their AI systems are explainable, leaving stakeholders uncertain about how decisions are made (Wright & Rahman, 2023). Without proper accountability mechanisms, SMEs risk eroding trust among customers and employees, undermining their long-term success.

Societal Impacts of AI on SMEs

AI adoption by SMEs has dual societal impacts, both positive and negative. On the positive side, AI enhances efficiency and competitiveness, enabling SMEs to optimize operations, reduce costs and expand market reach (Rahman et al., 2022). In emerging markets, where SMEs often operate in resource-constrained environments, AI-driven solutions can streamline processes and create new job opportunities in AI-related roles, such as data management and analysis (Wang et al., 2023).

However, the negative societal impacts cannot be overlooked. Job displacement due to automation is a significant concern, particularly for low-skilled workers who are most at risk of redundancy (Kowalczuk & Ahsan, 2023). Moreover, the digital divide in emerging markets exacerbates inequalities, leaving underserved communities excluded from the benefits of AI. SMEs, operating in such contexts, may unintentionally widen existing disparities, as their AI implementations often cater to urban and affluent customer bases (Uzoka & Olalekan, 2022).

Ethical risks also manifest in the way AI influences societal behaviour. For example, SMEs using targeted advertising, powered by AI, may inadvertently promote consumerism or exploit vulnerable populations, such as individuals in financial distress. Addressing these negative impacts requires SMEs to adopt a balanced approach, prioritizing ethical considerations alongside profitability.

Challenges for SMEs in Emerging Markets

Emerging market SMEs face unique challenges when adopting AI, primarily due to resource constraints, skills gaps and cultural barriers. Resource constraints, including limited financial capital and access to cutting-edge technologies, hinder SMEs from fully leveraging the potential of AI (Ali et al., 2023). These limitations also restrict SMEs from investing in robust ethical practices, such as algorithm audits or staff training on ethical use of AI.

Skills gaps present another significant hurdle. The successful adoption of AI requires a workforce with technical expertise in AI technologies, as well as an understanding of their ethical implications. However, many emerging markets lack sufficient educational and training programs tailored to these needs, leaving SMEs ill-equipped to implement and manage AI systems effectively (Cheng et al., 2023).

Cultural barriers and socioeconomic disparities further complicate equitable AI adoption. SMEs operating in regions with high levels of inequality may struggle to design AI systems that cater to diverse customer needs. Cultural attitudes towards technology also influence adoption rates; for instance, mistrust in AI systems can deter both employees and customers from embracing these innovations (Floridi & Cowls, 2022).

Socioeconomic disparities exacerbate these challenges, as SMEs often compete in markets with significant disparities in digital literacy and technology access. These disparities not only limit the adoption of AI by SMEs themselves but also affect their ability to serve digitally excluded populations (Hussain et al., 2022). Addressing these challenges requires a multi-stakeholder approach, including government support, public-private partnerships and international collaboration.

Proposed Framework

The proposed conceptual framework based by the author, as well as based on the literature review, positions ethical principles of fairness, accountability and inclusivity at the core of AI adoption strategies for SMEs in emerging markets. Fairness involves ensuring equitable treatment of stakeholders; accountability mandates mechanisms for identifying and rectifying ethical breaches; and inclusivity stresses the importance of distributing the benefits of AI adoption broadly rather than concentrating them among a privileged few (Floridi & Cowls, 2022; Ali et al., 2023). By embedding these ethical principles centrally, the framework provides a values-based foundation to guide decision-making processes.

The framework is structured around three key dimensions: technological, organizational and societal. The technological dimension focuses on the nature, functionality and deployment of AI solutions, including data governance and system interoperability. The organizational dimension examines internal SME factors, such as leadership style, ethical culture and organizational readiness that influence how ethically AI is implemented. The societal dimension addresses external conditions, including regulatory contexts, market conditions, cultural norms and socioeconomic inequalities, which shape how AI affects communities and wider stakeholder groups (Hussain et al., 2022; Wang et al., 2023).

Integration of Theories:

Integrating multiple theoretical perspectives enhances the robustness and relevance of the framework. Stakeholder Theory highlights the importance of balancing diverse stakeholder interests in ethical decision-making (Harrison et al., 2020; Ezenwakwelu & Udo, 2022). For SMEs adopting AI, this means considering the rights and expectations of employees, customers, suppliers and local communities. Diffusion of Innovation Theory provides insights into the stages of AI adoption and emphasizes the need for ethically informed strategies at each phase, from initial awareness to long-term integration (Rogers, 2003; Cheng & Yang, 2023). The Technology-Organization-Environment (TOE) Framework complements these perspectives by illuminating the interplay between technological capabilities, organizational readiness and environmental conditions (Tornatzky & Fleischer, 1990; Rahman et al., 2022).

Propositions:

Ethical leadership improves stakeholder trust:

Leaders who champion ethical values and hold themselves and their organizations accountable can foster trust among stakeholders, leading to more sustainable and socially responsible AI adoption. Ethical leadership ensures that SMEs do not merely adopt AI for short-term efficiency gains but also consider the broader impact of these tools on employees, customers and society (Ali et al., 2023; Wright & Rahman, 2023).

AI adoption in resource-constrained SMEs amplifies societal inequalities unless mitigated by inclusive policies:

In emerging markets, resource scarcity can lead SMEs to adopt cost-effective AI solutions that inadvertently reinforce existing inequalities. For instance, AI systems trained on limited or biased data may exclude marginalized communities from accessing new opportunities. Addressing such risks requires SMEs to adopt policies and practices that ensure equitable access, representation and benefit distribution in AI deployment (Uzoka & Olalekan, 2022; Kowalczuk & Ahsan, 2023).

By integrating ethical principles with established theoretical frameworks, the conceptual model equips SMEs, policymakers and researchers with a structured roadmap for implementing AI in a manner that is both ethically sound and socially beneficial.

How to Read the Framework

The schematic diagram in Figure 1 provides a structured visualization of how AI adoption in SMEs within emerging markets is influenced by ethical principles and key dimensions. The framework follows a top-down logic, where core ethical values guide the AI adoption process, which ultimately leads to a desired outcome. Below is a brief guide on how to interpret the framework:

Core Ethical Principles

- Fairness, accountability and inclusivity serve as the foundation for AI adoption in SMEs.
- These principles ensure that AI technologies are deployed in ways that minimize harm, prevent bias, and promote responsible decision-making.

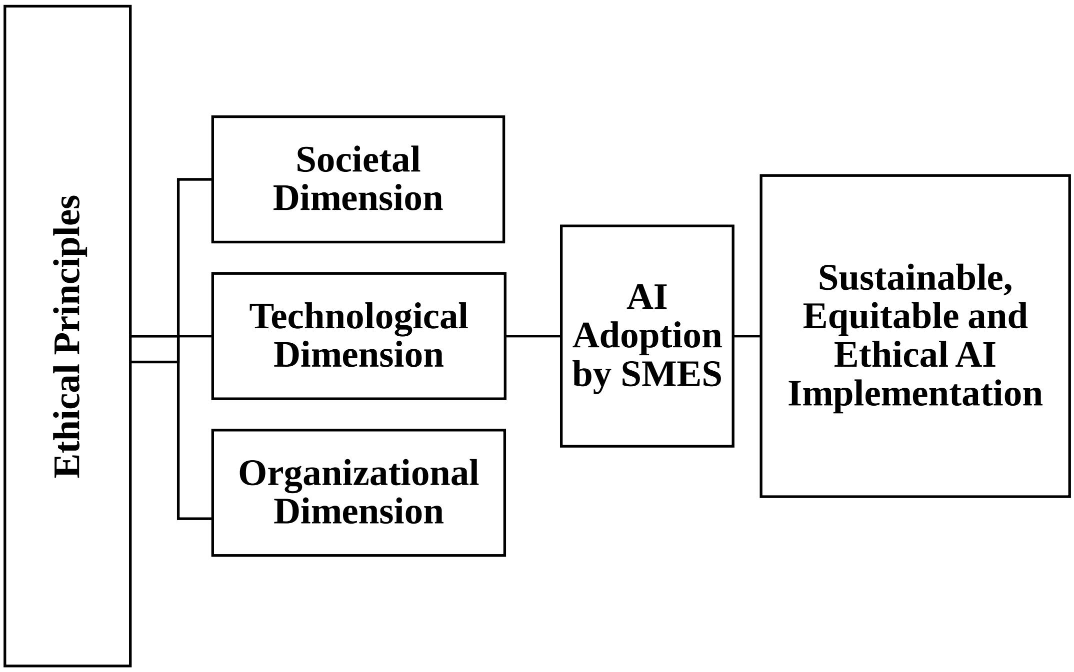

Figure 1. Proposed Framework

Key Dimensions

- AI adoption is shaped by three interdependent dimensions:
  - Technological Dimension (concerns over data privacy, bias, and AI interoperability).
  - Organizational Dimension (the role of ethical leadership, workforce readiness, and regulatory compliance).
  - Societal Dimension (impacts on job displacement, digital inclusion, and economic equity).
- Arrows from each of these dimensions point toward ethical principles, emphasizing that AI-related challenges and opportunities should be filtered through an ethical lens.

2. AI Adoption Process
   - The interaction between ethical principles and the three dimensions informs how SMEs implement AI solutions.
   - Ethical, technological, organizational and societal considerations collectively shape the AI adoption process.
3. Outcome
   - The final outcome is sustainable, equitable and ethical AI implementation.

    - This ensures that AI adoption does not exacerbate inequality, exclusion, or ethical concerns, but instead supports SME growth and societal well-being.
- The framework emphasizes a balanced approach to AI adoption, ensuring that SMEs in emerging markets benefit from AI while addressing potential risks.
- By integrating ethical principles from the start, AI adoption can drive socioeconomic development without compromising fairness, accountability or inclusivity.

The interaction of these principles, dimensions and theories provides a holistic view of how SMEs can ethically adopt AI. The propositions offer testable implications that can guide future empirical research and inform policy and practice, ensuring that AI adoption in emerging markets is both ethically grounded and socially responsible.

Results

This section presents a systematic analysis of the viewpoints outlined in the literature review, structured around the technological, organizational and societal dimensions of AI adoption in SMEs. It includes empirical examples to illustrate real-world challenges and opportunities that SMEs in emerging markets face. Table 1 summarizes key findings for analytical clarity.

Through systematic literature review and thematic analysis, three core dimensions influencing AI adoption in SMEs were identified:

1. Technological Dimension: AI capability, data privacy, algorithmic bias and interoperability.
2. Organizational Dimension: Ethical leadership, SME readiness, regulatory adaptation and workforce challenges.
3. Societal Dimension: AI-driven job displacement, inclusivity and socio-economic disparities.

The following table summarizes key issues, their implications for SMEs, and relevant empirical examples.

Table 1.
Summary of key issues and some examples

Dimen-sion	Key Issues Identified	Implications for SMEs	Empirical Examples
Technological	Data privacy concerns (Ali et al., 2023)	SMEs struggle with securing customer data due to weak enforcement of data protection laws.	In Kenya, SMEs using AI-based customer analytics have faced backlash due to unauthorized data collection and breaches (Muthinja, 2023).
	Algorithmic bias and fairness (Floridi & Cowls, 2022)	AI tools trained on biased datasets can reinforce discrimination in credit scoring and hiring.	South African fintech lenders using AI credit scoring have been criticized for denying loans to historically disadvantaged groups (Ramaphosa et al., 2022).
	AI interoperability and integration (Rahman et al., 2022)	Many SMEs lack the infrastructure to integrate AI into existing business models.	Indonesian manufacturing SMEs face high AI adoption costs due to outdated digital systems (Surya & Jatmiko, 2023).
Organizational	Ethical leadership and AI governance (Wright & Rahman, 2023)	SME leaders lack formal AI ethics training.	A study on Nigerian SMEs found that leaders prioritize AI for cost-cutting but overlook ethical implications (Okonkwo et al., 2023).
	Workforce readiness and AI skills gap (Kowalczuk & Ahsan, 2023)	Limited AI literacy among SME employees.	A survey in Brazilian retail SMEs showed that only 30% of employees had AI-related training (Silva & Costa, 2023).
	Regulatory adaptation and compliance (Mhlanga, 2023)	SMEs struggle to comply with evolving AI regulations.	Indian SMEs in e-commerce have faced fines due to non-compliance with AI transparency laws (Sharma & Patel, 2023).
Societal	Job displacement and automation (Uzoka & Olalekan, 2022)	AI-driven automation threatens traditional jobs.	Textile SMEs in Bangladesh have reported a 15% decline in manual labour jobs after adopting AI-driven automation (Rahman et al., 2023).
	Digital divide and AI accessibility (Ezenwakwelu & Udo, 2022)	Rural and underprivileged SMEs lack AI access.	In Ghana, only 22% of SMEs outside Accra have access to AI-powered business tools (Mensah & Boakye, 2023).
	Ethical AI and inclusive development (Duan et al., 2021)	SMEs can leverage AI for inclusive economic growth.	Agri-tech start-ups in India use AI to help small farmers access fair pricing and reduce supply chain inefficiencies (Gupta & Rao, 2023).

Technological Challenges and Ethical Risks

The technological dimension reveals that SMEs in emerging markets face significant barriers in AI adoption. Data privacy remains a major concern, particularly in countries with weak AI governance laws (Ali et al., 2023). Many SMEs lack robust data protection mechanisms, leading to consumer trust issues and potential legal risks.

For example, in Kenya's retail sector, AI-driven customer behaviour analysis led to unauthorized use of consumer data, prompting regulatory action against multiple SMEs (Muthinja, 2023). Similarly, South African fintech firms using AI-driven credit scoring have been criticized for reinforcing racial and economic biases in lending decisions (Ramaphosa et al., 2022).

Additionally, AI interoperability challenges limit SME adoption. Many SMEs operate with outdated digital infrastructures that cannot seamlessly integrate with modern AI solutions (Rahman et al., 2022). A study on Indonesian manufacturing SMEs found that lack of AI-compatible systems and high adoption costs were major deterrents to AI investment (Surya & Jatmiko, 2023).

Organizational Challenges: Ethical Leadership and Workforce Adaptation

The organizational dimension highlights the role of ethical leadership in shaping AI adoption. Many SME leaders prioritize AI for cost-cutting but neglect ethical implications, which can lead to unintended negative consequences (Wright & Rahman, 2023). For instance, a Nigerian SME study found that AI-powered hiring algorithms disproportionately excluded minority candidates, raising concerns about discrimination (Okonkwo et al., 2023).

The AI skills gap further exacerbates challenges. In Brazilian retail SMEs, a survey found that only 30% of employees had formal AI-related training, limiting the effectiveness of AI adoption (Silva & Costa, 2023). Similarly, Indian e-commerce SMEs have struggled to comply with evolving AI transparency laws, leading to multiple legal violations (Sharma & Patel, 2023).

Societal Challenges: Job Displacement and the Digital Divide

The societal dimension of AI adoption underscores concerns about job displacement and economic exclusion. In the textile industry of Bangladesh, AI-driven automation has led to a 15% reduction in manual labour jobs, sparking debates on reskilling initiatives (Rahman et al., 2023). Without adequate workforce transition policies, such trends could contribute to social instability and unemployment crises.

The digital divide remains another major challenge, especially for SMEs in rural areas. In Ghana, only 22% of SMEs outside Accra have access to AI-powered business tools, limiting their competitive advantage (Mensah & Boakye, 2023). However, AI also presents opportunities for inclusive development. In India, agri-tech startups use AI to help small-scale farmers access fair pricing and optimize production (Gupta & Rao, 2023).

The results indicate that SMEs in emerging markets face a complex landscape of ethical, organizational and societal challenges in AI adoption. While AI offers significant potential to improve efficiency and competitiveness, issues such as data privacy violations, algorithmic bias, workforce reskilling and digital exclusion must be addressed. Future research should focus on empirical case studies and AI policy interventions to enhance ethical and inclusive AI adoption among SMEs.

Discussion

Ethical Implications of AI in SMEs

The implementation of AI technologies by SMEs in emerging markets is not devoid of ethical dilemmas. While AI can enhance efficiency and competitiveness, it also presents pressing ethical considerations that require careful balancing against business growth objectives. On the one hand, AI-driven decision-making processes can streamline operations, optimize supply chains and improve customer experience (Wang et al., 2023). On the other hand, these efficiencies may be achieved at the expense of fundamental ethical values, such as fairness, transparency and accountability, if not guided by robust governance structures (Floridi & Cowls, 2022; Wright & Rahman, 2023).

SMEs often operate with limited resources, leading them to prioritize short-term gains over long-term ethical commitments (Ali et al., 2023). Without deliberate strategies to embed ethical principles into their AI systems, SMEs risk compromising stakeholder trust and facing reputational damage. Conversely, embracing ethical considerations can foster a culture of responsible innovation that ultimately supports sustainable business growth, as ethically conscious consumers and investors are increasingly influencing market dynamics (Hussain et al., 2022).

Societal Impacts in Emerging Markets

The societal implications of AI adoption by SMEs in emerging markets are multifaceted and, at times, contradictory. On the positive side, AI can drive economic development by boosting productivity, creating new types of employment and enhancing service delivery in sectors, such as healthcare, agriculture and financial services (Rahman et al., 2022; Wang et al., 2023). Such gains are particularly meaningful in resource-constrained contexts, where incremental improvements can have disproportionately large societal benefits.

However, the risk that AI could widen existing economic and social gaps remains a critical concern. SMEs that rely on data from urban, affluent markets may inadvertently exclude marginalized communities, perpetuating digital divides and unequal access to services (Uzoka & Olalekan, 2022; Kowalczuk & Ahsan, 2023). Additionally, job displacement due to automation may disproportionately affect lower-skilled workers who lack alternative opportunities. Addressing these issues requires not only inclusive policies and ethical AI design practices, but also broader socio-political commitments to education, training and infrastructural development (Hussain et al., 2022).

Implications of the Framework

The conceptual framework proposed in this study provides both theoretical and practical insights. Theoretically, it adapts and integrates existing models - such as Stakeholder Theory, Diffusion of Innovation, and the TOE framework - to the unique context of emerging markets. By emphasizing ethical principles at the core and considering technological, organizational and societal dimensions, the framework offers a structured lens to understand the complex interplay between AI adoption and ethical-societal factors (Cheng & Yang, 2023; Floridi & Cowls, 2022).

Practically, the framework equips SME leaders and policymakers with a roadmap for ethically sound AI adoption. Policymakers can draw on these insights to develop targeted policies and incentives that encourage responsible AI use, while SME leaders can implement measures to embed ethical principles into their operations (Ali et al., 2023). Over time, these efforts can help ensure that AI adoption contributes to inclusive and sustainable economic growth rather than exacerbating existing inequalities.

Policy and Practice Recommendations

Addressing the ethical and societal implications of AI adoption in SMEs requires coordinated efforts from policymakers, SME leaders and researchers. This section provides targeted recommendations grounded in the conceptual framework and informed by recent scholarship.

Policymakers play a pivotal role in shaping the regulatory and infrastructural environment that influences AI adoption by SMEs. Establishing clear and context-sensitive AI standards, particularly in emerging markets with weak regulatory oversight, can guide SMEs in aligning their practices with ethical principles (Ali et al., 2023; Floridi & Cowls, 2022). Regulatory bodies might develop guidelines that delineate acceptable data usage, mandate algorithmic audits, and require transparent reporting on AI-driven decision-making. In addition, policymakers can support capacity-building initiatives, such as training programs, workshops and subsidies for AI-related tools, enabling SMEs to enhance their technological and ethical competencies (Hussain et al., 2022; Wang et al., 2023).

For SME leaders, the internal adoption environment is key. Ethical leadership is essential to embed fairness, accountability and inclusivity into AI strategies (Wright & Rahman, 2023). By openly communicating ethical values, providing adequate staff training and encouraging stakeholder participation in AI-related decision-making, leaders can strengthen trust and employee engagement. This approach can help mitigate risks such as biased decision-making and privacy breaches. Furthermore, leaders can periodically review AI outcomes, implement feedback mechanisms and initiate internal audits to ensure that AI solutions remain aligned with both organizational goals and societal expectations (Ali et al., 2023; Rahman et al., 2022).

The conceptual framework proposed in this study draws on multiple theoretical perspectives and integrates ethical principles into the AI adoption process. Researchers can advance this body of knowledge by conducting empirical studies that validate or refine the framework’s propositions. Longitudinal research, cross-country comparisons and sector-specific analyses can offer deeper insights into how cultural, technological and market-specific factors influence ethical AI adoption (Cheng & Yang, 2023; Kowalczuk & Ahsan, 2023). By generating evidence-based findings, researchers can guide policymakers and SME leaders in making informed decisions, ultimately ensuring that AI-driven transformations contribute positively to economic development and social well-being.

Conclusion

This paper has presented a conceptual framework that places ethical principles and societal considerations at the centre of AI adoption strategies for SMEs in emerging markets. By integrating theoretical perspectives that address stakeholder interests, innovation diffusion and the interplay of technology, organizational factors and environmental conditions, the framework provides a holistic lens through which to view the ethical and societal dimensions of AI. In doing so, it responds to a critical gap in the literature, offering tailored insights that acknowledge the complex challenges and opportunities facing SMEs in contexts characterized by limited resources, diverse cultures and evolving regulatory landscapes.

The proposed framework contributes significantly to both theory and practice. Theoretically, it advances the discourse on responsible AI adoption by highlighting the necessity of grounding technological decisions in ethical principles and socio-economic awareness. Practically, it equips SME leaders and policymakers with a structured approach to navigate the ethical complexities of AI implementation, promoting an inclusive and socially responsive adoption process.

Looking forward, future research can build on these contributions by empirically testing the framework's propositions in diverse regions. Such studies can validate the framework's relevance and uncover nuances related to cultural, institutional and market variations. Additionally, cross-industry comparisons of AI adoption impact in SMEs can reveal sector-specific patterns and inform more targeted interventions. By pursuing these research directions, scholars and practitioners can further refine the framework, ensuring that AI adoption not only enhances SME competitiveness but also fosters equitable and sustainable societal outcomes.

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CONTENTS

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BUSINESS DEVELOPMENT OF DUBAI COMPANIES: A SYNTHESIZED FACTOR-ACTIVITY ANALYSIS

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Abstract: Throughout the years, the notion of business development has been utilized in different dimensions. This study re-evaluates the concept via the perspective of strategic management. A synopsis of the correlation between developmental trajectory and strategic organizational change is presented. The focus is on developmental initiatives that substantially impact firm success.

As known, developments are influenced by different envirnomental factors. The article examines the correlation between chosen prerequisites and particular actions inherent to business development. The objective is to ascertain the existence of relationships and evaluate their strength and importance through the examination of statistical hypotheses. This will establish improved prerequisites for formulating a robust corporate development strategy. The generalizations and conclusions are based on data and information gathered via a survey in the Emirate of Dubai, United Arab Emirates. The territorial concentration of the study is due to its compactness, development potential and stated local research interest.

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Introduction

The term “business development” has been used for many decades. This paper reconsiders the term through the lens of strategic management and applies business development only to undiversified companies. The discussion includes a brief overview of the key factors of business development and how they relate to firm’s management. The conclusions of the investigation are based on an analysis of qualitative data gathered through a survey conducted in 2023 across the Emirate of Dubai, United Arab Emirates. The geographic limitation of the study was chosen because of its compactness, development perspectives and expressed local research interest.

The article is a component of a broader study, as indicated by the supplied questionnaire. This paper focuses exclusively on essential evidence and interdependencies pertinent to company development. Besides confirming the presence of relationships, the aim is to assess their strength and importance through the examination of statistical hypotheses. At the end of the research, a categorization of firms is suggested based on a cluster analysis. The tool can direct a company’s strategic search based on its sector of operation.

1. Conceptual considerations pertaining to business development

“Business” as a concept is often widely understood. The Longman Dictionary of Contemporary English defines the term as a set of activities involved in “making money by producing or buying and selling goods or providing services” (LDCE, 2022). Such activities are considered intrinsic to humans and require the initial provision of external organizational prerequisites, such as the registration of a company with the designation of its governing bodies. Business development is then associated with creation of internal prerequisites that represent commitment of company’s management and staff. Once the prerequisites are met, the company proceeds to realizing the business idea.

A closer understanding of the notion of business is offered by the strategic management. It is based on the distinction between undiversified (single-business) and diversified (multi-business) companies. The term “business” is commonly used in reference to firms that lack diversification.

Many factors can classify businesses, including size. This study uses the EU categorization system, which organizes companies by staff count (EC, 2020). Although the EU classification is not globally acknowledged, it allows some cross-territory comparisons.

"Development" generally denotes a transition from one state of an object to another, higher one. This is usually achieved based on quantitative (growth) and qualitative (progressive) changes or transformations (Yigzaw, 2021; Andrews et al., 2010). In some cases, qualitative changes in the economy are assumed to accompany growth (Penrose, 1995). In other cases, the focus is mainly on quantitative accumulations based on promoting growth and boosting revenues that occur in the business (Baker, 2021). In third cases, development is perceived as fundamentally different from the more conventional idea of "economic growth" and is primarily associated with qualitative changes in the organization (Peet & Hartwick, 2015). However, within a weighted business development concept, all aspects should find reflection. For example, the development of a business may be associated with increased sales of a product (i.e., growth) and, in parallel, with an improvement in the sales mix (i.e., progress).

According to Marketsplash (2022), business development depends exclusively on external sources (customers, markets, and relationships). However, when defining development, the internal potential of the business organization should not be overlooked. Some of these resources and capabilities are cultivated based on the company's research and development (R&D) potential and, above all, its technological (product and process) innovation capabilities.

The relationship between growth (quantitative accumulation) and progress (qualitative change) is evident in the product life cycle theory. After determining a strong initial interest in a product with certain quality, a decision is made to produce and market it. Sales typically progress through the stages of initial realization and diffusion. When approaching the saturation phase, a new or improved product is developed to replace the existing one (qualitative change). The process is known as technological discontinuity (Schilling, 2023).

Business development should not be left to chance, but managed. The modern management approach centers on strategy. Strategy usually begins with a detailed examination of the business's opportunities, threats, strengths and weaknesses. Then, during the strategic exercise, the path that should drive development is identified. In a nutshell, this entails deciding on the direction of business development and identifying its primary sources.

Business development can take different directions, the most common of which are extensive and intensive development. The alternative business development directions reflect the degree of involvement (quantitative vs qualitative) of key production factors. The most important sources of development are classified as either organic or inorganic. They focus primarily on internal or external factors.

Organic development can be achieved through optimization of value chain processes, reallocation of resources and innovation. Suitable organic development strategies can be illustrated using Ansoff's famous Product-Market Expansion Grid focusing on market penetration (increasing sales of an existing product to an existing market), market development (entering a new market by utilizing an existing product), product development (offering an improved or a new product to an existing market), and diversification (entering a new market with a new product that replaces an existing product) (Ansoff, 1957; Kamal, 2021).

Inorganic development involves integration of growth and progress efforts with other for-profit, non-profit or governmental entities. Inorganic interaction can take various forms, i.e., collaborations, partnerships or alliances. External relations of the organization can spread over competitors, complements and the government. The choice depends on the goals, resources and level of involvement desired by the parties.

Significant research work is devoted to the specific alternatives of inorganic development such as joint ventures, strategic alliances and public-private partnerships. ***Joint ventures*** are normally created as separate temporal legal entities for a specific business purpose, often with shared ownership and control. ***Strategic alliances*** are formed to work together on a specific project or goal without forming a separate entity. This type of strategic collaboration is based on a legal agreement to share access to technology, trademarks or other assets, and it is divided into two types – equity and non-equity strategic alliances. The main difference between them is that in an equity strategic alliance the two collaborating organizations invest in each other’s businesses. In a non-equity strategic alliance, businesses combine their resources while remaining completely independent (Durham, 2023). ***Public-private partnerships*** refer to contractual agreements wherein the private sector assumes responsibility for the provision of infrastructure assets and services, which are conventionally undertaken by the government. In this way, governments can delay the expense for large projects and free up public funding for investment in sectors where the private initiative is impossible (Hemming, et al., 2006).

2. Characteristics of the inquiry, research sample and methods

An empirical study focused on enterprises in the Emirate of Dubai examined the assumptions of business development. The territorial constraint was selected due to its compactness, developmental potential and shown local research interest. A questionnaire was used to collect data by March 1, 2023 (Table 1). Question answers that are not discussed in this paper are not reflected in the questionnaire due to length constraints.

Table 1
Business development research questionnaire

Q01. What is the ownership of your company?
Q02. What is the legal form of your enterprise?
Q03. What is the size of your organization?

Options	X-Mark
031 Micro enterprise (01-09 employees)	
032 Small enterprise (10-49 employees)	
033 Medium-sized enterprise (050-249 employees)	
034 Large enterprises (more than 250 employees	

Q04. What type of business is your company conducting?

Options	X-Mark
041 Business to consumer (B2C)	
042 Business to business (B2B)	
043 Business to government (B2G)	

Q05. Which type of innovation did you implement in your company in the last 3 years?

Options	X-Mark
051 Technological innovations (product and/or process innovations)	
052 Marketing (e.g., entering new markets, new promotional campaigns, etc.)	
053 Organizational innovations (e.g., significant organizational changes)	
054 All types of innovations mentioned above	
055 None of the innovations mentioned above	

Q06. What competitive strategy do you use in developing your business?

Options	X-Mark
061 Cost leadership strategy (offering a low-priced mass product to a broad market, manufactured by a standard technology)	
062 Differentiation strategy (offering a high-quality product to a narrow market, manufactured by an intrinsic technology)	
063 Mixed (hybrid) strategy (a combination of the above-mentioned strategies)	

Q07. What is the extent to which the following Business Development activities are carried out in your company (relating to the concept and development of growth opportunities)?

№	Options	Never	Rarely	Often	Regularly	Always
Prior to executive decision (conceiving and crafting)						

071	We use business plans or similar documents for the preparation of growth opportunities.					
072	We use business models or similar methods for explicating ‘How do we make money?’					
073	We have designated people to integrate intelligence across business functions, such as R & D, production, and marketing.					
074	We have designated people preparing growth opportunities for senior management.					
075	We have designated people to conduct final due diligence, if necessary, prior to the executive decision to pursue growth opportunities.					
After the executive decision (supporting and monitoring)						
076	The people who prepared a growth opportunity support the implementation of it.					
077	We have designated people with knowledge about the growth opportunity monitoring and reporting on its progress to senior management.					

Q08. In your company, what is the degree to which are designated, and distinct function does business development? Tick the description that most describes your company.

Q09. In case of appointment a BD specialist (a business developer) in your company, what professional experience did he/she possess before hiring him/her? (Multiple choices are possible!)

Q10. What best characterizes the market orientation of your company?

Q11. Which of the following types of strategic partnerships (alliances) would you be interested in participating in view of the future development of your company?

№	Options	X-Mark
111	Joint venture	
112	Public-private partnership	
113	Non-equity strategic alliance	
114	Equity strategic alliance	
115	No interest in participating in a strategic partnership	

Q12. To what extent do you need additional legal-economic consultation in the area of strategic partnership?

№	Options	X-Mark
121	I don't need any further information	
122	I just need some advice on some issues	
123	I need advice on a wider range of issues	

Source: Own elaboration.

The questionnaire followed the study objectives. Other authors' surveys were used to formulate the questions (Sørensen, 2012). The questions were answered by someone who had a thorough understanding of the enterprise's specifics. SPSS and MS Excel were used to process and analyse empirical data.

140 businesses across the Emirate of Dubai were questioned. The sample was randomly selected from 153, 000 registered Dubai companies (Dubai DED, 2021). Given the nature of the investigation and the experience gained in studying socioeconomic phenomena, the sample may be considered representative: although the empirical survey shows error magnitude of 0.7 and accuracy rate of 90%, in the absence of a previous survey, similar parameters for determining sample representativeness seem acceptable (Subedi, 2021). The relative share of all answers corresponded to a valid percentage.

The following analysis will concentrate on some of the collected data, specifically the answers to the questions Q03, Q04, Q05, Q06, Q07, Q11, and Q12. These questions are directly related to the theoretical overview provided in the first section of this paper.

3. Results of the empirical study on business development

This section starts with analysis of the distribution of the surveyed firms based on their responses to the preceding questions. Table 2 shows frequency and percentage, i.e. ***univariable*** distributions. In the first column, the question number corresponds to number initially presented in the questionnaire. The second column lists possible answers to each question. The last two columns show distributions. Using colour shading, the answers are graded according to their frequency and percentage: the darker the colour, the higher the frequency and percentage it represents. It is important to note that most of the responses generated for each question can be considered factors related to business development.

Some of the forthcoming tables incorporate abbreviations to facilitate convenient referencing: F – Frequency; Df – Degree of Freedom; Rr – Rarely; Of – Often; Rg – Regularly; Al – Always; Tt – Total; Symm. – Symmetric; AsS – Asymptotic significance (2-Sided); ApS – Approximate significance; C – Count; MEC – Minimum Expected Count.

Table 2
Distributions of responses received from the surveyed enterprises to questions Q03, Q04, Q05, Q06, Q11 and Q12 from the questionnaire

Question	Answers	Frequency	Percent
Q01	...	...	...
...	...	...	...
Q03	Micro enterprise (01-09 employees)	4	2.9
	Small enterprise (10-49 employees)	37	26.4
	Medium-sized enterprise (050-249 employees)	41	29.3
	Large enterprises (more than 250 employees)	58	41.4
	Total	**140**	**100.0**
Q05	Business to consumer (B2C)	78	55.7
	Business to business (B2B)	62	44.3
	Business to government (B2G)	0	0
	Total	**140**	**100.0**
Q05	Technological innovations	45	32.1
	Marketing	35	25.0
	Organizational innovations	18	12.9
	All types of innovations mentioned above	36	25.7
	None of the innovations mentioned above	6	4.3
	Total	**140**	**100.0**
Q06	Cost leadership strategy	47	33.6
	Differentiation strategy	66	47.1
	Mixed (hybrid) strategy	27	19.3
	Total	**140**	**100.0**
Q07	In a separate table		
...	...	...	...
Q11	Joint venture	24	17.1
	Public-private partnership	54	38.6
	Non-equity strategic alliance	17	12.1
	Equity strategic alliance	28	20.0
	No interest in participating in a strategic partnership	17	12.1
	Total	**140**	**100.0**
Q12	I don't need any further information.	18	12.9
	I just need some advice on some issues.	30	21.4
	I need advice on a wider range of issues.	92	65.7
	Total	**140**	**100.0**

Source: Own elaboration.

The results of question Q07 are presented in a separate table. It lists the frequency and percentage distribution of the responses based on the respondents' expressed preference for a specific level of importance like rarely, often, regularly, and always (Table 3).

Table 3
Extent to which BD activities are carried out

R-No.	Answer	F/%	Rr	Of	Rg	Al	Tt
071	We use business plans or similar documents for the preparation of growth opportunities.	F	28	34	52	26	140
		%	20.0	24.3	37.1	18.6	100.0
072	We use business models or similar methods for explicating ‘How do we make money?’.	F	33	33	53	21	140
		%	23.6	23.6	37.9	15.0	100.0
073	We have designated people to integrate intelligence across business functions, such as R & D, production and marketing.	F	30	30	56	24	140
		%	21.4	21.4	40.0	17.1	100.0
074	We have designated people preparing growth opportunities for senior management.	F	31	27	59	23	140
		%	22.1	19.3	42.1	16.4	100.0
075	We have designated people to conduct final due diligence, if necessary, prior to the executive decision to pursue growth opportunities.	F	26	33	59	22	140
		%	18.6	23.6	42.1	15.7	100.0
076	The people who prepared a growth opportunity support the implementation of it.	F	18	38	50	34	140
		%	12.9	27.1	35.7	24.3	100.0
077	We have designated people with knowledge about the growth opportunity monitoring and reporting on its progress to senior management.	F	20	31	29	60	140
		%	14.3	22.1	20.7	42.9	100.0

Source: Own elaboration.

The values obtained after summarizing the answers allow for the elaboration of a profile of the business practices in Dubai: prevalence of large and medium-sized companies (70.7%); most of the companies (66.4%) adhere to the pursuit of a differentiation or mixed strategy; as business type, most of the companies (55.7%) are B2C oriented; most of the enterprises (57.8%) are pursuing technological innovation; 87.9% of enterprises are interested in cooperation with other companies; most companies carry out business development activities regularly.

The extent to which the above factors appear significant to business development can be determined by conducting dependency research using the ***bivariate distribution*** method. Bivariate distributions arise when the responses to one question (factor) are cross-referenced with the responses to another question (result). The comprehension of bivariate distributions is facilitated by χ (chi)-square tests. The reason for selecting the latter is that both the factors and the outcomes possess a qualitative variable typological characteristic.

Table 4
Size of the organization / Business development activities

Q03	**Q071**					**Q03 ↔ Q071**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					25%	.74
031	2	1	1	0	4	Pearson	58.842	9	.002			
032	14	15	7	1	37	Likelih. Ratio	57.729	9	.002			
033	3	5	30	3	41	Symm. Meas.						
034	9	13	14	22	58	Phi	.648			.002		
Tt	28	34	52	26	140	Cramer's V	.374			.002		
Q03	**Q072**					**Q03 ↔ Q072**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					25%	.60
031	2	1	1	0	4	Pearson	65.505	9	.002			
032	20	10	6	1	37	Likelih. Ratio	66.706	9	.002			
033	1	9	30	1	41	Symm. Meas.						
034	10	13	16	19	58	Phi	.684			.002		
Tt	33	33	53	21	140	Cramer's V	.395			.002		
Q03	**Q073**					**Q03 ↔ Q073**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					25%	.70
031	2	1	1	0	4	Pearson	63.079	9	.002			
032	18	10	7	2	37	Likelih. Ratio	63.882	9	.002			
033	1	8	31	1	41	Symm. Meas.						
034	9	11	17	21	58	Phi	.671			.002		
Tt	30	30	56	24	140	Cramer's V	.388			.002		
Q03	**Q074**					**Q03 ↔ Q074**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	Of	Rg	Al	Tt	χ-square test					25%	.67

031	2	0	2	0	4	Pearson	52.920	9	.002			
032	19	10	7	1	37	Likelih. Ratio	55.462	9	.002			
033	1	7	29	4	41	Symm. Meas.						
034	9	10	21	18	58	Phi	.615			.002		
Tt	**31**	**27**	**59**	**23**	140	Cramer's V	.355			.002		
Q03	**Q075**					**Q03 ↔ Q075**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					25%	.65
031	2	0	2	0	4	Pearson	66.921	9	.002			
032	17	12	7	1	37	Likelih. Ratio	66.296	9	.002			
033	3	6	31	1	41	Symm. Meas.						
034	4	15	19	20	58	Phi	.691			.002		
Tt	26	33	59	22	140	Cramer's V	.400			.002		
Q03	**Q076**					**Q03 ↔ Q076**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					31%	0.52
031	2	0	2	0	4	Pearson	71.935	9	.002			
032	12	17	7	1	37	Likelih. Ratio	72.703	9	.002			
033	1	4	29	7	41	Symm. Meas.						
034	3	17	12	26	58	Phi	.720			.002		
Tt	18	38	50	34	140	Cramer's V	.415			.002		
Q03	**Q077**					**Q03 ↔ Q077**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					25%	0.58
031	0	2	1	1	4	Pearson	46.407	9	.002			
032	13	15	2	7	37	Likelih. Ratio	49.638	9	.002			
033	3	1	15	22	41	Symm. Meas.						
034	4	13	11	30	58	Phi	.576			.002		
Tt	20	31	29	60	140	Cramer's V	.332			.002		

Source: Own elaboration.

The size of the organization (Q03, answers 031-034) and the business development activities (Q07, answers 071 to 077) have, albeit with some conditionality, a relationship with each other as the significance level of Pearson's χ-square is 0.002 and is less than the error α=0.05. The conditionality results from the fact that the conditions (i.e. MEC>1 and at the same time C<5 – less than 20%) for the application of the χ-square are not met. The relationships appear average in strength, because the Cramer's coefficients fall between 0.3 and 0.7 and are statistically significant. The result is logical since, as a rule, larger firms have more capacity to undertake diverse business development activities.

Table 5
Type of business / Business development activities

Q04	**Q071**					**Q04 ↔ Q071**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	11.51
041	23	7	35	13	78	Pearson	28.105	3	.002			
042	5	27	17	13	62	Likelih. Ratio	29.628	3	.002			
043	0	0	0	0	0	Symm. Meas.						
Tt	28	34	52	26	140	Phi	.448			.002		
						Cramer's V	.448			.002		
Q04	**Q072**					**Q04 ↔ Q072**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	9.31
041	20	14	36	8	78	Pearson	8.527	3	.036			
042	13	19	17	13	62	Likelih. Ratio	8.591	3	.035			
043	0	0	0	0	0	Symm. Meas.						
Tt	33	33	53	21	140	Phi	.247			.036		
						Cramer's V						
Q04	**Q073**					**Q04 ↔ Q073**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	10.31
041	20	17	32	9	78	Pearson	4.743	3	.192			
042	10	13	24	14	62	Likelih. Ratio	4.763	3	.190			
043	0	0	0	0	0	Symm. Meas.						
Tt	30	30	56	21	140	Phi	.184			.192		
						Cramer's V	.184			.192		
Q04	**Q074**					**Q04 ↔ Q074**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	10.19
041	20	12	39	7	78	Pearson	10.900	3	.12			
042	11	15	20	16	62	Likelih. Ratio	10.999	3	.12			
043	0	0	0	0	0	Symm. Meas.						
Tt	31	27	59	23	140	Phi	.279			.12		
						Cramer's V	.279			.12		
Q04	**Q075**					**Q04 ↔ Q075**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	9.74
041	18	15	36	9	78	Pearson	5.960	3	.114			

042	8	18	23	13	62	Likelih. Ratio	6.007	3	.111			
043	0	0	0	0	0	Symm. Meas.						
Tt	26	33	59	23	140	Phi	.206			.114		
						Cramer's V	.206			.114		
Q04	**Q076**					**Q04 ↔ Q076**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	7.97
041	14	14	32	18	78	Pearson	10.534	3	.15			
042	4	24	18	16	62	Likelih. Ratio	10.805	3	.13			
043	0	0	0	0	0	Symm. Meas.						
Tt	18	38	50	34	140	Phi	.274			.15		
						Cramer's V	.274			.15		
Q04	**Q077**					**Q04 ↔ Q077**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					0%	8,86
041	17	11	16	34	78	Pearson	12.120	3	.007			
042	3	20	13	26	62	Likelih. Ratio	13.017	3	.005			
043	0	0	0	0	0	Symm. Meas.						
Tt	20	31	29	60	140	Phi	.294			.007		
						Cramer's V	.294			.007		

Source: Own elaboration.

The data reveal that business type (Q04, answers 041-042) and business development activities (Q07, answers 071–077) are not necessarily related. The type of business is moderately or weakly associated with the use of business plans, models and people for monitoring and reporting business development progress (071–077). Other combinations indicate no correlation between business type and development activities (Pearson's χ-square significance level > α=0.05). Although the findings are unconditional (MEC>1 and C<5 – less than 20%), they are weak (Cramer's coefficients < 0.3 and is statistically significant). The differences between the last two cross-examinations are obvious. Most businesses are C2C, which may explain the above fact.

Table 6
Type of innovation / Business development activities

Q05	**Q071**					**Q05 ↔ Q071**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					35%	1.11
051	2	11	28	4	45	Pearson	65.578	12	.002			
052	12	15	8	0	35	Likelih. Ratio	72.537	12	.002			
053	4	6	3	5	18	Symm. Meas.						
054	9	2	8	17	36	Phi	.684			.002		
055	1	0	5	0	6	Cramer's V	.395			.002		
Tt	28	34	52	26	140							
Q05	**Q072**					**Q05 ↔ Q072**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					35%	**.90**
051	3	10	28	4	45	Pearson	52.943	12	.002			

052	12	15	8	0	35	Likelih. Ratio	57.909	12	.002			
053	8	2	3	5	18	Symm. Meas.						
054	9	6	9	12	36	Phi	.615			.002		
055	1	0	5	0	6	Cramer's V	.355			.002		
Tt	33	33	53	21	140							
Q05	**Q073**					33	53	21	140	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					35%	1.03
051	2	9	30	4	45	Pearson	58.620	12	.002			
052	12	8	15	0	35	Likelih. Ratio	68.494	12	.002			
053	7	3	3	5	18	Symm. Meas.						
054	8	10	3	15	36	Phi	.647			.002		
055	1	0	5	0	6	Cramer's V	.374			.002		
Tt	30	30	56	24	140							
Q05	**Q074**					**Q05 ↔ Q074**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					35%	1.00
051	3	10	29	3	45	Pearson	37.459	12	.002			
052	12	9	11	3	35	Likelih. Ratio	39.797	12	.002			
053	7	3	3	5	18	Symm. Meas.						
054	8	5	11	12	36	Phi	.517			.002		
055	1	0	5	0	6	Cramer's V	.300			.002		
Tt	31	27	59	23	140							
Q05	**Q075**					**Q05 ↔ Q075**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					35%	0.94
051	2	12	28	3	45	Pearson	59.517	12	.002			
052	8	16	11	0	35	Likelih. Ratio	67.322	12	.002			
053	7	3	3	5	18	Symm. Meas.						
054	9	1	12	14	36	Phi	.652			.002		
055	0	1	5	0	6	Cramer's V	.376			.002		
Tt	26	33	59	22	140							
Q05	**Q076**					**Q05 ↔ Q076**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					45%	.77
051	2	11	27	5	45	Pearson	59.741	12	.002			
052	5	17	7	6	35	Likelih. Ratio	50.412	12	.002			
053	3	6	4	5	18	Symm. Meas.						
054	8	3	7	18	36	Phi	.696			.002		
055	0	1	5	0	6	Cramer's V	.345			.002		
Tt	18	38	50	34	140							
Q05	**Q077**					**Q05 ↔ Q077**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	Rr	**Of**	**Rg**	**Al**	**Tt**	χ-square test					35%	.87
051	0	12	11	22	45	Pearson	27.377	12	.007			
052	6	13	7	9	35	Likelih. Ratio	37.777	12	.002			
053	4	5	3	6	18	Symm. Meas.						
054	9	1	7	19	36	Phi	.444			.007		
055	1	0	1	4	6	Cramer's V	.277			.007		
Tt	20	31	29	60	140							

Source: Own elaboration.

The innovation activities of the organization (Q05, answers 051-055) and the development activities (Q07, answers 071 to 077) have, albeit with some conditionality, a relationship with each other as the significance level of χ-square is 0.002 and less than the error $\alpha = 0.05$. The conditionality results from the fact that the conditions (i.e. MEC>1 and at the same time C<5 is less than 20%) for the application of the χ-square are not met. The relationships appear average in strength, because the Cramer's coefficients fall in the range 0.3–0.7 and are statistically significant.

Since innovation is essential to organic development, the surveyed organizations' inventions and business development activities are linked. The bivariate link is particularly clearer because 58% (Table 2) of the organizations surveyed practice technological innovation, which greatly influences the latter. The relationship is medium-strong, presumably because non-technological innovation is still too high and small enterprises do not regularly engage in company development. Regarding interest in inorganic forms of business development, most companies surveyed express such. Only 12.1% of respondents say they don't want strategic partnerships (Table 2). Table 7 shows a correlation between innovation and strategic alliance or partnership interest. Furthermore, the desire for legal-economic consultancy in strategic partnership sectors supports this idea. The cross-tabulation association is considerable (Table 8).

Table 7
Type of innovation / Type of strategic collaboration interest

Q05	**Q11**						**Q05 ↔ Q11**	**Value**	**Df**	**AsS**	**ApS**	**C<5**	**MEC**
	1	**2**	**3**	**4**	**5**	**Tt**	χ-square test					52%	.75
051	9	18	2	8	8	**45**	Pearson	68.273	16	.001			
052	3	20	7	2	3	**35**	Likelih. Ratio	65.043	16	.001			
053	9	5	0	2	2	**18**	Symm. Meas.						
054	2	10	8	16	0	**36**	Phi	.698			.001		
055	1	1	0	0	4	**6**	Cramer's V	.350			.001		
Tt	**24**	**54**	**17**	**28**	**17**	**140**							

Source: Own elaboration.

Table 8
Type of strategic collaboration interest / Needs for additional consultancy

Q11	Q12				Q05 ↔ Q12	Value	Df	AsS	ApS	C<5	MEC
	111	**112**	**113**	**Tt**	χ-square test					39.8%	2.19
111	0	0	24	**24**	Pearson	167.527a	8	.002			
112	1	8	45	**54**	Likelih. Ratio	131.490	8	.002			
113	0	12	5	**17**	Symm. Meas.						
114	0	10	18	**28**	Phi	1.094			.002		
115	17	0	0	**17**	Cramer's V	.774			.002		
Tt	**18**	**30**	**92**	**140**							

Source: Own elaboration.

The topic's importance prompts search for additional dependencies to broaden Dubai's business development perspective. This search suggests clustering *n* objects (enterprises) into *k* clusters using *p* properties (variables). The method also needs pre-determining the number of clusters (e.g. two) and applying the K-Means Cluster Technique with default iteration of 10 and similarity threshold of 0.02 (SPSSanalysis, 2024).

Different criteria can be used for the purpose of grouping of the enterprises surveyed. The choice falls on the following questions: Q05 "Which type of innovation did you implement in your company in the last 3 years?"; Q06 "What competitive strategy do you use in developing your business?"; and Q04 "What type of business is your company conducting?". SPSS makes it easier to derive the two clusters. Their centres are based on the three criteria that were chosen (Table 9).

Table 9
Initial cluster centres

	Cluster	
	1	**2**
Which type of innovation did you implement in your company in the last 3 years?	5	1
What competitive strategy do you use in developing your business?	1	3
What type of business is your company conducting?	1	2

Source: Own elaboration.

Three iterations are performed, the results of which are presented in Table 10.

Table 10
Iteration History

Iteration	Change in Cluster Centres[a]	
	1	2
1	1.488	1.413
2	.136	.144
3	.000	.000

a. Convergence achieved due to no or small change in cluster centres. The maximum absolute coordinate change for any centre is .000. The current iteration is 3. The minimum distance between initial centres is 4.583.
Source: Own elaboration.

The computational procedures find reflection in the following final cluster centers (Table 11).

Table 11
Final cluster centres

	Cluster	
	1	2
Which type of innovation did you implement in your company in the last 3 years?	4	1
What competitive strategy do you use in developing your business?	2	2
What type of business is your company conducting?	1	2

Source: Own elaboration.

The distribution of the surveyed enterprises by clusters is presented in Table 12.

Table 12
Number of Cases in each Cluster

Cluster	1	60
	2	80
Valid		140
Missing		0

Source: Own elaboration.

After applying the cluster procedure, generalizations can be made. As final cluster centres for the ***first cluster*** the following are formed: to the question “Which type of innovation did you implement in your company in the last 3 years?” – position 4 “All types of innovations mentioned above”; to the question "What competitive strategy do you use in developing your business?" – position 2 "Differentiation strategy"; to the question "What type of business is your company conducting?" – position 1 "Business to consumer (B2C)". As final cluster centres for the ***second cluster*** the following are formed: to the question "Which type of innovation did you implement in your company in the last 3 years?" – position 1 "Technological innovations"; to the question "What competitive strategy do you use in developing your business?" – position 2 "Differentiation strategy"; to the question "What type of business is your company conducting?" – position 1 "Business to business (B2B)".

The results of the cluster analysis indicate that there are two clusters that can be distinguished: (1) An "Innovation Diversity Cluster". This cluster is geared toward the B2C business type and employs the differentiation strategy. It represents 60 of 140 businesses; (2) A "Technology Innovation Cluster". This cluster is oriented toward B2B business type and prefers the differentiation strategy. It includes 80 of the 140 companies.

The emergence of innovation-driven clusters to shape corporate differentiation strategy development attitudes is not surprising. A recent study (Mihaylova et al., 2022) has concluded that the Dubai government is actively working to enable locally registered enterprises to confidently supply high-quality goods and services. This was achieved by supporting in-house or collaborative R&D in the Dubai Science Park, greenfield investments in high-tech companies and free zones for quick business expansion.

Summary and conclusion

Business development is often employed in a broad context. The strategic management theory restricts the use of the phrase to single-business enterprises exclusively. To delineate the strategic features of corporate development, it is essential to evaluate not just growth and progress but also its modalities and origins, along with certain organizational activities.

This study focuses primarily on organic and inorganic sources of business development, looking for a link between firm size, business types, innovation and business development activities. Based on data collected from a specific region (the Emirate of Dubai), it has been concluded that the regular implementation of business development activities is more related to the size of the firm and the type of innovation it conducts, given that most of the surveyed companies regularly carry out business development activities. The relationship between business type and business development activities appears to be less pronounced.

For firms operating in the Emirate of Dubai, differentiation is the most common competitive strategy. It is pursued by both B2C and B2B oriented firms. This is primarily due to the local business environment, which promotes and encourages modern and high-quality businesses.

Organic business development has strong support among the surveyed businesses. However, there is also interest in inorganic business development strategies, particularly public-private partnerships.

The findings can assist in identifying, selecting and implementing specific advantages, as well as revealing opportunities to enhance strategic management frameworks and systems. This will lay the groundwork for more effective strategic decision-making in business development.

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CONTENTS

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IMPACT OF BUSINESS INTELLIGENCE ON THE ECONOMIC GROWTH OF SMALL AND MEDIUM-SIZED ENTERPRISES (SMES): EVIDENCE FROM THE LIMA PLAZA CASE

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Abstract: In a context where digital transformation is a strategic challenge for the sustainability of small and medium-sized enterprises (SMEs), business intelligence (BI) emerges as a key factor in improving decision-making and fostering competitive advantages. This study analyzed the relationship between the implementation of BI solutions and business growth in 525 SMEs located in Lima, Peru. A quantitative, applied, and experimental methodology was used with descriptive and inferential analysis (Spearman Rho and Wilcoxon tests). The results reveal a strong positive correlation (Rho = 0.775, $p < 0.01$) between BI adoption and SME growth. Limitations include the geographic concentration of the sample, which may affect generalizations to other areas of the country. Nevertheless, the findings contribute to the literature by offering empirical evidence of the real-world effects of BI tools in emerging business environments and suggest that digital capabilities should be promoted in micro and small enterprises. Future research is encouraged to expand the scope to other economic sectors and regions.

Key words: Business intelligence; SME; economic; growth
JEL: M1, M2, O3, O4.
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Introduction

In today's business world, business intelligence plays a vitally important role both in visualizing an overview and obtaining competitive advantages, and in surviving in such a rigorous and changing market. In this context, the author Al Daabseh et al. (2023) indicated that it is necessary to understand the crucial activities for an appropriate approach to its application. The objective of this study was to determine the prevalence between the implementation of a business intelligence solution and the growth rate of an SME. The authors Ogliastri et al. (2023) considered it important to evaluate and identify the factors that have the greatest influence on the continuity of business operations. To develop the study, the following actions were carried out: descriptive and inferential analyses were generated for all the variables considered in the study, and hypothesis tests were used to determine the relationship between the implementation of a SBI and growth. This study contributed to the scientific literature by collecting various sources, analyzing them, and generating a body of knowledge on the relationship between business intelligence and the business environment. The business growth index was also identified as a key factor for business development. The study also identified variables that require certain conditions to influence business growth. The following hypothesis was considered: The implementation of a business intelligence solution is significantly related to the growth of SMEs. Furthermore, despite technological advances and the availability of data analysis tools, many small and medium-sized enterprises (SMEs) in Metropolitan Lima still exhibit poor economic growth. This raises the central question of this research: How does the implementation of business intelligence solutions significantly influence the growth of SMEs located in the Lima metropolitan area? This question guides the study to find evidence that allows us to establish a relationship between the two.

1. Related Research

Several recent studies have explored the influence of business intelligence (BI) and digital transformation on the performance of small and medium-sized enterprises (SMEs). Al Daabseh et al. (2023) investigated the relationship between BI capabilities and business outcomes in SMEs, concluding that the strategic use of competitive intelligence improves decision-making and boosts organizational growth. The author emphasizes that BI enables data to be transformed into competitive advantages, especially in highly uncertain environments. Meanwhile, Azevedo and Almeida (2021) designed a course aimed at decision-makers in SMEs to strengthen their digital capabilities. Their study showed that technical training in BI and data analytics facilitates a better digital transition and promotes a culture of evidence-based decision-making. Along the same lines, Lada et al. (2023) identified the key factors for the adoption of artificial intelligence in Malaysian SMEs, highlighting that technological knowledge, perception of usefulness, and environmental pressure are key determinants. The study suggests that BI adoption is influenced by organizational and attitudinal variables.

Diaz and Galdon (2023) developed a group decision-making model based on ambiguous logic and AHP, applied to measuring the level of digital maturity. They concluded that organizations with greater BI use tend to have higher levels of digital maturity, which translates into better adaptive capacity and strategic vision. Mero et al. (2022) addressed agile logic for the implementation of marketing automation software in startups, finding that the use of advanced digital solutions (including BI platforms) accelerates technological adoption and improves alignment with business objectives. Pelekamoyo and Libati (2023) proposed a smart geolocated application for market predictions aimed at SMEs. They concluded that the use of mobile BI tools improves the accuracy of processed information and strengthens real-time data visualization, contributing to more efficient decisions. Kamaruddin et al. (2020) analyzed data visualization-based business solutions for SMEs, demonstrating that these tools improve financial and operational analysis. The study highlights the need to modernize traditional methods with digital solutions that integrate BI.

Authors Lundberg and Oberg (2021) mention that the concept of business intelligence has been integrated into an organization's control systems in the competitive environment, and its functions include identifying and monitoring the various changes in said environment. Hassani and Mosconi (2021) consider that SMEs have financial limitations, are conservative, inflexible, and do not consider innovation. Likewise, Strilets et al. (2022) consider that SMEs are not prepared for new technologies and expectations. Furthermore, barriers were identified that could limit SMEs when implementing new technologies, including a lack of digital capabilities, skilled labor and financial resources, standardization issues, and cybersecurity issues. Cuevas-Vargas et al. (2016) indicated that the identified factors must be related to the company's specific context. The following factors were identified: year of operation, initial capital, personnel, sector, partnerships, and access to the international market.

Finally, Silva de Mattos et al. (2023) conducted a systematic review of technological transformation in SMEs, focusing on the adoption of emerging technologies and business model innovation. The authors concluded that the effective integration of tools such as BI is essential to strengthen competitiveness, improve adaptation to the environment, and sustain business growth. Overall, the reviewed studies highlight the positive impact of using business intelligence on organizational performance. There is a lack of applied research in the Latin American context, especially with representative samples of formal SMEs in shopping centers. This research aims to fill this gap by providing empirical evidence on a specific segment of Peruvian SMEs with technological growth potential.

2. Methodology

2.1 Population and Study

The target population for this study is made up of small and medium-sized enterprises (SMEs) located in the Lima Plaza in Metropolitan Lima, Peru. These companies operate in diverse sectors such as technology (IT), services, clothing, and crafts, which allows for capturing a diverse range of levels of digitalization, technological adoption, and organizational performance. A total of 525 active SMEs were identified within the shopping center at the time of the study, which constituted the population framework. To ensure the statistical representation of the results, simple probability sampling with a finite population formula was applied. A confidence level of 95% and a margin of error of 5% were established. The formula used was the following:

$$n = \frac{z^2 Npq}{e^2(N-1)+z^2 pq} \qquad (1)$$

n=sample, z=desv , p=prob. (0.5), q=0.5, N=525, e=0.05 máx. Error

Applying the formula, a sample size of 159 SMEs was obtained, allowing for statistical analyses with sufficient inferential power. The selected units of analysis met the following inclusion criteria: (i) Having been in continuous operation for at least one year, (ii) Belonging to one of the four previously defined sectors, and (iii) Voluntarily agreeing to participate through informed consent. It should be noted that the sample consists exclusively of SMEs located in the Lima Plaza shopping mall, which could limit the study's external validity. Although it allows for controlling for certain contextual variables at the Lima regional level due to the common nature of SMEs, this geographic delimitation restricts the generalization of the results to other areas of Peru. Further studies in various urban and rural contexts are recommended to compare the findings.

2.2 Types of variables

Eight variables were identified for the study:

Dependent Variable. Growth Index (GI): This variable was composed of business development indicators: revenue growth, staff expansion, product diversification, brand consolidation, infrastructure growth, and management improvement.

Main Independent Variable. Business Intelligence Solutions (SBI): This variable was dichotomous (0 = does not implement / 1 = does implement) and refers to the use of technological tools and platforms for data analysis, visualization, report generation, and evidence-based decision-making.

Complementary or Moderating Variables. These variables were used to characterize the companies and evaluate their potential moderating or controlling effect on the main relationship between SBI and CI. The terms 'business intelligence solution' (SBI) and 'business intelligence' are considered interchangeable, both referring to the set of technological tools and processes that enable the collection, analysis, and visualization of data to support strategic decision-making.

Table 1
Variables, type and scale

Variable	Abreviatura	Tipo	Escala
Business Intelligence Solution	SBI	Qualitative	Nominal
Growth Rate	IC	Quantitative	Ratio
Years of Operation	AO	Quantitative	Ratio
Initial Capital	CI	Qualitative	Nominal
Amount of Personnel	CP	Qualitative	Ratio
Industry	R	Qualitative	Nominal
Partner CC	SCC	Qualitative	Nominal
Activities Abroad	AE	Qualitative	Nominal

Source: The authors

For the study, an instrument was used to collect data using specific questions and according to the context of the variables under study. These questions are necessary for statistical calculations and obtaining results. To validate the instrument, Cronbach's alpha coefficient was calculated, obtaining a value of 0.84, indicating a high level of reliability. The authors Tavakol y Dennick (2011) specified that it is a statistical measure used to evaluate the internal reliability of a set of questions. The study was developed in an applied way, the authors González and Rojas (2020) specified that the applied type of study emphasizes the theoretical framework and with this, theories or updates to what was presented can be formulated. Likewise, according to the authors Hernández-Sampieri and Mendoza (2018), the applied type study aims to incorporate new knowledge and grow scientific understanding. The study was developed with an experimental design, and a quantitative approach.

3. Results

3.1 Data organization

Data analysis was carried out in two fundamental stages. The first consisted of preliminary information processing. Authors Romero et al. (2022) argue that adequate structuring of the database is essential to ensure the internal validity of the study. The second stage included statistical analysis, applying descriptive and inferential tests. This phase incorporated nonparametric tests such as Spearman's correlation and the Wilcoxon test, which are appropriate for non-normal data and distributions, as proposed by Tavakol and Dennick (2011) in studies that seek to guarantee the reliability of measurements. Furthermore, scientific rigor is not limited to technical or statistical aspects but must be accompanied by fundamental ethical principles. In this sense, Hirsch and Navia (2018) emphasize that all research must respect the principles of responsibility, confidentiality, informed consent, and accuracy of sources. Toro et al. (2023) identified specific ethical challenges in the Latin American context, such as the need to strengthen institutional capacities. Finally, Gamboa-Bernal (2024) highlighted the relevance of bioethics in scientific research, proposing an approach that integrates human rights and social justice as fundamental pillars for ethical conduct in research.

The results were processed and analyzed using the SPSS tool, which is precise for the analysis of data and complex processes. Thus, descriptive and inferential results were obtained, with correlations generated using the Spearman's Rho test. Osada et al. (2021) emphasize the importance of establishing correlations between variables to identify causes and trends related to a specific topic. Similarly, Palella and Martins (2006) indicate that relevant variables are those that guide the development of a study, but they depend greatly on the environment where it was executed and the way in which the data were obtained. In this sense, it is specified that it is a nonparametric test with abnormal distribution. Likewise, the Kolmogorov-Smirnov normality test was considered, which was indicated by Landa and Arriaga (2017) to verify whether the sample scores do not follow a normal distribution, proceeding to be considered a nonparametric test. Authors Lee and Jeong (2016) specified that it allows for the identification of significant deviations from the normal distribution. The results indicated that they did not meet the assumption of normality, so nonparametric statistical tests such as Spearman and Wilcoxon were used for inferential analysis.

Table 2
Normality test

	Kolmogorov-Smirnov		
	Estadistico	gl	Sig.
SBI	,445	159	,000
Indice	,219	159	,000

Source: The authors, SPSS Output

Table 2 presents the results obtained using the Kolmogorov-Smirnov normality test, considering the variables: SBI and CI. The p-value was = 0.000 < α = 0.05. Furthermore, a normal distribution was not obtained, so the Spearman correlation coefficient (nonparametric) was used, which is recommended by Lamb et al. (2017).

3.2 Descriptive results

Authors Torrico et al. (2018) specify that descriptive statistics allow a data set to be described and managed to facilitate the analysis of the results obtained from the application of the instruments. The study was conducted with data from a sample of 159 SMEs. To identify descriptive results, 5 reference variables were used: SBI, AO, IC, R and SCC.

Table 3
Results SBI

		Frecuencia	%	% valido	% acumulado
Válido	0	47	29,6	29,6	29,6
	1	112	70,4	70,4	100,0
	Total	159	100,0	100,0	

Source: The authors, SPSS Output

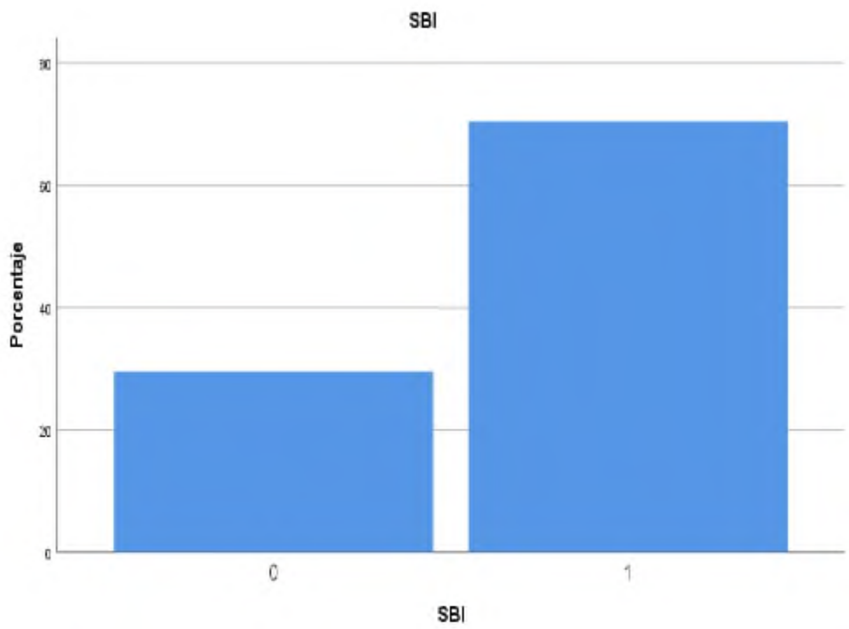

Figure 1. Results SBI
Source: The authors, SPSS Output

Table 3 shows results obtained for the SBI variable, showing that 112 SMEs (70.4%) had implemented a business intelligence solution, while 47 SMEs (29.6%) stated that they had not. The total was 159 SMEs (100%).

Table 4
Results AO

		Frecuencia	%	% valido	% acumulado
Válido	1	46	28,9	28,9	28,9
	2	32	20,1	20,1	49,1
	3	1	,6	,6*	49,7
	4	9	5,7	5,7	55,3
	5	34	21,4	21,4	76,7
	6	26	16,4	16,4	93,1
	10	11	6,9	6,9	100,0
	Total	159	100,0	100,0	

Source: The authors, SPSS Output

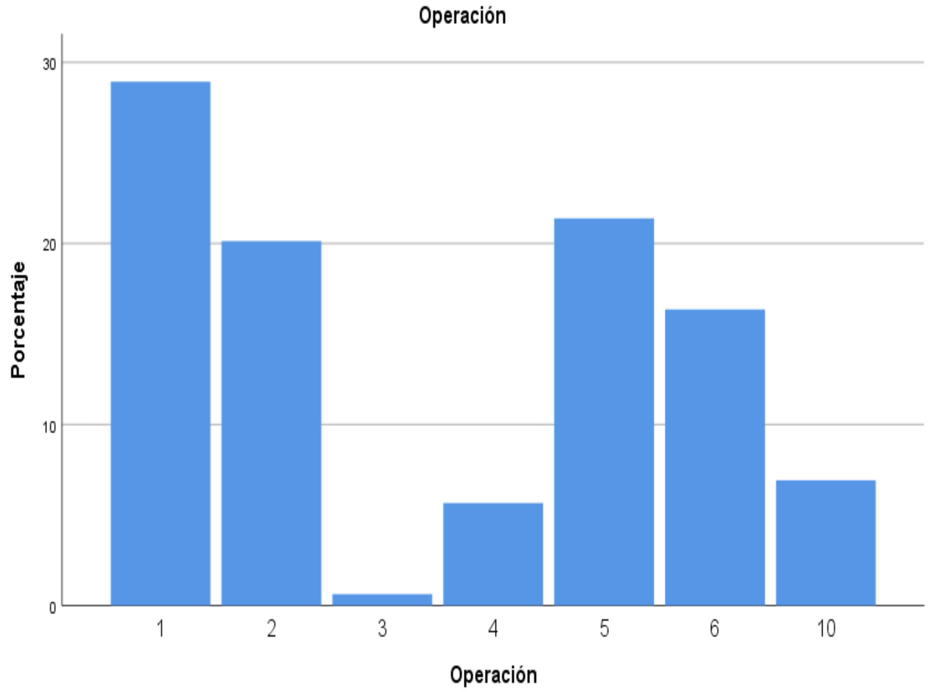

Figure 2. Results AO
Source: The authors, SPSS Output

Table 4 presents results for the variable AO, showing that the largest number was 46 SMEs (28.9%), which had been in commercial operation for 1 year, and the smallest number was 1 SME (0.6%), which had been in commercial operation for 3 years. The total was 159 SMEs (100%).

Table 5
Results IC

		Frecuencia	%	% valido	% acumulado
Válido	0	24	15,1	15,1	15,1
	1	4	2,5	2,5	17,6
	2	33	20,8	20,8	38,4
	3	30	18,9	18,9	57,2
	4	10	6,3	6,3	63,5
	5	56	35,2	35,2	98,7
	6	2	1,3	1,3	100,0
	Total	159	100,0	100,0	

Source: The authors, SPSS Output

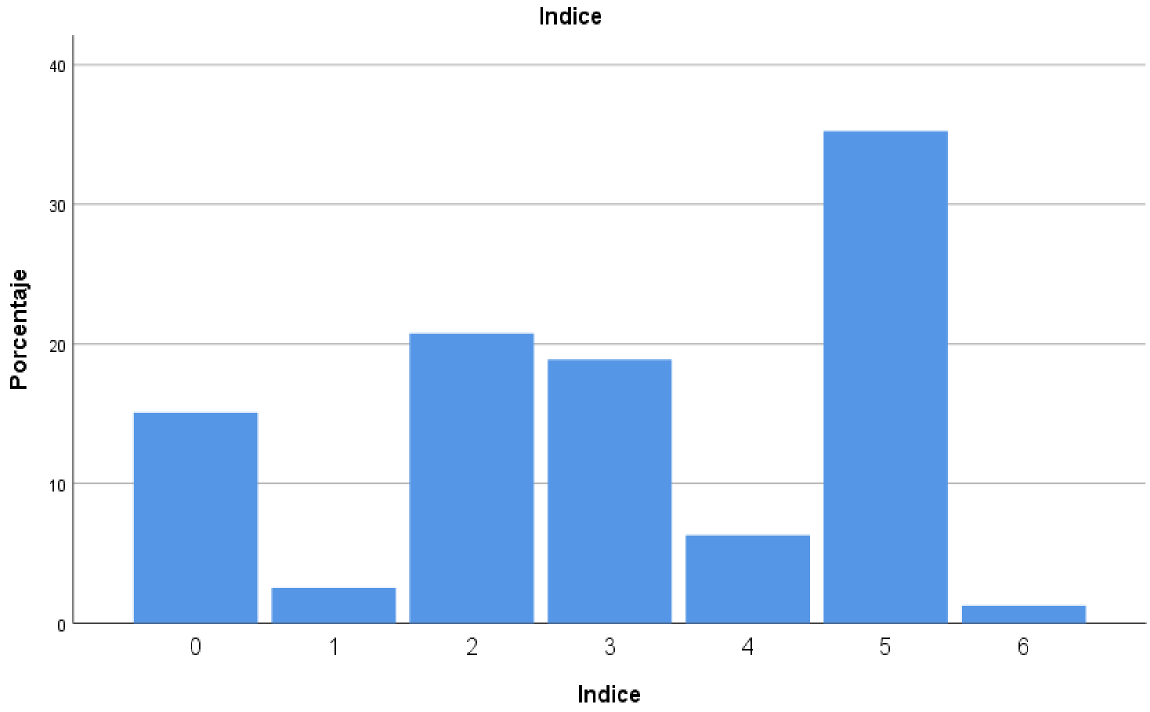

Figure 3. Results IC
Source: The authors, SPSS Output

Table 5 presented descriptive results of the IC variable, where it was found that the largest number was 56 SMEs (35.2%), which have an IC of 5 and the smallest number was 2 SMEs (1.3%), which have an IC of 6. Likewise, it was identified that 26 SMEs (15.1%) have an IC of 0. The total was 159 SMEs (100%).

Table 6
Results R

		Frecuencia	%	% valido	% acumulado
Válido	Artesanía	28	17,6	17,6	17,6
	Ropa	34	21,4	21,4	39,0
	Servicios	17	10,7	10,7	49,7
	TI	80	50,3	50,3	100,0
	Total	159	100,0	100,0	

Source: The authors, SPSS Output

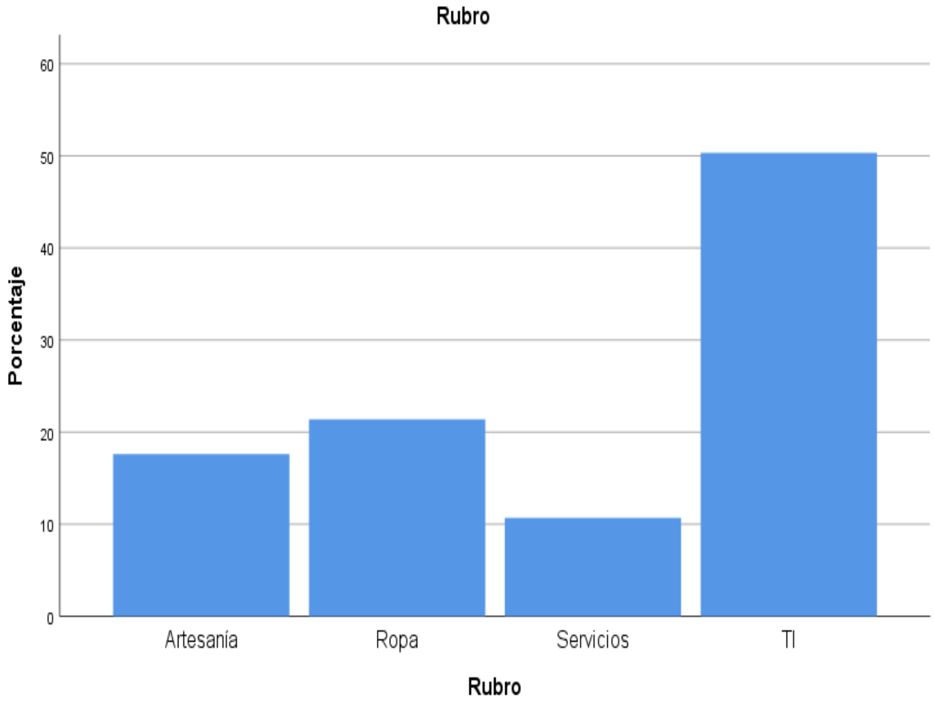

Figure 4. Resultads R
Source: The authors, SPSS Output

Table 6 presented descriptive results of the variable sector (R), where it was found that the largest number is 80 SMEs (53.3%), which are in the IT sector and the smallest number is 17 SMEs (10.7%), which are in the services sector. The total sample under study was 159 SMEs (100%).

Table 7
Results SCC

		Frecuencia	%	% valido	% acumulado
Válido	0	54	34,0	34,0	34,0
	1	105	66,0	66,0	100,0
	Total	159	100,0	100,0	

Source: The authors, SPSS Output

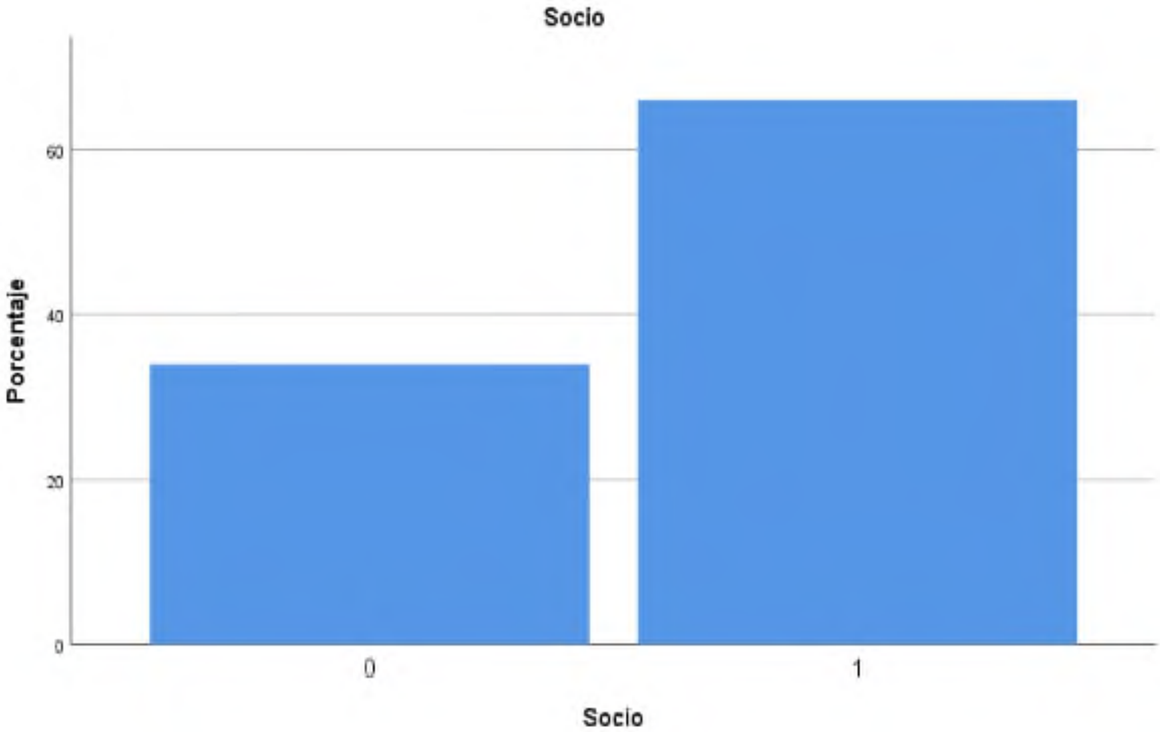

Figure 5. Resultados SCC
Source: The authors, SPSS Output

Table 7 presents descriptive results for the variable "chamber of commerce member" (CCM), showing that 105 SMEs (66.0%) belong to the chamber of commerce, and 54 SMEs (34.0%) state that they do not. The total sample size studied was 159 SMEs (100%).

The sample consisted of 159 companies located in the Lima Plaza gallery, with sectoral diversity represented mainly in the technology (IT), services, clothing, and crafts sectors. Regarding company age, 42.8% reported being between 5 and 10 years old, while 33.1% reported less than 5 years, demonstrating a prevalence of young or emerging businesses. In terms of initial capital, 55.3% had investments of up to S/10,000, suggesting a predominance of micro-business-based ventures.

Regarding the adoption of business intelligence (BI) solutions, it was found that 70.4% of the surveyed SMEs implemented some type of BI tool, ranging from advanced spreadsheets to commercial platforms such as Power BI, Tableau, or custom developments. Of these companies, 35.2% had a growth index of 5 out of 6, indicating outstanding business performance. This trend supports the hypothesis of a potential positive association between BI use and organizational growth. Regarding the growth index (GI), a positive skewed distribution was observed. The majority of companies (61.6%) had scores between 4 and 6, reflecting a favorable perception of business expansion and sustainability, particularly in areas such as increased sales, employee recruitment, process digitization, and brand recognition. Regarding staff size, 67.9% of companies operate with fewer than 10 employees, while only 4.4% have more than 50 employees. This structure confirms that Lima Plaza's business network is mostly made up of micro and small businesses, with limited scaling capabilities but with growth potential through technological innovation.

Another relevant finding was the limited institutional ties: only 38.4% of companies are affiliated with a chamber of commerce or trade association, which could limit access to support programs, business training, or financing. Furthermore, only 9.4% reported participating in international fairs or export activities, reflecting a low level of internationalization.

Finally, the analysis identified a gap in technical training in business intelligence, as more than 40% of respondents indicated they lack formal knowledge in the strategic use of data. This represents a key opportunity for training, incubation, and competitive development programs.

3.3 Inferencial results

Procedures for Hypothesis Testing

To test the hypothesis, it was necessary to verify whether there was a relationship between an SBI and the CI. In this regard, the analysis was resolved by calculating the values of the aforementioned variables and performing a test on them, which was performed using a statistical measure to identify whether there were significant differences between the variables. Likewise, the Kolmogorov-Smirnov normality test determined that the data did not have a normal distribution; therefore, the Spearman correlation was used to determine the relationship between the variables and the nonparametric Wilcoxon statistical measure.

Hypothesis Formulation

H0: There is no relationship, H1: There is a relationship

Statistical Significance Level and Confidence Level

The Kolmogorov-Smirnov normality test determined that a p-value of $0.000 < \alpha = 0.05$ for the SBI and IC variables. The results showed that the distribution was not normal. Resolving this test, the results obtained are:

Table 8
*Cross table SBI * IC*

		Indice							Total
		0	1	2	3	4	5	6	
SBI	0	24	4	19	0	0	0	0	47
	1	0	0	14	30	10	56	2	112
Total		24	4	33	30	10	56	2	159

Source: The authors, SPSS Output

Table 9
Wilcoxon nonparametric test

	SBI * IC
Z	$-1{,}886^{b}$
Sig.	.045

Source: The authors, SPSS Output

The results confirm that the Wilcoxon statistical test yielded a significance level of 0.045, which is <0.05. This test confirms that the null hypothesis is rejected and the alternative hypothesis is accepted. In this sense, it is confirmed that there is indeed a relationship between the implementation of a SBI and the CI.

Table 10
Correlation between SBI – IC variables

			SBI	Indice
Rho de Spearman	SBI	Coef correl	1,000	,775
		Sig. (bilateral)	.	,000
		N	159	159
	Índice	Coef correl	,775	1,000
		Sig. (bilateral)	,000	.
		N	159	159

Source: The authors, SPSS Output

Spearman's rho test and the correlation coefficient identify the degree of relationship between the implementation of a SBI and CI. It was found that $p = 0.000 < \alpha = 0.01$, thus rejecting the null hypothesis and accepting the alternative hypothesis. A rho value of 0.775 indicates a strong positive correlation. Furthermore, a significant relationship between SBI implementation and CI is confirmed. In other words, the implementation of a business intelligence solution in SMEs at the Lima Plaza gallery will increase the growth rate of SMEs at the Lima Plaza gallery.

4. Discussion

The results obtained in this study reveal a significant and positive correlation between the implementation of business intelligence (BI) solutions and the growth index (GI) in the SMEs analyzed. This relationship was demonstrated by the Spearman correlation coefficient (Rho = 0.775, $p < 0.01$) as well as the nonparametric Wilcoxon test ($p = 0.045$), which reinforces the hypothesis regarding the favorable impact of BI on business performance.

These findings are consistent with previous studies such as that of Al Daabseh et al. (2023), which highlights that BI capabilities, when well aligned with strategic objectives, allow for the generation of actionable information that drives competitive advantage. Likewise, Azevedo and Almeida (2021) conclude that the effective adoption of BI depends on the training of decision-making personnel, a critical variable also observed in this study, where many companies without BI exhibit low growth and weak institutional connection. Regarding the company profile, it was observed that 70.4% of the surveyed SMEs have implemented BI solutions, while 35.2% of them have a growth index of 5 out of 6, indicating favorable business performance.

The economic sector also had an impact: companies dedicated to the technology (IT) sector, which represented 50.3% of the sample, had a higher proportion of BI adoption and a higher growth rate. This pattern suggests that affinity with technology is a factor facilitating adoption. Similar results were reported by Lada et al. (2023), who found that the technological environment and prior knowledge influence the decision to implement advanced analytics tools. However, barriers to BI implementation were also evident. Companies that have not adopted these solutions reported financial limitations, poor training, and limited membership in chambers of commerce, factors that have also been identified as obstacles in the literature (Pelekamoyo & Libati, 2023). This highlights the need for public policies aimed at strengthening technological access and digital literacy to reduce the competitive gap between digitalized and non-digitalized companies.

Regarding the analysis of complementary variables such as initial capital, years of operation, and international participation, it was found that companies with more than five years of experience and with higher-than-average initial capital have a higher adoption of BI. This pattern corroborates what Díaz and Galdon (2023) indicated, who assert that companies with greater structural maturity are more likely to invest in digital transformation. Finally, it is noteworthy that the growth index used as a dependent variable allows for the evaluation of not only sales or personnel expansion but also the operational improvement perceived by business owners. Therefore, the positive correlation between SBI and IC provides relevant evidence on the real usefulness of business intelligence in emerging business environments.

5. Conclusions

Data analysis required statistical processing and analysis. The Kolmogorov-Smirnov normality test was used, and the Spearman's Rho test was used to correlate variables. The descriptive results for the SBI variable are presented in terms of 112 SMEs (70.4%) that implemented a SBI and 47 SMEs (29.6%) that did not. For the CI variable, the highest number was 56 SMEs (35.2%), with a CI of 5, and the lowest was 2 SMEs (1.3%), with a CI of 6. Likewise, 26 SMEs (15.1%) were identified as having a CI of 0. The total was 159 SMEs (100%). The Wilcoxon statistical test yielded results with a significance level of 0.045, <0.05. In this sense, it is confirmed that there is a relationship between the implementation of a SBI and the CI. The results obtained with the Spearman's Rho test were $p = 0.000$; the null hypothesis was rejected, and the alternative hypothesis was accepted. The value of Rho = 0.775 indicates a strong positive correlation. In this sense, it is confirmed that there is a highly significant relationship between the implementation of a SBI and the CI. The study confirms that the implementation of a business intelligence solution in the SMEs of the Lima Plaza gallery will increase the growth rate of SMEs in the Lima Plaza gallery.

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CONTENTS

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PROFITABILITY PUZZLES: INSIGHTS FROM NORTH MACEDONIAN BANKS

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Ilija Hristoski[2],
Dragica Odzaklieska[3]

Abstract: The profitability of banks, as an indicator of financial performance, reflects their ability to generate substantial revenues. Bank profits are essential for their survival, solvency, and growth in a highly competitive environment. Moreover, even if solvency is high, poor profitability undermines a bank's capacity to absorb negative shocks, which can eventually impact its solvency. Consequently, profitability has a vital role in maintaining the stability of the banking and financial system as a whole. Hence, the primary goal of this research is to analyze several important bank-specific and macroeconomic determinants of bank profitability in the Republic of North Macedonia. More specifically, the paper investigates how the Return on Average Assets (ROAA) is influenced by the Capital Adequacy Ratio (CAR), Credit Risk (CR), Cost to Income Ratio (CIR), Loan to Deposit Ratio (LDR), Inflation Rate (IR), and Gross Domestic Product Rate (GDPR). Using secondary data from trustworthy sources, covering quarterly time series from 2005:Q1 to 2023:Q4, we employ correlation analysis, Granger causality tests, as well as the Auto-Regressive Distributed Lag (ARDL) approach to examine both short- and long-run dependencies among the variables. The results confirm the statistically significant impact of the regressors on the dependent variable and provide a solid foundation for the successful management and enhancement of banks' profitability.

Keywords: banks, profitability, bank-specific determinants, macroeconomic determinants, ARDL methodology, North Macedonia

JEL: G21, C32, C52.

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Introduction

Banking institutions are the backbone of a nation's financial system, allocating resources by channeling surplus funds to deficit entities, enabling investments, and driving economic growth. Their role is crucial in developing economies, where weak capital markets make banks the main financing source. Enhancing banking efficiency boosts economic growth by reducing loan costs, increasing corporate investment, and stimulating consumption. Profitable banks also support monetary policy and ensure financial stability.

Bank profitability, a financial key performance indicator, reflects managerial efficiency and significantly impacts banking sector stability. Sustainable profitability is crucial for sector stability, while a sound, profitable banking system is better equipped to withstand shocks, enhancing overall financial stability (Athanasoglou et al., 2022). To enhance profitability, bank management must analyze its determinants to better understand financial performance and mitigate future shocks. These factors fall into two categories: internal and external. Internal, or bank-specific, factors include size, capital adequacy, risk exposure, and operational efficiency. External factors, beyond the bank's control, stem from the macroeconomic environment and influence financial performance.

This paper examines the impact of internal and external factors on bank profitability in North Macedonia from 2005 to 2023. It focuses on Return on Assets (ROA) as the key measure, analyzing its relationship with internal factors such as capital adequacy, credit risk, management efficiency, and liquidity risk, alongside macroeconomic determinants like inflation and GDP growth.

The paper is structured as follows: The first section reviews prior research on bank-specific and macroeconomic determinants of profitability across various contexts. Section 2 presents the data and methodology, while Section 3 discusses the ARDL analysis results, their economic significance and implications. The final section concludes with recommendations.

1. Related Research

The profitability and liquidity of banks are essential for maintaining financial stability. A large body of empirical research has explored the influence of various determinants on bank profitability. For instance, Caliskan & Lecuna (2020) analyzed the impact of bank-specific and macroeconomic factors on Turkey's banking sector profitability (1980–2017) using regression analysis. Their findings indicate that macroeconomic indicators, such as inflation, interest rates and exchange rates, significantly influence ROA and ROE.

Bank-specific factors such as liquidity, asset management and operational efficiency are also crucial for enhancing profitability. Similar conclusions were drawn by Karadzic & Dalovic (2021), Zampara *et al*. (2017), Lutf & Omarkhil (2018), and Ćurak *et al*. (2012), who highlighted the positive impact of macroeconomic factors on bank profitability. Using a balanced panel data model on annual data from 47 large banks across 14 European countries (2013–2018), Karadzic & Dalovic (2021) found that GDP growth, inflation and market concentration significantly impact bank profitability. Athanasoglou *et al*. (2022) analyzed profitability determinants in South Eastern European (SEE) credit institutions (1998–2002) using fixed effects (FE) and random effects (RE) panel data models, finding that macroeconomic factors had varying effects, with inflation exerting a strong positive influence on bank profitability. Supiyadi *et al*. (2019), along with Hasanov *et al*. (2018) and O'Connell (2023), found that inflation had a significant positive impact on bank profitability, as evidenced in their studies. Lutf & Omarkhil (2018) confirm these findings in their study on Pakistani banks, revealing a positive long-term relationship between inflation, GDP, and both Return on Assets (ROA) and Return on Equity (ROE). They also highlight the significant influence of bank-specific factors, including capital adequacy, interest income, bank size, costs and non-interest income, on profitability.

A large number of studies emphasize the dominant influence of bank-specific factors on the profitability of banks. Qehaja-Keka *et al*. (2023) found that non-performing loans, loan interest rates, and total loans significantly influenced bank profitability in Albania and Kosovo from 2010 to 2020. Durguti *et al*. (2020) emphasized the dominance of bank-specific factors over macroeconomic ones in determining bank profitability. Their analysis of Kosovo's banking sector (2006–2019), using OLS regression and the Arellano-Bond GMM estimator, found that non-performing loans, capital adequacy, efficiency, the real exchange rate, and inflation significantly influenced profitability indicators. Almaskati (2022) similarly found that bank-specific factors primarily drove profitability. His research highlighted that market power and bank size significantly influenced both profitability and risk exposure. The research by Athanasoglou *et al*. (2022) and O'Connell (2023) further verified that all bank-specific factors significantly influenced bank profitability. Durand (2019), Batten & Vo (2019), and Yang (2019) highlighted the positive influence of capital structure on bank profitability. Specifically, banks with higher capital adequacy ratios are better equipped to absorb potential loan losses, thereby mitigating credit and insolvency risks, which ultimately enhances profitability. Sheaba Rani & Zergaw (2017) and Bucevska & Hadzi Misheva (2017) emphasized the significant and positive impact of management efficiency on bank profitability, highlighting its role in optimizing operational performance and enhancing financial outcomes. Adelopo *et al*. (2018) analyzed the influence of bank-specific and macroeconomic factors on bank profitability in Africa across three financial crisis periods (1999−2006, 2007−2009, and 2010−2013) using West African bank panel data and fixed effects models. Their findings indicated that costs, liquidity, and bank size consistently impacted return on assets (ROA) across all three periods, whereas no definitive conclusion was reached regarding the effects of capital adequacy, market power, credit risk, GDP, or inflation. Using panel data analysis, Ercegovac *et al*. (2020) examined the impact of bank-specific factors on ROA and ROE among 22 major EU banks from 2007 to 2019, in the aftermath of the financial crisis. Their findings highlighted that the cost-to-income ratio and the ratio of non-performing loans significantly influenced bank profitability. Zahariev *et al*. (2022) explored the relationship between key bank profitability indicators, ROE and ROA, and nine macroeconomic factors, including inflation, GDP, foreign exchange rates, and interest rates, within the Bulgarian banking system from 2014 to 2020. Employing descriptive statistics, a correlation matrix, VECM (Vector Error Correction Model), and SVR (Support Vector Regression), their analysis found that these macroeconomic variables did not significantly impact bank profitability. Consequently, they emphasized the need for banks to enhance management efficiency and drive innovation in interbank competition. Bucevska & Hadzi Misheva (2017) found that macroeconomic factors, particularly inflation and economic growth, had no significant impact on bank profitability.

The research conducted by various authors highlights the negative impact of certain bank-specific factors on banks' profitability indicators. For example, Kosumi & Zharku (2024) analyzed the effects of bank-specific, macroeconomic, and legal factors on bank profitability in North Macedonia from 2007 to 2022 using the VECM method. Their findings indicated that credit risk, liquidity, banking sector size, and non-performing loans significantly but negatively impacted ROA, while capital adequacy, GDP, interest rates, and operational efficiency positively influenced profitability. They suggested that Macedonian banks should enhance asset management and diversify income structures to mitigate credit risk and non-performing loans while maintaining strong liquidity ratios. Aspinmaa (2019), in her bachelor's thesis covering the period 2003–2017, found that bank-specific factors including bank size, credit risk, capital adequacy, and management efficiency significantly but negatively impacted on ROA and ROE in Central and Eastern European (CEE) countries. Karadzic & Dalovic (2020) found that bank-specific factors did not significantly influence banking profitability. This aligns with other studies that suggest varying impacts of internal factors on profitability, depending on the region or economic environment.

The literature review has provided valuable insights into bank profitability determinants across different contexts, emphasizing both internal and external factors. Nonetheless, North Macedonia's bank profitability remains underexplored, offering an opportunity to analyze these factors within its distinct economic and regulatory framework. This study addresses this gap by providing empirical evidence from North Macedonia, enriching the broader understanding of profitability drivers in emerging markets.

2. Data and Methodology

2.1 Data

Since the study includes more regressors, the number of observations in a dataset should account for the increased complexity due to the higher number of parameters to be estimated, so the statistical power, accuracy, and reliability of econometric results is maintained. Therefore, the data utilized comprises quarterly time series, spanning from 2005:Q1 to 2023:Q4, resulting in a total of 76 observations (19 years $\times$ 4 quarters/year = 76 quarters). Our analysis focuses on a single dependent variable and six independent variables, outlined as follows:

- *Dependent variable*
    - Return on Average Assets (*ROAA*), in percentages [%], as an indicator used to assess profitability and gauge financial performance of banks;

- *Independent variables*
  - Capital Adequacy Ratio (*CAR*), expressed in percentages [%], is computed as a ratio between the bank's equity and risk weighted assets; it is a measure of how much capital a bank has available and expresses the ability of the bank to deal with unexpected losses due to availability of adequate capital;
  - Credit Risk (*CR*), given in percentages [%], is a ratio between impaired loans (i.e., non-performing loans) and gross loans; it is a measure of the possibility of a loss resulting from a borrower's failure to repay a loan or meet contractual obligations;
  - Cost to Income Ratio (*CIR*), in percentages [%], as a measure of management efficiency, comparing the bank's operating costs to its operating income;
  - Loan to Deposit Ratio (*LDR*), in percentages [%], as a measure of the liquidity risk; it assesses a bank's liquidity by comparing a bank's total loans to its total customer deposits for the same period;
  - Inflation Rate (*INFLR*), in percentages [%], as a measure of macroeconomic stability;
  - Gross Domestic Product Rate (*GDPR*), in percentages [%], as a measure of the economic activity in the country;

The first four independent variables (*CAR*, *CR*, *CIR*, and *LDR*) belong to the group of banking system specific factors, whilst *IR* and *GDPR* are macroeconomic determinants.

All the data used in this research have been exploited from secondary sources only. The data for the dependent variable *ROAA*, including the first four independent variables (*CAR*, *CR*, *CIR*, and *LDR*), have been obtained from the Statistical Web Portal of the National Bank of Republic of North Macedonia containing indicators relevant for the Macedonian banking system (NBRNM, −b), whilst the data for the last two independent variables (*INFLR* and *GDPR*) have been taken from the same web portal containing data for the basic economic indicators (NBRNM, −c).

2.2 Methodology

First, we assessed the strength and direction of the linear relationship between the dependent variable and six independent variables by computing the Pearson correlation coefficient. The resulting correlation matrix summarizes the data and serves as input for further analysis. This step is essential for identifying relationships, detecting multicollinearity, and ensuring model stability.

The integration order of each variable was determined using the Augmented Dickey-Fuller (ADF) (Dickey & Fuller, 1979) and Phillips-Perron (PP) (Phillips & Perron, 1988) tests. Both tests were applied based on the Akaike (AIC) and Schwarz (SIC) Information Criteria.

The mix of I(0) and I(1) variables justified using the Auto-Regressive Distributed Lag (ARDL) methodology, also known as the Bound cointegration technique, a widely used econometric tool for analyzing dynamic relationships (Nkoro & Uko, 2016). Based on Pesaran & Shin (1998) and Pesaran et al. (2001), ARDL models captured both contemporaneously and lagged effects of dependent and independent variables. Their reliance on least squares estimation makes them particularly useful when the integration order of variables is uncertain, highlighting the flexibility that sets this approach apart (Giles, 2017a, 2017b, 2017c).

Optimal lag order selection, a crucial step in ARDL modeling, was determined by estimating an unrestricted Vector Auto-Regressive (VAR) model using five criteria: the LR test statistic, Final Prediction Error (FPE), Akaike Information Criterion (AIC), Schwarz Information Criterion (SIC), and Hannan-Quinn Information Criterion (HQ).

Before analysis, the Granger causality test (Granger, 1969) was conducted to identify potential predictors of the target variable. Applied in the preliminary stage, it helped uncover predictive causal links, aiding in understanding variable dynamics and informing policy and strategy formulation.

To determine the impact and the magnitude of the set of six independent determinants on the chosen dependent variable (i.e., ROAA), we employed the ARDL methodology. In general, the ARDL(p, q_1, …, q_k) model, including a target variable Y and k regressors (X_1, X_2, …, X_k), can be specified as Eq. (1) suggests (Pesaran & Shin, 1999; Pesaran *et al*., 2001). Equation (1) represents an ARDL model that has been transformed into an Error Correction Model (ECM) to capture both the short-run dynamics and the long-run relationship among the variables.

$$\begin{aligned}\Delta Y_t = \gamma_0 + \sum_{i=1}^{p-1} \phi_i \cdot \Delta Y_{t-i} + \sum_{j=1}^{k} \sum_{m=0}^{q_j-1} \lambda_{j,m} \cdot \Delta X_{j,t-m} + \\ + \psi \left(Y_{t-1} - \sum_{j=1}^{k} \delta_j \cdot X_{j,t-1} \right) + \varepsilon_t \end{aligned} \quad (1)$$

where:

k is the number of independent variables, X_k;

$p \geq 1$ is the optimal number of lags for the dependent variable, Y;

$q_j \geq 0$, $j = 1, \ldots, k$; are the optimal number of lags for each of the k independent variables, X_k;

Δ is the first-differencing operator;

$\Delta Y_t = Y_t - Y_{t-1}$ and $\Delta X_{j,t} = X_{j,t} - X_{j,t-1}$ represent the first differences of the variables, capturing short-run changes;

γ_0 is a constant term (intercept);

ΔY_{t-i}, $i = 1, \ldots, p-1$; are the differenced lagged values of the dependent variable Y at times $t-1, \ldots, t-p-1$;

ϕ_i, $i = 1, 2, \ldots, p-1$; are the short-run coefficients for the differenced lagged dependent variable, ΔY_{t-i};

$\Delta X_{j,t-m}$, $j = 1, \ldots, k$; $m = 0, 1, 2, \ldots, q_j-1$; are the differenced lagged values of the k independent variables X_k at times t (current value for $m = 0$), and $t-1, \ldots, t-q_j-1$ (lagged values for $m = 1, 2, \ldots, q_j-1$);

$\lambda_{j,m}$, $j = 1, \ldots, k$; $m = 0, 1, 2, \ldots, q_j-1$; are the short-run coefficients for the differenced lagged independent variables, $\Delta X_{j,t-m}$;

ψ is the coefficient of the error correction term (ECT), representing the speed of adjustment back to the long-run equilibrium;

$\left(Y_{t-1} - \sum_{j=1}^{k} \delta_j \cdot X_{j,t-1} \right)$ is the error correction term (ECT), capturing the long-run relationship between the variables;

ε_t is the disturbance (white noise) term.

The Wald test was used to assess whether the short-run coefficients of lagged independent variables are jointly zero, indicating their significance in influencing the dependent variable. The Bounds Cointegration Test (F-Bounds Test) examined the presence (H_1) or absence (H_0) of cointegration among variables, using the “5. Const. & Trend” specification based on prior integration tests.

The ARDL(1, 4, 4, 1, 0, 0, 0) model was estimated with an optimal lag length of L = 4. Following the Bounds Cointegration Test results, the Error Correction Model (ECM) was employed to estimate long-run equilibrium relationships within a VAR framework with 4 lags.

Diagnostic tests included the Breusch-Godfrey LM Test for autocorrelation, the Breusch-Pagan-Godfrey Test for heteroscedasticity, and the Jarque-Bera Test for normality. Model stability was validated through CUSUM and CUSUM of Squares Tests.

All analyses were performed using EViews v10 and IBM SPSS Statistics v20.

3. Results and Discussion

Table 1 presents the correlation matrix with Pearson coefficients, where the lower left half visualizes a heat map and the upper right half displays significance levels. *ROAA* shows a positive correlation with *CAR*, *INFLR*, and *GDPR*, while its negative correlation with *CR*, *CIR*, and *LDR* is statistically significant. The positive correlation between *ROAA* and *INFLR* is also significant. High correlations between *CIR* and *CR* (positive, significant) and *LDR* and *CAR* (negative, significant) suggest potential multicollinearity, which may impact ARDL estimates. However, most independent variable correlations remain low to moderate, indicating minimal multicollinearity concerns.

Table 1.
Correlation matrix and heat map of the observed variables

	ROAA	CAR	CR	CIR	LDR	INFLR	GDPR
ROAA	1	0.15993	-0.42035**	-0.63919**	-0.45768**	0.28639*	0.20804
CAR	0.15993	1	0.39948**	0.28288*	-0.77336**	0.06719	0.17246
CR	-0.42035	0.39948	1	0.74141**	-0.24183*	-0.44400**	0.20146
CIR	-0.63920	0.28288	0.74141	1	-0.07341	-0.21581	0.08707
LDR	-0.45768	-0.77336	-0.24183	-0.07341	1	-0.04773	-0.27234*
INFLR	0.28639	0.06719	-0.44400	-0.21581	-0.04773	1	0.04978
GDPR	0.20804	0.17246	0.20146	0.08707	-0.27234	0.04978	1

Note on the level of significance (2-tailed):
(*) Significant at the 0.05 level;
(**) Significant at the 0.01 level;
Source: The authors, IBM SPSS v20 output

Table 2 summarizes the ADF and PP test results for the variables' order of integration, considering AIC, SIC, and the 'With Constant and Trend' option.

Table 2.
Summary of the ADF and PP tests (option 'With Constant & Trend')

		Variables						
		Dependent	Independent					
Test	Criterion	*ROAA*	*CAR*	*CR*	*CIR*	*LDR*	*INFLR*	*GDPR*
ADF	AIC	I(1)***	I(0)**	I(1)***	I(1)***	I(1)*	I(1)**	I(0)***
	SIC	I(1)***	I(0)**	I(1)***	I(1)***	I(1)***	I(1)***	I(0)***
PP	AIC	I(1)***	I(0)**	I(1)***	I(1)***	I(1)***	I(1)**	I(0)***
	SIC	I(1)***	I(0)**	I(1)***	I(1)***	I(1)***	I(1)***	I(0)***

Note on the level of significance: (*) Significant at the 10%; (**) Significant at the 5%; (***) Significant at the 1%;

Source: The authors, EViews v10 output

CR is stationary at level (I(0)) at a 1% significance level (ADF test) and 5% (PP test) under the 'Without Constant & Trend' option. *LDR* is I(0) with 'With Constant' (ADF) but I(1) otherwise. *INFLR* is I(0) with 'With Constant' and 'Without Constant & Trend' (PP test) but I(1) otherwise. To ensure consistency, these variables are considered I(1) under 'With Constant & Trend,' as shown in Table 2.

Since the variables of interest have different integration orders (i.e., some are stationary at level and others become stationary after being first-differenced), the ARDL modeling approach is appropriate for analysis.

The analysis of the resulting estimated optimal number of lags, L, for various criteria (LR, FPE, AIC, SC, and HQ) and various maximum number of lags, M, suggests considering optimal lag lengths of $L = 2$ for $M = 4$ and $L = 5$ for $M = 5$. Since $L = 4$ is commonly chosen for quarterly data, we have also considered it to obtain an alternative model to be compared with other ARDL specifications. The goal was to evaluate multiple models based on overall statistics, residual diagnostics, and stability. The comparison of these ARDL models is shown in Table 3.

Table 3.
Comparison of the vital statistics regarding three ARDL models specified using Option 5. 'Const. & Trend'

	ARDL models		
	M = 4, L = 2	M = 5, L = 5	L = 4
	ARDL (1, 0, 1, 0, 0, 0, 0)	ARDL (1, 0, 1, 0, 0, 0, 0)	ARDL (1, 4, 4, 1, 0, 0, 0)
R-squared	0.849911	0.849911	0.883885
Adj. R-squared	0.829129	0.829129	0.847330
Durbin-Watson stat	2.094409	2.094409	2.059332
Akaike info criterion	0.498076	0.498076	0.511835
F-statistic	40.89731	40.89731	24.17974
Prob (F-statistic)	0.000000	0.000000	0.000000

Source: The authors, EViews v10 output

Table 3 results show that the ARDL model (1, 4, 4, 1, 0, 0, 0) with L = 4 outperforms the L = 2 and L = 5 models. The high R-squared value (0.883885) indicates that 88.39% of the variation in ROAA is explained by the regressors. The Adjusted R-squared (0.847330) confirms strong explanatory power, while the Durbin-Watson statistic (2.059332 ≈ 2.00) suggests no autocorrelation. The Akaike Information Criterion (AIC) value (0.511835) is the lowest among 62,500 models, indicating optimal model fit considering accuracy and parameter count. The F-statistic (24.17974) is statistically significant (p-Value = 0.000000 < 0.05), indicating that the independent variables jointly explain the dependent variable and the model is significant. Hence, *L* = 4 was selected as the optimal lag length for subsequent analyses.

The Granger causality test does not imply true causality but assesses whether past values of one variable can predict another. The test results for *L* = 4 show that only *CIR* (F-statistic = 2.82428, p-Value = 0.0321 < 0.05) Granger-causes *ROAA*, implying that only *CIR*'s past values contribute to forecasting *ROAA*. Despite this, in ARDL modeling, all independent variables remain valid as the method captures both short-term dynamics and long-term relationships, which may not be fully captured by pairwise Granger causality tests.

The specification of the chosen ARDL(1, 4, 4, 1, 0, 0, 0) model in a short-run includes two fixed regressors (i.e., C and @TREND) (Table 4).

Table 4 reveals that:

- The first lag of *ROAA* shows a positive but insignificant effect (+0.067841, p-Value = 0.5878 > 5%);
- *CAR*'s coefficient at level is positively correlated (+0.118457) but insignificant (p-Value = 0.2813 > 5%);
- The first and third lags of *CAR* are insignificant, with negative and positive effects (−0.048979 and +0.079409, p-Values > 5%);
- The second and fourth lags of *CAR* show significant positive (+0.238782) and negative (−0.232541) influences (p-Values < 10% and < 5%);

Table 4.
Short-run coefficients of the ARDL(1, 4, 4, 1, 0, 0, 0) model

Variable	Coefficient	Std. Error	t-Statistic	Prob.
ROAA(−1)	0.067841	0.124427	0.545224	0.5878
CAR	0.118457	0.108847	1.088288	0.2813
CAR(−1)	−0.048979	0.126063	−0.388525	0.6992
CAR(−2)	0.238782	0.127620	1.871042	0.0668*
CAR(−3)	0.079409	0.123266	0.644208	0.5222
CAR(−4)	−0.232541	0.096605	−2.407128	0.0195**
CR	−0.184296	0.063864	−2.885753	0.0056***
CR(−1)	0.123611	0.078374	1.577198	0.1206
CR(−2)	−0.098410	0.076643	−1.283992	0.2046
CR(−3)	−0.013789	0.075232	−0.183282	0.8553
CR(−4)	0.131529	0.056755	2.317483	0.0243**
CIR	−0.071739	0.016659	−4.306367	0.0001***
CIR(−1)	−0.031401	0.020168	−1.556964	0.1253
LDR	0.003876	0.016469	0.235334	0.8148
INFLR	0.025886	0.011108	2.330259	0.0236**
GDPR	−0.000769	0.009765	−0.078788	0.9375
C	5.210434	2.530943	2.058692	0.0444**
@TREND	−0.028577	0.005643	−5.064172	0.0000***

Note on the level of significance: (*) Significant at the 10%; (**) Significant at the 5%; (***) Significant at the 1%;

Source: The authors, EViews v10 output

- *CR* has a significant negative impact at level (−0.184296, p-Value = 0.0056 < 5%);
- *CR*'s first and fourth lags show positive influences, with the latter being statistically significant (p-Value = 0.0243 < 5%);
- The second and third lags of *CR* exhibit negative but insignificant effects (−0.098410 and −0.013789, p-Values > 5%);

- *CIR* has a significant negative effect at level (−0.071739, p-Value = 0.0001 < 5%);
- The first lag of *CIR* is insignificant with a negative impact (−0.031401, p-Value = 0.1253 > 5%);
- *LDR*'s level effect is positive but insignificant (+0.003876, p-Value = 0.8148 > 5%);
- *INFLR*'s level coefficient is positively significant (+0.025886, pValue = 0.0236 < 5%);
- *GDPR* shows a negative but insignificant effect (−0.000769, p-Value = 0.9375 > 5%);
- The fixed regressors, C (+5.210434) and @TREND (−0.028577), are both significant (p-Values < 5%). Their inclusion in the ARDL model is justified and therefore they cannot be omitted.

The Wald test results in Table 5 confirm that all four lags of *CAR* and *CR* significantly influence *ROAA* in the short run at the 5% level, indicating short-run causality. However, the lags of *ROAA* and *CIR* do not impact *ROAA* in the short-run.

The F-Bounds Test results in Table 6 show that the calculated F-statistic (9.082199) exceeds the critical values for I(1) at all significance levels (10%, 5%, 2.5%, and 1%), i.e. 3.59, 4.00, 4.38, and 4.90.

Table 5.
Results of the Wald Test

Variable	Lags	Chi-square statistics	df	Prob.	Null hypothesis
ROAA	1	0.297269	1	0.5856	Accepted
CAR	4	10.37584	4	0.0346*	Rejected
CR	4	11.89726	4	0.0181*	Rejected
CIR	1	2.424137	1	0.1195	Accepted

Note on the level of significance: (*) Significant at the 5% level;
Source: The authors, EViews v10 output

The F-Bounds Test results allow rejection of the null hypothesis (no cointegration), confirming that *ROAA* is cointegrated with *CAR*, *CR*, *CIR*, *LDR*, *INFLR*, and *GDPR* in the long-run. This indicates a shared stochastic trend, meaning the variables move proportionally over time. While short-term shocks may cause deviations, they tend to converge in the long term.

The long-term relationship enables both short-term ARDL and long-term Error Correction Model (ECM) estimations.

The coefficient of the cointegrating equation (−0.932159) is negative and significant (p-Value = 0.0000 < 5%), indicating a long-run Granger causality running from all regressors to *ROAA*. The system adjusts toward long-run equilibrium at a rate of 93.22% per period, i.e. quarter (Table 7).

Table 6.
Results of the F-Bounds Test

Test Statistic	Value	Signif.	I(0)	I(1)
F-statistic	9.082199	10%	2.53	3.59
k	6	5%	2.87	4.00
		2.5%	3.19	4.38
		1%	3.60	4.90

Source: The authors, EViews v10 output

Table 7.
Statistics of the Cointegration Equation Coefficient

Variable	Coefficient	Std. Error	t-Statistics	Prob.
CointEq(−1)	−0.932159	0.110909	−8.404720	0.0000

Source: The authors, EViews v10 output

Table 8 shows both the long-run coefficients and the Error Correction (EC) term.

Table 8.
Long-run coefficients and the Error Correction (EC) term

Levels Equation Case 5: Unrestricted Constant and Unrestricted Trend				
Variable	Coefficient	Std. Error	t-Statistics	Prob.
CAR	0.166418	0.076131	2.185945	0.0332**
CR	−0.044365	0.026448	−1.677415	0.0992*
CIR	−0.110647	0.011692	−9.463640	0.0000***
LDR	0.004158	0.017584	0.236450	0.8140
INFLR	0.027770	0.012170	2.281828	0.0265**
GDPR	−0.000825	0.010492	−0.078670	0.9376
EC = ROAA − (0.1664*CAR −0.0444*CR −0.1106*CIR + +0.0042*LDR +0.0278*INFLR −0.0008*GDPR)				

Note on the level of significance: (*) Significant at the 10%; (**) Significant at the 5%; (***) Significant at the 1%;

Source: The authors, EViews v10 output

From Table 8 it is obvious that, in a long-run:

- Three regressors (*CR*, *CIR*, *GDPR*) negatively affect *ROAA* with coefficients −0.044365, −0.110647, and −0.000825, respectively;
- Three regressors (*CAR*, *LDR*, *INFLR*) positively affect *ROAA* with coefficients +0.166418, +0.004158, and +0.027770, respectively;
- The effects of *CAR*, *CR*, *CIR*, and *INFLR* on *ROAA* are statistically significant (p-Values of 0.0332 < 5%, 0.0992 < 10%, 0.0000 < 5%, and 0.0265 < 5%, respectively);

Applying the *ceteris paribus* principle, the coefficients in Table 8 also suggest that:

- A 1 percentage point [pp] increase in *CAR* raises *ROAA* by 0.166418 [pp] (statistically significant, p-Value = 0.0332 < 5%);
- A 1 percentage point [pp] increase in *CR* reduces *ROAA* by 0.044365 [pp] (statistically significant, p-Value = 0.0992 < 10%);
- A 1 percentage point [pp] increase in *CIR* decreases *ROAA* by 0.110647 [pp] (statistically significant, p-Value = 0.0000 < 1%);
- A 1 percentage point [pp] increase in *LDR* increases *ROAA* by 0.004158 [pp] (statistically insignificant, p-Value = 0.8140 > 10%);
- A 1 percentage point [pp] increase in *INFLR* raises *ROAA* by 0.027770 [pp] (statistically significant, p-Value = 0.0265 < 5%);
- A 1 percentage point [pp] increase in *GDPR* reduces *ROAA* by 0.000825 [pp] (statistically insignificant, p-Value = 0.9376 > 10%).

The residuals diagnostic tests have led to the following findings:

The correlogram of residuals (Q statistics) for the ARDL(1, 4, 4, 1, 0, 0, 0) model shows no autocorrelation or partial correlation up to 16 lags, with p-values exceeding 0.05 for all lags. This supports the acceptance of the null hypothesis (H_0) stating that there is no autocorrelation within the specified lag range. The Breusch-Godfrey Serial Correlation LM Test (Obs*R-squared = 12.34273, Prob. Chi-Square(8) = 0.1366 > 10%) suggests that the null hypothesis of no serial correlation in the residuals up to eight lags can be accepted at the 10% significance level. The Breusch-Pagan-Godfrey test shows that the ARDL(1, 4, 4, 1, 0, 0, 0) model's residuals are free from heteroskedasticity (Obs*R-squared = 26.67423, Prob. Chi-Square(17) = 0.0630 > 5%). Therefore, the null hypothesis of no heteroskedasticity can be accepted at the 5% significance level. The Jarque-Bera test (Jarque-Bera = 0.501784, Prob. = 0.778106 > 10%)

confirms that the residuals are normally distributed, allowing acceptance of the null hypothesis at the 10% significance level.

Residual diagnostics confirm the ECM's suitability for hypothesis testing and forecasting. Additionally, CUSUM and CUSUM of Squares plots remain within the 5% critical bounds, verifying the structural stability of the ARDL(1, 4, 4, 1, 0, 0, 0) model coefficients.

4. Conclusions

This paper investigates the impact of various bank-specific and macroeconomic factors on the profitability of Macedonian banks from 2005 to 2023. The study is focused on Return on Average Assets (*ROAA*) as the dependent variable, which serves as a measure of bank profitability. Specifically, the analysis explores how profitability is influenced by the Capital Adequacy Ratio (*CAR*), Credit Risk (*CR*), Cost-to-Income Ratio (*CIR*), Loan-to-Deposit Ratio (*LDR*), Inflation Rate (*IR*), and Gross Domestic Product Rate (*GDPR*). The study employs correlation analysis, Granger causality tests, and the Auto-Regressive Distributed Lag (ARDL) approach to examine these relationships.

The study reveals that Return on Average Assets (*ROAA*) is significantly correlated with several key determinants. Inflation Rate (*IR*) has a positive, yet statistically significant impact on bank profitability in both the short- and long-term. Similarly, the Capital Adequacy Ratio (*CAR*) positively influences *ROAA* in the long-run, though its short-term effect is positive but not statistically significant. The Loan-to-Deposit Ratio (*LDR*) also shows a positive influence on profitability, but this effect is not statistically significant. On the other hand, Credit Risk (*CR*) and the Cost-to-Income Ratio (*CIR*) have a negative and statistically significant impact on *ROAA* across both timeframes. The Gross Domestic Product Rate (*GDPR*) shows a negative, but insignificant effect on bank profitability.

The findings of this study align with those of Kosumi & Zharku (2024), especially regarding the significant, yet negative impact of Credit Risk (*CR*) on profitability and the positive influence of Capital Adequacy Ratio (*CAR*) on profitability indicators. Additionally, Kosumi & Zharku's study extends the analysis by highlighting the positive impact of GDP, interest rates and operational efficiency on Return on Average Assets (*ROAA*), thus providing a broader perspective on the factors influencing bank profitability. On the other hand, the findings of Curak *et al*. (2012) highlight a significant positive impact of macroeconomic factors, especially GDP growth, on bank profitability in North Macedonia. This contrasts with the conclusion of this study, which finds the GDP growth rate (*GDPR*) to have a statistically insignificant effect on Return on Average Assets (*ROAA*). This difference suggests that while GDP growth is widely seen as a key driver of bank profitability in many contexts (e.g., Kosumi & Zharku, 2024), its influence on Macedonian banks during the analyzed period appears to be more intricate and may be shaped by other unique economic or structural factors within the country.

Given that both Return on Average Assets (*ROAA*) and Return on Average Equity (*ROAE*) are widely accepted measures of bank profitability, future studies could explore the relationship between ROAE and the same bank-specific and macroeconomic determinants analyzed in this research. This would offer a more comprehensive understanding of the factors influencing profitability in the Macedonian banking sector.

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CORPORATE SOCIAL RESPONSIBILITY ASPECTS WITHIN THE PUBLICISED CORPORATE CULTURE OF BULGARIAN COMPANIES

Ilian Minkov[1],
Milen Dinkov[2],
Petya Dankova[3]

Abstract: In recent years, the notion that business has a broad responsibility to society as a whole has gained significant momentum and is reflected in various academic theories, business practices and regulations. The concept of corporate social responsibility (CSR) has been interwoven with various academic fields, including strategic management and corporate culture. In order to establish a reputation as a socially responsible business entity, it is necessary for companies to adequately disclose CSR activities and events in the virtual space. The aim of this paper is, therefore, to examine the scope of CSR disclosure on the Internet by organisations from different sectors of the Bulgarian economy and its role in the portfolio of publicised corporate culture. The research methods used to achieve this aim include establishing relative values, comparative analysis and content analysis. The findings of the study demonstrate that there is an ambiguous attitude towards CSR in the companies surveyed, as in certain sectors it has a high degree of disclosure and is leading in rank, while in others it is present on the web pages of a small number of companies and has a rather peripheral role among the other elements in the portfolio of publicised corporate culture. To improve the communication effectiveness of the CSR component within the publicised corporate culture, it is essential that the information, initiatives, and priorities disclosed on the company web page are relevant and consistent with the fundamental aspects of the culture, specifically the mission, vision and values/principles.

Key words: corporate social responsibility (CSR), publicised corporate culture, strategic management, business organisations

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1. Introduction

The concept of corporate social responsibility (CSR) has witnessed a notable rise in prominence in recent decades, attaining a pivotal position within both academic discourse and business practice. A retrospective analysis of the conceptualisation of the term 'CSR' reveals that its first appearance in the economic literature was in 1953, when Bowen defined it as “the obligation of businessmen to pursue those policies, make those decisions, or follow those lines of action that are desirable in terms of objectives and values of our society” (Bowen, 1953). In 1971, the Committee for Economic Development asserted that “business is being asked to assume broader responsibilities to society than ever before and to serve a wider range of human values. […] Since business exists to serve society, its future will depend on the quality of management’s response to the changing expectations of the public” (Committee for Economic Development, 1971). Consequently, the concept of CSR has been interwoven with various academic fields, including corporate governance (Elkington, 2006; Jamali et al., 2008; Tandoh et al., 2022; Xu et al., 2022), innovation management (Halkos & Skouloudis, 2018; Santos-Jaen et al., 2021; Yang et al., 2024), corporate branding (Crosby & Johnson, 2002; Vallaster et al., 2012; Maon et al., 2021; Raj et al., 2022), total quality management (Waddock & Bodwell, 2004; Zink, 2007; Abbas, 2019), etc. The considerable amount of empirical research conducted on the relationship between the CSR level of companies and their financial performance (McGuire et al., 1988; Pava & Krausz, 1996; Barauskaite & Streimikiene, 2020; Coelho et al., 2023) has prompted business organisations to explore ways to implement CSR in their practice.

The implementation of CSR in the practice of business organisations has been largely the result of the work of Edward Freeman, who published his seminal book “Strategic Management: A Stakeholder Approach” in 1984 (Freeman, 1984). Since then, CSR has been recognised as a prerequisite for the long-term success of any business organisation in the contemporary business landscape. Furthermore, CSR has been progressively acknowledged as a fundamental element of corporate strategic management (Baric, 2022; Mitra, 2021) and is intricately woven into the essential strategic planning and operational frameworks of numerous organizations. CSR is often incorporated into companies' strategic planning processes, influencing mission statements, stakeholder analysis, resource allocation, and the pursuit of competitive advantage. Porter and Kramer assert that "CSR can be much more than a cost, a constraint, or a charitable deed – it can be a potent source of innovation and competitive advantage" (Porter & Kramer, 2006).

Both CSR and strategic management involve decision-making that affects a company's long-term success. Incorporating CSR into strategic planning enhances a corporation's brand identity and overall reputation. A growing number of consumers and stakeholders are drawn to enterprises that exhibit a dedication to social and environmental accountability, resulting in enhanced customer allegiance and favorable public perception. As of today, a significant number of business organisations have incorporated CSR into their strategic management processes, regarging it as a key component of a company's long-term strategy (Caroll et al., 2015; Vitolla, et al., 2017; Nicole, 2022). This integration entails the alignment of social, environmental and financial objectives of the company. The incorporation of CSR into a company's strategy has been demonstrated to contribute to sustainable value creation, through the cultivation of trust and loyalty among stakeholders, and the enhancement of employee morale (Dankova, 2008; Strukelj et al., 2023).

With reference to the essence of CSR in the practice of business organisations, in pursuing its economic goals a socially responsible company shall: (1) respect the legal rights of its stakeholders, which include its staff members, customers, suppliers, creditors, and the local community; (ii) provide its employees with healthy and safe working conditions, fair pay, and equal opportunities for development; (iii) provide its customers with high-quality and safe products and services; (iv) have a strong commitment to fostering collaborative relationships with its business partners; (v) endeavor to mitigate the adverse environmental consequences of its operations, while promoting the sustainable utilization of natural resources; (vi) contribute to the economic stability and sustainable development of the municipality in which it operates (Dankova, 2012).

In the contemporary business environment, effective strategic management entails the analysis of how CSR initiatives can result in sustainable competitive advantages, such as enhanced brand reputation and customer loyalty. Thus, the disclosure of CSR policies, objectives and activities occupies an important place in the communication policy of business organisations. In this respect, it plays a certain role in the structure of a company's online publicised culture. Our previous theoretical research (Minkov, 2022) has found that CSR is one of the secondary elements of the corporate publicised culture, whose main task is to support the core elements – mission, vision and values/principles, in building a more complete strategic message to the company’s stakeholders. However, there is a lack of sufficient empirical research into the actual role of CSR in the portfolio of corporate culture elements disclosed on the Internet and thus reaching the various company stakeholders.

The aim of this paper is to examine the scope of CSR disclosure on the Internet by organisations from different sectors of the Bulgarian economy and its role in the portfolio of publicised corporate culture. The study is based on the official web pages of companies from five sectors of the Bulgarian economy, including beer production, courier services, construction, meat processing and insurance services.

2. Methods

The specific research methods used to achieve the aim of this paper included establishing relative values, comparative analysis and content analysis. The study focused on the following aspects:

- Establishing the relative proportion (in %) of companies that disclose CSR information on their web pages as an element of their online publicised corporate culture.
- Determining the rank of CSR among the other elements of the online publicised corporate culture.
- Identifying the main CSR priorities and initiatives of companies in each of the five sectors surveyed, including beer production, courier services, construction, meat processing and insurance services.

The research was conducted under the following limitations:

- It was based on a study of data available from the official web pages of the business organisations in the sample.
- The sample consisted exclusively of business organisations that had disclosed at least one element of official publicised culture on their web pages. These comprised 12 beer producers, 8 courier service companies, 23 construction companies, 37 meat processing companies and 22 insurance companies.

- The study encompassed the sectors of the Bulgarian economy for which publicised official culture constituted a component of the companies' online communication policy. Consequently, the findings and conclusions of the study are only valid to these sectors and do not reflect the broader context of Bulgarian business as a whole.

3. Results and Discussion

The findings of the study demonstrate that Bulgarian business organisations' senior management does not perceive the CSR policy to be of significant importance. This is evidenced by the average level of disclosure of this policy online, which stands at 37.45%. Concurrently, it has been documented that the discrepancy between the sectors demonstrating the highest (courier services) and the lowest (meat processing) levels of CSR disclosure is nearly 65 percentage points, thereby underscoring the presence of substantial inter-sectoral disparities, as illustrated in Figure 1.

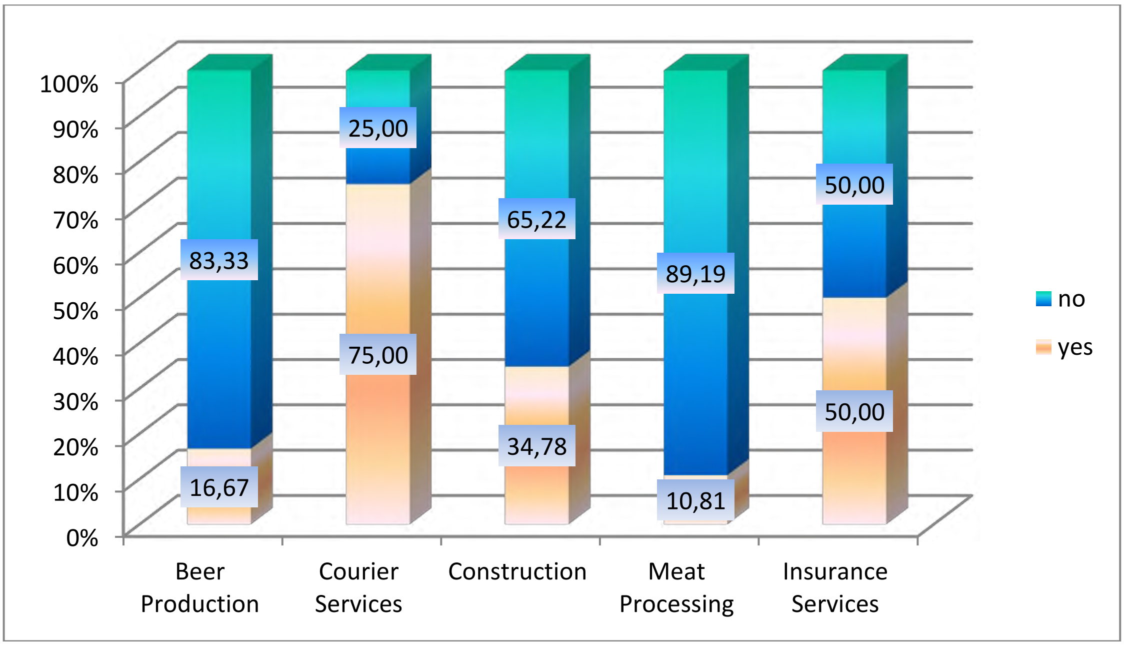

Figure 1. Relative share (in %) of companies disclosing CSR on their web pages

In consideration of the findings, it is possible to categorise the sectors of the Bulgarian economy into three groups, predicated on their apparent propensity to promote CSR initiatives.

The first group encompasses sectors of the economy in which a minimum of 50% of the organisations disclose CSR initiatives on their web pages. These sectors are courier services (75%) and insurance services (50%). This corporate behaviour can be attributed to the proactive adoption of the Corporate Sustainability Reporting Directive (CSRD) by companies in these sectors. The CSRD promotes transparency and environmental, social and governance (ESG) reporting, leading to enhanced ESG ratings and a positive image and reputation among partners and stakeholders.

The second group comprises sectors with a medium level of CSR disclosure, in which no less than one-third of firms include it in the portfolio of elements of publicised corporate culture. The construction industry can be identified as such a sector, with a CSR disclosure rate of 34.78%.

The third group encompasses sectors for which the promotion of socially responsible activities and initiatives is currently of peripheral interest. The survey indicates that these businesses include beer production (16.67%) and meat processing (10.81%). It is noteworthy that in these sectors, only the large companies – the leading firms in the market – disclose CSR, which demonstrates their desire to build and maintain an image of socially responsible business entities.

In the context of this discussion, it is relevant to determine the rank of CSR disclosure among other elements of the online publicised corporate culture (table 1).

Table 1.
Rank of CSR disclosure among the elements of online publicised corporate culture

Rank / Sector of economy	Relative to all the elements of corporate culture	Relative to all the secondary elements of corporate culture
Beer Production	5	3
Courier Services	1	1
Construction	1	1
Meat Processing	5	3
Insurance Services	4	2

As demonstrated by the data presented in table 1, it can be concluded that, in certain sectors of the Bulgarian economy, CSR is a leading element in the portfolio of publicised corporate culture, while in others it is of average importance. The analysis indicates that the disclosure of CSR policy on official web pages is a priority for companies in the courier services and construction sectors, where it ranks first among other elements of culture. The observed rank of CSR for courier companies is consistent with the identified high degree of its disclosure. The present study also draws attention to the curious situation observed in the construction sector, where CSR is ranked highest despite displaying an average disclosure rate. This phenomenon may be attributed to the under-disclosure of the official corporate culture within this sector, resulting in a relatively low proportion of firms disclosing even leading items.

CSR occupies a median position in the rankings of the elements in the beer production, meat processing and insurance services sectors, where it ranks fourth or fifth (second or third among the secondary elements). For the first two sectors, this position is pertinent to the results obtained on the relatively low level of CSR disclosure. However, a discrepancy emerges when examining the insurance services sector, where the average rank does not align with the high degree of CSR disclosure. The explanation for this discrepancy is analogous to the one previously outlined, with the opposite sign in this case, because not only do the leading elements of corporate culture exhibit a high level of disclosure, but indeed the majority of elements in this sector do so too.

In this regard, the difference between the level of CSR disclosure and the first-ranked element of corporate culture in each sector is of interest for this study (fig. 2).

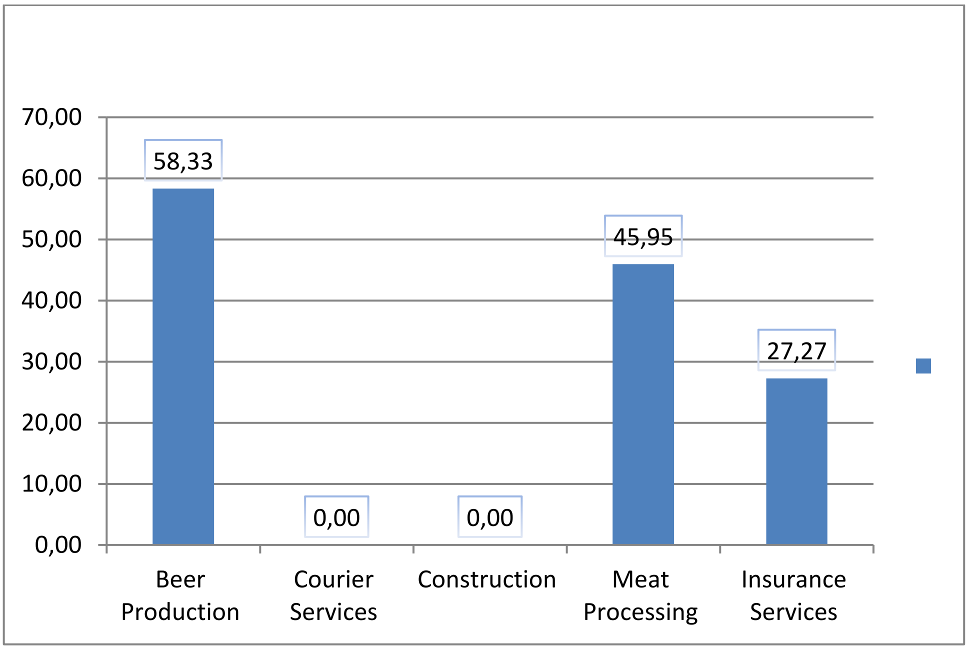

Figure 2. Difference (in percentage points) between the level of CSR disclosure and the first-ranked element of corporate culture

The findings indicate a significant lag behind the leading element in the meat processing sector[4] (45.95), and a more pronounced lag is observed in the beer production sector[5] (58.33). This corroborates the conclusions concerning the function of CSR in the publicised corporate culture portfolio of these two business sectors. Consistent with the findings above, the gap with the leading element in the insurance services sector[6] is significantly smaller (27.27).

A content analysis of the CSR policy aspects disclosed on companies' web pages was conducted with the aim of deepening the study and facilitating better understanding of the content of the CSR activities of companies in various sectors. The findings are summarised in Table 2.

Table 2.
CSR Policy Priorities of Bulgarian Business Organizations

Sector of Economy	**CSR Policy Priorities**
Beer Production	- Protection of water resources; - Reducing carbon emissions; - Caring for the community; - Safety and health, etc.
Courier services	- Nature conservation / environment protection; - Socially responsible initiatives; - Employee care and safe working conditions; - Human capital and talent development; - Charity, etc.
Construction	- Environment protection; - Creating healthy and safe working conditions for employees, etc.
Meat Processing	- Environment protection; - Philanthropy and social activities; - Sustainable development; - Consumer responsibility, etc.
Insurance Services	- Environment and climate protection; - Societal sustainable development; - Supporting sport and healthy lifestyles; - Care for children and young people, etc.

[4] The leading elements of the online publicised corporate culture in this sector are the company story and the company motto.

[5] The leading element of the online publicised corporate culture in this sector is the company story.

[6] The leading element of the online publicised corporate culture in this sector is the company story.

The results presented in table 2 allow for the identification of three key CSR policy priorities for Bulgarian business organisations:

- Protection of the environment, climate and water resources;
- Sustainable community development and care;
- Ensuring safe working conditions and care for employees.

It is important to acknowledge that, in addition to the aforementioned priorities, companies in each sector incorporate other significant elements stemming from the inherent nature and particularities of their operations.

4. Conclusions

The research findings indicate an ambivalent stance towards CSR within Bulgarian business organisations, as reflected in its position within the context of publicised corporate culture. In certain sectors, it is characterised by a high level of disclosure, while in others, it is present on the web pages of a small number of companies and has a rather peripheral role among the other elements in the corporate culture portfolio. Concurrently, a substantial consensus among managers from diverse sectors has been identified concerning the priority areas of CSR policy.

In order to enhance the communication capabilities of the CSR element within the publicised corporate culture, it is imperative that the measures, activities and priorities disclosed on the internet are pertinent and aligned with the core elements of the culture, namely the mission, vision and values/principles. This would result in a complete overall strategic message to partners and stakeholders, with a positive effect on the image of companies.

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CLUSTER ANALYSIS OF E-COMMERCE IN THE EUROPEAN UNION COUNTRIES

Zoya Ivanova[1]

Abstract: E-commerce is the fastest growing segment in the trade sector and is of ever-increasing importance to the economy of the European Union. However, huge imbalances occur between individual countries when measuring the intensity of this type of trade. In the context of the new realities, the aim of the present study is, based on cluster analysis, to segment and identify the differences between the EU Member States in the field of e-commerce, and on this basis to formulate corresponding conclusions. Cluster analysis is applied to selected indicators (I) for measuring e-commerce in the European countries—I1. "E-commerce sales of enterprises"; I2. "Value of e-commerce sales"; I3. "Internet purchases by individuals". The clustering of the studied data refer to 2024. Technical processing and calculations are made using IBM SPSS Statistics. As a result of the study conducted, the 27 Member States are differentiated into four homogeneous clusters, which allows them to be interpreted as: leading countries in the e-commerce segment; countries catching up with the leaders; countries moderately advancing in e-commerce; countries in an unenviably lagging position.

Keywords: cluster analysis, clusters, e-commerce, EU Member States.
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Introduction

The revolutionary development of the Internet and digital technologies is fundamentally changing and transforming the way goods and services are sold and bought. Commercial transactions are increasingly taking place online. In view of this, conventional trade is rapidly being displaced by e-

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commerce, which is becoming an integral part of the global trade network for buying and selling goods and services. Moreover, this is the “new trade” in the European Union, which creates opportunities for increasing the level of customer satisfaction and expanding the scope and efficiency of purchase and sales channels. In economic terms, due to the activities of enterprises selling online and the change in consumer behaviour, significant progress has been made in the development of e-commerce in the European Union. Despite the positive aspects, there are huge imbalances and asymmetries occurring between the individual Member States.

The aim of the present study is to segment and identify the differences between the EU Member States in the field of e-commerce, based on cluster analysis, and to formulate corresponding conclusions thereof. Essentially, this study aims to answer the following research questions (RQ):

RQ1: What is the number of clusters from the 27 EU Member States generated based on the indicators characterizing e-commerce?

RQ2: What is the grouping of the 27 EU Member States in the relevant clusters according to the level of the determined quantitative indicators?

RQ3: What are the disparities and differences between the European Union countries by cluster in implementing e-commerce and how can they be identified and measured?

1. Literature Review

E-commerce leads to radical and qualitative changes in commercial processes based on increased digitalization and accelerated technological development. The literature review shows the existence of numerous studies highlighting the advantages of e-commerce compared to the traditional method of buying and selling (Gupta et al., 2023; Jain et al., 2021; Rizaldi & Madany, 2021; Gu et al., 2021; Taher, 2021; Li & Zhang, 2021; Brzozowska & Bubel, 2015; Qin, Chang, Li, & Li, 2014).

In essence, the focus is on opportunities for businesses to sell online by adopting a customer-centric approach and thus demonstrating “intelligent” market behaviour (Shah & Murthi, 2021; Hunter & Perreault Jr, 2006; Petrova & Tairov, 2022). The paper emphasizes the importance of integrating and using a wide range of digital technologies in the commercial activities of enterprises ensuring a continuous cycle of data processing and exchange, through which online sales are optimized, reducing the time and costs for their implementation (Nicola & Setiawan, 2024; Hagsten, 2022; Lu, 2019; Goyal, et al., 2019; Ahearne, et al., 2013; Kuruzovich, 2013; Marshall et al., 2012). Besides, the benefits are indicated as a result of improving the orientation towards consumer demands, through the implementation of new models for managing the strategic and operational aspects of sales processes, which increases the competitiveness of enterprises and multiplies the beneficial effects (Pascucci, et al., 2023; Jain, et al., 2021). Despite the diversity of positions, researchers believe that e-commerce creates a new perspective for carrying out commercial transactions, through which enterprises generate higher economic and financial results, achieve better market positioning and realize greater efficiency of the overall commercial process.

On the other hand, there are a number of studies proving that modern consumers increasingly prefer to purchase goods and services for personal purposes over the Internet (Ren et al., 2025; Figueiredo et al., 2025;Yang, et al., 2023; Balacescu, et al., 2023; Zhang, et al., 2022). In this context, online consumers who participate in electronic transactions are provided with opportunities for global accessibility, a wide selection of goods, and parallel comparison of many goods and prices (Shah & Murthi, 2021). E-commerce is considered to be the more convenient and attractive segment for consumers seeking practicality and flexibility (Duch-Brown et al., 2017; Yu & Wu, 2007). Emphasis is placed on the role of e-commerce in transforming consumer behaviour and increasing satisfaction and commitment at every touchpoint in the customer experience lifecycle (Petrova, et al., 2022; Gimpel & Roglinger, 2015).

The literature review confirms the existence of numerous views, opinions and discussions on the impact, influence and significance of e-commerce. Research interest in this direction is constantly expanding. This is not accidental, since the relevance of the issue under consideration is undeniable.

2. Research Methodology

The systematic review of the various formulations in the outlined research area is a basis for adopting the position that e-commerce is becoming an indispensable part of the global trade network, which is developing extremely rapidly. At the same time, it should be emphasized that significant variations in the main indicators characterizing e-commerce are manifested, especially between the EU countries.

In view of this, in the present paper, the focus of the study is placed on the segmentation and identification of differences in terms of e-commerce in the EU Member States using cluster analysis. For the purposes of the analysis, three indicators are used, which are included in the main sub-indicators of the Digital Economy and Society Index—DESI[2], characterizing the penetration of digital technologies into the economy and society. Specifically, these are the indicators (I) for measuring e-commerce in the EU countries: *I1. "E-commerce sales of enterprises"; I2. "Value of e-commerce sales"; I3. "Internet purchases by individuals".* In addition, the first two indicators characterize the relative share of enterprises with 10 or more employees that have sold goods online and generated turnover in the previous 12 months. The third indicator is brought to the relative share of individuals aged 16 to 74 who have purchased goods and services for personal purposes over the Internet in the last 12 months. It is important to note that the main source of statistical data used to conduct the study is Eurostat. The statistical data are the result of business surveys reported in 2024.

In methodological terms, the analysis is based on the application of clustering methods. In this regard, the three basic methods and the sub-methods included in them are implemented—Hierarchical Cluster Analysis, Non-Hierarchical Cluster Analysis and Two Step Cluster Analysis. With the help of the IBM SPSS Statistics, a preliminary measurement of the distances between the clusters is performed, using all alternative clustering methods (Ganeva, 2016). In this specific study, the Squared Euclidean Distance is taken as a measure. For this purpose, the values of the variables are standardized with the z-scores function (i.e. they have a mean value of 0 and a standard deviation of 1) (Ganeva, 2016). The most appropriate hierarchical clustering method Average Linkage (Between Groups) is selected, forming an optimal number of clusters for the current study. The selection is made based on the highest measured value of the contingency coefficient (0.866) compared to the approved methods.

[2] The Digital Economy and Society Index (DESI) is an annual report published by the European Commission that monitors the progress of the EU Member States in the field of digital technologies.

3. Results and Discussion

E-commerce is developing and establishing itself as one of the main trade segments within the European Union. The relative share of enterprises in the EU countries that sell goods and services online in 2024 amounted to 23.8%, and the annual turnover from online sales amounted to 19.1%. The European Union is the third largest online market after China and the United States, with 77.0% of internet users having purchased goods and services online for personal purposes (Eurostat, 2024).

The data studied show that enterprises in the European Union are increasingly participating in digital markets and are striving to satisfy diverse preferences and increase added value for consumers. In this context, enterprises offer direct and fast access to goods and services, without geographical and time constraints, provide multiple choices, implement special services and personalized campaigns that meet the unique requirements of individual online consumers. On the other hand, consumer behaviour in the European Union is changing towards increasing the use of the Internet, as well as online ordering or purchasing of goods or services. Due to greater flexibility, convenience and confidentiality in online transactions, more than 7 out of 10 internet users make online purchases (Eurostat, 2024).

Despite the positive changes reported, there are ***huge differences in intensity and significant variations in the level of the main indicators for measuring e-commerce between the individual EU Member States of the European Union.***

In view of this circumstance, cluster analysis is implemented, which allows to identify the differentiation of the studied objects into separate groups (clusters) in terms of e-commerce.

For the specific study, the optimal grouping based on the identified similarities and differences between the selected indicators allows for the generation of a total of ***four clusters***, consisting of a different number of Member States, in 2024.

The clustering is visualized using the dendrogram graphical tool. Each Member State of the Community is assigned a number ranging from 1 to 27, which facilitates the graphical representation: see Figure 1.

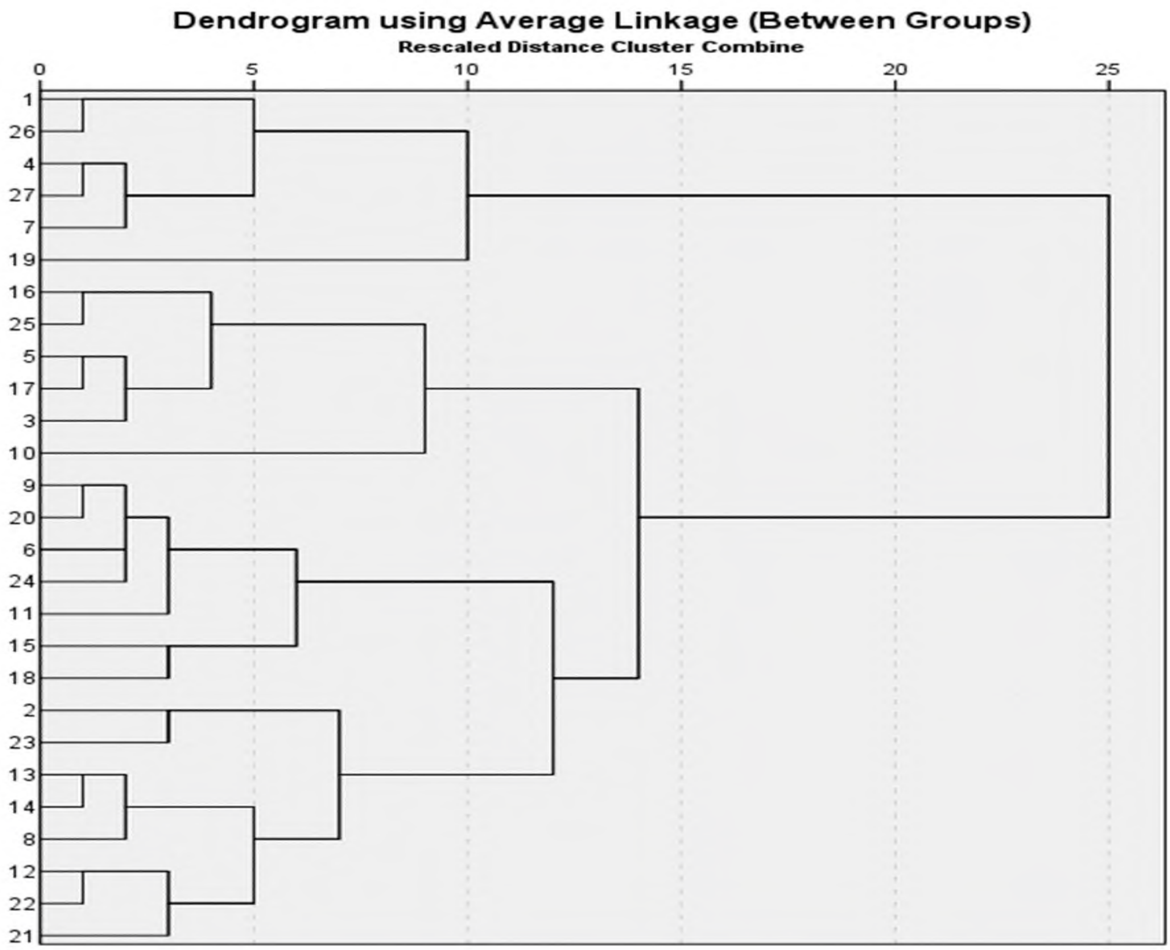

Figure 1. Dendrogram for generating clusters of the 27 EU Member States based on data on the three indicators characterizing e-commerce in 2024

Source: Eurostat and author's own calculations.

Figure 1 presents the four clusters generated by the applied software method. The further interpretation of the data for the three examined indicators characterizing e-commerce allows for the grouping of the individual Member States of the Community. In this analytical context, and in accordance with the attained level of e-commerce, it is appropriate to distinguish the following groups of countries: First cluster "Leading countries in the e-commerce segment"; Second cluster "Countries that are in an unenviably lagging position"; Third cluster "Countries catching up with the leaders"; Fourth cluster "Moderately advanced countries in e-commerce". The representation of the thus identified clusters is provided in Table 1.

Table 1
Clustering of the 27 EU Member States based on three indicators characterizing e-commerce in 2024

Cluster name	Countries
First cluster “Leading countries in the e-commerce segment”	Belgium (1), Finland (26), Denmark (4), Sweden (27), Ireland (7), Netherlands (19)
Second cluster “Countries that are in an unenviably lagging position”	Bulgaria (2), Romania (23), Cyprus (13), Latvia (14), Greece (8), Italy (12), Portugal (22), Poland (21)
Third cluster “Countries catching up with the leaders”	Luxembourg (16), Slovakia (25), Germany (5), Hungary (17), Czech Republic (3), France (10)
Fourth cluster “Moderately advanced countries in e-commerce”	Spain (9), Austria (20), Estonia (6), Slovenia (24), Croatia (11), Lithuania (15), Malta (18)

Source: Eurostat and author’s own calculations.

As can be seen from the table, the leading countries in the e-commerce segment are included in the first cluster, among them Ireland, Denmark, Sweden, and others. These are countries where 9 out of 10 internet users purchase goods and services online, and a significant part of enterprises are very active in e-sales and generate cumulatively high turnover through internet transactions.

It is observed that the second cluster is the most numerous, consisting of countries in a rather disadvantaged position, such as Bulgaria, Romania, Italy, and others. Primarily businesses in these countries cannot take advantage of the opportunities offered by various online sales methods. As a result, they have a harder and slower time influencing the behaviour of active online users. Online interactions are still very weak.

The third cluster includes the countries that are catching up with the leaders, Czech Republic, Germany, Slovakia, and others. It should be emphasized that in these countries e-commerce is developing extremely rapidly. There is a constant positive trend of growth of enterprises with online sales and consumers making online purchases, which opens opportunities for further convergence with the leading countries.

The fourth cluster comprises the moderately advancing countries in e-commerce, such as Austria, Spain, Slovenia, and others. These countries have seen a gradual change in e-commerce, with a degree of variation around the average values for the EU-27. In fact, the online sales enterprises from the countries in the cluster have a development potential that should be built on and expanded. Overall, this will affect the commercial relations with consumers and will lead to an increase in their online consumption.

At the same time, an important part of the cluster analysis algorithm is to find out the role of individual indicators in the formation of clusters. In this aspect, the value of the significance level (Sig.) for each of the determined quantitative characteristics should be tracked. As a rule, a check is performed that provides the ability to control the risk of error based on a certain acceptable level, most often assuming α = 0.05. In cases where the values of Sig. are less than the selected risk of error, the feature is considered significant in distinguishing the units, conversely—with values of Sig. higher than the risk of error, the variable is not important for the formation of clusters.

The results of testing the three indicators, the values of which for 2024 are included in determining the clusters, show that all, without exception, have statistical significance, which is confirmed by the data in Table 2.

Table 2
Determining the participation of indicators in the distribution of the 27 EU Member States by cluster in 2024

ANOVA Table

			Sum of Squares	df	Mean Square	F	Sig.
Value of e-commerce sales* Average Linkage (Between Groups)	Between Groups	(Combined)	605.587	3	201.862	13.797	<0.001
	Within Groups		336.500	23	14.630		
	Total		942.087	26			
E-commerce sales of enterprises* Average Linkage (Between Groups)	Between Groups	(Combined)	1366.663	3	455.554	23.950	<0.001
	Within Groups		437.478	23	19.021		
	Total		1804.141	23			
Internet purchases by individuals* Average Linkage (Between Groups)	Between Groups	(Combined)	2939.194	3	979.731	30.151	<0.001
	Within Groups		747.377	23	32.495		
	Total		3686.571	26			

Source: Eurostat and author's own calculations.

The data presented in Table 2 show that the significance level of the three indicators (Sig.) is less than the selected risk of error ($\alpha = 0.05$), therefore their role in distinguishing the four clusters is significant.

In parallel, it is necessary to determine the degree of significance of the individual indicators in clustering. For this purpose, the coefficient of determination (Eta Squared) is used, which is shown in Table 3. According to the data in Table 3, reported in 2024, the one with the greatest contribution to the formation of the clusters is I3. "Internet purchases by individuals". Second comes I1. "E-commerce sales of enterprises". With the smallest contribution is I2. "Value of e-commerce sales".

Table 3
Determining the contribution of indicators in the distribution of the 27 EU Member States by cluster in 2024

Measures of Association

	Eta	Eta Squared
Value of e-commerce sales* Average Linkage (Between Groups)	0.802	0.643
E-commerce sales of enterprises* Average Linkage (Between Groups)	0.870	0.758
Internet purchases by individuals* Average Linkage (Between Groups)	0.893	0.797

Source: Eurostat and author's own calculations.

In the context of the analysis conducted, it is found that the average values reported in 2024 for the individual indicators that form the four clusters vary widely. In addition, huge differences in the average values of the indicators between the individual clusters are found, which is evidence of major imbalances. Specific data in this regard are presented in Table 4 (See Table 4).

As can be seen from the table, the indicator with the highest average value in the first cluster "Leading countries in the e-commerce segment" is indicator I3. "Internet purchases by individuals" with 87.23%. I1. "E-commerce sales of enterprises" comes second with 35.39%. I2. "Value of e-commerce sales" is third with 26.18%.

The results illustrated in Table 4 show that in 2024 the indicators have a similar structure within the second, as well as within the fourth cluster. The highest average value in the second cluster "Countries that are in an unenviably lagging position" is taken by I3. "Internet purchases by individuals"—60.10%, followed by I1. "E-commerce sales of enterprises"—

19.44%. The lowest value is for I2. “Value of e-commerce sales”—13.41%. In the fourth cluster “Moderately advanced countries in e-commerce”, the average values are as follows: I3. “Internet purchases by individuals”—67.41%, I1. “E-commerce sales of enterprises”—31.83% and I2. “Value of e-commerce sales”—17.18%.

Table 4
Determining the average values and standard deviations of the indicators in the distribution of the 27 EU Member States by cluster in 2024

Report

Average Linkage (Between Groups)		Value of e-commerce sales	E-commerce sales of enterprises	Internet purchases by individuals
First cluster “Leading countries in the e-commerce segment”	Mean	26.1750	35.3867	87.2300
	N	6	6	6
	Std. Deviation	4.06869	3.55228	7.77607
Second cluster “Countries that are in an unenviably lagging position”	Mean	13.4137	19.4413	60.1025
	N	8	8	8
	Std. Deviation	4.16883	3.37233	6.40396
Third cluster “Countries catching up with the leaders”	Mean	21.0017	19.4550	78.6900
	N	6	6	6
	Std. Deviation	4.52393	4.53250	2.57261
Fourth cluster “Moderately advanced countries in e-commerce”	Mean	17.1786	31.8300	67.4143
	N	7	7	7
	Std. Deviation	2.22653	5.65771	4.56205
Total	Mean	18.9119	26.1996	72.1570
	N	27	27	27
	Std. Deviation	6.01948	8.33007	11.90761

Source: Eurostat and author’s own calculations.

Only in the third cluster “Countries catching up with the leaders”, the average values of the indicators are in a different structural relation. The average level of I3. “Internet purchases by individuals” is the highest—78.69%, I2. “Value of e-commerce sales” takes the second place with 21.00%, and I1. “E-commerce sales of enterprises” ranks third with 19.46%.

In fact, comparing the values of the indicators for the individual clusters in Table 4 proves the presence of huge differences and disproportions in the development of e-commerce. For example, the data for I1. “E-commerce sales of enterprises” show that the countries in the first cluster are the undisputed leader, since in 2024 they have a level 15.95 percentage points higher than the countries in the second cluster, 15.93 percentage points higher than those in the third cluster and 3.56 percentage points higher than the countries in the fourth cluster. From the point of view of the parameters of the indicator, there is an imbalance, since the average value of the indicator for the countries in the first cluster is 9.19 percentage points higher than the average level for the EU-27, and for the fourth cluster it is 5.63 percentage points higher. While in the second and third clusters, the level of the indicator is below the EU-27 average, by 6.76 and 6.74 percentage points, which is an indication of significant lagging behind.

There is also a large difference in I2. “Value of e-commerce sales” The countries in the first cluster are ahead of those in the second cluster by 12.77 percentage points, in the third cluster by 5.18 percentage points and in the fourth cluster by 9.00 percentage points. It should be emphasized that the average values of the indicator for the first and third clusters are above the average level for the EU-27, respectively by 7.27 and 2.09 percentage points. While for the second and fourth clusters they are below the average level for the EU-27 by 5.50 and 1.73 percentage points.

Regarding I3. “Internet purchases by individuals” an identical situation is observed, but with a more pronounced difference. The analysis of the data in Table 4 shows that in 2024 the indicator for the countries in the first cluster is 27.13 percentage points higher than the second cluster and 19.82 percentage points higher than the fourth cluster, while for the third cluster it is 8.54 percentage points higher. It is striking that the imbalances indicated are the largest, since the countries in the first cluster are 15.07 percentage points and the third cluster are 6.53 percentage points above the average level for the EU-27. The negative fact is that the indicator for the countries in the second and fourth clusters is 12.06 and 4.75 percentage points below the average for the EU-27.

To support the interpretation of the research results and to provide clearer visualization, Figure 2 is presented below.

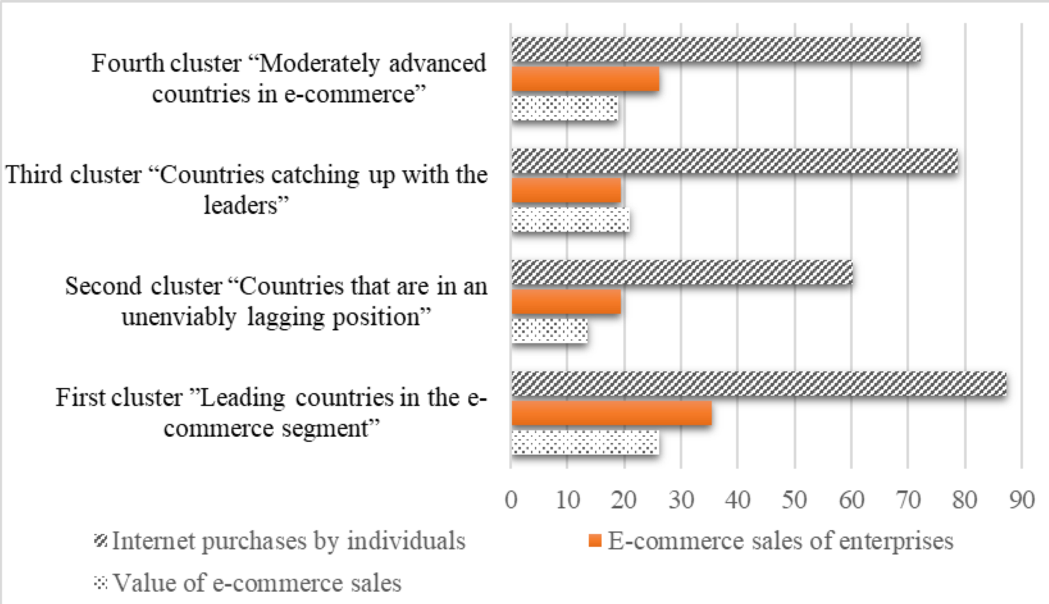

Figure 2. Average values of the indicators measuring e-commerce, by cluster, in 2024

Source: Eurostat and author’s own calculations.

The figure shows the existing differentiation among the four clusters in terms of the average values of the measured indicators characterizing: e-commerce sales of enterprises; value of e-commerce sales generated by the enterprises; and internet purchases by individuals in the European Union.

In the outlined research area, there is an opportunity for discussion and comparison of the results of other researchers, including at the European level. The conducted cluster analysis reveals a clear distinction between countries with a high level of digitalization and a well-developed online markets—such as Sweden, Denmark, Finland, Ireland, the Netherlands, and Belgium—and countries with lower levels of e-commerce development, predominantly from Southern and Eastern Europe. Similarly, the European E-commerce Report highlights significant regional disparities in online trade within the EU, despite an overall steady growth trend (European E-Commerce Report, 2024). Analogous observations have been made by researchers in their empirical studies using the cluster approach, highlighting, on the one hand, the qualitative differences in e-commerce practices among European enterprises (Zoroja et al., 2020; Scutariu et al., 2022), and on the other hand, the existence of distinct clusters based on consumer behaviour (Brcanov, et al., 2025). In this context, the present study confirms and builds upon existing research, offering new perspectives for reducing disparities among European Union Member States in the field of e-commerce.

Conclusions

E-commerce is developing extremely dynamically within the European Union. Regarding individual Member States, the conducted cluster analysis confirms the huge differences in intensity and significant variations in the level of the main indicators for measuring e-commerce.

In view of the results obtained from the analysis and the ensuing findings, it should be emphasized that the research questions posed are largely answered. Specifically:

First. Four homogeneous clusters are generated from the 27 Member States of the European Union, because of the identified similarities and differences between the selected indicators characterizing e-commerce.

Second. Grouping the countries in the European Union into clusters, according to the level of the determined quantitative indicators, allows them to be interpreted as: leading countries in the e-commerce segment; countries catching up with the leaders; moderately advancing countries in e-commerce; countries in an unenviably lagging position. The highest measured values of the analysed indicators are reached by countries such as: Ireland, Denmark, and Sweden, etc.

Third. There are huge differences in the average values of the indicators between the individual clusters. The countries in the second cluster lag significantly, especially in I2. “Value of e-commerce sales” and I3. “Internet purchases by individuals”. Countries such as Bulgaria, Romania, Latvia, Greece have unrealized opportunities in the area under study.

Fourth. The average values reported for the individual indicators that form the four clusters vary widely, which is evidence of large imbalances. Although the countries in the third cluster indicate a convergence to the leaders in two of the indicators, the leading countries in the first cluster are significantly differentiated from those in the fourth cluster and especially from the countries in the second cluster.

Therefore, the focus should be on accelerating the digitalization of trade processes and increasing the implementation of electronic transfers of goods and services, especially in the countries of the second cluster. Otherwise, the imbalances would worsen and deepen to the point of insurmountability. Continuing research in this area allows us to reach relevant and practically significant economic instruments for overcoming the differences between the European countries studied.

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CONTENTS

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THE PRACTICE OF WEBROOMING IN GENERATION Z: EMPIRICAL ANALYSIS ON PERUVIAN BUYERS

Aitana Maria Gonzalez Caceres[1], Maria Fe Portocarrero Mendoza[2]

Abstract: This article focuses on the practice of webrooming among Generation Z youth residing in metropolitan Lima. The research seeks to answer the following question: How do convenience orientation and impulse buying propensity influence the practice of webrooming, and how does this practice impact channel integration and purchase preferences in physical stores? To address this, a conceptual model was developed based on relevant academic literature, which was validated through field research. An online survey with Likert-scale questions was conducted with a non-probability sample of 384 young individuals with webrooming experience. Data analysis was carried out using Structural Equation Modeling (SEM), a technique appropriate for the type of hypotheses and sample size. The main results reveal that: 1) convenience orientation negatively influences webrooming; 2) impulse buying in digital media is driven by factors such as website design, product features, and retailer strategies; and 3) webrooming moderates the relationship between channel integration and purchase intention in physical stores, particularly among Generation Z consumers. These findings contribute both to the academic field and to omnichannel retail management by providing a better understanding of the behavior of a key segment entering the Economically Active Population.

Key Words: Webrooming, Retail Channel, Channel Integration, Generation Z

JEL: M31, D91, C38

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INTRODUCTION

Digital transformation, accelerated by the pandemic, boosted e-commerce and led companies to develop omnichannel strategies by integrating technologies (Abu et al., 2024; Hardi et al., 2024; Huang, 2021). To optimize these strategies, retailers analyze purchasing behavior and aim to integrate online and physical channels (Halibas et al., 2023; Bansal et al., 2024). As a result, showrooming and webrooming have become common practices (He et al., 2024). Webrooming linked to department stores with omnichannel strategies involves researching a product online and purchasing it in a physical store, which increases consumer trust (Chung et al., 2022; Kim & Han, 2022). Given the limited number of studies on this practice, its growing relevance has motivated further research (Wu et al., 2023; Xiang et al., 2024).

Causal relationships were identified in the causality model. Price consciousness and shopping enjoyment orientation promote webrooming, while convenience orientation reduces it (Kim et al., 2019). Additionally, online shopping is associated with promotions, advertising, impulsive purchases, and product exploration (Zhao & Wan, 2017). On the other hand, webrooming influences channel integration by generating patronage (Shakir et al., 2022), which affects the consumer's choice of which channel to use (Schiessl et al., 2023). Given the limited number of studies on these relationships, this model contributes to a better understanding of the behavior of consumers who engage in webrooming.

This research explores how the pursuit of convenience and impulsive buying influences webrooming, as well as the impact this behavior has on channel integration and purchase preferences. The study focuses on Generation Z in Lima, Peru who are known for being constantly connected, multitasking, and navigating the digital world (Geck, 2006). Given their potential as consumers, brands aim to attract them through personalized marketing strategies, relevant content, and constant interaction on social media (Ponomarenko et al., 2022). In 2023, it was estimated that 7 out of the 20 million people in Peru’s economically active population belonged to this generation (El Peruano, 2015), highlighting the importance of adapting retail strategies to webrooming a behavior now even more common than showrooming (Guo et al., 2022).

Webrooming is driven by purchase motivation, perceived channel benefits, and costs (Aw, 2019). For retailers, understanding why Generation Z research products online before buying them in physical stores is key to adjusting their strategies (Seemiller & Grace, 2018). Despite its growing relevance, there is still a gap in the literature on this phenomenon within the retail channel. This study delves into its impact on channel integration and offline patronage, offering a clearer understanding of the motivations behind this behavior (Kleinlercher et al., 2020).

1. Literature review

1.1. Webrooming

Flavián, et al. (2019) indicates that webrooming is a way of shopping where consumers research the product online and purchase it in the physical store. As a result, they perceive greater security and perceive themselves as intelligent buyers. Webrooming varies depending on the type of product and the personality of the consumer. For example, consumers tend to use webrooming more for high-risk information products (such as computers, laptops or cell phones) and experiential products (such as clothing or cosmetics) (Guo, et al. 2022). On the other hand, Bell et al. (2018) explains that webrooming economically benefits retail channels by increasing demand, generating indirect operational effects on other channels, and improving operational efficiency.

1.2. Purchase reasons

Purchase motives drive consumer behavior by fulfilling the needs they seek to satisfy when buying products (Noble et al., 2006). In the retail channel, the buying process involves gathering information, comparing prices, and looking for variety (Noble et al., 2005). In this context, there are two types of motivation. On one hand, hedonic motivation focuses on pleasure and enjoyment, and includes factors such as adventure, gratification, and role-playing (Arnold et al., 2003). It is also linked to impulsive buying behavior (Heitz-Spahn, 2013; Childers et al., 2002; Kim et al., 2019). On the other hand, utilitarian motivation seeks efficiency, prioritizing saving money, time, and effort (To et al., 2007). Likewise, price consciousness reflects the desire to save money (Heitz-Spahn, 2013; Noble et al., 2005).

1.3. Channel integration and its influence on sponsorship

Retail channel integration is the process of integrating multiple channels into a single business model (Cai, et al. 2020). Implementation increases efficiency and drives innovation in retail companies, generating better performance (Oh, et al. 2012). In customers, channel integration empowers consumers, increasing their trust and satisfaction and improves their patronage intention in an omnichannel retail environment (Zhang, et al. 2018). According to Kim, et al. (2019), patronage is consumers' preference and loyalty toward a specific retailer including making recommendations about that retailer. Herhausen, et al. (2015) and Li, et al. (2019) mention that the integration of online and offline channels creates a competitive advantage benefiting retailers.

1.4. CONCEPTUAL FRAMEWORK AND HYPOTHESIS DEVELOPMENT

1.4.1. Convenience Orientation

According to Kim, et al. (2019), convenience orientation is the ease and speed with which consumers can make their purchases. Haryanto, et al. (2019) mention that consumers consider convenience when choosing between product options, prioritizing saving time and effort in their purchases. Therefore, it is proposed that Generation Z consumers prefer to practice webrooming, because it allows them to make an effective purchase, obtaining information online and obtaining the product physically. So, the hypothesis is presented:

H1: Convenience orientation impacts positively webrooming behavior in Generation Z consumers.

1.4.2. Impulsive buying orientation

Impulse buying orientation refers to an individual's propensity to make unplanned purchases driven by an intense desire to purchase a product immediately. (Verhagen & Van Dolen, 2011). These buyers make immediate purchase decisions, so the time they research their purchases is short and limited (Skallerud, et al. 2009). This behavior is unexpected and emotional, and the purchasing environment acts as a stimulus (Ahrholdt, et al. 2017). For this reason, the hypothesis was proposed:

H2: Impulsive purchasing orientation negatively impacts webrooming in Generation Z

1.4.3. Channel and sponsorship integration in the physical store

Digital channels have transformed retail commerce and shopper behavior, driving the transition to omnichannel retail (Verhoef, et al. 2015). This strategy creates an improved experience through technological advances (Asmare & Zewdie, 2022). Channel integration seeks to provide optimal customer experience across online, physical and mobile channels (Piotrowicz & Cuthbertson, 2014). This is why channel integration positively influences consumers' channel preferences (Goraya, et al. 2020). Likewise, Kang (2018) states that webrooming in omnichannel consumers impacts the consumer experience and the patronage generated by the experience in different channels. Therefore, the following hypotheses were proposed:

H3: Webrooming impacts positively the relationship between channel integration and consumer patronage intention for the physical store for Generation Z consumers.

H4: Channel integration positively influences consumer patronage intention for physical stores and for Generation Z customers.

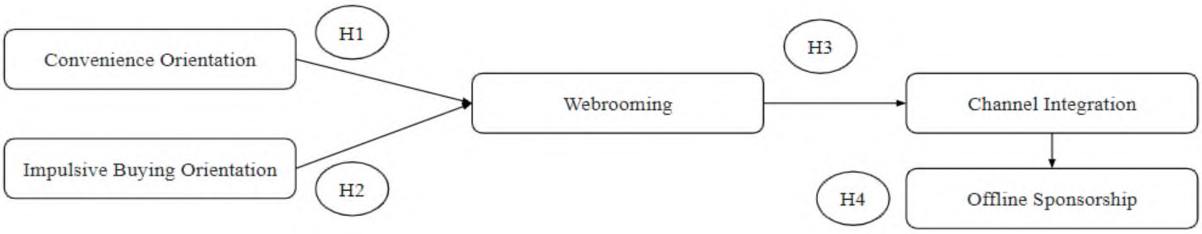

Figure 1. Causality model

Note: Own elaboration

2. Methodology

2.1. Research design and measurement tool

The study uses a non-probability research design, and the finite population formula was applied to determine the sample size. Its objective is to analyze webrooming behavior among Generation Z consumers in Metropolitan Lima, who combine retail and online channels. Both qualitative and quantitative methods were used, and the data were analyzed statistically. A questionnaire was administered online, targeting individuals who engage in webrooming. It began with demographic questions, followed by nine Likert-scale questions addressing five key webrooming factors, such as convenience orientation, impulsive buying, and satisfaction with the physical channel. The questionnaire was validated by five experts, and a pilot test was subsequently conducted to correct potential errors.

2.2. Population and sample

This study focuses on Generation Z in metropolitan Lima, Peru, as this generation is beginning to enter the economically active population and is expected to become the main group of consumers in the future (Kusa et al., 2024; Jung et al., 2024). The choice of this generation is based on four main reasons: their impact on economic growth and retail competitiveness (Baykal, 2020), the importance of omnichannel strategies for integrating sales channels (Bansal et al., 2024), and their high use of digital technology. Meanwhile, Metropolitan Lima was selected due to its significance in the retail sector and its sustained growth in sales (Peru Retail, 2024).

Table 1
Obtaining sample data

n = Sample size =		**nd**
N = Potential Market Size		**756,154**
P = Probability		**0.5**
Q = (1 – P) =		**0.5**
Confidence Level		
	Z = 1.96	1.96
	E = 0.05	0.05
n = $\dfrac{N \times Z^2 \times P \times Q}{(Z^2 \times P \times Q + (N-1) \times E\hat{}2)}$		726,210
		1,892
Sample size = n =		**383.8**

Note: Own preparation

A potential market of 756,154 Generation Z individuals in metropolitan Lima was estimated, based on the assumption that 70% of the total population shops online (Gamboa, 2023) and that this generation represents 20% of that group (Lujan & Polo, 2023). To determine the sample size, a 50% probability of online purchasing was assumed, along with a confidence level based on a statistical parameter of 1.96 and a maximum margin of error of 0.05. Using the finite population formula, a sample size of 384 people was obtained, following the Krejcie and Morgan method (Rehman, 2021).

2.3. FACTORIAL ANALYSIS APPLIED IN THE PILOT TEST

Factor analysis allows for the identification of factors within a population and the establishment of relationships between variables and their underlying causes (Stephenson, 1935). This study focuses on exploratory factor analysis; therefore, the Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy and Bartlett's Test of Sphericity were applied during the pilot test.

According to Hair et al. (2019), a KMO value above 0.75 is considered good, between 0.5 and 0.75 is acceptable, and below 0.5 is unacceptable. In all cases, the KMO was greater than 0.5, indicating that the test was valid. Additionally, Bartlett's Test values were below 0.05, confirming the significance of correlations among variables. Therefore, it was concluded that factor analysis was appropriate for application.

Table 2
Bartlett test and KMO test

Constructs and Items	Min	Max	KMO	Barllet Test
Webrooming adapted from Rap et al. (2015)	0.505	0.557	0.529	< .001
Convenience Orientation adapted from Kim et al. (2019)	0.632	0.790	0.693	< .001
Impulsive Buying Orientation adapted from Kim et al. (2019)	0.686	0.80	0.738	< .001
Channel Integration adapted from Schiessl et al. (2023)	0.459	0.569	0.533	< .001
Offline Sponsorship from Schiessl et al. (2023)	0.50	0.50	0.5	0.002

Note: All items with collinearity greater than 5 were eliminated. CO: Convenience Orientation, IBO: Impulse Buying Orientation, CI: Channel Integration, OFPI: Offline Sponsorship.

3.RESULTS

Table 3 presents the demographic data obtained from the surveys conducted.

Table 3
Demographic data of the respondents

Vari-ables	***Sex***	Feminine	51%
		Male	49%
	Education	No Education/ Incomplete Primary Education	0%
		Complete Primary / Incomplete Secondary School	18%
		Complete Secondary/ Incomplete Technical Higher Education	5%
		Complete Technical Higher Education	0%
		Incomplete/Complete University Higher Education	54%
		Postgraduate Education	26%
	Year of Birth	2010 - 2004	30%
		2003-1999	40%
		1998 - 1994	30%
	Marital Status	*Single*	80%
		Married	20%
		Divorced	0%
		Widow/ Widower	0%
	Currently Employed	*Yes*	70%
		No	30%

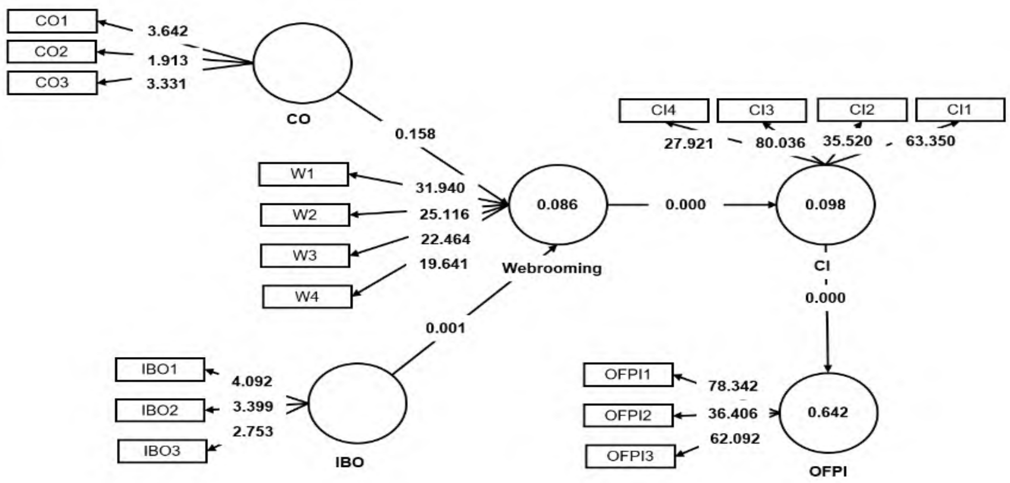

Figure 2. Results of the Structural Model

Note: All items with collinearity greater than 5 were eliminated. CO: Convenience Orientation, IBO: Impulse Buying Orientation, CI: Channel Integration, OFPI: Offline Sponsorship.

Cronbach's Alpha and AVE were used to assess the reliability and convergent validity of the instrument. An alpha greater than 0.80 indicates high reliability (Ruiz Bolívar, 2002). In this analysis, all indicators exceeded 0.5, demonstrating that the constructs explain at least 50% of the variance in their items. Additionally, three constructs showed strong reliability (Christmann & Aelst, 2006). All cases yielded an AVE above 50%, classifying the items as acceptable (Hair et al., 2019). Composite Reliability confirmed internal consistency, and according to Fornell and Larcker (1981), a factor must explain more than 50% of the variance to be considered valid, with an ideal threshold of 70%. This analysis meets that criterion, reinforcing the convergent validity of the instrument.

Table 4
Convergent Validity

	Cronbach's Alpha	**Composite reliability (rho_a)**	**Composite reliability (rho_c)**	**Average variance extracted (AVE)**
CO	0.747	1.21	0.834	0.634
IBO	0.621	0.557	0.781	0.545
IC	0.926	0.929	0.948	0.819
OFPI	0.929	0.931	0.955	0.875
W	0.915	0.936	0.939	0.794

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CO: Convenience Orientation, IBO: Impulse Buying Orientation, IC: Channel Integration, OFPI: Offline Sponsorship.

Discriminant validity is assessed by comparing indicator loadings on their respective constructs with cross-loadings on other constructs. According to Henseler et al. (2009) and Cohen (1988), validity is confirmed when the loadings on the intended construct are significantly higher, and the effect size exceeds 0.5. The results shown in Table 5 meet these criteria, ensuring strong discriminant validity. Additionally, Fornell and Larcker (1981) state that the average variance extracted (AVE) for each construct should be greater than the shared variance between construct pairs, ensuring that each construct captures distinct aspects (see Table 6).

Table 5
Cross Loading

	CI	**OFPI**	**CO**	**IBO**	**W**
CI1	**0.913**	0.761	-0.186	0.24	0.35
CI2	**0.9**	0.753	-0.255	0.098	0.284
CI3	**0.932**	0.684	-0.247	0.13	0.249
CI4	**0.875**	0.696	-0.044	0.151	0.231
OFPI1	0.77	**0.929**	-0.086	0.207	0.296
OFPI2	0.704	**0.938**	-0.23	0.098	0.31
OFPI4	0.772	**0.939**	-0.072	0.076	0.214
CO1	-0.132	-0.072	**0.833**	0.04	-0.055
CO2	-0.017	0.084	**0.574**	-0.124	-0.054
CO3	-0.238	-0.195	**0.938**	0.14	-0.162
IBO1	0.087	0.054	0.261	**0.825**	0.201
IBO2	0.228	0.199	-0.261	**0.696**	0.203
IBO3	-0.052	-0.055	0.451	**0.685**	0.063
W1	0.318	0.337	-0.067	0.298	**0.916**
W2	0.279	0.24	-0.246	0.137	**0.893**
W3	0.267	0.246	-0.066	0.304	**0.883**
W4	0.221	0.178	-0.141	0.066	**0.872**

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CO: Convenience Orientation, IBO: Impulse Buying Orientation, IC: Channel Integration, OFPI: Offline Sponsorship.

Table 6
Average variance extracted

	CO	IBO	IC	OFPI	W
CO	0.796				
IBO	0.079	0.738			
IC	-0.204	0.173	0.905		
OFPI	-0.135	0.136	0.802	0.935	
W	-0.14	0.244	0.311	0.291	0.891

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CO: Convenience Orientation, IBO: Impulse Buying Orientation, IC: Channel Integration, OFPI: Offline Sponsorship.

Table 7 presents the HTMT test, which is used to assess discriminant validity in structural equation models. According to Cepeda et al. (2016), an HTMT value below 0.90 indicates satisfactory discriminant validity. In this analysis, all values are below the 0.90 threshold, confirming adequate discriminant validity among the constructs.

Table 7
Heterotrait–monotrait ratio (HTMT)

	CO	IBO	IC	OFPI	W
CO					
IBO	0.54				
IC	0.194	0.225			
OFPI	0.174	0.189	0.86		
W	0.142	0.273	0.328	0.306	

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CO: Convenience Orientation, IBO: Impulse Buying Orientation, IC: Channel Integration, OFPI: Offline Sponsorship.

According to Hair et al. (2019), the p-value test shows that all items are significantly related to their corresponding variable, and if the p-value is greater than 0.05, the hypothesis is rejected. In Table 8, the hypothesis regarding convenience orientation is rejected. Additionally, the T-value is used to determine the statistical significance of the coefficients; when the T-value exceeds 1.96, the coefficient is considered significantly different from zero. Table 8 shows that most variables have a T-value above this threshold, except for convenience orientation.

Table 8
Test P y T

	Origin sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (|O/STDEV|)	P values	Results
CI -> OFPI	0.801	0.802	0.038	21.228	0	Is accepted
CO -> Webrooming	-0.159	-0.154	0.113	1.413	0.158	It is rejected
IBO -> Webrooming	0.262	0.27	0.08	3.269	0.001	Is accepted
Webrooming -> CI	0.313	0.308	0.079	3.964	0	Is accepted

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CO: Convenience Orientation, IBO: Impulse Buying Orientation, IC: Channel Integration, OFPI: Offline Sponsorship.

VIF is a metric used to assess collinearity among items. According to Tsagris and Pandis (2021), collinearity occurs when two or more predictor variables in a regression model are highly correlated, meaning they are not independent from one another. This can inflate the variance of the estimated coefficients, making it difficult to determine which variables are hugely significant in the model. Collinearity is considered a severe issue when the VIF value ranges between 5 and 10. Therefore, items with VIF values within that range were removed.

Table 9
Multicollinearity

	VIF
CI1	3.387
CI2	3.546
CI3	4.894
CI4	3.228
CO1	2.254
CO2	1.394
CO3	1.787
IBO1	3.117
IBO2	1.056
IBO3	3.026
OFPI1	3.36
OFPI2	4.221
OFPI3	3.908
W2	3.769
W3	2.782
W4	3.554
W1	3.267

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CO: Convenience Orientation, IBO: Impulse Buying Orientation, IC: Channel Integration, OFPI: Offline Sponsorship.

According to Hair et al. (2014), the coefficient of determination (R^2) is a measure of the predictive accuracy of the model, where values of 0.75, 0.50, and 0.25 indicate substantial, moderate, or weak levels of accuracy, respectively. Additionally, R^2 values equal to or greater than 0.90 may suggest overfitting or data manipulation. In the model, the dependent variable *webrooming* shows low dependence on the analyzed orientations (Table 10), such as convenience orientation and impulse buying. In other words, the predictor variables described explain only 8.6% of *webrooming* behavior. Hair et al. (2019) state that the higher the R^2 value, the greater the explanatory power. However, it is important to consider the context in which R^2 is assessed, as in some cases, low values may still be acceptable. Table 10 shows that convenience orientation is a predictor variable that may be influencing the R^2 value; when it is excluded, R^2 decreases. Therefore, in the presented model, this orientation is key to explaining *webrooming*. Similarly, channel integration shows an R^2 of 0.01, indicating that *webrooming* explains only 1% of this dependent variable. In contrast, *offline sponsorship* has the highest R^2 (0.642), with 64% of its variance explained by channel integration.

Khan et al. (2021) state that Q^2 values above 0% and 26% are significant and reflect the predictive capability of the model. While R^2 measures determination, Q^2 indicates certainty. Hair et al. (2019) establish that values greater than 0, 0.25, and 0.50 represent low, medium, and important levels of predictive relevance in PLS, respectively. Table 10 shows medium Q^2 values, confirming the model's predictive capacity. Additionally, metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are considered to evaluate prediction accuracy.

Table 10
R Square

	R^2	**Q^2predict**	**RMSE**	**MAE**
CI	0.098	0.033	0.994	0.878
OFPI	0.642	0.017	1.001	0.887
Webrooming	0.086	0.04	0.989	0.779

Note: All items with collinearity greater than 5 were eliminated. W: Webrooming, CI: Channel Integration, OFPI: Offline Sponsorship.

4. Discussion

4.1. Theoretical Contributions/Hypothesis Testing

This research article analyzed the practice of webrooming among Generation Z in Metropolitan Lima within the retail channel, given that this purchasing model has been underexplored, as outlined in the introduction. In this regard, the need arose to explore how variables such as convenience orientation and impulse buying propensity positively influence the practice of webrooming. In turn, this purchasing behavior promotes channel integration, enabling a smooth transition between digital and physical environments, which facilitates offline sponsorship. The decision to focus the study on Generation Z is based on their characteristics as digital natives who are highly influenced by technology in their purchasing decisions (Espejo et al., 2024). According to Flavián et al. (2019), webrooming involves researching products online before purchasing them in physical stores, which requires companies to efficiently integrate all customer touchpoints. In this context, omnichannel integration emerges as a key strategy to enhance consumer experience, as noted by Asmare and Zewdie (2022). Furthermore, considering that as of 2023 this generation has become an active part of Peru's Economically Active Population (EAP) (El Peruano, 2015), understanding their buying behavior and potential impact on the retail sector is crucial.

4.1.1. Validation of the convenience orientation hypothesis

Haryanto et al. (2019) point out that consumers consider convenience when choosing between product options, prioritizing saving time and effort in their purchases. In this sense, webrooming is presented as an ideal purchasing model, as it reduces time and cost effort by allowing consumers to research the product and obtain detailed information to make more informed decisions. However, the statistical results obtained in this study suggest that this hypothesis does not hold. As seen in table 8, the p-value obtained is greater than 0.05, indicating that the hypothesis must be rejected, according to the criteria of Hair et al. (2019).

Gottschalk (2018) also mentions that convenience orientation has a negative influence on webrooming, which is consistent with the findings of this study. Furthermore, Aw et al. (2021) reinforce this idea by stating that, in a webrooming model, the need to touch the product is essential for the consumer, especially when they are in a retail sector context. Products require thorough inspection, which is why Generation Z consumers do not necessarily seek to save time when researching before making a purchase in a physical store.

4.1.2. Validation of the compulsive buying orientation hypothesis

According to Febrilia et al. (2024), impulse buying on digital media is influenced by the website, product features, and retailer strategies that motivate individuals to make the purchase. Furthermore, impulse buying is done to obtain instant gratification (Einarsson et al., 2024). Thus, as seen in Table 8, the hypothesis is validated using the P and T tests. Furthermore, according to Figure 1, this variable moderates webrooming by 26%. Likewise, compulsive buying behavior increases retail sales and allows consumers to be guided by their emotions. On the other hand, webrooming has a negative influence on the purchasing model because consumers are responsible for making smart purchases. This is supported by the fact that in this purchasing model, individuals have the ability to research the product to obtain detailed information and then go to the store to validate this information. Furthermore, with the help of salespeople who assist consumers, individuals can perceive added value. (Flavian et al. 2020)

4.1.3. Validation of the offline sponsorship and channel integration hypothesis

Technological advances and the growth of points of purchase have impacted omnichannel purchasing behavior (Sharma et al., 2024). Furthermore, according to Nguyen and Tran (2023), the experience of customers using omnichannel positively influences satisfaction, and webrooming improves the customer experience. Given this, webrooming moderates the relationship between channel integration and purchase patronage intention in physical stores. Given this, the hypothesis presented in Table 8 is accepted. Furthermore, the hypothesis that channel integration positively influences consumer patronage intention for physical stores and for Generation Z customers is accepted. This is supported by Shakir et al. (2022), who mention that webrooming positively moderates the impact on channel integration and is linked to offline patronage intention.

4.2. Theoretical Implications

The importance of studying webrooming stems from the technological advances that arose from COVID-19, which brought with it the need for retailers to adapt to these changes and implement omnichannel strategies (Kohli et al., 2024). COVID-19 prompted retail channels to reinvent themselves, quickly adapting to changes in consumer behavior, forcing them to use webrooming. This is also due to the need for personalized assistance from salespeople with greater product knowledge and capable of answering questions. (K., 2024)

This scientific article explores webrooming in the retail channel, delving into the variables that impact its implementation. It also affirms the importance of Generation Z interacting with online and offline retail environments. It also highlights the importance of implementing an omnichannel strategy, which allows for channel integration to boost consumer purchase intent (Chen & Chi, 2024).

4.3. Practical Implications

This research is useful for retailers facing the challenge of attracting Generation Z, a group that is entering EAP. Understanding their purchasing behavior is essential to attracting them with products and services tailored to their preferences. Therefore, retailers can benefit from research findings showing how webrooming moderates the relationship between channel integration and in-store purchase intention. Akter et al. (2019) point out that today's customers use multiple channels simultaneously in their purchasing process. For example, they shop online and then pick up their products in-store and search for better prices online while in-store or switch channels.

Akter et al. (2019) and this research highlight the importance of retailers understanding omnichannel, as this will allow them to identify ways to create and capture greater value for consumers. The integration of technologies in retailers facilitates seamless experiences in both online and offline environments that meet the expectations of Generation Z (Thaichon, 2023).

On the other hand, the impulse buying hypothesis negatively impacts webrooming among Generation Z. These results provide retailers with valuable information to adapt to this behavior. This research suggests that promotions and discounts could be implemented at the physical point of sale. As mentioned above, Generation Z makes more informed purchases. Therefore, these strategies can encourage impulse purchases. However, as Gupta and Cooper (1992) point out, consumers' response to promotions will depend on the discount, the store's image, and the type of product (whether a well-known brand or the store's own).

In this sense, retailers can strengthen Generation Z's offline loyalty and patronage through incentives such as promotions or loyalty programs. As Liu (2007) points out, these programs can increase purchasing and loyalty levels with low or moderate levels of patronage. Furthermore, it is essential for retailers to align with Generation Z's values so that the customer experience reflects and addresses their preferences for convenience and impulse buying in an omnichannel environment. This could be achieved through shopping experiences that offer efficient navigation and a seamless transition between online and physical channels.

Conclusion

The findings of this study provide retailers with key information for developing effective marketing strategies in the digital environment. Understanding webrooming as a predominant behavior among Generation Z, along with channel integration, allows companies to create more personalized and consistent shopping experiences. This integration, complemented by an omnichannel approach, can significantly improve commercial effectiveness and strengthen customer loyalty in an increasingly hybrid market.

Contrary to initial assumptions, the convenience orientation does not have a positive impact on webrooming behavior within Generation Z. This can be explained by the consumers' need to physically interact with the products and receive assistance from salespeople, which remains essential to their shopping experience. Interaction with salespeople builds trust, influences the purchasing decision, and ensures that the acquisition process is not necessarily quick, which has relevant implications for retailers to adapt their marketing strategies.

On the other hand, the hypothesis regarding impulse buying in digital media was accepted, showing that this type of buying significantly influences consumer behavior, as it is linked to strategies that prioritize instant gratification. The results indicate that impulse buying moderates webrooming by 26%, highlighting the positive impact of this behavior on the hybrid shopping model. This underscores the importance of emotions in consumers' purchasing decisions.

Additionally, the hypothesis concerning channel integration and offline sponsorship was validated. Channel integration enhances the consumer shopping experience and has a direct impact on their intention to purchase in physical stores. Webrooming facilitates the transition between online and physical channels, motivating consumers to complete their purchases in physical stores. This highlights the importance of developing effective omnichannel strategies to optimize the experience and foster customer loyalty.

Finally, limitations were identified in the methodology, especially due to focusing solely on online surveys, which may generate bias and limit the generalization of results. The study is also limited to a single generation and specific hybrid shopping behavior. These limitations open new avenues for research, such as expanding the sample to other generations, analyzing the impact of geographic context, and comparing webrooming with showrooming. These future investigations will contribute to a deeper understanding of consumer dynamics in the omnichannel environment.

Although the study focuses on Generation Z residing in metropolitan Lima, the results obtained have implications that can be applied in broader contexts. The variables analyzed reflect behavioral patterns that have been identified in various international studies. In this sense, Generation Z's behavior regarding webrooming is part of a global trend linked to the evolution of retail and digital transformation. Therefore, the results of this research can serve as a starting point for comparative studies in other countries with similar socioeconomic characteristics, where the transition toward an omnichannel model is also underway. Moreover, by challenging the assumption that convenience orientation drives webrooming, this study opens a relevant line of analysis into the underlying motivations of young consumer behavior.

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CONTENTS

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MANAGERIAL COMPETENCES IN REGULATED INDUSTRIES

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Olga Niemi [2]

Abstract: Background: Managerial competencies are vital in regulated industries, where organizations must balance strict legal compliance with competitiveness. **Methods:** This study applies Quinn's managerial competency model to a Latvian pharmaceutical company, focusing on three factors: Processes and Results (F1), Human Relations and Innovation (F2), and Public Interest (F3). Data from 55 employees were collected using a validated Likert-scale questionnaire. Reliability was confirmed with Cronbach's Alpha ($\alpha > 0.7$). Data analysis included descriptive statistics, correlation, and regression testing. **Results:** Correlation analysis revealed strong positive links between F1 (.821**) and F2 (.819**) with company performance, while F3 (.808**) showed weaker significance. Regression results confirmed that F1 ($t = 3.759$, $p < 0.001$) and F2 ($t = 3.632$, $p = 0.001$) significantly predicted performance ($R^2 = 0.742$), while F3 was excluded. **Conclusions:** Findings emphasize that process management and human relations are the most critical managerial competencies in regulated pharmaceutical contexts. Regulated industries demonstrate stronger alignment with public-sector management logic, where compliance, innovation, and stakeholder engagement drive performance.

Key words: managerial competences, regulated industry, pharmaceutical industry
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Introduction

Meeting performance expectations in regulated sectors poses a significant challenge for companies. On one hand, businesses must adhere to various legal requirements and standards (Bondarenko et al., 2021; Koval et al.,2019; Gryshova et al., 2019). On the other hand, they need to maintain their developmental momentum and competitiveness within the market (Anjoran, 2022; Deloitte, 2023). The pharmaceutical sector stands out as one of the most heavily regulated industries, irrespective of whether it involves pharmaceutical production or operations within the supply chain. Within the European Union, pharmaceutical supply chain processes, encompassing storage and distribution, are subject to rigorous regulations. At the same time, in the modern business scope there is a heightened importance of managerial competencies as in the pursuit of successful performance results in the face of dynamic market demands and the complexities of modern workplaces (Koval et al., 2020). Competent leaders are essential for empowering teams and driving organizational success (Martinkiene & Modestas, 2019; Petrova et al., 2020; Odinokova et al., 2018).

Managerial competences are exceptionally crucial for regulated sectors. Regulated sectors often have intricate and evolving regulatory frameworks. Managers need a deep understanding of these regulations to ensure the company operates within the legal boundaries (OECD, 2010). Competent managers can interpret and implement these regulations effectively, ensuring the organization remains compliant and avoiding legal complications. Their attention to detail is crucial in preventing errors that might lead to costly legal consequences or reputational damage (Memon et al., 2022). In regulated sectors, there is a delicate balance between adhering to regulations and fostering innovation, while regulated industry players often balance between the public and private companies (Petrova et al. 2025). Moreover, in regulated sectors, stakeholders, including customers, investors, and regulatory authorities, closely monitor an organization's compliance record.

In the realm of research concerning managerial competencies within regulated industries, as in sectors that integrate aspects of both private and public enterprises, there exists a notable knowledge gap. Existing studies have predominantly focused on analysing competencies within private or public companies or have compared these competencies between different sectors. However, there has been a relative lack of research concerning regulated industries. Only in recent years has attention been drawn to this specific area. This paper aims to address this knowledge gap by analysing the significance and impact of specific managerial competences within regulated industries, with a particular focus on the pharmaceutical sector. The study will concentrate on a pharmaceutical company X (Latvia) that operates not only in the segments of production and supply chain management, but also engages actively in registration and pharmacovigilance processes. Furthermore, the chosen company not only demonstrates financial success but also operates under strict legal obligations from various dimensions. By analysing this case, this research is going to shed light on the connection between competences in a comprehensive manner, thus contributing to the existing body of knowledge.

1. Overview

The essence of regulated industries

Regulated sectors of the economy or so-called regulated industries are sectors of the market where the government actively intervenes and controls the economic activities of companies and individual industries. This type of regulation is a key aspect of the political economy of many countries and is driven by the desire to achieve certain socio-economic goals (Awasthi et al., 2019).

Researchers working on this topic are faced with the lack of a single, comprehensive, and relevant tool for determining the level and nature of regulation in various industries. Firstly, many studies focus on analysing specific aspects of regulation or use a limited set of parameters to determine the extent of regulation in an industry. Secondly, existing tools for classifying regulated industries, such as indices, have their limitations as may be outdated. Iannarelli & O'Shaughnessy discuss the complexity of data management in highly regulated industries. The problems of tracking data storage rules, as well as organizing and structuring large volumes of information are highlighted. Authors caution organizations in highly regulated industries against insufficient engagement with legislators (Iannarelli & O'Shaughnessy, 2015). The importance of maintaining constant communications with legislators for active participation in bills and ensuring the organization's influence on legislative changes is highlighted.

Snedaker & Rima emphasize the need to recognize the limitations and have a legal team in highly regulated industries (2014).

Pharmaceutical manufacturing challenges

Pharmaceutical manufacturing and distribution are among the most regulated industries in the world. This industry is subject to a high level of control and regulation by various government agencies and international organizations. Governments closely monitor the production and distribution of medicines and medical devices to ensure safety, effectiveness and quality of products. Additionally, pharmaceutical companies are also subject to strict monitoring by medical research organizations and regulatory agencies such as the FDA (Food and Drug Administration) in the US and the EMA (European Medicines Agency) in Europe (FDA, 2023; EMA, 2022). These agencies are responsible for assessing the safety and effectiveness of new medical products and ensuring that they are registered and licensed for use on the market. Regulation in the pharmaceutical industry is imperative due to the many stakeholders involved and the high risk of ethical and legal conflicts when promoting and distributing new patented medicines (Petrova & Tairov, 2022). History shows past instances of illegal activity, including the actions of physicians, scientists, medicine regulators, and patient organizations (Kontoghiorghes, 2021). Regulatory affairs play a key role in the pharmaceutical industry, developing and implementing strategies to efficiently develop medicines. One of the main tasks is to ensure that the medicines being developed comply with global standards and recommendations. Regulatory authorities are important players in this context, ensuring the competitiveness of new medicines in the pharmaceutical industry (Majumder, 2020).

Managerial competences in regulated industries

Managers working in regulated industries face unique challenges that require balancing the competencies of those managers in private and public companies. One of the main reasons is the need to comply with strict laws and regulations set by the government. Regulated industries such as pharmaceuticals, healthcare, telecommunications and energy have stringent rules to ensure consumer safety, the environment and social justice. Managers in these areas must be aware of all current laws and standards, their changes and updates, to effectively manage the company and avoid possible legal problems. In addition to strict compliance with laws, managers in regulated industries must consider the needs and expectations of society. Unlike private companies, their activities are often subject to public scrutiny and criticism. In addition, managers in regulated industries must be flexible and adaptable in the face of rapidly changing regulatory environments and technological innovation (Schiavone & Simoni, 2019). These industries often face new challenges, such as the introduction of new technologies, changes in medical standards or data protection requirements. Managers must be able to respond quickly to these changes so that the company can remain competitive and comply with all regulations (Moura et al., 2022).

In the context of the regulated sector, specifically within regulated industry companies, the emphasis on managerial competences presents a unique challenge due to the presence of professionals who often resist managerial intervention in their work. This resistance has historical roots and has made it difficult for traditional private sector practices to be effective within regulated or public organizations.

Development and trends in managerial competences

The trend of managerial competences in regulated sectors, particularly in fields like healthcare and pharmacy, is moving toward a more tailored approach that recognizes the specific challenges posed by the presence of professionals within bureaucratic frameworks. This trend suggests a departure from the one-size-fits-all mentality often associated with private sector practices. Instead, organizations within regulated sectors are increasingly recognizing the need to balance managerial requirements with the unique characteristics and expectations of professionals, such as clinicians in healthcare settings (Fanelli et al., 2019). Managers in regulated sectors are expected to develop competences that not only encompass traditional managerial skills but also demonstrate a deep understanding of the professionals' expertise and a respect for their autonomy. In the post-COVID era, several trends are shaping the landscape of managerial competencies, reflecting the evolving needs of businesses in response to the challenges posed by the pandemic and the changing global environment. Basic managerial competencies are shifting towards non-specialized cognitive skills such as critical thinking, creativity, and innovativeness. At the same time, managerial competencies, specifically in the areas of communication, leadership, teamwork, and planning, have a significant influence on business continuity. Consequently, the trend in managerial competences appears to be moving towards an approach that combines traditional skills like communication and leadership, with modern requirements such as adaptability, strategic thinking, and a deep understanding of market dynamics. Managers are expected to possess a diverse skill set that equips them to navigate complex and ever-changing business environments, ensuring the continuity and success of their organizations (Kabii & Kinyua, 2023).

Moreover, in today's digital economy, managers are confronted with a transformative movement that significantly impacts the way businesses operate. One of the key challenges faced by managers in this digital era is the need for making strategic decisions that are not only fundamental and proficient, but also timely (Armusevica et al., 2019). This shift towards a digital economy underscores the importance of strategic decision-making based on a foundation of well-informed and relevant knowledge. It is also being noted that leadership skills, while undoubtedly crucial, do not directly influence firm performance in the digital economy (Heubeck, 2023).

2. Methodology

Methodological background

To determine and develop managerial competencies, various assessment methods are used, which allow not only to measure the level of professionalism, but also to adapt management strategies to the constantly changing requirements of the business environment. In recent years, based on research and fruitful collaboration with the business sector, several key tools for measuring and analysing managerial competencies have been developed. As per the literature (Gerhart et al, n.d; Roszyk-Kowalska, 2016; Lawicka & Sitko-Lutek, 2017; Raisiene, 2014) the division is as follows: Assessment centers are special programs in which managers are assessed on a range of competencies while completing a variety of tasks and scenarios. These programs typically include role-playing games, business simulations, and group exercises. The assessment is carried out using structured questionnaires or interviews, which allow for different perspectives on the manager's competencies from different groups of people in the organization. The MSAI (Managerial Skills Assessment Instrument) questionnaire is designed to assess managerial skills and abilities. The SHL (Saville & Holdsworth Limited) Manager Test aims to assess various aspects of managerial ability, including analytical thinking, problem solving, communication and time management. Advanced Managerial Tests are more complex and specialized methods for assessing managerial competencies. These may include various case studies, simulations of business situations, or analysis of strategic decisions. Psychometric tests are aimed at assessing the psychological characteristics of a manager, such as personality traits, motivation, stress resistance, etc. Private companies can adapt these assessment methods to the specific requirements and strategies of the company, which allows them to obtain a more nuanced and targeted assessment of the competencies of managers.

For the public sector, these assessments of managerial competencies are less suitable since they are usually subject to strict government control and regulation. The use of complex evaluation methods may be more difficult to implement in an organization where there exist a more rigid hierarchy and bureaucracy. As per the literature (Freitas & Odelius, 2022; Paz & Odelius, 2021) in the public sector, there is often a simplification of scales for assessing managerial competencies. Unlike the private and corporate spheres, where more complex methods are used, the public sector, due to its specifics, usually resorts to more universal and simplified assessment models. This is because state and public organizations are often focused on meeting national needs and social functions, and not on competition in the open market. Three main categories of managerial competencies in the context of the public sector are identified. The first category includes people skills such as empathy, communication, and teamwork. The second category is business skills, including budget management, planning and project coordination. The third category is self-management skills, which involve the ability to adapt to change, manage stress, and use time effectively. Quinn proposed a managerial competency model that is often used in the public sector. According to this model, successful managers in government and public organizations must possess four key characteristics: internal stability, external focus, flexibility, and systems thinking.

Regulated industries are often a unique mix of private and public components that arises from the need to meet government standards and regulations. While maintaining the competitiveness and efficiency associated with the private sector, there is a challenge in assessing management competencies in such industries is balancing government expectations with private business requirements, which requires a unique and flexible approach to assessment. Based on a review of the literature, regulated industries do not have a single rating scale that satisfies all needs. In recent years researchers (Sacre et al., 2023; Iam, 2019; Kakemam & Liang, 2023) are actively creating their own scales, considering the specifics of their industry. However, often these scales are either too broad and include too many parameters, or, conversely, too narrow and do not cover the full complexity of the managerial tasks.

Methodology of the research

To assess managerial competencies in a regulated industry, particularly in a pharmaceutical company, the author turned to Quinn's model of managerial competencies. This model is appropriate for the study because Quinn identifies the key characteristics of successful managers as being internally resilient, externally focused, flexible, and exhibit systems thinking. In the pharmaceutical industry, where government standards must be met while remaining competitive, these qualities are especially important. To conduct the survey, it was decided to use an already proven scale for assessing managerial competencies. The already tested scale has high validity and reliability, which will allow us to obtain accurate and reliable results. This choice will provide us with the opportunity to conduct a comparative analysis of data and highlight the main trends in managerial competencies in a pharmaceutical company, taking into account the specifics of the regulated industry and compliance with government standards. The choice of an already proven scale for assessing management competencies, intended primarily for public companies, is justified by a number of important factors that make it suitable in the context of a regulated industry, in particular in a pharmaceutical company. Since the scale covers various aspects of management and leadership, which is important in the pharmaceutical field, where it is necessary to manage both business processes and research projects. It covers not only technical skills, but also soft competencies such as communication, team motivation and strategic thinking.

To analyse the influence and importance of various competencies in the regulated pharmaceutical industry, a model proven of its significance was used, narrowed down to three main factors: Processes and results (F1), Human Relations and innovation (F2) and Public Interest (F3). Each of these factors included several subfactors. To assess the influence of each factor and their importance for the operation of an enterprise in a regulated industry, two key questions were proposed: how strongly a managerial competence influences the company's performance in this industry, and how important this managerial competence is. The average score was calculated by multiplying the responses to both questions. To conduct the survey, a Likert scale was used, consisting of five levels, where each level represents a specific answer to the question. The scale covers a range of responses from “absolutely no” to “absolutely yes.” The use of the Likert scale is justified by its ability to provide survey participants with the opportunity to express their opinions, considering various nuances of opinion, which allows us to obtain a more detailed and accurate picture of the assessment of each of the questions (Iam, 2019). Each question has its own Likert scale, where respondents can choose a response level from 1 to 5 depending on their opinion. After collecting data for each factor, the average indicator (mean) was calculated for all subfactors within each of the dimensions of managerial competences (F1, F2 and F3).

To determine the overall impact and importance of each competency in the regulated pharmaceutical industry, the impact and importance scores obtained from the responses to the corresponding questions were multiplied. In addition, the questionnaire contains questions related to assessing the performance of both the enterprise, department, and individual productivity of respondents. The ratings obtained in answers to these questions is subject to calculation of the mean, which will form an indicator of the performance of the enterprise. The object is a pharmaceutical company operating as a comprehensive distributor of pharmaceutical goods and medical devices in Latvia. The company under study employs a total of 59 employees excluding the Board, with four new additions to the team. For the purpose of this research, the main focus will be on the 55 existing employees, excluding the recent hires. This group forms the core population that will be analyzed and studied in detail to gain insights into various aspects of the company's operations, procurement processes, and compliance with regulations.

3. Results

First, the reliability analysis assesses the internal consistency and stability of the measurement scales used in the research.

Table 1
Reliability results

Dimension	Managerial competence factor	Cronbach's Alpha
Influence	Processes and results	0.902
	Human Relations and innovation	0.889
	Public Interest	0.804
Importance	Processes and results	0.878
	Human Relations and innovation	0.923
	Public Interest	0.892
Performance	Performance	0.828

Source: Authors' development

A commonly accepted rule of thumb is that a Cronbach's Alpha value above 0.7 is considered acceptable for research purposes, all scales demonstrated the required results above threshold of 0.7.

Within the Influence dimension, "Processes and Results" received a moderate score of 3.23. This suggests a collective belief that managing processes and achieving specific outcomes holds a moderate level of importance for the company's success in the regulated industry. Similarly, "Human Relations and Innovation" garnered a mean score of 3.32, indicating a perceived moderate importance placed on fostering positive relationships and encouraging innovation within the organization. The factor "Public Interest" received a relatively higher score of 3.52, signaling a consensus that active engagement in public interest initiatives holds a moderately significant impact on the company's overall performance. Moving on to the Importance dimension, "Processes and Results" were rated slightly above moderate at 3.49, indicating a belief that effective management of processes and the attainment of desired outcomes carry some significance for the company's performance. "Human Relations and Innovation" received a moderately higher importance score of 3.55, suggesting that respondents consider cultivating positive relationships and promoting innovative thinking as moderately important factors influencing the company's success in the regulated industry. Lastly, "Public Interest" was perceived as relatively more important, receiving a score of 3.67. This signifies that respondents view the company's engagement in public interest initiatives as moderately significant, believing that societal contributions play a role in enhancing overall performance.

The multiplied scores for each factor of managerial competences' influence and impact provide a comprehensive perspective on their combined significance for the company's performance in the regulated industry. With a moderate combined score of 11.36, effective management of processes and specific outcomes showcases a notable impact, emphasizing its role in shaping the company's overall success. Fostering positive relationships and encouraging innovation, represented by a combined score of 11.88, are also moderately significant, highlighting the importance of a collaborative and innovative work culture. The factor with the highest combined score, 13.27, belongs to engaging in public interest initiatives, underscoring the considerable impact of the company's societal contributions and community engagement on its overall performance.

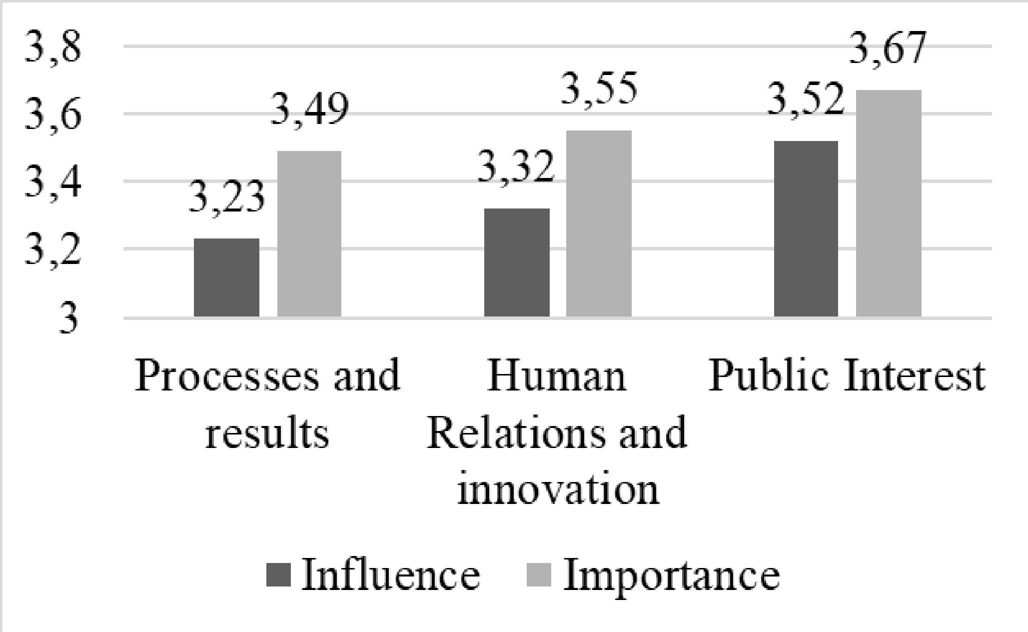

Figure 1. Comparison of the dimensions
Source: Authors' development

Next, authors follow a two-step process: first, checking for normality using the Kolmogorov-Smirnov test with p>0.05, meaning the null hypothesis of normal distribution is not rejected, and then using the Pearson method to analyze the data correlation assuming it is normally distributed. The correlation analysis revealed the significant impact of different managerial competences on overall company performance within the regulated industry context. Management of Processes and Results (F1) demonstrated the strongest positive correlation with Performance (.821**), indicating that companies excelling in process optimization and desired outcomes tend to achieve higher performance levels in the regulated sector. Similarly, fostering positive Human Relations and Innovation (F2) (.819**) within the organization showed a robust positive correlation with Performance, underlining the importance of cultivating positive relationships and encouraging innovative thinking in enhancing overall company success. Active engagement in Public Interest initiatives (F3) also exhibited a substantial positive correlation with Performance, emphasizing the positive influence of social contributions on a company's overall performance (.808**) in the regulated industry.

Table 2
Linear regression results

Model		Unstandardized Coefficients	
		B	Std. Error
1	(Constant)	2.105	0.156
	Processes and results	0.079	0.021
	Human Relations and innovation	0.077	0.021
a. Dependent Variable: Performance			

Source: Authors' development

The results indicated that while Processes and Results (F1) and Human Relations and Innovation (F2) significantly influenced company performance, Public Interest initiatives (F3) did not show a significant impact and were consequently excluded from the regression model. In the regression analysis conducted to examine the impact of different managerial competences on company performance within the regulated industry, the model presented explanatory power, with an R-squared value of 0.742. The regression coefficients further illustrate the relationship between the variables. The coefficients represent the change in the Performance score for each one-unit increase in both managerial competence's types. The t-values for both Processes and Results (F1) (t = 3.759, Sig. = 0.000) and Human Relations and Innovation (F2) (t = 3.632, Sig. = 0.001) were significant, highlighting the robustness of these relationships.

4. Discussion

The strong correlation between Processes and Results (F1) and Human Relations and Innovation (F2) with company performance in the regulated industry can be attributed to several key factors. In regulated industries, adherence to processes and regulations is paramount. Companies that excel in managing their processes and achieving desired outcomes demonstrate a high level of operational efficiency and compliance. The correlation between Human Relations and Innovation (F2) and performance highlights the importance of fostering a culture of innovation within the company. Innovations in products, services, or operational methods can give a company a competitive edge, allowing it to adapt to changing market demands and regulatory landscapes. A workforce that encourages and implements innovative ideas can navigate regulatory challenges more effectively, leading to improved performance outcomes.

At the same time, Effective management of processes (F1) is directly linked to the quality and safety of products and services. In regulated industries, ensuring the highest quality standards and product safety is essential to compliance. Companies that excel in these areas not only meet regulatory requirements but also build customer trust.

The results of this study fit well with previously reviewed management competency models and literature, highlighting the importance of certain aspects of management in regulated industries, especially the pharmaceutical industry. The first key factor highlighted in this study is effective process management and compliance, which highlights the importance of strong compliance in regulated industries. This aspect is consistent with the literature analysis, which highlights people skills, business skills, and self-management skills. The second important aspect highlighted in the results of this study is the relationship between human relations and innovation and overall company performance in a regulated industry. This aspect also corresponds to the ideas of the literature that highlighted the importance of human capital and innovation for the successful development of companies. Finally, ensuring product quality and safety, as well as maintaining ethical standards, are key aspects of management highlighted in the work of Quinn et al., where public interest factors are highlighted. These aspects are also consistent with the model examined in our study.

Based on the discussed models and research results, it can be concluded that aspects of managerial competencies in regulated industries are closer to the analysis of public companies than to private ones. In summary, the managerial competency models and research findings confirm that regulated industries have more in common with state-owned companies in terms of managerial skills and competencies.

Conclusions

The results of the study allow us to draw several key conclusions regarding managerial competencies in this area. First, effective process management and regulatory compliance prove to be fundamental aspects in regulated industries. Positive relationships within the team and external innovation contribute to adaptability to changing regulations and market demands. The strong correlation between Processes and Results, Human Relations and Innovation, with company performance in the regulated industry underscores the importance of effective operational management, a culture of innovation, positive stakeholder relationships, assurance of quality and safety, and adherence to ethical standards. These competences not only facilitate regulatory compliance but also contribute significantly to the overall success of companies in regulated sectors. Companies that excel in these areas are better positioned to navigate complex regulatory environments, meet industry standards, and deliver value to their stakeholders, ultimately leading to improved performance outcomes.

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D. A. Tsenov Academy of Economics, Svishtov

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CONTENTS

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CHALLENGES AND SOLUTIONS FOR SUSTAINABLE TRANSPORT IN THE ERA OF NEW ENVIRONMENTAL REQUIREMENTS IN BULGARIA AND TÜRKIYE

Antoaneta Kirova[1],
Selahattin Kosunalp[2]

Abstract: The background of the research is based upon the conclusions of the International Transport Forum (OECD, 2-21) that one of the big challenges for the environment comes from the transportation sector, which is responsible for approximately 23% of the discharged CO2 emissions; this quantity is expected to increase to 40% by 2030 and to 60% by 2050. The already expressed negative tendency is calling for new models for freight transport represented in the form of inter-modality. This means the logistics side of international economic relations, represented by land mode of transport (road and rail), inland waterways, short-sea shipping, and even airborne transport at short distances, should make use of viable combinations to decrease the harmful substances; it also concerns domestic transport policy which should stimulate non-road transportation solutions. The European Green Deal practically outlines the plan for the decarbonization of the economies of the member states of the union until 2050, with the aim of improving the well-being and health of European citizens and future generations with an accent over transport activities. Usually, logistics providers offer transport solutions, combining several modes, led by stakeholders' expectations for reduced time and costs of transferring goods, reliability, and frequency of services. The biggest challenge is to design a resilient transport network, where the users can easily switch between transport modes in real time, while there is enough evidence that combined transport solutions are in accordance with the contemporary requirements towards freight transport. Infrastructure investments, easing transborder connections and interoperability in general have a positive effect over the development of combined transport. This paper is addressing the opportunities and difficulties in employing combined transport solutions using the methodology SWOT-TOWS and applying the research results to the existing possibilities in Bulgaria and Turkey as a neighbor non-EU country in the role of advanced competitor in the development of

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combined transport solutions. The conclusions confirm that economic development and its interaction and dependence on international logistics development is the most important factor of the environment encouraging or detaining the combined transport solutions. It also affects investments in transport infrastructure, the application of ITS and AI, the technological factors influencing the general trends of transport development and in particular, freight transport.

Key words*:* combined transport, “green” international logistics, transport policy, strategic analysis

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Introduction

According to the research of ITF about the Decarbonizing Transport Initiative, one of the big challenges for the environment is that the transportation sector is responsible for approximately 23% of energy-related CO2 emissions, which are expected to increase to 40% by 2030 and to 60% by 2050 (International Transport Forum, 2019). This negative tendency requires multimodal freight transport models in international logistics, represented by land mode of transport (road and rail), inland waterways, and also transport at short distances, both sea and air. Transport policy at the national level should stimulate non-road transportation solutions, a part of which are the new approaches in last-mile logistics.

Usually, logistics providers offer multi-transport solutions, combining several modes, led by stakeholders’ expectations for reduced time and costs of goods transferring, reliability, and frequency of services. The difficulties arise from the already existing modal split due to political, economic and technological issues. The biggest challenge is to design a resilient transport network where users can easily switch between transport modes in real time. Still, in our opinion, logistics providers are already offering “green” solutions, creating a favorable environment for the choice made by clients.

The European Green Deal practically outlines the plan for the decarbonization of the economies of the member states of the union until 2050, with the aim of improving the well-being and health of European citizens and future generations. Part of it is the provision of efficient, safe, and environmentally friendly transport (European Council). This paper aims to discuss challenges and solutions for green freight transport based on multimodal solutions.

1. Literature review and methodology

Reducing the share of road transport on the way towards less polluting and more energy-efficient modes of transport is the main aim of European transport policy due to the negative consequences, due to the expected increase of road freight transport by 40% until 2030 and by over 80% until 2050 (Common Transport Policy). There is a strong support to intensify the use of multimodal solutions by multiple actions, which have proven their importance over time, such as:

1) The internalization of external costs in view that all the environmental outcomes connected with the surplus of transportation services should be covered in line with the polluter pays principle (OECD, 1975). A lot of measures have been performed towards internalization of external costs throughout the years, such as increasing taxes on heavy road vehicles, fuel taxes, etc. (OECD, 2022) The new trends in this field are linked to the effects of digitalization, aiming at highlighting the potential for making the mobility system more sustainable, showing the risks that could occur in cases of mismanagement (European Environment Agency, 2023).
2) More targeted infrastructure investments, aimed at better interconnections between the single modal networks. For instance, new projects envisaged by the CEF financial instrument of the EU will contribute to transport policy objectives such as enhancing major interoperable transport axes along the Core Network Corridors, improving safety and resilience of the network and increasing the capacity and performance of the rail, inland waterways and short sea shipping infrastructure, within an integrated multimodal transport system, with a view of obtaining the interconnections referred to;
3) Better use of information concerning traffic, capacities, availability of infrastructure, cargo and vehicle positioning.

In our opinion, combined transport is an adequate general response to the needs of resilient transport in the era of new environmental requirements. The definition of intermodal transport by OECD (OECD, 2006) is “movement of goods (in a single loading unit or a vehicle), by successive modes of transport without intermediate handling”. Within ASEAN, combined transport is considered “the carriage of goods by more than two modes of transport without any handling of the freight when changing the modes through an intermodal transport chain with one single contract of carrier”. In the USA, it refers to “Containerized Rail Transport”. The definition by the EU Commission from the past century is “a characteristic of a transport system that allows at least two different modes to be used in an integrated manner in a door-to-door chain" (1977). Combined transport (CT) is key to achieving a carbon-neutral transport sector in the context of the European Green Deal. Combined Transport, with a range of benefits, contributes to a better quality of life and proposes a seamless transport solution, improving the productivity of the entire chain. (UIC, n.d.).

An extended view over the definition, market structure, and key elements of combined transport is presented by the 2020 Report on Combined Transport in Europe (UIC, 2020), based mainly on containerization, accompanied or unaccompanied trucks, with an accent on the possibilities of extended use of railway transport. The geographic scope is also discussed, dividing the combined freight into internal (domestic) and international. The European freight transport market is envisaged, with proof of the role of international combined transport. The topic is further followed in the UIRR Report of 2023 (Patru).

The authors accept the definition for combined transport of Eurostat: "multimodal (combined) transport of goods in the same intermodal transport unit by successive modes of transport without handling when changing modes" which is reflecting, to the full extent, the role of transport in the logistics chain. The modal split is visualized by Fig.1.

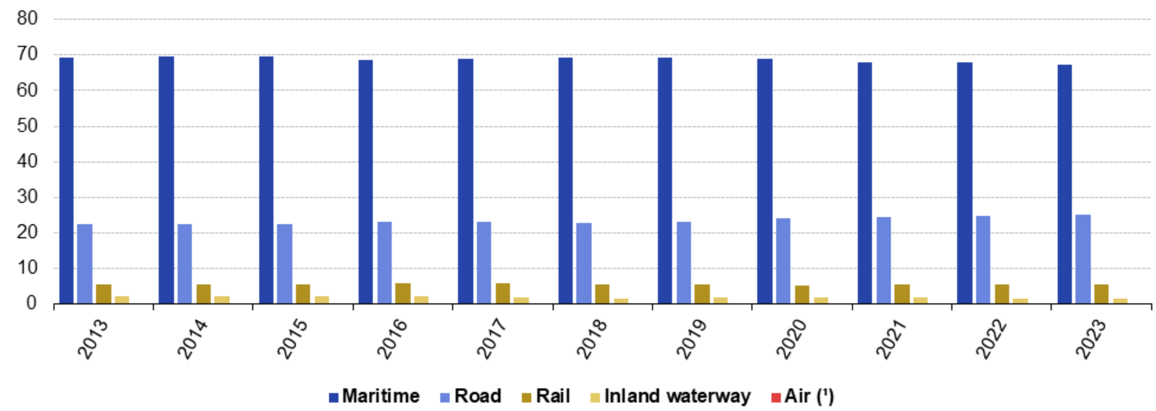

Figure 1. Freight transport modal split (Eurostat)

The data proves the role of maritime transport, with the biggest share in the international trade realization, followed by road transport, and still an insufficient participation of rail and inland waterways. Bearing in mind the opportunities of maritime transport to participate in international logistic chains with combined transport, the above-shown tendency is positive, but on the other hand, due to the limited share of rail and inland waterways, there is free potential for further development.

Due to the dynamics and urgent transition to digitalization in transport, the expectations are for the provision of better synchronized, smooth, and efficient transport services. The on-time information exchange is a further step towards cost optimization, energy savings, and above all, raising service quality.

Further, the big share of road transport with proposals for technological changes and transition to “low carbon combined transport” is discussed in a study of UIRR Zero Carbon Combined Transport (UIRR, 2025). Research on combined transport is done within UN ESCAP, by generating a multimodal transportation concept and a framework, proposing different definitions of combined transport. In fact, the number of definitions is one of the issues about combined transport in general.

In our opinion, the basic strength of combined transport is that it unites positive features of different modes, while eliminating the negative, thus cutting delivery costs, an important issue in international logistics chains. Combined transport becomes more important in view of the European Green Deal and the general tendency to diminish the destructive effects of freight transport on the environment.

To assess the situation about the combined transport from the point of view of its positive influence on business and the environment, the method SWOT-TOWS (Gonan Bozac, 2008) is applied, based on the results of secondary research. The outcomes are focused on strategic issues concerning the future of combined transport in the Republic of Bulgaria and the Republic of Turkey.

2. Situation analysis of the attitudes toward combined transport

As a result of deregulation, leading to the necessity of restructuring, refocusing, and changing business models, the private sector companies have taken the initiative in the field of inter-modality. Combined Transport is promoted within the EU, both legally and financially, providing support for projects relating to combined transport. The aim is to enable modal shift and reduce long-distance road freight transport through encouraging cooperation with rail transport and the other ecologically friendly modes; also, the achievement of the goal for transferring 30% of road freight over 300 km to other modes by 2030 and over 50% by 2050, outlined by EC transport policy in 2011 White Paper on Transport, is dependent on the good cooperation between different modes of transport and the seamless door-to-door freight services. The following types of actions support greater use of multimodal solutions:

1) Internalized external costs in all modes of transport, with a view to distributing them among the users, operators, and investors according to the “polluter pays” principle.
2) More targeted infrastructure investments, aiming at better interconnections between the single modal networks.
3) Better use of information (on traffic, capacities, availability of infrastructure, cargo, and vehicle positioning).

The public consultation carried out by the European Commission, based on an online questionnaire about the application of EU regulations over combined transport (Figure 2) showed different advantages and shortcomings, which are indicated in the SWOT table below.

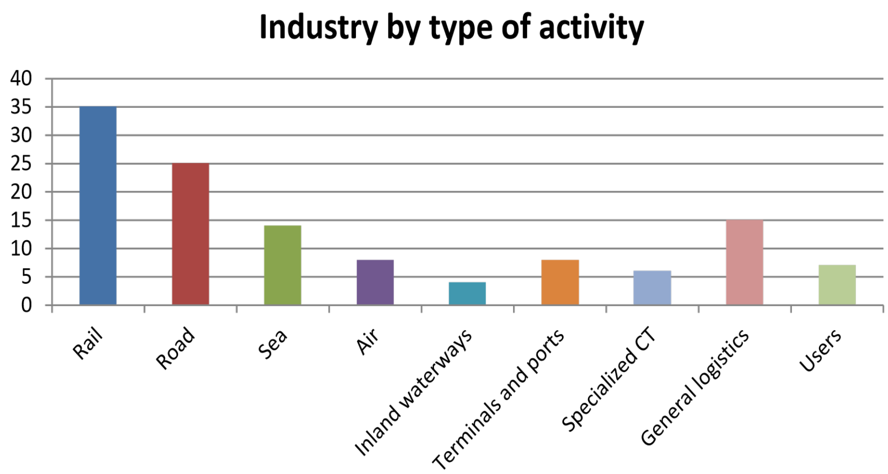

Figure 2. Interested parties in Combined Transport (individuals, NGOs, public authorities, business associations, large enterprises, industry, SME ((KombiConsult GmbH, 2915)

As far as the organizational forms of combined transport are concerned, the coverage “land transport”, followed in the transportation chain either by maritime transport or a combination “maritime/air transport”, was supported. An interesting result came from the assessment of the short-sea shipping (SSS), which is generally encouraged by European transport policy (SSS) (European Union, Document 51998PC0414(01)) and the distance at which it will be applied. Also, tri-modal (or more) combinations, in particular road with rail + SSS and road with rail + inland waterway, are considered relevant.

The research results clearly determine that attitudes toward combined transport on the side of the interested parties listed above are positive overall, though still initiatives are more policy-driven than market-driven, with expressed preferences for logistics chains including land, sea, and air modes of transport.

3. SWOT-TOWS analysis of the combined transport in international logistics

The analysis starts with a SWOT Matrix for combined transport. Each of the quoted strengths, weaknesses, opportunities, and threats is carefully derived from secondary literature research and is evaluated by number 1. For quality analysis, all four categories are divided into three subcategories, namely organizational, economic (business) and legal, and the overall impact of them is calculated as a sum of the elements included in them. The results are also presented by a SWOT Graph.

Table 1

SWOT Matrix for combined transport

Strengths	Score	Weaknesses	Score
Organizational	**4**	**Organizational**	**4**
1) Technologies without transshipment for better protection of the transported goods.	1	1) Reduction of the average speed of movement, due to the different possibilities and requirements in the individual modes of transport, included in a combined transport chain.	1
2) Standard, closed, sealed, and bar-coded packaging reduces the risks of mis-sending goods.	1	2) Lack of sufficient capacity for carrying out combined transport (for example, block trains) as well as appropriate infrastructure (logistics centers, cargo villages for combined transport, offering conditions for different loads).	1
3) The simplification of loading speeds up the turnover, hence shorter delivery periods, and increases the commercial security of transport.	1	3) Insufficient administrative capacity of all interested parties to use the possibilities of financial instruments, grants, and state aid.	1
4) Fewer inspections since containers are subject to pre-control and accordingly sealed.	1	4) Insufficient personnel with appropriate education and qualifications for the needs of combined transport.	1
Economic (business)	**3**	**Economic (business)**	**2**
1) Further cost reductions (no vehicle detention, decreased insurance fees) due to the lack of transshipment operations	1	1) As per calculations in the National plan for development of combined transport in Bulgaria until 2023, the direct costs of combined door-to-door transport, incl. overloading, on rail distances below 550 km are higher than those for road transport.	1
2) Delivery costs optimization through fuel savings and fewer loading and unloading operations	1	2) Lack of equal and transparent access to private railroad terminals.	1
3) Possibility of carrier choice to obtain the best rates for each route stretch	1	**Legal**	**1**
Legal	**3**	Outdated bilateral border agreements that hinder the effective and efficient reception and transfer of freight trains and especially block trains, between container hubs	1
1) Important issues solved, increasing the possibility of using the ecological transport modes (rail and waterborne transport), and reducing road accidents	1		
2) Increased ability to negotiate terms per stage or stretch of the route, since each supplier is responsible for its service.	1		
3) Single Contract of Carriage, though the carrier's liability is divided per stage of services.	1		
Overall score	**10**	**Overall score**	**7**

Opportunities	Score	Threats	Score
Organizational	**6**	**Organizational**	**3**
1) CT operations bring new ecologically acceptable and sustainable alternatives; increasing the use of rail transportation is expected to lessen the congestion levels and the harmful emissions of noise and poisonous gases, such as CO_2.	1	1) Delays in the implementation of projects for the modernization of port and rail infrastructure	1
2) Rail interoperability and deployment of ITS contribute to reducing costs for rail carriers.	1	2) Part of the functioning industrial and newly constructed logistics areas are not connected to the railway network.	1
3) Modernization of rail and port infrastructure, thus increasing the competitiveness of CT.	1	3) Technological and IT security inconsistencies (for example, regarding the capacity of ports for simultaneous handling of container ships, automation, working information channels, such as EDI and blockchains between ports, intermodal terminals, container lines, customs, and other government institutions, warehouses, carriers, and use of Intelligent transport systems (ITS)	1
4) Introduction of an electronic document exchange system to replace paper documents and stamping.	1		
5) Establishment of an electronic information exchange system at seaports, which will further bring an increased efficiency in containerships' processing.	1	**Economic (business)**	**1**
		Difficulties in the predictability of the costs and time for the passage of containers through the ports, due to the existing customs system	1
Economic (business)	**3**	**Legal**	**2**
1) The existing road, rail, and port infrastructure for combined transport provides good coverage of major production centers and urbanized areas, with assured connections to sea and river ports and airports	1	1) Multiple definitions of combined transport leading to deviations between the interested parties.	1
2) International trade relations, served by developed container shipping over the past decade, with generally available land and port infrastructure capacity.	1	2) The different rate of accepting and implementing the control authorities' rules by the EC members (the "transposition act"); issues arising about the quantitative aspects of combined transport (in the quantitative field, concerning weights and measures of the transportation process components)	1
3) Liberalized rail and road freight transport with multiple licensed carriers providing services in a highly competitive environment.	1		
Legal	**1**		
Solving legal problems within the EU (the European Commission revised the Directive on CT in 2023 for a stronger support to the shift from road freight to lower emission transport modes such as inland waterways, maritime transport, and rail) (European Commission, 2023)	1		
Overall score	**10**	**Overall score**	**6**

The SWOT Graph clearly indicates that the right side, formed by the weaknesses and threats, is prevailing. This means that despite the political efforts and the general interest concerning combined transport, there are still issues for its application from the point of view of finding resilient solutions for international logistics chains.

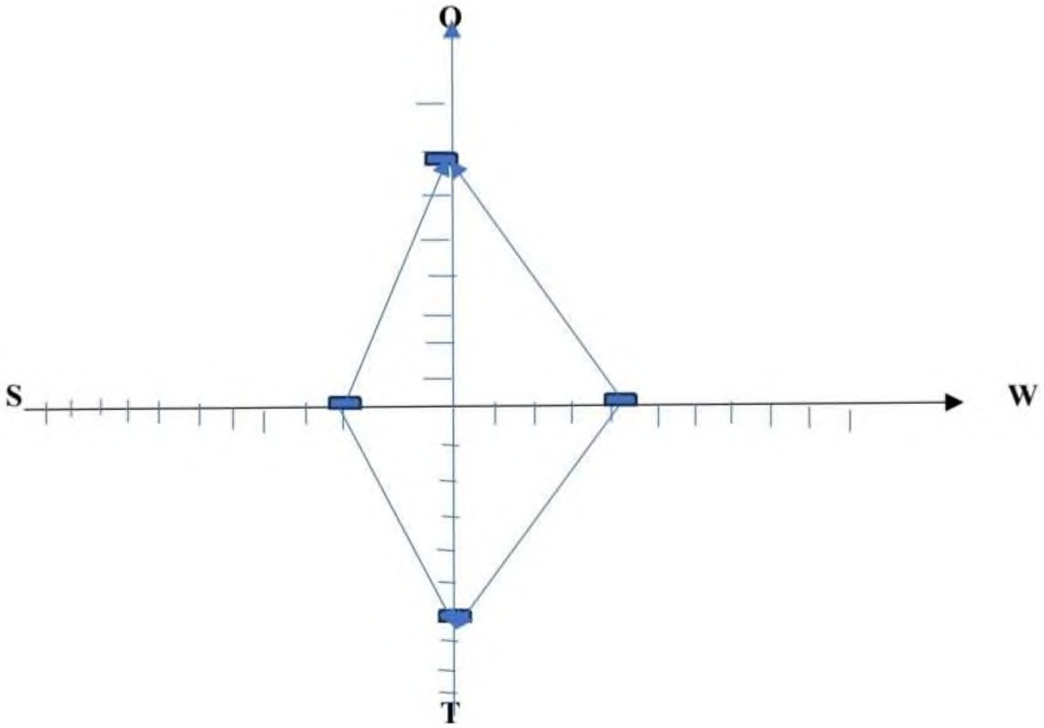

Figure 3. SWOT Graph

Further, the combinations between weaknesses/threats (W/T), strengths/threats (S/T), weaknesses/opportunities (W/O) and strengths/opportunities are developed with suggestions for strategic measures aimed at improving the existing situation and drive attention towards successful efforts for applying combined transport solutions in international logistics.

Table 2
TOWS Matrix for combined transport (suggestions for change management)

	Weaknesses	Strengths
Opportunities	**W/T** **Comparing weaknesses/threats in organization**, we may propose a strategy to speed up the realization of investment projects about the transport infrastructure to overcome the speed reduction, enable more capacity for carrying out combined transport, etc. Despite the efforts in this field, there are still issues, so, incentives to use the possibilities of financial instruments, grants and state aid should be made aware to the interested parties, like different forms of European financing, through proper orientation. Also, the lack of specific infrastructure, such as logistics centers, cargo villages for combined transport and new container terminals in ports should be considered. The training of personnel for the needs of combined transport is another issue, bearing in mind the general withdrawal of trainees for the transport sector. Also, technological barriers connected with IT and digitalization should be removed as a part of the general picture. **Comparing weaknesses/threats in the economic (business) field**, we have to propose combined door-to-door transport to be included mainly in international logistics chains. Clients are easily discouraged if they find discrepancies or unequal treatment in terminals, so measures should be directed to encourage private terminal operators to do their business transparently. As far as customs operations are concerned, it is difficult to make suggestions or recommendations, only that everything connected with border-crossings should be prepared correctly beforehand. **Comparing the legal weaknesses/threats** we recommend an urgent revision of the outdated bilateral border agreements to make the process of transferring block trains between container hubs more efficient. As far as the multiple definitions of combined transport are concerned, it is enough if in practical adaptation of combined transport, the interested parties agree on one of the existing. The transport policy of the different Member States should speed up the process of transposition and implementation of rules on combined transport since only unified rules would motivate the implementation.	**S/T** **Comparing strengths/threats in organizational**, there is a good overall progress determined by high quality and reliability of service, due to improved packaging, simplified loading operations and no trans bordering. Policies should speed up infrastructure renovation as well as the establishment of railway connections to newly constructed logistics areas so that the strengths could be realized. The service reliability will be further improved through future digitalization and automation of processes. **Comparing strengths/threats in the economic (business) field**, the prospects for cost reductions are real, due to speeding up the turnover of vehicles and rolling stock, less insurance payments because of the reduces risks while completing trans boarding and loading. The choice of operators and rates should also be considered. Hence, the only recommendation is for the improvement of the existing customs system, which is currently slowing down the passage of containers through the ports. **Comparing the legal strengths/threats** we focus on the ecological side and safety of combined transport, particularly if rail and waterborne transport comprise parts of the logistics chains, while limiting the participation of road transport, to reduce road accidents. Overall, there are positive signals for the users in terms of negotiating terms per stage of the logistics chain, while they can make a Single carriage contract. The existence of multiple definitions would not be considered in case of urgent transposition and implementation of technical rules by the control authorities of the different Member States

Threats	**W/O** **Comparing weaknesses to opportunities in the field of organization,** we can recommend furthering the achievement of rail interoperability, deployment of ITS, while putting strong efforts in the modernization of rail and port infrastructure, as well as of specific infrastructure (such as terminals) to increase the competitiveness of combined transport. The positive outcomes will solve the issues of average speed reduction, the insufficient capacity for carrying out combined transport. The introduction of electronic document exchange is another positive measure to ease the formalities. The issues connected to insufficient administrative capacity and personnel in general with appropriate education and qualification for the needs of combined transport must be solved first of all by presenting a vision, in view of the increased digitalization and IT applications. **Comparing weaknesses to opportunities in the economic (business) field,** there are stimuli for combined transport solutions on the demand side, such as developed international trade relations backed up by the supply side, represented by container shipping, licensed road carriers, etc., with numerous options to select carriers. Also, the transport infrastructure to be used by combined transport is comparatively well-developed. Further, the proposed business solutions should be based on lower costs due to less detention, lack of transshipment operations, and general savings on loading operations and fuel costs optimization. **Comparing the legal weaknesses to the legal opportunities,** we can suggest encouraging the shift from road freight to lower emission transport modes such as inland waterways, maritime transport, and rail by suitable policy measures. Also, to form better prerequisites for the use of railway transport in combined transport logistics chains, the outdated bilateral border agreements should be renewed, and the transfer of block trains, as well as other freight trains between container hubs made easier.	**S/O** **Comparing strengths to opportunities in the field of organization,** we can clearly identify the opportunities for a higher quality and security of transportation services, due to reduced risks in connection with loading/transshipment and introducing the electronic exchange of documents and information, which also contributes to speeding up deliveries. The favorable process is further backed up by infrastructure modernization and particularly by reaching railway interoperability. Besides the demand side, which expects better efficiency of operations, the combined transport alternatives, if applied, are completely in accordance with the requirements for a sustainable and environmentally friendly transport process. **Comparing strengths to opportunities in the economic (business) field** indicates clearly that there are enough foundations to expect cost optimization on the side of clients, backed up by the rights to choose suitable carriers in a developed market (due to liberalization of trade and transport offerings), according to the needs as well as comparatively good infrastructure coverage of the transportation process. **Comparing the legal strengths to the legal opportunities,** we can identify the Single Contract of Carriage, divided into different stages, decreasing the role of road travel, and besides the increased role of ecological transport modes, such as rail and waterborne transport, a real intention to fight road accidents. A positive outcome linked with the shift from road freight to lower emission transportation modes such as inland waterways, maritime transport and rail is to be expected after the revised version of the Directive on CT is approved, maybe by the end of 2023.

4. Transferring the analysis results to the situation of combined transport in Turkey and Bulgaria

4.1. Combined transport in the Republic of Turkey

Located at the crossroads of Europe and Asia, the Republic of Turkey is one of the most important trading partners for countries on both continents, giving it a great advantage in both importing and exporting goods. As a peninsula which has coastal lines in Mediterranean, Aegean and Black Sea, the transport policy of Turkey is based upon the idea of a balanced transport system, integrating road, rail and maritime transport and providing opportunities for combined transport decisions.

The Republic of Turkey is among the most important trading partners of the Republic of Bulgaria outside the EU. In this regard, the Bulgarian-Turkish border, which is external to the Community, is among the busiest, and this leads to delays in supply chains and additional costs for businesses. In previous periods, in the field of Ro-La transportation, Holding Bulgarian State Railways developed technology for the transportation of freight cars by rail between the Halkali (Turkey) - Dragoman (Bulgaria) terminals, but the initiative never started.

Bearing in mind the role of combined transport for a resilient and sustainable freight transport system, the following recommendations are relevant:

- Further investments in renovation and modernization of railway infrastructure for improved access and connectivity of different types of transport infrastructure, and for the construction of new terminals for combined transport.
- Investments in rolling stock, such as general implementation of innovative technologies for combined transport, from organizational, such as block trains, to the equipment applied, including containers, swap bodies, semi-trailers, etc. For instance, benchmarking shows a widespread measure aiming at specialized rolling stock (wagons) for the carriage of intermodal transport units (such as Ro-La schemes).
- Tax relief for combined transport when the towing vehicle is transported by train.
- Liberalization of specific road corridors for the initial and final stages of Ro-La connections between terminals and nearby border stations (i.e. no bilateral authorization is required for the cargo transportation on these

corridors, starting and ending between the nominated terminals of the Ro-La connections).

- Schemes and measures to reduce costs in combined transport (such as concessions on infrastructure charges, subsidization, and compensation for congestion costs).
- Introduction of fiscal incentives to reduce or exempt taxes, which should further motivate interested parties to use combined transport.

As far as the role of railway transport and the possibilities for inclusion in the international logistics chain, the Connecting Europe Facility (CEF), Cohesion Policy, and the Recovery and Resilience Facility (RESF) are suitable funding opportunities that can drive the much-needed modernization of the rail sector. In our opinion, in connection with the negotiations for future accession to the Community and in accordance with the White Paper on European Transport Policy from 2010, the Republic of Turkey actively supports initiatives in the field of combined transport, due to which there are considerable outcomes. First, "freight villages" were developed (Aksoy Okan, 2015) to ensure the enhanced attractiveness of combined transport, an increase in customer satisfaction, and the share of freight transportation, as well as the prevention of pollution. This is an example of good international practices, with potential to be applied in Bulgaria as well.

In Turkish national and international operations, different forms of intermodal transport can be distinguished:

(a) Road-rail operations for the transport of containers, swap-bodies, semi-trailers, or trucks that are carried on Ro-La wagons.

(b) Block train services and international container train services in Turkey are also developing in international logistics. Several national and international partnerships have been established, with the participation of operators, such as BALCO and the Turkish Chambers and Commodity Exchange Association. A "National Combined Transportation Strategy Document Draft" has been prepared under the EU Twinning Project for Strengthening Intermodal Transport in Turkey. In this context, it aims to establish a safe, balanced, convenient, sustainable, and environmentally friendly transportation infrastructure in Turkey. Considering the goals of Turkey in the period after 2023, the development of the railway network in the direction of block train transportation is very promising for the future in the name of increasing the cargoes of countries such as Turkey and China.

(c) Roll-on-roll-off (RO-RO) maritime operations for the transport of lorries, semi-trailers, containers, or swap-bodies to ships on their own wheels or on wheels attached to them for this purpose. This also covers national and international ferry services, including railway ferries and national short-sea shipping. Also, the importance of sea–rail inter-modality was studied based upon a case between Turkey and Italy, indicating that Turkey needs to invest in the rail connections of the ports.

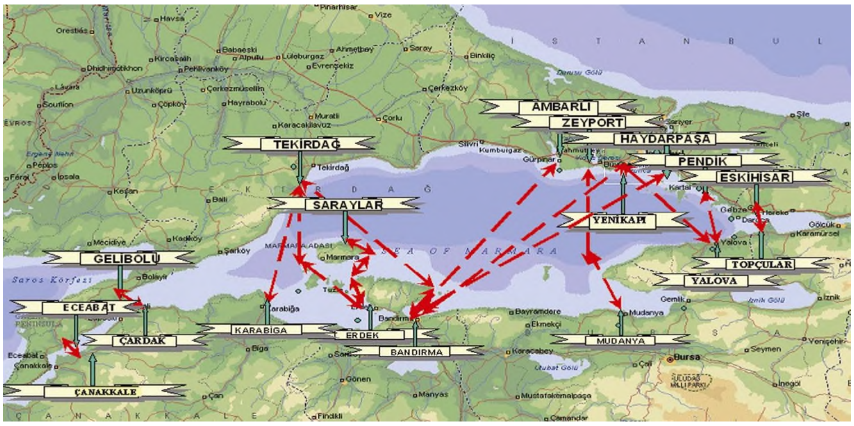

Figure 4. RO-RO services in Marmara region
(Source: Turkish Undersecretariat for Maritime Affairs)

The Republic of Turkey and the Republic of Bulgaria have been initiators of international RO-RO ferry boat services to Western Europe in the 1980s and the early 1990s. Consequently, there are already traditions established, and again, this is an example of good practice about the development of international logistics in times of political turbulence, leading to transportation issues, as was the case for road operators in the Balkan area at that time.

Figure 5. Combined transport of Turkey in the Black Sea region

4.2. Combined transport in Bulgaria

As a member of the EU, Bulgarian transport and forwarding companies, supported by the state authorities, are further exploring the opportunities of combined transport. Since the beginning of Black Sea rail ferry services in 1978, for over 39 years the service has been operated by the Bulgarian shipping company Navigation Maritime Bulgaria as a liner operator working within the framework of a triple intergovernmental agreement between Bulgaria, Ukraine and Georgia regarding operation of direct rail ferry service between ports of Varna (Bulgaria), Chornomorsk (Ukraine) and Poti/Batumi (Georgia). However, since the beginning of the Russian-Ukrainian war, this opportunity has been practically eliminated.

Figure 6. Rail ferry service of Bulgaria

The Ministry of Transport and Communications has launched a procedure to support intermodal operators and the development of existing terminals. The total amount of funding is 34.5 million leva, provided by the Transport Connectivity Program 2021-2027 (Ministry of transport and communication of Bulgaria, 2022). The program aims to support combined transport terminal operators by providing funds for the modernization and expansion of existing infrastructure. The grant for each project is from 1.5 million leva to 4 million BGL, with applicants receiving financial support from the Transport Connectivity Program up to 50% of the eligible implementation costs.

The funds are provided for the purchase of new zero-emission shunting equipment, the implementation of modern IT systems for transport management, and the improvement of the railway and road infrastructure of the terminals. The investments are aimed at all regions of the country, except for the South-West region (Ministry of transport and communications of Republic Bulgaria, 2025).

Conclusions

After presenting evidence that combined transport solutions are in accordance with the contemporary requirements regarding new models of freight transport at an organizational level, it is necessary to overcome geographic obstacles (land-sea-river connections) or to provide opportunities for an increased use of rail transport, combined transport has its potential for wider application in the international logistics decisions.

Infrastructure investments, easing trans-border connections and interoperability, in general, have a positive effect on the development of combined transport. These, however, must be supported by political decisions and economic measures directed both to the operators and the users to encourage the development of new alternatives and their urgent acceptance and implementations.

The decisive factor is economic development, and its interaction and dependence on international logistics, simultaneously encouraging combined transport solutions to protect the environment. The second important factor, closely following the first, is linked to the investments in transport infrastructure, the application of ITS and AI, and the technologies influencing the general trends of transport and particularly, freight transport.

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CONTENTS

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CORPORATE TRANSPARENCY AND DISCLOSURE: METRICS OF MACEDONIAN JOINT STOCK COMPANIES

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Tatjana Spaseska[3],
Meri Boshkoska[4],
Violeta Gligorovski[5]

Abstract: The disclosure and transparency of joint stock companies is fostering safe and sound governing of joint stock companies and reduces the risks of corporate crises and scandals.

The purpose of the research is to identify the transparency level of the Macedonian listed companies that are obligated to comply with the Corporate Governance Code of the Macedonian Stock Exchange, according to 20 indicators related to the publication of information on the company's website and in their annual reports. The study adopts a mixed methods approach, combining qualitative and quantitative techniques. Qualitative component involved collecting and examination of data for 20 corporate governance indicators obtained from reliable sources. Quantitative analysis entailed assessing and scoring each indicator based on predefined criteria, followed by statistical processing and graphical representation of the results using Microsoft Excel.

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The research has shown that the average transparency level of analysed joint stock companies belongs to the "good" level. Considering the findings, the study provides recommendations aimed to enhance the transparency practices of Macedonian joint stock companies.

Key words: corporate governance, transparency, listed joint stock companies, transparency level

JEL: K22, C81, C38, G38

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1. Introduction

Transparency and disclosure (T&D) become a *conditio sine qua non* in contemporary business. A series of events has contributed to the growing attention on corporate governance elevating it to a central issue that is now widely discussed in corporate boardrooms, academic meetings, and policy circles worldwide. During the wave of financial crises around the world, from US and Europe to North East and Southeast Asia, corporate governance (CG) has emerged as a critical issue not just at the national level, but also within regional and international forums. In the aftermath of these events, not only has the phrase corporate governance become more of a household term, but researchers, the corporate world, and policymakers everywhere recognize the potential macroeconomic, distributional and long-term consequences of weak corporate governance systems (Claessens & Yurtoglu, 2012). The main objective of corporate governance is to protect the shareholders' interest by pursuing the management to govern the firm in the direction which is favourable for shareholders (Mohamad et al., 2020). Therefore, the information users such as investors, rulers, academics, creditors, workforces, customers and other stakeholders increasingly demand more transparent and reliable information (Ratnayake & Rajakulanajagam 2022). Mckinsey (2002) found that over 60% of investors are depending on good corporate governance practices as a key factor in decision-making. Therefore, transparency must be a fundamental factor of a corporate governance system.

This paper explores the various dimensions of corporate governance within firms by examining extensive literature that highlights its importance and impact. The rest of the study is organized as follows: In the second section, the authors present a review of the relevant literature and brief analysis of the legal framework regarding the transparency of the joint stock companies. Third, the data and the methodology are presented. Fourth, the main results of the study are discussed. In the final section, the conclusion of the paper is given.

2. Literature review

Corporate transparency is known as one of the pillars of a good corporate governance system (Aras & Crowther, 2008). According to Rastogi and Srivastava (2010) corporate governance ensures the proper functioning of the company in terms of disclosing financial information, insurance of the company against any potential fraud, thus safeguarding the interests of both existing and potential investors. Consequently, corporate governance plays a vital role in creating a corporate culture of consciousness, transparency, and openness. Therefore, corporate transparency is recognized as a key element of corporate governance because it provides facts in relation to capital allocation, financial performance and monitoring and corporate transactions which are indeed for an effective decision making (Ratnayake & Rajakulanajagam, 2022).

The most famous definition was given for corporate transparency as availability of firm specific information to those outside publicly traded firms and viewed as the joint output of multi-faceted systems whose components collectively produce, gather, validate and disseminate information to market participants (Bushman et al., 2004). Transparency is a business enterprise's most significant untapped competitive benefit (Gorsht, 2014) since it increases customer pride or value, client enchantment, and competitive position, i.e., transparent agencies are covered by clients, while those that are less transparent are not (Singh, Islam, Ahmed & Amran, 2019). One of the expected benefits of transparency for shareholders is increased share liquidity i.e. higher firm disclosure, higher overall liquidity of the share (Farvaqueet al., 2011).

The significance of transparency is increasing day by day, as stakeholders and investors begin to strongly request quality information about the company activities (Bhimavarapu & Rastogi, 2021) as a result of many corporate failures which had been experienced even under good governance, especially in 2008, when many companies fell down as a result of the global financial crisis. The presentation of financial statements in a more transparent and user-readable format by an organization helps in building investor‘s trust and attracts higher attention. Transparency also leads to increased market capitalization and has a competitive edge through enhancement of the public value of the firm.

The OECD Principles of Corporate Governance play a leading role in shaping and promoting good corporate governance practices globally. These principles highlighted the influence of the principle of transparency and disclosure in section IV, which states that timely and accurate disclosure on all material matters regarding the corporation, including the financial situation, performance, sustainability, ownership, and governance of the company is essential (OECD Principles of Corporate Governance, 2023). OECD Principles of Corporate Governance emphasized that easily accessible, user friendly, timely and cost-efficient access to information is the only acceptable way for informing all stakeholders. One of the meeting criteria solutions for better transparency level is setting up a comprehensive website. According to Yasan (2019), in the UK and German company law, setting up a website and publishing relevant information is considered one of the most important requirements of the corporate governance principle. The leading role of OECD Principles of Corporate Governance is confirmed in the analysis of Bosakova, Kubicek and Strouhal (2019), where they came to a conclusion that the code contents and approaches regarding the disclosure and transparency of the 11 developing and emerging countries is well covered and has a high level of compliance with OECD principles.

On the ground in Europe there are a lot of organisations and networks whose aim is to promote the corporate governance among EU members companies such as: the European Corporate Governance Institute, European Confederation of Directors' Associations etc. They accept by acclamation the principle of accountability and transparency as: "frequently voluntarily disclose more information than required by law as a means of gaining the confidence and commitment of investors and other external stakeholders" (IFC & EcoDa, 2015). The role of the EU as a stimulator of corporate transparency is undoubted, especially in the adoption of a numerous acts such as: The Shareholder Rights Directives, Directive regarding the encouragement of long-term shareholder engagement, Directive regarding the use of digital tools and processes in company law, Transparency Directive, Directive on corporate sustainability due diligence, Directive on the annual financial statements, consolidated financial statements and related reports of certain types of undertakings, etc.

Many studies highlight the significance of corporate transparency and good governance policies of the firm on a firm's financial performance in both developed and developing countries. A part of them confirmed the positive relationship between the corporate transparency and the financial performance of the firms. For example, the study conducted by Mohammed Osman Gani et al. (2021) confirmed that corporate transparency is positively related to performance, enhancing managerial accounting measurements. Therefore, corporate transparency is an essential tool towards maintaining a competitive advantage and enhancing a company's financial performance. Similarly, Amman et. al. (2011) investigate the relation between firm-level corporate governance and firm value based on a large and previously unused dataset from Governance Metrics International (GMI) comprising 6,663 firms' annual observations from 22 developed countries over the period from 2003 to 2007. They found a strong and positive relation between firm-level corporate governance and firm valuation. In addition to these results is the study of Abed Al-Nasser Abdallah and Ahmad K. Ismail (2017) who found that the positive relationship between governance quality and firm performance is maintained and is stronger at low levels of concentrated ownership. Rink (2020) posited that transparency significantly improves company performance and increases financial liquidity.

The approaches of reactive and proactive transparency in the public sphere, where the proactive approach means deliberation of information by initiative of the information holder, without a request being filed (Darbishire, 2010), starts to be recognized among the companies. Today, transparency is taking on a new meaning of more comprehensive and proactive disclosures instead of the release of corporate governance details or policies in a 'reactive' fashion (Fung, 2014).

The requirement to include a corporate governance statement in the annual management report of listed joint stock companies in North Macedonia, as stipulated in Article 20 of the Directive on the annual financial statements, consolidated financial statements and related reports of certain types of undertakings, was transposed into national legislation through the 2020 Amendment to the Law on Trade Companies. The Corporate Governance Code (hereafter abbreviated as CGC) as a soft law mechanism was introduced by the Securities and Exchange Commission and the MSE, with the support of the EBRD in October 2021. This code has demonstrated an unwavering commitment for promoting good corporate governance and strengthening the confidence of stakeholders in the securities market. From a market perspective, the CGC serves as a development tool by advancing business transparency and improving shareholder protection through more flexible and efficient corporate governance systems. Among the list of benefits of the implementation of the CGC are increased openness of companies to the public, a proper balance between management bodies and shareholders, and the overcoming of information asymmetry between management and other stakeholders (Gjorgjioska, et al., 2024)

The level of the corporate governance and transparency in the Republic of North Macedonia have been an object of analysis in only a few studies. Arsov (2012) conducted a survey on 29 Macedonian listed companies based on the companies' websites, but also on the documents they have made available through the Stock exchange's website, which is mandatory for them. The results were rather disappointing. Namely, the overall calculated transparency and disclosure (T&D) score is 35 which is at the lower end of the respective scores calculated as averages for the six continents one decade ago. To follow the improvement of the corporate governance practices by companies that should apply it, MSE issued two annual reports on the compliance of listed companies with the CGC in 2023 and 2022. The main shortcoming of the reports is that, as stated at the outset, the data were collected from listed companies based solely on their responses to the Corporate Governance Questionnaire, thus in analysing these responses, the assessment relied on the companies' self-reported compliance with specific provisions of the Code (MSE, 2024).

Considering the fact of limited studies on these issues in North Macedonia, our study aims to fill the gap in the literature review by conducting an examination of the transparency level of listed companies that are obliged to comply with CGC.

3. Data and Methodology

The prerequisites for listed companies that are obligated to comply with this code are specified in the Listing Rules of the MSE and the list of the companies is published on the website of the MSE every year. The structure of the questionnaire which listed companies should respond to is given in the Manual for the Application of the CGC of the MSE. For the purposes of this paper, data is taken from relevant segments (sub-sections 7.1, 7.2 and 7.3.) of Corporate Governance Questionnaires of analysed 26 companies that are obliged to comply with CGC, their annual reports for operation in 2023 (at the time of writing the paper they were the latest released) and the content of their websites.

A descriptive statistical analysis of these companies was conducted according to the listing subsegment they are listed, industry sector they are operating and their governance system. The fulfilment of the requirements for reactive transparency of the listed companies on the MSE in 2023 was measured according to 20 indicators, following the segments of the Governance Questionnaires. The first indicator relates to sub-section 7.1, indicators 2–11 relate to the publication of information on the company's website (sub-section 7.2), while indicators 12–20 pertain to the disclosure of mandatory information in the annual report (sub-section 7.3). The indicators include:

1. Published Annual Report and audited financial statements, as well as other mandatory information related to the Company's business operations, financial condition and ownership structure on the Company's website.
2. Special section on the Company's website (published information describing the rights attached to each class of shares, Company's Statute and other internal acts).
3. Published internal acts that regulate the manner of proposing agenda items, asking questions and proposing decisions by shareholders.
4. Published Shareholders' Assembly decisions and shareholders Questions and Answers for a period of at least five years.
5. Published details, email address and telephone number of the person appointed to contact shareholders.
6. Published internal acts that determine powers and responsibilities of the Supervisory and Management Boards/Board of Directors.
7. Published Supervisory Board profile on the Company's website.
8. Published Rules of Procedure for each Committee of the Supervisory Board/ Board of Directors.
9. Published Code of Ethics of the Company.
10. Published a procedure for protected reporting by a whistleblower.
11. Internal acts that address its responsibility for the environment and social issues.
12. The number of held meetings of the Supervisory Board/ Board of Directors and attendance by the members of the Board.
13. Activities undertaken to achieve gender representation in the Supervisory and Management Boards/the Board of Directors.
14. The succession plan of the Supervisory Board/ Board of Directors.

15. The composition of the Committees of the Supervisory Board/ Board of Directors, the number of meetings and the attendance of the Committee members.
16. Data on the remuneration of individual members of the Supervisory and Management Boards/ Board of Directors.
17. Data regarding membership in other management bodies of other companies of the members of the Management Board/Board of Directors.
18. The name of the external auditor and details of any other services that the external auditor provides to the company.
19. A summary of the communication with stakeholders undertaken during the year.
20. Information on environmental issues and issues of public interest.

The measurement was carried out individually for each indicator and for each listed company, followed by an assessment of the average transparency level of the analysed companies. Companies received a maximum of 5 points for full compliance with each individual indicator, partial compliance awarded 2.5 points, while no points were given in cases of non-compliance.

The methodology used to measure active transparency of institutions by the Center for civil communications (2022) was used for assessing the level of transparency. The final ranking of the listed companies was based on the degree (percentage) of fulfilment of obligations, which was obtained as the ratio between the number of points awarded and the total number of possible points. Here, 0 is the lowest rank, and 100 is the highest. The transparency level is divided into five groups, depending on the degree of fulfilment of obligations. Listed companies with 80 - 100% fulfilment are ranked in the "very good" group, those with 60 - 80% in the "good" group, companies with 40 - 60% fulfilment are ranked in the "average" group, with 20 - 40% fulfilment - in the "poor" group, and with 0 - 20% - in the "very poor" transparency group.

To draw a general conclusion about the transparency level of companies, the results are presented according to different parameters, while the individual results for each analysed company, as annexes to this research, can be made available to interested business entities. The graphs and tables generated by the application Microsoft Excel enable clear visualization and presentation of the results.

4. **Results and discussion**

The number of companies that are obligated to report on compliance with the Code is 26, or 30% of the total number of listed joint stock companies (Macedonian Stock Exchange, Report on the compliance of the listed companies with the CGC in 2023). Their number by listing segments is presented in the Table 1:

Table 1
Total number of companies required to report compliance with the Code, categorized by listing segment

	Super Listing	Exchange Listing	Mandatory Listing
Total Number of companies that are required to report compliance with the Code	**1**	**15**	**10**

Source: (Macedonian Stock Exchange)

This means that around 38% of the analysed companies are in the mandatory listing segment, while around 58% are in the exchange listing segment. Out of the companies analysed, 11 (42%) have a one-tier governance system, while 15 (58%) have a two-tier governance system.

Specifically, out of the 15 companies that fall under the stock exchange listing, 7 have a one-tier governance system, while 8 have a two-tier governance system. Out of the 10 companies that fall under the mandatory listing, 4 have a one-tier system and 6 have a two-tier system. The company that falls under the super listing has a two-tier governance system. Figure 1 shows the activities performed by these companies.

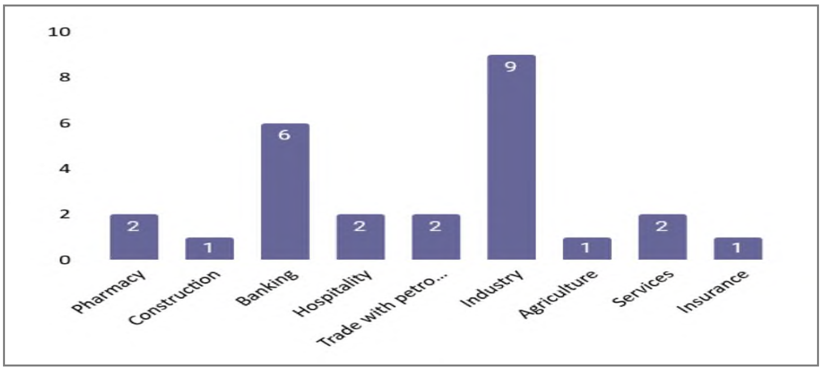

Figure 1. Number of companies by sector
Source: (authors' compilation from data published by MSE)

Figure 1 shows that the largest number of companies analysed (about 35%) belong to the industrial sector, followed by banking companies (23%), and the remaining companies are distributed across various sectors: pharmacy, catering, trade of oil derivatives, services, construction, agriculture and insurance.

The relationship between the number of companies by sector, listing segments and type of governance system is presented in the following Figure:

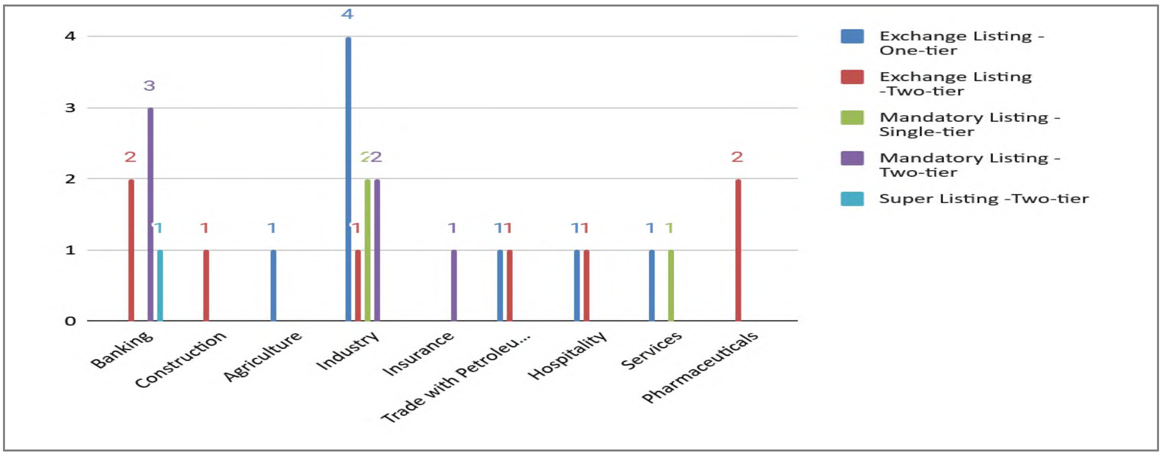

Figure 2. Distribution of companies by industry, listing segment, and governance system

Source: (authors' compilation from data published by MSE and web sites of listed companies on MSE)

The average fulfilment of obligations by individual indicators of both groups together is presented in Figure 3:

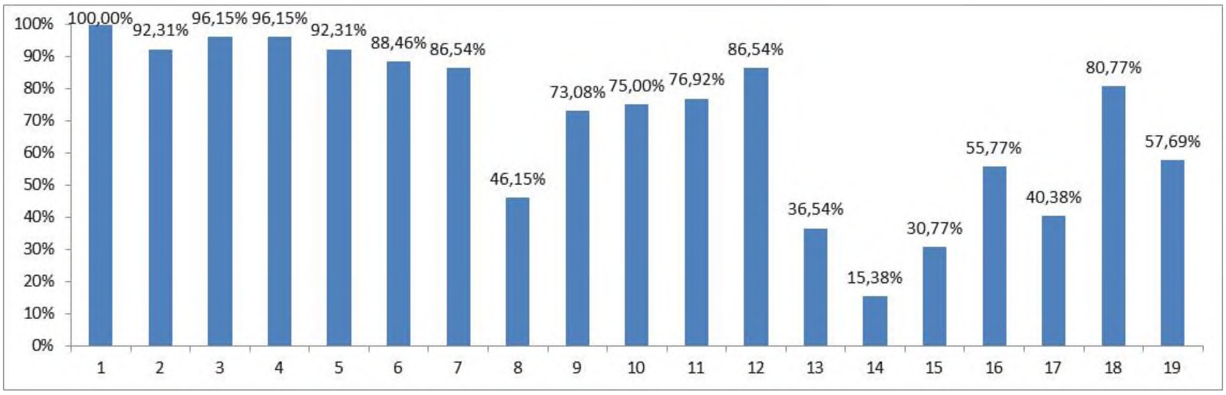

Figure 3. Average fulfilment of the transparency obligations measured by each indicator

Source: (authors' calculation)

From Figure 3, it can be concluded that the lowest percentage of fulfilment of obligations by listed companies is in the following areas: Publishing data on the succession plan of the Supervisory Board/ Board of Directors in their Annual Report (15.38%), Publishing data about the composition of the Committees of the Supervisory Board/ Board of Directors, the number of meetings and the attendance of the Committee members in its Annual Report (30.77%) and publishing data on activities undertaken to achieve gender representation in the Supervisory and Management Boards/Board of Directors (36.54%). While, the highest percentage of fulfilment of obligations by listed companies is in the following areas: publishing Annual Report and audited financial statements, as well as other mandatory information related to the Company's business operations, financial condition and ownership structure on the Company's website (100%); publishing internal acts that regulate the manner of proposing agenda items, asking questions and proposing decisions by shareholders on the Company's website (96.15%); publishing on the Company's website Shareholders' Assembly decisions and shareholders questions and answers for a period of at least five years (96.15%); availability of information describing the rights contained in each type and class of shares, as well as publication of the statute and other internal acts regulating the rights of shareholders on the Company's website (92.31%), and publishing details, email address and telephone number of the person appointed to contact shareholders on the Company's website (92.31%). The high transparency level for certain indicators is largely due to the requirements of the Law on Trade Companies, whose non-compliance is subject to misdemeanour sanctions.

The average fulfilment of all obligations is 71%. However, as illustrated in Figure 3, the average level of obligation fulfilment in sub-section 7.1 is 100%, indicating full compliance. In sub-section 7.2 (indicators 2–11), the average fulfilment rate is 82%, reflecting a generally high level of information disclosure on company websites. In contrast, the lowest average is observed in sub-section 7.3 (indicators 12–20), with a fulfilment rate of only 54.49%, suggesting significant room for improvement in the disclosure of mandatory information within annual reports. In contrast, the Report on the compliance of listed companies with the CGC in 2023 issued by MSE shows that the average compliance of companies with obligation for transparency and disclosure is 86%, the average fulfilment of the indicators related to publication of information on the Company's website (sub-section 7.2) is 62% and average fulfilment of the indicators related to the incorporation of information in the Annual Report (sub-section 7.3) is 81%. The discrepancies in the results arise from differences in the data sources used. Unlike the MSE report, which relies solely on information provided by companies through the "comply or explain" principle in their questionnaires, this paper cross-verified those responses with the content published in each company's annual reports or on their official websites. The results of the indicators 2-11 (sub-section 7.2), show that only one indicator percentage is lower than 50%, while in areas 12-20 (sub-section 7.3), four indicators have a percentage lower than 50%.

The representation of listed companies in the sub-levels of transparency is presented in the following figures:

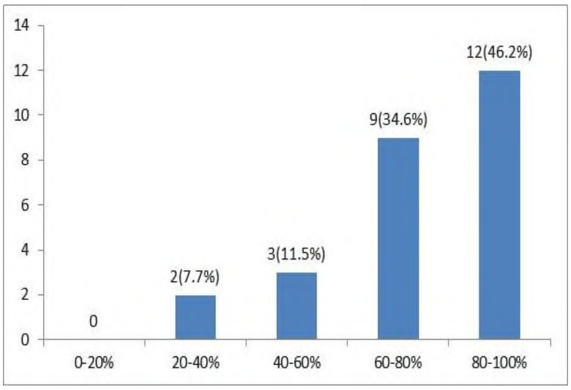

Figure 4. Representation of listed companies in the sub-levels of transparency
Source: (authors' calculation)

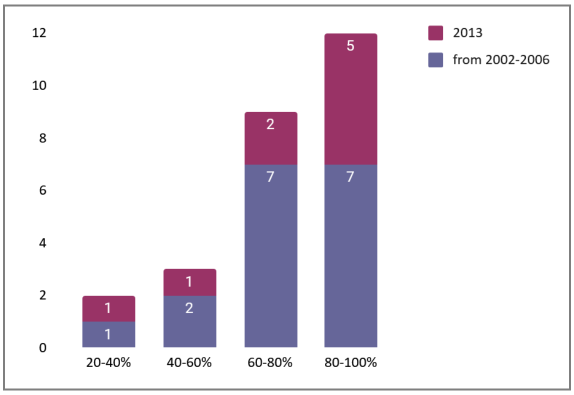

Figure 5. Representation of listed Companies by years of listing on the Macedonian Stock Exchange in each transparency levels
Source: (authors' calculation)

The average transparency of listed companies with respect to these indicators is 73%, which may be interpreted as a percentage of the reactive transparency of listed companies in North Macedonia. 12 out of 26 analysed companies belong in the "very good" transparency level category (46.2%), 9 (34.6%) of the companies belong in the "good" level of transparency, i.e. 80,7% of the companies have reached the best two transparency levels, and neither of the companies have a "very poor" transparency level.

Based on the previously conducted descriptive statistics of listed companies, several analyses were performed to determine which types of listed companies (classified by governance system, listing sub-segments, sector, etc.) belong into each transparency level.

Figure 6 presents the representation of listed companies by sector in each transparency level:

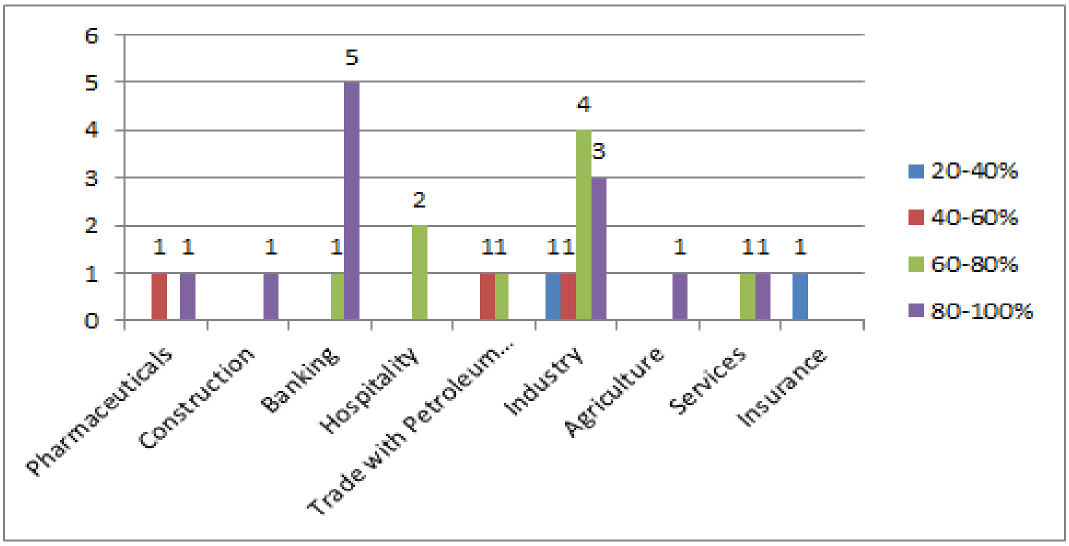

Figure 6. Representation of listed companies by sector in each transparency level
Source: (authors' calculation)

From Figure 6 can be concluded that out of the six banking institutions, five belong to the highest transparency level, one company from the construction industry belongs to the highest transparency level and one company from agriculture belongs to the highest transparency level. The average transparency level of analysed banks is higher than other sectors since banks have an additional obligation in accordance with Central's banking requirements. A positive side must be stressed out that none of the analysed companies belong in the lowest transparency level, while only one (insurance) company falls into the 20-40% level. In the service sector, one company belongs to the "good" transparency level, one to the "very good" level. In the industrial sector, one company is in the "weak" level, one in the "average" level, four in the "good" level and three in the "very good" transparency level. In the oil derivatives trade sector, one company is at the "average" level, while the other is at the "good" transparency level.

The following graphs present the representation of listed companies by listing sub-segments and the transparency levels:

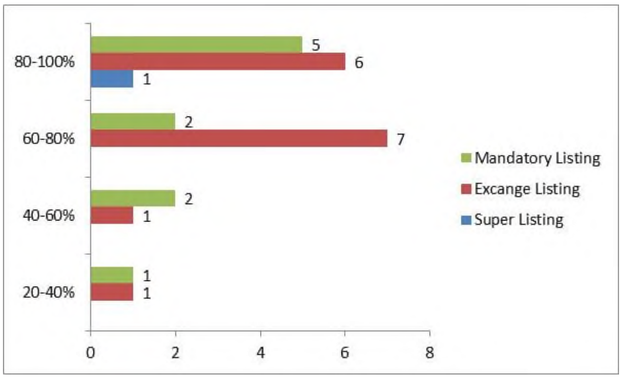

Figure 7. Representation of listed companies by listing sub segments and the transparency levels
Source: (authors' calculation)

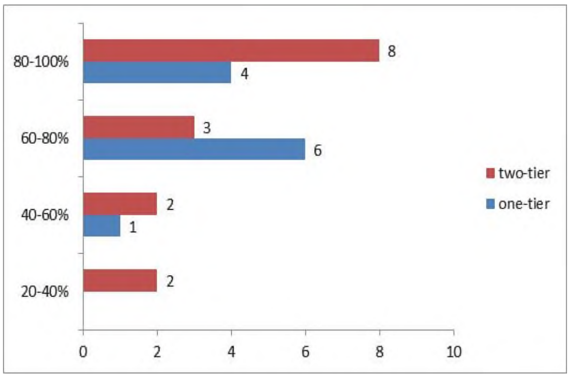

Figure 8. Representation of listed companies by governance system at different transparency levels
Source: (authors' calculation)

Figure 7 shows that the listed companies that fall into the highest transparency level are almost evenly distributed, i.e. 5 are in mandatory (50% of the total that are in mandatory), 6 in stock exchange (40% of the total that are in stock exchange), and 1 (out of 1) in super listing sub-segments. The sub-listing segment has not had a significant impact on the transparency level of companies. Due to the low number of companies representing super listing, any conclusions drawn would lack reliability.

From Figure 8 can be seen that 8 listed companies that fall into the highest transparency level have a two-tier (53% of the total with a two-tier), and 4 have a one-tier (36% of the total with a one-tier) board.

Figure 9 presents the representation of listed companies combined by governance system and sub-listing segments in each transparency level, and Figure 10 presents representation of companies with most liquid, most traded and highest share growth by transparency level:

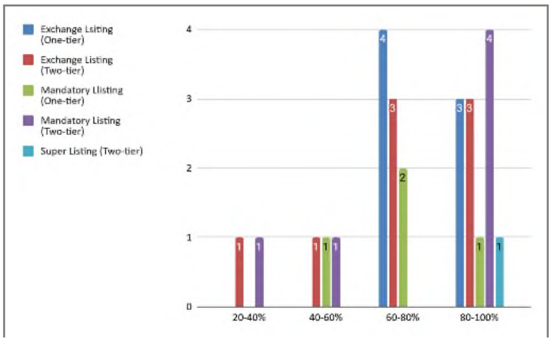

Figure 9. Representation of listed companies combined by governance system and sub-listing segments categorized by transparency level
Source: (authors' calculation)

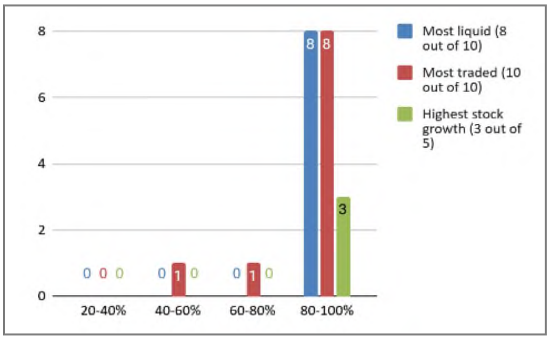

Figure 10. Representation of companies with the most liquid, most traded and highest share growth categorized by transparency levels
Source: (authors' calculation)

Figure 9 shows that the highest level of transparency is most commonly associated with the following categories: stock exchange-listed companies with a one-tier management system (43% of such companies), stock exchange-listed companies with a two-tier management system (38%), mandatory-listed companies with a one-tier system (25%), mandatory-listed companies with a two-tier system (67%), and super-listed companies with a two-tier system (represented by one company).

Based on cross-tabulation of the data from the MSE website-specifically the lists of the most liquid, most traded companies and those with the highest share price growth-with the assessed transparency levels of each analysed company, Figure 10 reveals the following: all 8 analysed companies that appear on the list of the 10 most liquid and 80% of most traded companies demonstrate the highest level of transparency. All 3 analysed companies among the 5 with the highest stock price growth also exhibit the highest transparency level (the other 2 most liquid companies and 2 with highest stock price growth do not have an obligation to comply with the CGC). The results confirmed that these companies fully meet the transparency requirements set by the CGC. The strong transparency practices observed among the most liquid, most traded and listed companies with the highest stock price growth should serve as motivation for other companies aiming to be included in top-tier rankings.

Conclusions

The benefits of corporate transparency for both the companies and investors are well known, thus the latest reforms in corporate governance emphasized the importance of transparency.

The results of the research based on comprehensive analysis of the transparency level of all joint-stock companies that are obligated to report on compliance with the CGC by cross-verifying the answers on the Questionnaires and the content published in each company’s annual reports or on their official website, have shown that the average fulfilment of all obligations is 71%. The obligations of the first group indicators related to publication of information on the Company's website (sub-section 7.2) shows an average fulfilment in 2023 of 82% and indicators related to the incorporation of information in the Annual Report (sub-section) shows fulfilment of an average 54,49%. The best scores of transparency by all analysed companies are shown in points 1, 3 and 4 of the Questionnaire

(100%, 96.15% and 96.15% respectively) that are in the domain of mandatory transparency-stipulated obligations with Law on trade companies. The lowest percentage of obligations fulfilled by listed companies are shown in the points 14, 15 and 13 of the Questionnaire (15.38%, 30.77% and 36.54% respectively).

The average transparency level of all analysed companies for all indicators is 73%, which means that the Macedonian companies can be categorized in companies with “good level” of transparency and almost half of analysed companies belong in "very good" transparency level.

The crossed analysis showed that the best score of transparency by industry or sector was reached by the banking institutions. A conclusion may be drawn that the governance system of the companies, the sub-listing segment and the time lapsed since the companies were listed on the MSE have no significant influence on the transparency level.

A limitation of the research that must be emphasized is that the CGC is currently mandatory only for listed companies that meet certain criteria and according to the North Macedonia 2024 Report, the Code should apply to all listed companies (European Commission, 2024). The mandatory implementation of the Code across all listed companies will enable more comprehensive assessments of transparency in future research.

Strengthening the quality of disclosed information would enhance access to qualitative data and support a more proactive approach to corporate transparency. Accordingly, the following recommendations are proposed:

- A standardized template for annual reports should be introduced to ensure consistency and comprehensive coverage of all required elements from the Corporate Governance Questionnaire;
- Listed companies should place greater emphasis on providing comprehensive explanations related to data disclosure in their annual reports;
- The use of the "Not Applicable" option in the Questionnaire should be clearly defined to prevent misuse, particularly in cases where companies may attempt to avoid certain obligations;
- Minimum web design standards should be implemented for corporate websites, specifically regarding the presentation and accessibility of transparency-related information.

The managing bodies have the final authority on what information is disclosed and how it is made publicly available. Transparency should not be viewed merely as a compliance tool - it must be recognized as a strategic asset.

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CONTENTS

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- **JEL** classification code for papers in Economics (http://ideas.repec.org/j/index.html);
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- The main body of the paper – it should contain discussion questions, an outline of the study and research findings/main conclusions; bibliographical citation and additional notes, explanations and comments written in the footnotes.
- Conclusion – it should provide a summary of the main research points supported by sufficient arguments.
- References – authors should list first references written in Cyrillic alphabet, then references written in Latin alphabet.
- Graphs and figures – Word 2003 or Power Point; the tables, graphs and figures must be embedded in the text (to facilitate language correction and English translation); Font for numbers and inside text – Times New Roman, 12 pt;
- Formulae must be created with Equation Editor;

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SUPPLY CHAINS OF PERISHABLE AGRI-FOOD PRODUCTS IN THE EAEU: CONCEPTUAL APPROACHES AND DEVELOPMENT OF LOGISTICS INFRASTRUCTURE

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Diana Madiyarova [3]

Abstract: The purpose of this study is to develop a conceptual approach to the formation of an integrated logistics platform for perishable agri-food products within the Eurasian Economic Union (EAEU). The methodological framework is based on systems and scenario analysis, SWOT assessment, comparative analysis of the logistics potential of EAEU member states, and modeling of logistics chain efficiency under different levels of integration using panel empirical data for the period 2015–2023. Selected indicators of logistics infrastructure are employed to provide a descriptive assessment of the current state of agrologistics as of 2023.

The modeling results demonstrate that coordinated investment policies and regulatory harmonization can significantly enhance the efficiency of agrologistics chains in the context of regional integration. The findings confirm that digitalization and the development of cold chain infrastructure are key determinants of logistics sustainability, while the coordination of regulatory and investment mechanisms substantially improves logistics efficiency across the EAEU member states. Based on the estimated parameters, a scenario-based forecast of agrologistics development up to 2030 is proposed. The study introduces a scalable model of an integrated agrologistics platform incorporating a cluster-based approach, a unified digital framework, and public–private partnership mechanisms adaptable to the specific conditions of EAEU countries. The scientific

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novelty of the research lies in substantiating a platform-based approach to cross-border agrologistics grounded in the synergy of digital and physical solutions, institutional coordination, and targeted infrastructure policy.

Keywords: agri-food products, cold chain logistics, logistics, infrastructure, digitalization, supply chain

JEL: O18, Q13, R42, F15, L91

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1.Introduction

The growing global emphasis on food security reinforces the importance of efficient supply chains for perishable agri-food products. Their reliable delivery is critical not only for product quality and economic sustainability, but also serves as the core of regional integration strategies (FAO, 2021). Within the EAEU, the problem is exacerbated by spatial, climatic and regulatory differences between the member countries.

Existing literature predominantly focuses on technical aspects (temperature control, transport efficiency) or economics at the national level (Kumar & Budin, 2006; Aung & Chang, 2014), ignoring cross-border institutional and infrastructural barriers. In contrast, the EU and ASEAN demonstrate efficiency through harmonized legal and digital systems (Rodrigue et al., 2016; Kampan, 2017). Current research highlights the need for holistic logistics models that take into account digitalization, resilience and institutional alignment, especially in the face of climate and geopolitical challenges (Ivanov & Dolgui, 2020). However, such models are rarely adapted to decentralized systems as in the EAEU.

This study proposes a conceptual model of an integrated EAEU agro-logistics platform that synthesizes digital, institutional and infrastructural elements into a single framework. Based on the theoretical framework of supply chain management (Christopher, 2016) and inter-organizational coordination (Zajac & Olsen, 1993), it is argued that digital interoperability, institutional alignment and strategic investments are critical for the sustainability of perishable goods logistics in the region.

A literature review on sustainable agro-logistics, digitalization and institutional coordination (Aung & Chang, 2014; UNEP & FAO, 2022; Hassini et al., 2012; Raimbekov et al., 2017; World Bank, 2023) highlights the need for a unified platform to improve the transparency and reliability of supply chains in the EAEU (Raimbekov et al., 2017).

The proposed model includes institutional mechanisms, digital traceability and infrastructure clusters, providing an adaptive, sustainable and integrated logistics system.

The key research question is: how can EAEU build a coherent and sustainable logistics network for perishable products that ensures economic integration and food security of the region? To address this objective, the study sets the following tasks: (1) analyze current logistics systems in the EAEU member states and identify bottlenecks and development opportunities in the cold chain infrastructure; (2) assess sustainability factors and scenarios of agro-logistics development until 2030, (3) identify investment priorities (4) develop a model of the EAEU integrated logistics platform.

2. Methodology

The methodological framework of this study integrates both qualitative and quantitative approaches to assess the resilience and efficiency of perishable agri-food logistics across EAEU member states. The research employs a combination of systematic literature review, statistical data collection, semi-structured expert interviews, SWOT analysis, and econometric modeling.

The empirical component includes interviews with 15 experts from all EAEU countries—comprising representatives from logistics businesses, government agencies, and industry specialists—which enabled the identification of institutional barriers and region-specific characteristics of agro-logistics development. SWOT analysis was applied to evaluate country-level strengths, weaknesses, opportunities and threats.

Selected tables reflect the state of agrologistics infrastructure as of 2023 and are used for a descriptive assessment of the current level of development, whereas the econometric analysis is based on data for the period 2015–2023.

The econometric model is based on multiple linear regression, where the dependent variable is an integrated logistics efficiency index (Y) capturing the reliability, speed and adaptability of supply chains. Independent variables include the level of logistics digitalization (X_1), cold storage capacity per capita (X_2), the logistics infrastructure index (X_3) (World Bank, 2023), the volume of logistics-related investments (X_4) (Eurasian Economic

Commission [EEC], 2021), and the share of intra-union trade (X_5; Eurasian Economic Commission [EEC], 2023).

Scenario analysis explores three development trajectories through 2030: a baseline (business-as-usual), an integration-driven (enhanced coordination), and a technology-driven (prioritizing digitalization and sustainability) scenario. It accounts for political, economic and climatic factors, as well as international recommendations on sustainable cold chains (UNEP & FAO, 2022; Burnasov et al., 2020).

The conceptual model was developed using a systems approach and aims to establish an integrated logistics platform for the EAEU by unifying digital, institutional, and infrastructural components (Hassini, 2012; Christopher, 2016). The methodology specifically considers the decentralized institutional structure of the EAEU, distinguishing it from most existing studies focused on more standardized integration blocs such as the EU or ASEAN (Aung & Chang, 2014; Burnasov et al., 2020).

3. Results of the study

3.1 Logistics infrastructure of EAEU countries: current state and limitations

Analyses of EAEU logistics indicators (in terms of LPI, cold infrastructure and total logistics base) revealed significant disproportions in institutional, digital and infrastructure development (Tables 1-3), which causes heterogeneous conditions for sustainable supply chains of perishable products.

Table 1.
Logistics performance of EAEU countries by LPI sub-indices, 2023

Country	Rank	Score	C	I	IS	CL	T&T	TML
Kazakhstan	79	2.7	2.6	2.5	2.6	2.7	2.9	2.8
Russia	88	2.6	2.4	2.7	2.3	2.6	2.9	2.5
Belarus	79	2.7	2.6	2.7	2.6	2.6	3.1	2.6
Armenia	97	2.5	2.5	2.6	2.2	2.6	2.7	2.3
Kyrgyzstan	123	2.3	2.2	2.4	2.4	2.2	2.4	2.3
Average	–	–	2.46	2.58	2.58	2.42	2.54	2.80

Note: compiled based on (World Bank, 2023), where T – Customs, I – Infrastructure, ***IS*** *– International Shipments, CL – Logistics Competence, T&T – Tracking and Tracing, TML – Timeliness.*

Table 1 reveals that Kazakhstan and Belarus lead the EAEU in terms of logistics infrastructure development, demonstrating the highest scores across key indicators. Armenia stands out for its relatively high technological adoption despite weaker overall logistics performance. Russia shows average results, while Kyrgyzstan significantly lags behind in all components. These disparities suggest the need to strengthen service quality, foster technological integration, and enhance supranational coordination to improve the efficiency of perishable goods supply chains across the Union.

Table 2 presents an assessment of cold chain infrastructure deficits and identifies key barriers to its development in each member state.

Table 2
Cold Chain Coverage in EAEU Countries, 2023 (% of Demand)

Country	Cold storage capacity (thousand tons)	Estimated deficit (%)	Key barriers
Russia	7,500	15%	Ageing infrastructure
Kazakhstan	1,200	35%	Lack of investment
Belarus	1,800	20%	Tariff policies
Armenia	450	50%	Geographic constraints, high energy consumption
Kyrgyzstan	300	60%	Absence of rural storage facilities

Source: Data on cold storage capacity and logistics barriers for 2023 are based on an analysis of national and regional logistics strategies of EAEU member states, as well as reports by international organizations (FAO, World Bank).

The assessment of cold chain logistics coverage in the EAEU countries reveals pronounced disproportions (Table 2). Russia has the largest capacity (7.5 million tonnes), where the deficit is 15%, but development is constrained by deteriorating infrastructure. Kazakhstan demonstrates a significant gap (35%) against the background of limited investment inflow. In Belarus, despite a moderate deficit (20%), tariff policy remains the main constraint. Armenia and Kyrgyzstan have the most acute deficits (50% and 60% respectively), due to geographical fragmentation, energy intensity and lack of storage facilities in rural areas. These factors highlight the need for targeted logistics modernization, taking into account national conditions and barriers.

A comparative analysis of logistics digitalization in the EAEU countries shows that Russia (65%) and Belarus (58%) lead the way in implementing digital solutions such as automation, digital routing and supply chain management. Kazakhstan (45%) shows moderate progress with high growth potential, while Armenia (30%) and Kyrgyzstan (25%) lag significantly behind, especially in terms of IoT, blockchain and digital platform adoption.

Accelerating digital transformation in the EAEU requires differentiated support for less developed countries - through investments, infrastructure measures and adaptation of external IT solutions to regional specifics. For spatial analysis of the logistical accessibility of agro-food flows in the EAEU, a cartogram of agro-logistic hubs was developed, classified into central (Moscow, Minsk, Almaty), local (Yerevan, Bishkek) and potential (Karaganda, Osh, Gyumri) based on infrastructure development, institutional connectivity and role in interstate supply chains. The analysis has shown the need for targeted upgrading of local centers and support for prospective nodes, which will diversify routes, enhance regional connectivity and improve the sustainability of supply chains.

Comparative analysis of the logistics infrastructure of the EAEU countries (Table 3) revealed institutional advantages and infrastructural constraints hindering the development of agro-logistics.

Table 3
Results of the Analysis of Logistics Infrastructure Indicators in EAEU Countries, 2023

Indicator	Russia	Kazakhstan	Belarus	Armenia	Kyrgyzstan
Cold Storage Capacity (thousand m^3)	12,000	3,500	2,000	450	300
Logistics Performance Index (LPI)	2.6	2.7	2.7	2.5	2.3
Road Network Quality (scale 1–7)	3.1	2.8	3.6	2.4	2.2
Digital Tracking Infrastructure (out of 10)	7.5	5.0	6.5	4.0	3.5

Source: Compiled by the author based on data from the World Bank (Logistics Performance Index, 2023), FAO, the Eurasian Economic Commission, and national statistical agencies

Russia and Belarus are leading the way with developed cold and digital infrastructure, while Kazakhstan needs modernization and Armenia and Kyrgyzstan need to overcome systemic barriers. Improving efficiency requires coordinated integration policies to address imbalances and strengthen supply chain resilience.

The SWOT analysis confirms the heterogeneity of logistics development in the EAEU: the strengths of individual countries (geopolitical position, agricultural potential, market access) are combined with common challenges such as lack of cold storage capacity, digital solutions and regulatory coherence. However, there are prospects for sustainable growth through digital transformation and the development of regional logistics clusters.

These results form the basis for targeted national policies and harmonized integration mechanisms to address logistics imbalances and increase the resilience of agri-food chains in the EAEU.

Effective development of agro-logistics in the EAEU requires a coherent integration strategy focused on the expansion of cold chain infrastructure, digitalization and institutional harmonization. The key conditions are investment coordination, development of logistics corridors taking into account regional demand and activation of PPP mechanisms, which ensures the formation of a sustainable and adaptive logistics system of the Union.

3.2 Sustainability factors and scenarios for agro-logistics development

The empirical assessment of agrologistics sustainability factors in EAEU countries is based on panel data, which makes it possible to capture the dynamics of key indicators and smooth the effects of short-term shocks. The use of a multi-year time horizon ensures the robustness of correlation and regression analyses and enhances the reliability of scenario-based estimates. Correlation analysis based on panel data for the period 2015–2023 revealed a strong positive relationship between the sustainability of logistics chains and factors such as digitalization ($r = 0.85$), the level of infrastructure ($r = 0.79$), and the logistics performance index (LPI, $r = 0.88$). The resulting regression equation has the following form:

$$Y = 12{,}3 + 0{,}34 \cdot X_1 + 0{,}29 \cdot X_2 + 0{,}41 \cdot X_3 + 0{,}18 \cdot X_4 + 0{,}14 \cdot X_5 + \varepsilon$$

The multiple linear regression model ($R^2 = 0.89$, $p < 0.05$) confirms the statistical significance of all included variables. Logistics infrastructure exerts the strongest influence on the dependent variable, followed by digitalization, underscoring the critical role of transport accessibility and digital technologies in supply chain resilience. The impact of investment and integration variables also proves significant, highlighting the necessity of a coherent logistics policy at the EAEU level.

To assess the efficiency of agrologistics for perishable agri-food products in EAEU countries over the medium-term time horizon, a multiple regression model was developed incorporating the key determinants (X_1–X_5). Based on these parameters, three development scenarios were developed: inertial, integration-driven, and technological—each differing in digital maturity, institutional coordination, and investment dynamics.

The estimated coefficients of the regression model, based on data for the period 2010–2023, were used to construct a scenario-based forecast of agrologistics development in EAEU countries up to 2030.

Simulation results indicate that the highest values of the composite logistics index are achieved under the scenario that combines digital transformation with infrastructure modernization, confirming the strategic value of a coordinated approach to agro-logistics development in the EAEU (Table 4).

Table 4
Scenario-Based Forecast of Logistics Efficiency in the EAEU up to 2030 (Integrated Index Y)

Scenario	Digi-talisation (X_1)	Cold Chain Capacity (X_2)	LPI Growth (X_3)	Investment (% of GDP) (X_4)	Intra-EAEU Trade Share (X_5)	Projected Integrated Index (Y)
Baseline (Status Quo)	+10%	+15%	+0.2	1.2%	22%	0.598
Integration (Moderate)	+25%	+35%	+0.5	1.8%	30%	0.726
Technological Breakthrough	+40%	+50%	+0.9	2.5%	38%	0.831

Note. Figures for 2024–2030 are authors' forecasts/estimates. Source: Authors' calculations based on data from the Eurasian Economic Commission (2023a, 2023b)

The 'technology leap' scenario shows the best sustainability and efficiency indicators, emphasizing the role of digitalization, cold chain development and active investment policies.

According to calculations, the Status Quo scenario has a logistic efficiency index of 0.598, reflecting fragmented systems and limited interstate interaction.

Moving to the Moderate Integration scenario, which includes standardization, targeted investment and partial digital transformation, leads to an increase in the index to 0.726. The highest efficiency (0.831) is achieved in the deep integration scenario, which provides for the coordination of infrastructure solutions, the formation of a unified digital environment and institutional convergence.

Thus, the modelling results confirm the key dependence of the efficiency of EAEU agro-logistic systems on the degree of digital maturity, infrastructural coherence and institutional coordination. Deep integration forms the basis for sustainable logistics chains, reduces losses and contributes to the creation of a unified agri-food environment in the region.

3.3 Investment priorities and integration effects

The analysis of investment strategies of the EAEU countries and international recommendations (United Nations Environment Programme & Food and Agriculture Organization of the United Nations [UNEP & FAO], 2022) allowed to identify priority areas for sustainable development of agro-logistics and improving the efficiency of supply chains of perishable goods. The main areas are the expansion of the cold chain, digitalization of logistics processes, modernization of cross-border infrastructure, development of agro-logistics clusters and strengthening of public-private partnership mechanisms. Modernization of cold chain infrastructure will reduce losses and extend the shelf life of products, while digitalization through IoT, blockchain and sensor systems will increase traceability and reduce transaction costs. Upgrading customs terminals and streamlining procedures will accelerate cross-border trade, while agro-logistics clusters will promote SME development and regional economic activity. Involvement of private capital through PPPs will ensure sustainable financing and improved operational efficiency.

In the short term (2024–2026), the development of the cold chain delivers the greatest impact; in the medium term (2026–2030), sustainable outcomes are achieved through the implementation of digital technologies, while the long-term multiplicative effect is generated by the development of agrologistics clusters and public–private partnership mechanisms. The main barriers, such as infrastructural and digital asymmetry, lack of unified standards and limited interstate coordination, require a transition to a single integrated logistics platform based on a network of agro-logistics clusters, a single digital infrastructure (IoT, blockchain, AI) and harmonization of regulatory requirements (packaging, labelling, temperature control). The most important implementation mechanism remains public-private partnership, which will create a coordinated and technologically advanced logistics system in the EAEU.

3.4 Model of an integrated logistics platform

The conceptual model of an integrated agro-logistics platform of the EAEU is a framework for the formation of a unified digital-physical infrastructure that ensures efficient processing and cross-border movement of perishable agro-food products. The model includes institutional and technological levels, integrating national logistics nodes into a common

Eurasian network. Its architecture is based on digitalization (IoT, blockchain), synchronization of cold chains and harmonization of regulations between participating countries, which allows for the formation of an adaptive and compatible ecosystem (Figure 1). Integration of digital and physical infrastructure, institutional interoperability of standards and consideration of regional specificities contribute to reducing losses and costs, increasing supply sustainability and investment attractiveness.

The model creates a basis for the development of high value-added logistics services, interstate co-operation and increased regional trade.

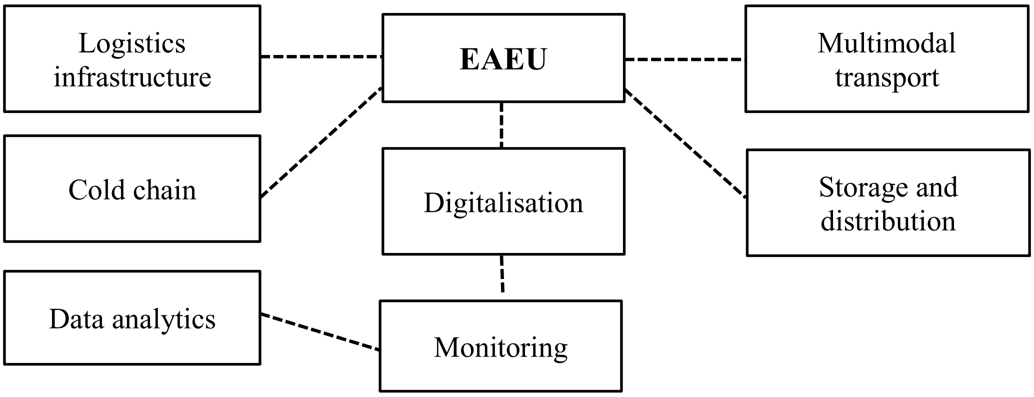

Figure 1. Model of the EAEU integrated logistics platform
Source: Compiled by the authors

The integrated development of agro-logistics in the EAEU requires taking into account the differences in infrastructure and the level of digital maturity of the countries (Raimbekov et al., 2023). For states with a predominance of small farms (Armenia, Kyrgyzstan), agro-logistics clusters are a priority, while digitalization and regulatory harmonization are the main factors of transformation. The development of a single digital platform, the EAEU AgriLogistics Hub, ensures transparency, adaptability and sustainability of supply chains.

The model includes intergovernmental coordination, environmental solutions and involves a governance framework, investment support, competence development and the use of IoT, blockchain and AI technologies.

Simulations show that synchronized investment and digital harmonization significantly improve logistics performance. Underestimated advantages of individual countries, such as Kazakhstan, were identified, and the impact of infrastructure modernization was higher than expected.

The novelty of the model lies in the end-to-end cross-border integration of digital and physical logistics, which was previously limitedly addressed in the academic literature (Palazzo & Vollero, 2022; Bandaranayake et al., 2024). The presented results confirm the feasibility of platform solutions and the need for a supranational agro-logistics strategy in the context of regional integration.

Conclusion

The study confirmed the key role of logistics in ensuring the sustainability of agri-food chains in the EAEU. Digitalization and the development of cold infrastructure act as the main drivers of efficiency improvement, demonstrating a high correlation with logistics maturity. Scenario analysis has shown that the maximum effect is achieved when implementing a strategy of technological transformation based on digital integration, regulatory harmonization and coordinated investments. At the same time, institutional barriers and cross-country disparities remain significant challenges.

The developed conceptual model of the EAEU agro-logistics platform offers an adaptive solution to these problems through the development of clusters, digital services and PPP mechanisms. Its implementation requires coordinated policy, priority investments and large-scale digitalization, which will ensure the sustainability and competitiveness of regional agro-logistics.

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D. A. Tsenov Academy of Economics, Svishtov

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CONTENTS

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MEASURING THE LEVEL OF ORGANISATIONAL AMBIDEXTERITY OF MUNICIPAL ADMINISTRATIONS

Borislav Borisov[1],
Yuliyan Gospodinov[2]

*'Science begins where measurement begins.
Exact science is unthinkable without measurements.'
D. I. Mendeleev*

Abstract: The research on organisational ambidexterity, understood as an ability to effectively solve current problems while ensuring long-term strategic development, is relatively recent. Very limited number of studies focus on ambidexterity in public organisations, or if any, it is considered primarily as an outcome rather than a strategy for achieving long-term success. This study aims to assess the organisational ambidexterity of a group of municipalities in two ways: by answering questions posed to municipal representatives and by calculating financial indicators that provide evidence of local governments' preferences for funding current needs and investments for the future. The results were tested for statistically significant difference using the t-test. It is concluded that it is evident from their responses that municipal officials understand the need to achieve organisational ambidexterity, but the data from the analysis of the financial indicators of the studied municipalities clearly indicate a preference for spending on current needs over spending on investment and development.

Keywords: organisational ambidexterity, municipal administration, strategic development

JEL: H70, L38

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1. Introduction

Like many other concepts, the term *'ambidexterity'* was brought into management science from other sciences, in this case - from medicine. Ambidexterity is the ability of people to serve themselves equally with both hands. For most people, one hand, especially the right hand, dominates most activities. However, there is a small group of people whose both hands can do equally well as 'primary' hands. This condition is called ambidexterity and has some interesting manifestations.

The term *'Organisational ambidexterity'* has entered into the management science from other fields such as medicine and anthropology, like the terms 'diagnosis', 'prevention', 'stress', 'motivation', 'ethics', 'leadership and empathy', 'organisational behaviour', etc. According to Irena Mladenova, 'Organisational ambidexterity denotes the dynamic ability of organisations to combine exploration (experimentation, new knowledge creation, innovation) and application (improvement, use of existing knowledge) activities in order to be successful in a dynamic environment' (Mladenova, 2022). She links the organisational ambidexterity to the ability of organisations to simultaneously engage in routine activities and development-oriented change (innovation).

O'Reilly and Tushman also claim that the term 'ambidexterity' is borrowed from psychology and medicine, where it denotes the disposition of a subject to serve equally well and, with the same skill, with both the left and the right hands, and they emphasize that organisational ambidexterity seeks to explain how organisations deal with two, largely conflicting goals - *exploration* and *exploitation*. The same authors define it as 'the ability to simultaneously pursue incremental and revolutionary innovation and change that results from maintaining multiple conflicting structures, processes, and cultures within the firm' (O'Reilly, 2013).

Early research on the topic supports the notion that exploration and exploitation are distinct, opposing activities in nature. This suggests they should be separated in time (alternating periods of exploration and exploitation) or in space (different departments engage in each of these two activities). More recent developments attempt to reconcile this contradiction.

Similar is the understanding of the organisational ambidexterity of Kortmann, Gelhard, Zimmerman, and Piller, who consider it as a link between strategic flexibility and operational effectiveness of organisations (Kortmann, 2014).

In other words, organisational ambidexterity requires organisations to be strategically agile to adapt to unforeseen situations and rapidly changing environments while optimizing their business processes to achieve operational efficiency. It suggests that **balancing strategic agility and operational efficiency maintains sustainability by combining long-term and short-term goals**.

A number of scholars have contributed to the development of the theory. Among them, the following should be noted:

- J. G. March who introduced the fundamental exploration–exploitation dilemma, on which the concept of ambidexterity later rests. (March, 1991)
- Raisch, Birkinshaw, Probst, and Tushman who conducted a systematic review and created an integrative model that unites the structural, contextual, and leadership perspectives (Raisch, 2009)
- Gibson and Birkinshaw who published the seminal article on contextual ambidexterity, highly relevant to the public sector. (Gibson, 2004)
- O'Reilly and Tushman, whose monograph study deepens the role of the top management ('tone at the top') in maintaining ambidexterity in the long term. (O'Reilly, 2016)
- Turner, Swart, and Maylor who provide a useful overview of management mechanisms (processes, structures, incentives) that can support the discussion and conclusions (Turner, 2013).

Alexandra Sidorova draws attention to the change in approach to understanding organisational ambidexterity: from the recognition of the irreducibility of the conflict between exploration and exploitation and, as a consequence, the need to separate these types of activities in time and space, to the contemporary strategic adoption of incremental and fundamental innovations in terms of their complementarity and reinforcement (Sidorova, 2023). She considers **four types of organisational ambidexterity**:

1. *Sequential ambidexterity* involving alternating periods of study and periods of operation over time.
2. *Structural ambidexterity*, which is the structural separation of exploration and exploitation through the creation of autonomous units.
3. *Contextual ambidexterity* that involves resolving the tension between exploration and exploitation at the individual level, playing an important role in creating optimal organisational environment with a social context and an efficiency management context.

4. *Inter-organisational ambidexterity* involving combining the efforts of different organisations to carry out research and exploitation activities (mergers and acquisitions, creation of alliances).

Sequential ambidexterity is more effective in a stable external environment and suitable for organisations with limited budgets that do not have the resources to do both exploration and exploitation. It is not suitable for public sector organisations that are constantly engaged in delivering public services, regulations and addressing other ongoing issues, which does not allow them to step back from these at the expense of devoting more attention to innovation processes.

The basic idea of **Structural ambidexterity** is the structural separation of research and exploitation through the creation of autonomous divisions, with each type of activity developing its own set of competencies and incentive system (O'Reilly, 2013). This approach helps to reduce tensions between them and increase productivity due to increased concentration and specialization of employees in these departments. A prerequisite for the success of structural ambidexterity is the active and consistent actions of the organisation's management aimed at uniting the efforts of the research and operations departments to achieve a common strategic goal based on shared corporate values. An important task is also the creation and maintenance of mechanisms for knowledge sharing and sharing organisational resources. The success of structural ambidexterity therefore depends to a large extent on effective leadership designed to alleviate the tensions that arise in relation to multi-tasking as well as to define the overall development vector of the organisation.

In contrast to structural ambidexterity, which involves resolving the contradictions between exploration and exploitation at the organisational level, **Contextual ambidexterity** involves resolving this problem at the individual level. It implies that authorized employees of the organisation can independently determine and change the ratio between exploration and exploitation in their work. Individuals should be encouraged to make their own decisions about the optimal allocation of work time.

Inter-organisational ambidexterity involves collaboration between different organisations to carry out research and operational activities. It requires creation of research alliances between different organisations to solve problems by exploiting and extending existing knowledge, each bringing new knowledge to its activities. The advantage of inter-organisational ambidexterity is the distribution of resources (financial, human, material) for internal operational activities and external research through collaboration (Stettner & Lavie, 2014).

Although the concept of organisational ambidexterity can be easily transferred from the business sphere to the public context, many authors point out that this requires more research focusing on its application in public organisations, taking into account the uniqueness of public management (Kobarg, 2015).

Very limited amount of studies focus on ambidexterity in public organisations, or if any, it is considered primarily as an outcome rather than a strategy for achieving long-term success, although it is argued that in order to perform better governance, public organisations need to optimize but also innovate their services and processes (Gieske, 2019).

As Cannaerts, Segers, and Warsen argue, what is missing, however, is a sense of how ambidexterity can be achieved in public organisations, or under what conditions. So far, there has been little discussion of how public organisations can simultaneously balance efficiency and innovation (Cannaerts, 2020).

Currently, the activities of the administration are accompanied by numerous challenges, that attract the attention of managers, researchers, practitioners and society, in general, increasingly being directed towards the opportunities and potential dangers that the introduction of artificial intelligence can bring.

Artificial intelligence (AI), and in particular the advance of large language models (LLMs), is becoming increasingly important to public administration management because it can improve core administrative processes while enabling more systematic oversight of performance and risk. In routine operations, AI can support high-volume service delivery, improve internal coordination through faster information retrieval and synthesis, and strengthen strategic management by surfacing patterns from administrative data that are difficult to detect. The value of these systems in the public sector depends not only on their raw capabilities but on whether they reliably advance specific institutional objectives such as timeliness, consistency, procedural fairness, transparency, and compliance. This makes governance-oriented evaluation essential - rather than relying solely on generic model metrics, administrations increasingly need aligned benchmarking that tests AI behaviour against operational goals and embeds ongoing evaluation into daily management routines. In this sense, a modular benchmarking approach, such as the hybrid methodology proposed in (Andonov, 2025), which integrates structured goal-aligned test design with automated LLM-as-a-Judge evaluation illustrates how public organizations can move from simple, generic adoption of AI models to continuous performance management of AI-enabled processes, supporting accountable and sustainable modernization.

2. Methodology

Like any characteristic of an organisation, its organisational ambidexterity would remain an imaginary concept if it were not measurable. This means assigning quantitative values to its qualitative characteristics, in this case its ambidexterity. Moreover, the quantitative measures must cover all the main aspects of the studied issue. Metrics of object properties would be meaningless if they were not used for comparisons with parameters of other similar objects, for their ranking, for tracking trends, and for making predictions.

An evaluation is much more than a simple analysis ending with conclusions and recommendations. It is an opportunity to answer the question: ‘Where are we among the others?’ In order to answer this, we need to be clear about which areas, aspects and elements of the object we will be assessing and what criteria we will be using to make the assessment. Like other characteristics of organisations, such as ‘administrative capacity’, ‘sustainable development’ and ‘good governance’, organisational ambidexterity cannot be observed directly, so it is called a latent characteristic. Information about it can only be obtained by looking at the characteristics or values of manifested (observable) factors that indicate its hypothetical or real presence. The two options for assessing a latent construct are to evaluate the factors that shape it, i.e. the conditions for its presence or absence, on the one hand, and to evaluate the results of its presence or absence, on the other. We call the former *formative indicators* and the latter *reflective indicators*.

The model for measuring latent characteristics of public organisations through the evaluation of formative and reflective indicators was first applied in Bulgaria by B. Borisov in determining the levels of the Index of Administrative Capacity of Public Administration in the Republic of Bulgaria, which has been calculated since 2017 (Borisov, 2019). This approach to measuring organisational ambidexterity using formative and reflective indicators was also applied to the study of ambidexterity of municipalities in the Danube region of Bulgaria.

For the application of the model for measuring the organisational ambidexterity of municipal administrations, 16 Danube river municipalities were selected, representing 6% of all municipalities in Bulgaria. There are several reasons for this:

1. These municipalities have relatively similar infrastructural, social, environmental and other problems that they have to solve both on a daily basis and in the future;
2. The Danube river municipalities are different in size and territory, as well as in their economic opportunities, which allows to draw conclusions about which groups of municipalities, formed by some of their common features, apply to a greater extent the policy of organisational ambidexterity and which - not.
3. The Danube river municipalities are united in their regional association - Association of Danube River Municipalities ‘Danube’, which enables them to exchange experience and good practices, to conduct joint initiatives and trainings. On this basis, an assessment of the ambidexterity can reveal the real differences and priorities in their policies aimed at achieving operational efficiency, on the one hand, and research and innovation aimed at future development, on the other;
4. The existence of a common governing body, the Association's Board of Directors, facilitated the collection of data that was needed to assess the organisational ambidexterity.

The questionnaire for the assessment of the organisational ambidexterity of the Danube river municipalities contained 15 questions, the answers to which were evaluated according to a Likert scale from 1 to 5, where 1 means no ambidexterity, 2 - low ambidexterity, 3 - medium ambidexterity, 4 - good ambidexterity, and 5 - high ambidexterity. These scores act as *formative factors*.

As *reflective indicators* of the level of organisational ambidexterity, the main financial indicators of the assessed municipalities were set to assess the commitment of the municipal administrations to finance current activities and investments for the future. Their number was also 15, and their values were divided into pentiles[3] (5 groups from the minimum to the maximum recorded value), which allows to make a statistical analysis of the correlation between formative and reflective indicators of the organisational ambidexterity.

[3] From Latin „penta“ – five.

3. Results

The first group of questions, the answers to which were perceived as formative indicators of the organisational ambidexterity, are presented in the table below.

Table 1
Questions included in the survey on organisational ambidexterity of municipalities

1. The administration deals easily with external challenges such as pandemics, climate change, changes in resource prices, political changes, etc.
2. The administration has developed effective and efficient business processes with designated accountable persons (process owners) who are responsible for monitoring and controlling their compliance.
3. The administration regularly analyses business processes and updates them if necessary.
4. The organisation has entered a system of key performance measurement indicators through which processes are monitored and controlled.
5. There is a clear strategic view in the administration of the importance of both forward-looking research and operational effectiveness in addressing current tasks.
6. The municipality's strategic development documents (Integrated Municipal Development Plan and others) set targets that ensure an equal balance of exploitation and exploration efforts.
7. The administration monitors data on the dynamics of the National Statistical Institute's sustainable development indicators, which are directly relevant to its work and the state of the municipality.
8. The administration encourages the introduction of environmentally sound energy and transport systems.
9. The administration develops or assigns to external specialists the development of balances of water resources on the territory of the municipality, of the land fund, forests, air, population, municipal property and/or others.
10. All strategic planning documents for the development of the municipality, developed by the municipal administration and adopted by the Municipal Council, have a list of indicators for achieving each set goal, with specific values for the beginning of the planning period.
11. To achieve the priorities, objectives and measures set out in all strategic planning documents of the municipality, developed by the municipal administration and adopted by the Municipal Council, the necessary funds are indicated by sources and by year.
12. When developing draft annual budgets and three-year forecasts, the goals set in the strategic planning documents for the development of the municipality are always taken into account.
13. When determining the amount of local fees and service prices, the basic principle is that their amount should correspond to the direct and administrative costs that the municipal administration incurs for their provision, determined through calculation.
14. The municipal administration monitors the condition of municipal property and can claim that it is increasing at a faster rate than the rate of its physical and moral wear and tear.
15. During the annual presentation of the draft budget and the three-year budget forecast, the municipal administration also presents a report on the main financial indicators that shows the financial health of the municipality.

The following ratings were given to the respondents' answers: 1 - completely disagree; 2 - disagree; 3 - neither agree nor disagree; 4 - agree; and 5 - completely agree. Accordingly, low ratings indicate a low level of organisational ambidexterity, and high ratings indicate a high level.
Based on official data from the municipal budget reports for 2024, 15 financial ratios that play the role of reflective indicators of the organisational ambidexterity were calculated. The lowest levels of the values of each financial indicator falling in the first pentile were rated 1, and the highest levels falling in the fifth pentile - 5. Here again, low ratings indicate a low level of organisational ambidexterity, and high ratings indicate a high level of ambidexterity. The list of financial indicators that were calculated and evaluated are presented in the following Table 2.

Table 2
Financial ratios indicating the level of organisational ambidexterity of municipalities

Own revenues/All budget revenues (%)
Total income per capita (BGN)
Own revenue per capita (BGN)
Own revenues/Co-financing of state-delegated activities from local revenues
Capital expenditure/Total expenditure (%)
Capital expenditure/Current expenditure (%)
Own revenues/Capital expenditures
Capital expenditure per capita (BGN)
Capital expenditure/Debt amount
Own revenues/Debt amount
Total budget revenues/ Debt amount
Total budget revenues/Debt payments
Own revenues/Debt payments
Total budget revenues/Administrative expenses
Number of people served by one municipal employee

The calculated Likert scale ratings of the forming indicators in the range from 1 to 5 and the ratings of the financial indicators, acting as reflecting indicators, also in the range from 1 to 5, as well as the difference between them, are shown in the following Table 3 and visualized in Figure 1.

Table 3
Ratings of the organisational ambidexterity of the studied municipalities, given by the representatives of the municipal administrations (forming indicators) and ratings of the financial results of the municipalities (reflecting indicators)

Municipalities	Self-assessment of ambidexterity by municipalities	Assessment of ambidexterity based on results	Difference in ratings
Ruse	4.73	2.20	2.53
Levski	3.53	2.67	0.87
Vidin	3.53	1.73	1.80
Tutrakan	4.00	1.47	2.53
Cherven bryag	3.60	1.40	2.20
Dolni Dabnik	3.40	2.47	0.93
Glavinitsa	4.00	1.47	2.53
Dolna Mitropoliya	3.87	1.40	2.47
Byala Slatina	3.60	1.87	1.73
Bregovo	3.67	167	2.00
Lom	3.27	2.73	0.53
Pleven	3.93	2.00	1.93
Oryahovo	3.93	2.33	1.60
Nikopol	3.67	1.53	2.13
Svishtov	4.73	1.80	2.93
Belene	4.13	1.73	2,40

Source: Authors' own calculations

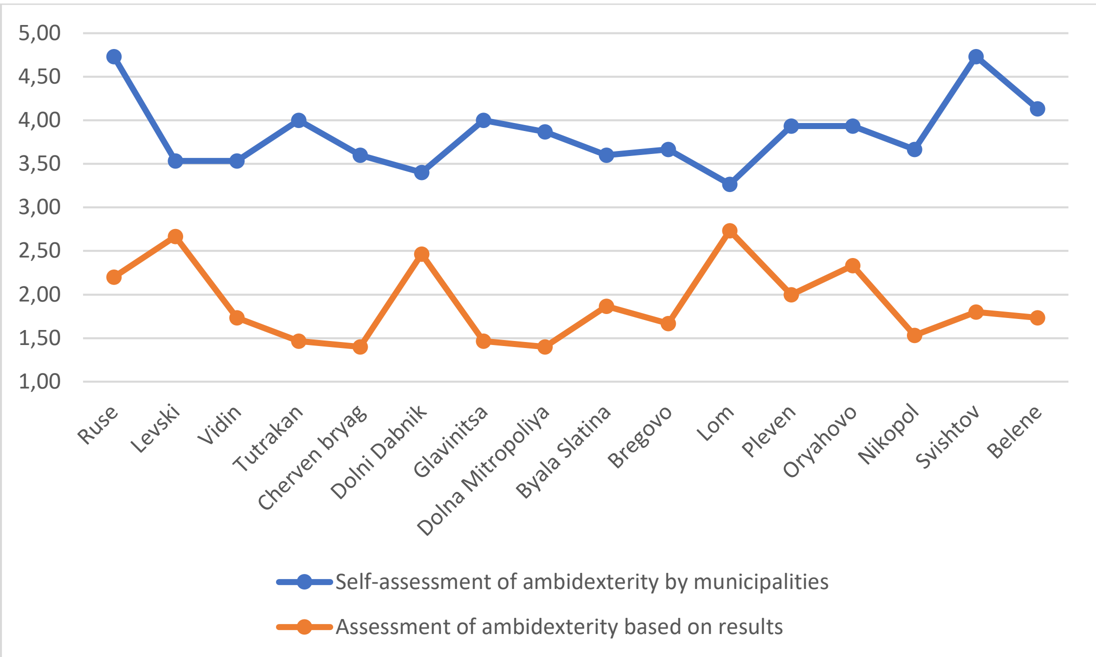

Figure 1. Assessments of the organisational ambidexterity given by municipal representatives (forming indicators) and assessments of their financial results (reflecting indicators)

To test whether there is a statistically significant difference between the two samples – that of the answers to the questions given by the representatives of the municipalities and that of the values of their financial results, the IBM SPSS Statistics 27 software product was used to analyse the t-test for related samples (independent-samples t-test). The results are presented in the following Table 4.

Table 4
Results of the t-test of the organisational ambidexterity assessments given by the representatives of the municipal administrations and the assessments calculated based on their financial indicators

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	VAR00017	3,8494	16	,41784	,10446
	VAR00018	1,9044	16	,44867	,11217

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
					95% Confidence Interval of the Difference				
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			
Pair 1	VAR00017 - VAR00018	1,94500	,68086	,17022	1,58219	2,30781	11,427	15	<,001

Source: Authors' own calculations

The table shows that the average ratings of the organisational ambidexterity given by the representatives of the municipal administrations is 3.8494, and the average ratings of the values of the financial indicators for organisational ambidexterity is 1.9044. This is a significant difference that is subject to verification whether it is random or not.

Two hypotheses are formulated - H0, which states that there is no statistical difference between the samples, and H1, which states that there is such a difference. The value displayed in the Sig. (2-tailed) column shows whether there is a statistically significant difference between the samples of the two groups at a 95% confidence interval. If it is more than 0.05, it cannot be concluded that there is a difference between them and vice versa, if it is less than 0.05, this means that there is a difference between the two arithmetic mean values of the two samples. In this case, it turns out that it is less than 0.001, therefore it is confirmed that there is a statistically significant difference between the assessments of the representatives of the municipal administrations and the financial results achieved by them, testifying to the level of the organisational ambidexterity. In fact, this means that the representatives of the municipalities are aware of the need to achieve organisational ambidexterity, but they do not apply it in practice in their financial policy.

4. Discussion

Although the concept of the organisational ambidexterity can easily be transferred from the business world to the public context, many authors point out that more research is needed to address its application in public organisations, taking into account the uniqueness of public management (Gieske, 2019). Very limited amount of studies focus on ambidexterity in public organisations, or if any, it is considered primarily as an outcome rather than a strategy for achieving long-term success, although it is argued that in order to perform better governance, public organisations need to optimize, and also innovate their services and processes. As Nele Cannaerts, Jesse Segers and Rianne Warsen argue, what is missing, however, is an idea of how ambidexterity can be achieved in public organisations or under what conditions. So far, there has been little discussion of how public organisations can simultaneously balance efficiency and innovation (Cannaerts, 2020).

The main question that needs to be answered when assessing the organisational ambidexterity of public organisations is which form of ambidexterity will be assessed – sequential, structural, contextual, or inter-organisational? In our opinion, *sequential ambidexterity* is not suitable for public sector organisations, since they cannot afford to devote more attention and resources to solving current tasks or only to future development within a certain period of time. *Inter-organisational ambidexterity* is a possible form and implies cooperation of municipalities with various external organisations in the implementation of both routine activities and research and innovation. Such forms are public-private partnerships, contracts with guaranteed final results, awarding public procurement contracts, partnerships with scientific institutes and higher education institutions, etc. They develop in parallel with the main activity and do not provide an unambiguous characteristic of ambidexterity, understood as a balance of exploitation and innovation processes.

Structural ambidexterity is implemented through specialization of the structural administrative units into those that are primarily responsible for operational activities, and those that are responsible for research, innovation and development. According to Art. 21, para. 1, item 2 of the Bulgarian Law on Local Self-Government and Local Administration, the Municipal Councils adopt the so-called staffing schedule of the municipal administrations, approving their total number and structure. According to the Law on Administration, depending on the distribution of activities, the administration is divided into general and specialized. The *general administration* mainly carries out technical activities for administrative services and creates conditions for the work of the *specialized administration*, which carries out specific functions in the various areas of competence of the relevant state authority. We can speak of structural ambidexterity when the structure of an administration in its set of directorates, departments or sectors is strictly specialized in two groups – those that are engaged in current activities of serving the population and providing public services, and others that are engaged in investments and development planning. There is no such practice in Bulgaria, although there are directorates and departments that deal primarily with spatial planning, but they are not financially separate as independent units, as is the case in other countries. The figure below presents the structure of the municipal administration of the American municipality of Rockville, from which it is evident that the administrative units are financially and organisationally separated into those responsible for current activities and those dealing with future development.

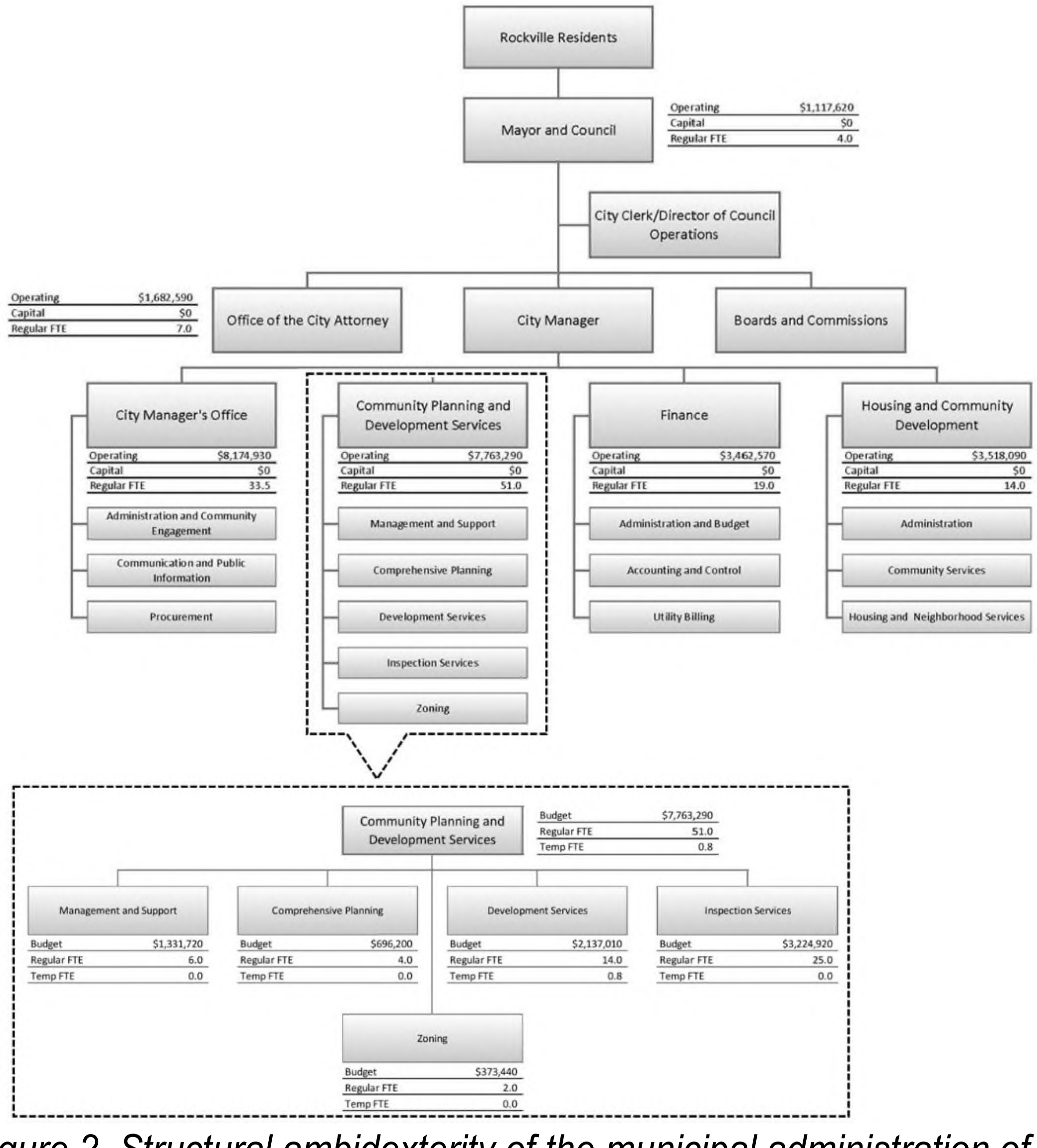

Figure 2. Structural ambidexterity of the municipal administration of the municipality of Rockville, USA

Source: Adopted operating budget and capital improvements program July 1, 2024-June 30, 2025

Unlike structural ambidexterity, in *contextual ambidexterity* each administrative unit can determine and change the ratio of exploration and exploitation in its activities as appropriate and depending on the specific situation. Individual administrative units are not separated as teams with independent funding. This form does not require undertaking organisational reengineering, but largely depends on the given ‘tone at the top’, i.e. on the understanding of the management that its responsibilities exceed its mandate.

Since **contextual ambidexterity** is the only one possible for application under the current legal, political and historically inherited conditions in Bulgaria, it was precisely this that was the subject of the study to establish its levels in Bulgarian municipalities.

5. Conclusions

Organisational ambidexterity is a new management concept that refers to the ability of organisations to balance between exploitation (optimization of existing processes and products) and exploration (search for new opportunities and innovations). The results of the empirical study conducted in 16 municipalities - members of the Association of Danube River Municipalities in Bulgaria, show that municipal employees understand the need to be guided by both the need to meet the current needs of the population and the concern for ensuring future development in their functions, although they do not formulate this feeling as organisational ambidexterity. The analysis of the financial indicators of the studied municipalities, however, clearly indicates a preference for spending on covering current needs over those for investment and development. This can be explained by a number of circumstances, including the following:

- limited financial resources for long-term investments of strategic importance;
- public pressure from citizens and organisations to solve current problems;
- desire of municipal leadership to gain public trust from voters;
- conservative structure of municipal administrations, not allowing sufficient flexibility;
- routine performance of activities by employees and lack of incentives for innovations;

- old, outdated job descriptions, manuals, methodologies and work regulations for employees;
- lack of administrative capacity to effectively combine operational with strategic activities;
- failure to implement program budgeting, which would allow for better allocation of funds to achieve current and long-term goals, etc.

Overcoming these weaknesses would allow local authorities to successfully implement the principles of sustainable development, effectively solving current problems, and considering the long-term consequences.

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CONTENTS

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ANALYSIS OF EMPLOYEE SATISFACTION IN THE NIGERIAN HEALTHCARE AND IT SECTORS

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Abstract: This study explores the determinants of job satisfaction among professionals in Nigeria's healthcare and information technology sectors, with particular emphasis on the roles of supervisor support, compensation, benefits and workload. Data was collected through a questionnaire survey administered to a sample of 195 professionals, and analysed using Pearson correlation coefficients to assess the strength and direction of relationships among the examined variables. The findings reveal a significant positive correlation between supervisor support and job satisfaction ($r = 0.540$), underscoring the critical role of effective leadership in enhancing employee morale. Furthermore, satisfaction with salary and benefits exhibits a strong positive association with overall job satisfaction ($r = 0.629$), highlighting the importance of offering competitive remuneration packages. Contrary to conventional assumptions, the study also identifies a positive correlation between increased workload and job satisfaction ($r = 0.572$), suggesting that higher job demands may, under certain conditions, contribute positively to employee engagement and fulfilment. These insights offer valuable implications for organisational policy and human resource management within high-demand professional environments.

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Introduction

Job satisfaction constitutes a foundational construct within the fields of organisational psychology and human resource management, recognised for its critical influence on employee performance, motivation, and overall well-being. In contemporary organisational research, it is widely acknowledged that satisfied employees are more productive, committed, and resilient, contributing positively to organisational outcomes. While global studies have extensively explored the antecedents and consequences of job satisfaction, context-specific investigations remain essential for understanding how local sociocultural and economic conditions mediate these dynamics.

In the Nigerian context, job satisfaction assumes particular significance due to the country's complex interplay of cultural diversity, economic volatility, and evolving workplace structures. Nigeria's labour market is characterised by significant heterogeneity—ethnically, culturally and regionally—which shapes both employee expectations and organisational practices. Within this milieu, factors such as leadership style, interpersonal relations, compensation structures and perceived fairness are likely to exert varying degrees of influence on job satisfaction and, by extension, employee performance.

Moreover, relational dimensions—such as managerial recognition, mutual respect, and communal values—may hold greater importance in Nigerian organisations compared to those in more individualistic cultures. This context-sensitive understanding is essential, as conventional models developed in Western settings may not fully capture the intricacies of job satisfaction in Nigeria.

In this context, the purpose of this study is to empirically examine the key factors shaping job satisfaction among professionals in Nigeria's healthcare and information technologies sector. In order to ground this analysis conceptually, the following section presents a review of the relevant literature on job satisfaction, with particular emphasis on research conducted in comparable socio-economic and cultural settings.

1. Literature review

Job satisfaction is a complex and abstract construct, making its measurement inherently challenging. It is closely intertwined with personal life satisfaction, such that significant changes in an individual's personal life are likely to influence job satisfaction, at least in the short term. The factors that contribute to job satisfaction are multifaceted and vary in significance across individuals (Schein, 2017). Employee satisfaction surveys serve as a crucial tool for organisations seeking to enhance their operational effectiveness. These surveys aim to identify areas in which working conditions can be improved, thereby increasing employee engagement and productivity (Suta, 2023; Baquero, 2022). The primary objective of such surveys is to gain a deeper understanding of employees' needs and expectations, facilitating the creation of a conducive and productive work environment (Chikhaoui, 2022).

In Herzberg's (1959) Two-Factor Theory, job satisfaction is shaped by a combination of intrinsic factors, such as the nature and outcomes of the work, and extrinsic factors, such as recognition and reward. Maslow's (1943) hierarchy of needs further highlights the importance of fulfilling higher-order needs, including self-actualisation and esteem, as key drivers of overall satisfaction (Laksana et al., 2024). An employee's sense of well-being within an organisation significantly influences their job performance, creativity, adaptability to change, and both life and job satisfaction. Furthermore, these factors contribute to their organisational commitment. Higher levels of employee commitment correlate with improved organisational performance and greater overall satisfaction. In turn, organisations can foster this commitment by providing various forms of support and benefits to their employees (Nguyen & Tran, 2021). Key aspects such as an employee's role within the workplace, their work hours, and the satisfaction of fundamental needs are crucial elements that contribute to overall job satisfaction (Hussein et al., 2022).

The factors that influence workplace satisfaction are multifaceted, encompassing aspects such as the nature of the work, working conditions, compensation, and corporate social responsibility initiatives (Robbins & Judge, 2021). Employee satisfaction is also significantly shaped by management support, the clarity of organisational goals, the workplace community, the quality of the working environment, and the effectiveness of the remuneration system (Al-Ansi et al., 2023). A leader’s commitment, decisiveness, competencies, and leadership style play a crucial role in shaping employees' job satisfaction (Alboliteeh, 2022). Employees often generalise their interactions with managers to their perceptions of the entire organisation. Therefore, when employees trust their managers, their overall perception of the organisation tends to be more positive, resulting in higher levels of job satisfaction (Yue et al., 2022). Trust between individuals, particularly within the workplace, has been shown to enhance employee satisfaction (Basiswanto & Elmi, 2023). Budiono (2024) highlights that a supportive work community and an innovative management style positively influence employee satisfaction. Furthermore, the quality of working conditions remains a significant determinant of job satisfaction (Putra et al., 2023). Persistent negative feedback from management or integration into a new workgroup, with which the employee feels disconnected, can substantially affect job satisfaction in the long term (Matzler & Renzl, 2006). Organisational changes, lack of recognition from management, and unprofessional behaviour from colleagues are among the most common causes of job dissatisfaction (Andrade & Westover, 2018). Farkas et al. (2013) identified that employees expressed the highest levels of dissatisfaction with their compensation packages, followed in decreasing order of significance by rewards, lack of motivation, limited opportunities for career development, management style, and responsibility. In a study by Garai-Fodor and Jäckel (2024), employee performance and satisfaction were found to be significantly influenced by strong motivation and adequate compensation. Additionally, Elrayah and Semlali (2023) investigated the effects of both monetary and non-monetary benefits, revealing a positive correlation between these benefits and employee performance, with a particular emphasis on the link between monetary rewards and employee satisfaction. Perkasa et al. (2023) conclude that a satisfied employee is primarily influenced by appropriate remuneration, opportunities for professional development, and positive interpersonal relationships within the workplace community. Similarly, Lin and Wang (2025) indicate that the good working atmosphere and its associated satisfaction are key in reducing employee turnover. Furthermore, Belás et al. (2015) emphasise that employee satisfaction plays a critical role in ensuring long-term organisational success and sustainability.

In the context of Nigeria, job satisfaction is deeply shaped by the nation's distinct cultural and economic dynamics. Nigeria, with its rich cultural diversity and complex socio-economic conditions, provides a unique framework for examining job satisfaction. Adebayo (2006) underscores the influence of cultural factors, such as collectivism and power distance, which significantly affect employees' perceptions of job satisfaction, differentiating them from those observed in more individualistic Western cultures. Moreover, Nigeria's economic challenges, including volatile employment rates and the shifting industrial landscape, as highlighted by Okpara (2004), are pivotal in influencing job satisfaction levels.

When comparing job satisfaction across various nations, Nigeria offers distinct insights. Studies like that of Oluwatunmise et al. (2020) illustrate how factors such as job security and working conditions, which may be considered secondary concerns in more developed economies, assume critical importance in the Nigerian context. These comparative analyses emphasise the need to approach job satisfaction with a clear understanding of the specific socio-economic and cultural contexts within which it occurs. Research conducted within Nigeria, including Adebayo’s (2006) work, demonstrates a positive correlation between job satisfaction and employee productivity. Further studies reveal additional facets of job satisfaction within Nigeria. Adeoye and Fields (2014) found that insurance workers reported dissatisfaction with their compensation, which was associated with lower job satisfaction, increased absenteeism, and higher turnover rates. This dissatisfaction was partly due to the misalignment between remuneration and the tasks performed, as well as concerns regarding workplace safety. Ojha et al. (2025) found a positive link between job satisfaction and emotional intelligence, suggesting that employees with higher emotional intelligence tend to report greater satisfaction with their roles. Additionally, Fu (2019) established a connection between job satisfaction and employee commitment among hospital workers in Nigeria. Akinwale and George (2020) highlight that nurse satisfaction not only positively influences work productivity and quality of care but also improves patient safety, reduces turnover, and enhances overall commitment. Ugochukwu et al. (2024) demonstrated a positive correlation between job satisfaction, job enrichment and organisational citizenship behaviour, further underscoring the importance of job satisfaction in promoting favourable organisational outcomes. However, not all sectors show positive outcomes. In Jegede's (2024) study, Nigerian pharmacists were found to be dissatisfied with their work environments, suggesting that sector-specific factors also play a critical role. Asekun (2016) further corroborates these findings by highlighting the strong relationship between salary and job satisfaction, which remains a central issue for many workers in Nigeria.

These studies collectively emphasise that while certain general trends, such as the relationship between job satisfaction and productivity, are observed globally, local socio-economic and cultural factors play a significant role in shaping the determinants and outcomes of job satisfaction in Nigeria. Understanding these nuances is crucial for effectively addressing job satisfaction and its impact on employee performance within the Nigerian context.

In alignment with the principles of deductive research, three hypotheses were developed based on the insights derived from the existing body of literature:

H1: There exists a positive correlation between satisfaction with salary and benefits and overall job satisfaction.

H2: A positive relationship is observed between job satisfaction and the perceived support from supervisors.

H3: A negative correlation is present between workload intensity and job satisfaction.

2. Methodology

This research employs a quantitative research design to investigate the factors influencing job satisfaction within Nigerian workplaces, specifically targeting employees in the health and information technology (IT) sectors. These sectors were chosen due to their dynamic work environments and their significant roles in the Nigerian economy. The research aims to explore job satisfaction through a structured survey administered to employees from four private organisations in the health and IT industries.

A stratified random sampling technique was employed to ensure a representative sample of the workforce across the selected organisations. The population was divided into subgroups or strata based on specific characteristics such as department, job role and tenure. Participants were randomly selected from each subgroup, ensuring that the sample captured a wide range of perspectives within the organisations. The inclusion criteria for the study were clearly defined: only full-time employees who had been employed for a minimum of six months were eligible to participate. This criterion was established to ensure that participants had adequate experience in their current roles and could provide informed insights into their job satisfaction levels.

A structured questionnaire survey was used as the primary data collection method. The questionnaire was designed to collect responses on various dimensions of job satisfaction, including supervisor support, salary, benefits and workload. A 5-point Likert scale was utilised for all questions regarding job satisfaction, where 1 represents ‘Very Dissatisfied,’ 2 denotes ‘Dissatisfied,’ 3 indicates ‘Neutral,’ 4 represents ‘Satisfied,’ and 5 corresponds to ‘Very Satisfied.’ This scale was chosen to capture the varying degrees of satisfaction across different respondents and facilitate statistical analysis of the collected data.

The survey was administered confidentially to protect the anonymity of the respondents, ensuring that their identity was not disclosed at any point during the study. Data collection took place in the spring of 2025, with a total of 195 responses gathered.

The collected data were entered into IBM SPSS Statistics 23 for analysis. The data were first subjected to a normality test, specifically using the Kolmogorov-Smirnov and Shapiro-Wilk tests. The results of both tests indicated that the data did not violate the assumption of normality, as the p-values were greater than 0.05 ($p > 0.05$), confirming that the data were normally distributed. To test the research hypotheses, Pearson correlation analysis was conducted. This method was chosen due to its ability to detect statistical relationships between two continuous variables, making it suitable for examining the associations between the different factors influencing job satisfaction. The results of the Pearson correlation analysis provided insights into the strength and direction of the relationships between supervisor support, compensation, workload, and overall job satisfaction.

3. Results and Discussion

Demographic data are presented in Figure 1. 62.1% of the respondents were women, 36.9% were men, and the remaining 1% did not wish to state their gender. The age distribution of the respondents is crucial as it reflects the different career stages and generational perspectives within the workforce. The survey included a diverse age range, categorised into several groups: the 18-25 age group, 26-35 age group, 36-45 age group, 46-55 age group, and 56+ age group. Young Adults (18-25 years) - this group, while typically having less work experience, provides insights into the aspirations and early career satisfaction levels of younger employees entering the workforce. Early Career (26-35 years) respondents in this category are often in the growth phase of their career, looking to establish stability and advancement opportunities. Mid-Career (36-45 years) - this segment tends to hold more managerial positions and might evaluate job satisfaction in light of family responsibilities and long-term career achievements. Experienced (46-55 years) - these respondents are likely to have significant experience and may focus on job satisfaction factors related to job security, recognition and preparation for retirement. Senior (56+ years), although smaller in number, this group's perspectives are crucial as they reflect on satisfaction towards the end of their career trajectory. A significant percentage of respondents have a bachelor's degree or work full-time.

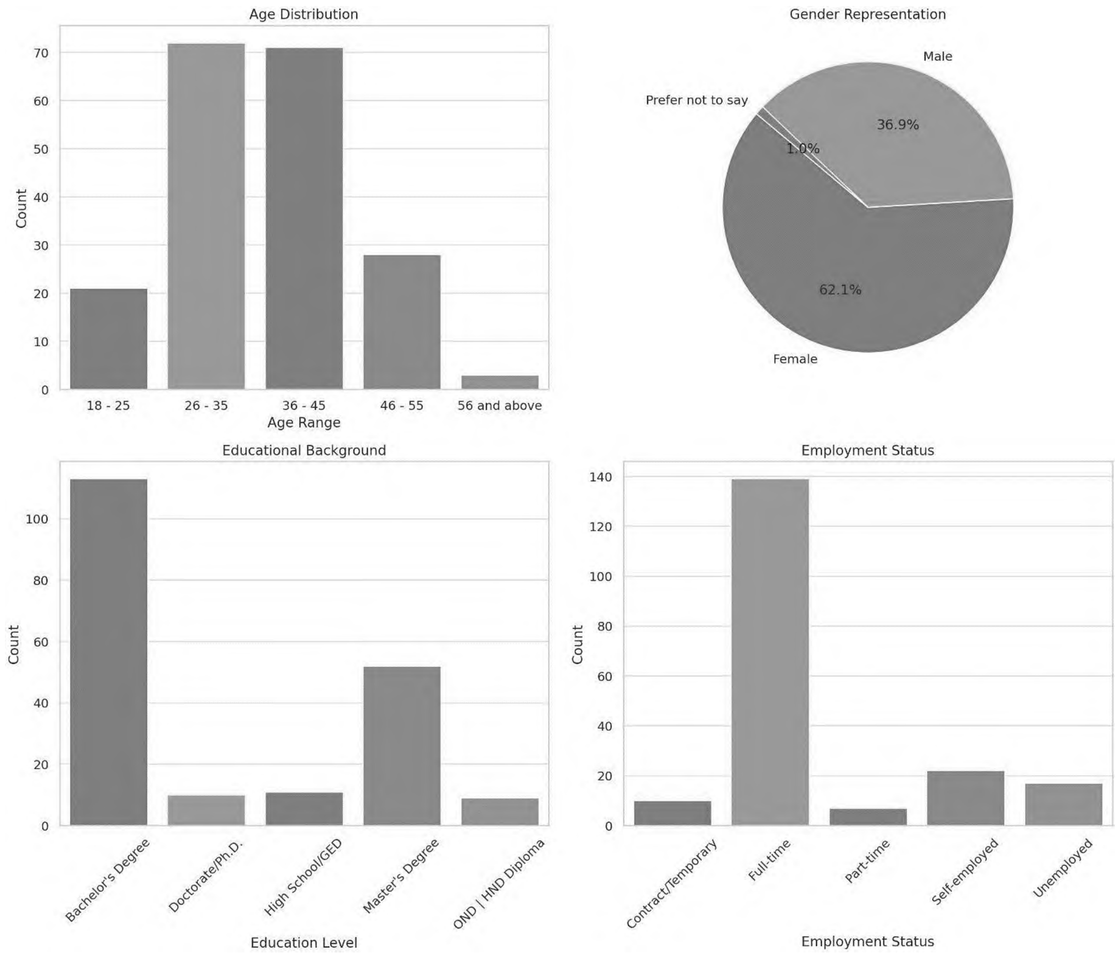

Figure 1. Demographic data
Source: Own editing (Questionnaire survey, 2024)

Salary and benefits are frequently mentioned as crucial elements in research on job satisfaction. These factors encompass not just the tangible remuneration for completed tasks, but also encompass the subjective notions of justice and equality, which play a crucial role in determining employee contentment. This investigation examines the correlation between these elements and work satisfaction levels among employees in the target sectors. Pearson correlation coefficients was utilised to compute in order to assess the magnitude and direction of the association between income and benefits and overall work satisfaction. This statistical technique is very valuable for measuring the extent of linear correlation between two continuous variables. Pearson Correlation Coefficient (r) value ranges from -1 to +1. A value closer to +1 indicates a strong positive correlation, suggesting that as salary and benefits increase, so does job satisfaction. Conversely, a value closer to -1 would indicate a strong negative correlation, where higher salary and benefits might unexpectedly correlate with lower job satisfaction. A value around 0 suggests no linear correlation. Significance (p-value) indicates whether the observed correlation is statistically significant. Typically, a p-value less than 0.05 is considered statistically significant, meaning there is less than a 5% chance that the correlation observed is due to random variation in the sample data. In Table 1 the Pearson Correlation Coefficient (r) 0.629 indicates a strong positive correlation between overall job satisfaction and satisfaction with salary and benefits. A value of 0.629 suggests that as satisfaction with salary and benefits increases, overall job satisfaction also tends to increase. This is a substantial correlation, indicating a significant relationship between these two aspects of job experience. The Significance (p-value) is less than 0.001, which is highly significant statistically. This indicates a very low probability that the observed correlation is due to random chance. Thus, we can be very confident in the robustness of the relationship between these two variables. The robust positive association between job satisfaction and contentment with income and benefits highlights the crucial significance of remuneration in shaping employees' perception of their overall job experience. The correlation implies that efforts focused on enhancing remuneration packages may result in increased work satisfaction among employees.

Table 1
Pearson correlation between compensation (salary and benefits) and job satisfaction

		Salary and benefits	Overall job satisfaction
Salary and benefits	Pearson Correlation	1	.629**
	Sig. (2-tailed)		<.001
	N	178	178
Overall job satisfaction	Pearson Correlation	.629**	1
	Sig. (2-tailed)	<.001	
	N	178	178
** Correlation is significant at the 0.01 level (2-tailed)			

Source: Own editing (Questionnaire survey, 2024)

H1 hypothesis posits that there is a positive correlation between contentment with wage and perks and work satisfaction. The study reveals a Pearson correlation coefficient of 0.629, demonstrating a robust positive link between contentment with salary and perks and overall work satisfaction. The relevance of competitive remuneration packages as a significant driver of job satisfaction is emphasised by the significance of this result (p-value <0.001). This discovery highlights the crucial significance of both monetary and non-monetary advantages in influencing employees' opinions of the worth of their job. Organisations should evaluate their compensation packages by comparing them to industry standards and improving benefit schemes in order to increase satisfaction and retention. Our findings agree with international results, i.e. Marni and Adi (2022), Elrayah and Semlali (2023), Perkasa et al. (2023), Prianto and Handayani (2024) findings, at the same time, with the research carried out Asekun (2016).

In Table 2 the Pearson Correlation Coefficient (r) 0.540 indicates a strong positive correlation between overall job satisfaction and support from supervisors. The Significance (p-value) is less than 0.001, which is highly significant statistically. This indicates a very low probability that the observed correlation is due to random chance. Thus, we can be very confident in the robustness of the relationship between these two variables.

Table 2
Pearson correlation between job satisfaction and support from supervisors

		Overall job satisfaction	Support from supervisors
Overall job satisfaction	Pearson Correlation	1	.540**
	Sig. (2-tailed)		<.001
	N	178	178
Support from supervisors	Pearson Correlation	.540**	1
	Sig. (2-tailed)	<.001	
	N	178	178
** Correlation is significant at the 0.01 level (2-tailed)			

Source: Own editing (Questionnaire survey, 2024)

H2 hypothesis proposes that there exists a direct and positive relationship between job satisfaction and the perceived support from supervisors. The Pearson Correlation Coefficient (r) for the analysis is 0.540, which indicates a moderate to high positive link between job satisfaction and the level of support from supervisors. The observed association is statistically significant and not attributable to chance, as indicated by the significance level of less than 0.001. The results confirm the modified hypothesis H2, showing that a greater perception of support from supervisors is linked to increasing levels of job satisfaction. This corroborates the hypothesis that proficient administration and encouraging leadership play a crucial role in augmenting employee morale and contentment, and corresponds with results from international research by Budiono (2024), Lin and Wang (2025).

The impact of workload on job satisfaction is commonly acknowledged across diverse industries. It is essential to comprehend the influence of workload on job satisfaction in the health and IT sectors in Nigeria, given the possibility of high-stress conditions in these industries. The analysis employs data obtained from a survey that inquired about respondents' ratings of their workload satisfaction and overall job satisfaction.

Table 3
Pearson correlation between job satisfaction and workload and demands of job

		Overall job satisfaction	Workload and demands of job
Overall job satisfaction	Pearson Correlation	1	.572**
	Sig. (2-tailed)		<.001
	N	178	178
Workload and demands of job	Pearson Correlation	.572**	1
	Sig. (2-tailed)	<.001	
	N	178	178
** Correlation is significant at the 0.01 level (2-tailed)			

Source: Own editing (Questionnaire survey, 2024)

H3 hypothesis has initially suggested that workload has a detrimental effect on job satisfaction, however the data contradicted this assertion, revealing a positive association instead. Contrary to the initial premise that higher workloads have a negative impact on job satisfaction, the data in Table 3 shows a positive correlation ($r = 0.572$) between workloads and job satisfaction. This correlation is strongly supported by a p-value of <0.001. This result implies that in specific situations, having more work to complete does not have a negative impact on job satisfaction. Instead, it may have a positive effect, possibly reflecting better levels of involvement and accountability. The positive connection suggests that employees who are more engaged and stimulated by their work tend to have higher levels of satisfaction. A positive connection indicates that when the workload grows, there is a tendency for total job satisfaction to increase as well, which may initially seem contrary to what one may expect. This finding suggests that in the population studied, a greater workload may be linked to work that is more stimulating or satisfying. It is also possible that individuals with higher workloads have more responsibilities and, maybe, hold higher positions and receive greater incentives in their jobs. The Significance (p-value) being less than 0.001 indicates that the correlation between workload and overall job satisfaction is statistically significant, with a very low probability that this result is due to random chance. The observed positive association is both statistically and practically significant. It indicates that in Nigeria's health and IT industries, people who have heavier workloads are likely to have greater job satisfaction. There are multiple possible explanations for this. Employees who exhibit greater levels of involvement and stimulation in their work and are faced with demanding tasks may have heightened levels of job satisfaction, even in the presence of a heavy workload. This suggests that higher workloads may be linked to good challenges, engagement, and a sense of success. Our results can be supported by Putra et al. (2023), Perkasa et al. (2023), Okpara (2004), Oluwatunmise et al. (2020).

Conclusion

This study provides an empirically grounded analysis of the determinants of job satisfaction within Nigeria’s healthcare and information technology sectors, focusing specifically on the roles of supervisor support, financial compensation and workload. The results demonstrate statistically significant positive correlations between job satisfaction and each of the examined variables, underscoring the multidimensional nature of employee contentment in professional environments marked by rapid transformation and high demands.

Supervisor support has emerged as a key predictor of job satisfaction ($r = 0.540$), reaffirming the crucial role that effective, empathetic and communicative leadership plays in shaping positive workplace experiences. Similarly, the strong correlation between satisfaction with salary and perks and overall job satisfaction ($r = 0.629$) highlights the importance of competitive remuneration strategies in retaining talent and enhancing organisational commitment. Perhaps most notably, the study reveals a counterintuitive positive relationship between workload and job satisfaction (r

= 0.572), suggesting that in certain contexts, increased responsibilities may be perceived as indicators of trust, engagement and professional growth rather than as sources of stress.

Collectively, these findings suggest that job satisfaction in the Nigerian context is influenced not only by material rewards but also by relational and psychological factors. They underscore the need for a nuanced, context-sensitive approach to employee well-being that accounts for the complex interplay between leadership quality, compensation structures and job design.

Limitations, Practical Implications and Future Research Directions

The outcomes of this study suggest that organisational policies should emphasise the development of robust internal support systems that foster transparent communication, structured mentorship and inclusive professional development. Specific interventions may include the implementation of regular, skills-based training programs for supervisory staff, the provision of clearly defined and achievable career advancement pathways and the establishment of transparent, merit-based reward systems. These strategies have the potential to significantly enhance employee satisfaction and, by extension, organisational performance.

Despite its contributions, the study is subject to certain methodological limitations. The sample was limited in size and confined to specific sectors (healthcare and IT) within Nigeria.

Future research should aim to address this limitation by incorporating a more extensive and diverse sample encompassing multiple industries and geographical regions. Moreover, the cross-sectional nature of the research design precludes the establishment of causal inferences.

In light of these findings, several practical recommendations are proposed to enhance job satisfaction and organisational effectiveness. First, organisations are encouraged to invest in leadership development initiatives that equip supervisors with the interpersonal and managerial competencies necessary to motivate and support their teams. Second, regular assessments and adjustments of compensation structures should be undertaken to ensure alignment with industry standards and employee expectations. Third, organisations must strive to calibrate workloads in a manner that is both challenging and sustainable, thereby maximising employee engagement while mitigating the risk of burnout. Importantly, the study highlights that increased workload, when accompanied by recognition and autonomy, may contribute positively to employees' sense of purpose and professional fulfilment.

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CONTENTS

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A DUAL-LAYER HYBRID SECURITY FRAMEWORK: ANALYZING DIGITAL MODELS TO ENHANCE CYBERSECURITY MANAGEMENT FOR START-UPS AND SMES

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Nomeda Dobrovolskienė[4],
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Abstract: The rapid evolution of generative AI is transforming cybersecurity practices by providing advanced capabilities for threat detection and response. For start-ups and SMEs, these technologies offer a cost-effective way to defend against sophisticated online threats without the need for massive capital investment. While AI facilitates the identification of anomalies and malicious behavior, it also introduces new risks, such as deepfakes, zero-day exploits, and the potential for attackers to manipulate AI models themselves. The study examines the impact of large language models and artificial intelligence on the software and architecture of cybersecurity systems. The main types of attacks on artificial intelligence systems and the changes in the software development paradigm that result from them are being considered. To address the evolving threat landscape, this study proposes a Dual-Layer Hybrid

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Security Framework that integrates high-speed algorithmic detection with cognitive analysis. We present an algorithm for building a security system using LLM models. We also propose a software application tailored explicitly for the comprehensive analysis of network traffic while detecting cybersecurity incidents using Python and the Pandas library. Furthermore, it leverages artificial intelligence to provide a comprehensive approach to detecting and mitigating cybersecurity threats. This will enable start-ups and small businesses to manage their cybersecurity in-house.

Keywords: Cybersecurity, AI, LLM, Smart Analytics, Hybrid Security Architecture, Start-ups, SMEs

JEL: M15, L26, O32

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Introduction

Generative artificial intelligence (AI) technologies, particularly generative pre-trained transformers (GPT, Gemini, etc.), are changing the field of security. AI provides advanced capabilities for detecting, preventing, and responding to threats, which significantly improves cybersecurity practices. This gives companies the ability to defend against more sophisticated online threats and is especially crucial for start-ups and newly established businesses, which may lack the financial resources to make substantial cybersecurity investments and are often required to cut costs.

This is also linked to the widespread development of innovation in enterprises (especially start-ups) implementing new technologies (Pukala, 2016). It should also be emphasised that innovative enterprises are very important for the country's economy and are its driving force (Ishchuk et al., 2021) as well as for the sector in which they operate (Pukala & Petrova, 2019).

Moreover, as the renewable energy sector continues to expand, the supporting infrastructure has become a prime target for cyberattacks. This is due to the increasing interconnectivity of technologies and systems within the energy sector, making them more susceptible to cyber threats. Attackers are exploiting vulnerabilities in smart grids, solar and wind farms, and other renewable energy facilities, posing significant risks to the stability and security of our energy supply.

DNV (2023) highlighted in "Energy Cyber Priority: Closing the Gap Between Awareness and Action" the increasing awareness of cybersecurity threats within the energy industry. The report points out that companies are ramping up their investments to address these threats, driven by factors such as heightened geopolitical tensions, emerging compliance requirements, and the rapid adoption of digitally connected infrastructure.

There is a pressing need to place a stronger focus on securing safety-critical systems in light of these developments. Moreover, as cybersecurity's role in enabling digital transformation and the energy transition becomes more recognized, DNV (2023) emphasizes the importance of delving into the state of cybersecurity within the energy sector. This involves understanding the evolving landscape of cyber threats and the measures being taken to mitigate risks and ensure the resilience of critical energy infrastructure.

Artificial intelligence systems are able to sift through huge amounts of data and identify trends that may be signs of online threats. Anomalies, malicious behaviour, and well-known attack patterns can be detected using trained machine-learning models. Because of this, security services can detect and eliminate threats in advance.

However, it is important to note that artificial intelligence in cybersecurity does not come without challenges. Attackers can also potentially exploit AI systems to their advantage or avoid detection through anti-AI techniques. Artificial intelligence use can lead to cybersecurity issues in several ways, namely (Yermachenko et al., 2023):

• Emerging new types of cyber threats: Hackers can use artificial intelligence to create more sophisticated cyberattacks (or cyberattacks of entirely new types), such as creating fake content (e.g., deepfakes), bypassing technical security measures (e.g., voice biometrics) or exploiting vulnerabilities (e.g., zero-day exploits).

• Compromise of artificial intelligence systems: Artificial intelligence systems themselves may be vulnerable to cyber-attacks by tampering with their data, algorithms, or results. This can bias AI systems, which in turn can cause errors.

• Ethical and social issues: artificial intelligence systems may violate ethical and social issues related to cybersecurity, such as the privacy of private information, transparency, and reliability of data.

• Justice, human dignity, etc.: These issues can affect and/or violate the safety and well-being of individuals and society.

The research addresses these vulnerabilities by proposing a bifurcated framework where machine learning detection and Large Language Model interpretation operate as a single, cohesive unit.

AI-based programs differ from classical programs since they integrate the large language models (LLMs) and their key components, such as training data, models, policies, questions, and actions. While the functional components of classical applications (interface, backend, infrastructure, database, and standard functions) remain an integral part of AI-based applications, they will be modified.

1. Architecture of AI-based software

1.1. Background

Noever (2023) carried out an in-depth comparative analysis to evaluate the performance of state-of-the-art artificial intelligence-based systems, such as machine learning and deep learning algorithms, in comparison with a range of traditional static analysers. Including data flow analysis and abstract interpretation, for the purpose of detecting and rectifying various types of software vulnerabilities such as buffer overflows, SQL injection, and cross-site scripting. GPT4, in particular, has proven to be very effective, especially in terms of its capability to provide detailed explanations of vulnerabilities in code and to offer effective solutions to patch these vulnerabilities. The success of prediction can be significantly increased by integrating advanced natural language processing models with traditional static code analysis techniques. This integration allows for the thorough identification and elimination of security vulnerabilities within software systems, thereby improving the overall reliability and security of the predictive models. By leveraging the strengths of both paradigms, organizations can effectively fortify their systems against potential security threats and ensure the robustness of their predictive capabilities.

Koch (2023) demonstrated that GPT-3 outperformed a state-of-the-art static code analyzer in predicting security vulnerabilities. This highlights the potential of advanced language models such as GPT-3 in the cybersecurity and code analysis fields.

Bakhshandeh et al. (2023) found that the utilization of artificial intelligence significantly enhanced the capability of developers to identify and localize vulnerable code. This was particularly evident when artificial intelligence was combined with static analyzers, resulting in more accurate and efficient detection of vulnerabilities within the codebase.

Kalinin (2024) concluded that the rapid pace of global digital transformation significantly impacts the economic security of enterprises. The high level of openness exhibited by companies is identified as a contributing factor to the emergence of diverse threats and risks that directly affect their operational activities (Demianchuk et al., 2021; Lelyk et al., 2022; Kaminsky et al., 2023).

Therefore, it is important to study the problems associated with the risks of digital transformation and AI development (Popova et al., 2024; Popova et al., 2023; Popova & Petrova, 2024; Tairov et al., 2024; Popov et al., 2024).

1.2. Benchmarking the Circuit-Based and Artificial Intelligence Architecture

Software applications based on schemes (circuit-based) have a clear and rigid structure. Inputs and outputs must be explicitly established, routed, and maintained. Any deviation from this structure leads to errors, and adding new features requires a linear effort on the part of the organization's developers.

The new software will be based on intelligence. Such programs are based on natural language, in which requests are sent to a system that actually understands what the person is asking. Adding new features to such a program would be as simple as asking different questions and/or providing different commands. Comprehension-based applications adapt to the questions that are asked of them. To add new functionality, you ask different questions and ask for different actions on the results.

The new AI-based software architecture is expected to be much more elegant and based on AI systems, including a four-component structure: state, policy, question, and action. In essence, this is a transition from circuit-based application architecture to an intelligence-based architecture (Table 1).

Table 1
Circuit-Based Architecture vs Artificial Intelligence Architecture

Structure elements	Circuit-Based	AI-Based
input	Rigid – programme runs must be explicitly coded by the development team using explicit queries against a structured database.	Unlimited – programs run using natural language, so there is no need to explicitly code new requests to the server when other questions need to be asked.
data	Structured – data repositories have well-defined schemas that require specific ways of adding, retrieving, changing, and deleting data.	Dynamics – system's backend is a set of models that understand the world and your business. To upgrade the data repositories, get bigger/better data and save your models.
output	Stativity – program output returns the specific information requested by the specific query. To do something else, one needs to write separate code for it.	Flexibility – the output of the program is as flexible as the questions asked of the models. And questions and answers can be directly related to actions.
action	Isolation – actions are separated from query results. For example, after querying a database for housing prices, one or more additional steps must be taken to make them useful.	Coherence – actions can be linked to an outcome because models understand how they are related. This means that there is no need to create explicit bridges from each result to new actions.
understanding	Missing – the program does not understand the data, the queries being made against it, or the result it returned. It is a simple mechanism for storing and retrieving data.	Deep – models understand the data they are dealing with, which means that improving the models simultaneously improves most aspects of the application.
flexibility	Linear - to add more features to a program, you need to explicitly expand the number of encoded inputs, the number of output operations, the data structure, or all together.	Explicit – the main concepts: 1) ensuring the quality and freshness of the models; 2) ensuring the correctness of the questions.

1.3. Attacks on Artificial Intelligence Systems

In LangChain, agents are intelligent entities designed to achieve specific objectives. They are equipped with a variety of tools and resources that enable them to carry out their tasks effectively. Each agent operates with a clear purpose, utilizing its capabilities to navigate challenges and implement solutions efficiently.

A coordinated attack on the agents would have serious repercussions for entrepreneurs, potentially disrupting their operations and hindering business growth. For AI agents to be effective, they must be empowered with two key capabilities (Bondarenko et al., 2022):

1) They should possess extensive knowledge about the relevant objects, including highly personal and sensitive information.

2) They need the ability to act like the user in various tasks, such as sending money, posting on social networks, writing letters, and sending messages. An attacker who gains this knowledge and access will hence gain significant leverage over the target (Figure 1).

Tools refer to the resources available to Agents for performing their tasks. These tools can include web search capabilities, document scanning and analysis software, plagiarism detection systems, and other similar resources.

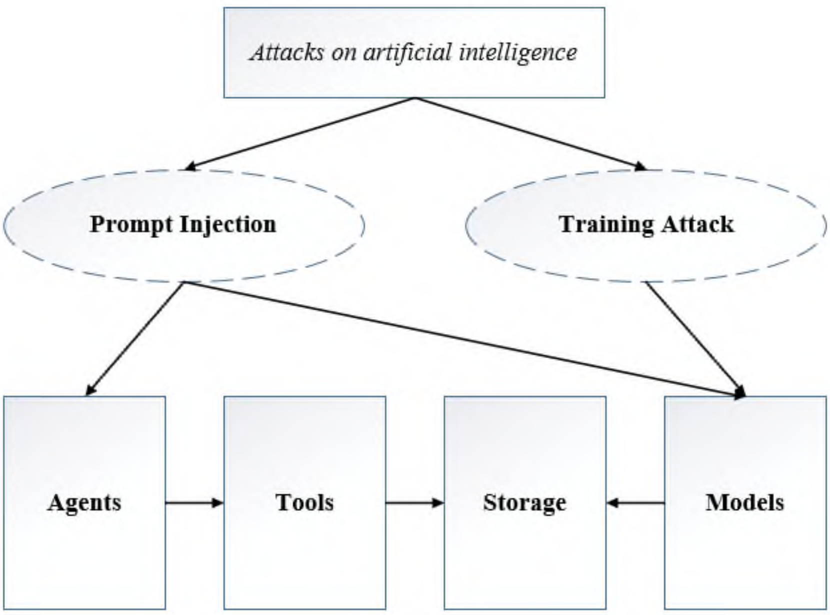

Figure 1. Attacks on artificial intelligence systems

Models are essential components of AI security, serving as the foundational frameworks that enable effective machine learning applications. In recent years, numerous business applications have adopted these advanced techniques, enhancing their operational efficiency and decision-making processes. By leveraging sophisticated algorithms and data analysis, organizations can better detect potential security threats, optimize resources, and improve overall performance.

Attacks on artificial intelligence systems are designed to manipulate models in such a way that they begin to exhibit undesirable behaviors. The goal is to undermine their reliability, making them less trustworthy in their outputs. Additionally, these attacks can result in models generating content that is more toxic and biased, further distorting their intended purpose and functionality.

Table 2
Basic types of attacks on artificial intelligence systems

Components of AI systems	Attack types
Agents	• alter the agent’s routing • send commands to unknownsystems
Tools	• execute arbitrary commands • go through injection on connected tool systems • code execution in the agent's system
Data repositories	• attack embedded databases • extract confidential data • change the data, which leads to fake results of the models
Models	• bypassing the protection model • make the model exhibit bias • extraction of data of other users and/or server data • forcing the model to exhibit intolerant behavior • poison the results of other users • violate the trust/reliability of the model • access to unpublished models

The inherent redundancy of the Dual-Layer Hybrid Security Framework addresses these vulnerabilities by ensuring that Layer 1 detection results are validated against Layer 2 policy and context, thereby reducing the success rate of model manipulation and poisoned outputs.

AI implementation is associated with several challenges, including data preparation, model building, scalability, and ethical issues. Companies need a systematic methodology to leverage several core business and technology capabilities to implement AI and machine learning technologies in the enterprise.

The traditional approach to automating the detection of malicious code involves using static analysis tools such as SonarQube, CodeQL, Snyk, and Coverity. These tools can be used at any time but are typically used to scan code for vulnerabilities after the code freeze phase and before the formal release processes are complete. They work by parsing the source code into an abstract syntax tree or control flow graph that shows how the code is organized and how all the components (classes, functions, variables, etc.) are related to each other. Rule-based analysis and pattern matching are then used to identify a wide range of problems.

Static analysis tools can be integrated with IDEs and CICD systems throughout the development cycle, and many offer customizable configuration, query, and reporting options.

Static code analysis takes input and asks two things: What's wrong? How to fix it? Analyse this approach from the point of view of software transformation under the influence of artificial intelligence, presented in Figure 2.

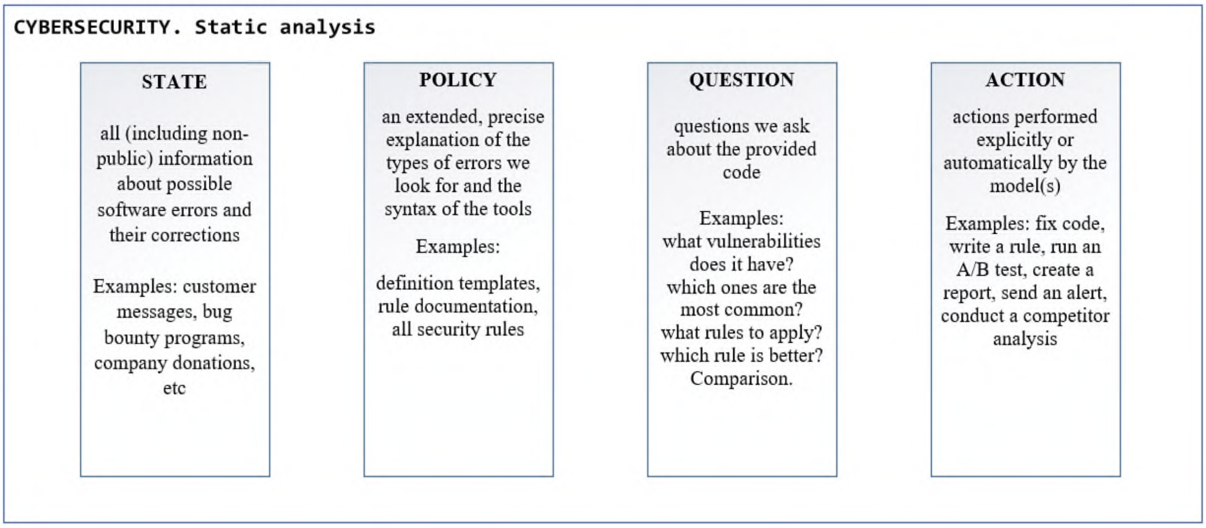

Figure 2. Static analysis in LLM-based applications

Artificial intelligence will transform all existing software into one that is based on understanding. Once it is sufficiently exposed to the problem through one's "state" and understands what one is trying to do through one's "policy", it can do much more than just find code problems and fix them.

It will be able to perform the following actions: find the problem; show how to fix it in any language (coding or human); write an instruction that will avoid these mistakes; write a rule in the tool's technology that would detect this; will provide you with a fixed code; and make sure the code works. This cognitive ability is achieved through the integration of the dual-layer approach, which enables the system to interpret errors within a specific organizational context rather than merely identify them.

In addition, it is possible to build multiple versions of a cyber defence system to see how each version will respond to the most common attacks and then make recommendations based on those results.

2. Building a Security System

2.1. Algorithm for Building a Security System Using LLM Models

The proposed Dual-Layer Hybrid Security Framework operationalizes the transition from circuit-based to intelligence-based architectures. This framework bifurcates security operations into two specialized tiers. Layer 1 serves as the detection engine, utilizing a Random Forest algorithm to provide high-speed, real-time anomaly detection within network traffic. Layer 2 acts as the interpretation engine, where a Large Language Model ingests the alerts from the primary layer to generate human-readable incident summaries, map threats to specific security policies, and provide actionable remediation steps.

1. Choose a base model. Start with the latest general GPT model from OpenAI or alternatives like Gemini from Google and Meta. Many companies are anticipated to implement advanced language models, which may be designated as OpenAI GPT-4. It has an impressive breadth of knowledge across various fields, including security protocols, biotechnological advancements, project management strategies, effective planning, meeting facilitation, budget management, incident response tactics, and audit preparedness. This expertise positions it as a crucial resource for companies looking to navigate complex challenges and ensure operational efficiency. However, to fully leverage its capabilities, organizations need to provide a more personalized context that aligns with their specific industry needs and operational goals. This tailored approach will allow companies to maximize the potential benefits of GPT-4's insights and recommendations.

2. Develop a custom machine learning model that leverages company-specific data relevant to GPT-4 queries. This data set should encompass a variety of sources, including the status section, telemetry information, company context details, firewall logs, financial documents, chat conversations, email communications, and transcripts from meetings.

Make sure to thoroughly train the model on this diverse dataset to enhance its accuracy and relevance.

3. Train your policy model. A separate model is meticulously crafted to ensure alignment with your company's mission and objectives. This model takes into account the challenges you face and integrates your strategic business priorities (Klochan et al., 2021), creating a cohesive framework that supports your organization's vision and enhances overall effectiveness. We use the guidance we get from people to drive part of the action architecture. Artificial intelligence will do this with the help of rules formed in the policy model.

4. Instruct the system to execute the following action: the models have now been merged. We have integrated GPT-4 with both our state model and policy model. This powerful combination allows them to analyze and interpret our behavior and preferences in ways that often surpass our own understanding. Together, they create a comprehensive picture of who we are, delving into the nuances of our thoughts and actions with remarkable insight.

2.2. Model task

We are developing a comprehensive software application designed for network traffic analysis and anomaly detection, leveraging advanced artificial intelligence techniques. The following software application demonstrates the practical execution of Layer 1 within the proposed framework, focusing on the technical requirements for robust, automated detection of suspicious traffic patterns. It will utilize synthetic data to train the model, ensuring it can accurately identify unusual patterns and potential threats within network traffic. By simulating various traffic scenarios, we aim to enhance the model's robustness and effectiveness in real-time monitoring and security assessment. This project will provide critical insights into network behavior, helping organizations proactively address anomalies and safeguard their digital infrastructure. All information transmitted over the Internet is divided into smaller units known as packets. Each packet contains a specific set of data that is addressed to a designated destination, much like how physical mail is sent to a particular address.

1. Installing and importing libraries. The code is shown in Fig. 3:

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
import gradio as gr
import warnings
warnings.filterwarnings('ignore')
```

Figure 3. Linking libraries

2. Creating a training data set. A software module is being developed to create a data set that represents a given network. This module will feature several key functions, including:

- ‘packet_size’: This function will determine the size of individual packets as they travel through the network, providing insights into traffic patterns.
- ‘duration’: This function will measure the time it takes for packets to traverse the network, helping to identify latency issues.
- ‘protocol_type’: This function will categorize the protocols used within the network, allowing for analysis of traffic types and their respective behaviors.
- ‘number_of_packets’: This function will count the total packets transmitted, enabling an understanding of network load and performance.
- ‘port_number’: This function will identify the specific ports in use, facilitating the analysis of communication endpoints and potential vulnerabilities.

Together, these functions will create a detailed representation of network activity, supporting performance analysis and optimization efforts. The values for these functions will be randomly generated using the NumPy library. The target variable ‘traffic_type’ indicates "normal" or "suspicious" traffic. The goal is to train a model for smart data analysis. The code is shown in Fig. 4:

```python
# Simulated dataset
np.random.seed(0)
n = 100
features = ['packet_size', 'duration', 'protocol_type', 'num_packets', 'port_number']
target = 'traffic_type'
# Protocol
protocols = ['TCP', 'UDP', 'ICMP']
# Generate random data
df = pd.DataFrame({
    'packet_size': np.random.randint(100, 1500, n),
    'duration': np.random.uniform(0, 120, n),
    'protocol_type': np.random.choice(protocols, n),
    'num_packets': np.random.randint(1, 100, n),
    'port_number': np.random.randint(1, 65535, n),
    'traffic_type': np.random.choice(['normal', 'suspicious'], n)
})
```

Figure 4. Creating a training data set

3. Creating a model. The code converts the DataFrame column named "protocol_type" into a numeric format, then splits the data into separate sets of functions and targets, and subsets for training and testing. Based on the input training data, the RandomForestClassifier model is trained. The code defines a prediction function that takes network traffic characteristics as input and converts them into the expected format and uses an already trained model to predict and determine the type of traffic ("normal" or "suspicious"), as shown in Fig. 5.

```python
# Encoding data
df['protocol_type'] = df['protocol_type'].astype('category').cat.codes

# Split the data
X = df[features]
y = df[target]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train a simple model
model = RandomForestClassifier()
model.fit(X_train, y_train)

# Define the prediction function
def predict(packet_size, duration, protocol_type, num_packets, port_number):
    input_data = pd.DataFrame({
        'packet_size': [packet_size],
        'duration': [duration],
        'protocol_type': [protocols.index(protocol_type)],
        'num_packets': [num_packets],
        'port_number': [port_number]
    })
    prediction = model.predict(input_data)[0]
    return prediction
```

Figure 5. Data transformation and traffic forecasting

We anticipate that the proposed software application will be effective for monitoring and diagnosing network issues, and it may be enhanced with a graphical interface in the future.

Conclusions

Incorporating artificial intelligence (AI) alongside large language models (LLM) demands a comprehensive analysis of the system's architecture, as well as the implementation of robust security measures to mitigate potential challenges. It may also be worth exploring the concept of LLM pen-testing, where security professionals test applications to discover vulnerabilities against their LLM models.

Integrating AI-powered applications into cybersecurity is a significant shift that has the potential to revolutionize the industry in the foreseeable future. By leveraging AI, cybersecurity systems can better detect and respond to advanced threats, predict potential vulnerabilities, and adapt to evolving attack techniques. This integration also enables the automation of routine tasks, allowing cybersecurity professionals to focus on more complex and strategic initiatives.

Using the Dual-Layer Hybrid Security Framework, SMEs can achieve robust cybersecurity by combining the efficiency of high-speed statistical models with the contextual depth of Large Language Models, thereby enhancing their overall security posture and staying ahead of sophisticated cyber threats. It becomes obvious that AI-powered applications offer ample opportunities for threat detection, risk assessment, and general intelligence enhancement. However, they also have potential cybersecurity issues that need to be addressed.

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CONTENTS

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DIGITAL TRANSFORMATION OF REPORTING-RELATED PROCESSES IN BULGARIAN ENTERPRISES (CONCEPTS AND REALITIES)

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Marin Marinov[3],
Galina Chipriyanova[4],
Ralitsa Dancheva[5]

Abstract: The study presents the use of digital technologies to improve the accuracy, efficiency, and accessibility in the context of the digital transformation of reporting-related processes in enterprises in the Republic of Bulgaria. This transformation includes automating data entry, using advanced analytics, and leveraging artificial intelligence-driven tools to optimize workflows. This changes both the way enterprises collect and analyze data and the way they report, which in turn leads to improved decision-making. The aim of the study is to examine the current state of the use of digital technologies in reporting activities of Bulgarian enterprises and the attitudes towards the implementation of new technologies in their reporting activities. The methods of analysis and synthesis, scientific observation, induction, deduction, structured interview, Crosstabs, SWOT Analysis, etc. were used in conducting the study. The strengths, weaknesses, opportunities and threats of digital transformation of processes related to reporting are highlighted. Based on the analysis of the results of structured interviews, the problems in the implementation and use of digital technologies in reporting activities and the attitudes of small, medium and large enterprises in Bulgaria are outlined.

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Introduction

Digital transformation is a recognized mechanism for economic growth due to the ability of digital technologies to positively influence the quality of economic activities, efficiency and costs. The development of digital technology has had a significant impact on various areas of human activity, and accountability is no exception to this. They are treated not only as a modern trend, but also as a factor to ensure high competitiveness of enterprises. The main objective of the study is to examine the current state of the use of digital technologies in the reporting activities of Bulgarian enterprises and the attitudes regarding the implementation of new technologies in their activities. Tasks to achieve the objective are: 1) to provide a critical analysis of digital transformation of processes related to reporting; 2) to argue the need for digital transformation of processes related to reporting; 3) to identify the key digital technologies; 4) to identify the strengths, weaknesses, problems and challenges of the use of digital technologies in the reporting activity; 5) to determine the current status of the use of digital technologies in the accounting activities of Bulgarian enterprises and to outline the directions for future development. This study argues that the use of digital technologies is changing the way information is collected, stored, analyzed, disseminated, used, etc., and this is reflected in the formation/imposition of new ways/directions for organizing reporting activities. The existence of various barriers prevents the realization of the potential of digital technologies in the processes related to the reporting of Bulgarian enterprises and improving their competitiveness. The object of the study is small, medium-sized and large enterprises in the Republic of Bulgaria, and the subject – digital transformation of the processes related to reporting.

Materials and methods

To achieve the research tasks, a practical study has been conducted through the methodology of structured interview among small, medium-sized and large enterprises in Bulgaria. The sample was formed by the method of random non-recurrent selection (01 January 2023 - 31 December

2023). The sample includes enterprises that have their own accounting department. Enterprises with outsourcing reporting service are excluded. The formula for calculating the sample size is:

$$n = \frac{p(1-p)z^2N}{\mathrm{p}(1-p)z^2 + \Delta^2 N}$$

where: n – sample size; p – relative proportion of units possessing the studied characteristic; z – confidence coefficient

The number of non-financial enterprises with more than 9 employees according to the NSI data for 2023 is 30, 053. Therefore, for the survey results to be reliable, the number of enterprises in the sample should be at least 76. In the course of our research, we requested 87 enterprises, but 80 enterprises actually responded and participated in the structured interview.

Digital transformation of the economy requires a change of business approaches in order to keep enterprises competitive. Enterprises are increasingly using digital technologies in their activities and in this regard it is quite logical to explore the nature and possibilities of using digital technologies in reporting activities. Reporting in Bulgaria is going through several stages of information technology adoption – use of Excel spreadsheets; specialized software products and integration of accounting programs into Enterprise Resource Planning (ERP) system. In the following years, reporting has undergone even greater changes, catalysed by modern digital technologies. They offer great opportunities compared to traditional accounting. The research is also divided into three categories for diagnosing the economic, social and environmental sustainability of the respondents in accordance with the two parallel processes of digitalization and sustainable development. For the purpose of the research, businesses are classied as profit-minded, eco-minded and open-minded, because this will allow to link the use of digital technologies to the strategic orientation of the business. In this regard, we define the classification criteria (enterprise profile) as follows: 1/ Profit-minded. Enterprises that are primarily focused on financial efficiency and profit. They invest in modern digital technologies only if they bring direct economic benefits. 2/ Eco-minded. Businesses that focus on sustainability, environmental initiatives and corporate social responsibility. They view digitalization as a way to optimize resources and reduce their carbon footprint. 3/ Open-minded. Innovative enterprises open to technological change, experimentation and new business models. They are expected to be most willing to adopt cloud technologies, AI and big data analytics. The methods of analysis and synthesis, scientific observation, induction, deduction, structured interview, Crosstab, SWOT Analysis, etc. to identify the distinguishing characteristics of the enterprises to the highest possible extent were also used.

1. Literature review

Based on a critical analysis of existing research in the works of M. K. Cenzarik, et al. (Cenzarik et al., 2020), J. Bumann and M. Peter, Viacheslav Saprykin digitization is defined as the transfer of information from analog to digital form, its transfer from physical (paper) media to digital. From the point of view by L. Chyzhevska et al. (Chyzhevska, 2021), digitalization is related to the use, along with traditional IT tools, of new technologies that have been widespread not so long ago: artificial intelligence, robotics, cloud technologies, blockchain.

Digital transformation is the complete transformation of an enterprise's operations, business processes, competencies and business models, maximizing the full use of digital technologies to enhance competitiveness, create value and scale in the digital economy (Cenzarik et al., 2020). From the point of view by economists V. G. Halin and G. V. Chernova (Halin & Chernova, 2018), digital transformation is defined as a process of gradual change that begins with the introduction of digital technologies and then grows into a comprehensive transformation of all spheres. It is the adoption of new technologies to improve productivity, value creation and social welfare. The concept includes four main components without change, of which its implementation is impossible: introduction of new technologies; changes in value creation; structural and financial changes (Matt & Hess, 2015).

Digital transformation, as explained by H. Demirkan et al. (Demirkan, et al., 2016), represents a major transformation of business models and methods, and even of business processes to fully use the potential provided by digital technologies, also the impact on society in the long term. O. B. Lukina (Lukina et al., 2020) (Verhoef, et al., 2021) defines digital transformation not only as the implementation of digital technologies, but also as the transformation of multiple horizontal and vertical business processes, the optimization of operational processes, the modification of established models and formats of interaction of actors and the value chain.

Moses Alabi (Alabi, 2024) provides a more global view by defining digital transformation as the integration of digital technologies into all areas of business, fundamentally changing the way businesses operate and deliver value to their customers. It involves transforming business models, optimizing operations and driving cultural change across enterprises to leverage digital capabilities and adapt to a rapidly evolving business environment. A universal definition of digital transformation is defined by Gregory Vial (Vial, 2019), describing it as a process aimed at improving an object by initiating significant changes in its properties through a combination of information, computer, communication and network technologies. Modern digital technologies used in reporting range from cloud technology, blockchain technology, technology for contactless identification of accounting information (barcoding, etc.), big data, advanced analytics, artificial intelligence and machine learning, mobile applications, the Internet of Things and others to the introduction of innovations such as platform-based products facilitating interactions and transactions between different user groups (Lazarova, 2019).

Based on the above, we consider the digital transformation of processes related to reporting to be the strategic integration of digital technologies into all aspects of enterprise's accounting and financial activities. It is not limited to the automation of individual operations, but includes the optimization of business processes, changes in established models of interaction between internal and external participants, and a cultural transformation of the enterprise, with the aim of more efficient generation, processing, and provision of reporting information. In this way, the digital transformation of reporting-related processes supports adaptation to the rapidly changing business environment, improves the quality and transparency of reporting, and creates added value for all stakeholders.

2. Results and discussion

2.1. SWOT analysis of the use of digital technologies in reporting activities

With the development of new technologies, the digital transformation of processes related to reporting has become an integral part of some businesses' operations. The introduction of digital technologies in reporting activities opens new horizons for reporting. Systematically assessing strengths, weaknesses, opportunities and threats allows businesses to better integrate digital technologies into reporting activities, maintaining a balance between traditional methods and the potential that digital technologies hold, and to make more informed and strategic decisions. The main guidelines for the digital transformation of reporting-related processes are:

2.1.1. Cloud technologies

Cloud computing is provided as the delivery of dynamically scalable and virtualized resources over the Internet in the form of services (Mummigatti, 2010). They involve the use of one of three service delivery models according to functional criteria: Infrastructure as a Service (IaaS); Platform as a Service (PaaS) and Software as a Service (SaaS) made available for use over the Internet. In the following lines, when describing cloud technologies in reporting, the SaaS model of applications and business tasks performed as a service is implied, which can be particularly useful as they provide off-the-shelf accounting software. As explained by A. P. Krulikowski et al. (Krulikowski & Korelev, 2019), cloud technologies provide the necessary infrastructure that can support all the accounting activities of an enterprise. Our research shows that cloud accounting platforms provide secure and scalable access to financial data from anywhere. These tools support real-time collaboration, automated updates, and seamless integration with other financial systems.

The advantages of using cloud technologies in reporting over traditional accounting software (used only locally) are: 1) cost reduction; 2) elimination of the need to purchase accounting software; 3) improves business efficiency – facilitates the generation of real-time reports, adapting to changing business conditions in a short time; 4) saves time; 5) saves maintenance costs (software updates, fixing various failures, e.g. on the server, etc.), which are entirely at the expense of the provider; 6) easy access and use – no need to install a program, only a device (computer, phone or tablet) and an internet connection, remote access, no need to access the local network; 7) unlimited data backup at any time; 8) remote maintenance is possible; 9) etc. (Krulikowski & Korelev, 2019). Complementing these benefits is the ability to analyse in real time; effortless document management – maintaining all primary documentation electronically and being able to access it at any time and from any location; scalable file storage – cloud technologies allow for expanded data storage space at significantly lower costs; increasing the accuracy of accounting operations (Berdichevskaya, 2023).

The disadvantages of using cloud technologies are as follows: 1) the possibility of data leakage; 2) the possibility of attacks; 3) the problem of choosing a reliable provider; 4) in the event of a denial of service, there is no guarantee that the data will be destroyed; 5) technological risks - internet outages, lack of stable connection for enterprises located in smaller settlements; 6) the vulnerability of the server (Astakhova & Kochetova, 2015; Hasan & Petrova, 2023; Berdichevskaya, 2023). The lack of a stable internet connection is a relatively small percentage, as in our country the enterprises with a fixed internet connection for 2024 are 89.2%, as 54.1% of the maximum download speed data (Institute, 2024).

2.1.2. Blockchain

Blockchain technology according to A. V. Basova (Basova, 2022) can become both the basis of accounting operations and a system of internal control of business entities. The positive aspects of using blockchain technology in reporting, she points out, are: the reliability of information, as no changes can be made to the blocks; the transparency of information and thus increasing its value, due to the participation of a large number of users and control activities; the efficiency of information processing, as all participants in the system have equal rights; can accelerate the development of the industry by reducing the costs associated with the maintenance and consistency of accounting data. The risks associated with blockchain technology in reporting are the risk of system failures, information may be partially lost which may affect the logical consistency and completeness of information; the biggest risk is the lack of a regulatory framework for the use of blockchain technology.

Blockchain technologies open up great opportunities (Azimov & Petrova, 2022). Our research shows that it provides secure and immutable records of financial transactions. It improves transparency and trust while simplifying audit trails and compliance processes. Accountants are experts in the field of accountability and blockchain technology can effectively help them to archive documents in a defined compliant system that auditors and authorized third parties can access safely and securely (Mirsodikova, 2023). The main advantages of blockchain in reporting are: 1) openness of transactions – the user of the network is able to freely trace the history of transactions on the site; 2) the presence of a digital record – makes it impossible to falsify, change and delete data; 3) each participant in the blockchain has a permanently up-to-date copy of the database; 4) high speed and accuracy of transactions, which provide speed and reliability of transactions; 5) decentralized communication between servers, which provides high security for users and operations (Mirsodikova, 2023).

2.1.3. Artificial intelligence and machine learning

Artificial intelligence (AI) and machine learning (ML) are used together in many applications. ML allows AI to learn by gathering information and putting the knowledge into practice, thus optical character set systems together with artificial intelligence generate information about invoices, etc. (Dydina & Kondakova, 2024). AI has increased the efficiency and accuracy of accounting work by: automating routine tasks – thus reducing the risk of human error, providing accountants with time for analysis and strategic decisions; analysing financial information – AI can be programmed to search and process financial information from a variety of sources; building models and forecasts – can be used to create and forecast based on historical information and indicate different opportunities; improving communication - can be used for training. (SV Smart Consulting) The application of artificial intelligence in reporting also allows the processing and analysis of large amounts of accounting information, verification of accounting data, error detection and fraud prevention, increasing the efficiency of the accounting process. The risks are related to the redundancy of a significant number of jobs, complexity of implementation, lack of regulations, etc. (Pravdiuk & Pravdiuk, 2024). Based on the above, we consider that accounting technologies powered by artificial intelligence enhance accuracy by detecting anomalies, automating reconciliations, and forecasting financial trends. These capabilities are transforming the way businesses approach tasks such as financial forecasting and compliance checks.

2.1.4. Big Data

In modern conditions, intelligent decisions are based not only on historical data but also on the results gathered from big data analysis and the information or trends that such analyses indicate about the forthcoming future (Rijal, 2023). Using big data in reporting helps to optimize business processes, generate predictive analytics and manage business risks, as well as helping to identify inefficient processes and optimize activities. Using these technologies requires significant investment and data analytics skills (Dydina & Kondakova, 2024) (Petrova & Popova, 2022; Petrova et al., 2022; Petrova & Popova, 2024; Popova et al., 2023; Popova et al., 2024).

Based on the above information, a SWOT analysis of the use of digital technologies in reporting was conducted (Table 1).

Table 1
SWOT analysis of the use of digital technologies in reporting

INTERNAL FACTORS	
STRENGTHS	WEAKNESSES
- improving the efficiency and accuracy of accounting information; - enhancing analytical capabilities through deeper and more multidimensional analysis and presentation (through visualisations) of valuable information for management decision-making; - the ability for employees at different levels to work remotely and access from anywhere with Internet; - more effective interaction between departments of the enterprise, which significantly improves the quality of reporting activities; - increased speed of data and information processing; - no need to install additional software as updates are performed automatically.	- the need for investment to adapt and expand employees' digital skills, knowledge and capabilities, including understanding the basics of information security, etc; - the need to establish security systems; - significant implementation costs.
EXTERNAL FACTORS	
OPPORTUNITIES	THREATS
- more effective communication between stakeholders; - optimization of document flow in the overall digital exchange of documents between enterprises, contractors, state administration, banks, NRA, NSSI, Trade Register, etc; - increased competitiveness; - Our country's commitment, through the Ministry of Innovation and Growth, to provide grants by the end of 2024 for the implementation of digital technologies in small and medium-sized enterprises.	- risks of cyber-attacks and leaks of confidential information; - the possibility of job losses related to routine reporting tasks; - mobile network or internet connection failure; - ethical issues from the use of artificial intelligence and big data; - continuous emergence of new digital technologies; - Bulgarian low level of digital maturity.

Source: Authors' own interpretation.

Synthesizing the above, it can be concluded that the digital transformation of reporting-related processes undoubtedly has enormous potential to improve reporting activities, but its implementation must be carried out carefully, regarding all possible threats.

2.2. Digital transformation of reporting related processes in Bulgarian enterprises – level of usage, views and strategic directions

This section presents the results of the research related to the level of usage of digital technologies in reporting activities, the attitude of businesses towards digital transformation and the level of awareness of the benefits and challenges they provide. It also analyses the readiness of enterprises to adopt new technologies, the main barriers to this process, and the impact of business strategy and policy on the speed of digital transformation. In regard to the current status and extent of use of digital technologies in reporting activities, it should be noted that Bulgarian enterprises do not make full use of digital tools and thus do not benefit from the opportunities they offer.

Table 2

Degree of use of digital technologies in reporting activities

Using digital technologies in reporting activities	Categories of businesses						Total	
	Employees 10-49		Employees 50-249		Employees more than 250			
	Count	%	Count	%	Count	%	Count	%
Cloud technologies	2	3.13	3	23.08	3	100	8	10
Artificial intelligence	1	1.56	1	7.7	3	100	5	6.25
Big data	-	-	1	7.7	3	100	4	5
Blockchain technologies	-	-	-	-	-	-	-	-

Source: Authors' own interpretation.

The survey found that the digital technologies used in reporting are mainly cloud technologies, big data, artificial intelligence. All the surveyed enterprises use specialized accounting software. The proportion of respondents using cloud accounting solutions, artificial intelligence and big data analytics was generally low - about one tenth. All enterprises with more than 250 employees use digital technologies in their accounting activities.

Blockchain technologies are not recognised by respondents. Also, none of the surveyed enterprises use the new cloud-based platform for finance and accounting management in our market. This is an innovative service on our market, which is well known in developed countries, where there are many providers of the service, while currently on the Bulgarian market the service is offered by only one fintech company.

Further, our task is to analyze how the extent of use of cloud technologies, AI, big data and blockchain matches the enterprise profile. For example, if an enterprise only uses AI for financial analysis, it is more likely to fall into profit-minded. If it uses cloud technologies with an ESG focus, it is more likely to be eco-minded. Eighty enterprises from the sample are grouped, and the most pronounced profile in their use of digital technologies is tracked. It will be interesting to see which profile predominates, if there is a group that does not use any technology.

Based on the data about the use of digital technologies, we can derive the profile of enterprises by focusing on:

1. Classification criteria – using the degree of digital transformation of processes related to reporting, we classify enterprises into three groups of profiles: a) Profit-minded. The enterprise only uses AI, therefore focuses on cost optimization, efficiency and profit. b) Eco-minded. The enterprise uses cloud technologies, including for environmental sustainability (e.g. to optimize resources). c) Open-minded. The enterprise uses a combination of technologies, so it is oriented towards innovation and future trends.

2. Enterprise clustering. From 76 enterprises, only 8 use some of the technologies. Blockchain is not used by any enterprise. Their distribution by enterprise size: 2 (10-49 employees), 3 (50-249 employees), 3 (250+ employees). Based on this, we can draw the following conclusion: 1/ Enterprises that make limited use of the modern digital technologies, discussed above, fall into the profit-minded profile (conservative enterprises, minimizing their costs). 2/ Enterprises that use cloud technologies are eco-minded. 3/ Enterprises that use AI, cloud technologies and big data analytics in combination are open-minded.

Analysis in such a cross-section reveals the relationship between business category and enterprise profile. We are able to trace whether the size of the enterprise influences its belonging to one of the profiles and to what extent. Furthermore, to what extent the lack of blockchain confirms that enterprises in Bulgaria are not yet ready for high-tech transformations.

The degree of digital transformation of reporting-related processes is not a matter of technology, but of organisational culture and strategic thinking. Profit-minded enterprises focus on short-term efficiency, eco-minded balance sustainability and digitalization, and open-minded ones rely on modern technology as a means of strategic leadership.

Table 3
Attitude of the business towards the digital transformation of processes related to reporting

The attitude of the business towards the digital transformation of reporting-related processes	Categories of businesses						Total	
	Employees 10-49		Employees 50-249		Employees more than 250			
	Count	%	Count	%	Count	%	Count	%
High level of interest	54	84.38	12	92.31	3	100	69	86.25
Low level of interest	5	7.81	1	7.69	-	-	6	7.5
Lack of interest	5	7.81	-	-	-	-	5	6.25
Total	64	100	13	100	3	100	80	100

Source: Authors' own interpretation.

The attitude of the majority of enterprises towards the digital transformation of reporting-related processe is positive, with over 85% of respondents showing a strong interest in them. Our research shows that some of the factors driving the digital transformation of reporting processes and leading to increased interest are: 1/ Regulatory complexity – the growing complexity of financial reporting regulations, including various documents, requires accurate, timely, and transparent reporting. Modern digital tools streamline compliance processes, reduce errors, and ensure compliance with standards; 2/ Stakeholder expectations – investors and stakeholders require access to real-time financial data. Digital accounting solutions provide immediate information on financial results, enabling better decision-making; 3/ Competitive pressure – to stay ahead, businesses must embrace digital technologies to improve efficiency, reduce costs, and improve the quality of their financial reporting; 4/ Data explosion – the volume of financial data continues to grow, making manual accounting processes untenable. Big data tools enable companies to manage, analyze, and efficiently use vast amounts of financial data.

Table 4
Level of business awareness of the strengths, weaknesses, opportunities and threats of using digital technologies in reporting

Level of business awareness of the strengths, weaknesses, opportunities and threats of using digital technologies in reporting	Categories of businesses						Total	
	Employees 10-49		Employees 50-249		Employees more than 250			
	Count	%	Count	%	Count	%	Count	%
High level of awareness	34	53.13	9	69.23	3	100	46	57.5
Medium level of awareness	20	31.25	3	23.08	-	-	23	28.75
Low leverl of awareness	5	7.81	1	7.69	-	-	6	7.5
Lack of awareness	5	7.81	-	-	-	-	5	6.25
Total	64	100	13	100	3	100	80	100

Source: Authors' own interpretation.

More than half of the enterprises interviewed have a high level of awareness of the strengths, weaknesses, opportunities and threats of using digital technology in accounting. Their systematic assessment allows enterprises to better integrate digital technologies into accounting activities, maintaining a balance between traditional methods and the potential that digital technologies have, and to make more informed and strategic decisions. Our research shows that awareness of digital technology varies depending on the scale and business orientation of enterprises. Open-minded enterprises are fully informed and ready for digitalization, while eco-minded ones balance between traditional and new practices. Profit-minded enterprises are aware of the need for digitalization, but often lack the resources for rapid implementation. These trends define different approaches to innovation and competitiveness in the future (see table 5).

Table 5
Business strategy, policy and vision according to enterprise profile

Enterprise profile	Awareness	Business strategy	Policy	Vision
Profit-minded	Diverging awareness	Flexible adoption of digital technologies	Testing and adaptation of innovative accounting models	Integration of new technologies through experimentation
Eco-minded	Adequate but not maximum awareness	Balance between digitalization and sustainability	Using cloud solutions for environmental reporting	Compliance with ESG regulations
Open-minded	High level of awareness	Automation to reduce costs and increase efficiency	Investment in ERP systems, AI and blockchain	High technology competitiveness

Source: Authors' own interpretation.

The majority of the surveyed enterprises are not ready for digital transformation of reporting-related processes (see table 6).

Table 6
Degree of readiness to implement new technologies in reporting activities

Degree of readiness to implement new technologies in reporting activities	Categories of businesses						Total	
	Employees 10-49		Employees 50-249		Employees more than 250			
	Count	%	Count	%	Count	%	Count	%
High level of readiness	2	3.13	3	23.08	3	100	8	10
Medium level of readiness	12	18.75	5	38.46	-	-	17	21.25
Low level of readiness	8	12.5	2	15.38	-	-	10	12.5
Lack of readiness	42	65.62	3	23.08	-	-	45	56.25
Total	64	100	13	100	3	100	80	100

Source: Authors' own interpretation.

The research in accordance with the enterprise profile shows:

First, among Profit-minded enterprises, only 2 enterprises have high readiness for digital transformation. The majority (42) have no readiness at all, indicating low investment capacity and possibly a lack of skilled staff for it. Although 12 enterprises have medium readiness. The strategy of this profile of enterprises is reflected in no major investment in modern technologies with quick returns, flexibility and adaptation to competition, and their policy on operational efficiency and regulatory compliance rather than innovation. In terms of leadership, there is a conservative management style, avoiding high-risk investments. In regard to competitiveness – it is low, these enterprises are vulnerable to larger competitors. Their mission is process optimization, market sustainability.

Second, Eco-minded enterprises are striving to balance efficiency, sustainability and technological modernisation. Three enterprises have high readiness for digitalization; five of them have medium readiness, suggesting a gradual transition to new technologies; two enterprises have low readiness and three are not ready, indicating some resistance to change. The strategy of this profile of enterprises is to implement digital solutions with sustainability as a strategic factor (e.g. ESG reporting standards, carbon reporting) and their policy is a regulation and standards oriented approach to maintain competitive advantage. In terms of leadership – a collective management style is observed, with a focus on sustainability and stakeholder engagement. Regarding their competitiveness – it is moderate, indicating their ability to adapt but dependence on the market environment and regulations. The mission of this profile of enterprises – long-term stability, sustainability and social responsibility.

Third, Open-minded enterprises are characterized by a focus on innovation, experimentation with new technologies, strategic leadership. All large enterprises show some degree of digital readiness – 3 at a high level, with no enterprises with low or no readiness. This means that these enterprises are the drivers of innovation, have the resources to experiment and are actively developing new digital models. Their strategy is leading-edge technologies such as AI, blockchain, automated accounting systems and data analytics, and their policy is to develop internal innovation programmes, R&D (Research and Development) investments and technology partnerships. This profile of enterprises is distinguished by a visionary approach, a high degree of experimentation, the implementation of new business models, and their competitiveness is high as they shape the market rather than adapt to it. Their mission is leadership in digital transformation, sustainable growth through innovation.

Or, open-minded enterprises are innovators and the main drivers of digital transformation. Eco-minded enterprises are balancing sustainability and digitalisation, but some are not yet fully ready for large-scale digital change. And profit-minded enterprises are the least prepared, with a significant proportion lacking the resources or motivation for digital transformation.

Table 7
The most frequently mentioned reasons for the slow rate of digital transformation of reporting-related processes

Reasons for the slow rate of digital transformation of processes related to reporting	**Categories of business**						**Total**	
	Employees 10-49		**Employees 50-249**		**Employees more than 250**			
	Count	**%**	**Count**	**%**	**Count**	**%**	**Count**	**%**
Lack of resources to invest in the digital transformation of reporting-related processes	43	67.19	5	38.46	-	-	48	60
Insufficient staff training	51	79.69	7	53.85	-	-	58	72.5
Need to invest in acquiring new knowledge, skills and competences	51	79.69	7	53.85	-	-	58	72.5
Confidentiality of reporting information	38	59.38	10	76.92	-	-	48	60
The vulnerability of companies to cyber attacks	52	81.25	11	84.62	1	33.33	64	80

Source: Authors' own interpretation.

One of the main reasons is the lack of resources invested in the digital transformation of reporting-related processes. Bulgaria, through the Ministry of Innovation and Growth, has provided grants for the implementation of digital technologies in SMEs by the end of 2024, as a result of which digital technologies are expected to be increasingly integrated into business. Another problem businesses face is the mismatch between the qualifications and skills of the labor force, in terms of both general skills and specific skills. This, in itself, raises an even more serious issue of improving the training, education and retraining of people in line with new market requirements. This requires cooperation between all stakeholders – the state, business and academic community. Inadequate preparation of the personnel in the enterprises, in itself, requires investments in trainings to acquire new knowledge, skills and competences. Regarding the other main reason given for the slow speed of digital transformation of processes related to reporting – the vulnerability of enterprises to cyber-attacks, in 2023 Bulgaria has opened the first Cyber Security Centre, which aims to improve the cyber security capability. This will provide the skills needed to identify, perceive and conduct defensive operations against cyber attacks.

The study in accordance with the enterprise profile shows:

- the barriers faced by Profit-minded enterprises are indicated as: a) *lack of resources to invest in the digital transformation of processes related to reporting*. Small enterprises have limited budgets and prioritize expenditures that bring direct and quick results. Investment in modern digital technology is often seen as a secondary cost. b) *insufficient staff training*. These enterprises often lack a dedicated IT department. What is the reason? Profit-minded enterprises survive through efficiency and pragmatism. They are cautious about taking financial risks, and the lack of guaranteed returns from digital transformation make them skeptical of rapid change.
- the main barriers for Eco-minded enterprises are: a) *the need to invest in acquiring new knowledge, skills and competences*. Medium-sized enterprises are aware of the need for digital transformation but often face challenges in transforming their staff. They understand that modern technology is inevitable but prefer a step-by-step and sustainable approach. b) *confidentiality of reporting information*. These enterprises are sensitive to the security of their data, especially if they are engaged in international regulations or sustainable business models that require accountability and transparency. What is the reason? Eco-minded enterprises balance innovation and sustainability. They are investing in new technologies, but with caution, to ensure their operations remain stable and meet environmental, social and governance (ESG) standards.
- barriers to open-minded enterprises include: a) *vulnerability of enterprises to cyber-attacks*. Large enterprises have the resources and vision to deploy modern digital technologies, but they are aware that large-scale transformations also carry serious risks. They place cyber security as a top priority. b) *privacy of reporting information*. Due to the complexity of their financial operations, open-minded enterprises are wary of digital

technologies, especially when it comes to cloud solutions and external software providers. What is the reason? Open-minded enterprises are innovative but not “reckless”. They carefully assess risks and opportunities, placing security and stability first.

Conclusions

Based on the data collected and the analysis, it can be concluded that the digital transformation of reporting-related processes in Bulgaria is still at an early stage, but is gradually gaining speed. Although enterprises are aware of the need for digital transformation, a number of factors prevent this process. *First*, digital transformation requires a comprehensive strategic vision and long-term planning, which are not yet sufficiently widespread in practice. This means that many enterprises have taken their first steps toward introducing automated processes, but the overall integration of digital solutions has yet to be developed more actively. *Second*, the main technology solutions used in reporting activities include ERP systems, cloud technologies and big data, but their share among enterprises remains low. Adoption of innovative solutions such as blockchain technologies is facing difficulties due to lack of knowledge and security concerns. *Third*, the main barriers to digital transformation include inadequate staff training, high startup costs for technology deployment, and cybersecurity and data privacy concerns. In addition, psychological barriers among staff also slow down the process of adapting to new technologies. Despite these challenges, there is a growing interest in the application of artificial intelligence in reporting activities. More and more enterprises are considering the possibility of using AI to automate routine tasks. This would allow accountants to focus on higher value-added activities such as analysis, interpretation, and strategic management support.

In conclusion, different profiles of enterprises perceive digital transformation through the prism of their strategic priorities – small (profit-minded) enterprises face difficulties due to lack of resources and trained staff, eco-minded ones seek a balance between investment and sustainability, and large and innovative (open-minded) enterprises are focused on long-term transformations. To accelerate the digital transformation of processes related to reporting, targeted investments in training, clear strategies and support from public policies are needed.

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CONTENTS

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DIGITAL DISTANCE AND TRADE DYNAMICS: THRESHOLD EFFECTS ON TURKIYE'S EXPORT PERFORMANCE

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Abstract: This study investigates whether digital distance—defined as disparities in digital and technological capabilities between countries—constitutes a critical determinant of Turkey's export performance in the evolving architecture of international trade. Moving beyond traditional gravity-based explanations centered on geographical distance, the article develops a multidimensional digital distance index derived from the Frontier Technological Readiness Index (FTRI) and the Economic Complexity Index (ECI). Using annual panel data for 108 trading partners over the period 2008–2022, the analysis integrates GDP and exchange-rate controls within a panel threshold regression framework, complemented by panel causality tests. The findings reveal a statistically significant single-threshold structure, demonstrating that digital distance exerts nonlinear and asymmetric effects on Turkey's exports. While convergence in ICT infrastructure and financial access tends to enhance export performance, narrowing gaps in certain dimensions—particularly skills and R&D—does not uniformly yield positive trade outcomes. In fact, larger disparities in selected technological capabilities may generate temporary competitive advantages, consistent with technology-gap theory. Robustness checks incorporating economic complexity distance confirm these threshold dynamics. Moreover, causality results indicate predominantly bidirectional relationships between exports and digital dimensions, underscoring mutual reinforcement between trade and digital transformation. Overall, the results challenge the presumption that digital

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convergence universally promotes trade. Instead, they highlight the strategic importance of selective digital specialization and innovation-driven competitiveness. By conceptualizing and empirically validating digital distance as a structural trade determinant, the study contributes to the emerging literature on digitalization and international trade and offers policy insights for adaptive, data-driven export strategies in an era of technological disruption.

Keywords: Digital distance, export performance, frontier technological readiness index (FTRI), economic complexity index (ECI), panel threshold regression.

JEL: F10, O32, O33

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Introduction

Recent advancements in trade, logistics, finance and the economy raise the question if the struggles or achievements in these fields are deriving from a geographical distance or are the result of a "digital distance". Historically, not only goods but also technologies are transferred through trade routes (Christian, 2000: 5). The effect of digitalization can be categorized as structural and competitional (Ahmedov, 2020).

Distance is also an obstacle to tackle in trade. Huang (2007) states that trade volumes with distant partners are not as high as the ones that are closer. The low volumes have been analyzed considering the effects of transportation costs (Obstfeld & Rogoff, 2000), obstacles in flow of capital, and informational barriers (Rangan & Lawrence, 1999). In a digitalized economy, discussions on the subject can be shifted towards the level of digitalization, in other words, how trade partners can be seen very close geographically but enormously distant digitally. To visualize this phenomena, Figure 1 illustrates the Differences from Turkey in Economic Complexity Index (ECI) by The Observatory of Economic Complexity and Frontier Technological Readiness Index (FTRI) by UN Trade and Development (UNCTAD).

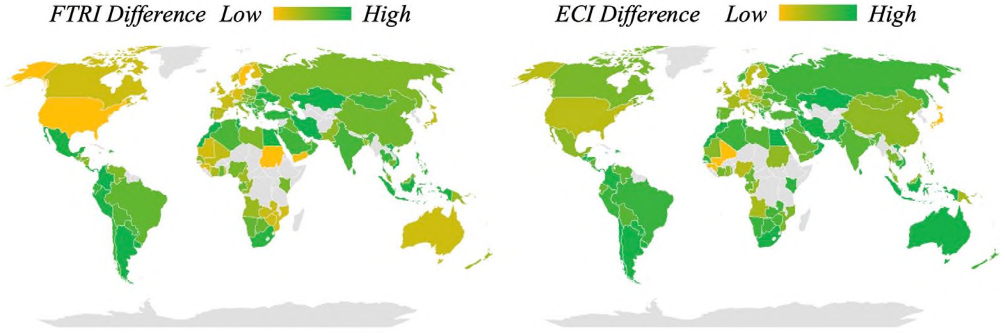

Figure 1. ECI and FTRI differences from Turkey

In Figure 1, those that are closest to Turkey are shown in green and the furthest ones are represented in yellow. The figure helps understanding how economically close countries can in fact be digitally distant. For instance, some Balkan countries seem to be not differing from Turkey at an economic development level. At the same time, the same countries change color to a more yellowish tone in terms of differences from Turkey in readiness to frontier technologies. The intuitive results correspond to the aim of this study in investigating the results of digital distance between Turkey and its trade partners and effects on international trade.

The new era of digitalization brings new challenges to international trade. In regard to studies such as Cermeno and Vazquez (2009), technological capabilities are linked to activities in R&D, human capital, openness and investment levels. Since these capabilities are highly related to bilateral trade flows as stated by Esfahani and Tayebi (2016), some certain digital distances may affect exports more than others both positively and negatively. Hence, this research contributes to the literature by analyzing how digital distance can be a major performance issue between trade partners. This research also aims to determine the impact of digital distance on Turkey's export dynamics, and reveal its implications on trade competitiveness and economic policies.

1. Literature Review

1.1. Historical View

The impact of technological differences on *trade* between economies, nations, firms and even sectors has been an important research topic for decades. As technological innovations have accelerated as a result of the Industry 4.0 revolution, recent publications have focused on *digitalization* in *trade* activities. While the subject was explained with the *technological gap* in the early 1980s, today we view the emergence of the concept of *digital distance* between countries. Not limited to these, many terms have been added to the literature over time, such as *technological level, technological distance, digital gap, digital divide and digitalization level.* Figure 2 illustrates how frequently these concepts have been used in publications over time. The dataset was extracted from Web of Science (WOS) database by searching the key words "*technological gap" and "trade"; "technological level" and "trade"; "technological distance" and "trade"; "digital gap" and "trade";*

"digitalization level" and "trade"; "digital distance" and "trade"; "digital divide" and "trade"; "digitalization" and "trade"; "digital technology" and "trade". The publication period covers the period until October 04, 2025. The search field of *TOPIC* has been selected.

It is viewed that the concept of technological gap has still been examined since 1981, although a new concept of digital gap emerged after the 2000s. Moreover, the concept of technological level has been analyzed since the early 1990s, the number of studies has increased over time, and it still stays popular even today. Moreover, after 2019, the concept of digitalization level began to appear in literature. In addition to this, it was noticed that the concept of digital divide was also widely studied as of 2001. Contrary to this, a new concept, digital distance, was limitedly studied in 2024 and 2025.

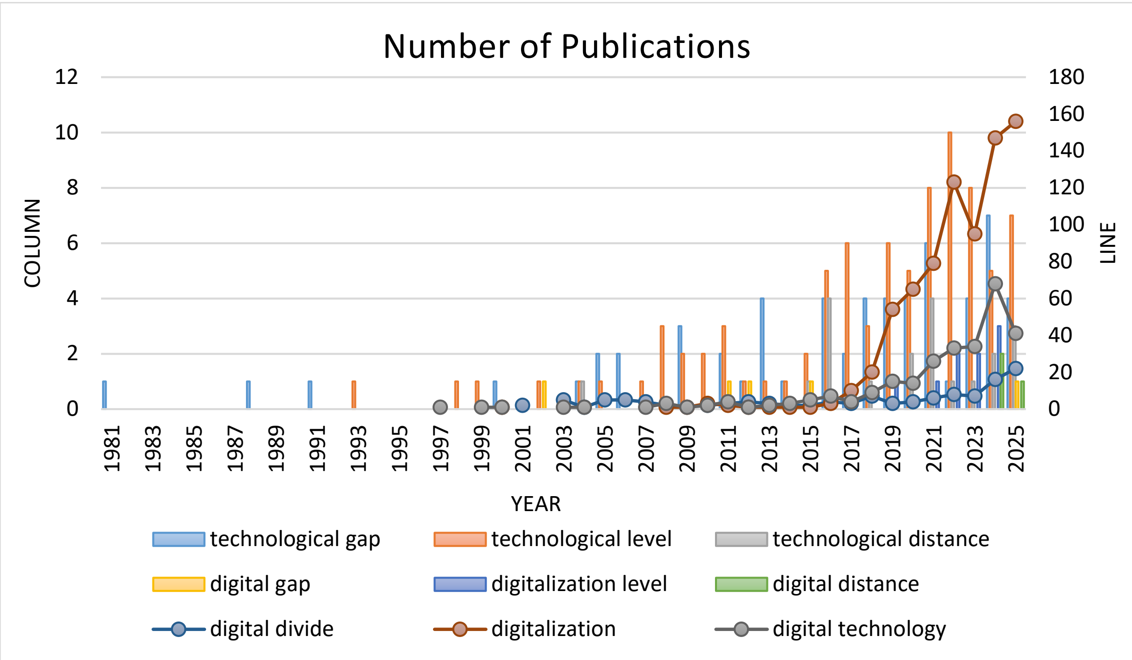

Figure 2. Number of publications related to concepts by year
Source: Prepared by authors based on data extracted from WOS database, 2025

Particularly, since the 2000s, digitalization and related concepts have been studied more frequently in literature and there has been a significant increase in the number of publications today. More than 750 publications related to digitalization and trade have been yielded since 2000. Findings show that publications on the related subjects have increased significantly since 2018, and their number nearly doubled after 2021.

1.2. Conceptual View

Posner's (1961) technology-gap framework suggests that when technological progress occurs unevenly across countries—affecting certain industries but not others—temporary advantages arise that can generate trade until innovations diffuse or are imitated abroad, thereby altering relative production costs. Building on this idea, Soete (1981) empirically evaluates the technology-gap hypothesis and shows that innovation across a broad set of industries plays a decisive role in shaping international trade patterns. Antimiani and Costantini (2013) further demonstrate that the European Union's enlargement narrowed technological disparities between existing and new member states, with particularly strong export gains for newcomers and technology-intensive sectors. Cermeño and Vazquez (2009) report that the agricultural technology gap between developed and developing economies widened from 1961 to 1991, noting that technological capabilities are closely linked to human capital, R&D activity, openness, and investment levels, even if this relationship does not always hold for every individual group. Hawash and Lang (2020) examine ICT adoption and its influence on total factor productivity, finding that although digital disparities matter, reducing them alone is insufficient to generate substantial productivity improvements. Dung (2024) argues that trade liberalization can actively reshape the technological gap in developing economies by allowing them to import intermediate goods from technologically advanced partners. Additional empirical work measures technological distance at the firm level (Bar & Leiponen, 2012; Gilsing et al., 2008; Patel & King, 2016), within industries (Vom Stein et al., 2015), and across countries (Fukuchi and Satoh, 1999). Nonetheless, an unresolved question in the literature concerns whether cross-country technological distance ultimately enhances or restricts bilateral trade flows (Esfahani & Tayebi, 2016).

Recent research consistently shows that technological advancement enhances export performance at both the firm and country levels (Ortigueira-Sánchez et al., 2022; Hong and Nam, 2021; Lopez et al., 2022; Marquez-Ramos and Martínez-Zarzoso, 2010; Ozsoy et al., 2022; Asghar et al., 2024). Numerous firm-level studies focus specifically on digital transformation. Denicolai et al. (2021) report that SMEs' international outcomes improve with stronger artificial intelligence capabilities, noting that sustainability and digitalization are mutually reinforcing. Abendin et al. (2022) demonstrate that digitalization significantly strengthens bilateral trade within the Economic Community of West African States. Likewise, Cao and Hsu (2025) use the

Digital Economy and Society Index to show that digitalization mitigates the adverse effects of country differences, helping firms overcome administrative, linguistic, and economic barriers that constrain exports. In contrast, Wang et al. (2024) examine the geographic patterns of emerging digital cross-border mergers and acquisitions and find that greater digital distance between countries can discourage such transactions.

In sum, the studies conducted so far have addressed the concept of technological distance in a narrower scope compared to the concept of digitalization, and the concept of digital distance has been rarely studied with different perspectives. This study differs from the above-mentioned by developing an index to measure how countries' digitalization levels affect bilateral export activities and by proposing a new term, *"digital distance".*

2. Data and Methodology

2.1. Data Specification

Contemporary scholarship underscores that global trade is shaped by multiple determinants, including technological progress (Manickam et al., 2021), economic complexity (Dar et al., 2020; Ajide et al., 2025), exchange-rate movements (Mehtiyev et al., 2021; Kang and Dagli, 2018; Nguse et al., 2021; Asteriou, 2016), and macroeconomic expansion (Nguyen, 2020; Ji et al., 2022; Akhter et al., 2022). In this context, the present study evaluates whether digitalization gaps between Turkey and its trade partners meaningfully influence trade performance, given the centrality of technological capability in advancing sustainable economic development. Focusing on export behavior, the analysis extends recent contributions on technology–trade linkages (Dung, 2024; Cao & Hsu, 2025). The empirical framework employs annual data for 2008–2022 from four databases, incorporating Turkey's bilateral exports, GDP indicators for all countries, exchange rates, the frontier technological readiness index (FTRI), and the economic complexity index (ECI). Table 1 provides variable definitions and econometric specifications.

Table 1
Description and measurement of variables

Variables	Definitions and Measures	Data Source
$Export_{tr}$	Exports of Turkey	Turkey Statistical Institute
GDP_{tr}	Gross Domestic Product of Turkey (Constant 2015 $)	World Bank
GDP_{k}	Gross Domestic Product of Partner Countries (Constant 2015 $)	World Bank
$Exchange_{tr}$	Official Exchange Rate of Turkey (LCU per US$, period average)	World Bank
$Exchange_{k}$	Official exchange rate of Partner Countries (LCU per US$, period average)	World Bank
$DigiD$	Difference of Turkey's FTRI from Partner Country's FTRI	UNCTAD
$DigiD_{ict}$	Difference of Turkey's FTRI sub-index ICT from Partner Country's FTRI sub-index ICT	UNCTAD
$DigiD_{skills}$	Difference of Turkey's FTRI sub-index Skills from Partner Country's FTRI sub-index Skills	UNCTAD
$DigiD_{rd}$	Difference of Turkey's FTRI sub-index Research and Development from Partner Country's FTRI sub-index Research and Development	UNCTAD
$DigiD_{ia}$	Difference of Turkey's FTRI sub-index Industry Activity from Partner Country's FTRI sub-index Industry Activity	UNCTAD
$DigiD_{af}$	Difference of Turkey's FTRI sub-index Access to Finance from Partner Country's FTRI sub-index Access to Finance	UNCTAD
$EciD$	Difference of Turkey's ECI from Partner Country's ECI	The Observatory of Economic Complexity

This study provides valuable insights for policymakers by examining how Turkey’s digital distance with its trading partners influences its export performance. The selection of countries includes both developing and developed economies to which Turkey exports, ensuring data availability for the specified period. The primary objective is to assess the impact of digital distance on Turkey’s export dynamics, shedding light on its implications for trade competitiveness and economic policy.

2.2. Empirical Model

This research investigates how the digital gap between Turkey and its trade partners functions as a central factor shaping Turkey's export performance. The empirical model includes Turkey's GDP, partner-country GDP, and exchange rates as control variables. While Turkey's GDP reflects domestic production capacity, partner-country GDP captures external demand, and exchange-rate movements jointly influence relative export prices. Although Ozsoy et al. (2022) emphasize that national digitalization significantly explains high-technology export flows, no prior study has incorporated digital distance as an explicit predictor within a comparable econometric setting. Here, digital distance represents the difference in digital advancement between Turkey and its partners; positive values denote Turkey's superior digital capabilities, whereas negative values indicate lagging readiness. Equation 1 includes GDP, exchange-rate, and digital distance variables, and Equations 2–6 extend the model with the UNCTAD (2025) FTRI sub-indices—ICT deployment, skills, R&D activity, industrial capacity, and financial access—alongside the Economic Complexity Index (ECI) to strengthen empirical robustness (Harvard Growth Lab, 2025).

Utilizing these variables, we develop an econometric framework based on equations (1,2,3,4,5,6) to examine the impact of digital distance on export performance:

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 DigiD_{i,t} + \varepsilon \quad (1)$$

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 DigiD_{ict_{i,t}} + \varepsilon \quad (2)$$

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 DigiD_{skills_{i,t}} + \varepsilon \quad (3)$$

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 DigiD_{rd_{i,t}} + \varepsilon \quad (4)$$

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 DigiD_{ia_{i,t}} + \varepsilon \quad (5)$$

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 DigiD_{af_{i,t}} + \varepsilon \quad (6)$$

Here, *i* and *t* represent country and time, respectively. The stochastic error term is denoted by ε, while the parameters β_1, β_2, β_3, β_4, and β_5 capture the long-run elasticities of the variables in the equations. Additionally, the model (7) used for robustness checks is presented below.

$$lnExport_{tr_{i,t}} = \beta_0 + \beta_1 lnGDP_{tr_{i,t}} + \beta_2 lnGDP_{k_{i,t}} + \beta_3 lnExchange_{tr_{i,t}} + \beta_4 lnExchange_{k_{i,t}} + \beta_5 EciD_{i,t} + \varepsilon \qquad (7)$$

2.3. Panel Threshold Regression

In threshold analysis, selected threshold variables are integrated into the regression framework, based on the premise that explanatory variables affect the outcome variable in a nonlinear manner. Threshold models use a threshold indicator to divide observations into relatively homogeneous regimes, allowing the relationship between variables to shift as the threshold level changes (Li et al., 2020).

Hansen (1999) proposed an estimation technique for balanced panel datasets that incorporates individual effects and identifies structural breaks, enabling the determination of the true threshold value. This approach employs asymptotic distribution and relies on bootstrap procedures to improve inference accuracy. In this study, a panel threshold regression model following Hansen's methodology is applied to assess the nonlinear influence of digital distance on Turkey's export performance (12):

$$e_{it} = \{\alpha_i + \delta' x_{it} + \beta_1 d_{it} + \varepsilon_{it} \quad if \quad d_{it} \leq \gamma \; \alpha_i + \delta' x_{it} + \beta_2 d_{it} + \varepsilon_{it} \quad if \quad d_{it} > \gamma \qquad (12)$$

$$\delta = (\delta_1, \delta_2, \delta_3)', \quad x_{it} = (k_{it}, l_{it}, m_{it}, n_{it})'$$

Here, e_{it} denotes Turkey's exports to its trading partners, serving as the dependent variable. The explanatory variable and threshold variable, d_{it}, is represented by DigiD, while γ is the estimated threshold value. In order to control for additional factors that may systematically influence Turkey's exports, four control variables, x_{it}, are included: k_{it} (Turkey's GDP), l_{it} (trading partners' GDP), m_{it} (Turkey's exchange rate), and n_{it} (trading partners' exchange rate). The fixed effect, α_i, accounts for cross-country heterogeneity arising from differences in digitalization levels. The error term, ε_{it}, is assumed to be independently and identically distributed with a mean of zero and finite variance, $\sigma^2(\varepsilon_{it} \sim iid(0, \sigma^2))$. Here, *i* represents trading partner countries, and *t* corresponds to different time periods.

The refined threshold regression model (1) can be reformulated as follows (13):

$$e_{it} = \mu_{it} + \delta' x_{it} + \beta_1 d_{it} I(d_{it} \leq \gamma) + \beta_2 d_{it} I(d_{it} > \gamma) + \varepsilon_{it} \qquad (13)$$

Here, I $I(\blacksquare)$ denotes the indicator function. Equation (13) can alternatively be expressed as follows (14):

$$e_{it} = u_i + \delta' x_{it} + \beta' d_{it}(\gamma) + \varepsilon_{it}, \qquad \beta = (\beta_1, \beta_2)' \tag{14}$$

$$e_{it} = \mu_{it} + [\delta', \beta'] \left[\frac{x_{it}}{d_{it}(\gamma)}\right] + \varepsilon_{it}$$

$$e_{it} = \mu_{it} + \omega' d_{it}(\gamma) + \varepsilon_{it}$$

$$d_{it}(\gamma) = \left[\frac{d_{it} I(d_{it} \leq \gamma)}{d_{it} I(d_{it} > \gamma)}\right]$$

Here, $\omega = (\delta', \beta')'$ and $d_{it} = (x_{it}', d_{it}'(\gamma))'$. The observations are categorized into two distinct regimes based on whether the threshold variable d_{it} is below or above the threshold value (γ). These regimes are characterized by different regression slopes, β_1 and β_2. The parameters $(\gamma, \beta, \delta\ and\ \sigma^2)$ are estimated using the known values of e_{it} and d_{it}.

2.4. Panel Threshold Regression

Identifying causal linkages among variables is vital for designing effective policies. To explore these linkages, the study applies the Dumitrescu–Hurlin (2012) panel causality test, which assumes that causal patterns evident in one country may extend to others, with reliability increasing as sample size grows. The associated linear model is shown in Equation 15:

$$Y_{it} = \beta_i + \sum_{j-1}^{J} \delta_i^j Y_{i(t-j)} + \sum_{j-1}^{J} \eta_i^j X_{i(t-j)} + \varepsilon_{it} \tag{15}$$

In this framework, δ_i^j and η_i^j represent the autoregressive coefficients and elasticity, respectively, with both assumed to exhibit variation across distinct cross-sectional units. The null and alternative hypotheses are defined as follows (16, 17):

$$h_0\colon \mu_i = 0 \quad \text{for } \forall \text{I} \tag{16}$$

$$h_1\colon (\mu_i = 0\ for\ all\ i = 1,2,3,4, \dots, N_1\ \mu_i \neq 0\ for\ all\ i = N_1 + 1,2,3,4, \dots, N\ ) \tag{17}$$

The hypotheses are tested using Wald statistics, which are computed as outlined in equation (18).

$$W_{N,T}^{HNC} = N^{-1} \sum_{i=1}^{N} W_{i,T} \tag{18}$$

Here, $W_{i,T}$ signifies the Wald test statistic, which is calculated for each individual unit *i*.

3. Empirical Findings

Table 2 summarizes the descriptive statistics for twelve variables. Mean values show central patterns, with average exports at 1367444063 and a digitalization score of 0.049. Standard deviations reveal substantial dispersion in exports, GDP measures, partner-country exchange rates, digital distance, and economic complexity. Jarque–Bera results confirm non-normal distributions ($p<0.05$), underscoring the need for robust or non-parametric analytical techniques.

Table 2
Preliminary statistics

Variables	Mean	Std. Deviation	Maximum	Minimum	Prob.
$Export_{tr}$	1367444063	2556329671	21141783053	154635	0.000
GDP_{tr}	854043829491	188716078094	1194401442556	566434966772	0.000
GDP_k	649573589601	2153876852198	21443388432051	99130304	0.000
$Exchange_{tr}$	4.281	3.939	16.549	1.302	0.000
$Exchange_k$	2316.219	23732.128	386444	0.214	0.000
$DigiD$	0.049	0.256	0.54	-0.51	0.000
$DigiD_{ict}$	0.017	0.255	0.56	-0.62	0.000
$DigiD_{skills}$	0.022	0.234	0.55	-0.54	0.000
$DigiD_{rd}$	0.168	0.241	0.57	-0.62	0.000
$DigiD_{ia}$	-0.002	0.187	0.58	-0.48	0.000
$DigiD_{af}$	0.044	0.161	0.63	-0.29	0.000
$EciD$	0.379	0.958	2.899	-1.927	0.000

Table 3 reports correlations between economic variables and digital distance, showing mixed directional links. Turkey's exports and partner-country GDP display weak to moderate negative associations with digital distance, while Turkey's GDP and both exchange-rate measures exhibit positive correlations, suggesting these indicators rise as digital disparities between Turkey and its partners increase.

Table 3
Correlation matrix

Variables	$Export_{tr}$	GDP_{tr}	GDP_k	$Exchange_{tr}$	$Exchange_k$	$DigiD$	$DigiD_{ict}$	$DigiD_{skills}$	$DigiD_{rd}$	$DigiD_{ia}$	$DigiD_{af}$	$EciD$
$Export_{tr}$	1											
GDP_{tr}	0.119	1										
GDP_k	0.415	0.035	1									
$Exchange_{tr}$	0.115	0.835	0.029	1								
$Exchange_k$	-0.034	0.028	-0.023	-0.003	1							
$DigiD$	-0.393	0.087	-0.351	0.059	0.102	1						
$DigiD_{ict}$	-0.322	-0.112	-0.185	-0.034	0.072	0.861	1					
$DigiD_{skills}$	-0.326	0.193	-0.186	0.125	0.115	0.902	0.749	1				
$DigiD_{rd}$	-0.512	0.082	-0.551	0.071	0.069	0.872	0.654	0.699	1			
$DigiD_{ia}$	-0.227	0.022	-0.294	-0.006	0.079	0.799	0.623	0.603	0.681	1		
$DigiD_{af}$	-0.181	0.256	-0.305	0.169	0.096	0.765	0.589	0.653	0.641	0.601	1	
$EciD$	-0.359	0.086	-0.322	0.079	0.069	0.898	0.777	0.779	0.793	0.812	0.672	1

Selecting a suitable estimator is essential for managing cross-sectional dependence and slope heterogeneity in panel datasets, particularly given the interconnected nature of global markets. To evaluate these issues, the Breusch–Pagan LM (Breusch & Pagan, 1980), Pesaran scaled LM, and Pesaran CD tests (Pesaran, 2004) were applied, all indicating significant dependence across countries. Swamy's (1970) and Pesaran–Yamagata's homogeneity tests (2008) further reveal notable slope differences, demonstrating that country-specific dynamics are present and must be incorporated through methods accommodating structural heterogeneity.

Table 4
Cross-sectional dependence and homogeneity tests

Variables	CD_{BP}	CD_{LM}	CD	$\tilde{\Delta}$	$\tilde{\Delta}_{adj}$
Model (1)	15582.561***	91.206***	23.023***	14.147***	19.372***
Model (2)	14521.675***	81.337***	48.664***	14.146***	19.37***
Model (3)	16111.104***	96.123***	23.006***	14.083***	19.284***
Model (4)	15835.426***	93.559***	26.931***	13.917***	19.057***
Model (5)	16877.234***	103.249***	33.819***	14.849***	20.333***
Model (6)	16805.562***	102.583***	34.338***	14.358***	19.661***
Model (7)	17226.587***	106.499***	34.239***	14.528***	19.893***

Note: ***, ** and * indicate significance at 1%, 5% and 10% levels, respectively.

Table 5 reports first-generation unit root results from Im et al. (2003) and Levin et al. (2002), along with second-generation tests by Pesaran (2007). These procedures assess stationarity while accounting for cross-sectional features. All variables are found to be level-stationary, supporting the suitability of panel threshold estimation for subsequent empirical analysis.

Table 5
First- and Second-generation unit root test results

Variables	First-Generation Unit Root Test Results				Second-Generation Unit Root Test Results	
	Im, Pesaran, and Shin		**Levin, Lin and Chu**		**PESCADF**	
	Intercept	Intercept and Trend	Intercept	Intercept and Trend	Intercept	Intercept and Trend
$Export_{tr}$	-5.342***	-10.438***	-3.635***	-11.474***	-2.226***	-2.871***
GDP_{tr}	1.861**	-9.979***	17.071***	-12.353***	-2.612***	-2.853***
GDP_{k}	-2.691***	-7.324***	-5.172***	-5.893***	-1.925***	-3.019***
$Exchange_{tr}$	-4.521***	-8.242***	-2.729***	-4.185***	-2.235***	-2.692***
$Exchange_{k}$	-3.839***	-8.423***	-2.953***	-14.033***	-2.281***	-2.457**
$DigiD$	-4.086***	12.962***	-8.571***	-6.714***	-2.293***	-3.085***
$DigiD_{ict}$	-5.405***	-9.464***	-9.051***	-10.267***	-1.878**	-2.593***
$DigiD_{skills}$	-3.619***	-6.478***	-18.881***	-2.741***	-2.671***	-2.483**
$DigiD_{rd}$	-2.966***	-13.683***	-3.458***	-22.184***	-1.991***	-2.648***
$DigiD_{ia}$	-5.508***	-11.309***	-8.735***	-9.638***	-2.156***	-2.887***
$DigiD_{af}$	-3.899***	-7.917***	-16.005***	-8.895***	-2.108***	-2.507***
$EciD$	-3.008***	-4.196***	-3.326***	-12.888***	-1.876**	-2.529***

Note: ***, ** and * indicate significance at 1%, 5% and 10% levels, respectively.

Before estimating the panel threshold regression, the study first identified the appropriate number of thresholds. Using Hansen's (1999) panel threshold framework and a bootstrap procedure repeated 300 times, threshold values were estimated with digital distance serving as the threshold variable (Models 1–6). The results indicate a statistically significant single-threshold structure linking Turkey's exports with the GDP and exchange rates of Turkey and its trading partners, a pattern also confirmed in the robustness specification (Model 7). As shown in Table 5, all single-threshold models are significant at the 1% or 5% levels, verifying the presence of threshold behavior. The estimated threshold values—30.56, 17.16, 88.25, 39.41, 36.57, 38.71, and 39.83—demonstrate that digital distance generates asymmetric effects on export performance depending on its position relative to the threshold, as summarized in Table 6.

Table 6
Threshold existence tests

Variables	Threshold		F Stat.	Prob.	Critical Values		
					%10	%5	%1
Model (1)	Single	0.23	30.56**	0.026	21.503	23.923	33.339
	Double	0.36	13.22	0.331	21.757	26.361	34.603
Model (2)	Single	0.35	17.16**	0.031	14.602	16.852	22.629
	Double	0.31	7.65	0.422	13.692	18.148	22.501
Model (3)	Single	0.15	88.25***	0.000	25.878	30.631	46.067
	Double	0.21	23.81	0.113	25.311	30.281	34.493
Model (4)	Single	-0.07	39.41***	0.010	20.938	26.951	38.298
	Double	0.46	20.82	0.125	21.133	26.683	30.224
Model (5)	Single	0.27	36.57**	0.020	23.681	29.518	37.856
	Double	0.15	11.47	0.164	19.863	21.491	27.089
Model (6)	Single	-0.09	38.71***	0.003	24.051	32.507	38.292
	Double	-0.15	18.74	0.153	27.821	38.079	48.176
Model (7)	Single	0.69	39.83***	0.010	27.042	30.881	39.283
	Double	1.42	14.85	0.487	30.068	33.235	50.905

Note: ***, ** and * indicate significance at 1%, 5% and 10% levels, respectively.

The likelihood ratio function graphs in Figure 3 depict the process of identifying threshold values and constructing confidence intervals. The red dotted lines indicate the approximate 95% confidence level. Overall, the findings confirm the existence of a nonlinear threshold effect when digital distance serves as the threshold variable for the GDP and exchange rates of Turkey and its trading partners (See Figure 3).

To facilitate a comparative analysis within the panel threshold model, this study examines the impact of Turkey and its trading partners' GDP and exchange rates on Turkey's exports, considering the digital distance threshold. Table 7 presents the estimation results of the panel threshold regression. The findings reveal significant threshold effects of GDP and exchange rates on Turkey's exports, which vary across different levels of digital distance. The detailed results are as follows.

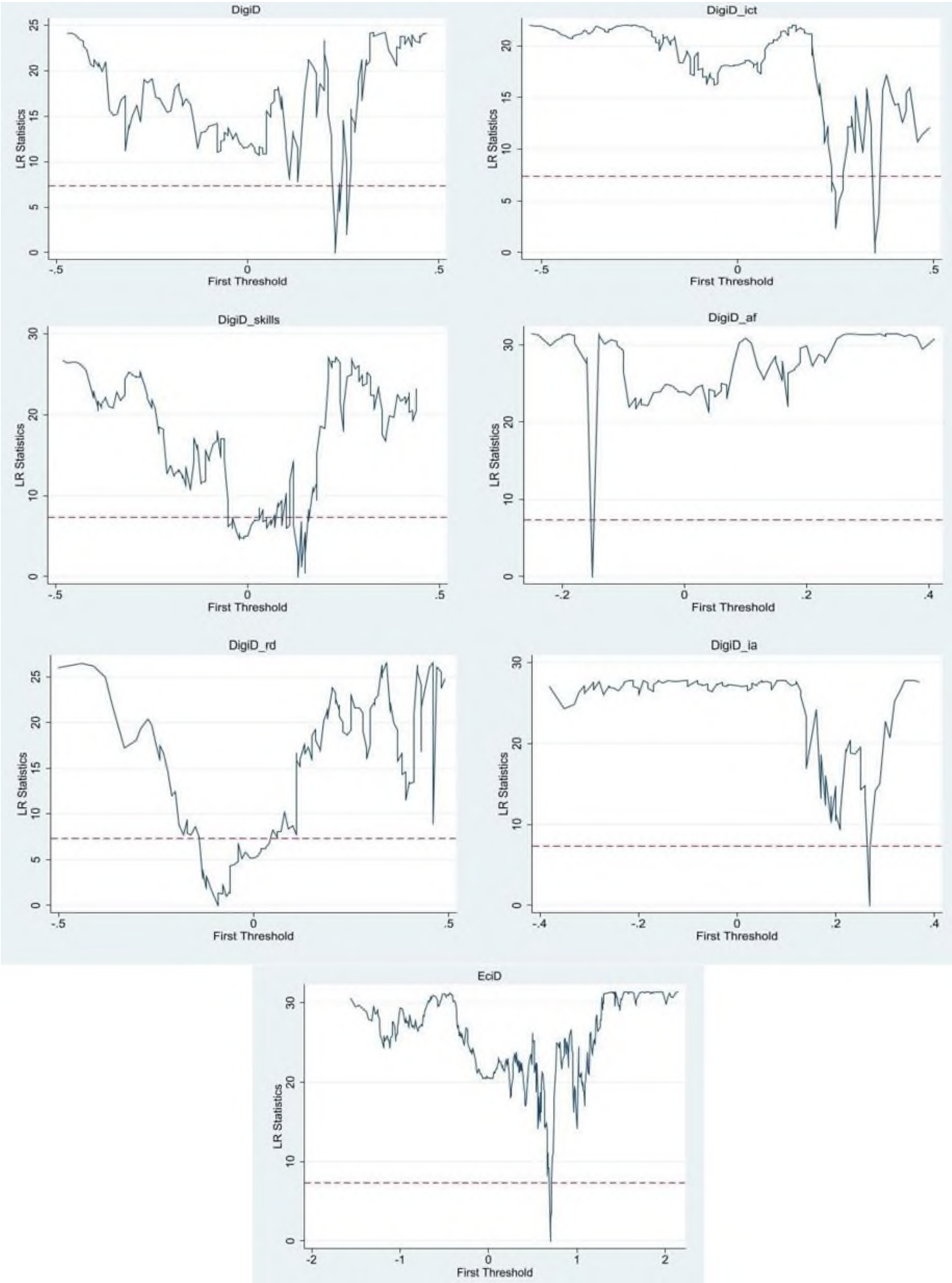

Figure 3. The Likelihood Ratio function graph of modes

Table 7
Estimation results of Threshold models

	Variables	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Model (6)
Control Variables	GDP_{tr}	1.329***	1.323***	1.263***	1.373***	1.247***	1.651***
	GDP_k	0.221***	0.229***	0.165***	0.224***	0.252***	0.219***
	$Exchange_{tr}$	0.069**	0.058	0.081**	0.065*	0.069**	0.015
	$Exchange_k$	-0.105***	-0.091***	-0.104***	-0.118***	-0.095***	-0.115***
	Constant	-22.364***	-22.372***	-19.248***	-23.744***	-20.829***	-31.034***
Threshold Variables	$DigiD$	-0.808*** (π<0.23)	-	-	-	-	-
		0.316 (0.23<π)	-	-	-	-	-
	$DigiD_{ict}$	-	0.112 (π<0.35)	-	-	-	-
		-	-0.277*** (0.35<π)	-	-	-	-
	$DigiD_{skills}$	-	-	-0.676** (π<0.15)	-	-	-
		-	-	1.642*** (0.15<π)	-	-	-
	$DigiD_{rd}$	-	-	-	-1.483*** (π<-0.07)	-	-
		-	-	-	0.852*** (-0.07<π)	-	-
	$DigiD_{ia}$	-	-	-	-	0.898*** (π<0.27)	-
		-	-	-	-	0.223 (0.27<π)	-
	$DigiD_{af}$	-	-	-	-	-	-1.917*** (π<-0.09)
		-	-	-	-	-	-0.454* (-0.09<π)
	R^2	0.49	0.48	0.51	0.49	0.48	0.49
	F-test	201.26***	198.77***	205.36***	184.75***	202.23***	211.36***
	Observations	1620	1620	1620	1620	1620	1620

Note: ***, ** and * indicate significance at 1%, 5% and 10% levels, respectively.

In Model 1, a single cutoff value separates digital distance into two intervals. When $DigiD$ is below 0.23, the coefficient of –0.808 is significant at the 1% level, indicating a strong positive association: decreases in digital distance correspond to reduced Turkish exports. Above the 0.23 threshold, the link becomes statistically insignificant, suggesting that digital distance influences Turkey's export performance only up to this cutoff, beyond which the effect diminishes.

In Models 2 through 6, the threshold indicators are linked to the sub-components of digital distance—ICT deployment, Skills, R&D activity, Industry activity, and Access to finance. Each specification identifies a single statistically valid cutoff point. The estimated thresholds—0.35, 0.15, –0.07, 0.27, and –0.09 for Models 2 to 6—are significant at either the 1% or 5% levels. In Model 2, ICT deployment ($DigiD_{ict}$) shows no meaningful effect on exports when its value is below 0.35; once this level is exceeded, the coefficient becomes –0.277 and is highly significant, indicating that a widening ICT gap reduces Turkey's export flows. Model 3 demonstrates that when Skills ($DigiD_{skills}$) falls below 0.15, the coefficient is –0.676 and significant at the 5% level. Beyond the threshold, however, the coefficient rises to 1.642 with strong significance, showing that larger disparities in Skills are associated with higher exports. Model 4 yields a similar pattern: when R&D activity ($DigiD_{rd}$) is below –0.07, the coefficient is –1.483, but it switches to 0.852 above the threshold. These outcomes imply that greater distance in Skills and R&D may support Turkey's export performance. In Model 5, lower Industry activity distance ($DigiD_{ia}$) below 0.27 increases exports, while no significant relationship appears above this level. Finally, Model 6 indicates that greater distance in Access to finance ($DigiD_{af}$) consistently depresses exports, though the adverse effect diminishes once the threshold is crossed.

Table 8
Robustness of threshold model

Variables		Model (7)
Control Variables	GDP_{tr}	1.324***
	GDP_k	0. 202***
	$Exchange_{tr}$	0.062*
	$Exchange_k$	-0.103***
	Constant	- 21.791***
Threshold Variables	$EciD$	-0.293*** (π<0.69)
		0.168 (0.69<π)
	R^2	0.49
	F-test	206.01***
	Observations	1620

Note: ***, ** and * indicate significance at 1%, 5% and 10% levels, respectively.

Table 8 provides a robustness assessment using a panel threshold model that incorporates the robustness variable. The estimated cutoff for the economic complexity index distance ($EciD$) is 0.69. Below this level, the coefficient is –0.293 and highly significant, while no significant effect appears above it. This implies that narrowing the economic complexity gap reduces Turkey's exports. The pattern mirrors the influence of digital distance, supporting the core model's conclusions.

Table 9
Dumitrescu & Hurlin panel causality test results

Null Hypothesis (h_0)	$Z^{HNC}_{N,T}$	Z^{HNC}_{N}	Direction of Causality
$Export_{tr}$ does not Granger-cause GDP_{tr}	4.526***	8.245***	Bi-directional
GDP_{tr} does not Granger-cause $Export_{tr}$	33.282***	50.253***	
$Export_{tr}$ does not Granger-cause GDP_k	2.284***	4.971***	Bi-directional
GDP_k does not Granger-cause $Export_{tr}$	14.728***	23.148***	
$Export_{tr}$ does not Granger-cause $Exchange_{tr}$	4.762***	8.591***	Bi-directional
$Exchange_{tr}$ does not Granger-cause $Export_{tr}$	32.108***	48.537***	
$Export_{tr}$ does not Granger-cause $Exchange_k$	0.875	0.932	Neutrality
$Exchange_k$ does not Granger-cause $Export_{tr}$	0.282	0.621	
$Export_{tr}$ does not Granger-cause $DigiD$	0.793	2.792***	Bi-directional
$DigiD$ does not Granger-cause $Export_{tr}$	2.511***	5.301***	
$Export_{tr}$ does not Granger-cause $DigiD_{ict}$	6.844***	11.632***	Bi-directional
$DigiD_{ict}$ does not Granger-cause $Export_{tr}$	1.316	3.556***	
$Export_{tr}$ does not Granger-cause $DigiD_{skills}$	2.234**	4.897***	Bi-directional
$DigiD_{skills}$ does not Granger-cause $Export_{tr}$	10.978***	17.671***	
$Export_{tr}$ does not Granger-cause $DigiD_{rd}$	12.941***	20.537***	Bi-directional
$DigiD_{rd}$ does not Granger-cause $Export_{tr}$	2.711***	5.594***	
$Export_{tr}$ does not Granger-cause $DigiD_{ia}$	5.288***	13.436***	Bi-directional
$DigiD_{ia}$ does not Granger-cause $Export_{tr}$	2.317***	7.833***	
$Export_{tr}$ does not Granger-cause $DigiD_{af}$	3.273***	6.414***	Bi-directional
$DigiD_{af}$ does not Granger-cause $Export_{tr}$	10.425***	16.863***	
$Export_{tr}$ does not Granger-cause $EciD$	5.277***	9.342***	Bi-directional
$EciD$ does not Granger-cause $Export_{tr}$	6.729***	11.462***	

Note: ***, ** and * indicate significance at 1%, 5% and 10% levels, respectively.

Table 9 reports the outcomes of the Dumitrescu–Hurlin heterogeneous panel causality test, highlighting notable causal interactions among the examined variables. The results show robust two-way causality between Turkey's export performance and nearly all variables, except the partner country's exchange rate. ICT investment, a key dimension of digitalization, exhibits reciprocal causality with exports, aligning with Goldar (2023). Likewise, R&D investment demonstrates mutual causality with exports, supporting evidence from Gocer (2013) and Altiner et al. (2022). A similar bidirectional link emerges between access to finance and exports, consistent with Tran et al. (2024). Overall, the findings underscore how economic and digital progress both shape and respond to Turkey's export activity.

4. Conclusion and Policy Recommendations

Extensive research has explored how distance shapes international trade. As technology accelerates digitalization, differences in countries' digital capacities have introduced a new form of separation: digital distance. This shift makes examining its influence on trade increasingly important. Traditional trade models emphasize geographic and cultural barriers but largely ignore digital gaps, despite substantial evidence on the trade effects of technological progress. Incorporating digital distance can therefore fill a notable void in existing trade theory.

The empirical findings of this study demonstrate that digital distance has become a critical determinant of Turkey's export performance, transcending the traditional notions of geographical and cultural distance emphasized in the gravity model of international trade (Huang, 2007; Obstfeld & Rogoff, 2000; Rangan & Lawrence, 1999). While geographical distance remains relevant, our analysis reveals that digital disparities—measured through ICT infrastructure, skills, R&D capacity, industrial application, and access to finance—play a far more complex role in shaping bilateral trade dynamics. This aligns with recent research emphasizing that technological differences significantly influence trade flows between countries (Posner, 1961; Soete, 1981; Esfahani & Tayebi, 2016).

An important insight from the analysis is that narrowing digital distance does not consistently enhance Turkey's export performance. Convergence in ICT infrastructure and R&D capabilities tends to weaken exports, while larger gaps in digital skills and R&D unexpectedly stimulate them. This pattern supports Dung's (2024) view that technological distance can have nonlinear effects, and aligns with Cao and Hsu (2025), who show that digitalization reshapes trade dynamics. Overall, digital distance creates asymmetrical, dimension-specific impacts.

Policy implications suggest Turkey should not pursue uniform digital convergence. Aligning with technological gap theory (Posner, 1961; Soete, 1981), maintaining selective digital advantages through frontier R&D and skill development can sustain temporary export benefits, rather than eliminating all digital disparities with trading partners.

Second, the finding that gaps in ICT capacity and financial access hinder exports highlights the need for stronger digital infrastructure and inclusive financing (Goldar, 2023; Manickam et al., 2021). Strengthening digital finance tools, supporting venture capital for tech-driven firms, and promoting international ICT partnerships would help Turkey improve export performance (Tran et al., 2024).

Third, the asymmetric role of digital skills and R&D capacity suggests that human capital formation and innovation ecosystems must remain central to Turkey’s trade strategy. As demonstrated by Altiner et al. (2022) and Gocer (2013), R&D expenditures directly boost high-tech exports, while Lopez et al. (2022) and Hong and Nam (2021) emphasize that firm-level innovations strengthen international competitiveness. Accordingly, Turkey should reinforce digital education, incentivize private-sector R&D investment, and create innovation clusters to sustain export dynamism.

Fourth, the findings highlight the need to embed digital distance considerations into Turkey’s broader trade policy and negotiation strategies. As Abendin et al. (2022) show for West Africa, digitalization significantly enhances bilateral trade, while Wang et al. (2024) caution that excessive digital distance can hinder cross-border M&As. Turkey must therefore negotiate digital trade agreements that reduce excessive asymmetries while safeguarding its comparative advantages in selected technologies.

Finally, given the nonlinear and threshold effects observed, trade policy must become more adaptive. Policymakers should employ continuous monitoring of digital distance indicators such as UNCTAD’s Frontier Technological Readiness Index and Harvard’s Economic Complexity Index to design dynamic interventions. This echoes the growing literature advocating for data-driven, adaptive policy frameworks in an era of technological disruption (Ajide et al., 2025; Ji et al., 2022).

In conclusion, the threshold-based analyses presented here offer a basis for future research to implement econometric approaches that mitigate endogeneity concerns, particularly by incorporating lagged values of the dependent variable through dynamic panel estimators such as the system generalized method of moments (system-GMM) and comparable methodologies.

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